

HEARTS & HYPERTENSION · COMBINED REFERENCE

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- HEARTS Technical Package for Cardiovascular Disease Management in Primary Health Care
- HEARTS Implementation Guide
- Global Report on Hypertension: The Race Against a Silent Killer (WHO, 2023)



Technical package for cardiovascular disease management in primary health care



Healthy-lifestyle counselling



WHO/NMH/NVI/18.1

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HEARTS Technical Package

More people die each year from cardiovascular diseases (CVDs) than from any other cause. Over three-quarters of heart disease and stroke-related deaths occur in low- and middle-income countries.

The HEARTS technical package provides a strategic approach to improving cardiovascular health. It comprises six modules and an implementation guide. This package supports Ministries of Health to strengthen CVD management in primary care and aligns with WHO's Package of Essential Noncommunicable Disease Interventions (WHO PEN).

HEARTS modules are intended for use by policymakers and programme managers at different levels within Ministries of Health who can influence CVD primary care delivery. Different sections of each module are aimed at different levels of the health system and different cadres of workers. All modules will require adaptation at country level.

The people who will find the modules most useful are:

- **National level – Ministry of Health NCD policymakers responsible for:**
 - developing strategies, policies and plans related to service delivery of CVD
 - setting national targets on CVD, monitoring progress and reporting.
- **Subnational level – Health/NCD programme managers responsible for:**
 - planning, training, implementing and monitoring service delivery
- **Primary care level – Facility managers and primary health care trainers responsible for:**
 - assigning tasks, organising training and ensuring the facility is running smoothly
 - collecting facility-level data on indicators of progress towards CVD targets.

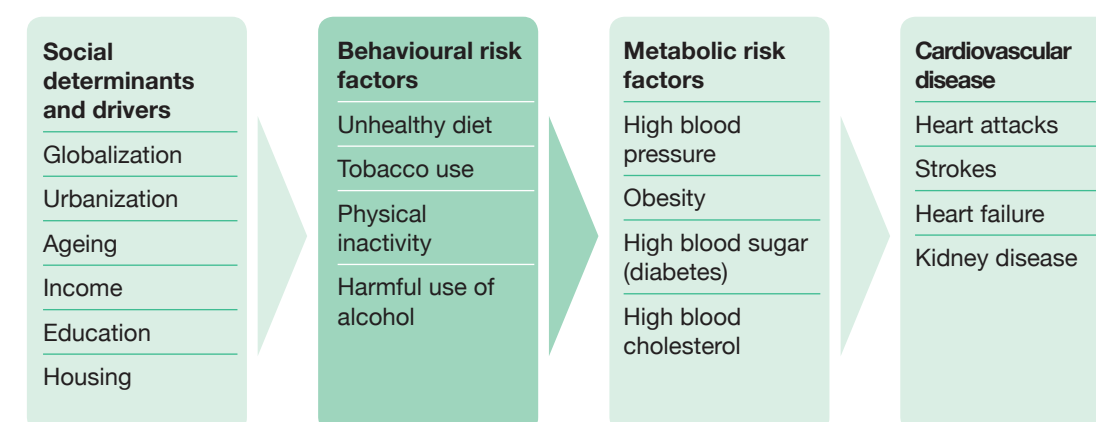
Target users may vary, based on context, existing health systems and national priorities.

MODULES OF THE HEARTS TECHNICAL PACKAGE				
Module	What does it include?	Who are the target users?		
		National	Subnational	Primary care
H ealthy-lifestyle counselling	Information on the four behavioural risk factors for CVD is provided. Brief interventions are described as an approach to providing counselling on risk factors and encouraging people to have healthy lifestyles.		✓	✓
E vidence-based protocols	A collection of protocols to standardize a clinical approach to the management of hypertension and diabetes.	✓	✓	✓
A ccess to essential medicines and technology	Information on CVD medicine and technology procurement, quantification, distribution, management and handling of supplies at facility level.	✓	✓	✓
R isk-based CVD management	Information on a total risk approach to the assessment and management of CVD, including country-specific risk charts.		✓	✓
T eam-based care	Guidance and examples on team-based care and task shifting related to the care of CVD. Some training materials are also provided.		✓	✓
S ystems for monitoring	Information on how to monitor and report on the prevention and management of CVD. Contains standardized indicators and data-collection tools.	✓	✓	✓

Introduction

The risk factors for CVD include behavioural factors, such as tobacco use, an unhealthy diet, harmful use of alcohol and inadequate physical activity, and physiological (metabolic) factors, including high blood pressure (hypertension), high blood cholesterol and high blood sugar or glucose. Both kinds of factor are linked to underlying social determinants and drivers (Fig. 1). (1)

Figure 1: Factors contributing to the development of cardiovascular disease and related complications



This module focuses on the behavioural risk factors and provides information on:

- tobacco use, unhealthy diet, insufficient physical activity and harmful use of alcohol as important contributors to CVDs
- behavioural change, brief interventions for counselling and key points for motivational interviewing
- the theory of the 5As for brief interventions, as well as sample brief interventions for each risk factor, using the 5As.

Target users of this module

This module is intended for trainers of primary health care workers, including physicians, nurses, and other health workers.

Primary care workers should be trained on the risk factors and counselling approaches, adapting to local customs, culture and context.

NCD programme managers may also use it for planning purposes.

1 The four main behavioural risk factors for CVD

Unhealthy diet

The risk of CVD is increased by an unhealthy diet, characterized by a low intake of fruit and vegetables and high intakes of salt, fats (described in further detail in Box 1) and sugars. An unhealthy diet contributes to obesity and overweight, which are also risk factors for CVD. Unhealthy eating habits can also result in a range of other conditions, including cancer, diabetes and micronutrient deficiencies. (2)

What is a healthy diet?

A healthy diet is important to address hypertension, diabetes and obesity and help prevent heart attack and stroke. The composition of a healthy diet depends on individual needs (for example, age, sex, lifestyle, degree of physical activity), cultural context and locally available foods. However, the basic elements of a healthy adult diet remain constant (2):

a) a variety of foods

- o a range of fruits, vegetables, legumes (for example, lentils, beans) and nuts
- o whole grains such as unprocessed maize, millet, oats, wheat and brown rice, and starchy tubers or roots such as potato, yam, taro or cassava
- o foods from animal sources (for example, meat, fish, eggs and milk).

b) at least 400 g (five portions) of vegetables and fruits per day

- o one portion is equivalent, for example, to a single orange, apple, mango, banana, or 3 tablespoons of cooked vegetables. (Potatoes, sweet potatoes, cassava or other starchy tubers or roots do not count as one of these portions.)

c) less than 5 g of salt per day (equivalent to approximately 1 level teaspoon) (3)

- o including salt added while cooking or eating, as well as salt contained in foods such as processed foods and bread.

d) a total daily energy intake from fats of less than 30%

- o unsaturated fats are preferable to saturated fats (Box 1)
- o less than 10% of total energy intake should be from saturated fats

Box 1: What are saturated fats, trans-fats and unsaturated fats?

Saturated fats are found mainly in animal products such as meat, milk, butter, cream, cheese, ghee and lard. They can also be found in palm and coconut oil. Many saturated fats are solid, such as the fat in meat. Consuming saturated fats in unhealthy amounts can lead to raised cholesterol levels and can increase the risk for heart attack and stroke.

Trans-fats (also called hydrogenated or partially hydrogenated vegetable oils) are liquid vegetable oils that have been processed to make them solid. This makes them unhealthy. Trans-fats are often found in processed food, fast food, snacks, fried food, frozen pizza, pies, cookies, margarines and spreads.

Unsaturated fats/oils are generally found in plant foods such as seeds, grains, nuts, vegetables (for example, avocado) and fruit, and also in fish. They can be either polyunsaturated (as in sunflower, soya, corn, and sesame oils), or monounsaturated (for example, olive and rapeseed oils). Consuming unsaturated fats/oils instead of saturated fats can help to control cholesterol levels and reduce the risk of heart attack and stroke.

- o trans-fats are not part of a healthy diet and should be avoided.

e) total daily energy intake from free sugars of less than 10% (Box 2) (4)

- o equivalent to 50 g (or approximately 12 level teaspoons) for a person of healthy body weight
- o for additional health benefits, less than 5% of total energy intake from free sugars is ideal.

Box 2: What are free sugars?

Free sugars are those added to foods such as cakes, cookies and sweets or drinks (for example, soda, sweetened milk, fruit juices). Free sugars are also naturally present in honey, syrups, fruit juices and fruit juice concentrates.

Insufficient physical activity

Insufficient physical activity is one of the leading risk factors for premature death worldwide. People who are insufficiently active are between 20% and 30% more likely to die prematurely than those who are sufficiently active. Insufficient physical activity is a key risk factor for CVD, cancer and diabetes.

What is physical activity?

WHO defines physical activity as any movement produced by the skeletal muscles of the human body that uses energy. It covers a range of bodily movements and activities of daily life, such as playing, working, walking, household chores and recreational activities. Physical activity includes exercise, a subcategory of physical activity that is planned, structured and repetitive, with the objective of improving or maintaining physical fitness. (5)

What are the benefits of physical activity?

Regular and adequate levels of physical activity help to reduce the risk of hypertension, heart attack, stroke, diabetes, various types of cancer (including breast and colon cancer) and depression. Physical activity also contributes to weight control, diabetes control, improved blood pressure and improved levels of cholesterol and other blood lipids. (5)

What are healthy levels of physical activity for adults?

Some physical activity is better than none. Inactive people can start with small amounts of physical activity (even as a part of their normal daily activities) and gradually increase duration, frequency and intensity. (5)

Adults should perform:

- at least 150 minutes of moderate physical activity (a mild increase in heart rate or breathing rate resulting from, for example, brisk walking, climbing stairs, dancing, gardening or doing household chores) spread throughout the week, or
- at least 75 minutes of vigorous physical activity (including vigorous gardening, running, fast cycling, fast swimming, or playing sport) spread throughout the week), or
- an equivalent combination of moderate and vigorous activity;
- muscle-strengthening activities involving major muscle groups on two or more days a week. (6)

For additional health benefits, moderately intense physical activity should be increased to 300 minutes per week, or equivalent. Those with reduced mobility should perform physical activity on three or more days per week in order to improve balance and prevent falls. These recommendations apply to all adults, including those with CVD

and diabetes, unless there is a specific medical recommendation to the contrary. If the recommended regime of physical activity cannot be undertaken because of health constraints, it should be adapted as abilities and conditions allow. (5)

Tobacco use

Tobacco kills nearly half of its users and causes 6 million deaths every year. Ten per cent of these deaths are the result of inhaling secondhand smoke in homes, restaurants, offices or other enclosed spaces. (7)

What is tobacco use?

Tobacco can be smoked, sucked, chewed or sniffed. Tobacco products can generally be divided into two types:

- smoked tobacco (in cigarettes, cigars, pipes and water pipes)
- smokeless tobacco (in chewing tobacco and snuff).

All tobacco products contain the addictive substance nicotine that is absorbed into the bloodstream when a tobacco product is used.

Consequences for health

All tobacco products are harmful. Tobacco smoke contains at least 250 harmful chemicals, of which at least 69 are known to cause cancer. Tobacco use can damage every part of the body and is one of the main risk factors for CVD. It causes an estimated 10% of all CVD deaths. (7)

What are the benefits of quitting tobacco use?

Quitting is the best action tobacco users can take to improve their health. Quitting has immediate and long-term health benefits, including living up to 10 years longer.

a) Benefits to health (7)

Within a few months of quitting smoking, coughing and shortness of breath decrease. After a year of quitting, the risk of heart attack and angina is about half that of a smoker. Fifteen years after quitting, the risk of heart attack and angina is the same as that of a non-smoker. Additionally, quitting smoking after a heart attack reduces the chances of having another heart attack by 50%.

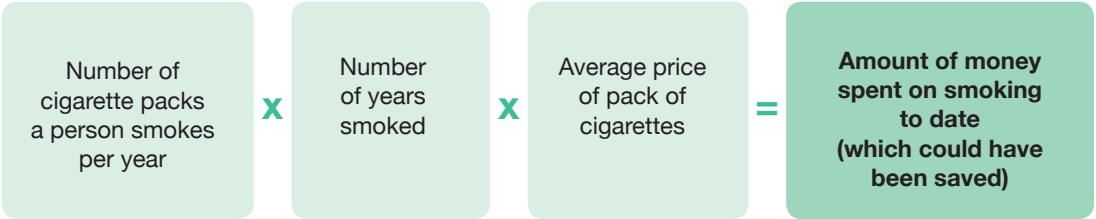
Quitting tobacco use reduces the likelihood of problems such as impotence, difficulties in getting pregnant, premature births, low birth weight and miscarriage. It also reduces the risk of childhood diseases related to secondhand smoke, such as asthma and middle-ear disease.

Other benefits of quitting smoking include: better ability to taste food, improved sense of smell, greater physical well-being, better performance in physical activities, and improved appearance, including reduced wrinkling/ageing of skin and whiter teeth. Quitting also improves the health of family members. Quitting tobacco use is also likely to slow progressive kidney function loss among those with chronic kidney disease.

b) Financial benefits (7)

At an individual level, there are substantial savings to be made on money previously spent on tobacco (see Figure 3). At a national level, potential health care costs (related to illnesses caused by tobacco) are prevented.

Figure 2: Calculation of money spent on tobacco



c) Social benefits (7)

Someone who quits tobacco use will find that their personal environment (home, car), clothing and breath will smell better. Their self-esteem is likely to increase and, as tobacco-free zones become accessible, social isolation they might have experienced will reduce. A person's productivity, too, will be greater as a result of quitting tobacco use, since smoking breaks will no longer be necessary. Once a parent has stopped smoking, their children become less likely to start the habit and more likely to quit if they smoke already.

Harmful use of alcohol

The harmful use of alcohol is a risk factor for CVD and has many negative health and social consequences. Harmful use of alcohol is a causal factor in more than 200 diseases and injuries. Beyond health consequences, the harmful use of alcohol brings significant social and economic losses to individuals and society at large. Like tobacco, alcohol can have a marked impact on the health of people other than the drinker. (8)

What is harmful use of alcohol?

Harmful use of alcohol, when defined as a public health problem, refers to “drinking that causes detrimental health and social consequences for the drinker, the people around the drinker and society at large, as well as patterns of drinking associated with increased risk of adverse health outcomes.” (9)

Harmful use includes high-level drinking each day, as well as single or repeated episodes of drinking to intoxication.

The risks related to alcohol are linked to the pattern of drinking, the amount of alcohol consumed and, sometimes, the quality of the alcohol consumed. There is no definition for safe or non-harmful use of alcohol. However, individuals are generally considered to be at lower risk for negative consequences of alcohol use if they **drink no more than two units of alcohol per day and do not drink on at least two days of the week.** (8, 9)

The alcohol content of a drink depends on the strength (alcohol percentage) of what is consumed and the volume of the container. There are wide variations from country to country in the strength of alcoholic drinks and in the sizes of drinks served. Therefore, when providing health information on alcohol use, it is essential to define “a drink” according to what is most common in the local context. (9)

What are the risks of harmful use of alcohol?

High levels of alcohol consumption and regular heavy episodic drinking are clearly associated with increased risk of CVD, since harmful use of alcohol damages the heart muscle, increases the risk of stroke and can cause cardiac arrhythmia. There may also be an increased risk among light-to-moderate drinkers who have irregular heavy drinking episodes. Whilst low-level consumption without any episodes of heavy drinking may be associated with a reduced CVD risk in some cases, overall alcohol consumption is associated with multiple health risks that may outweigh potential benefits. (9)

Brief interventions in primary care have been shown to be useful in identifying and managing people with alcohol problems. (10) However, the 5As brief intervention on harmful use of alcohol in this module should be used only to screen for potential harmful use and to provide basic information to patients on the harmful effects of alcohol. Detailed guidance for alcohol problems is outside the scope of this module and requires further training and resources. Some tools and guidance are referenced below:

- ASSIST: The Alcohol, Smoking and Substance Involvement Screening Test. Manual for use in primary care. Geneva: World Health Organization; 2010 (http://www.who.int/substance_abuse/publications/assist/en/, accessed 31 March 2017)
- AUDIT: The Alcohol Use Disorders Identification Test. Guidelines for use in primary care. 2nd edition. Geneva: World Health Organization; 2001 (http://www.who.int/substance_abuse/activities/sbi/en, accessed 31 March 2017)
- Brief intervention for Hazardous and Harmful Drinking. A manual for use in primary care. Geneva: World Health Organization; 2001 (http://www.who.int/substance_abuse/publications/audit_sbi/en/, accessed 31 March 2017)

2 Behaviour change

Understanding behaviour change

Reducing health risks may require changes to be made in some behaviours. Behaviour change is a complex issue and is not a one-off event but rather a set of different stages through which a person moves.

Behaviour change takes time, and different people go through the stages at various speeds. Some may remain indefinitely at one stage: for example, they may recognize and think they need to change but not be ready to make the change. Even though a person intends to start or maintain a new behaviour, for a range of reasons they may move back to an earlier stage. A relapse into an old behaviour can occur at any point and does not necessarily mean a failure to change. Many people who eventually adopt a new behaviour make several attempts before it is maintained over the long term. A person may feel disappointed, frustrated or tired, and relapse as a consequence. (11)

Understanding behaviour change is important for primary health care providers as they usually provide the first point of contact for patients accessing the health system. It is therefore necessary for the primary health care worker to assess and understand the different stages of readiness of the patient to make the required change in behaviour. This allows for more contextual, relevant and patient-centred counselling. Every visit or interaction with a healthcare provider is an opportunity for providing information and counselling on the risk factors, particularly for people with existing NCDs.

Counselling for behaviour change through brief interventions

Counselling can be described as professional guidance and support to help a person to solve a problem. Counselling for healthy lifestyles involves guiding and supporting patients toward making changes in certain behaviours to reduce the risk of heart attack, stroke and other conditions (diabetes, lung disease and cancer). A brief intervention is a short interaction of between three and 20 minutes between health worker and patient. It aims to identify a real or potential problem, provide information about it and then motivate and assist the patient to do something about it. A brief intervention involves two important aspects: how and what.

“How” refers to the communication style that the health worker uses to interact with the patient during the brief intervention. Communications techniques (Box 3) and motivational interviewing (Box 4) can be used as approaches to deliver the brief interventions.

Box 3: Communication techniques – “OARS”

- Open-ended questions
- Affirm
- Reflective listening
- Summarize

“**What**” refers to the structure of the counselling process and the content that should be covered during the brief intervention. An example of a brief intervention that can be used for healthy-lifestyle counselling is the 5As (see Section 3).

Motivational interviewing

Motivational interviewing is a style of talking with a person that can help in motivating them to change. In motivational interviewing, the counsellor does not try to convince the person to change, but instead guides them to reach conclusions themselves, and draws out the internal motivations unique to the person.

A central idea in motivational interviewing is that most people experience some ambivalence or mixed feelings about change. They have some reasons to change and also some reasons for staying the same, and may remain caught in the middle, unable to change. Motivational interviewing encourages people to explore these mixed feelings so that they can move toward positive change.

Box 4: Key principles of motivational interviewing

- Don't tell the person what to do
- Listen and show empathy
- Help the patient see the gap between where they are and where they want to be
- Let the patient tell you they need to change
- Help the patient to feel confident about changing
- Roll with resistance

Boxes 3 and 4: Adapted from: Latchford G, Duff A. Motivational interviewing: a brief guide. Leeds: University of Leeds; 2010.

3 Brief interventions to encourage behaviour change

General theory of the 5As brief interventions

Primary health care workers play an important role in helping patients to change their unhealthy behaviours and maintain healthy behaviours. Short interactions of between three and 20 minutes, called brief interventions, aim to identify a real or potential problem, provide information about it and motivate and assist the patient to do something about it. (10, 12)

The 5As is a tool used for brief interventions. It summarizes what a health worker can do to help someone who is ready to change. (10, 12) This can be integrated into regular visits by health providers of any level, either at community or facility level. Risk-factor counselling should be integrated with existing programme delivery at all levels of care.

Table 1 provides a general theoretical framework and further details on the 5As and how to use this tool. This framework can be adapted for each CVD behaviour risk factor, and samples are provided in Tables 2 to 5. (The sample for unhealthy diet focuses on fruit and vegetables, but can be expanded to include salt, fats and oils and sugars.)

Table 1: The 5As: General theoretical framework for how to do it

5As	What to say/do and how to say/do it									
Ask	Ask the patient about the relevant risk factor(s) at every visit. Ask in a friendly way, without being judgmental. Keep the questions simple. Record the information in the patient's medical record/notes. ▼									
Advise	Health workers have special authority because of their training. Patients usually respect this expertise. Provide information, key messages and advice in a clear, simple, and personalized manner. Link the advice to something that is relevant for the person. For example: • a person with hypertension may be interested in the benefits of reducing salt intake • people with young children may be concerned about the effects of secondhand smoke. ▼									
Assess	Assess the patient's readiness to start making a change by asking two questions: 1. Are you ready to have a diet that includes more healthy options? Be more physically active? Be a non-smoker? Be a lower-risk drinker? 2. Do you think you will be able to make the change? <table><tr><td>Question 1</td><td>Yes</td><td>Not sure</td><td>No</td><td rowspan="2">Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change.</td></tr><tr><td>Question 2</td><td>Yes</td><td>Not sure</td><td>No</td></tr></table> Answers in the white area suggest that you and the patient can move on to the next step. ▼	Question 1	Yes	Not sure	No	Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change.	Question 2	Yes	Not sure	No
Question 1	Yes	Not sure	No	Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change.						
Question 2	Yes	Not sure	No							
Assist	Help the person to develop a plan that can increase the chance of success. Provide practical counselling that focuses on: • provision of basic information about the risk factor • identification of situations that could trigger relapse • ways of coping with trigger situations. Provide social support including: • providing encouragement • communicating interest and concern • encouraging the person to talk about the change process with family and friends. Provide and ensure availability of health education materials and details about additional resources, such as support groups, quit lines, etc. ▼									
Arrange	Arrange a follow-up contact, by phone or in person. Discuss timing of follow-up with patients. At follow-up for all patients: • identify problems already experienced as well as new ones that could arise • remind them of the additional support that is available • schedule next follow-up visit. Refer to specialist services if needed and available. For those who have made the planned changes: • congratulate them on their success. For those who have challenges: • remind them to view this as a learning experience • review their circumstances and motivate them to re-commit • link to more intensive support, if available.									

Table 2: 5As brief intervention for a healthy diet – example fruit and vegetables

5As	Fruit and vegetables								
Ask	How many portions of fruit and vegetables do you eat each day? 1 portion = 1 orange, apple, mango, banana, or 3 tablespoons of cooked vegetables*. Provide local examples and equivalent serving sizes. *Potatoes, sweet potatoes, cassava or other starchy tubers or roots do not count as one of these portions. ▼								
Advise	Eat at least 5 portions of fruit and vegetables per day. Eat a variety of fruits, vegetables, legumes (lentils, beans), nuts and whole grains (unprocessed maize, millet, oats, wheat, brown rice), starchy tubers or roots (potato, yam, taro or cassava) and foods from animal sources (meat, fish, eggs and milk). Provide local examples. Advantages: - Eating a variety of these foods every day helps you to take in the right amounts of essential nutrients. - Eating enough healthy food helps to avoid unhealthy foods that can lead to overweight and obesity, and diseases such as hypertension, diabetes, heart attack and stroke. ▼								
Assess	1. Are you ready to make some changes to your diet in order to include more healthy food options? 2. Do you think you will succeed in making the changes? <table><tr><td>Question 1</td><td>Yes</td><td>Not sure</td><td>No</td></tr><tr><td>Question 2</td><td>Yes</td><td>Not sure</td><td>No</td></tr></table> Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change. Answers in the white area suggest that you and the patient can move on to the next step. ▼	Question 1	Yes	Not sure	No	Question 2	Yes	Not sure	No
Question 1	Yes	Not sure	No						
Question 2	Yes	Not sure	No						
Assist	Help the patient to set goals and make a plan to start introducing some changes to their eating habits. Provide practical counselling about unhealthy foods and healthier choices. For example: - Avoid deep fried foods. - Eat fresh vegetables and fruit that are in season. Have fresh fruit available and in plain sight. Engage the patient in the conversation and allow time for them to share ideas: - Can you think of ways to increase the amount of fruit and vegetable you eat every day? - Can you think of healthier types of food that you enjoy and that you could eat instead of the less-healthy option? Provide social support: - Invite the patient to bring family members to the next visit in order to discuss healthier diet options for the whole family. - Provide health and nutrition education materials. ▼								
Arrange	Refer to specialist support services (dietician, nutritionist) if needed and available Follow-up: decide the timeline and method and schedule the next appointment. Ask about successes and challenges. For those of have made the planned changes to their eating habits: - Congratulate them on their success. For those experiencing challenges: - Remind them to view the process as a learning experience and that it takes time to establish new habits - Review circumstances, discuss ways to address challenges and encourage recommitment to their plan. Link with more intensive support, if available. Remind all patients of any additional support and resources that are available.								

Table 3: 5As brief intervention to increase physical activity

5As	Physical activity									
Ask	In the past week, on how many days have you been physically active for a total of 30 minutes or more? For example: walking, cycling, cleaning, gardening, climbing stairs, dancing or playing sport. Adapt examples to local context. ▼									
Advise	All adults should do at least 2½ hours (150 minutes) of physical activity per week. This can be spread over short sessions throughout the day and week, starting from as little as 10 minutes per session. Being more active can start in small ways which are part of daily life. This can include going for a walk, playing with children, gardening and domestic chores. Adapt examples to local context. Advantages of physical activity: • Reduces the risk of heart attack and stroke or of developing hypertension, diabetes and cancer • Can help to control blood pressure, cholesterol and diabetes • Helps with weight loss and weight control • Helps to prevent and manage depression. • Some physical activity is better than none. ▼									
Assess	1. Are you ready to start being more physically active? 2. Do you think you will be able to succeed in increasing your activity levels? <table><tr><td>Question 1</td><td>Yes</td><td>Not sure</td><td>No</td><td rowspan="2">Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change.</td></tr><tr><td>Question 2</td><td>Yes</td><td>Not sure</td><td>No</td></tr></table> Answers in the white area suggest that you and the patient can move on to the next step. ▼	Question 1	Yes	Not sure	No	Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change.	Question 2	Yes	Not sure	No
Question 1	Yes	Not sure	No	Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change.						
Question 2	Yes	Not sure	No							
Assist	Help the patient to develop a plan to start increasing physical activity. Provide practical counselling. • Help the patient to identify areas of their daily life where they could start to increase their activity levels. • Help to identify activities that they would enjoy doing. • Help to identify possible challenges and suggest how to overcome them. • Provide social support. • Encourage the patient to talk with family, friends and work colleagues about their efforts to increase activity levels. • Provide health education materials and information on additional resources. • These could include contact details for organizations such as walking groups and activity clubs Adapt to local context. Provide (if available) or advise on devices to help motivate or monitor activity e.g. a pedometer. ▼									
Arrange	Refer to specialist support services if needed and available. Follow-up: decide the timeline and method and schedule the next appointment. Ask about successes and challenges. For those who have become more physically active: • Congratulate them on their success. For those experiencing challenges: • Remind them to view the process as a learning experience and that it takes time to establish new habits. • Review circumstances, discuss ways to address challenges and encourage recommitment to their plan. • Link with more intensive support if available. Remind all patients of any additional support and resources that are available.									

Table 4: 5As brief intervention to quit tobacco (12)

5As	Tobacco											
Ask	Have you smoked or used any other tobacco product in the past 12 months? (for example, cigarettes (including home-made), cigars, pipe, water-pipe, chewing tobacco, snuff):											
	Yes		No									
	Do you currently smoke or use any other tobacco product?:											
	Yes		No									
	Does anyone smoke around you at home or at work, or do you often go to places where there is a lot of smoke such as restaurants or bars?											
	Yes		No									
	If No to all these questions:											
	• Advise not to start tobacco use smoke and to avoid secondhand exposure											
	If Yes to any questions:											
	• Advise on risks of exposure to secondhand											
	▼											
Advise	Quitting tobacco is the most important thing you can do to protect your health now and in the future.											
	Advantages:											
	• Tobacco use is a major cause of heart attack and stroke, of serious lung problems and certain cancers. • Tobacco can damage every part of the body. • Secondhand smoke damages the health of your family and others around you.											
	▼											
Assess	1. Are you interested in quitting tobacco use?											
	2. Do you think you will succeed in quitting?											
	<table><tr><td>Question 1</td><td>Yes</td><td>Not sure</td><td>No</td><td rowspan="2">Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change.</td></tr><tr><td>Question 2</td><td>Yes</td><td>Not sure</td><td>No</td></tr></table>				Question 1	Yes	Not sure	No	Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change.	Question 2	Yes	Not sure
Question 1	Yes	Not sure	No	Any answer in the shaded area indicates that the person is not yet ready to change. In this case, effort needs to be made to increase motivation for change.								
Question 2	Yes	Not sure	No									
	Answers in the white area suggest that you and the patient can move on to the next step.											
	▼											
Assist	Help the patient to develop a quit plan using the STAR method:											
	Set a quit date, ideally within 2 weeks.											
	Tell family and friends about quitting and ask for support.											
	Anticipate challenges in the quit attempt.											
	Remove tobacco products from personal environment and make home smoke-free.											
	Provide practical counselling:											
	• Provide basic information about tobacco use and quitting. • Help the patient to identify situations (e.g. feelings, places, activities) that could increase the risk of smoking or relapse. • Help to identify and practise ways of coping with these situations.											
	Provide social support.											
	• Provide encouragement in the quit attempt by showing care and concern. • Encourage the patient to talk about the quitting process.											
	Provide health education materials and information on additional resources, e.g. support groups, quit lines.											
	Recommend the use of medications if indicated and available, e.g. nicotine replacement therapy.											
	▼											

5As	Tobacco
Arrange	Refer to specialist support services if needed and available.
	Follow-up: - Decide the timeline and method and schedule the next appointment. - Ask about successes and challenges.
	For those who have quit: Congratulate them on their success.
	For those who have used tobacco again: - Remind them to view any failures as a learning experience. - Review circumstances and encourage them to recommit to quitting. - Link with more intensive support if available.
	For all patients: - Identify problems and discuss ways to address them. - Remind them of additional support and resources that are available. - Assess use of medications and any problems experienced.

Please also refer to the WHO publication *Strengthening health systems for treating tobacco dependence in primary care. Part III: Training for primary care providers: Brief tobacco interventions*, listed in Selected further reading.

Table 5: 5As brief intervention to screen for harmful use of alcohol

5As	Alcohol
Ask	Do you ever drink alcohol? Yes No
	If Yes: - How often do you have an alcoholic drink? - How many alcoholic drinks do you have on a usual day when you are drinking? ▼
Advise	For people drinking fewer than 2 units (according to local drink strengths and commonly available sizes) per day and drinking on 5 or fewer days per week, inform them of the following: - Use of alcohol can increase the risk of having a heart attack or stroke. - It also increases the risks of getting certain cancers and can cause damage to other parts of the body. - Overall, the best way to avoid the health risks of alcohol use is to abstain. - If you do drink alcohol, keep in mind that “less is better”. - Avoid having more than two units on any single day and do not drink any alcohol on at least two days per week. - Do not drink alcohol for “health” reasons. - Do not use alcohol when you are: - driving - operating machinery - pregnant or breastfeeding - taking medications that interact with alcohol. - living with medical conditions that are made worse by alcohol - having difficulties controlling how much you drink. ▼
Assess	For people who drink two or more units per day, who drink on more than five days per week and/or have any indication that alcohol could potentially be a problem, say: “Your drinking habits could be harmful to your health. May I ask you a few more questions to have a better idea of the possible risks?” ▼ Yes No
Assist and Arrange	If No: - Give brief advice. End the discussion positively by saying, “This can be a difficult issue to discuss but I am here to help you.” - Provide health education materials and information about additional resources such as help lines, counselling, support groups. - At the next visit, repeat the brief intervention.
	If Yes: Give brief advice and refer for further counselling and tests (for which further training, tools and resources are required). See AUDIT test (link below). The score will determine next steps.

4 Next steps

The theoretical information on the four risk factors (unhealthy diet, physical inactivity, tobacco use and harmful use of alcohol) has been provided. Similarly, the theory on behaviour change and sample frameworks using the 5As brief interventions has also been presented. The next step will entail adaptation of this information and frameworks to support local implementation.

This should be done in the country by relevant working groups who are knowledgeable about the topic, country dynamics, culture, locally appropriate languages as well as what has worked in other similar areas of health.

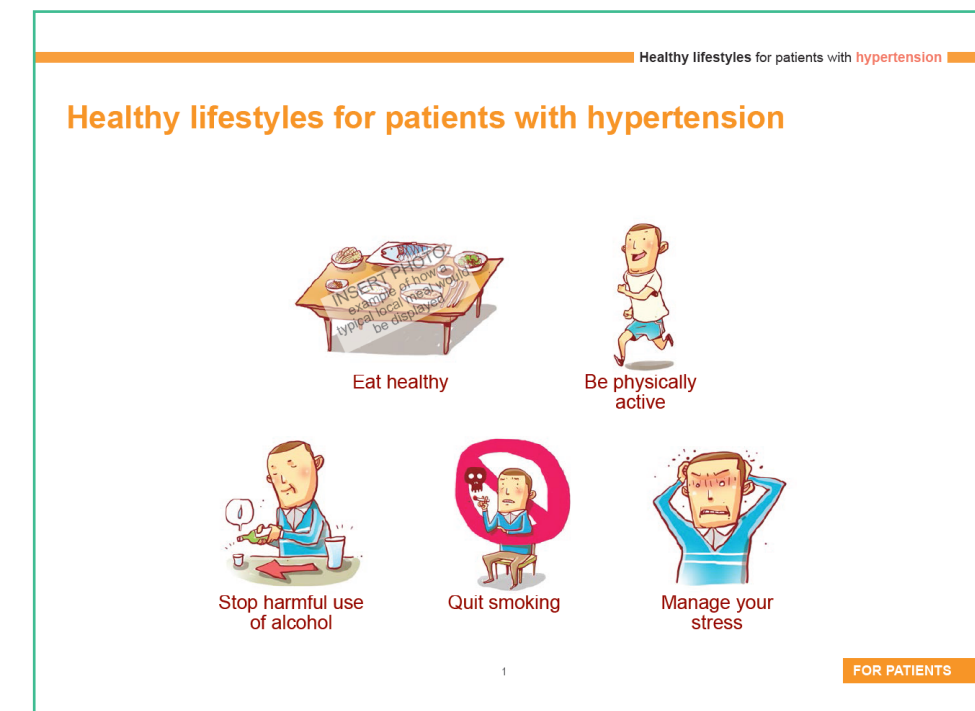
A variety of supporting materials can be developed at the local level including but not limited to:

- job aids
- flip charts
- patient information leaflets
- posters
- mobile apps

Examples of these materials are provided in the Annex.

Annex: Material supporting healthy lifestyles

Examples from *Healthy lifestyles for patients with hypertension* (13)



Eat healthy (low-salt diet)

Patient education

- A balanced diet is very important for people with hypertension. It is also vital to eat less salt.
- For example, instead of drinking the soup broth, eat only the ingredients in the soup.
- Also cut back on pickled and processed foods like ketchup, sausage and ham.
- It is highly recommended to eat fresh and healthy local food.

- Eat less soup broth
- Eat less pickled and processed food
- Eat fresh, local food



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Healthy lifestyles for patients with hypertension

Good exercises for patients with hypertension

Aerobic exercise

- Strengthens heart and lung function
- Strengthens joints and increases body flexibility
- Strengthens bones and muscles

Examples of recommended exercises



Walking/
Jogging



Running



Cycling



Swimming

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FOR PATIENTS

Healthy lifestyles for patients with hypertension

Smoking and hypertension

Patient Education

- It is best to stop smoking.
- Smoking increases the risk of lung diseases, cancer and cardiovascular diseases such as angina and stroke.
- If you wish to stop, but have failed, you can visit any clinic or doctor to get help.

Smoking is

the most important risk factor for chronic lung disease, cancer, cardiovascular disease, stroke and peripheral vascular disease.

- If you quit smoking, antihypertensive drugs become more effective.
- The risk of cardiovascular disease halves after a year of not smoking.
- The risk of cardiovascular disease becomes similar to non-smokers after 15 years of not smoking.



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FOR PHYSICIANS

Healthy lifestyles for patients with hypertension

Alcohol and hypertension

Patient education

- Drinking alcohol lowers the effect of antihypertensive drugs and causes more side-effects.
- Alcohol also supplies many calories without any nutrition, disturbing body-weight management, so cutting alcohol intake is important.
- It is recommended that men limit drinking to less than two glasses a day and women limit drinking to less than one glass a day.

Harmful use of alcohol could cause:

- blood pressure to rise
- drug effectiveness to weaken
- unwanted side-effects of medication to increase
- malnutrition
- body weight to increase



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FOR PHYSICIANS

Healthy lifestyles for patients with hypertension

Take-home message
Healthy lifestyles



INSERT PHOTO
example of how
typical local healthy food
would be displayed

Eat healthy
Eat less salt
Eat a low-fat diet



Be physically active
More than 150 minutes
a week walking, cycling,
swimming



Quit smoking



Stop harmful
use of alcohol
Maximum of
one or two
drinks per day



Manage your
stress

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FOR PATIENTS

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For more information

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Technical package for cardiovascular disease management in primary health care



Evidence-based treatment protocols



WHO/NMH/NVI/18.2

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Introduction

In many low- and middle-income countries, there is a wide gap between evidence-based recommendations and current practice. Treatment of major CVD risk factors remains suboptimal, and only a minority of patients who are treated reach their target levels for blood pressure, blood sugar and blood cholesterol.

In other areas, overtreatment can occur with the use of non-evidence-based protocols. The aim of using standard treatment protocols is to improve the quality of clinical care, reduce clinical variability and simplify the treatment options, particularly in primary health care. Standard treatment protocols can be developed by preparing new national treatment guidelines or by adapting or adopting international guidelines.

The *Evidence-based protocols* module uses hypertension and diabetes screening and treatment as an entry point to control cardiovascular risk factors, prevent target organ damage, and reduce premature morbidity and mortality. A comprehensive risk-based approach for integrated management of hypertension, diabetes, and high cholesterol is included in the *Risk-based CVD management* module.

This module includes clinical practice points and sample protocols for:

1. hypertension detection and treatment
2. type 2 diabetes detection and treatment
3. identifying basic emergencies – care and referral.

HEARTS emphasizes adaptation, dissemination, and use of a standardized set of simple clinical-management protocols, which should be drug- and dose-specific, and include a core set of medications. The simpler the protocols and management tools, the more likely they are to be used correctly, and the higher the likelihood that a programme will achieve its goals.

The use of a standardized algorithm is critical to success because it:

- enables task-sharing, with the entire health care team able to support patients
- increases ease of logistics in terms of drug inventory, drug forecasting, and quality monitoring
- enables large reductions in cost of medication
- enables evaluation of impact
- simplifies implementation of changes to protocols, if needed.

Use of module

This module is intended for national policymakers, subnational programme managers and primary health care facility managers who are in the position to adapt these protocols and align to the local context. This ensures that just one protocol is used at the national or in some instances subnational level.

1 Hypertension detection and treatment

All adults should have their blood pressure checked. The treatment protocols provided can be used if the patient is known to have hypertension or is newly diagnosed with hypertension when presenting at the health care facility.

When to measure blood pressure

Measuring blood pressure is the only way to diagnose hypertension, as most people with raised blood pressure have no symptoms.

Blood pressure measurements should be conducted on adults during routine visits to primary health care facilities, including all adults at first presentation to the facility, and, if normal, periodically thereafter (e.g., every 1 to 5 years). Every patient with elevated blood pressure readings requires immediate follow-up, according to the protocol.

Blood pressure measurement and control is particularly important in adults who:

- have had a prior heart attack or stroke
- have diabetes
- have chronic kidney disease (CKD)
- are obese
- use tobacco
- have a family history of heart attack or stroke.

How to measure blood pressure

Effective treatment algorithms for hypertension are dependent on accurate blood pressure measurement. The following advice should be followed for measuring blood pressure (1):

- Use the appropriate cuff size, noting the lines on the cuff to ensure that it is positioned correctly on the arm. (If the arm circumference is >32 cm, use large cuff.)
- Although at the initial evaluation it is preferable to measure blood pressure in both arms and use the arm with the higher reading thereafter, this may not be practical in a busy primary care environment.
- The patient should be sitting with back supported, legs uncrossed, empty bladder, relaxed for 5 minutes and not talking.
- For persons who are getting their blood pressure measured for the first time, it is preferable to take at least two readings and to use the second reading.

Blood pressure can be measured either by a conventional sphygmomanometer, using a stethoscope, or by an automated electronic device. The electronic device, if available, is preferred because it provides more reproducible results and is not influenced by variations in technique or by the bias of the observers.

If the primary health care facility has electricity or regular access to batteries, then consider an automated validated blood pressure device with a digital reading. If the primary health care facility has no electricity or batteries, then a manual BP cuff will have to be used with a stethoscope.

Diagnosing hypertension

The diagnosis of hypertension should be confirmed at an additional patient visit, usually 1 to 4 weeks after the first measurement. In general, hypertension is diagnosed if, on two visits on different days:

- systolic blood pressure on both days is ≥ 140 mmHg and/or
- diastolic blood pressure on both days is ≥ 90 mmHg.

Hypertension treatment

Who should receive hypertension treatment?

Hypertension treatment is indicated for adults diagnosed with hypertension, as defined above (SBP ≥ 140 mmHg and/or DBP ≥ 90 mmHg). Patients with SBP ≥ 160 mmHg or DBP ≥ 100 mmHg may be indicated for immediate treatment based on one assessment.

Lifestyle counselling (healthy diet, physical activity, tobacco use, and harmful use of alcohol) is a critical component of good hypertension management and is often recommended as a first step for patients with blood pressure of SBP 130–139 mmHg and/or DBP 80–89 mmHg (2) who do not have other CVD risk factors. However, in settings where people do not regularly visit the doctor, people who are recommended *only* lifestyle modification may not return for re-evaluation and needed treatment, resulting in uncontrolled hypertension and associated complications. (See *Healthy-lifestyle counselling* module for more information on lifestyle counselling.)

What medications should be used to treat hypertension?

There are four main classes of antihypertensive medications: angiotensin converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARB), calcium channel blockers (CCB), and thiazide and thiazide-like diuretics. Any of these four classes of antihypertensive medication may be used unless there are specific contraindications. Proper treatment of hypertension usually requires a combination of hypertension medications.

Notes on specific hypertension medications

- Pregnant women and women of childbearing age not on effective contraception should *not* be given ACE inhibitors, ARBs, or thiazide/thiazide-like diuretics; CCBs should be used. If not controlled with intensification dose of medication, refer to specialist.
- Beta blockers are not recommended as first-line therapy. If a heart attack has been diagnosed within the previous three years, or there is atrial fibrillation or heart failure, then a beta blocker should be added to the starting dose of antihypertensive medication. Patients with angina may also benefit from treatment with a beta blocker.

Treatment targets

For most patients, blood pressure is considered controlled when SBP <140 mmHg and DBP <90 mmHg. However, for patients with diabetes or a high risk of CVD, certain guidelines recommend lower targets: SBP <130 mmHg and DBP <80 mmHg.

Other treatment considerations

- If there is a prior heart attack or stroke, or the person is otherwise at high risk of CVD, start a statin at the same time as starting antihypertensive medication. (Statins should not be used in women who are or who may become pregnant.)
- If there is a prior heart attack or ischemic stroke, start low-dose aspirin.
- The hypertension protocols included in this module serve well for initiation and maintenance of successful treatment. If there are serious adverse events, lack of control of blood pressure, or if a major medical event intervenes, then referral to a specialist will be needed.
- If the patient is already on another medication regimen, blood pressure is controlled to the target level, and the medications the patient is taking are accessible and affordable, there is no reason to change the regimen.
- If the patient feels faint on standing, check blood pressure while standing. If the systolic blood pressure is consistently less than 110 mm Hg in a patient on medical treatment, consider reducing the dosage or number of medications used.

Treatment adherence

Adherence to treatment is critical for blood pressure control. If antihypertensive medication is being prescribed, the following are critical to ensuring adherence:

- Teach the patient how to take the medications at home.
- Explain the difference between medicines for long-term control (for example, of blood pressure) and medicines for quick relief (such as for headaches).
- Explain the reason for prescribing the medicine(s).
 - Explain the diagnosis of hypertension.
 - Discuss the asymptomatic nature of hypertension and explain that medications must be taken even if there are no symptoms.
 - Inform patient of the complications of untreated hypertension, including stroke, heart attack, kidney failure.
 - Explain the disability and economic and family burden these preventable complications cause.
- Show the patient the appropriate dose.
- Explain how many times a day the patient should take the medication and at what time, and adopt the following simple steps to help them to adhere to the guidelines:
 - Label and package the tablets.
 - Check the patient's understanding before the patient leaves the health centre.
 - Wherever possible, use once-daily dosages of all medications, to be given at the same time each day.

- Explain how important it is for the patient to:
 - Keep an adequate supply of medications safely at home.
 - Take the medicines regularly as advised, even if there are no symptoms.
- Explain potential adverse effects of the medications and what to do if the patient experiences them.

Sample hypertension treatment protocols

The following pages outline sample protocols for managing hypertension, each with a different starting medication. Each programme/country should select and then adapt the option that best suits their circumstances.

This series of sample protocols represents a range of possible approaches to managing hypertension. Depending on the local preferences, influenced by usual practice, resources and demography, countries may select one or more protocols from this sample to adapt.

To assist in the selection of the most suitable option, a non-comprehensive list of advantages and disadvantages is presented in a box relating to each protocol. Selection of a single option greatly facilitates logistics, training, supervision, evaluation, and overall programme implementation. The simpler the protocol, the more likely it is to be followed and to achieve the programme objective.

The following sample hypertension protocols have been endorsed by the:

World Hypertension League and the International Society of Hypertension

1. Diuretic as first-line treatment (3–11)
2. CCB as first-line treatment (5, 6, 9, 12–17)
3. ACE-I or ARB as first-line treatment (5, 6, 9, 10, 12, 13, 18–21)
4. ACE-I or ARB + CCB as first-line treatment (5, 6, 12, 13, 15, 22)
5. CCB + diuretic as first-line treatment (5, 6, 13, 14)
6. ACE-I or ARB+ diuretic as first-line treatment (5–7, 13, 14, 22–25)

The following sample protocol has been endorsed by the:

World Heart Federation

7. Use of BP-lowering drugs in patients with ischaemic CVD (5, 6, 9, 10, 12, 13, 18–21)

The following sample hypertension protocols have been adapted and endorsed by:

Resolve to Save Lives

8. Adapted example: CCB as first-line treatment (26)
9. Adapted example: telmisartan 40 mg/amlodipine 5 mg single-pill combination regimen (26)

Box 1: Advantages and disadvantages of diuretics

Advantages of diuretics

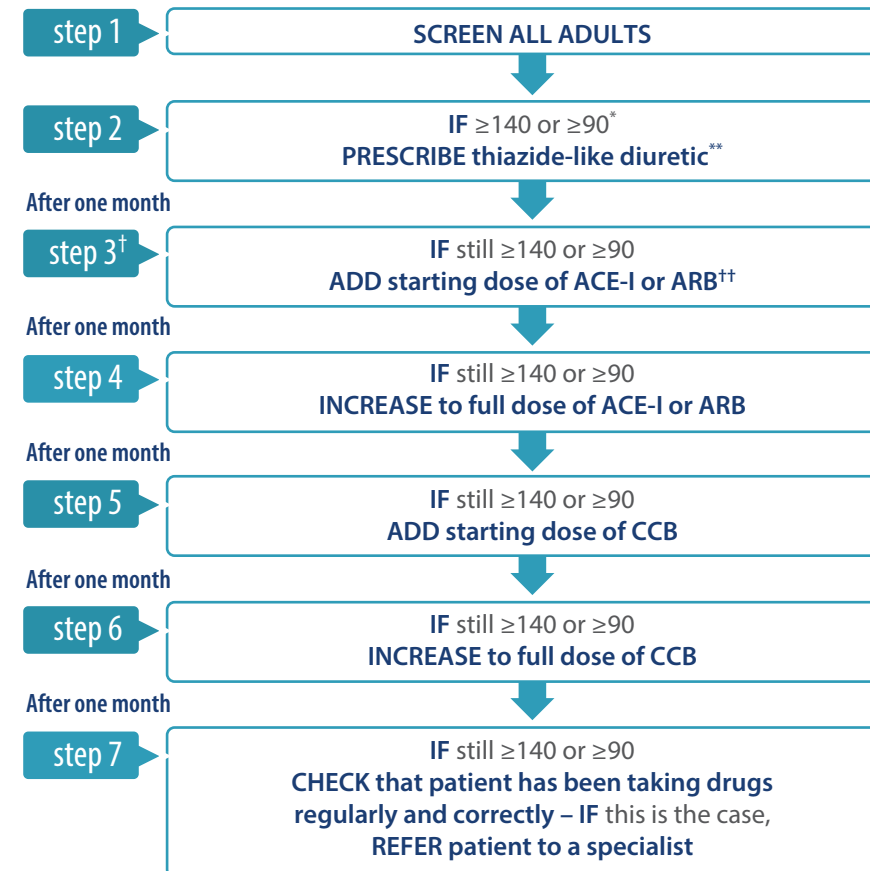
- May be less expensive than other hypertension medications;
- Probably effective for all races.

Disadvantages:

- Risk of hypokalemia;
- Has unfavourable effects on lipid and glucose measurements; clinical significance unclear.

HYPERTENSION PROTOCOL

Diuretic as first-line treatment



PROVISION FOR SPECIFIC PATIENTS

► THIS PROTOCOL IS CONTRAINDICATED FOR WOMEN WHO ARE OR COULD BECOME PREGNANT.

- Manage diabetes as indicated by national protocol.
- Aim for BP $<130/80$ for people at high risk, such as individuals with diabetes, CAD, stroke, or CKD.

LIFESTYLE MANAGEMENT ADVICE FOR ALL PATIENTS

- Stop all tobacco use, avoid secondhand tobacco smoke.
- Drink no more than two units of alcohol per day and do not drink on at least two days of the week.
- Increase physical activity to equivalent of brisk walk 150 minutes per week.
- If overweight, lose weight.
- Eat heart-healthy diet:
 - Eat a low-salt diet.
 - Eat ≥ 5 servings of vegetables/fruit per day.
 - Use healthy oils (e.g. olive, safflower).
 - Eat nuts, legumes, whole grains and foods rich in potassium.
 - Limit red meat to once or twice a week at most.
 - Eat fish or other food rich in omega 3 fatty acids (e.g., flax seeds) at least twice a week.
 - Avoid added sugar from cakes, cookies, sweets, fizzy drinks and juice.

DRUGS AND DOSES[¥]

Class	Medication	Starting dose	Intensification dose
diuretic [‡] thiazide-like	chlorthalidone [§]	12.5 mg	25 mg
	or indapamide SR [§]	1.5 mg	stay at 1.5 mg
ACE inhibitor [‡] (angiotensin-converting-enzyme inhibitor)	lisinopril	20 mg	40 mg
	ramipril	5 mg	10 mg
	perindopril	4–5 mg	8–10 mg
ARB [‡]	losartan	50 mg	100 mg
	telmisartan	40 mg	80 mg
CCB (calcium channel blocker)	amlodipine	5 mg	10 mg

* Or other BP target, as determined by clinical factors. If BP ≥ 160 or ≥ 100 , start same day. If 140–159 or 90–100, check on a different day and if still elevated, start.

** Consider statin use. Consider increase to intensification dose diuretic. Hypokalaemia more common using intensification dose diuretic – consider increased lab monitoring.

† Consider optional switch of steps 3 and 4 (ACE-I) with steps 5 and 6 (CCB).

†† ACE-I or ARB according to local guidelines/costs/intolerance to ACE-I. ACE-Inhibitors cause chronic cough in approximately 10% of patients. Neither ACE-I nor ARBs should be given to pregnant women.

¥ These are suggested examples of medications based on scientific evidence, once-daily suitability, common usage, and availability.

‡ Before initiating and several weeks after starting ACE-Is, ARBs or diuretics, consider checking serum creatinine and potassium.

§ If neither diuretic agent is available, hydrochlorothiazide can be used (25 mg starting, 50 mg full) or indapamide (1.25 mg starting, 2.5 mg full) can be used.

Box 2: Advantages and disadvantages of CCBs

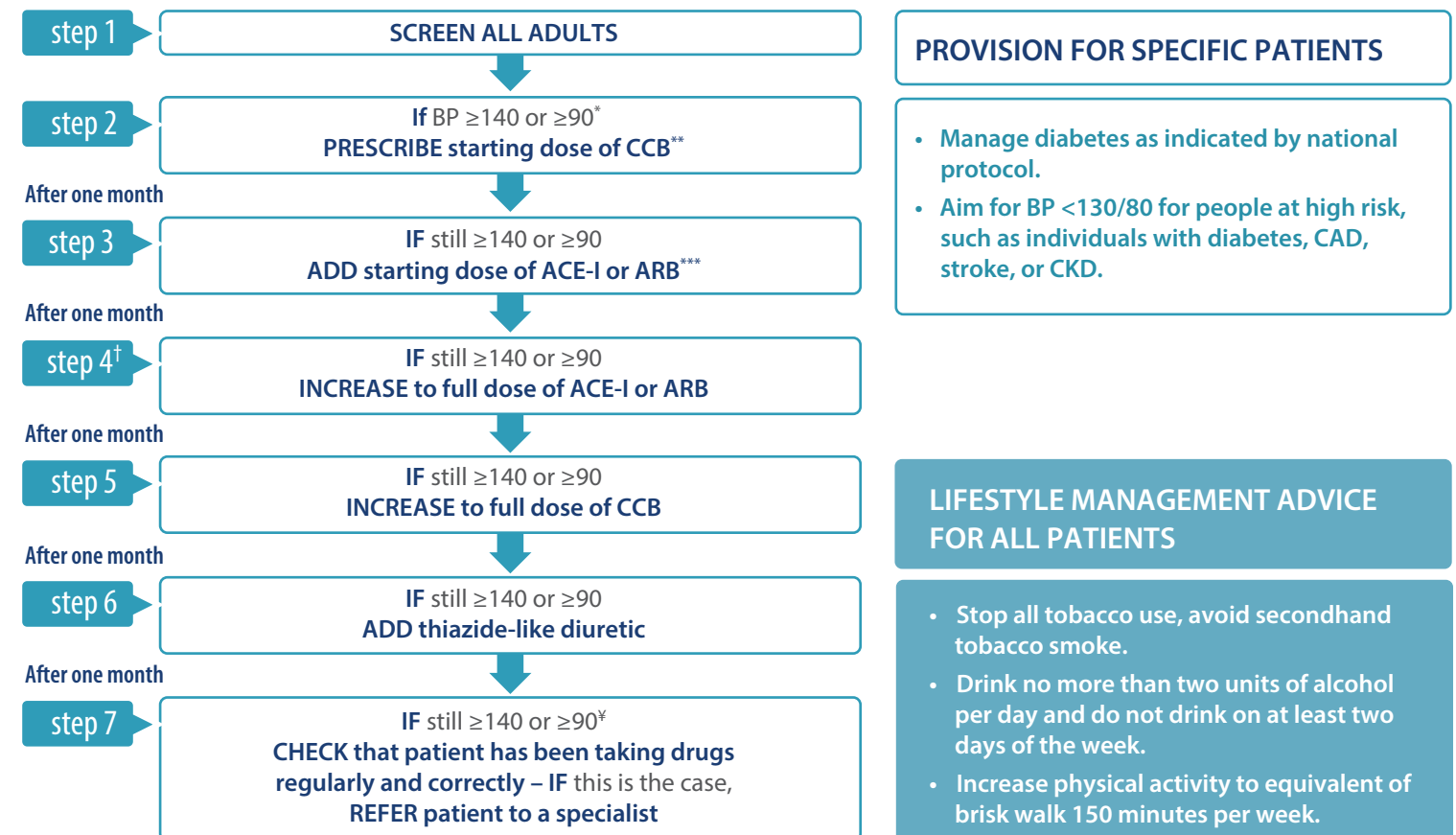
Advantages:

- Probably effective for all races;
- Reduces need for monitoring of electrolytes and renal function;
- Avoids need for different treatment for women of childbearing age who may become pregnant.

Disadvantages:

- May be more expensive than diuretics;
- Ankle edema may occur in up to 10% of patients, particularly with intensification dose in the absence of an ACE-inhibitor or ARB.

HYPERTENSION PROTOCOL CCB as first-line treatment



DRUGS AND DOSES[‡]

Class	Medication	Starting dose	Intensification dose
CCB (calcium channel blocker)	amlodipine	5 mg	10 mg
ACE inhibitor [§] (angiotensin-converting-enzyme inhibitor)	lisinopril	20 mg	40 mg
	ramipril	5 mg	10 mg
	perindopril	4–5 mg	8–10 mg
ARB [§]	losartan	50 mg	100 mg
	telmisartan	40 mg	80 mg
diuretic [§] thiazide-like	chlorthalidone [¶]	12.5 mg	25 mg
	or indapamide SR [¶]	1.5 mg	stay at 1.5 mg

* Or other BP target, as determined by clinical factors. If BP ≥ 160 or ≥ 100 , start same day. If 140–159 or 90–100, check on a different day and if still elevated, start.

** Consider statin use. Consider increasing to intensification dose CCB before introducing ACE-I/ARB.

*** ACE-I or ARB according to local guidelines/costs/intolerance to ACE-I. ACE-Inhibitors cause chronic cough in approximately 10% of patients. Neither ACE-I nor ARBs should be given to pregnant women.

† Consider optional switch of steps 3 and 4 (ACE-I) with step 6 (thiazide-like diuretic).

‡ Consider increase to full dose diuretic. Hypokalaemia more common using full dose diuretic – consider increased lab monitoring.

§ These are suggested examples of medications based on scientific evidence, once-daily suitability, common usage, and availability.

§ Before initiating and several weeks after starting ACE-Is, ARBs or diuretics, consider checking serum creatinine and potassium.

¶ If neither diuretic agent is available, hydrochlorothiazide can be used (25 mg starting, 50 mg full) or indapamide (1.25 mg starting, 2.5 mg full) can be used.

ACE-I or ARB* as first-line treatment



Box 3: Advantages and disadvantages of ACE inhibitors

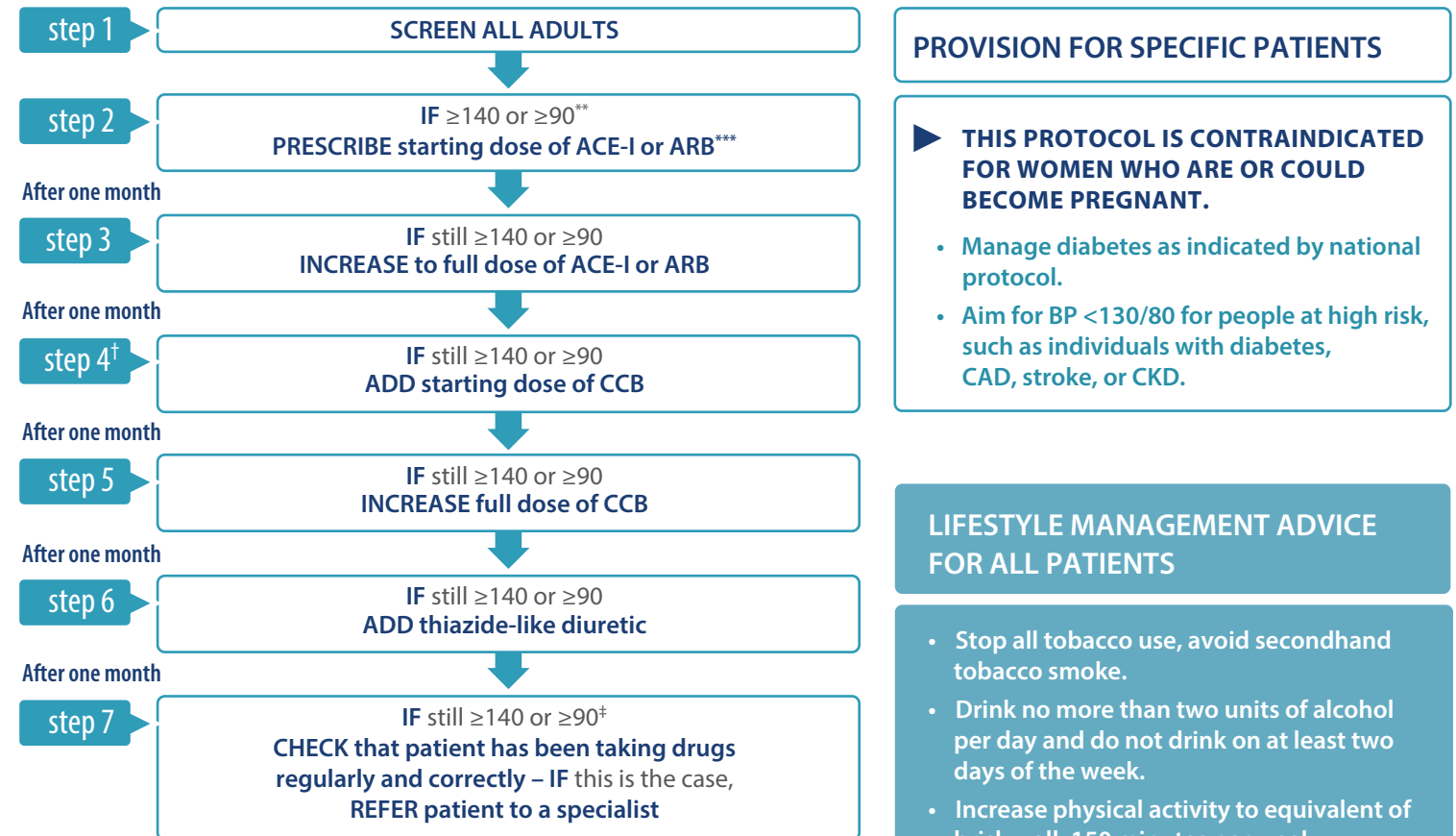
Advantages:

- Benefits some patients with kidney disease, prior myocardial infarction, and low ejection fraction.

Disadvantages:

- May be more expensive than other hypertension medications;
- A persistent cough is experienced by up to 10% of patients treated with an ACE inhibitor (not observed with ARBs); this risk is higher in people of African descent;
- Less effective as single medication in people of African descent;
- Small risk of angioedema; risk is higher in people of African descent (not observed with ARBs);
- ACE inhibitors (and ARBs) should NOT be given to women who are or who may become pregnant;
- Risk of hyperkalemia, especially if patient has CKD.

Note: Research suggests that ARBs may have effectiveness similar to ACE inhibitors.

DRUGS AND DOSES[¥]

Class	Medication	Starting dose	Intensification dose
ACE inhibitor [§] (angiotensin-converting-enzyme inhibitor)	lisinopril	20 mg	40 mg
	ramipril	5 mg	10 mg
	perindopril	4–5 mg	8–10 mg
ARB [§]	losartan	50 mg	100 mg
	telmisartan	40 mg	80 mg
diuretic [§] thiazide-like	chlorthalidone [‡] or indapamide SR [‡]	12.5 mg 1.5 mg	25 mg stay at 1.5 mg
CCB (calcium channel blocker)	amlodipine	5 mg	10 mg

* ACE-I or ARB according to local guidelines/costs/intolerance to ACE-I. ACE-Inhibitors cause chronic cough in approximately 10% of patients. Neither ACE-I nor ARBs should be given to pregnant women.

** Or other BP target, as determined by clinical factors. If BP ≥ 160 or ≥ 100 , start same day. If 140–159 or 90–100, check on a different day and if still elevated, start.

*** Consider statin use.

† Consider optional switch of steps 4 and 5 (CCB) with step 6 (thiazide-like diuretic).

‡ Consider increase to intensification dose diuretic. Hypokalaemia more common using intensification dose diuretic – consider increased lab monitoring.

¥ These are suggested examples of medications based on scientific evidence, once-daily suitability, common usage, and availability.

§ Before initiating and several weeks after starting ACE-Is, ARBs or diuretics, consider checking serum creatinine and potassium.

⌘ If neither diuretic agent is available, hydrochlorothiazide can be used (25 mg starting, 50 mg full) or indapamide (1.25 mg starting, 2.5 mg full) can be used.

ACE-I or ARB* + CCB as first-line treatment



Box 4: Advantages and disadvantages of ACE inhibitors plus CCBs

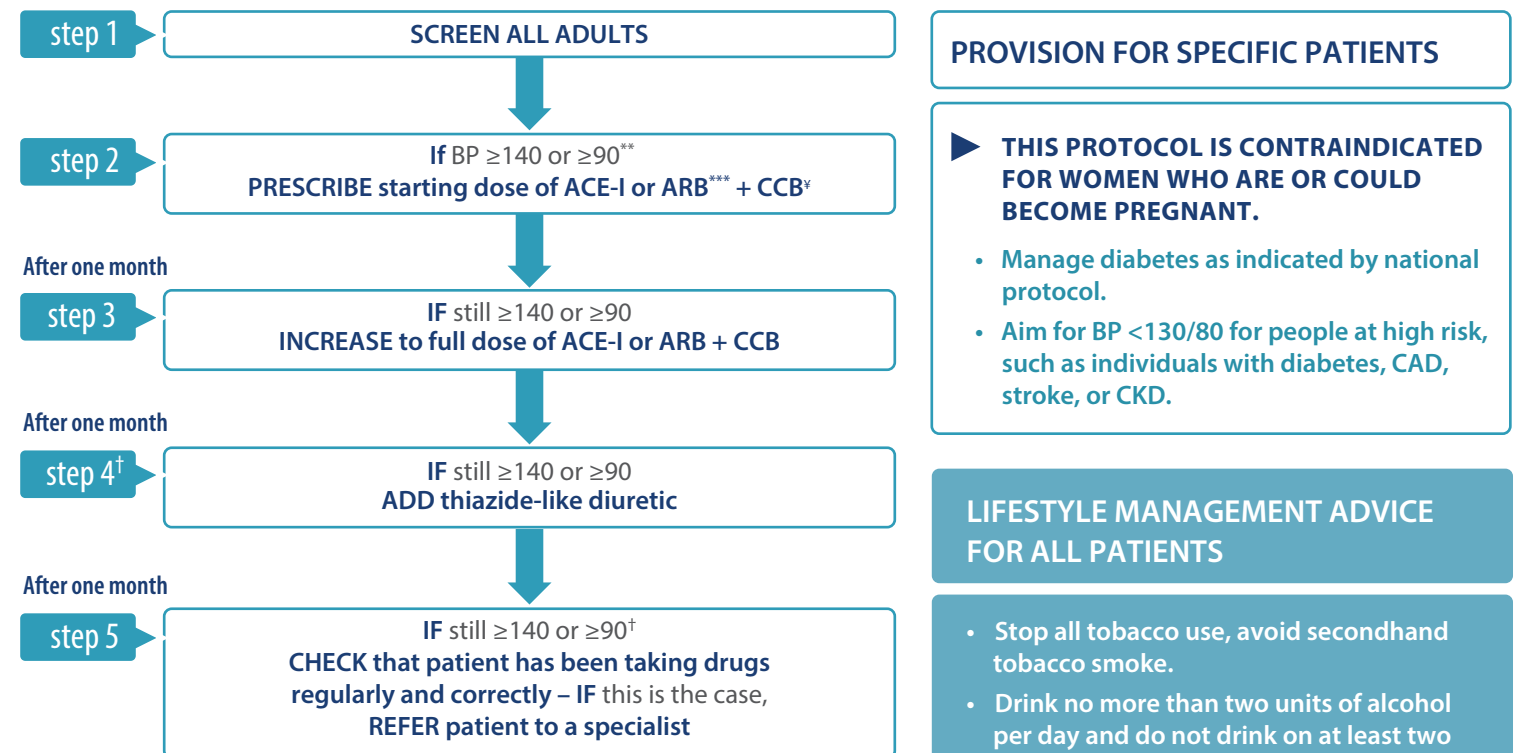
Advantages:

- Single-pill combination medication increases ease of logistics and reduces patient barriers;
- Reduces the number of steps in the protocol, increasing the likelihood of control of blood pressure;
- Reduces number of pills and increases adherence as most patients will require multiple anti-hypertensive medications;
- Benefits some patients with kidney disease, prior myocardial infarction, and low ejection fraction;
- Reduced incidence of ankle edema compared to CCB use without ACE-inhibitors or ARBs.

Disadvantages:

- Single-pill combination therapy may be more expensive than other hypertension medications given as single agent drugs;
- Not all patients would have required both medications;
- A persistent cough is experienced by up to 10% of patients treated with an ACE inhibitor; risk is higher in people of African descent;
- Small risk of angioedema; risk is higher in people of African descent;
- ACE inhibitors (and ARBs) should NOT be given to women who are or who may become pregnant;
- Risk of hyperkalemia, especially if patient has CKD.

Note: Research suggests that ARBs may have effectiveness similar to ACE inhibitors.



DRUGS AND DOSES‡

Class	Medication	Starting dose	Intensification dose
ACE inhibitor [§] (angiotensin-converting-enzyme inhibitor)	lisinopril	20 mg	40 mg
	ramipril	5 mg	10 mg
	perindopril	4–5 mg	8–10 mg
ARB [§]	losartan	50 mg	100 mg
	telmisartan	40 mg	80 mg
CCB (calcium channel blocker)	amlodipine	5 mg	10 mg
diuretic [§] thiazide-like	chlorthalidone [¶]	12.5 mg	25 mg
	or indapamide SR [¶]	1.5 mg	stay at 1.5 mg

* ACE-I or ARB according to local guidelines/costs/intolerance to ACE-I. ACE-Inhibitors cause chronic cough in approximately 10% of patients. Neither ACE-I nor ARBs should be given to pregnant women.

** Or other BP target, as determined by clinical factors. If BP ≥ 160 or ≥ 100 , start same day. If 140–159 or 90–100, check on a different day and if still elevated, start.

*** Consider statin use.

† The two medications can be used as two free agents or as a single pill combination (SPC), accordingly.

‡ Consider increase to intensification dose diuretic. Hypokalaemia more common using intensification dose diuretic – consider increased lab monitoring.

§ These are suggested examples of medications based on scientific evidence, once-daily suitability, common usage, and availability.

§ Before initiating and several weeks after starting ACE-Is, ARBs or diuretics, consider checking serum creatinine and potassium.

¶ If neither diuretic agent is available, hydrochlorothiazide can be used (25 mg starting, 50 mg full) or indapamide (1.25 mg starting, 2.5 mg full) can be used.

HYPERTENSION PROTOCOL

CCB + diuretic as first-line treatment



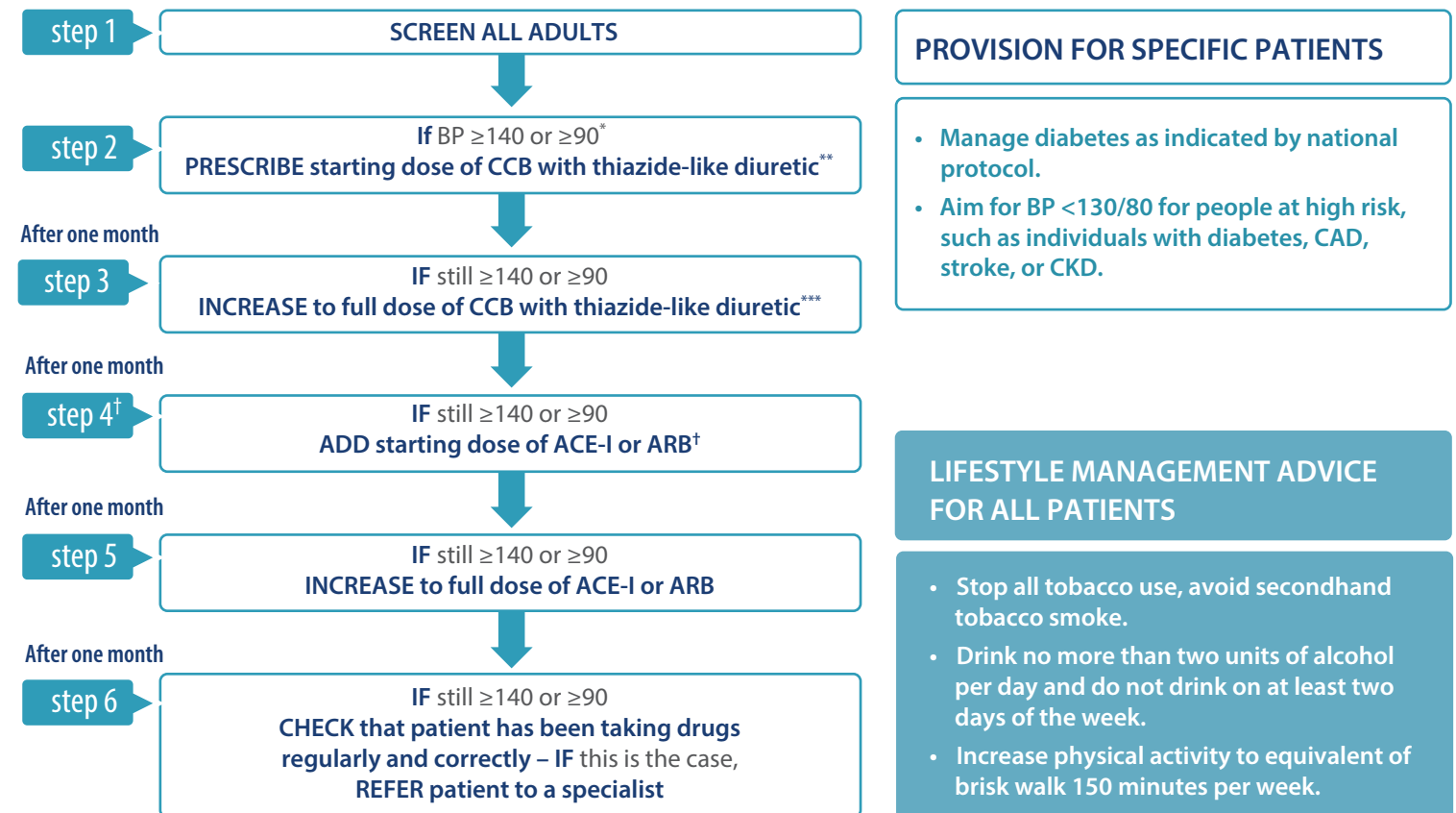
Box 5: Advantages and disadvantages of CCBs and diuretics

Advantages:

- Single-pill combination medication increases ease of logistics and reduces patient barriers;
- Reduces the number of steps in the protocol, increasing the likelihood of control of blood pressure;
- Reduces number of pills and increases adherence as most patients will require multiple anti-hypertensive medications;
- Probably effective for all races; may be more effective than other combinations in people of African descent.

Disadvantages:

- Single-pill combination therapy may be more expensive than other hypertension medications given as single agent drugs;
- Not all patients would have required both medications;
- Risk of hypokalemia;
- Diuretics have unfavourable effects on lipid/glucose measurements; clinical significance unclear;
- Ankle edema may occur in up to 10% of patients, particularly with intensification dose in the absence of ACE-inhibitors or ARBs.



DRUGS AND DOSES[‡]

Class	Medication	Starting dose	Intensification dose
CCB (calcium channel blocker)	amlodipine	5 mg	10 mg
diuretic [§] thiazide-like	chlorthalidone [‡]	12.5 mg	25 mg
	or indapamide SR [‡]	1.5 mg	stay at 1.5 mg
ACE inhibitor [§] (angiotensin-converting-enzyme inhibitor)	lisinopril	20 mg	40 mg
	ramipril	5 mg	10 mg
	perindopril	4–5 mg	8–10 mg
ARB [§]	losartan	50 mg	100 mg
	telmisartan	40 mg	80 mg

* Or other BP target, as determined by clinical factors. If BP ≥ 160 or ≥ 100 , start same day. If 140–159 or 90–100, check on a different day and if still elevated, start.

** Consider statin use.

*** Consider increasing to intensification dose of thiazide-like diuretic. Hypokalaemia more common using intensification dose diuretic – consider increased lab monitoring.

† ACE-I or ARB according to local guidelines/costs/intolerance to ACE-I. ACE-Inhibitors cause chronic cough in approximately 10% of patients. Neither ACE-I nor ARBs should be given to pregnant women.

‡ These are suggested examples of medications based on scientific evidence, once-daily suitability, common usage, and availability.

§ Before initiating and several weeks after starting ACE-Is, ARBs or diuretics, consider checking serum creatinine and potassium.

⌘ If neither diuretic agent is available, hydrochlorothiazide can be used (25 mg starting, 50 mg full) or indapamide (1.25 mg starting, 2.5 mg full) can be used.

ACE-I or ARB* + diuretic as first-line treatment



Box 6: Advantages and disadvantages of ACE inhibitor plus diuretic

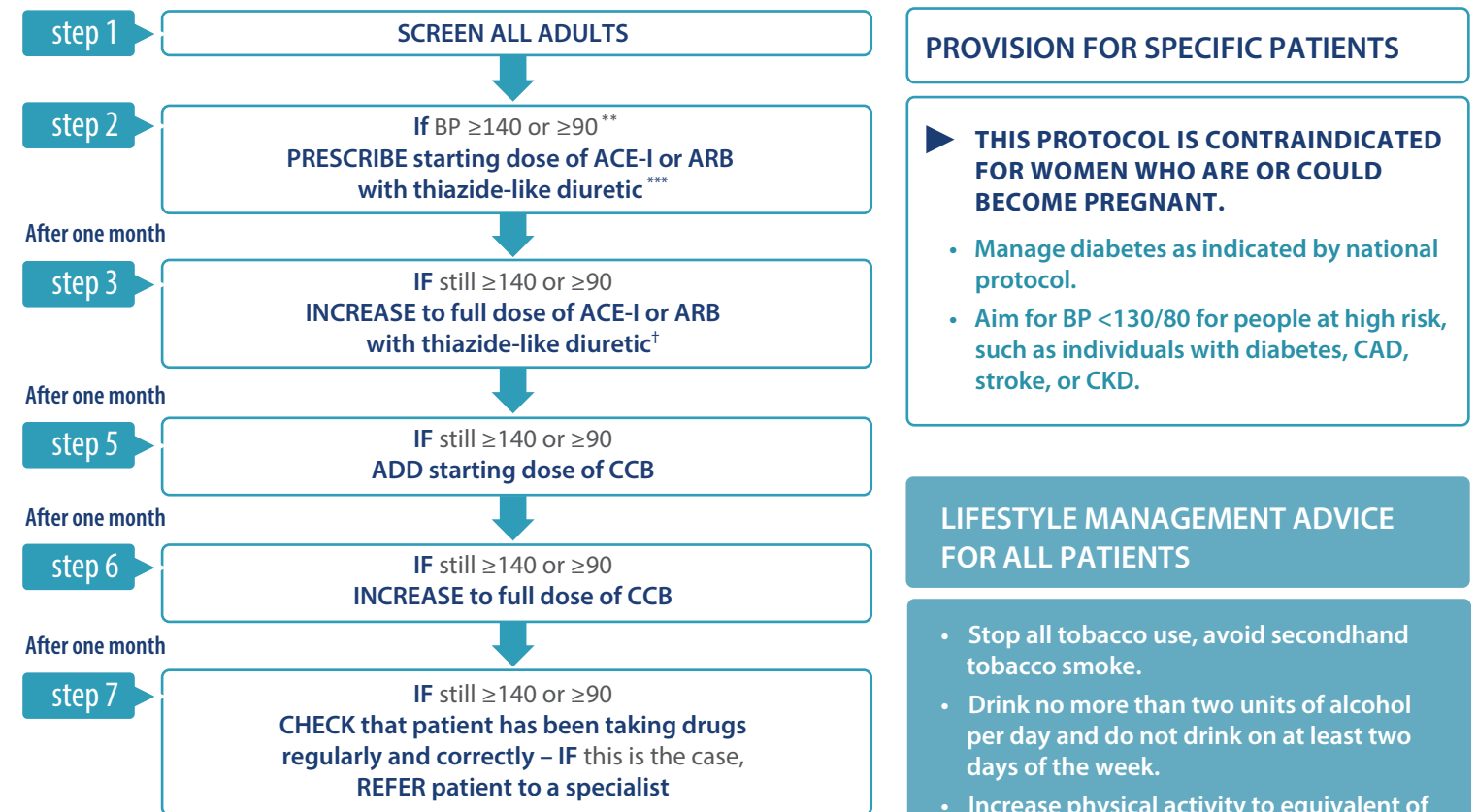
Advantages:

- Single-pill combination medication increases ease of logistics and reduces patient barriers;
- Reduces the number of steps in the protocol, increasing the likelihood of control of blood pressure;
- Reduces number of pills and increases adherence as most patients will require multiple anti-hypertensive medications;
- Likely to be effective for all races;
- Use of ACE inhibitor reduces risk of hypokalemia with diuretic;
- ACE inhibitors benefit some patients with kidney disease, prior myocardial infarction, and low cardiac ejection fraction.

Disadvantages:

- Single-pill combination therapy may be more expensive than other hypertension medications given as single agent drugs;
- Not all patients would have required both medications;
- Up to 10% of patients treated with an ACE inhibitor experience a persistent cough; risk is higher among people of African descent;
- There is a small risk of angioedema with an ACE inhibitor; this risk is higher among people of African descent;
- Risk of hyperkalemia, especially if patient has CKD, but less than when ACE inhibitor or ARB used without diuretic;
- ACE inhibitors (and ARBs) should NOT be given to women who are or who may become pregnant;
- Diuretic has unfavourable effects on lipid/glucose measurements; clinical significance unclear.

Notes: Research suggests that ARBs may have effectiveness similar to ACE inhibitors.



DRUGS AND DOSES †

Class	Medication	Starting dose	Intensification dose
ACE inhibitor [§] (angiotensin-converting-enzyme inhibitor)	lisinopril	20 mg	40 mg
	ramipril	5 mg	10 mg
	perindopril	4–5 mg	8–10 mg
ARB [§]	losartan	50 mg	100 mg
	telmisartan	40 mg	80 mg
diuretic [§] thiazide-like	chlorthalidone [‡]	12.5 mg	25 mg
	or indapamide SR [‡]	1.5 mg	stay at 1.5 mg
CCB (calcium channel blocker)	amlodipine	5 mg	10 mg

* ACE-I or ARB according to local guidelines/costs/intolerance to ACE-I. ACE-Inhibitors cause chronic cough in approximately 10% of patients. Neither ACE-I nor ARBs should be given to pregnant women.

** Or other BP target, as determined by clinical factors. If BP ≥ 160 or ≥ 100 , start same day. If 140–159 or 90–100, check on a different day and if still elevated, start.

*** Consider statin use.

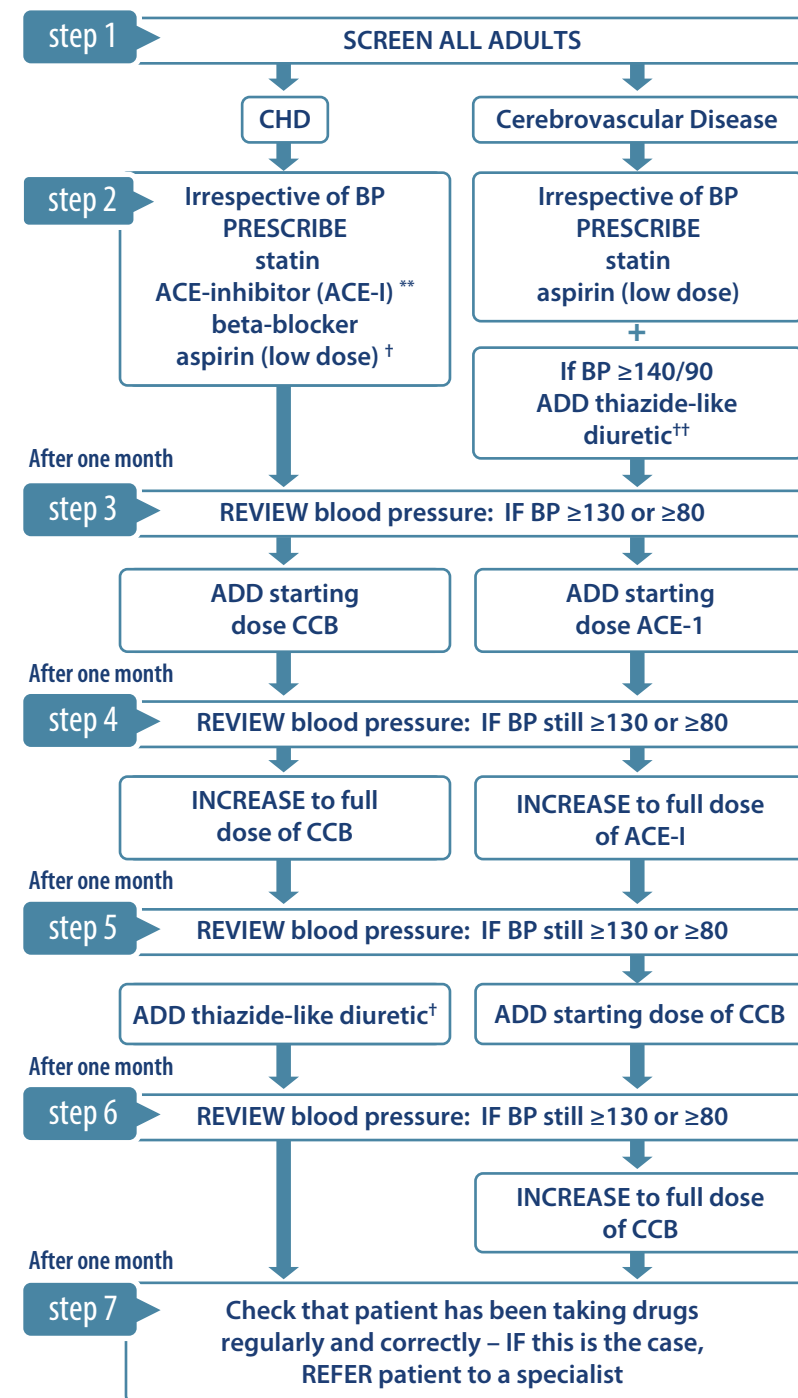
† Consider increasing to intensification dose of thiazide-like diuretic. Hypokalaemia more common using intensification dose diuretic – consider increased lab monitoring.

‡ These are suggested examples of medications based on scientific evidence, once-daily suitability, common usage, and availability.

§ Before initiating and several weeks after starting ACE-Is, ARBs or diuretics, consider checking serum creatinine and potassium.

‡ If neither diuretic agent is available, hydrochlorothiazide can be used (25 mg starting, 50 mg full) or indapamide (1.25 mg starting, 2.5 mg full) can be used.

Use of BP-lowering drugs in patients with ischaemic CVD*



PROVISION FOR SPECIFIC PATIENTS

- Manage diabetes as indicated by national protocol.
 - Aim for BP <130/80 for people at high risk, such as individuals with diabetes, CAD, stroke, or CKD.
- People with history of atrial fibrillation or heart failure and cerebrovascular disease:
- Add beta blocker with initial treatment

LIFESTYLE MANAGEMENT ADVICE FOR ALL PATIENTS

- Stop all tobacco use, avoid secondhand tobacco smoke.
- Drink no more than two units of alcohol per day and do not drink on at least two days of the week.
- Increase physical activity to equivalent of brisk walk 150 minutes per week.
- If overweight, lose weight.
- Eat heart-healthy diet:
 - Eat a low-salt diet.
 - Eat ≥5 servings of vegetables/fruit per day.
 - Use healthy oils (e.g. olive, safflower).
 - Eat nuts, legumes, whole grains and foods rich in potassium.
 - Limit red meat to once or twice a week at most.
 - Eat fish or other food rich in omega 3 fatty acids (e.g., flax seeds) at least twice a week.
 - Avoid added sugar from cakes, cookies, sweets, fizzy drinks and juice.

DRUGS AND DOSES †

Class	Medication	Starting dose **	Intensification dose
ACE inhibitor § (angiotensin-converting-enzyme inhibitor)	lisinopril	20 mg	40 mg
	ramipril	5 mg	10 mg
	perindopril	4–5 mg	8–10 mg
ARB §	losartan	50 mg	100 mg
	telmisartan	40 mg	80 mg
diuretic § thiazide-like	chlorthalidone [‡] or indapamide SR [‡]	12.5 mg 1.5 mg	25 mg stay at 1.5 mg
CCB (calcium channel blocker)	amlodipine	5 mg	10 mg

* Ischaemic Stroke/TIA/Myocardial Infarction ≥1 month ago.

** ACE-Inhibitors cause chronic cough in approximately 10% of patients. If not tolerated, give an ARB. Neither ACE-I nor ARBs should be given to pregnant women.

† When BP <140/90, ACE inhibitor and beta-blocker should be added unless symptoms preclude.

†† Consider increasing to intensification dose of thiazide-like diuretic before further titration. Hypokalaemia more common using intensification dose diuretic – consider increased lab monitoring.

‡ These are suggested examples of medications based on scientific evidence, once-daily suitability, common usage, and availability.

§ Before initiating and several weeks after starting ACE-Is, ARBs or diuretics, consider checking serum creatinine and potassium.

‡ If neither diuretic agent is available, hydrochlorothiazide can be used (25 mg starting, 50 mg full) or indapamide (1.25 mg starting, 2.5 mg full) can be used.

Box 7: Advantages and disadvantages of CCB as first-line treatment

Advantages:

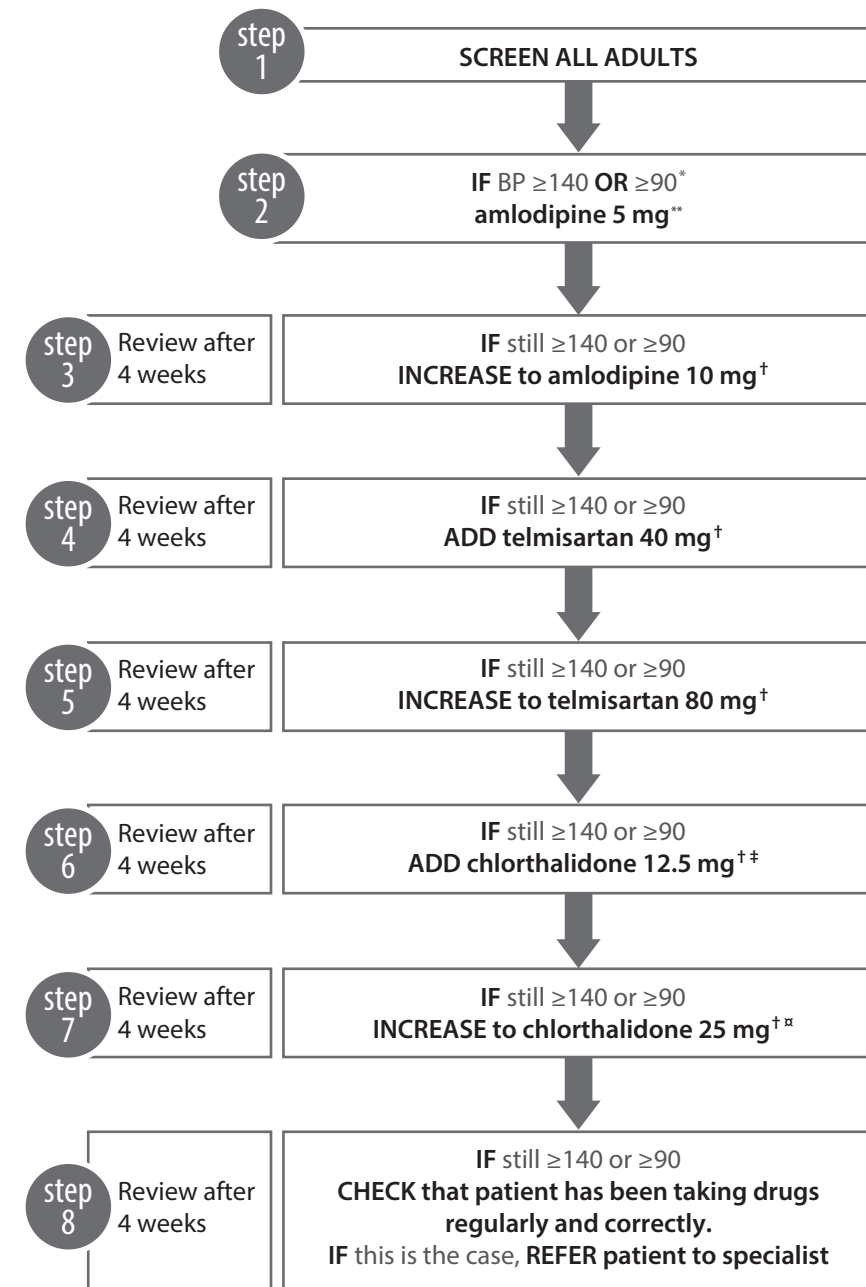
- Probably effective for all races;
- Reduces need for monitoring of electrolytes and renal function;
- Avoids need for different treatment for women of childbearing age who may become pregnant.

Disadvantages:

- May be more expensive than diuretics;
- Ankle edema may occur in up to 10% of patients, particularly with intensification dose in the absence of ARBs or ACE-inhibitors.

HYPERTENSION PROTOCOL

Adapted example: CCB as first-line treatment



PROVISION FOR SPECIFIC PATIENTS

- **Manage diabetes as indicated by national protocol**
- **Aim for BP <130/80 for people with diabetes or otherwise at high risk**
- **Start statin and aspirin in people with prior heart attack or ischemic stroke**
- **Start beta blocker in people with heart attack in past 3 years**
- **Consider statin in people at high risk**

LIFESTYLE MANAGEMENT ADVICE FOR ALL PATIENTS

- Stop all tobacco use, avoid secondhand tobacco smoke.
- Avoid unhealthy alcohol intake.
- Increase physical activity to equivalent of brisk walk 150 minutes per week.
- If overweight, lose weight.
- Eat heart-healthy diet:
 - Eat less than 1 teaspoon of salt per day.
 - Eat ≥5 servings of vegetables/fruit per day.
 - Use healthy oils.
 - Eat nuts, legumes, whole grains and foods rich in potassium.
 - Limit red meat to once or twice a week at most.
 - Eat fish or other food rich in omega 3 fatty acids at least twice a week.
- Avoid added sugar.

* If BP ≥160 or ≥100, start same day. If 140–159 or 90–99, check on a different day, and if still elevated, start.

** Smaller, fragile patients should be started on 2.5 mg per day. Alternatively, amlodipine can be replaced with a thiazide diuretic (e.g., chlorthalidone 12.5 mg, indapamide 1.25 mg, or indapamide SR 1.5 mg; if neither chlorthalidone nor indapamide is available, hydrochlorothiazide 25 mg) or a once-daily angiotensin receptor blocker (ARB) (e.g., telmisartan 40 mg or losartan 50 mg) or once daily angiotensin converting enzyme inhibitor (ACE-I) (e.g., lisinopril 20 mg, ramipril 5 mg, perindopril 4 mg). ACE-I and ARB should NOT be given to women who are or who may become pregnant. Before initiating and several weeks after starting a thiazide diuretic, ACE-I, or ARB, check serum creatinine and potassium if possible.

† If systolic BP repeatedly ≤110, consider going to prior, less intensive regimen.

‡ Indapamide can be used if chlorthalidone is not available (1.25 mg starting dose, 2.5 mg intensification; for indapamide SR 1.5 mg, do not increase dose at Step 7). Hydrochlorothiazide can be used if neither of the other diuretic agents is available (25 mg starting dose, 50 mg intensification).

‡ Hypokalemia is more common using full-dose diuretic – consider regular lab monitoring. If a diuretic is used instead of amlodipine in the initial treatment, this consideration would apply earlier in the protocol.

Box 8: Advantages and disadvantages of telmisartan 40 mg/amlodipine 5 mg single-pill combination regimen as first-line treatment

Advantages:

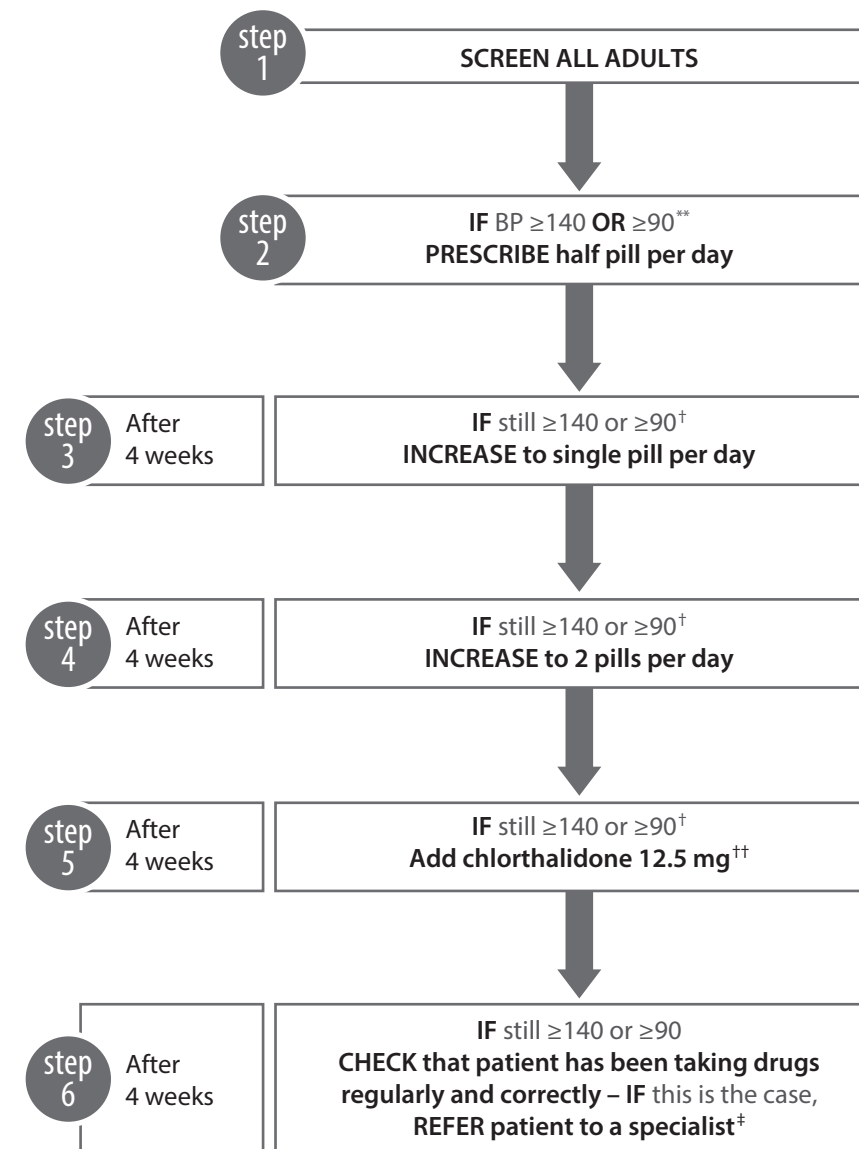
- Single-pill combination medication increases ease of logistics and reduces patient barriers;
- Reduces the number of steps in the protocol, increasing the likelihood of control of blood pressure;
- Reduces number of pills and increases adherence as most patients will require multiple anti-hypertensive medications;
- Benefits some patients with kidney disease, prior myocardial infarction, and low ejection fraction;
- Reduced incidence of ankle edema compared to amlodipine use without telmisartan (or other ARBs and ACE-inhibitors).

Disadvantages:

- Single-pill combination therapy may be more expensive than other hypertension medications given as single agent drugs;
- Not all patients would have required both medications;
- Telmisartan (as well as other ARBs and all ACE-inhibitors) should NOT be given to women who are or who may become pregnant;
- Risk of hyperkalemia, especially if patient has CKD.

HYPERTENSION PROTOCOL

Adapted example: telmisartan 40 mg^{*}/amlodipine 5 mg[§] single-pill combination[¥] regimen



PROVISION FOR SPECIFIC PATIENTS

- **Manage diabetes as indicated by national protocol**
- **Aim for BP <130/80 for people with diabetes or otherwise at high risk**
- **Start statin and aspirin in people with prior heart attack or ischemic stroke**
- **Start beta blocker in people with heart attack in past 3 years**
- **Consider statin in people at high risk**

LIFESTYLE MANAGEMENT ADVICE FOR ALL PATIENTS

- Stop all tobacco use, avoid secondhand tobacco smoke.
- Avoid unhealthy alcohol intake.
- Increase physical activity to equivalent of brisk walk 150 minutes per week.
- If overweight, lose weight.
- Eat heart-healthy diet:
 - Eat less than 1 teaspoon of salt per day.
 - Eat ≥5 servings of vegetables/fruit per day.
 - Use healthy oils.
 - Eat nuts, legumes, whole grains and foods rich in potassium.
- Limit red meat to once or twice a week at most.
- Eat fish or other food rich in omega 3 fatty acids at least twice a week.
- Avoid added sugar.

* Telmisartan 40 mg can be replaced with any once-daily angiotensin receptor blocker (ARB) (e.g., losartan 50 mg) or once daily angiotensin converting enzyme inhibitor (ACE-I) (e.g., lisinopril 20 mg, ramipril 5 mg, perindopril 4 mg). ACE-I and ARB should NOT be given to women who are or who may become pregnant. Before initiating and several weeks after starting ACE-Is or ARBs check serum creatinine and potassium if possible.

§ Amlodipine can be replaced with another once-daily dihydropyridine calcium channel blockers. Alternatively, amlodipine can be replaced with

chlorthalidone 12.5 mg, indapamide 1.25 mg, or indapamide SR 1.5 mg. If neither chlorthalidone nor indapamide is available, hydrochlorothiazide 25 mg can be used. If a diuretic is used instead of amlodipine, check serum potassium if possible and see ‡ below.

¥ Medications can be used as individual agents if single-pill combinations are not available.

** If BP ≥160 or ≥100, start same day. If 140–159 or 90–99, check on a different day, and if still elevated, start.

† If systolic BP repeatedly <110, consider going to prior, less intensive regimen.

†† If a diuretic is used initially instead of amlodipine, then amlodipine or another once-daily dihydropyridine calcium channel blocker would be used at this step.

‡ Consider increase to full-dose diuretic (chlorthalidone 25 mg or indapamide 2.5 mg; indapamide SR 1.5 mg is both the start and the full dose). Hypokalemia is more common using full-dose diuretic – consider regular lab monitoring. If a diuretic is used instead of amlodipine in the initial treatment, this consideration would apply earlier in the protocol.

2 Diabetes detection and treatment

Treatment protocols are for patients in the primary care setting with established or newly diagnosed type 2 diabetes and are derived from the WHO Package of Essential NCD interventions (27).

Diabetes risk factors and symptoms

Risk factors for type 2 diabetes include:

- overweight/obesity
- physical inactivity
- having a first-degree relative with diabetes
- history of gestational diabetes, or preeclampsia
- history of CVD, HTN, dyslipidaemia, or polycystic ovary syndrome.

On average, people have type 2 diabetes (with or without symptoms) for between five and seven years before it is diagnosed. Symptoms of diabetes include:

- polyuria (excessive passing of urine)
- polydipsia (excessive thirst)
- unexplained weight loss
- polyphagia (excessive hunger)
- vision changes
- fatigue.

Diagnostic testing for diabetes

Test adults who are 40+ years old and who are overweight (BMI >25) or obese (BMI >30)* (28), or follow national guidelines.

Fasting plasma glucose (FPG) is the most practical test for low-resource settings, given its low cost**. HbA1c can also be used, but it is considerably more costly.

Plasma glucose two hours after a 75 g oral glucose load (OGTT) can also be used to screen for and diagnose diabetes, but is less practical and more costly.

If patient is not fasting and has symptoms, a random plasma glucose (RPG) test can also be performed. It is the least accurate of the diagnostic tests. It is useful to confirm the diagnosis in person with symptoms; however, a negative test does not rule out the diagnosis of diabetes.

* Calculate body mass index (BMI, kg/m²) in all adults. For some populations (e.g., South Asian), weight-associated risks similar to those at a BMI of 25 in Caucasian populations are found at a BMI of approximately 23, and intensify at a BMI of approximately 28.

** Point of care devices can be used in diagnosing diabetes if laboratory services are not available.

Diagnostic values:

Test	mmol/l	mg/dl
Fasting blood sugar (FBS)*	≥7	≥126
Random plasma glucose (RPG)	≥11.1	≥200
Plasma glucose two hours after a 75 g oral glucose load-OGTT	≥11.1	≥200
	mmol/mol	%
Haemoglobin A1c	≥48	≥6.5%

Drug therapy

Control of blood glucose

Metformin is recommended as the first-line drug in the treatment of diabetes. Sulfonylurea is recommended as the second-line treatment, and human insulin as the third-line treatment.

Patients may require two or three drugs. Although there are other drug classes usually used as second- and third-line treatment, including thiazolidinediones (TZDs), DPP-4 inhibitors, SGLT2 inhibitors, and GLP-1 receptor agonists, these medicines tend to be more costly than metformin, sulfonylurea and insulin, with currently limited evidence of superior effectiveness. They may, however, be considered in the rare cases when treatment with metformin, sulfonylurea, and insulin is not possible. Insulin treatment should be introduced and monitored according to national practices.

Control of blood pressure and blood lipids

Hypertension treatment is indicated when SBP ≥130 and/or DBP ≥80). Statins are recommended for all people with type 2 diabetes older than 40 years, but only if this does not negatively impact access to glucose-lowering and blood pressure-lowering medication.

Monitoring glycaemic control

If diabetes is diagnosed, monitor glycaemic control every three months until diabetes is controlled, then every six months after that.

HbA1c is the most accurate measurement of long-term glycaemic control and represents the average blood glucose over the previous two to three months. HbA1c < 7% is generally considered to be adequate glycaemic control. In people with frequent severe hypoglycaemia, severe complications and low life-expectancy, the goal for HbA1c could be relaxed, e.g. to <8%.

Fasting plasma glucose (FPG) can also be used to monitor control.

Goal for glycaemic control	Plasma glucose
Fasting	≤7.0 mmol/l (126mg/dl)

* Fasting: no food and only water for 8–14 hours or more before the test.

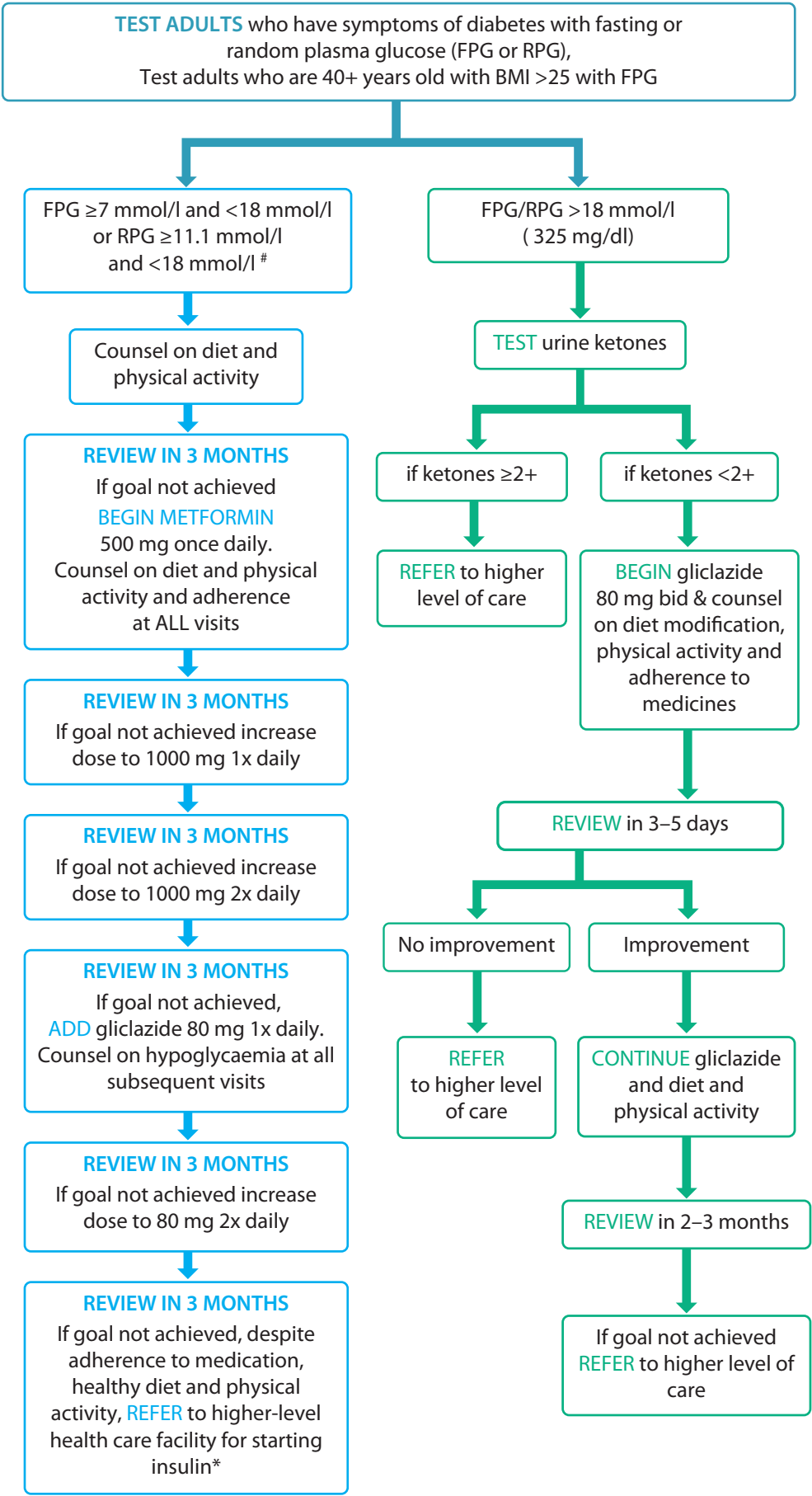
Clinical practice recommendations

- Provide counselling around lifestyle change, including diet, physical activity and smoking cessation (see *Healthy-lifestyle counselling*).
- Initiate diabetes self-management education to reinforce treatment goals.
- Prescribe aspirin for patients with CVD.
- Measure blood pressure at every visit. Treat as per hypertension protocol if $\geq 130/80$ mmHg. **Note:** if 130/80 mmHg is chosen as target, then treatment should start if over 130/80 mmHg.
- Measure weight and calculate BMI at every visit.
- Take A1c measurements every three to six months; every six months if stable on unchanging treatment.
- Arrange fasting lipid panel annually if available.
- Conduct foot exam for amputation risk annually, or every visit if high-risk.
- Conduct annual urine protein dipstick (microalbuminuria dipstick if available to calculate albumin to creatinine ratio) and serum creatinine measurement (GFR calculation) for CKD screening. CKD is defined by $\text{GFR} < 60 \text{ ml/min/1.73 m}^2$ OR the presence of moderate or severe albuminuria (albumin-creatinine ratio $\geq 30 \text{ mg/mmol}$).
- Conduct dilated pupils retinal exam every two years if treatment available.

Treatment adherence

- Explain the diagnosis of diabetes.
- Inform patient of the complications of untreated diabetes.
- Discuss the possible symptoms of diabetes.
- Show the patient the appropriate dose.
- Prescribe once-daily medications, less expensive generics, and longer-lasting supplies of medicine whenever possible.
- Explain potential adverse effects of the medications and what to do if the patient experiences them.
- Explain how many times a day the patient should take the medication and at what time, and adopt the following simple steps to help them adhere to the guidelines:
 - Label and package the tablets.
 - Check the patient's understanding before the patient leaves the health centre.
- Explain to patient how important it is to:
 - Keep an adequate supply of medications safely at home.
 - Take the medicines regularly as advised, even if there are no symptoms.
- Provide tools such as pill boxes and medication logs to help patients remember to take their medications.
- Assess adherence and discuss barriers at every visit.
- Reconcile clinician's medication list with patient's list, adjust dose, and eliminate unneeded medications.

DIABETES PROTOCOL
Type 2 diabetes management protocol derived from WHO-PEN (27, 29)



19 February 2018

SCREENING FOR CHRONIC COMPLICATIONS

- Measure blood pressure at every scheduled visit, review medication as per hypertension protocol
- REFER for dilated-pupil retinal exam upon diagnosis, and every two years thereafter, or as per ophthalmologist recommendation
- Examine feet for ulcers at every visit. REFER to higher level of care if ulcer present
- Assess risk of lower limb amputation annually (foot pulses, sensory neuropathy by monofilament, presence of healed or open ulcers, calluses). REFER to higher level of care if ulcer present or pulse absent
- Test for proteinuria annually. REFER to higher level of care if positive.

MANAGEMENT OF ACUTE COMPLICATIONS

- Severe hypoglycaemia** (plasma glucose $< 50 \text{ mg/dl}$ or 2.8 mmol/l) or signs:
- If conscious, give a sugar-sweetened drink
 - If unconscious, give 20–50 ml of 50% glucose (dextrose) IV over 1–3 minutes.
- Severe hyperglycaemia** (plasma glucose $> 18 \text{ mmol/l}$ (325 mg/dl) and urine ketone 2+) or signs and symptoms of severe hyperglycaemia:
- Set up intravenous drip 0.9% NaCl 1 litre in 2 hours; continue at 1 litre every 4 hours, REFER to hospital.

Goal for glycaemic control	Plasma glucose**
Fasting	$\leq 7.0 \text{ mmol/l}$ (126mg/dl) [†]

refer to table on diagnostic values for other tests which can be used to diagnose diabetes.
* If they are more affordable than insulin, DPP4-inhibitors, SGLT2-inhibitors or pioglitazone can be used before insulin in cases of treatment failure with metformin and gliclazide. Introduce and titrate insulin treatment according to local practices.
** HbA1c should be used where available.
† Consider less stringent glycaemic control in patients with frequent severe hypoglycaemia, advanced complications, serious comorbidities and/or limited life expectancy.

3 Identifying emergencies and need for referral

A primary care visit may also be for an emergency condition, or evolve to an emergency, and the provider can be equipped to identify and refer such patients to the nearest facility with the required capacity.

Morbidity from acute exacerbations of cardiovascular disease (including but not limited to hypertensive crisis, heart attack, stroke, and diabetic ketoacidosis) can be very time-dependent. Early recognition, resuscitation and referral of patients with acute cardiovascular emergencies is essential.

All providers delivering primary care for cardiovascular conditions should:

1. Know the nearest appropriate higher-level facilities for referral of acute cardiovascular emergencies.
2. Understand how to approach the person with acute difficulty in breathing, shock, or altered mental status (such as that taught in the WHO Basic Emergency Care course).
3. Screen each patient for danger signs that would suggest the need for immediate referral. (This screening may take 2–3 minutes in patients without acute symptoms. A simple list of danger signs is provided below, but facilities may want to use existing referral protocols).

Urgent referral criteria

- New chest pain, chest pain at time of assessment or change in severity or frequency of chest pain
- Symptoms/signs of transient ischemic attack (TIA) or stroke
- Symptoms/signs of heart failure (shortness of breath, difficulty breathing, leg swelling) or heaving cardiac apex
- Severe leg pain with symptoms of claudication
- Blood pressure >180/110 mmHg with severe headache, chest pain, shortness of breath, blurred vision, mental status changes, reduced urine output, nausea, vomiting, lethargy, seizures, papilloedema, focal neurologic signs or signs of heart failure
- Blood pressure >200/>120 mmHg
- Plasma glucose >18mmol/l (325mg/dl) and urine ketones 2+ or signs and symptoms
- Hypoglycaemia unresolved by intravenous glucose
- Suspected type 1 diabetes in a newly diagnosed patient
- Any severe infection or infected ulcer
- Recent deterioration of vision
- Anuria
- Symptomatic tachycardia.

Other referral criteria (non-emergency)

- Known prior heart disease, stroke, TIA, diabetes, kidney disease (if the patient has not had an initial assessment)
- Cardiac murmurs
- Irregular pulse

- Aged under 40 years with BP $\geq 140/90$ mmHg (to exclude secondary causes of hypertension)
- Persistent blood pressure $\geq 140/90$ (in diabetes $\geq 130/80$ mmHg) while on treatment with two or three anti-hypertensives
- Newly diagnosed diabetes with urine ketones 2+ or in lean persons of <30 years
- Diabetes with poor control despite maximal metformin dosing with or without sulphonylurea
- Diabetes with recent deterioration of vision or no eye examination in past two years
- Pregnant women with diabetes or hypertension
- Total cholesterol > 8 mmol/l (310 mg/dl)
- Any protein in the urine (proteinuria), or elevated creatinine
- Symptoms/signs of kidney problems
- Signs of peripheral vascular disease.

Annex: Example of essential CVD medications

Medicine class	Medicine	Usual starting dose (daily)	Usual intensification dose (daily)	Practice points
ACE-I (angiotensin-converting enzyme inhibitors)	lisinopril	20 mg	40 mg	If possible, check serum creatinine and potassium before starting an ACE Inhibitor Contraindications: - pregnancy - advanced CKD - aortic stenosis Common adverse effects: - cough - angioedema
	alternative: ramipril	5 mg	10 mg	
	alternative: perindopril	4–5 mg	8–10 mg	
	alternative: enalapril	20 mg	40 mg	
ARB (angiotensin receptor blockers)	losartan	50 mg	100 mg	ARBs cause cough less frequently than ACE inhibitors. Therefore, they are often substituted for ACE inhibitors when patients complain of cough with ACE inhibitors. Contraindications: pregnancy
	alternative: telmisartan	40 mg	80 mg	
Dihydropyridine calcium channel blockers (DHP CCBs)	amlodipine	5 mg	10 mg	May be more appropriate in certain ethnic groups, such as African origin Common adverse effects: ankle swelling
Thiazide/thiazide-like diuretics	chlorthalidone	12.5 mg	25 mg	Common adverse effects: - frequent urination - electrolyte imbalance
	alternative: indapamide SR	1.5 mg	1.5 mg (no increase)	
	hydrochlorothiazide (HCTZ)	25 mg	50 mg	

Medicine class	Medicine	Usual starting dose (daily)	Usual intensification dose (daily)	Practice points
Beta blockers	bisoprolol	2.5 mg	20 mg (increase 2.5 mg every 2–4 weeks)	Contraindications: - acute asthma - heart rate <55 beats/min
	alternative: atenolol	50 mg	100–200 mg after 1 week as needed	
	alternative: metoprolol	25–50 mg	100 mg	
	alternative: carvedilol	Start at 3.125–6.25 mg twice daily	50 mg	
Biguanides	metformin	500 mg with a meal	2000 mg in divided doses	Low risk of hypoglycaemia, but monitor especially in elderly patients Caution in renal impairment Contraindications: - renal failure - liver disease Common adverse effects: - nausea - diarrhoea
Sulphonylurea	gliclazide	40–80 mg with breakfast	320 mg Above 160 mg/day, take twice daily in divided doses	Gliclazide has a lower risk of hypoglycaemia compared to glibenclamide. Avoid glibenclamide if ≥60 years. Common adverse effects: - hypoglycaemia, especially in elderly people - weight gain
	glibenclamide	2.5 mg	5 mg twice daily before meals	
Lipid-lowering therapy	simvastatin	20 mg at night	40 mg at night	Common adverse effects: - muscle pain or myalgia - the maximum recommended dose for simvastatin when taken with amlodipine and diltiazem is 20 mg day
	alternative: atorvastatin	40 mg	80 mg	
Antiplatelet therapy	aspirin	75–100 mg	–	Avoid in individuals with history of major bleeding

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Technical package for cardiovascular disease management in primary health care



Access to essential medicines and technology



WHO/NMH/NVI/18.3

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Procurement and supply management activities are fundamental to consistent and reliable access to essential medicines and health products.

To reduce the impact of CVD, action needs to be taken to improve prevention, diagnosis, care and management of CVD diseases. Affordable essential medicines and technologies to manage CVD disease must be available where and when they are required.

Medicines and technologies need to be managed appropriately to ensure that the correct medicines are selected, procured in the right quantities, distributed to facilities in a timely manner, and handled and stored in a way that maintains their quality. This needs to be backed up by policies that enable sufficient quantities to be procured in order to reduce cost inefficiencies, ensure the reliability and security of the distribution system, and encourage the appropriate use of these health products.

In order to avoid stock-outs and the disruption of treatment, all related activities need to be conducted in a timely manner, with performance continually monitored, and prompt action taken in response to problems that may arise.

Additionally, medication must be dispensed correctly and used rationally by the healthcare provider and patient alike. The purpose of this guide is to explain the necessary steps.

This module focuses on access to medicines and basic equipment for CVD management, and includes:

- information on the pharmaceutical management cycle and policies
- selection of appropriate drugs and technologies
- supply-chain management, including quantification, forecasting, distribution, storage and handling
- ensuring supply and accountability
- rational use of medicines.

1 Framework and policy

Drug management requires continuous support, including policies and legislative frameworks, such as a country's national CVD programme and regulatory authority. If countries do not have a regulatory authority, the WHO List of Prequalified Medicinal Products can be consulted to ensure that quality-assured medication is being procured and utilised.

Access to essential medicines by policy makers depends on four factors:

- rational selection
- affordable prices
- sustainable financing
- reliable health systems.

Essential medicines are those medicines that satisfy the priority health care needs of the majority of a given population and are selected after consideration of their relevance to public health, their efficacy and safety, and their comparative cost-effectiveness. They are intended to be available within the context of functioning health systems and should be available at all times, in adequate amounts, appropriate dosages, with quality assurance and adequate information, and at a price that individuals and communities can afford. (1)

An essential medicines list (EML) gives priority status to medicines that address a country's most pressing public health programmes, and provides recommendations, based on quality and cost-effectiveness, to assist those tasked with purchasing medicines and products.

An essential medicines list, drawn up as part of the core principles of a national drug policy, helps to set priorities for all aspects of the pharmaceutical system. This is a crucial step in ensuring access to health care and in promoting rational use by health professionals and consumers. The implementation should be flexible and adaptable to address all levels of care:

- private and public sectors
- primary, secondary and tertiary levels
- in both rural and urban areas.

Political commitment is critical to ensuring a successful national drug policy and the selection of essential medicines.

Policy makers can do the following:

- Establish a consensus process to develop and use national EMLs (NEMs). (2)
- Use an objective, and transparent process, in conjunction with clinical standard treatment guidelines.
- Ensure funding for medicines and products on the NEM list.
- Use the NEM list to guide policies related to access.
- Regularly review and update the NEM list against the most recent WHO Model List. www.who.int/medicines/publications/essentialmedicines/20th_EML2017_FINAL_amendedAug2017.pdf?ua=1

2 Selection of drugs

Ensuring the availability of, and access to, essential medicines and health products for the management of hypertension, diabetes and increased lipids requires government commitment, guaranteed financing, and effective supply management, including procurement and service delivery.

The Hearts Technical Package lists the essential medicines for the management of hypertension, diabetes and increased lipids. (3)

- thiazide or thiazide-like diuretic
- calcium channel blocker (CCB) (long acting) (amlodipine)
- angiotensin converting enzyme inhibitor (ACE-I) (long acting) and angiotensin receptor blocker (ARB)
- statin
- insulin
- metformin
- glibenclamide
- beta-blocker
- aspirin.

When medications are not on a country EML, a country can adapt the protocols to address their specific needs. It is, however, recommended that every country's NEML reflects the list of essential CVD medicines consistent with protocols in the E module of the HEARTS technical package to ensure the effective management of CVD risk.

Addressing availability of these essential medicines within countries and at all levels of care and sectors is critically important. An analysis by Cameron and colleagues across 40 countries showed that chronic disease medicines were significantly less available than those for acute conditions, in both the public and private sectors, with this lack of availability even more evident in the public sector, as highlighted in Table 1.

Table 1: Availability of medicines for chronic and acute disease across 40 countries

Mean availability of lowest-priced generic medicines (% of facilities)		
	Chronic disease medicines	Acute disease medicines
Public	36	54
Private	55	66

Chronic Disease Medicines Acute Disease Medicines

N=2779 facilities (total of public and private outlets) across 40 countries. (4)

3 Selection of equipment

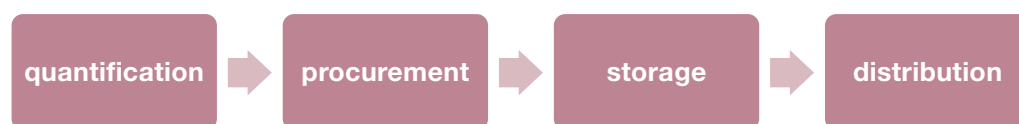
Monitoring technologies, such as glucometers and urine test strips, are excluded from most NEMLs, (5) and in many countries there is only limited availability. With the increasing burden of chronic diseases in LMICs, it will be important to include these essential technologies in lists of essential medical products, from national to global, if these diseases are to be managed effectively.

A simple essential list of technologies (6) for CVD screening and monitoring includes:

- stethoscope
- measuring tape
- weighing scale
- equipment and supplies for measuring urine albumin and ketones
- blood pressure measurement device
- equipment and supplies for measuring blood glucose and cholesterol.

4 Supply management

Figure 1: Components of a supply chain



Quantification

Quantification, comprising both forecasting and supply planning, is the process of determining the quantity of any medicine or technology to procure in a set timeframe. This includes forecasting future needs for the short-, medium-, and long term.

Effective forecasting for NCD medicines and technologies at the district and then national level starts with the number of patients currently being treated by facility for a given condition, current and anticipated burden of disease, and existing stock information. The district health management team and NCD programme coordinator should maintain estimates of the number of patients likely to come to health facilities within the district for these services (maybe based on the experiences of other health care facilities in the country).

Supply planning involves the prediction of future needs in order to ensure adequate supply, based on demand and available funding.

Information needed for supply planning includes:

- expiration dates of current available stock
- supplier details such as prices and lead times
- funding information such as source and disbursement schedule
- procurement details
- distribution details. (7)

Quantification can be challenging for cardiovascular diseases because of the lack of reliable and consistent data on the burden of disease and essential medical product (EMP) use. As the burden of disease grows, it will be even more challenging to develop accurate estimates.

Procurement

The national EML (NEML) list can serve as the basis for the training and education of health professionals and of public education about medicine use. It can also be the basis for public-sector procurement and distribution, and can influence practice in the private sector through education. The public sector or health insurance system cannot afford to supply or reimburse the purchase of all medicines available in the market, and may also struggle to know which products are qualified and are cost-effective. EMLs can be used to guide the procurement and supply of medicines in the public sector and by systems that reimburse medicine costs. Using the NEML as a starting point, care institutions such as district hospitals and primary health centres can develop their own list of medicines.

A cycle of procurement involves three interlinked activities:

- alignment of needs and procurement budget
- tendering and purchasing
- contract performance.

Procurement models commonly used

A model of centralized procurement usually leads to bulk purchasing and better prices.

Depending on the nature of health financing in a country, often a decentralized procurement model is used, whereby provinces, counties, districts or health facilities purchase directly from suppliers. While it does not have the benefits of bulk purchasing benefits found in centralized procurement systems, and prices can be high, it leads to more responsive procurement and supply.

Alternative procurement options are to use a UN agency (e.g. UNICEF or non-profit international procurement agents such as IDA, Mission Pharma and Crown Agents). In some regions, countries have joined pooled procurement groups, which buy collectively on behalf of member countries. (The Eastern Caribbean Pooled Purchasing, PAHO Revolving Fund and Gulf Coordination Council are notable examples.) Ultimately, countries will have to study which procurement model best serves the procurement of CVD products.

Tendering models used:

- International Competitive Bidding (ICB) and limited ICB
- National Competitive Bidding (NCB)
- direct purchase.

Pricing and market control

To allow for distribution, and health system and consumer confidence in the medicines and products distributed, it is recommended that countries facilitate the registration of quality and cost-effective EMPs. The WHO Model list of Essential Medicines can be a guide to the development of an NEML. Along with regulatory approval, an NEML is a key factor in determining procurement and hence supply and availability of EMPs at the country level as well.

The availability, price, and affordability of these EMPs are indicators of access to treatment for CVDs, so it is important to assert market control over these medicines and health products. Price is assessed as the median price ratio (MPR), which compares the local price for a product to the international reference price (IRP) for the product,

Box 1: Variations in pricing

In an analysis of affordability of cardiovascular medicines in 36 countries, mean MPR for public sector procurement of hydrochlorothiazide (HCT) was 9 times the IRP, revealing that governments in Africa procured HCT at prices, on average, 9 times that of the IRP, though this varied widely within the region. (8)

A diabetes-focused secondary analysis of 32 country surveys found that the median public sector MPR for glibenclamide was 1.1 times the IRP, with a range between 0.27 in Chennai, India to 17.37 times the IRP in Nigeria. (9) Median public-sector procurement price for metformin was below the IRP, with a median MPR of 0.72 across 21 surveys, ranging from 0.17 in Chennai, India to 8.17 times the IRP in Shandong, China. In another survey, of 36 countries, the mean MPR for public-sector procurement of glibenclamide was 2.15 times the IRP, with mean MPR the highest in the African region at 3.42 times the IRP. (10)

thereby demonstrating how much the procurer or end-user pays for the product as compared to the IRP.

Box 2: Case study of mark-ups

Mark-ups in price from procurement to the end-user purchase point vary widely and significantly affect affordability. An analysis of the affordability of metformin and captopril demonstrated the unaffordability of these essential medicines. In only 2 of the 25 countries surveyed did a month's course of these two medicines together cost less than a day's wage. The price in Ghana was the least affordable, with these two medicines costing 15 days' wages for a 30-day supply. (11)

In addition, limited public sector availability of EMPs often means that people have to purchase these at private retail stores, which, if unregulated, can lead to higher prices that make some essential products unaffordable. (10)

Mark-ups in price along the supply chain, from procurement to end-user purchase point, can impact affordability significantly, and wide variations in pricing reflects variations in regulation, procurement efficiency, and market conditions within countries. Governments can exercise market

control in order to control these mark-ups in price to allow for greater affordability of these EMPs.

Storage

Medicines and related supplies need to be stored correctly for optimal performance. A sound storage system preserves a product's physical integrity, stability and efficacy, ensuring it reaches the patient in usable condition while minimizing waste and loss.

The appropriate storage of EMTs include the following key steps:

- Maintaining a cold chain and proper storage temperatures when required.
- Maintaining a secure storage space that can be locked and is cool, dry, well maintained and well organized to act as a pharmacy store. The storage space should be separate from the area where medicines are dispensed. If a health care facility does not have a room to use as a pharmacy store, a lockable cupboard or cabinet with shelves should serve as the "store".
- Monitoring stock levels and their entry and exit at facility level (health facility and storage facility) through storeroom recordkeeping, using tools such as stock cards. Recordkeeping will aid understanding of the flow of supplies into and out of a health care facility and will also enable staff to know:
 - what items are in stock
 - the quantity of each item in stock
 - how much stock is used on a weekly, monthly and quarterly basis
 - when and how much of each item should be reordered.
- Ensuring traceability of medicines and equipment to prevent loss.

At the national level, it is important to ensure that processes for clearing customs are not onerous and that custom duties and value added tax imposed on EMPs do not create a barrier for these essential drugs to be procured. These fees increase the end price of the products for both the national procurers and the end user, and can discourage importation of these essential medicines and technologies.

Box 3: The importance of national and county-level commitment

The Kenya Health Sector Strategic and Investment Plan (KHSSP) of July 2013 to June 2017 identifies the Kenya Package for Health (KEHP) service package for providing essential health services by level of care. Under the KEHP service package, general outpatient services for the "Management of Endocrine and Metabolic Conditions (Diabetes mellitus, Hypothyroidism, Hyperthyroidism)" are to be provided at the primary care level. (12)

While the sampling size was limited, survey data found little to no availability of diabetes medicines and technologies at primary level health care facilities in the public health sector. The one exception was Nyeri County, where

county health care officials have chosen to promote the availability of diabetes medicines and technologies at primary-level health care facilities.

For example, for metformin, Nyeri county and Nairobi County had 100% availability, while Kisumu, Vihiga, and Homa Bay had 0% availability. For both amlodipine and blood glucose test strips Nyeri County had 100% availability and the other counties had 0% availability. These findings illustrate the impact that a commitment by county-level health officials can have on improving availability of essential medicines and technologies at primary-level health care facilities. (12)

Box 4: Distribution challenges

In Kyrgyzstan, one study found that insulin supplies were sufficient to meet country demand, but distribution processes were ineffective and inefficient. Distributors provided facilities with whatever insulin was available, regardless of whether or not it was the type ordered, forcing patients to switch their insulin regimens and sometimes use incorrect syringes for the type of insulin they were using. In addition, distribution only took place every three months, so facilities limited their distribution of supplies to avoid stock-outs. (13)

Distribution

Distribution can often be the most challenging stage of the supply chain, as it is dependent on local infrastructure. At the national level, distribution consists of transporting EMPs through the supply chain, from the arrival port and customs, to the central warehouse, the regional/ district warehouse, and onwards to the health facility or other final destination.

5 Ensuring a secure supply

Market dynamics is a set of skills and approaches used to evaluate, and thereby improve access to, products and services. It involves regular interactions that take place among key stakeholders (producers, distributors, purchasers and consumers) and are based on the effectiveness and efficiency of key attributes of market health: affordability, availability, assured quality, appropriate design, and awareness. Such an analysis might lead to inefficiencies being identified in the way the market operates, and potential interventions being devised that would ultimately lead to improved health outcomes.

Efforts to shape the market in global health care seek to support sustainable access to medicines and technologies by catalysing the development of new markets and/or improving existing ones. For essential CVD medicines and technologies, investments in market shaping may help to further catalyse access to existing products in low- and middle-income countries. The following recommendations use a market dynamics perspective to identify and address existing market shortcomings.

Improve demand visibility to ensure supply availability

Investing in routine collection of market information, and coordinating the messaging of demand expectations to members of industry (both distributors and manufacturers) early and often can help mitigate supplier risk as well as ensure consistent supply.

In the public health sector, manufacturers and distributors usually lack accurate forecasting information to effectively manage production for and sales to LMICs. Even with a number of generic CVD medicines available in the market, supply cannot be assured without better predictability of demand.

Conduct routine market analyses to improve local decision-making

Collecting broad market information can help ensure access to CVD medicines.

For instance, before procuring medicines or devices, it is a good idea to understand the supply landscape (e.g., available brands and/or models, manufacturing capacity, pricing and quality). Note that landscaping exercises can be resource intensive, particularly for individual countries to support. Therefore, before a separate process is initiated, it is worth making enquiries with global partners such as the Global HEARTS programme or UNICEF supply division, who often maintain such supply landscapes for global use.

Improved market information increases visibility throughout the supply chain, which can ensure proactive stock management at all levels of care, and enable informed decision-making for potential improvements.

For example, assessing import taxes and tariffs, the margins of local sales agents or distributors, and shipping costs may help identify pain points in the supply system for CVD medicines and technologies, as well as potential opportunities to improve the efficiency of local distribution.

An understanding of the market landscape may also help purchasing agents justify tenders as single or split awards, based on an understanding of immediate purchasing needs, minimum production requirements for manufacturers, and opportunities for price reductions.

Finally, improved market information not only serves domestic decision-making, but can also inform collaboration with a broader set of regional or global stakeholders. Well-articulated need and potential demand can contribute to global discussions to improve access to and further prioritize activities and investments for CVD medicines and technologies.

Improve confidence in the market by providing predictable financing

Making public-sector investments in longer-term financing strategies may help optimize and allocate resources most efficiently. Such investments will also establish demand expectations and improve predictability for suppliers, based on realistic budget constraints.

Limited and unpredictable financing of CVD medicines and technologies by governments can discourage suppliers from engaging in LMIC markets.

Strategically categorizing and prioritizing health technologies can help key decision-makers determine the most appropriate model for technology ownership (e.g., rent, lease, or buy), and the best strategies for sourcing maintenance (insourcing vs. outsourcing) and for contracting ongoing service.

Generate demand by increasing awareness

The generation of demand is an essential component in improving the market for essential CVD medicines and technologies. Increased incentives for healthcare providers, pharmacists and/or retail drug shop owners to screen and identify patients with symptoms, to stock the required screening tools and treatments, and to maintain affordable prices in delivery of these services will help drive awareness for the prevalence of CVD, and increase demand. In addition, awareness-raising for secondary prevention and key risk factors, adherence support services, and real-time medicines and technology alerts for patients are programmatic efforts that link closely to the effectiveness of efforts to increase demand for these services in this area.

6 Ensuring accountability

Maintaining good records for the steps involved in ordering, receipt, storage, inventory control and dispensation is critical to ensuring an overall accountability regarding drug supply at your health centre. This is especially important for CVD medicines and technologies, as they are expensive and at higher risk of pilferage and diversion.

Maintain all your documents and records

Apart from regularly updated stock-control cards, the following documents should be maintained and organized into files that can be easily accessed:

- daily use/dispensing register
- stock control register
- requisition/delivery form
- any other correspondence with District, Regional or Central Stores or other supply sources regarding drug supplies.

Track performance

When time permits, also try to estimate performance measurements for your health centre, such as:

- percentage of individual items ordered in each month or quarter that were delivered on time and in the full quantities requested;
- for each item, the quantity ordered/requested vs the quantity received in each month;
- the number of days when a product was not in stock in your health centre.

Discuss these metrics routinely with the Medical Officer in Charge (MOIC) of your health centre and also with the district or regional supplies officer/pharmacist.

Cycle counts

Cycle counts should be performed regularly to ensure that the quantity of medicines actually in stock matches up the quantity of a drug the health facility should have on the shelf, based on stock registers. Cycle counts can be done on randomly selected items, on items that are high cost or at higher risk of theft/diversion, or items where routine review by the MOIC reveals some discrepancies.

Conduct audits and maintain supervision

Every few months the MOIC should conduct a self-audit:

- Ensuring that drugs are properly arranged on shelves in the dispensary.
- Tallying records of receipts and cross-checking with stock-control cards.
- Tallying records from daily issues register and cross-checking with the stock-control cards/stock register.
- Recalculating at random a few of the requisition quantity figures to ensure that they are correct.
- Asking questions about the reasons for any items not being in stock.

Cold/refrigerated products

Insulin requires refrigeration to maintain stability. The MOIC should periodically review the temperature log book, which has records of temperature/refrigerator status. If it is found that a product that requires refrigeration was either out of the fridge for longer than the recommended amount of time or the refrigerator was not functioning, the product needs to be discarded. This can be discussed with the district or provincial pharmacist.

The MOIC should also ensure that adequate security measures for the drug store/dispensary are always followed: e.g. locking when staff leave for the day, vetting of substitute staff on days the main assigned person is on leave. They should also follow up regarding how to fulfil the training needs of the person responsible for drug supplies at the health facility.

Box 5: Maintenance of equipment

It is important to regularly check the devices and supplies used. For example, the appropriate maintenance of equipment is critical for accurate measurement of BP. The equipment – whether aneroid, mercury, or electronic – should be regularly inspected and validated to eliminate conditions that could cause the blood pressure measurement to give a reading that was erroneously high or low.

Likewise, glucose monitors should be regularly checked for device malfunctions. Temperature or environmental error codes on the glucose meter should alert the operator to check the test strips using control samples to verify reagent performance prior to further patient testing. Improper storage and use of urine and glucose testing, rough handling, and expired test strips are some of the most common sources of measurement error.

7 Rational medicine use and adherence

Rational use of medicines is when “patients receive medications appropriate to their clinical needs, in doses that meet their own individual requirements, for an adequate period of time, and at the lowest cost to them and their community”.

Irrational (inappropriate, improper, incorrect) use of medicines is when any of these conditions are not met.

While evidence-based treatment protocols exist to guide the rational use of medicines for the treatment of high blood pressure, diabetes, high blood cholesterol and the reduction of CVD risk, rational use can be affected by a number of factors, and influenced by health workers and patients. Given the chronic nature of these diseases, short-term changes in individual requirements affecting dosage, and long-term changes due to co-morbidities and advancement of disease mean that the rational use of medication must be regularly assessed.

There are several tools to promote the rational use of medicines in health facilities:

- **Clinical guidelines:** The HEARTS Technical Package’s *Evidence-based protocols* module provide clear guidance on treatment regimens for high blood pressure, diabetes, high blood cholesterol and various levels of risk and co-morbidities. These resources can be used to establish treatment protocols for CVD, if national protocols do not exist.
- **Essential Medicines List:** The essential medicines and technologies lists in this module are harmonized with WHO Model List of Essential Medicines and evidence-based recommendations for CVD management. An NEML and a national EMP list will provide a core selection of medicines and products for procurement and ordering.
- **Health worker education:** Training can be offered through HEARTS training modules or local programmes so that staff continuously update their knowledge base and practice.
- **Use of independent information on medicines:** Use reliable sources of information, such as WHO’s technical guides and national guidelines, to guide practice.
- **Public education about medicines and technologies:** Patient education is essential in order to achieve long-term rational use of medications. The role of clinical pharmacists in counselling and monitoring a person while dispensing their medication to them is invaluable in terms of making sure the medication has the maximum effect. Educate the public about good generic medications, which it is perfectly acceptable to use.

To support rational use, the dispenser is responsible for following protocol and for maintaining attention to detail in each step of the dispensing process, as described earlier in this section. Errors and problems in dispensing can occur, so it is important to keep these in mind when reviewing dispensing practices, and identify ways of preventing or resolving these issues.

Potential problems include the following (this list is not exhaustive):

- wrong interpretation of the prescription (or diagnosis)
- retrieval of the wrong drug from stock
- wrong dosages
- inadequate packaging/labelling of proprietary drugs
- inaccurate counting
- inadequate or absent labelling
- no knowledge of proper drug compliance
- poor patient understanding of the disease and its effect on their body
- insufficient time to talk with patients about their medication
- inability to communicate with patients about the treatment regimen.

Bulk dispensing should also be considered as a way of improving adherence in stable patients. This requires further investigation into your current dispensing system and ability to handle bulk dispensing.

It is important to note the additional requirements for proper storage, monitoring and patient education regarding insulin. Devising a system to distinguish between medication packages that are similar in appearance can help avert mistakes.

Rational use is influenced by availability, affordability, financial constraints, patients’ sociocultural backgrounds, co-morbidities, age, sex, drug allergies, personal preferences, and health beliefs such as “if I don’t have symptoms I am not sick”. It is important for the dispenser to be aware of these factors and be prepared to address rational use with patients, using an appropriately sensitive approach.

Pharmacovigilance

Pharmacovigilance is the science and activity relating to the detection, assessment, understanding and prevention of adverse effects or any other possible medicine-related problems. Pharmacovigilance aims to:

- improve patient care and safety in relation to the use of medicines and all medical interventions;
- improve public health and safety in relation to the use of medicines;
- detect problems related to the use of medicines and communicate findings in a timely way;
- contribute to the assessment of benefit, harm, effectiveness, and risk of medicines, leading to the prevention of harm and maximization of benefit;
- encourage the safe, rational and more effective (including cost-effective) use of medicines;
- promote understanding, education, and clinical training in pharmacovigilance and its effective communication to the public. (14)

Box 6: Adverse Drug Event (ADE)

The Adverse Drug Event Prevention Study Group reported that the odds ratio of a severe ADE in relation to cardiovascular medication was 2.4 times that of other medications.

An error at any step of the dispensing process, poor patient adherence to medication, and adverse drug events (ADEs) can negatively impact the efficacy and safety of a treatment regimen. Good pharmacovigilance requires adhering to good dispensing practices (14), as described in this section, namely:

- Assessing patient adherence to medication.
 - Provide patient education and/or referral as needed to establish and maintain proper adherence.
- Ensuring patients (and/or families/caregivers) are aware of possible adverse reactions to prescribed medications, and explaining what to do in the event of an adverse reaction.
 - Provide additional education as required. Refer patients to care for management of adverse reactions as needed. Document any relevant adverse reactions.
- Being alert to and advising patients on potential drug–drug or food–drug interactions. Additionally, being aware that comorbidities or acute illness (e.g., infection) can alter blood pressure or blood glucose and may require adjustment to treatment regimens.
 - Provide education on signs and symptoms of ADEs commonly caused by drug–drug or food–drug interactions and refer to care as needed.

Note: Attention to food–drug ADEs is especially important for people with diabetes requiring insulin injections. Assess patient understanding of the interactions between diet, dosage, and timing (time between insulin dose and food intake). Assess the patient’s food security and provide additional education as necessary and appropriate.

Record each patient’s medication history, including complementary therapies and current medications (obtained both at your facility and elsewhere) to enable identification of potential drug–drug interactions or duplication of medications.

If a potential drug–drug interaction or duplication is identified, contact the patient’s prescribing physician or refer the patient to care to re-evaluate the current medication regimen and prevent an ADR.

Aids such as posters, patient education materials, interview guidelines and materials for taking medication histories can support facility uptake and implementation of good pharmacovigilance efforts.

Annex A: Ordering supplies

Ensuring that a health facility is adequately stocked with medicines and health technologies to serve CVD patient needs must be balanced with preventing the overstocking of medicines that could expire before they are used. Over time, it will be essential to establish new data systems and automated decision-making to ensure that quantities ordered are “in sync” with a patient’s clinical stage and other information obtained from patient registers and electronic records. This section of the facility guide provides a step-by-step approach for such facilities to decide when to order and how much to order.

Step 1: Compile a list of essential medicines and technologies relevant to the HEARTS package

Compile a list of medicines supplied by the central/district agency in charge of medicines supply. Quantities cannot be calculated until you know which products are to be ordered.

Here is the list of medicines and products recommended as part of the HEARTS package:

- thiazide or thiazide-like diuretic
- calcium channel blocker (CCB) (long acting) (amlodipine)
- beta-blocker
- angiotensin converting enzyme (ACE) inhibitor (long acting) and angiotensin receptor blocker (ARB)
- statin
- insulin
- metformin
- glibenclamide
- aspirin.

Obtain a copy of the National Essential Medicines List (NEML) and the supply catalogue of the supply agency (central or regional store, or other supplier) from which your health facility orders its stock. Using the list in Table 1, develop a detailed list of products, including the following information.

- product generic name, or INN
- dosage form (tablet, dispersible tablet, test strip)
- strength or concentration
- basic units (tablet, pack)
- pack sizes available/to be stocked.

Treatment protocols and comorbidities for many of the conditions are such that they create strong interdependencies between the items on this list. For example, patients with hypertension and CKD will often be treated concomitantly with a thiazide-like diuretic or CCB; and an ACE inhibitor or ARB. Furthermore, if a tender process is used it would be important to obtain (as far as possible) the same medication (e.g. same CCB) to ensure that practitioners and patients can identify the medication easily.

Non-availability of any one of the three of four single drugs required by a hypertensive patient with, say, a comorbidity of diabetes or CKD, has consequences for the health outcomes, sometimes serious ones. Similarly, non-availability of complementary products such as glucometer strips can put an entire treatment programme in jeopardy.

In some cases, two, three or four medicines may be combined in one tablet, known as a fixed-dose combination (FDC), which eases this problem. But the availability and use of FDCs for most CVDs is currently very limited for a variety of reasons, including availability of these products, trial and research data, and the nature of customization in therapy required for each patient.

Step 2: Determine how often your health care facility receives deliveries

If your source of supply (central/regional store, district hospital, other source) routinely delivers supplies to your health facility, then how often do they deliver: monthly / 3-monthly?

If someone from your health facility travels to the regional/district to obtain supplies, how often do they go to collect supplies? weekly / monthly / every 2 months / every 3 months?

The delivery or collection of supplies may not be regular, but somewhat ad-hoc. Capture the most likely interval. Also, keep in mind the shortest and the longest interval between two resupplies. We will use this to determine safety stock later.

In most cases a **monthly** delivery or collection schedule is recommended, as it achieves a good balance between not having to order too far into the future and not making too frequent collection/delivery trips.

The reorder factor is a number that will help you calculate how much of each item you need to order. It includes the requirement to hold enough stock to cover demand up to your next reordering, and an additional buffer to protect against higher and anticipated demand or delays in delivery/pickup.

Box 7: Drawing down from framework contracts

In some countries, large health facilities can purchase some products directly from the suppliers under a framework contract. This establishes the product specifications, prices and delivery terms.

If your facility has such an option, consult with the department that has created the framework contract and confirm to your supplier your draw-down quantity accordingly.

This could be particularly useful for maintaining supplies of essential medicines and technologies on the national EML that are not in stock at the regional/central store. CVD medicines may require such an approach during the preliminary stages of the programme.

The following reorder factors are recommended (supply interval (month) x 2) for most primary-level health care facilities.

Box 8: Reorder factor

The reorder factor is 2 if supplies are delivered once a month ($1 \times 2 = 2$)

The reorder factor is 6 if supplies are delivered every 3 months ($3 \times 2 = 6$)

The reorder factor is 12 if supplies are delivered every 6 months ($6 \times 2 = 12$)

Step 3: If you are starting to offer new NCD services, estimate the quantity of medicines needed to start CVD services at your facility.

Effective forecasting for CVD medicines and technologies starts with the number of current patients on treatment for a given condition and projected patients to be enrolled. If your facility is starting to offer CVD services, quantities of CVD medications are determined by the recommended treatment guidelines for the agreed initial number of patients likely to come to your health care facility to start hypertension, diabetes, services. Consult your district health management team to obtain details of estimates they may have conducted of the number of patients likely to attend. These quantities may have been based on experiences from other health care facilities in your country. *If you are interested in learning more about how such estimation is carried out, see Annex B.*

Box 9: Why medicines should be ordered more frequently

Often, increased transport costs (delivery model) or the time spent in travelling and waiting to collect supplies is used to justify infrequent deliveries or pickups. This increases the need for stock at the health facility and increases the risks of stock-outs, as it is harder to anticipate demand for a few months into the future than for the next 15 days or month.

CVD medicines and technologies should be ordered as frequently as possible. During a programme scale-up or at

the start of a new programme, past consumption is often a poor predictor of future months' demand. Ordering enough stock to cover 2 to 3 months based on past consumption or an uncertain forecast would make stock-outs more likely. If drugs are not always available in the early days of the programme, patients may lose confidence and will be discouraged from continuing their care and ensuring proper adherence to their treatment.

Remember to plan for a small supply of buffer stock in case there are delays in deliveries. Also, make sure you consult with the district/provincial health management team and NCD programme coordinator for guidelines on determining initial stock levels.

Box 10: Simple method for calculating the initial supplies for patients with diabetes

The Taylor Clinic will start hypertension services for an initial number of **50 patients next month**.

Forty of them will be treated with amlodipine. They have recently been started on amlodipine (5 mg PO) at other facilities, but will be transferred to your facility. The remaining 10 patients have a contraindication to amlodipine and will be treated with lisinopril (20 mg PO).

Step 1: For the planned 50 patients the diabetes medications for one month's treatment will be:

40 patients: Amlodipine tabs: 40 x 1 (tabs/day) x 30 (days treatment) = **1,200 tabs**

10 patients: Lisinopril tabs: 10 x 1 (tabs/day) x 30 (days treatment) = **300 tabs**

Step 2: Check the unit sizes carefully for each medicine.

Amlodipine comes in boxes of 84 tabs. For the required 1,200 tabs, you need: $1,200 \div 84 = 14.3$ **~15 boxes** (As you cannot order less than a whole box, always round up to the nearest unit size.)

Lisinopril comes in boxes of 100 tabs. For the required 300 tabs, you need: $300 \div 100 = 3$ **boxes.**

Step 3: Calculate the amount to order for the first month. If you receive supplies once a month, the first time you order you will need to order for approximately 2 months. If you receive supplies every 3 months you will have to order for 6 months.

If the Taylor clinic receives supplies every month, the initial order of medicines will be:

	expected consumption	x reorder factor	= quantity to be ordered
Amlodipine	15	x 2	= 30 boxes
Lisinopril	3	x 2	= 6 boxes

Remember that after the initial few weeks of treatment, the patient dose and strength will be titrated to achieve the target levels of blood glucose, blood pressure and lipid control. Take this into account when ordering for supplies beyond the first month.

Step 4: Transition to a more routine requisitioning model

After the initial month of ordering supplies, the ordering system needs to stabilize to a more routine reordering system, based on past consumption.

Box 11: Top-up system of replenishment

Where demand is not seasonal and a treatment programme has reached equilibrium stage, a top-up system can be used. Once the initial maximum stock levels are determined, based on average monthly consumption, there is no need for health facility staff to reorder. Each month the stock that is used is replenished by a delivery vehicle, accompanied by a

pharmacy technician or a staff trained in restocking.

Variants of this model also exist where the pharmacy technician (or staff trained in restocking) determine the quantity to be replenished, based on more sophisticated replenishment algorithms using decision-support technology tools.

In a typical system, when stock of an item falls below a set minimum desired level, sufficient units of that item are ordered/requisitioned to raise the stock level to the maximum desired level. Typically, the reorder points and minimum stock levels for your health facility should be set up by a trained logistician. In estimating the re-order level, they will consider factors such as monthly drug consumption, as estimated earlier; ordering frequency (monthly, quarterly); time lag between placing orders and receiving the orders; and uncertainties in forecasted demand or reliability of delivery lead time.

A Projected monthly consumption (as determined in Box 10 above, Step 1)

B Lead-time + review period = say 1 month

C Safety stock (to be determined by logistician, use 1 month of supply when no other estimate available)

D Current stock-on hand (from stock-control card or stock register)

Quantity to be ordered = **(A x B) + C - D**

After the first 6 months of starting CVD services, as the system stabilizes, past monthly consumption should be collated with data obtained from *stock-control cards* and *stock register*. At this stage, monthly consumption also becomes a good way to approximate the relative ratio of patients with a condition and how they will be put on different products/dose/strength.

Compare the average monthly consumption with the estimates created using population and prevalence data (see above). Collaborate with the MOIC to assess large deviations between the average monthly consumption (obtained from stock-cards) and the estimated monthly consumption (obtained from prevalence data).

Over time, the right quantity to order will be based on estimates of average past consumption, adjusted for any patients expected to be enrolled based on subjective knowledge. See Annex B for details on forecasting method for monthly consumption.

Apart from its use in estimating the quantity to order, it is important to consider the average consumption rate as a guide-post for assessing stock-out and expiration risks. Pay close attention to the stock of items with a low average monthly consumption rate. Some CVD medicines are likely to be slow movers and it is important to ensure they are not overstocked and their stock is rotated quickly.

Quantification methods for single-use products such as laboratory reagents and consumables

Consistent availability of laboratory reagents and consumables is a critical component of a facility's ability to manage CVD risk. The quantification process detailed above for medicines can be applied to laboratory supplies. However there are a few additional factor to consider.

Box 12: Definition of terms (15)

Reagents: Reagents are defined as compounds such as sulphuric acid, hydrochloric acid, sodium hydroxide, etc. that are used in a chemical reaction to detect, measure, examine, or produce other substances. Reagents vary widely in cost, stability, cold-chain requirements,

availability, and associated hazards. Reagents can either be chemical or biological. The reagents used in a particular laboratory are determined by which tests the laboratory performs, the equipment used, and the standard operating procedures in place.

Consumables: You can define consumables for logistics purposes as items that are used once while performing a test and are not reused. Consumables can include test-specific items, such as microscope slides and cover slips. Other consumables cut across all testing services and are classified as general laboratory consumables, such as bleach, pipette tips, and gloves.

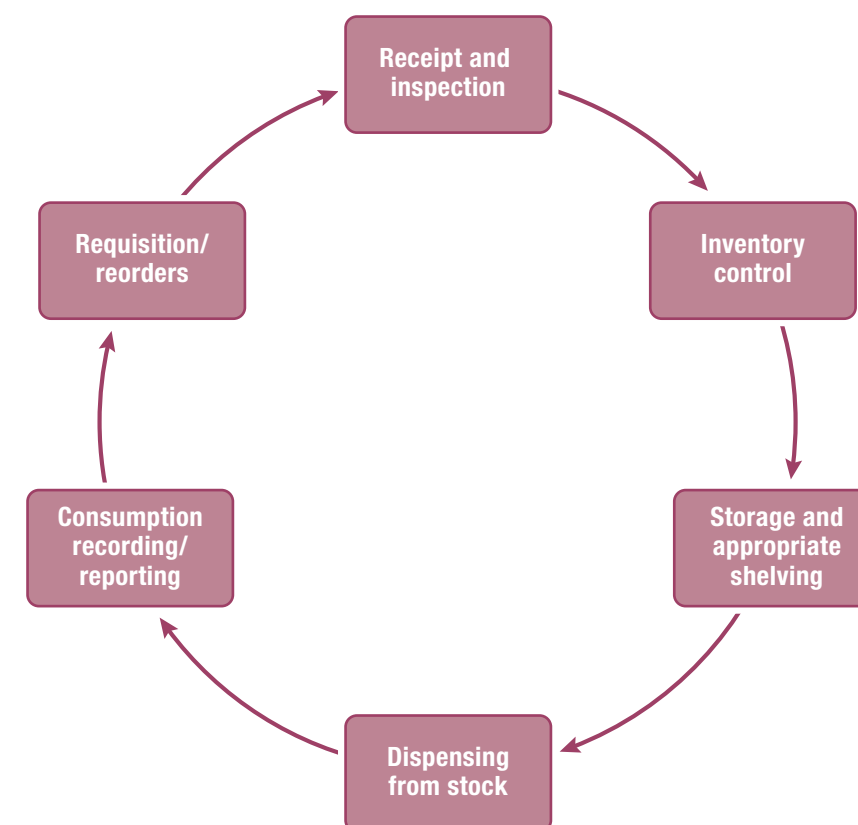
Some consumables are not used in the laboratory, but they are necessary to carry out testing and, therefore, you should include them in the quantification.

For example, a vacutainer needle is required to draw blood; it is used once and discarded. If the needle is not available, the specimen cannot be collected and no test can be carried out, and so it should be included when forecasting for testing needs. After blood is drawn for laboratory testing, it is placed in a specific container for preservation and storage, then it is sent to the laboratory. The testing protocols in the laboratory will determine what type of container and preservative you should use. Include these commodities in the forecast.

Key factors to consider in determining need:

- Testing protocols are specific for each test and for each laboratory.
- Some tests can be purchased as a pre-packaged kit, with all the reagents and consumables needed for a specific test.
- Some testing equipment requires proprietary reagents and consumables; this is a closed system. In open systems, the equipment is compatible with a larger variety of standard reagents and consumables available from a variety of sources.
- Consumption rates will vary by reagent. Some reagents may be used very frequently and for multiple tests, while use of others may be quite infrequent.
- Shelf life varies greatly between reagents.
- Ensure that a designated member of the laboratory staff is responsible for or directly involved in the quantification and ordering process for laboratory reagents and consumables.

Figure 2: The receipt-to-reorder cycle



Annex B: Estimating number of patients and treatment demand

It is good practice to develop as accurate an estimate as possible of the number of patients coming to a health facility for each condition. Such estimation is best carried out by the District Health Management Team. This Annex is intended to provide details of the data that feed into such estimates and typical methods used.

Factors that influence the number of patients for each condition are:

- the profile of the population your health facility serves
- disease pattern by age.

First, a preliminary assessment is conducted to establish what forms of data are available, and how recent and accurate these data are.

Useful ways to obtain such data are:

Health Management Information System (HMIS)

This collects data about patients, the reasons for their visit to health facilities and the services provided. In addition to health facility utilization data reported through the HMIS, there are often programme monitoring reports, which also capture such data at the facility level.

Morbidity data from surveys

Prevalence or incidence of each disease/condition, such as diabetes, can be obtained from surveillance or research study surveys done for a variety of reasons by national or local governments, foreign donors, or others. These may contain data from which it is possible to estimate national incidence or prevalence, usually expressed as episodes per 1,000 or 100,000 population.

Demographic data includes information about the population, such as the number of patients accessing the public sector compared with the private sector, population growth and demographic trends. Demographic data are collected through surveys and censuses. For example, Demographic and Health Surveys (DHS) are conducted every five years.

For some NCD conditions and risk factors, the WHO STEPwise approach to Surveillance (STEPS) (16) and DHS surveys (17), which have added NCDs, with measurements of fasting blood glucose and blood pressure can be a useful source.

Some districts have a demographic surveillance system that provides verbal autopsy and other specific data which can be used when other sources of information are not available or reliable. If nothing else works, a programme has to be started by relying on qualitative/subjective estimates by clinicians.

An initial survey (Table 2) is carried out to assess which of these data elements are available.

Table 2: Status of data sources

Data Source	Data element	Most recent date	Accuracy and completeness of reporting
HMIS	Number of treatment episodes for a given condition		
Prevalence survey	Percentage of population (stratified by age or other demographic) with disease/ condition		
Facility-level surveys	Facility catchment population		

Table 3: Places to look for data regarding population and disease prevalence

Data required	Where to find it	What to watch out for
Total population in health-facility catchment area	National census	Survey data may be outdated; limited subnational data available
Population stratified by age group	National census	Age breakdown may not be available for the groups needed or for the targeted population
Disease incidence	STEPS Survey, DHS	Data may be outdated; limited sub-national data
Number of cases treated by each type of health facility	HMIS reports	May be unreliable because of poor or incomplete reporting by health facilities.
STGs	Ministry of Health	Guidelines may propose different medicines for the same condition; STG is not always used by health providers

Estimating population-level need

Use the most recent STEPS survey (or other relevant survey or research data) from your country. Find the most recent population data from your district (See Table above for sources) or the catchment population for your health facility, with age breakdowns that match STEPS and with figures for men and women. List the population 18–29 years, 30–49 years, and 50–69 years. Important points to consider:

- It is better to use national data rather than a limited sample from participating districts only (due to small size). However, choose either the rural or urban estimates, depending on the characteristics of your health facility or what is most representative of your district.
- Use the detailed results with age breakdowns rather than the combined age category of 18 to 69 in the simplified fact sheets.
- Use the fasting blood glucose threshold for diabetes, not prediabetes.

Calculate approximate population-level need (**N**) by multiplying facility catchment population or district population figure for (men + women) x disease or risk factor percentage. An example of how to use STEPS to calculate **N** is presented below:

Table 4: Table for calculating population-level need

(1) List district population or health facility catchment population				(2) STEPS results in percent-- choose rural or urban (3) Calculate approximate number in district by multiplying district population figure for (men+women) times percent (except cervical cancer screening – women only)									
	Men	Women	Both	SBP ≥140 and/or DBP ≥ 90mmHG, excluding those on medication for raised BP	N	SBP ≥140 and/or DBP ≥ 90 mmHg or currently on medication for raised blood pressure	N	Raised fasting blood glucose >7 mmol	N	Diagnosed with diabetes, on medication	N	Raised blood glucose OR currently on medication for diabetes	N
18–29 years				%		%		%		%			
30–49 years				%		%		%		%			
50–69 years				%		%		%		%			
50–69 years				%		%		%		%			

Translating population-level need into health-facility demand

The STEPS survey data can help estimate the total population-level need as well as the current treatment rate for hypertension and diabetes. Knowing the current treatment gap is useful in estimating how many of those at risk will come to the facility to seek diagnosis and treatment.

Let us call **F** our estimate of the percentage of those in the catchment population with a condition requiring treatment who will seek care at the health facility or district. A good starting estimate for **F** is the percentage of the catchment population that visits the health facility for any health/treatment episode in each month.

Use **N x F** as an estimate of the number of patients with a condition who will visit the health facility.

At the end of the first few months of having started NCD services, revisit the calculation of **F** and update it, based on actual patient visits observed in the previous 6 months.

For each condition (e.g. diabetes, hypertension), use the standard treatment guidelines to determine the number of units of each item required by a patient in a month. Record monthly need estimates for each item on the list.

For most CVD health conditions, there is more than one treatment option on the EML. In order to determine the quantity of each drug required, consider the agreed upon protocol for the standard or average treatment for each health condition. Make a calculation of the frequency with which each product will be used for a given condition.

If there is an ideal treatment and a standard treatment guideline, then it should be assumed 100% of the time. In most cases, assumptions have to be made on the basis of empirical data from retrospective studies, expert opinion or anecdotal data.

Seek advice from your district or provincial team to estimate these figures. More detailed data on the distribution of patients between different treatment options will improve the accuracy of the forecasted order quantities.

When a programme has been ongoing for 3 to 6 months, and a sufficient number of patients are on treatment, the percentage of patients receiving each drug/specific product will approximate to a steady state. Data on consumption of products for all patients currently on treatment (just initiating, new and on-treatment for > 3 months) can then be used to estimate overall requirements.

Annex C: Forecasting future consumption

The choice of the right method to estimate your health facility’s expected monthly consumption depends on the type of data available and its accuracy. Inaccurate or incomplete reporting of stock and consumption data continues to be a major challenge.

Consumption method – uses past consumption of individual medicines or products (adjusted for stock-outs and projected changes in use) to project future need. When there are significant shortages/stock-outs, or medicines are not dispensed as per treatment guidelines, this method may significantly underestimate or overestimate need. The consumption-based approach has the potential to perpetuate the problems of the past into the future. Also, for new products and programmes there is no data on past consumption.

Morbidity method – estimates the need for specific medicines or products, based on disease incidence (using surveillance and demographic data), expected number of health care facility attendances (using service delivery data), and standard treatment patterns for the disease (STGs). Morbidity-based quantification is a complex and time-consuming method. Usually, several assumptions have to be made. In many countries, obtaining prevalence/incidence data by region is quite challenging. This method also assumes standard dispensing protocols.

Proxy consumption method – uses data on disease incidence, medicine consumption or use, and/or pharmaceutical expenditures from other geographic or product contexts to extrapolate the consumption or use rate to the target programme and product, considering population coverage or the service level needed, such as taking Denmark’s consumption data to estimate needs in South African health facilities. Proxy consumption is the most likely to yield accurate projections when used to extrapolate from one set of facilities to another set that serves the same type of population in the same type of geographic and climatic environment. If the data from another country is used, the results will be only a rough estimate of need.

Even when target and standard facilities are closely matched, quantification estimates are only approximate, because it is a big leap to assume that disease incidence, utilization patterns, and prescribing habits will be essentially the same in both settings. Still, this method may be the best alternative in the absence of suitable data required for the other two methods. The proxy consumption method is also useful for cross-checking projections made with other methods.

Table 5: Comparison of forecasting methods

Method	Strengths	Limitations
Consumption database	Most reliable if product has been used and is in mature stage of demand, there have been no large shortages or stock-outs.	Does not work when starting a new treatment programme. Any non-adherence to Standard Treatment Guidelines, under or overuse, gets carried forward.
Morbidity based	Estimating needs in new programmes. Estimating true underlying need.	Prevalence/morbidity not available or very outdated. Standard treatment guidelines are not followed.
Proxy consumption	Estimating needs in new programmes/products when there is no consumption data and morbidity estimates are not reliable.	Requires finding analogous products or best-case geographies. Analogues are always questionable in terms of how well they compare.

Annex D: Essential laboratory investigations and tools

Table 6: Essential products and investigations

Essential technologies/lab investigations	Tools
stethoscope blood pressure measurement device* measuring tape weighing machine (scale) glucometer blood glucose test strips urine albumin test strips urine ketones test strips	BMI chart
Add, when resources permit:	
blood cholesterol assay lipid profile serum creatinine assay serum potassium haemoglobin a1c urine microalbuminuria test strips	

*Validated blood pressure measurement devices with digital readings are preferable.

Annex E: NCD technology

Table 7: NCD technology specifications, storage and prescribing information (18)

Technology	Single use or multiple use	Storage conditions Complimentary parts	Specification ^a	Operation/ additional information ^b
Stethoscope	Multiple	Clean, dry, room temperature	The Y tube is treated rubber with large diameter of 10 mm. Arms with spring treated to give lasting spring and maximum reliability and comfort. Removable ear-pieces. Easy to dismantle to clean and disinfect.	
Sphygmomanometer (manual blood pressure measurement device)	Multiple	Clean, dry, room temperature Standard cuff (25cm x 12 cm) Alternative cuff (36 cm x 12 cm)	Inflatable rubber cuff surrounded by durable, flexible cover that can be easily fastened around upper arm. Aneroid pressure gauge displaying cuff pressure. Pumping bulb and valve allowing adjustment up and down for cuff pressure. Pressure gauge to allow reading of pressure to 2 mmHg accuracy. Maximum pressure to be at least 300 mmHg.	For facilities with non-physician health workers, a validated blood pressure measurement device with digital reading is preferable for accurate measurement of blood pressure.
Measurement tape	Multiple	Clean, dry, room temperature	Tape measure made of non-stretchable material, preferably fibreglass but not cloth or steel. The preferred tape is one that is self-retracting, with locking capability, which can accurately measure circumference in millimetres.	
Manual weighing machine	Multiple	Clean, dry, room temperature	Analogue scale, capacity up to 200 kg, graduation weight 100 g. Reading on both sides, robust lever system, height-adjustable rod. Height rod range 60–200 cm with 1 mm graduations. Anti-slip platform. Adjustable zero point. Weighing units: kg and/or lb. Transport castors.	Calibration should be tested by certified test weights. See manufacturer's instructions for recalibration as required.

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Technical package for cardiovascular disease management in primary health care



Team-based care



WHO/NMH/NVI/18.4

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Many low-resource settings have a shortage of physicians and health workers. (1) In order to provide patient-centred continuous care more effectively, primary care systems can include team-based care strategies in their clinic workflows and protocols.

Team-based care uses multidisciplinary teams (which may involve new staff, or the shifting of tasks among existing staff). Teams can include patients themselves, primary care physicians, and other allied health professionals, such as nurses, pharmacists, counsellors, social workers, nutritionists, community health workers, or others.

Teams reduce the burden on physicians by utilizing the skills of trained health workers. Strong evidence shows that team-based care is effective in improving hypertension control among patients in a cost-effective way. (2) Some amount of task shifting/team-based care is already taking place in many settings; this module provides further guidance on how to maximize this approach for greater impact.

What this module includes

- Description of team-based care and advantages and disadvantages of the approach;
- Suggested steps on how to implement team-based care;
- Case studies of team-based care in different countries;
- Sample workflow charts and tables that can be customized to implement team-based care in a specific facility.

1 About team-based care

The world is facing a chronic shortage of trained health workers. Recent statistics indicate that there is a global health workforce deficit of more than 4 million. At the same time, the demand for health care is rising and noncommunicable diseases (NCDs) are becoming more common globally. NCDs require health systems to treat large numbers of patients for long periods of time. A quality community-based health workforce is critical to addressing the NCD epidemic. Learning from experiences in treating HIV, team-based care and/or task shifting can improve management and control of NCDs. (3)

Task shifting is the reassignment of clinical and non-clinical tasks from one level or type of health worker to another so that health services can be provided more efficiently or effectively. For example, when medical officers are in short supply, some services can be effectively shifted to equipped and well trained non physicians such as clinical officers and nurses, while maintaining quality. This increases accessibility of health services to the community. Task shifting also can apply to laboratory functions, supply management, and pharmacy services. (4)

Team-based care is a strategic redistribution of work among members of a practice team. In the model, all members of the physician-led team play an integral role in providing patient care. The physician (or in some circumstances a nurse practitioner or physician assistant) and a team of nurses and/or medical assistants (MAs) share responsibilities for better patient care.

Team-based care and task shifting, or a combination of the two, are useful models that can be tailored to meet country needs. This module refers to both but will use the phrase “team-based care” as shorthand.

Advantages (5, 6)

- Expanded access to care (more hours of coverage, shorter wait times)
- Better patient support
- Team member collaboration
- Improved patient adherence to medications
- Better follow-up
- Improved BP control and other patient outcomes (CVD morbidity and mortality, and comorbid CVD risk factors such as diabetes and high cholesterol)
- Improved patient knowledge
- Better quality of life
- Time saving for patient and health care team
- Cost efficient
- Improved patient and physician satisfaction

Barriers

- Rapid staff turnover
- Retention of training
- Patient attitudes: perception by patients of being treated by non-physician health workers
- Physician attitude and reactions
- Legislation and policy

Requirements

Policy decisions are usually made nationally, but there are methods that health centre managers can use to help ensure successful implementation:

- Consult closely and coordinate with the physician.
- Train health care workers in new skills.
- Enable additional health workers to prescribe medications.
- Clearly define roles and responsibilities for different team members.
- Arrange close supervision, mentoring and support by experienced health centre staff.
- Schedule regular clinical team meetings and good communications between staff to discuss patient cases and issues, so that they can work together to solve problems.
- Facilitate regular dialogue between staff about how to improve tasks in order to increase service efficiency and quality.
- Devise measurable processes and outcomes.

Refer to Section 2, Implementation, for more detailed information and tools to put these methods into practice.

2 Implementing team-based care*

Implementing team-based care (7) is specific to country, regional and clinic contexts. These recommendations are meant to be adapted as needed. This section provides step-by-step implementation guidance and additional resources and examples that can assist with implementation.

Steps to implementing team-based care

Box 1: Implementation steps

1. Engage the team
2. Determine the team composition
3. Design workflows to reflect the new model of care
4. Increase communication among the team, practice and patients
5. Use a gradual approach to implement the model
6. Optimize the care model

1 Engage the team

Bring together a multi-disciplinary team of nurses, medical assistants (MAs), physicians, pharmacists, community health workers, nutritionists, administrators and information technology staff members with a leader who has enough authority within the practice or organization to empower the process. Consider involving patients on the team as well (see Table 1 overleaf).

2 Determine the team composition

Design the model of care that will meet the needs of your patients and team. Consider which current team members could learn a new skillset and fulfil a new role. Your team may include a counsellor, nutritionist, nurse practitioner, physician assistant, clerical staff or others (see Table 2 overleaf).

Ensure that the team(s) consists of physicians and supporting team members who are eager to transform the clinic into a team-based care model. They should be champions and good communicators who are willing to put in extra effort to prepare for the transition and continue to develop the new model once it is underway.

* The following team-based care framework is adapted with permission from the American Medical Association Steps Forward™ team-based care module. For more innovative practice transformation strategies that can help your organization achieve the quadruple aim, visit www.stepsforward.org.

Table 1: Team-based care for hypertension in Cuba

Activity	Task shift	New personnel in charge	Required activities	Lessons learned
Hypertension control	Measuring blood pressure	Health professionals: Nurses Dentists Physical therapists Health professionals in training: Medical students Nursing students In community settings: Community health workers Patients' relatives	Certification of blood pressure measurement. Introduction to digital blood pressure measuring devices.	There is a need to take advantage of all possible opportunities to promote hypertension control. Including community health workers and patients' relatives is possible due to the increased access to digital blood pressure measuring devices.
	Reporting consumption of anti-hypertensives	Pharmacists	Generation of monthly reports from the health area of the patients that are not refilling their hypertensive medications at the corresponding community pharmacy.	Pharmacies have the potential of becoming ideal health education and hypertension control points in the community.
Registry	Entering the data in the registry system	Nursing assistants	Appropriate training for data entry	Having the data entered in the system allows for easier access and visibility of the data.
Education	Self-management	Medical and nursing students Community health workers Relatives of patients Patients themselves	Training on self-management for patients and their relatives, and community health volunteers.	There is a need to increase access to digital blood measurement devices. Need to improve self-management education.

Table 2: Distribution of staff responsibility for hypertension management

Task	Doctor	Hypertension specialist	Nurse	Pharmacist	Counsellor	Nutritionist	Social Worker	Community health worker (CHW)	Clerical staff
Take patient history			✓						
Diagnosis	✓								
Regular evaluation for secondary causes, additional risk factors and organ damage	✓								
Highly complex patients*		✓							
Identify barriers			✓	✓		✓			
Take BP measurement				✓				✓	
Perform lifestyle counselling			✓	✓	✓	✓	✓	✓	
Refill medications			✓	✓					
Adjust medications			✓	✓					
Patient follow-up			✓			✓	✓	✓	
Refer patient			✓	✓				✓	
Data entry									✓
Appointment scheduling					✓		✓	✓	✓
Appointment reminders					✓		✓	✓	✓

* Defined as patients who do not improve despite multiple therapies, or patients with suspected secondary causable.

Table 3: Task reassignment table for clinic and staff management

Use the table below to identify the best approach to implement a team-based care strategy, given your current staff capacity.

A. Task	B. Who does it now?	C. In a perfect world, who would do it?	D. Does this person need additional training to complete this task?	E. If answer to D is yes, then what kind of training is best (training to role or training as a team)?
Take patient history				
Diagnosis				
Regular evaluation for secondary causes, additional risk factors and organ damage				
Highly complex patients				
Identify barriers				
Take BP measurement				
Arrange lab testing				
Perform lifestyle counselling				
Refill medications				
Adjust medications				
Refer patient				
Data entry				
Appointment scheduling				
Appointment reminders				

Module 19. Implementing Care Teams. Content last reviewed May 2013. Agency for Healthcare Research and Quality, Rockville, MD (<http://www.ahrq.gov/professionals/prevention-chronic-care/improve/system/pfhandbook/mod19.html>, accessed 17 October 2017).

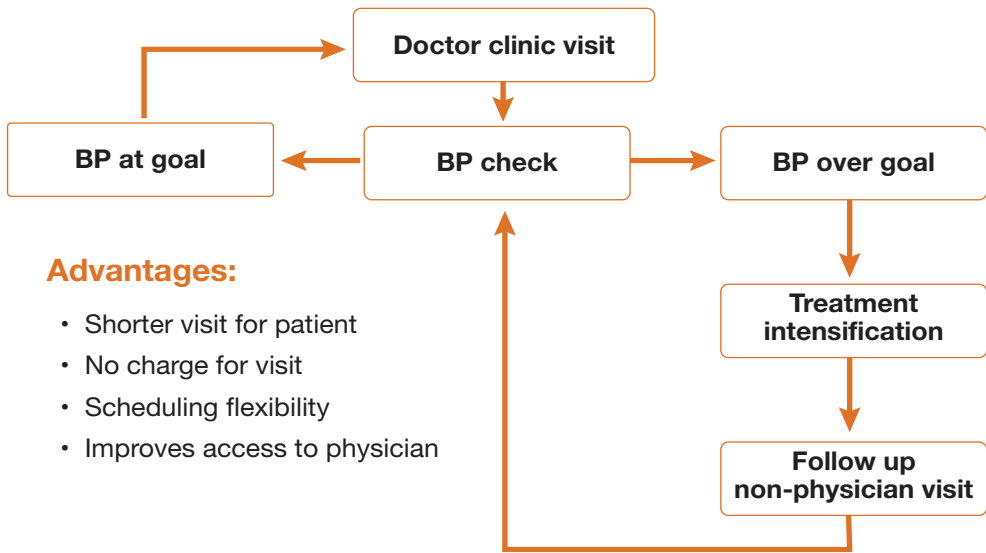
Institute for Healthcare Improvement, Partnering in Self-Management Support: A Toolkit for Clinicians, Self-Management Support Roles and Tasks in Team Care (<http://www.ihl.org/resources/Pages/Tools/SelfManagementToolkitforClinicians.aspx>, accessed 17 October 2017).

3 Design workflows to reflect the new model of care

Determine your new team-based care workflows. Remember, you are creating your ideal future state, so think outside the box when designing your dream team and ideal practice. If certain aspects of your current workflow function well, feel free to incorporate them into your future state! Try not to limit yourself; consider how an already great process can be made better.

You can design workflows for an entire clinical process (including pre-visit, visit, and post-visit), or for individual tasks that you want to assign to the team. Figure 1 is an example from Kaiser Permanente, California, of a simple workflow for a blood pressure check and follow-up process for nurses or healthcare workers.

Figure 1: Workflow for blood pressure check



- Advantages:**
- Shorter visit for patient
 - No charge for visit
 - Scheduling flexibility
 - Improves access to physician

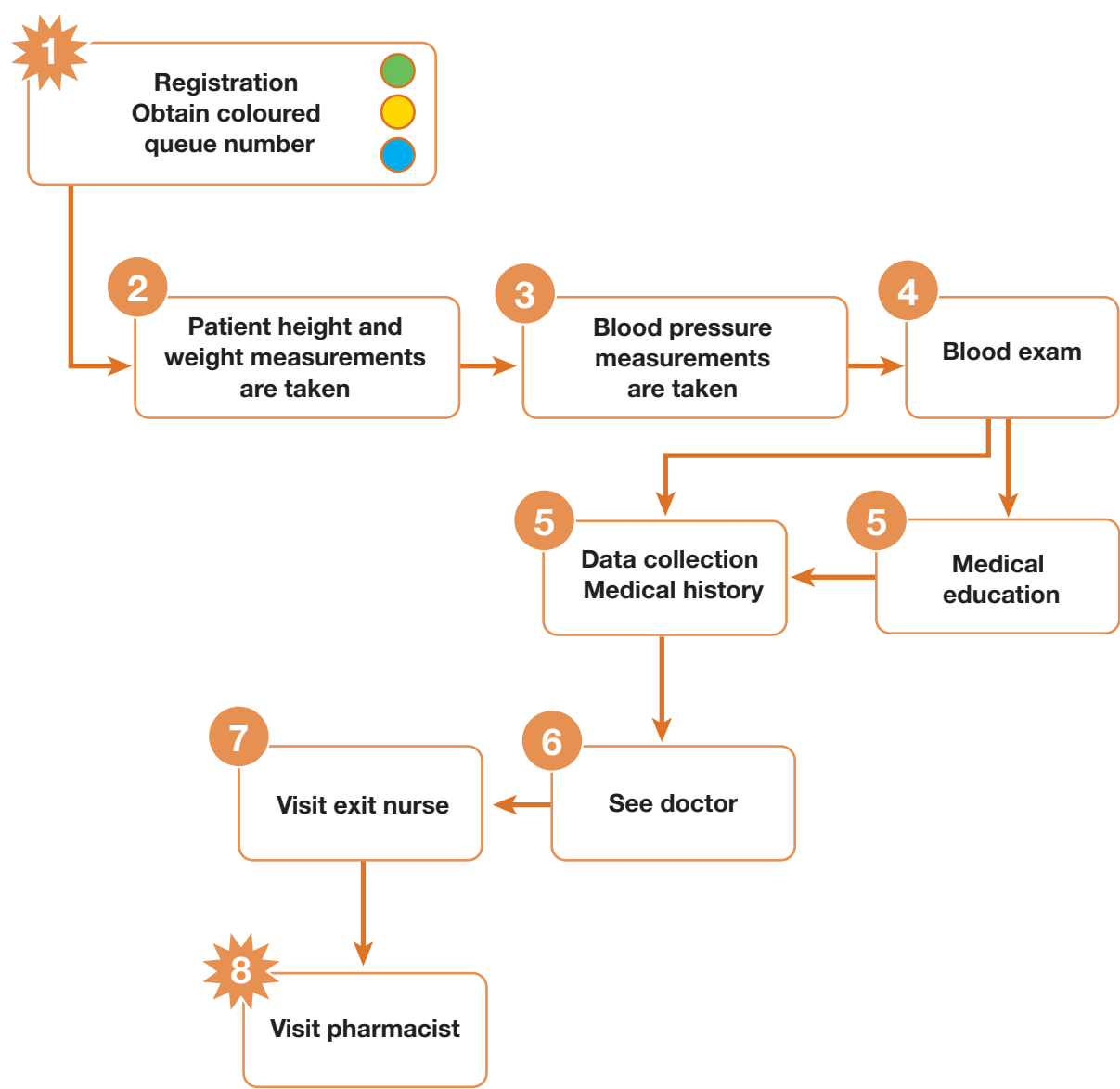
Jaffe, M. Care Delivery Models in Different Settings – The Kaiser Permanente Northern California Hypertension Control Project: How a large health care organization improved blood pressure control. March 7, 2013.

Figure 2 (overleaf) is an example of the clinical workflow for hypertension at the Ongkarak District Level community hospital in Thailand.

In Thailand, the system of patient flow and the logistical aspects throughout every step on the pathway are integrated (8):

Registration: The registration desk is the first contact point in the patient pathway continuum. Tuesdays and Thursdays are set days designated especially for hypertensive patients. Patients are sorted through a numbered colour scheme registration system: green for those with appointments, yellow for older individuals as a fast track option, and blue for general OPD patients.

Figure 2: Clinical workflow example: integrated patient pathway, Thailand



Height and weight: Body weight and height measurements are collected at every visit for every patient, along with blood pressure measurements.

Blood pressure: Blood pressure measurements are carefully collected and repeated if the first measurement is high. A second high indicator is followed up with a manual BP measurement. Patients are referred to the emergency room (ER) if the manual measurement is also high. ER patients are also measured for BP. Blood pressure measurement machines are calibrated on a yearly basis by the regional medical science office.

Blood exam: Patients coming in for blood work are referred to the lab.

Medical history and medical education: Patients are directed to either a nurse for a consultation on their current medical history before meeting with the doctor, or directed to a health and nutritional team before consulting with the nurse on their medical history. This medical history is entered into the client chart and hospital database. For high-risk patients requiring additional health and nutritional education, such as patients with Stage 3 chronic kidney disease (CKD), they are identified and referred to a community peer counselling group where nurses and pharmacists provide relevant information regarding eating habits, food and drug consumption, and current trends in the community. At the end of these sessions, a feedback system ensures that participants provide their opinions, ideas, and perspectives.

Doctor visit: Patient charts and registration numbers are assembled and prepared for the doctor. Patients are called through a televised screen that includes their name, registration number, and photograph. Patient charts include a star system that ranks their sugar or blood pressure levels for the visit. The doctor provides further consultation and adjusts and prescribes essential medicines as needed.

Exit nurse: Patients meet with an exit nurse to review their visit and prescribed medicine before visiting the pharmacy for their medicine. Additional critical information is collected by the nurse and added to the patient record.

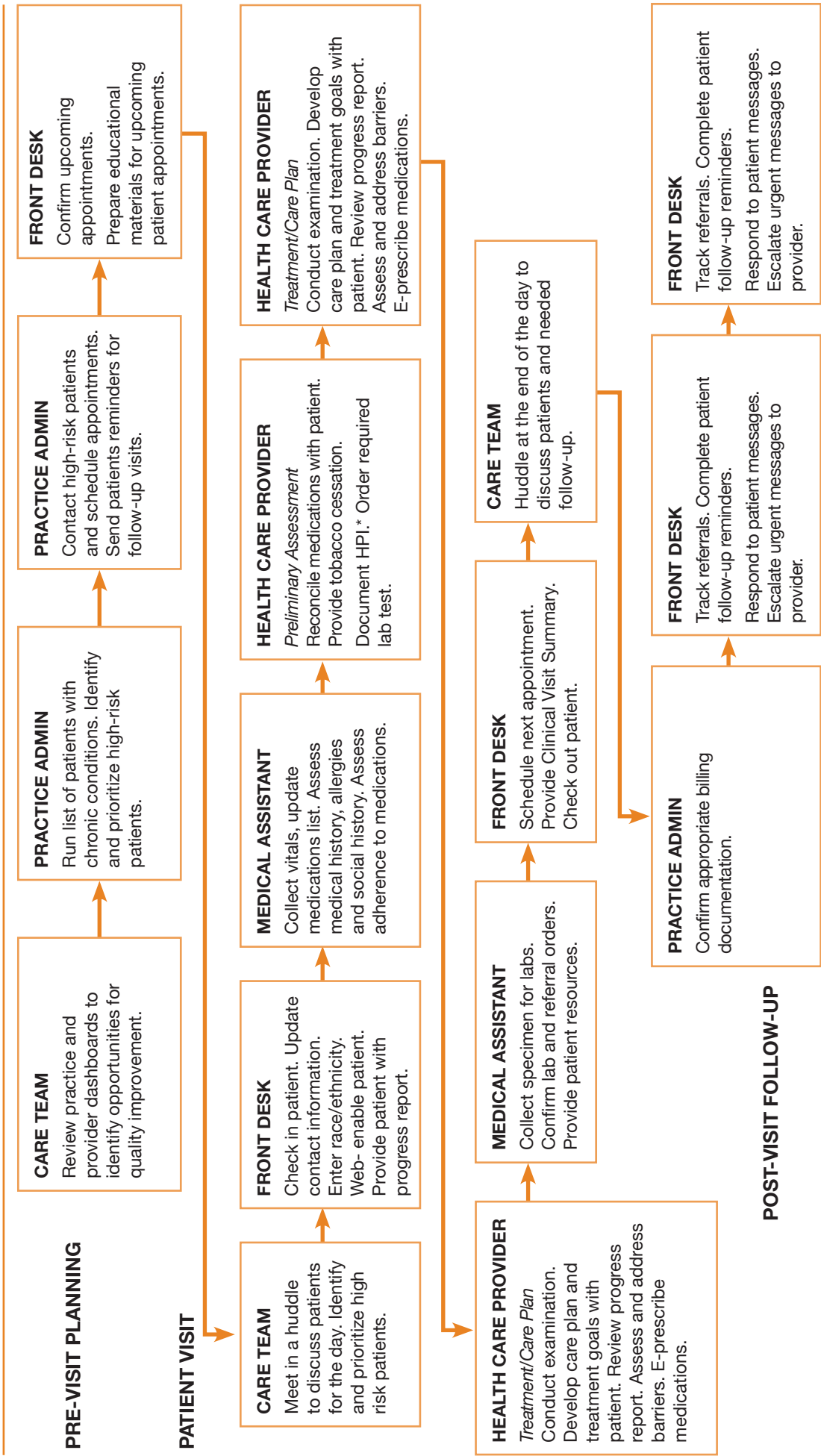
Pharmacist: Patients provide the pharmacy with their prescriptions and wait for their medicine. The pharmacy team reviews previous treatment history and ensure that patients do not have an excess of medicine left over from previous visits.

Highlights from Thailand

- Direct communication between the district hospital and the health promoting hospital is a consistent method that helps to expedite transfers for those transferred out.
- Proximity to services at every step on the pathway makes it easier for patients to navigate the health system.
- The star system regarding sugar and BP levels at every visit helps patients understand their current health state and reinforces health knowledge.
- The televised screen and photograph arrangement helps ensure patients who might not be able to read and/or have vision problems know when they are being called.
- Peer counselling and the inclusion of pharmacists in patient health promotion initiatives provide patients with a multifaceted team of experts.

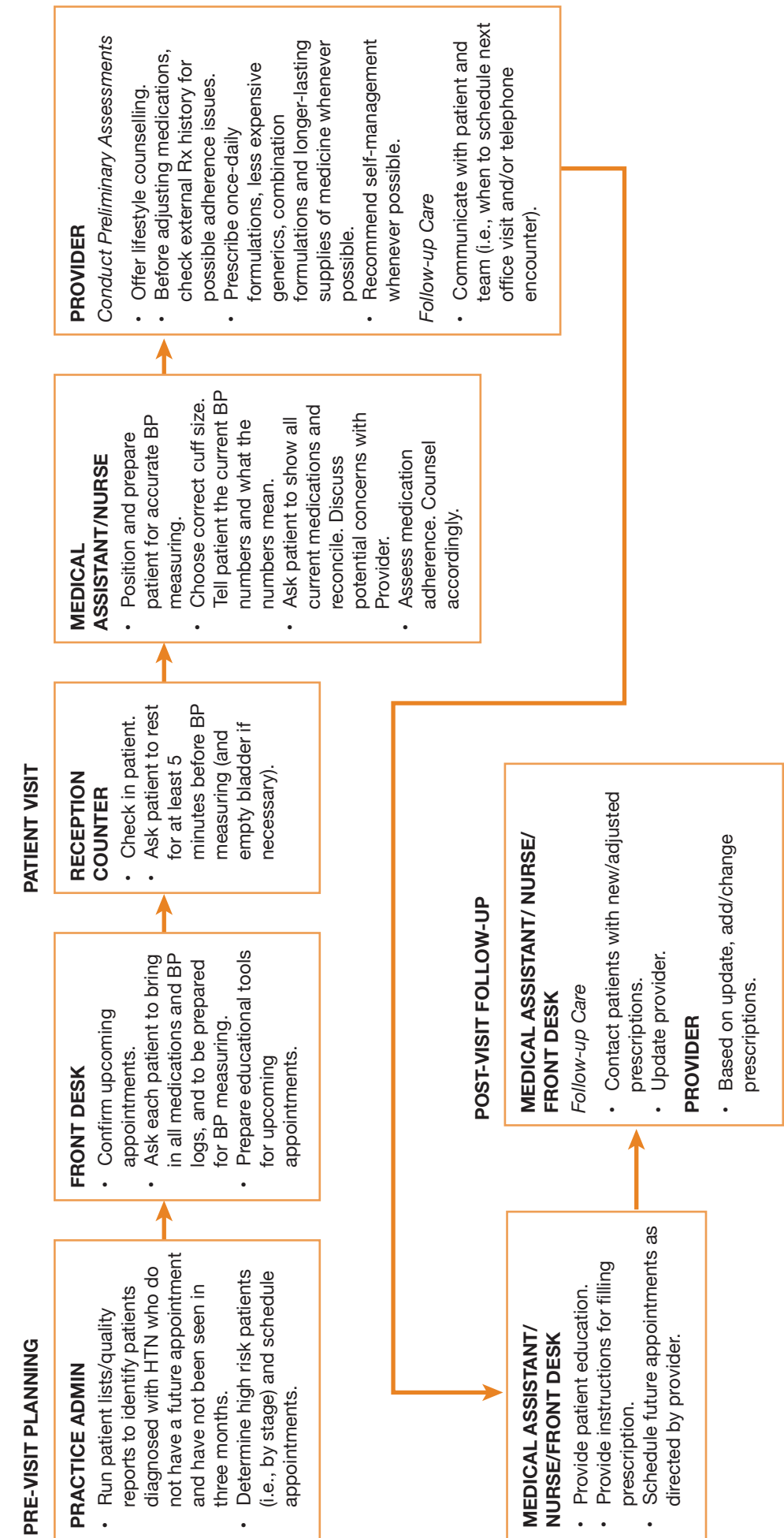
Figures 3 and 4 show some additional examples of office workflows to refer to when planning your clinic workflow. To help you plan comprehensive team-based care for patients before, during, and after their visit, refer to the workflow mapping worksheets in Annexes A to C.

Figure 3: Workflow mapping worksheet example: chronic care



Adapted from: ABCS Toolkit for the Practice Facilitator, HealthyHearts NYC. New York City Department of Health and Mental Hygiene. 2017.

Figure 4: Workflow mapping worksheet example: blood pressure control



4 Increase communication among the team, practice and patients

Start by keeping the practice aware of the team's work. Physicians and staff may feel out of the loop and disengage if they are not involved.

What kind of communication tactics can you use with your team?

- Include the team's task-shifting work as a standing agenda item at team meetings and department gatherings.
- Broadcast updates in a weekly meeting or call.
- Have doctors share space with the rest of their team in a common workspace to support team communication.

Box 2: Team-based care pilot

Some clinics in Ethiopia piloted a team-based care approach to improve control of diabetes and hypertension. Health care managers, administrators, laboratorians, pharmacists, and other team members met every two weeks during the early stages of the pilot, and monthly once the pilots were more established. The meeting objectives were to discuss patient screening, care retention, supplies, lab services, health education, community activities, challenges and reporting. (9)

Communicate the team's work to your patients as well. Pamphlets in the waiting and exam rooms could also be used to remind patients of the changes before their visit begins.

5 Use a gradual approach to implement the model

Team-based care implementation will be a gradual process. It will take time, and every day will not be perfect. Be patient; know that several months may go by before the team feels like they are really comfortable with the new system.

One physician who has implemented team-based care recommends that physicians who are considering implementation make sure that they are completely committed because it is not easy. He followed up with, "I cannot imagine practising any other way."

A medical assistant (MA) who works in a team-based care model said that it took her about two months to feel like she was really getting the hang of documenting patient visits for her physician. She worked very closely with him as he taught her his preferences and showed her how he edited every single patient note. This type of time commitment is necessary to successfully implement team-based care. As the model expands, experienced staff can mentor or assist with training new staff.

6 Optimize the care model

Shared work space

Teams that sit in closer proximity communicate with greater frequency and ease. Questions can rapidly be answered, reducing the time that someone may have to wait before completing a task or responding to a patient. Everyone will be aware of the work that their teammates are doing, enabling easier task-sharing and division

of work. Finally, after a busy clinic day, your inbox will not be filled with messages that could have quickly been triaged by another team member during the day.

Communication management

In a team-based care model, the number of phone and email messages that are sent to the team should decrease for several reasons:

- Lab results are discussed during the visit, so the number of messages sent back and forth to discuss results or set up a call is significantly reduced.
- Patients receive additional education at the conclusion of their visit, resulting in fewer questions after the visit.
- Care coordination is enhanced. Patients will leave with their follow-up appointments, corresponding labs and diagnostics scheduled, so they should have fewer requests after leaving the office.
- Referrals to supportive services such as behavioural health or to a health educator can be made during the visit. Involving additional team members in a patient's care provides them with a point of contact for follow-up questions regarding these specific services.

Of the questions that do come into the office, the team should be able to handle most of them. The physician may delegate most questions and concerns to the nurses or MAs who work with him or her. Their knowledge of each patient's case will be much greater in a team-based care model, and they will be able to answer most questions according to what was said during the visit or the plan of care that was determined. They will also build their skillset over time, further engaging them in this critical work.

Box 3: The benefits of team-based care

The benefits from team-based care have gone beyond what I envisioned. Originally, I simply hoped to regain eye contact with patients, as this is such an important assessment tool for me during visits. What I've seen has really gone way beyond that. Yes, my face time with patients is increased, but the visit is also more efficient and relevant. Since my documentation is now in "real time", my notes are better and timelier. Our clinical staff is learning so much more now that they feel like they are truly part of a team and they enjoy the added dimensions to their clinical practice. They have more confidence when teaching patients during office visits as well as when they're on telephone triage. Team care has been a win-win here. And best of all, families love it!

David Lautz, MD, Stanford Coordinated Care

3 Case studies

Hypertension care in Thailand

Diagnosis and treatment of hypertension is done at community hospitals at district level (one per 75,000) supported by a network of health-promoting hospitals (one per 5,000) at the sub-district level and village health volunteers at the village level.

At the community hospital, a senior clinical nurse is designated as the Case Manager (or clinic manager) for organizing services. For old and new patients, paramedical staff measure height, weight, and blood pressure using a digital device, and take the patient's history. For new patients, the doctor makes the diagnosis based on blood pressure readings and writes the prescription. Patients receive group counselling by a nurse or public health nutritionist (if available). The pharmacist dispenses the medicines and the data clerk enters paper-records into the hospital electronic registry.

At follow-up visits, patients who have not achieved blood pressure goal are again seen by the physician, who adjusts the drugs/dose. Follow-up patients with controlled blood pressure are seen by a nurse. Patients who do not attend the clinic on the date of the appointment are reminded by a phone call by data clerk and given another appointment.

Patients with controlled blood pressure are referred to a health-promoting hospital (HPH) closer to their homes. These are run by non-physicians (a team consisting of a clinical nurse and a public health nurse). The nurse provides continuing prescription/refills to controlled patients, or adjusts dose in consultation with a doctor on the phone, when blood pressure is not controlled.

Roles of various team members

Senior nurse: In charge of service organization (also referred to as Case Manager), prescribes continuing medicine (without change in drug/dose), counsels patients.

Doctor: Makes diagnosis, prescribes medicines and tests, titrates drug/dose.

Nurse: Measures height, weight, BP and takes history.

Counsellor: Provides nutrition counselling and lifestyle advice.

Pharmacist: Checks for adherence, dispenses medicines and advises on how to take medicines.

Data clerk: Enters patient BP and treatment details into the electronic registry; generates list of defaulters and makes calls, generates list of patients for appointments on next day.

Nurse at HPH: Provides refills, counsels patients and occasionally adjusts drug/dose in consultation with doctor on phone.

Team-based care for NCDs in Nepal

In order to address the growing burden of NCDs in Nepal, the Primary Health Care (PHC) division of the Ministry of Health has introduced the WHO PEN package in two pilot districts of Kailali and Ilam. Given the success of these pilots, MOH intended to scale up implementation in 20 districts in the following year.

Early detection, management and referral of hypertension, diabetes and cardiovascular disease risk (as a total-risk approach) is done at the primary health care level in the district.

At the district level, basic NCD services are provided by a network of primary health care centres and health post level and community health volunteers at the village level. At the primary health care level, a medical officer is designated as the manager and at the community level a health assistant is the manager.

For old and new patients, paramedical staff measure height, weight and blood pressure, using and assessing CVD risk using WHO tools at the health post level. For patients visiting primary health care centres, the medical officer makes the diagnosis based on blood pressure readings, blood glucose level, total cholesterol and risk exposures, and writes the prescription.

Patients receive counselling by either a medical officer, health assistant, staff nurse or auxiliary nurse midwife, as available, who also enter paper records that are sent to the district public health office and to the national health management information system.

At follow-up visits at health posts, patients who have not achieved their blood pressure goal are again referred to the medical officer, who adjusts the drugs/dose. Follow-up patients with controlled blood pressure and diabetes are managed by the paramedic health staffs at the health post level.

Patients who do not attend the clinic on the date of the appointment are reminded by a phone call to attend their next one.

Roles of various team members

Medical Officer (MO): In overall charge of service delivery for NCDs. MOs diagnose, manage patients and refer as needed, prescribe medicines and counsel patients.

Senior/Health Assistant: In overall charge of service delivery for NCDs. They are responsible for early detection, management and counselling patients. They do not prescribe medicines but refill medicines, as prescribed by the MO.

Staff Nurse (PHC): Measures height, weight and BP, counsels the patients and enters patient BP and treatment details in the NCD registry.

Assistant Health Worker: Measures height, weight and BP, counsels the patients and enters patient BP and treatment details in the NCD registry.

Senior ANM: Measures height, weight and BP, counsels the patients and enters patient BP and treatment details in the NCD registry.

Community Health Worker: Provides counselling on lifestyle and nutrition.

Pharmacist (PHC): Checks for adherence, dispenses medicines and advises on how to take medicines.

Task shifting in Ethiopia

Ethiopia has a relatively large number of health workers, but its large population makes for a very low ratio of health worker to population at 0.84/1000 population.

The shortage of health workers became perceptible in the early 2000s during the HIV/AIDS crisis. One of the policy reforms implemented by the Ethiopian government to address this was a task-shifting program, in addition to increasing pre-service education at different levels. A health extension program trained community health workers to provide health promotion and preventive services at the community level (health posts and home visits). Similarly, an Accelerated Health Officer Training Program was launched in 2005 to assist with the physician shortages. Furthermore, nurses also began in-service training to perform the additional tasks of prescribing medication for managing HIV patients.

Currently, because of the progressively increasing concern about rising NCDs, the country has adopted some task-shifting practices for NCD service integration at the PHC level in a pilot project. Primarily, the task-shifting strategy focuses on using nurses to manage chronic problems like hypertension, diabetes, and CVD risk assessment, using simple logical algorithms and job aids. Additionally, at the primary hospital level, general practitioners and health officers replace the work of specialists. Training for these professionals is designed as a cascade, involving specialists of regional hospitals training GPs and health officers; GPs and health officers training nurses, and nurses training health extension workers.

The pilot has shown promising results and it is planned that the MOH will scale up this model, ensuring that there is regional capacity to conduct training and strengthen undergraduate training to include NCD service integration.

Annex: Planning exercises

The following exercises are provided to help facility managers develop workflow plans for the process of preparing for a patient visit, the patient visit itself, and the post-visit follow-up.

Example workflow plans are given, and the user is invited to:

- compare processes in their own clinic against these examples of good practice
- use the template provided to map out workflow plans for the future.

As part of the exercise, the user is invited to work with members of staff through a series of questions relating to current processes, during which they may identify barriers or duplications that need to be overcome in order to arrive at the ideal process. The user is encouraged to think about ways that current practices can be adjusted to incorporate different members of the healthcare team.

Pre-visit planning

Workflow mapping worksheet
Assessment of current practice

Patient visit

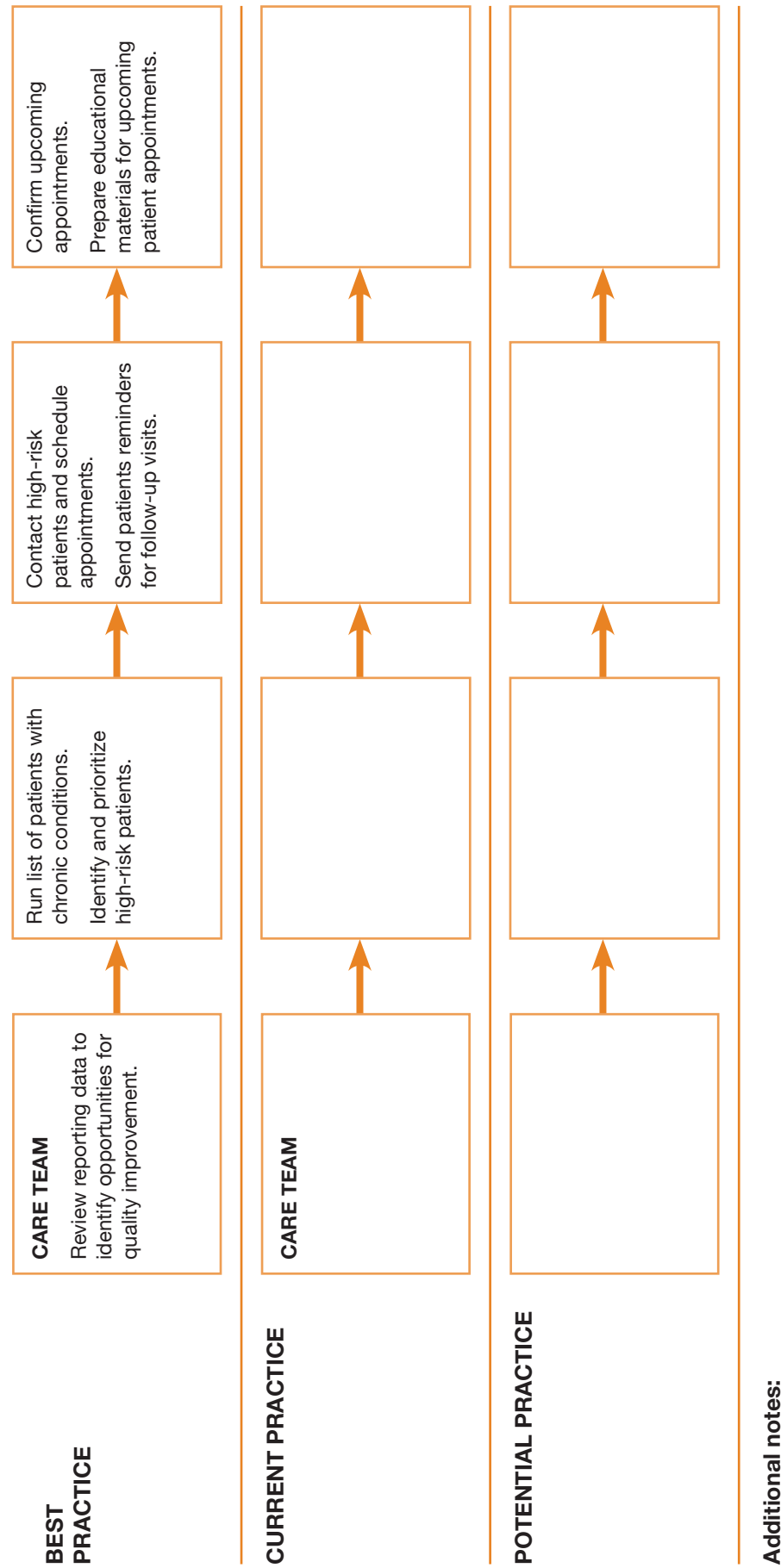
Workflow mapping worksheet
Assessment of current practice

Post-visit follow-up

Workflow mapping worksheet
Assessment of current practice

The following material has been adapted from: ABCS Toolkit for the Practice Facilitator, HealthyHearts NYC. New York City Department of Health and Mental Hygiene. 2017

Pre-visit planning – Workflow mapping worksheet

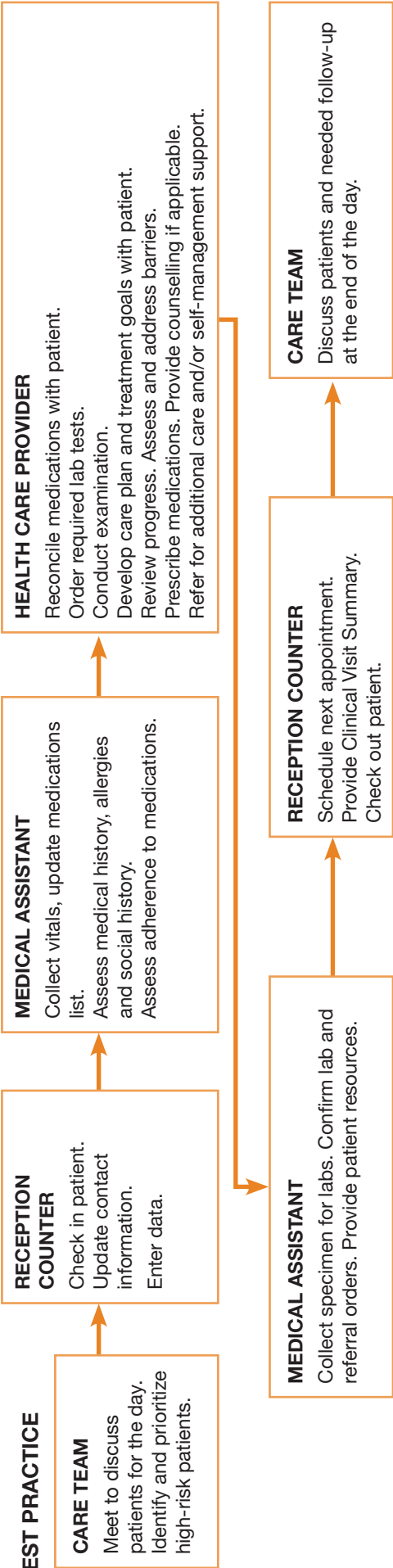


Pre-visit planning – Assessment of current practice

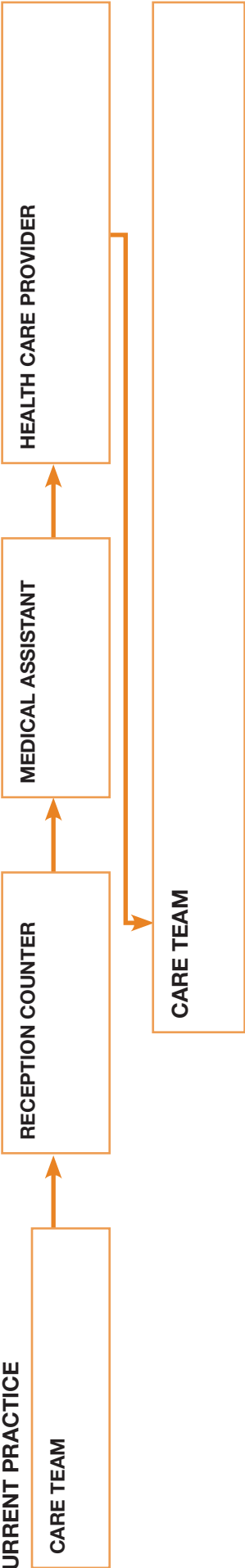
Process	Current process	Barriers/duplication	Ideal process
1. Does the practice conduct any pre-visit planning? Describe the process.			
Registry and reporting			
2. How does the practice monitor and track patients with chronic conditions?			
3. Does the practice monitor at-risk populations? How is the physician alerted?			
4. What are the current reporting requirements for the practice (e.g., patient-specific populations, time studies, clinical operating reports, incoming referrals, patient-care measures)? How frequently are these reports run?			
5. Does anyone at the practice conduct chart reviews? How frequently? Describe this process.			
Additional notes:			

Patient visit – Workflow mapping worksheet

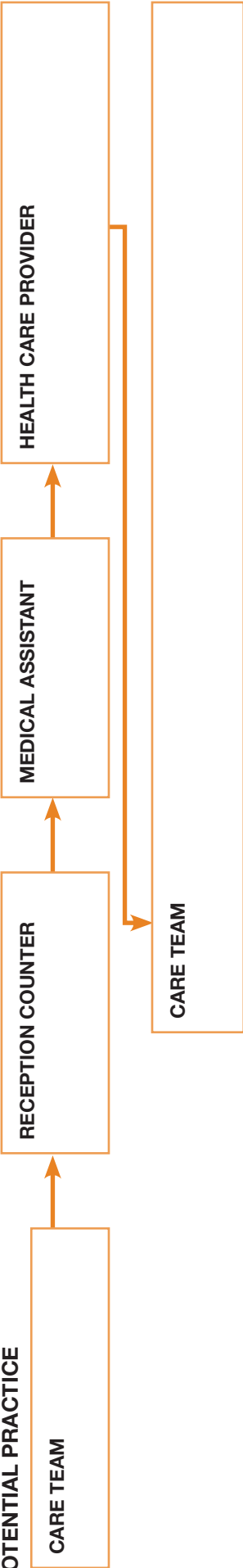
BEST PRACTICE



CURRENT PRACTICE



POTENTIAL PRACTICE



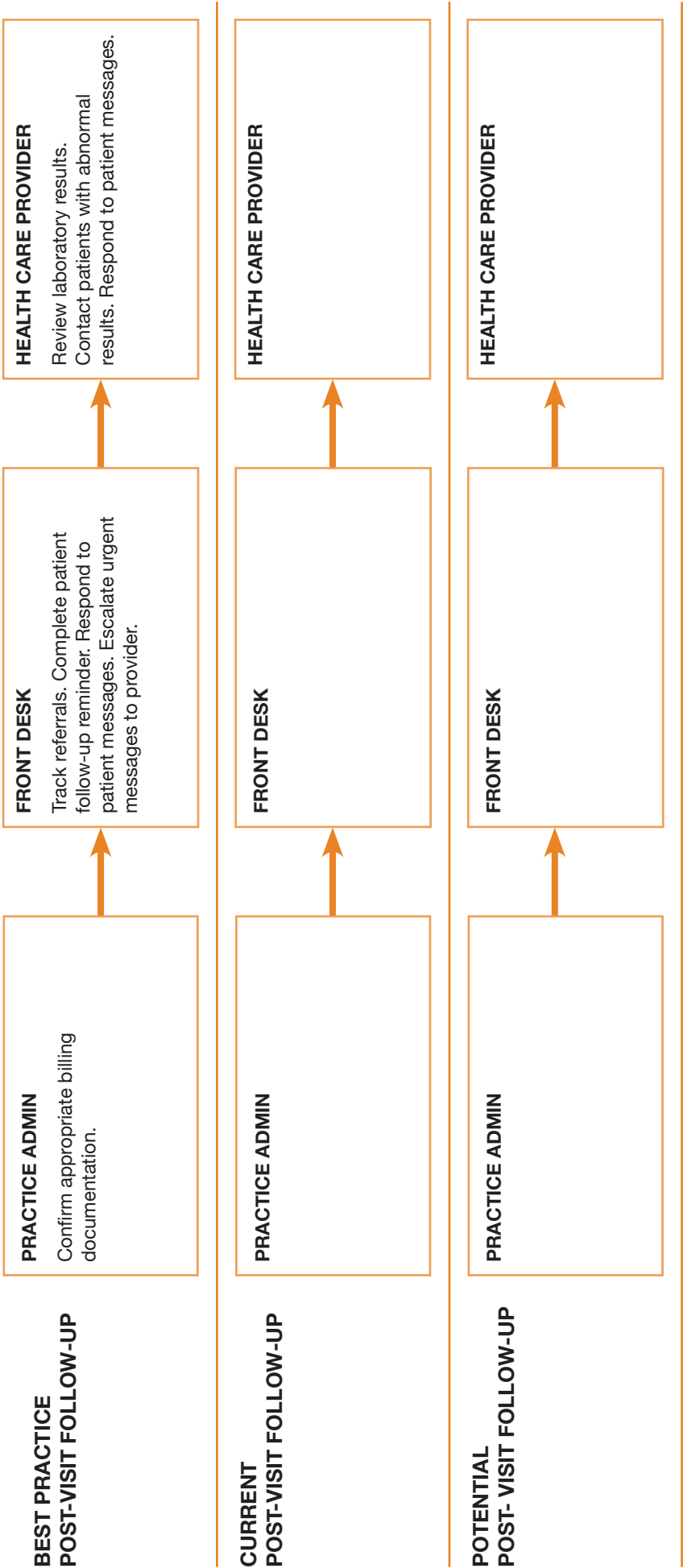
Patient visit – Assessment of current practice

Rooming the patient	Current process	Barriers/duplication	Ideal process
1. What is the average patient wait time to see the doctor?			
2. How does the front office notify the medical assistant that the patient is ready to be taken back?			
Vitals/intake	Current process	Barriers/duplication	Ideal process
3. Does the practice have a triage room/area? Who conducts triage, where is it conducted and how? What is measured? How is it recorded?			
4. Who conducts initial screenings, i.e., chief complaints, subjective history, etc.?			
5. Who reviews current medications in the medical record? Is it completed for every visit?			
6. Does the practice perform tobacco screening and cessation counseling for tobacco users?			
Provider	Current process	Barriers/duplication	Ideal process
7. What is the communication and handoff between the medical assistant and the provider (e.g., reviewing of vitals, concerns)?			
8. Review how the provider manages patients with chronic conditions, i.e., referrals, medication reconciliation and adherence, treatment procedures, etc.			
9. How does the provider communicate to the medical assistant that the patient is ready for check out? What action is taken, i.e., bill given, lab orders, vaccines, etc.?			
10. How long does it take the provider to write visit notes? Does the provider complete during or after the visit? When does the provider sign-off the chart?			

Patient visit – Assessment of current practice (continued)

Clinical discharge	Current process	Barriers/duplication	Ideal process
11. Are prescriptions provided to the patient or sent directly to the pharmacy?			
12. Who gives the patient their prescriptions and associated prescription education (i.e., pharmacist or physician during visit)?			
13. Who provides patient education? How (i.e., paper, electronic)?			
Referrals	Current process	Barriers/duplication	Ideal process
14. Do you have a list of providers you commonly refer patients to?			
Check out	Current process	Barriers/duplication	Ideal process
15. How does the patient get directed to check out? Describe the check-out process.			
16. Is a follow-up visit scheduled? Are instructions/education provided?			
17. Does the practice provide the patient with any forms during check-out?			
Additional notes:			

Post-visit follow-up – Workflow mapping worksheet



Referrals	Current process	Barriers/duplication	Ideal process
1. How is the loop closed to ensure continuity of care (e.g., nurse, medical assistant, practice administrator)?			
2. Does the practice commonly generate outgoing referrals? Receive incoming referrals? Which is more common? Describe overall processes.			
Clinical telephone encounters	Current process	Barriers/duplication	Ideal process
3. What are the most common telephone encounters (e.g., refills, test results, referrals)? Describe the process. Are there specific policies in place regarding turnaround time?			
4. How are incoming calls tracked (i.e., is a telephone encounter created for each call)?			
5. What is the process for returning calls? Outgoing calls (e.g., lab results)?			
Clinical telephone encounters	Current process	Barriers/duplication	Ideal process
6. Does the provider offer after-hours or weekend calls? How are these documented? Describe the process.			
Additional notes:			

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Technical package for cardiovascular disease management in primary health care



Systems for monitoring



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Introduction

Monitoring is a crucial element in any successful programme. It is important to know if health care facilities – and ultimately countries – are meeting the agreed goals and objectives for preventing and managing cardiovascular diseases (CVD).

Monitoring is the on-going collection, management and use of information to assess whether an activity or programme is proceeding according to plan and/or achieving defined targets. Not all outcomes of interest can be monitored. Clear outcomes must be identified that relate to the most important changes expected to result from the project and to what is realistic and measurable within the timescale of the project. Once these outcomes have been articulated, indicators can be chosen that best measure whether the desired outcomes are being met.

To allow progress to be monitored, this module provides a set of indicators on CVD management. Agreeing on a set of indicators allows countries to compare progress in CVD management and treatment across different districts or subnational jurisdictions, as well as at a facility level, identify where performance can be improved, and track trends in implementation over time. Monitoring these indicators also helps identify problems that may be encountered so that implementation efforts can be redirected.

This module starts from the collection of data at facility level, which is then “transferred up” the system: facility-level data are aggregated at subnational level to produce reports that allow tracking of facility and subnational performance over time and allow for comparison among facilities. National-level data are obtained through population-based surveys.

Implementing a monitoring system requires action at many levels. At national and subnational levels, staff can determine how best to integrate data elements into existing data collection systems – such as the routine service-delivery data that are collected through facility-level Health Management Information Systems (HMIS). In the facility setting, personnel must be aware of what data are needed. Sample data-collection tools are included, recognizing that countries use different data-management systems for HMIS, so the CVD monitoring tools will be adapted to work with the HMIS system being used by the country, such that the indicators can be collected with minimal disruption/work to existing systems and tools.

This module provides:

- indicators – including defining each indicator, the data elements that need to be collected for each indicator, and the data sources
- sample tools for data collection and reporting.

Use of module

This module is intended for facility managers and subnational and/or national-level staff involved in collecting, planning or adapting monitoring systems for CVD-management services in primary health care.

Choosing indicators

Indicators are the foundation of a monitoring system.

A monitoring system starts with defining the indicators needed to answer the most important monitoring questions. It is essential that indicators are standardized so that they can be used for comparisons across time, place, and populations.

Collecting and analysing data to calculate indicators involves time and resources. Any system will have limits on the amount of data it can obtain in a way that is timely and useful and that assures good data quality. A limited number of carefully selected indicators that are related to action and that can be relatively easily collected and integrated into existing country health management information systems (HMIS) is preferable to a large number of indicators that may end up being inconsistently collected, of poor quality, or not used.

The number and type of indicators that can feasibly be collected also depends on the type of HMIS used in a country. For example, a country with a mainly paper-based HMIS will be more limited in the indicators it can feasibly obtain compared with a country that has a system based on electronic health records. Indicators must be standardized so that they can be used for comparisons over time and across places and populations. This involves defining the way it is calculated, providing a definition for the numerator and the denominator, and suggesting the sources, tools, and frequency of collection.

Levels of monitoring

The CVD monitoring system consists of three types of monitoring that work together to provide the CVD management indicators.

Health-facility level

At the health-facility level, individual patient monitoring involves monitoring of the health status and the management of a single patient over time, using an individual CVD patient treatment card (see Annex). A facility-based register combines the details of all patients in the facility. Subsets of data from the CVD patient treatment card are extracted and used for programme monitoring. The elements of the CVD patient treatment card can also be used to monitor services such as quality of care for hypertension and diabetes, including adherence to medication, follow-up examinations, and end organ damage, depending on the local context.

Subnational level

At subnational level, aggregated data from health facilities can help to assess the outcomes within the programme and also monitor availability of medicines. Some elements of quality of care can be assessed at the subnational level using the checklists.

Population level monitoring

Population-level monitoring involves surveys in the population at the national or subnational level. Surveys use standardized tools and can provide an estimate of the prevalence of the condition and related parameters, such as the proportion of people receiving medication and the proportion with blood pressure at target, etc. Population-based indicators are a reflection of all interventions and programmes in the catchment area. Over time, they can provide trends and will serve as an overall indicator of the effectiveness and coverage of the programme.

Models of record keeping

Model 1: Paper-based system

The data collection described in this module represents a basic system that could be implemented in all contexts, including low-resource settings. This model is based on a system of paper-based individual health records, registers, and data-collection tools. Once collected, the data are entered into electronic databases for aggregation and analysis at subnational level.

Model 2: Hybrid paper and electronic system

A data-collection model using a longitudinal register (paper-based or electronic) may also be considered. In a longitudinal register, the patient's name is entered once in the register, but selected clinical information is entered at successive visits along the same line in the register. Immunization registers or TB treatment registers are examples of longitudinal registers. A sample longitudinal register (facility register) is provided in the Annex.

Model 3: Electronic record system

Where a system of electronic health records is operational, a CVD module could be included in the electronic individual record. The electronic system should collect and aggregate the facility data exactly as the paper-based system does, but with significantly more ease, particularly for the large volume of patients with CVD. This system enables tracking of key individual patient CVD parameters as well as regular, and potentially more frequent, automated extraction of data for aggregate reporting.

1 Indicators

This section outlines a list of indicators to be used for monitoring HEARTS implementation. Some of these data come from the health facilities providing the services, while other components will need to be collected by population-level surveys. Monitoring needs to take place in three settings to get the data needed for these indicators: at the health-care delivery (facility setting), the district or subnational level for aggregated indicators, and at the population level.

Table 1: Indicators

Health-facility level				
No.	Indicator	Source of data	Reporting frequency	Health system considerations
1	Six-monthly control of blood pressure among people treated for hypertension	Hypertension treatment register in the facility	Once in 3 months	Feasible in all settings in primary health care and a core indicator for quality of services
Subnational (district/province/state) level (aggregated from health facilities offering the services within the programme)				
	Indicator	Source of data	Reporting frequency	Considerations in the interpretation
2	Control of blood pressure among people with hypertension within the programme	Aggregated reports from all the health facilities reporting the hypertension indicator in a defined subnational area; estimation of hypertension prevalence	Once in 12 months	This will give estimated community control rates with the numerator coming from facilities reporting as part of the programme (in some instances patients may be receiving BP meds from private sector or other levels of care within the public system)
3	Availability of core cardiovascular disease/diabetes drugs	Aggregated reports from all the health facilities reporting drug availability indicators in a defined subnational area	Once in 3 months	This is for the programme quality control and will assist with forecasting of medicines and improvements in supply chain management
Population level (control of hypertension, diabetes and CVD risk)				
	Indicator	Survey method	Frequency	Other considerations
4	Hypertension control in the population	Population-based sample survey (STEPS or similar survey)	Once in 3–5 years	Population-level survey as part of national survey or a special survey for the programme
5	Proportion of eligible persons receiving drug therapy and counselling (including glycaemic control) to prevent heart attacks and stroke (1)	Population-based sample survey (STEPS or similar survey)	Once in 5 years	Population-based (preferably nationally representative) survey, including behavioural parameters with physical and biochemical measurements

Table 2: Six-monthly control of blood pressure among people treated for hypertension

Definition	Proportion of patients registered for hypertensive treatment at the health facility whose blood pressure is controlled 6 months after treatment initiation
Purpose	To measure the effectiveness of clinical services in the programme to control blood pressure among cohorts of treated patient
Method of calculation	A = Number of patients with controlled blood pressure (SBP <140 and DBP <90 mmHg) at the last clinical visit in the most recent quarter (just before the reporting quarter) out of B B = Number of patients registered for treatment of hypertension during the quarter that ended 6 months previously Calculation: A ÷ B
Source of data	Health facility register for hypertension
Recommended target	Fix a target as per the local context
Key data elements	Date of registration, date of last visit, systolic blood pressure, diastolic blood pressure
Frequency of reporting	Quarterly
Users of data	Facility managers: to understand what proportion of patients at their facility are achieving the blood pressure goal District-level manager: to assess the overall quality of hypertension treatment services, to identify poorly performing facilities and rectify problems at an early stage
Data collection tool	Facility register for hypertension (see Annex)

Table 3: Control of blood pressure among people with hypertension

Definition	The proportion of hypertensive people at health facilities in a given geographical area, such as a district, province, or state, with controlled blood pressure
Purpose	To measure the increase in coverage of the programme to treat and control hypertension in a given geographical area, such as a district, province, or state
Method of calculation	A = Cumulative number of registered patients with controlled blood pressure (SBP <140 and DBP <90) in the most recent quarter at all health facilities in a given geographical area, such as a district, province, or state B = Estimated number of people with hypertension at the subnational level Calculation: A ÷ B
Source of data	Numerator: registers from health facilities reporting in the given geographical area such as a district, province, or state Denominator: Prevalence of hypertension from population-based survey (STEPS or similar survey)
Disaggregated by	Health facility
Recommended target	Fix a target as per local context
Key data elements	Date of last visit, systolic blood pressure, diastolic blood pressure
Frequency of reporting	Annual
Users of data	District, province, or state programme managers to monitor increase in programme coverage of hypertension services within a geographical area National programme managers to monitor progress towards universal health coverage
Data collection tool example	Health facility register for hypertension (see Annex) Health facility report (see Annex)

Table 4: Availability of core CVD/diabetes drugs

Definition	The proportion of facilities in a given geographical area that have core CVD/diabetes drugs available (see list of drugs below)
Purpose	To ensure uninterrupted supply of essential drugs and thereby improve patient treatment adherence
Method of calculation	A = number of health facilities in the programme reporting “no stock-out” of core CVD/diabetes drugs in the last quarter B = Number of health facilities participating in the programme Calculation: A ÷ B
Source of data	Aggregated health-facility drug stock register; health facility report
Disaggregated by	Health facility
Recommended target	No stock-out
Key data elements	Count of number of facilities reporting “no drug stock-out” in the last quarter; number of days of drug stock-out of selected medicine at each health facility
Frequency of reporting	Quarterly
Users of data	District- and province-level managers to focus supervision on health facilities reporting drug stock-outs, prevent drug stock-out situations and strengthen health systems to ensure uninterrupted drug supply
Data collection tool example	Health facility report (see Annex)

Core CVD/diabetes drugs

- thiazide or thiazide-like diuretic
- calcium channel blocker (CCB) (long acting) (amlodipine)
- angiotensin converting enzyme inhibitor (ACE-I) (long acting) and angiotensin receptor blocker (ARB)
- statin
- insulin
- metformin
- glibenclamide
- beta-blocker
- aspirin.

Table 5: Hypertension control in the population

Definition	Proportion of all hypertensive people with controlled blood pressure in the population
Purpose	To measure population-level hypertension control, including trends over time
Method of calculation	A = Number of respondents with SBP <140 and DBP <90 who are EITHER being currently treated with medications for hypertension OR have been diagnosed with hypertension B = Number of survey respondents with SBP ≥140 or DBP ≥90 OR who are currently treated with medicines for hypertension Calculation: A ÷ B
Source/methodology	Population-based sample survey (national or subnational health survey)
Disaggregated by	Age, sex, socio-economic status
Frequency of reporting	Once in 3–5 years
Users of data	National policy makers to measure progress toward universal health coverage, formulate national health policies, allocate programmatic budget Global policy makers to compare progress in UHC across countries
Data collection tool example	http://www.who.int/ncds/surveillance/steps/en/

Table 6: Proportion of eligible persons receiving drug therapy and counselling (including glycaemic control) to prevent heart attacks and strokes

Definition	Percentage of eligible persons (defined as aged 40 years and older with a 10-year cardiovascular disease (CVD) risk ≥30%, including those with existing CVD) receiving drug therapy and counselling* (including glycaemic control) to prevent heart attacks and strokes
Purpose	To measure change in population-level CVD-risk management
Method of calculation**	A = Number of eligible survey participants who are receiving drug therapy and counselling*** B = Total number of eligible survey participants. (defined as aged 40 years and older with a 10-year cardiovascular risk ≥30%, including those with existing cardiovascular disease) Calculation: A ÷ B
Source/methodology	This is generated from population-based surveys such as a population-based sample survey (STEPS or similar survey)
Disaggregated by	Age, sex, socio-economic status
Recommended target	5% increase every year
Frequency of reporting	Once in 5 years
Users of data	National policy makers to measure progress towards NCD global action plan targets Global policy makers to compare progress in NCD global action plan targets across countries
Data collection tool example	http://www.who.int/ncds/surveillance/steps/en/

* Feasible in settings that have a comprehensive population-based survey with behavioural parameters along with physical and biochemical measurements.

** More information on the indicator is available at http://www.who.int/nmh/ncd-tools/indicators/GMF_Indicator_Definitions_Version_NOV2014.pdf

*** Use of the term “eligible persons” does not imply that others should not receive treatment. Countries may wish to consider analyses that include persons at high risk as defined by the national guidelines.

2 Data collection and reporting tools

1 Health-facility level

Table 7: Health-facility-level reporting

Tool	Use	Advantages
Treatment card	This captures data for the management of an individual patient being registered in the system. Each patient assessed and managed for hypertension, diabetes and/or cholesterol should have an individual record in which information is recorded at every follow-up visit. An example of a patient treatment card is shown in the Annex.	Highlights the key CVD-related information that the health worker should review and record at each visit Facilitates comparison of key clinical information over time Can serve as a clinical reminder of important aspects of care Can be used as a basis for supervision Serves as a primary source of data for the minimum indicators
Facility registers	A patient register to which selected patient data extracted from the individual patient treatment card are recorded. Data from an individual patient are entered in the register once only, when the patient is first registered at the facility. If the patient is started on treatment, this is indicated. If the treated patient is then transferred out, lost to follow-up or dies, then this information is also indicated.	Provides the health facility with a quick way of assessing the number of patients registered in the system, stratified by gender, age, and systems approach. Allows a quick assessment, each quarter, of the number cumulatively transferred-out, lost to follow-up, or who have died, enabling health facility staff to check on their preparation of quarterly reports Useful to maintain an out-patient register to fill in brief information (name, ID number, date of visit, hypertension status, diabetes status, and SBP/DBP) which can be compared to patient cards during clinical audits and regular compilation of indicators.
Facility-based quarterly report	The data for the quarterly report is compiled by using the patient treatment cards as well as the patient register. The register can be used to calculate the denominators. For some indicators, treatment cards may be needed to calculate the numerators. As the cumulative number of patients in the clinic increases, this task becomes more time consuming if reliant on paper-based systems. Movement to an electronic records system would considerably ease the task.	The review of aggregate patient data occurs at regular time intervals, such as every three months, and these are termed “quarterly reports”. These quarterly reports focus on the agreed minimum set of indicators that are used to monitor progress of the programme. Supervision and audit are also carried out at set intervals (monthly or quarterly) to ensure that guidelines and processes are being followed

2 Subnational level (aggregated data)

Table 8: Subnational-level reporting

Report	Process of compiling	Review
District-level quarterly report	Data collated from facility-level reports	The review of aggregate patient data from all facilities in the district occurs at regular intervals, usually quarterly, enabling district performance to be assessed, and comparisons made between facilities in the district. Challenges encountered during facility-based supervision visits, referring to use of supervision and audits tools, are discussed at district quarterly meetings. Annual programme reviews build on quarterly cumulative reports and assess progress made each year against the key indicators.

3 National level

Data collection for national indicators

Data for the population-level indicators can come from different sources. One example, the WHO STEPwise approach to Surveillance (STEPS), is a simple, standardized method for collecting, analysing and disseminating data in WHO member countries. By using the same standardized questions and protocols, all countries can use STEPS information not only for monitoring within-country trends, but also for making comparisons across countries. The approach encourages the collection of small amounts of useful information on a regular and continuing basis.

The STEPS survey, usually carried out every 3–5 years, provides population-level indicators on the prevalence of CVD risk factors, the proportion of people on treatment for diabetes and raised blood pressure, and the proportion of those who have it under control. Other global surveys, such as the Demographic and Health Surveys may also be able to provide this information. In addition, countries may have their own health surveys, which can be a good source of this information. Any survey is acceptable, provided the required data can be obtained.

STEPS

The tool used to collect data and measure noncommunicable disease (NCD) risk factors within the WHO STEPwise approach to surveillance is called the STEPS Instrument. The STEPS Instrument covers three different levels or “steps” of risk-factor assessment:

Step 1: questionnaire

Step 2: physical measurements

Step 3: biochemical measurements.

When undertaking national STEPwise surveys or other similar surveys, countries can use the indicators relevant for hypertension and also have additional sample size to get estimates from the population of interest (population covered by hypertension management). Standalone surveys can be conducted in the

population of interest using the STEPS approach, and the WHO STEPS manual provides the details of how to do this.

Population-based surveys are taken up through the national NCD surveillance focal point in the country and also with the support of academic institutions or health research institutions in the country.

For more information see:
<http://www.who.int/ncds/surveillance/steps/instrument/en/>

4 Supervision and clinical audit checklist

A clinical audit is a part of the continuous quality-improvement process. It consists in measuring a clinical outcome or a process against well-defined standards, established using the principles of evidence-based medicine. The Annex provides sample tools for supervision and clinical audit. The comparison between clinical practice and standards leads to the formulation of strategies, in order to improve the quality of daily care. (2)

The uses of a supervision and clinical audit:

- enables extraction of data from a random sample of patient records to measure the minimum indicators for quality of care and health outcomes
- can be used as a supervision tool to review a limited number of records for data quality (completeness and coherence with register) as well as quality of care
- can be used to review a limited number of records as part of internal facility quality-management processes.

3 Analysing and reviewing data

The purpose of collecting data is to improve patient care and service delivery. Analysis and review of data and indicators takes place at different levels of the health system to meet the monitoring and management needs at each level.

Table 9: Examples of data-review processes for CVD management at different levels of the health system

Level	Process
Individual patient monitoring	Review of patient data takes place at each visit of an individual patient.
Facility-level monitoring	Periodic internal quality review and/or supervision: • Review output/outcomes data as reported in the facility quarterly report. • Review sample of between 5 and 10 records using clinical audit tool; use as case studies for further discussion. • Review availability of resources using facility assessment tool as a checklist.
District-level monitoring	Quarterly district meeting: • Review output data from all facilities, including district totals and compare results across facilities. • Highlight challenges encountered during supervision visits, referring to use of clinical audit tool as a supervision aid. Annual programme review: • Review annual output data: - o Review annual trends of district totals. - o Compare facility performance. • Conduct clinical audit: - o Present district summary indicators. • If sample sizes permit, present facility comparisons.
National-level monitoring	Annual program review: • Review annual output data: - o Review annual trends of national totals. - o Compare district performance. • Compare district performance for outcomes.

Annex: Example recording tools

Examples of tools for recording and reporting treatment and services at all levels are provided as follows:

CVD patient treatment card

Facility register for hypertension and diabetes

Health facility report

Supervision and clinical audit tools:

- Treatment supervision / audit form
- Patient interviews
- Summary of supervision vision

[illegible]

6-9-month cohort monitoring – registration and corresponding reporting dates	
If patient registered between:	Report to district on:
1 January – 31 March	15 October
1 April – 30 June	15 January
1 July – September 30	15 April
1 October – 1 December	15 July

HEALTH FACILITY REPORT		
Sections A and B will be filled out by health facilities where HTN Facility Register is placed.		
Sections C and D will be filled out by all health facilities		
Name of health facility:	Name of district:	
Name of state:	Date of reporting (day/month/year)	
Quarter for which you are making the report:	Quarter:	Year:
	This is the 'Reporting Quarter'. Usually this is the most recent quarter that has just finished.	
Section A: Quarterly treatment enrolment and outcomes		Number of patients
A1: Number of patients registered two quarters earlier		
A2: Out of (A1), number of patients whose BP was documented to be <140/90 mmHg in the Reporting Quarter		
Section B: Annual treatment enrolment and outcomes (To be filled in only once a year, with Quarter 1 report)		Number of patients
B1: Number of patients whose BP is documented as <140/90 mmHg during Quarter 1. (If the patient made more than one visit in the quarter, use most recent reading.)		
B2: Estimated number of people with hypertension in the catchment population (only for district level).		
Section C: Drug consumption and availability		
Quarterly consumption of drugs (Give number of tablets)	Quantity of drugs available at the health facility (Give number of tablets)	Quantity of drugs requested for the next quarter (Give number of tablets)
calcium channel blocker		
angiotensin receptor blocker		
angiotensin converting enzyme inhibitor		
thiazide/thiazide like diuretic		
statin		
aspirin		
beta blocker		
Section D: Quarterly supervision		
Was there a supervision visit to this health facility by district staff during the reporting quarter?	Yes	No

TREATMENT SUPERVISION/AUDIT FORM		
Facility name:		District name:
Name of supervisor:		Date:
Name of medical officer:		
Has the facility started the hypertension treatment programme? Circle, as appropriate:		Y N NA
N°	Indicator	Circle any
1	Screening and BP measurement	
1.1	Is opportunistic screening done for all adults?	Y N
1.2	Is the BP measurement protocol displayed on the wall/desk?	Y N NA
1.3	Is there at least one functioning BP instrument in the facility?	Y N NA
1.4	Are all patients with BP ≥140/90 referred to the medical officer for treatment?	Y N NA
1.5	For how many patients was BP measured correctly? (Observe 5, >2 of each staff who measure BP.)	0 1 2 3 4 5
2	Treatment	
2.1	Is the treatment algorithm displayed on the wall/desk?	Y N NA
	Randomly audit 10 patient treatment cards (see Patient card audit form). Write for what proportion of patients:	Proportion:
2.2	BP was recorded at every visit for the last three visits	
2.3	Initial antihypertensive medication was given as per protocol	
2.4	Medication was intensified or added as per protocol if BP ≥140/90 (write NA if not applicable)	
2.5	Aspirin was given if patient had prior CVD (write NA if not applicable)	
2.6	Statin was given if patient >40 yrs with diabetes or if patient had prior CVD (NA if not applicable)	
2.7	Referral to a specialist was made if BP ≥140/90 after treating with three drugs (NA if not applicable)	
2.8	BP was <140/90 at last visit	
3	Counselling and follow-up	
3.1	Is there a staff assigned for patient counselling?	Y N NA
3.2	Are patient counselling tools/materials available?	Y N NA
3.3	Is there a system for counselling patients individually or as a group?	Y N NA
3.4	Is there a system for tracking initial defaulters?	Y N NA
3.5	Is there a functional system for patient reminder and follow-up?	Y N NA
4	Service delivery. Interview 5 patients and validate (see Patient interview report card). Circle number of positive responses	
4.1	Was BP measured at every visit?	0 1 2 3 4 5
4.2	Did the patient receive all prescribed medicines at this visit?	0 1 2 3 4 5
4.3	Did the patient ever have to pay for medicines in the past?	0 1 2 3 4 5
4.4	Does the patient have correct understanding of how to take medicines?	0 1 2 3 4 5
4.5	Does the patient know his/her BP reading at this visit?	0 1 2 3 4 5
4.6	Does the patient know the target BP?	0 1 2 3 4 5
5	Drug inventory system	
5.1	Is there a functioning drug inventory system in place?	Y N NA
5.2	Was there a stock-out of core drugs in the past quarter?	Y N NA
5.3	If there was a stock-out this quarter, which drugs were not available?	Y N NA
5.4	Is there enough buffer stock of core drugs for the next quarter?	Y N NA
6	Patient recording and reporting system	
6.1	Is there a functioning recording and reporting system in place?	Y N NA
6.2	Are there sufficient patient cards for next three months?	Y N NA
6.3	Is the facility register for follow-up available?	Y N NA
6.4	Is there a place to arrange/store patient cards?	Y N NA
6.5	Are the cards organized by serial number or other system so easily retrievable	Y N NA
6.6	Was last quarter's report sent on time?	Y N NA
6.7	Does the clinic in charge know the percentage of patients with BP <140/90 at the facility?	Y N NA
6.8	Is last quarter's 6-month BP control rate reported accurately? (check register from last quarter)	Y N NA

PATIENT INTERVIEW REPORT CARD							
Interview 5 patients and write yes or no for each question. To calculate the total, simply tally the number of yeses in each row. Copy the total into the clinical audit tool.							
N°	Items	Pt 1	Pt 2	Pt 3	Pt 4	Pt 5	Total
1	Did the patient receive all prescribed medicines at this visit?						
2	Did the patient ever have to pay for medicines in the past 6 months?						
3	Does the patient have correct understanding of how to take medicines?						
4	Did the patient know whether their BP was under control at the last visit?						
5	Does the patient know the target BP?						

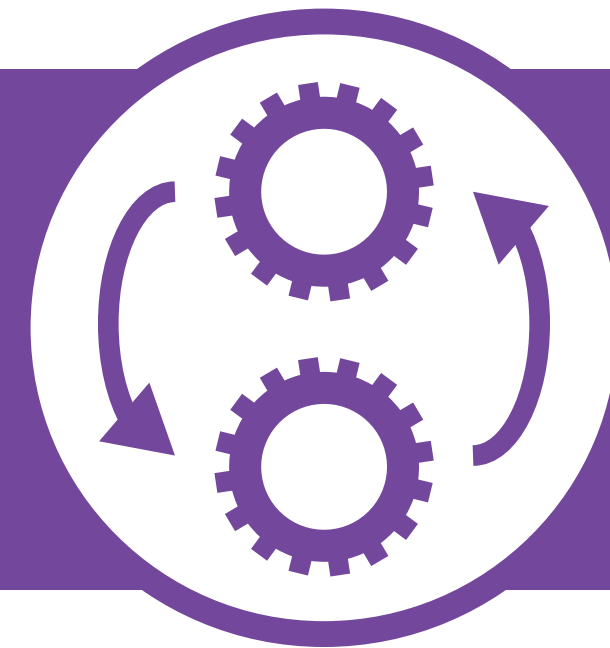
SUMMARY OF SUPERVISION VISITS		
Problem identified		Recommendations
1	Screening and BP measurement	
2	Treatment	
3	Counselling and follow-up	
4	Service delivery including costs to patient	
5	Drug inventory	
6	Recording and reporting	
7	Any other	

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Technical package for cardiovascular disease management in primary health care



Implementation guide

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HEARTS Technical Package

More people die each year from cardiovascular diseases (CVDs) than from any other cause. Over three-quarters of heart disease and stroke-related deaths occur in low- and middle-income countries.

The HEARTS technical package (HEARTS) provides a strategic approach to improving cardiovascular health. It comprises six modules and an implementation guide. This package supports Ministries of Health to strengthen CVD management in primary care and aligns with WHO's Package of Essential Noncommunicable Disease Interventions (WHO PEN).

HEARTS modules are intended for use by policymakers and programme managers at different levels within Ministries of Health who can influence CVD primary care delivery. Different sections of each module are aimed at different levels of the health system and different cadres of workers. All modules will require adaptation at country level.

The people who will find the modules most useful are:

- **National level** – Ministry of Health NCD policymakers responsible for:
 - developing strategies, policies and plans related to service delivery of CVD
 - setting national targets on CVD, monitoring progress and reporting.
- **Subnational level** – Health/NCD programme managers responsible for:
 - planning, training, implementing and monitoring service delivery
- **Primary care level** – Facility managers and primary health care trainers responsible for:
 - assigning tasks, organising training and ensuring the facility is running smoothly
 - collecting facility-level data on indicators of progress towards CVD targets.

Target users may vary, based on context, existing health systems and national priorities.

MODULES OF THE HEARTS TECHNICAL PACKAGE				
Module	What does it include?	Who are the target users?		
		National	Subnational	Primary care
H ealthy-lifestyle counselling	Information on the four behavioural risk factors for CVD is provided. Brief interventions are described as an approach to providing counselling on risk factors and encouraging people to have healthy lifestyles.		✓	✓
E vidence-based protocols	A collection of protocols to standardize a clinical approach to the management of hypertension and diabetes.	✓	✓	✓
A ccess to essential medicines and technology	Information on CVD medicine and technology procurement, quantification, distribution, management and handling of supplies at facility level.	✓	✓	✓
R isk-based CVD management	Information on a total risk approach to the assessment and management of CVD, including region-specific risk charts.		✓	✓
T eam-based care	Guidance and examples on team-based care and task shifting related to the care of CVD. Some training materials are also provided.		✓	✓
S ystems for monitoring	Information on how to monitor and report on the prevention and management of CVD. Contains standardized indicators and data-collection tools.	✓	✓	✓



Implementing HEARTS

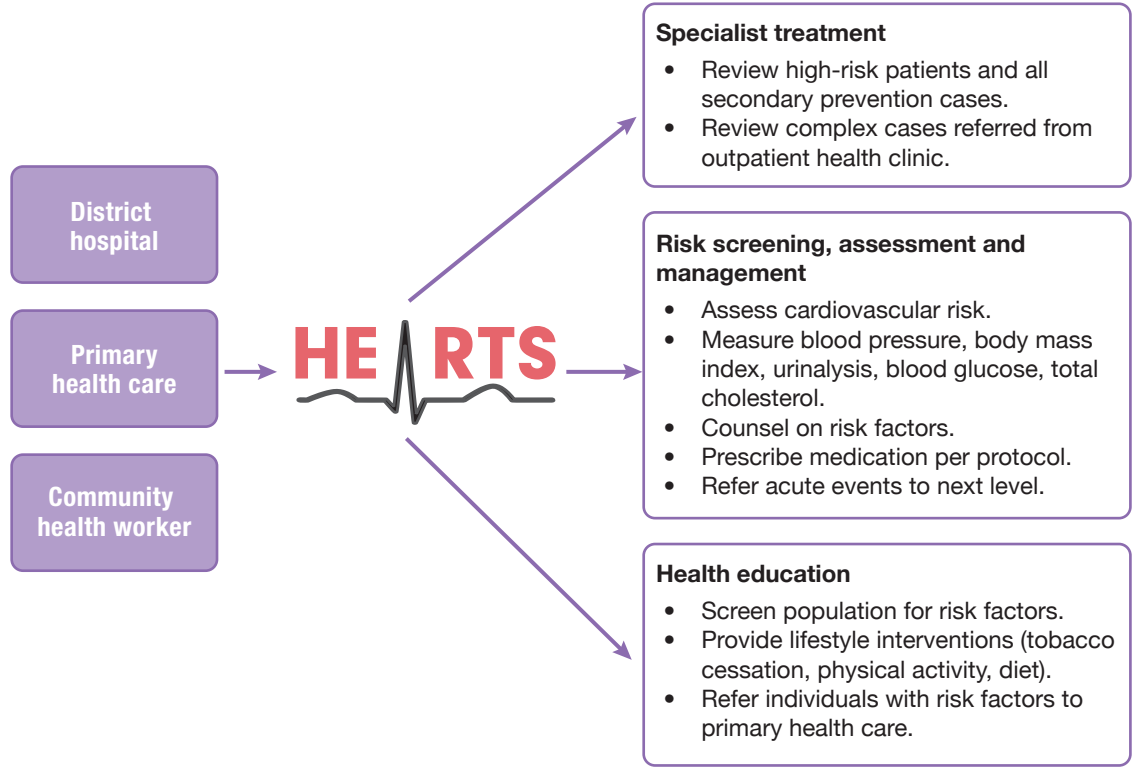
HEARTS provides a set of locally adaptable tools for strengthening the management of CVD in primary health care.

HEARTS is designed to enhance implementation of WHO PEN by providing:

- operational guidance on further integrating CVD management
- technical guidance on evaluating the impact of CVD care on patient outcomes.

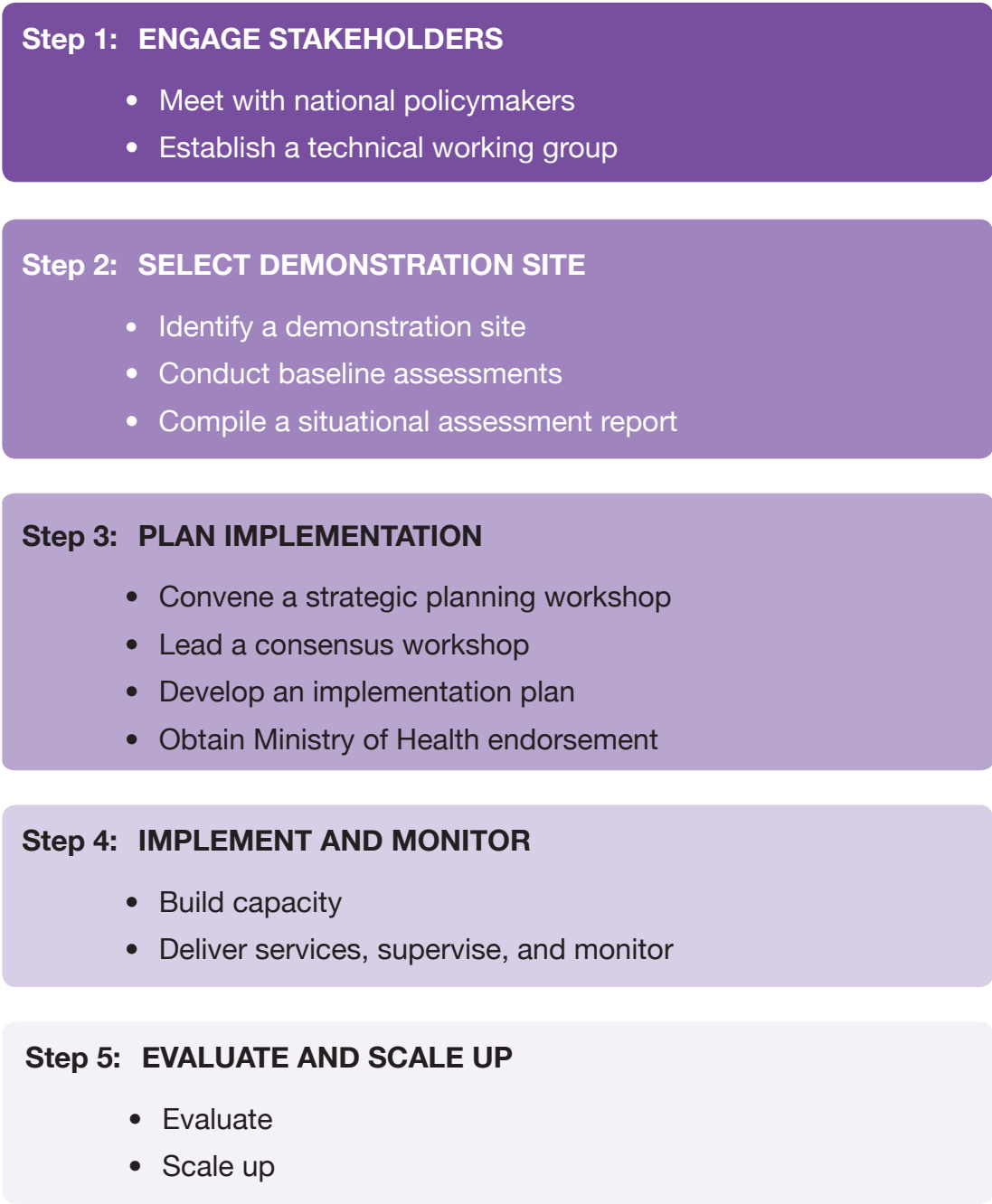
For countries not using WHO PEN, CVD management can still be integrated into primary health care. The process of implementing HEARTS will vary, depending on country context, and may require a significant reorienting and strengthening of the health system. At some sites, existing CVD management services may be reoriented toward a risk-based approach, while other sites may adopt a public health approach, strengthening management of particular risk factors such as hypertension. Whether or not introducing CVD management into primary care is a new intervention, successful implementation will require engagement with national and local health planners, managers, service providers, and other stakeholders.

Figure 1: Integrating HEARTS into primary care for CVD management



This guide provides a step-by-step approach to implementing HEARTS, as informed by the successful approaches used in various low- and middle-income countries. Effective implementation techniques may vary by country, and the process should be adapted to meet local needs. A sample of pre-implementation adaptation from the Pan American Health Organization can be found in Annex 1.

Figure 2: Step-by-step approach to implementing HEARTS



Step 1: Engage stakeholders

To successfully implement HEARTS, first establish buy-in from stakeholders. Engage key stakeholders through a meeting with national policymakers and through the creation of a technical working group.

(a) Meet with national policymakers

Implementing HEARTS in existing primary health care systems requires strong, high-level political commitment, leadership, and ownership by the Ministry of Health (MoH). MoH endorsement and involvement are essential to laying the necessary groundwork for successful implementation.

When meeting with policymakers, it is critical to raise awareness of the high burden and cost of CVDs, and to provide information about proven, feasible, cost-effective interventions that can reduce CVD burden in the country. Present the HEARTS technical package as a strategic approach to CVD management that allows for country-level customization and adaptation. The goal of the introductory meeting is for leaders to recognize the need to improve the quality of CVD care and treatment, and to commit to strengthening CVD management.

The following can be used as a reference for introductory meetings.

(i) Review the global targets to reduce CVD burden

In 2011, countries committed to taking action against NCDs in the Political Declaration of the High-Level Meeting of the General Assembly on the Prevention and Control of Noncommunicable Diseases. (1) Nine voluntary global NCD targets were defined, and their achievement will be assessed in 2025. Three of these targets relate directly to CVD management. CVD control efforts are also included in the Sustainable Development Goals (SDGs), for which countries will be assessed in 2030. It may be helpful to emphasize the important role that each country has in helping to meet these global targets. Table 1 outlines the different targets that HEARTS addresses.

Figure 3: Relevant global targets and indicators

Sustainable Development Goal targets (2)	 3.4 By 2030, reduce by one-third the premature mortality from non-communicable diseases through prevention and treatment and promote mental health and well-being 3.4.1 Mortality rate attributed to cardiovascular disease, cancer, diabetes or chronic respiratory disease
NCD Global Action Plan targets (3)	 A 25% relative reduction in risk of premature mortality from cardiovascular diseases, cancer, diabetes or chronic respiratory diseases  At least 50% of eligible people receive drug therapy and counselling (including glycaemic control) to prevent heart attacks and strokes  An 80% availability in both public and private facilities of affordable basic technologies and essential medicines (including generics) required to treat major noncommunicable diseases
WHO progress monitoring indicators (4)	 9 Member State has evidence-based national guidelines/protocols/standards for the management of major NCDs through a primary care approach, recognized/approved by government or competent authorities  10 Member State has provision of drug therapy (including glycaemic control) and counselling for persons at high risk in order to prevent heart attacks and strokes, with emphasis on the primary care level

(ii) Highlight statistics demonstrating the scale of CVD burden

Global and national data and statistics are important to provide background on the CVD burden in the global and local context. Use available local data to demonstrate relevance for policymakers. Possible data sources include:

- WHO Global status report on noncommunicable diseases 2014 (5)
- WHO's NCD country profiles and country capacity survey. (6)

(iii) Address the role of primary health care

Primary health care services can adapt operations to include management of CVD risk factors. Primary care services are more accessible than specialist care in most settings and are the most feasible, affordable, and equitable means of providing necessary care.

Conduct a rapid review of existing national and subnational CVD services to provide data to inform discussions with policymakers on primary health care adaptation. Potential information to cover in the rapid review includes:

- existing structure and political alignment
- coverage of a centralized or decentralized system
- existing patient pathways for CVD management
- staff responsibilities and scope of practice.

Additional guidance for conducting a rapid review is provided in Annex 2.

(iv) Discuss available resources and resource gaps

Support from policymakers will be essential in securing resources necessary for successful implementation and maintenance of HEARTS. Therefore, it is critical that policymakers understand the resources available within primary health care, as well as the potential resource gaps. Discuss resource availability relating to staff, laboratory tests, technologies and medicines with policymakers. Identify any legal concerns or restrictions.

HEARTS provides guidance on:

- enhancing staff efficiency and task-shifting (*Team-based care* module)
- identifying appropriate medicines and equipment (*Access to essential medicines and technology*)
- collecting data and implementing monitoring systems (*Systems for monitoring*).

(b) Establish a technical working group

A technical working group (TWG) can provide guidance throughout the planning, implementation, and evaluation of HEARTS. When forming a TWG, it can be helpful to identify and include local subject-matter experts and representatives from:

- national and subnational ministries of health
- academic institutions
- medical schools
- health institutes
- professional associations
- WHO and other international partners.

After a demonstration site has been selected (more details in Step 2), consider including:

- health facility management

- selected health facility staff, such as nurses, physicians, pharmacists, and other allied professionals.

Convening and coordinating the TWG will be critical to the success of the programme. For an effective TWG, it is important to establish clear roles and responsibilities for the participants. Roles will vary, depending on the expertise of the participants. If useful, establish sub-groups or committees to focus on particular aspects of implementation.

Step 2: Select demonstration site

After establishing buy-in from the MoH, the TWG, and other key stakeholders, identify a demonstration site for initial implementation, conduct baseline assessments, and compile a situational assessment report.

(a) Identify a demonstration site

Certain data will be required to make an informed selection and assist with planning. Implementation of HEARTS should be introduced into a demonstration district that ideally has:

- a population of approximately 100,000–200,000
- at least 8–10 functioning primary health care facilities
- sufficient primary health care staff to implement CVD management
- a referral facility accessible to primary health care patients (district or regional hospital)
- motivated district leadership
- a trained and motivated district-level primary health care or NCD officer who can serve as technical leader and advocate of the project.

(b) Conduct baseline assessments

Once a potential demonstration site is identified, collect baseline data and conduct necessary assessments to obtain the following:

- district population data
- brief description of district-level CVD services
- map of facilities within district
- facility-level questionnaire results (Annex 3)
- population survey results on hypertension awareness, treatment, and control (e.g., WHO STEPS).

(c) Compile a situational assessment report

Data from the previous steps can be compiled into a report to be shared with the TWG and other key stakeholders in advance of a strategic planning workshop. For the report, consider including:

- introductory meetings summary (Step 1a)
- rapid review results (Step 1a.iii)
- designated points of contact and technical working group members (Step 1b)
- selected demonstration site (Step 2a)
- key baseline data (Step 2b).

*A district is a defined geographic area with a specified catchment population, a number of primary health care facilities and at least one hospital to serve as a referral facility.

Step 3: Plan implementation

Planning the implementation of HEARTS at the selected demonstration site can be achieved through a staged approach, with input from key stakeholders integrated into the process. An initial strategic planning workshop can support the development of a service delivery model. Following that, a consensus workshop can generate agreement on standardized treatment protocols. Informed by these meetings, determine an implementation plan and arrange for its endorsement by the MoH.

(a) Convene a strategic planning workshop

After sharing the situational assessment report from Step 2c with the TWG and key stakeholders who contribute to the primary health care system, some of whom may have attended the introductory meetings in Step 1, convene a strategic planning workshop. The workshop can provide an opportunity for stakeholders to contribute their expertise. During the workshop, conduct a SWOT analysis and develop a preliminary service-delivery model.

(i) Conduct a SWOT analysis

A SWOT analysis is a strategic-planning technique that facilitates assessment of strengths, weaknesses, opportunities, and threats. Strengths and weakness are typically internal to an entity, whereas opportunities and threats are typically external. Understanding SWOT factors allows for more informed and effective programme planning. A SWOT analysis of the district-level health system can help identify necessary adjustments to the service-delivery model, assist in determining which HEARTS modules to implement, and guide implementation plans.

Figure 4: SWOT analysis template

Strengths	Weaknesses
1.	1.
2.	2.
3.	3.
Threats	Opportunities
1.	1.
2.	2.
3.	3.

Components to include under the SWOT:

- infrastructure
- personnel
- equipment
- service delivery
- medication
- monitoring
- supervision.

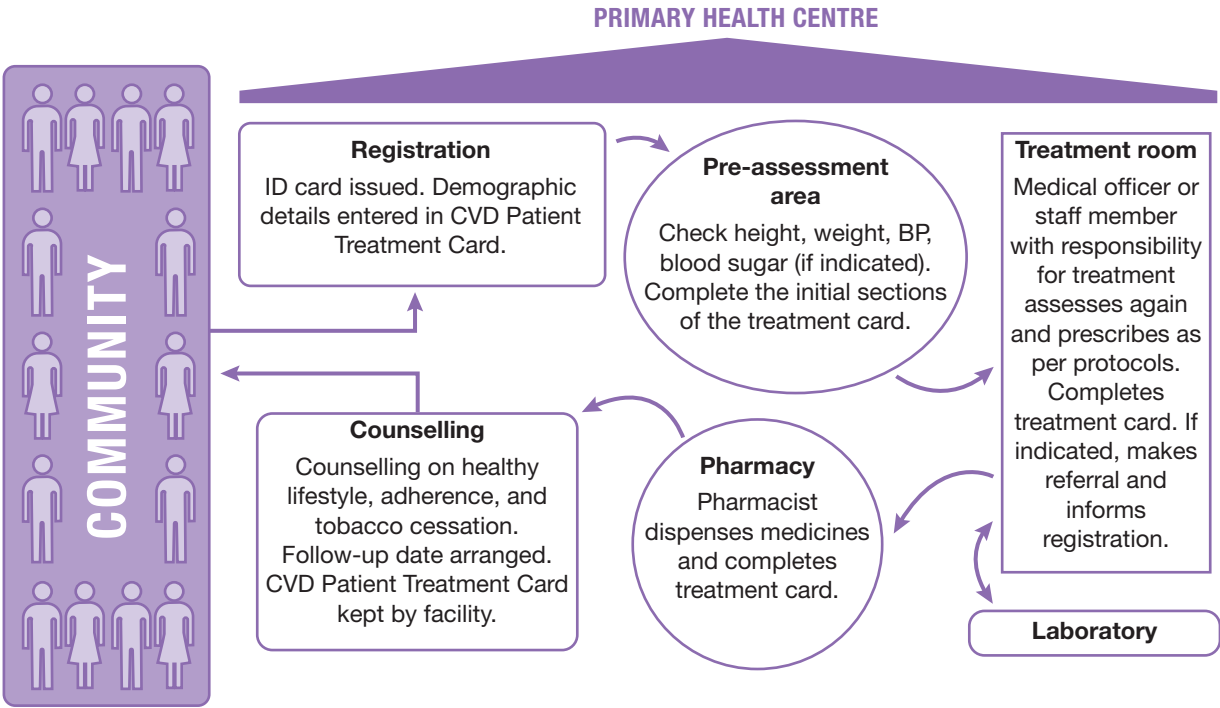
(ii) Develop a service-delivery model

Incorporating CVD management into primary care will require modification to the current service-delivery model and patient-flow pathway. When developing or adjusting the service-delivery model, consider all levels of care.

HEARTS provides guidance on:

- outlining clear roles for different health care workers (*Team-based care*)
- incorporating processes for ensuring access to core medicines and technologies (*Access to essential medicines and technology*)
- strengthening referral mechanisms to secondary and tertiary levels of care (*Evidence-based treatment protocols*)
- developing patient monitoring system (*Systems for monitoring*)
- preparing for future impact evaluations of patient outcomes (*Systems for monitoring*).

Figure 5: Example of a patient-flow pathway that supports HEARTS implementation



(b) Lead a consensus workshop

A workshop of local leaders, stakeholders, and national and international experts – including academic institutions and professional societies – can be critical to selecting and gaining consensus on standardized treatment protocols. The consensus workshop can also serve as an opportunity to present the proposed service-delivery model and invite additional feedback.

(i) Establish treatment protocols

Standardized treatment protocols are essential to improving adherence among health care facilities, producing clear data on patient outcomes, and comparing project sites. Formal endorsement of standard treatment protocols by academic societies, professional association, and key stakeholders can enhance protocol adoption and adherence.

HEARTS provides guidance on:

- selecting hypertension and diabetes management protocols (*Evidence-based treatment protocols*)
- implementing task-sharing that supports protocol adoption (*Team-based care*)
- identifying and tracking core indicators through measured impact (*Systems for monitoring*).

(c) Develop an implementation plan

An implementation plan can make broad programme goals achievable through identifying and organizing the specific steps that must be taken to accomplish the goals. The implementation plan can identify resources needed, outline measureable action steps, and provide accountability.

Informed by the situational assessment report, the strategic planning workshop and the consensus workshop, the TWG develops an implementation plan that includes:

- outlines of roles and responsibilities for all aspects of implementation
- timelines for internal review process and approval
- timelines for capacity building
- timelines for data analysis and feedback reports
- plans for monitoring and evaluation.

Implementation research can be incorporated into the plan to ensure the documentation of best practices. Further information on implementation research can be found in *A guide to implementation research in the prevention and control of noncommunicable diseases*. (7)

After the implementation plan has been drafted, ensure that key stakeholders have the opportunity to review and provide feedback. Stakeholders will be able to assist in identifying potential issues and improve the viability of the plan.

As part of the implementation planning process, develop a corresponding budget. The budget should estimate the cost of implementation activities.

HEARTS provides guidance on the added costs of implementing the HEARTS modules at the district level (*Costing Tool*).

(d) Obtain MoH endorsement

MoH support is critical for the success of the programme. The MoH should therefore endorse the implementation plan through a written agreement, and designate a point of contact. The MoH focal point coordinates with the implementing site(s) and the national level MoH NCD leadership to develop the work plan, track progress through regular reports, and communicate results.

Step 4: Implement and monitor

Once the implementation plan has been endorsed, arrange for training and capacity-building activities to be conducted prior to initiating delivery of new services. Monitoring and supervision should occur throughout the implementation process to troubleshoot any challenges and provide feedback on the effectiveness of the implementation.

(a) Build capacity

Comprehensive training for different health care teams should be provided. Training

can be provided through workshops or incorporated into regular classes or in-service training.

Trainings on local and global CVD burden, treatment protocols, service delivery, patient monitoring, reporting systems, assessment, and supervision should be provided, as appropriate, for:

- primary health care providers such as doctors, nurses, and other clinical staff
- health facility managers
- district supervisors.

Plan for additional training sessions throughout the implementation process. Additional training can both address difficulties that arise during implementation and maintain momentum.

Develop champions at different levels of the health care system (local, district, province levels); this can help mitigate the risk of intervention disruption due to turnover, and can improve continuity.

(b) Deliver services, supervise, and monitor

With staff trained, resources secured, and a plan in place, start the implementation of services in the demonstration site. During all stages of the implementation process, supervision and monitoring will be essential.

Supervision allows performance to be assessed and guidance provided so that staff are best-equipped to provide services effectively. Ensure that ongoing mentoring and capacity-building is provided, especially in the initial stages of implementation and service delivery.

Through monitoring, activities can be tracked to see if implementation is occurring as intended, or if adjustments are needed.

The Pan American Health Organization's Monitoring and Evaluation Framework for Hypertension Control Programs (8) provides a foundation for monitoring and evaluation that allows countries to select indicators based on their own resources and priorities. The framework is designed to be used at different intervention levels and includes five strongly recommended core indicators found in the *Systems for Monitoring* module, a PAHO-WHL recommended indicator, and optional indicators for consideration.

Monitoring activities should be built into the programme, and data should be reviewed regularly to identify unanticipated challenges, assess progress, and improve implementation.

Step 5: Evaluate and scale up

While monitoring ensures the programme is progressing as planned, evaluation uses monitoring data and other information to assess whether the programme is effectively and efficiently achieving its goals. Once the programme is able to successfully produce the intended results, scale-up can be planned.

(a) Evaluate

Programme evaluation is a systematic approach to collecting, analysing, and using data to examine the effectiveness and efficiency of the programme. Programme evaluations reveal why a programme may or may not be working, and can inform programme adjustments and improvements. Ensure that evaluation is ongoing and includes all programme stakeholders.

Programme evaluation frameworks can provide practical, nonprescriptive steps and standards to organize essential evaluation elements. Frameworks can provide guidance on:

- steps in evaluation practice
- standards for effective evaluation
- applying the framework as a communication and implementation roadmap
- integrating evaluation into the programme.

Programme evaluation resources include:

- PAHO Monitoring and Evaluation Framework for Hypertension Control Programs
- WHO Evaluation Practice Handbook (9)
- CDC Framework for Program Evaluation. (10)

Annual reporting on the programme's successes and challenges to stakeholders, including facility managers, relevant district personnel, and MoH contacts, can demonstrate programme impact, maintain momentum, and justify resource allocations.

(b) Scale up

After piloting the programme at the demonstration site and evaluating improvements in CVD risk management, the programme may be scaled up. A phased approach to scale-up is preferable as it allows for learning and adaptation. A cost estimate for country-level scale-up should be made to determine the necessary resources. To mitigate the risk of the project being adversely affected by elections or political and administrative changes, it is important to work with the MoH to:

- integrate the chosen model into the existing health care system to institutionalize the approach
- develop a strategic and operative plan for national expansion of the project
- develop and strengthen technical and managerial capacity to ensure sustainability.

Annex 1: Technical Note

PAHO-HEARTS-1-2018

Prerequisites and preparation for the implementation phase

Background

HEARTS is an initiative spearheaded by the World Health Organization, involving various global actors including the U.S. Centers for Disease Control and Prevention (CDC), International Society of Hypertension, International Society of Nephrology, International Diabetes Federation, Resolve to Save Lives, World Heart Federation, and World Hypertension League. PAHO's Department of Non-communicable Diseases and Mental Health coordinates HEARTS in the Americas, which disseminates best practices for the prevention and management of cardiovascular diseases, aiming to have a positive impact on the attributable burden of these diseases and to move towards the achievement of the Sustainable Development Goals 2030.

HEARTS in the Americas is an initiative of the countries, led by the Ministries of Health, with the participation of local actors and with the technical cooperation of PAHO. The initiative seeks to smoothly and progressively integrate HEARTS into existing health-delivery services to promote the adoption of global best practices in the prevention and control of cardiovascular diseases (CVD) and improve the performance of the services through better control of high blood pressure and the promotion of secondary prevention, with emphasis on primary health care.

The Global Hearts Initiative promotes the adoption of the MPOWER technical package for tobacco control, SHAKE for the reduction of salt intake, REPLACE for transfat elimination, and HEARTS for the clinical management of conditions related to CVD. These packages guide the implementation of the initiative and catalyze the fulfilment of the goals and targets related to non-communicable diseases.

During the first phase, the initiative completed the proof of concept in Barbados, Colombia, Chile, and Cuba (founding countries) and verified in practice that the model of HEARTS works, that it is acceptable to patients, providers, and funders, and that it improves the coverage and control of hypertension. During the second phase, the priority is scale-up in the founding countries and expansion to other countries in the region.

Principles

This methodology applies both to new countries and to the founding countries.

- With the understanding that the country has the political will to improve the prevention and control of non-communicable diseases, stakeholders formalize through written official communication their interest in the HEARTS initiative and their commitment to PAHO to implement the initiative gradually (phased approach), according to its resources and the characteristics of the health system, and to provide a strategic design and operational plan with a realistic timeline.

- PAHO will provide technical cooperation to the country in capacity building and in the implementation of HEARTS.
- HEARTS should build on what already exists, using available resources, continuously improving performance and results, taking advantage of the leadership, technical capabilities, and successful projects/programs which are already in practice to optimize resources and establish the necessary synergies to make the initiative work. Thus, HEARTS is not a vertical, isolated project or parallel to the existing health care system. On the contrary, its sustainability depends on the institutionalization of the model in the everyday practice of implementers.

Preparation for the implementation phase (4 months)

1. Prepare a plan of action based on a situational analysis and establishment of management/coordination teams. The constitution of the national and local coordinating teams led by the Ministry of Health and with the participation of the main actors defined by the country/territory should be as follows:
 - Define the mandate, operation, and the members of the management teams through the recommended administrative procedures for each country.
 - Each member of the management/coordination teams must have clearly defined responsibilities. In order to ensure the highest level of capacity, it is recommended to assign a specialist according to competencies and skillset to each of the HEARTS modules.
 - The lifespan of the management/coordination team must be subject to the institutionalization of the model. To the extent that the model becomes institutionalized, the management/coordination team will be phased out.
 - The management/coordination teams must distinguish clearly between those in charge of implementation and those responsible for the evaluation/research, ensuring that those selected have the competencies and skillset required for each function.
2. New areas of implementation of HEARTS must be jurisdictions or territories: at a minimum, a municipality that has a defined administrative and sanitary structure. The formalization of the areas of implementation of HEARTS requires a letter of commitment from the local authorities responsible for the implementation, in the same way that it proceeded at the national level (local authority concurrence).

In addition, the municipalities selected for implementation shall comply with the following requirements/attributes. Municipalities implementing HEARTS (demonstration sites) should have:

- A population of at least 100,000 people (catchment area population) and a network of primary care level services/centers.
- Basic information on the population served with the potential to build a clinical registry.
- Epidemiological information on the prevalence and control of high blood pressure or at least a reasonable estimate to apply to this population.
- A referral hospital of second/third level where the population presenting acute cardiovascular events are served, especially for cerebrovascular and coronary heart disease. The hospital should have the capacity to record these events.
- Motivated and trained personnel, including a leader of recognized technical and managerial leadership.
- The basic health structure to comply with the objectives and functions of

training and capacity-building activities.

- A network of pharmacies with potential to participate in the project in the roles indicated in HEARTS (Team-based care module) and to have a registry of patients.
3. It is important to have at least one academic institution willing to participate in the project and undertake the tasks of monitoring and evaluation as well as research.
 4. The members of management teams, both national and local, should familiarize themselves with the methodological documents of HEARTS available on WHO's website.
 5. In tandem, the management/coordination teams should arrange an initial assessment/situational analysis based on the framework of monitoring and evaluation recommended by PAHO-WHL. This evaluation should benefit the management team as an exercise to test its operational capacity and to define the overall strategy, the goals and the operational plan of implementation in the short-, medium-, and long-term, as suggested by the evaluation framework. The strategy and the plan of implementation resulting from the evaluation exercise must be approved by the appropriate administrative authority.
 6. The focal point of Non-communicable Diseases and Mental Health of the PAHO Office in the country will accompany the preparation phase and will coordinate the first field site visit of the technical advisors and experts from PAHO to formally assess and complete this preparation phase and to begin the implementation. The delivery of a report that reflects compliance with the steps outlined in this technical note is mandatory.

Annex 2: Rapid health system review template

Guidance for conducting a rapid review of the health system

GOVERNANCE AND LEADERSHIP	Is CVD risk management in primary health care included in the: • national/district health strategy • national NCD strategy • national operational plans • basic package of services? Is management of CVD/hypertension/diabetes included in national clinical guidelines for primary health care? Do national clinical guidelines for primary health care include evidence-based protocols for risk-based CVD management? Are there standardized systems/tools for mentoring and supervising primary health care staff? What is the frequency of district management meetings? Who attends?
HEALTH FINANCING	Is there a specific NCD budget within health financing? If yes, what is it? In those systems with health insurance, are CVD/NCD services and medicines included in benefit packages?
ACCESS TO ESSENTIAL MEDICINES AND TECHNOLOGIES (see <i>Access to essential medicines</i> module)	Are the nine minimum essential medicines (see section 4 of the Baseline Facility Assessment in Annex 3) for CVD risk management included in the National Essential Medicines List and the Minimum Primary Health Care Medicines List? Are the essential CVD and diabetes technologies included in minimum standards for primary health care facilities? Describe the national medicine-supply management system (selection, quantification, procurement, storage, distribution).

HEALTH WORKFORCE (see <i>Team-based care</i> module)	Are there dedicated management staff for CVD/NCD management at the national and district levels? Which staff have authority to prescribe and/or authorize medication refills? Have task-sharing approaches in primary health care been adopted or considered? Do in-service training packages exist for management of CVD, hypertension or diabetes in primary health care? Has any in-service training on NCD/CVD risk management occurred in last two years? If yes, who delivered it?
HEALTH INFORMATION SYSTEMS (see <i>Systems for monitoring</i> module)	Are there mechanisms for data feedback from national, to sub-national, to facility level? Are there dedicated staff to collect data at district level? Describe the district-level database for the routine health-management information system and other facility data. Are CVD/NCD management indicators included in a national minimum indicator set? Describe the type of individual patient record format used in public primary health care facilities.
ORGANIZATION OF SERVICE DELIVERY	Describe the facility levels within the public health system. Describe CVD risk-management (if applicable) services available at each level of care, including a healthy lifestyle counselling component. Are catchment populations defined for primary health care? What is the current service delivery model(s) in public primary health care facilities? For example, general outpatient services, where patients see any available provider; disease-specific clinics. Are there established national and/or district-level quality-improvement systems for primary health care?

Annex 3: Baseline facility assessment template

Facility information

Date:

Name of person completing the questionnaire:

Job title:

Facility name:

Rural ☐Urban ☐

1 Human resources

1.1 Availability of human resources for managing major noncommunicable diseases (NCDs):

	Number present today	Number full-time	Number part-time
Specialist doctor			
Generalist/family doctor/physician			
Nurse			
Community health worker/health educator			
Pharmacist			
Laboratory technician			
Registration/data clerk			
Other (specify)			

1.2 Are doctors, nurses and/or other health workers trained in CVD management?

	Doctors	Nurses	Other health workers
Yes, regularly trained			
Yes, one-off training			
No, not trained in CVD management			

2 Equipment

2.1 Availability of basic equipment for managing CVD

Equipment	Number of functional devices available	Number of devices out of service/awaiting repair
a) Blood pressure measuring devices (BPMD)	Total:	Total:
Breakdown	Mercury BPMDs: Aneroid BPMDs: Automatic BPMDs:	Mercury BPMDs: Aneroid BPMDs: Automatic BPMDs:
b) Measuring tape/height board/stadiometer		
c) Weighing machines		
d) ECG machines		
e) Glucometer		
f) Stethoscope		

2.2 How often are blood pressure measuring devices calibrated and checked for accuracy?

Once a year or more ☐ Less than once a year ☐ Never ☐ Don't know ☐

2.3 How is equipment usually repaired and maintained?

Repaired at the facility ☐ Sent back to manufacturer for repair ☐

Sent back to "government store" for repair ☐

Other (specify)

2.4 What, if any, are the difficulties in repairing equipment?

3 Infrastructure/services

3.1 Can the following investigations be carried out?

(Tick "Yes" only if the investigation can be done on the day of the survey)

Investigation	Yes, at this facility	Yes, at referral, but not at this facility	No
Urine dipstick testing – protein/ glucose/sugar			
Blood sugar			
Urine ketone bodies			
Urine microalbuminuria			
HbA1c			
Serum cholesterol			
Serum creatinine			

3.2 Do you use national guidelines for the diagnosis and management of CVD and diabetes in this facility?

Type of guideline	Yes, seen	Yes, not seen	No
Cardiovascular disease			
Diabetes			

3.3 Are the following services available at this facility?

	Yes	No	Don't know	Who provides (doctor, nurse, counsellor, etc.)
a) Patient counselling and education on smoking, diet, alcohol consumption, physical activity				
b) Counselling and education of family members on smoking, diet, alcohol consumption, physical activity				
c) Cardiovascular risk assessment				
d) Patient counselling for self-management of diabetes				
e) Patient education for self-administration of insulin				

3.4 Are health education materials available at this facility for:

	Yes	No	Don't know
a) Smoking			
b) Diet			
c) Alcohol consumption			
d) Physical activity			
e) Hypertension			
f) Diabetes			

4 Medicines

4.1 Are medicines purchased directly by the facility for distribution to patients?

Yes ☐ No ☐

4.2 If “yes”, is the purchase of medicines subsidized by the government?

Yes ☐ No ☐

4.3 Availability of medicines in the facility (tick only one box for each medicine):

Medication	Always available	Sometimes available	Never available	Stock-out in last 3 months
Aspirin				
ACE inhibitor (enalapril) or ARB				
Beta blocker (atenolol)				
Calcium channel blocker				
Thiazide				
Statin (lovastatin or simvastatin)				
Metformin				
Sulphonylurea (glibenclamide / gliclazide / glipizide)				
Insulin (injection)				
Statins				

5 Service utilization

5.1 Number of visits, patients, consultations

Utilization	Number	Based on register	Based on estimation
Total number of visits to the health facility for outpatient services last month			
Total number of visits to the health facility for outpatient services yesterday			
Average number of consultations per day			
Number of patients registered for hypertension			
Number of patients registered for diabetes			

6 Patient referral

6.1 Can you refer patients to another facility in the event of a CVD emergency?

Yes ☐ No ☐

If you answered “Yes”, go to Question 6.2. If “No”, go to Question 6.8.

6.2 How far from your facility is the nearest referral institution for a medical emergency (in minutes, hours, kilometres)?

Minutes ☐ Hours ☐ Kilometres ☐

6.3 Have you ever wanted to refer a patient with acute, severe symptoms or emergency related to heart disease, diabetes or asthma, but were unable to do so?

Yes ☐ No ☐

If yes, why?

6.4 Does your facility have an ambulance?

Yes ☐ No ☐

6.5 If the facility does not have an ambulance, can patient transfer by ambulance be arranged?

Yes ☐ No ☐ Don't know ☐

6.6 What means of transport is most frequently used to transfer emergency patients at your facility? (check only one)

Ambulance ☐ Public transport ☐ Commercial vehicle (e.g. taxi) ☐

Private vehicle ☐ Other (specify):

6.7 (a) Can you refer patients with CVD for a second opinion/specialist consultation?

Yes ☐ No ☐

If no, why not?

6.7 (b) If yes, the patients will usually be:

Referred back to you for follow-up ☐

Followed up at the upper-level (referral) facility ☐

6.7 (c) Can you refer patients with CVD to the nearest referral medical institution for additional tests?

Yes ☐ No ☐

If no, why not?

6.8 Approximately how long does it take to transfer a patient to the nearest referral medical institution?

Minutes ☐ Hours ☐ Days ☐

7 Record keeping/medical information system

General

7.1 Does the facility have the following?

Description	Yes, functional	Yes, awaiting repair	No
Computer			
Telephone			
Internet connection			

Patient records

7.2 How do patients access the facility?

Walk-in only ☐ By appointment only ☐

Combination of appointments and walk-ins ☐

7.3 Does the facility keep a record of patient visits?

Yes, records kept for all visits ☐

Yes, records kept for certain types of visit (specify):

No records kept ☐

If you answered "Yes", go to Question 7.4. If "No", skip to Question 7.5.

7.4 How are records kept?

a) Patient files ☐ Registry system ☐

Other (specify):

b) Paper records ☐ Electronic health records ☐

7.5 Are patient files retrieved and consulted each time they visit the facility?

Yes, patient files usually/always consulted ☐

Yes, patient files consulted, but only when necessary ☐

No, patient files not consulted ☐

Facility records

7.6 Does the facility have a stock card or log books for:

a) Medicine

Yes, but not used routinely ☐ Yes, used routinely and currently up to date ☐

No ☐

b) Consumables (e.g. syringes, bandages)

Yes, but not used routinely ☐ Yes, used routinely and currently up to date ☐

No ☐

8 Financing and administration

8.1 Do patients pay the facility for medicines?

Yes, full payment ☐

Yes, partial payment ☐ Proportion paid by patient: ____%

No, medicines are provided free ☐

8.2 If medicines are provided free or for partial payment, who subsidizes it (check all that apply)?

Central government ☐	Local government ☐	Private insurance ☐
Social assistance plans ☐	Other (specify):	Don't know ☐

8.3 Do patients pay the facility for consultations?

Yes, full payment ☐

Yes, partial payment ☐ Proportion paid by patient: ____%

No, consultations are provided free ☐

8.4 If consultations are provided free or for partial payment, who subsidizes it (check all that apply)?

Central government ☐	Local government ☐	Private insurance ☐
Social assistance plans ☐	Other (specify):	Don't know ☐

8.5 Do patients pay the facility for diagnostic tests?

Yes, full payment ☐

Yes, partial payment ☐ Proportion paid by patient: ____%

No, diagnostic tests are provided free ☐

8.6 If diagnostic tests are provided free or for partial payment, who subsidizes it (check all that apply)?

Central government ☐	Local government ☐	Private insurance ☐
Social assistance plans ☐	Other (specify):	Don't know ☐

9 Community links

9.1 Are there any community activities to support NCD services provided at primary health care facilities?

Yes ☐ No ☐ Don't know ☐

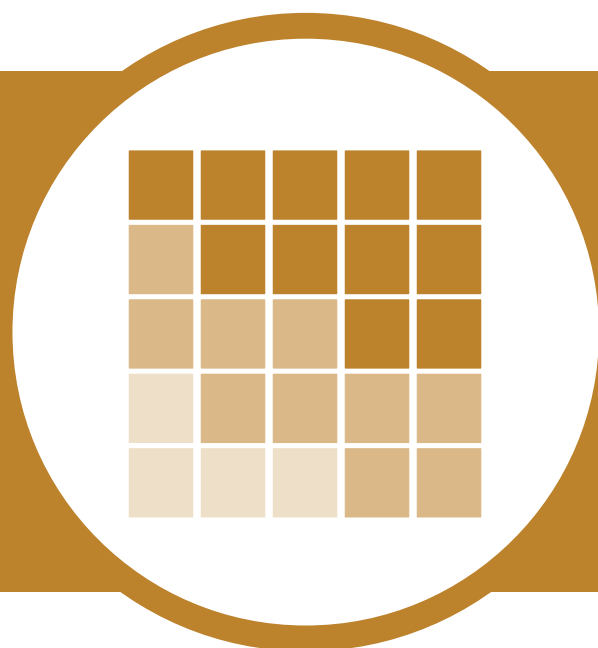
If “yes”, specify: (e.g. vehicle for patient transfer is provided free by the community, patient support groups):

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Technical package for cardiovascular disease management in primary health care



Risk-based CVD management



HEARTS technical package for cardiovascular disease management in primary health care: risk based CVD management

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Abbreviations

BMI	body mass index
CAD	coronary artery disease
CVD	cardiovascular disease
DM	diabetes mellitus
ESC	European Society of Cardiology
GBD	Global Burden of Disease
HTN	hypertension
HDL-C	high-density lipoprotein cholesterol
IHME	Institute for Health Metrics and Evaluation
IHRMS	INTERHEART Modifiable Risk Score
LMIC	low- and middle-income countries
MI	myocardial infarction
NCR-RisC	NCD Risk Factor Collaboration
PEN	Package of Essential Noncommunicable Disease Interventions
SBP	systolic blood pressure
TC	total cholesterol
WHO/ISH	World Health Organization / International Society of Hypertension



HEARTS Technical Package

More people die each year from cardiovascular diseases (CVDs) than from any other cause. Over three quarters of heart diseases and stroke-related deaths occur in low- and middle-income countries (1). The HEARTS technical package provides a strategic approach to improving cardiovascular health. It comprises six modules and an implementation guide. This package supports Ministries of Health to strengthen CVD management in primary care and aligns with World Health Organization's Package of Essential Non-communicable Disease Interventions (WHO PEN).

The WHO HEARTS modules are intended for use by policy-makers and programme managers at different levels within Ministries of Health who can influence CVD primary care delivery. Different sections of each module are aimed at different levels of the health system and different cadres of providers. All modules will require adaptation at country level.

The people who will find the modules most useful are:

- **National level** – Ministry of Health noncommunicable disease (NCD) policy-makers responsible for:
 - Developing strategies, policies and plans related to service delivery for CVD
 - Setting national targets on CVD, and monitoring and reporting progress
- **Subnational level** – Health/NCD programme managers responsible for:
 - Planning, training, implementing and monitoring service delivery
- **Primary care level** – Facility managers and primary health care providers and trainers responsible for:
 - Assigning tasks, organizing training and ensuring the facility is running smoothly
 - Collecting facility-level data on indicators of progress towards CVD targets.

Target users may vary, based on context, existing health systems and national priorities.



Introduction

Development of cardiovascular disease (CVD) is influenced by risk factors such as: tobacco use, an unhealthy diet, physical inactivity, obesity (which can result from a combination of unhealthy diet, physical inactivity, and other factors), elevated blood pressure (hypertension), abnormal blood lipids (dyslipidaemia) and elevated blood glucose (diabetes mellitus). Continuing exposure to these risk factors leads to further progression of atherosclerosis, resulting in clinical manifestations of these diseases, including angina pectoris, myocardial infarction, heart failure and stroke. Total CVD risk depends on the individual's overall risk-factor profile.

In 2007, the WHO published Guidelines for the assessment and management of cardiovascular risk (2) that provide guidance for reducing disability and premature deaths from CVD in people at high risk who have not yet experienced a cardiovascular event. These 2007 guidelines were used as a framework for the development of national guidance on CVD prevention, and provide the WHO and International Society of Hypertension (WHO/ISH) cardiovascular (CVD) risk prediction charts for regional CVD risk prediction. WHO updated the CVD risk charts in 2019 (3). This Risk-based CVD management module (HEARTS-R) will present the updated CVD risk charts for the assessment and management of CVD risk. Relevant information from the 2007 guidelines is reproduced.

Timely, affordable and sustained healthy-lifestyle interventions and, when needed, drug treatment will reduce the risk of heart attack and stroke in people with a high total risk of CVD, and hence will reduce premature morbidity, mortality and disability. Those who have their elevated blood pressure or cholesterol treated can reduce their risk by more than one-quarter to one-third (4) and have their risk reduced by half if both are treated (5). For many years we have had inexpensive medications for hypertension (HTN). More recently, generic statins have been added to the WHO Essential Medicines list and have become available in many more countries. Despite the increased availability of drugs, less than 10% of the global population has its blood pressure controlled (6).

Poor control of risk factors is in part a result of a lack of awareness among individuals of their risk status. The identification of risk level by health care providers is therefore a useful means of detecting those with high CVD risk and identifying who could benefit from treatment of high blood pressure, abnormal blood lipids and elevated blood glucose (7). Recommendations for assessment and management of CVD risk factors have evolved over time in CVD guidelines.

This evolution in CVD recommendations and guidelines is a response to:

- an improved ability to quantify and identify those at highest risk
- an increase in the population affected by these risk factors
- changes in generic therapies available
- a recognition that despite these changes we are failing globally to significantly prevent CVD mortality, particularly in low- and middle-income countries (LMICs).

One change in guidelines has been a greater use of overall CVD risk levels to determine intensity of treatments. The additional focus on overall risk has occurred for three main reasons. First, there was recognition that an individual who has multiple risk factors, even if each factor is only moderately elevated, may, when all risk factors are taken into account, be at an overall higher risk than someone with an elevated level for a single risk factor. For example, a middle-aged female with moderate levels of several risk factors could have a 5- to 10-fold increased risk of having CVD in the next 10 years, when compared to a young male with an isolated elevated risk factor.

Second, targeting of those individuals at highest risk was recognized to lead to greater efficiency in benefit in terms of number of events avoided, because the relative risk reduction would be applied to a higher baseline risk (8). Therefore, targeting patients with a high risk is the highest priority in a risk stratification approach (9).

Third, improved computing and statistical methods have allowed researchers to pool large global data sets to create more accurate risk prediction tools for various populations.

As the cost of medicines is a significant component of total preventive health care costs, it is particularly important to base drug treatment decisions principally on an individual's risk level rather than on criteria such as ability to pay, or on blanket preventive strategies. In addition, guidelines based on total CVD risk, which use risk scoring methods, have been shown to be both less expensive and more effective than guidelines based on single risk-factor levels to treat the same number of patients (10).

Drug therapy (including glycaemic control for diabetes mellitus, control of HTN and cholesterol, using a total risk approach) and counselling to individuals who have had a heart attack or stroke and to persons with high risk ($\geq 20\%$) of a fatal and non-fatal cardiovascular event in the next 10 years is one of the "best buys" for tackling NCDs (11). For those with a pre-existing condition such as prior CVD, risk identification is simple. For others, the use of clinical algorithms or "risk charts" can be used to further stratify patients.

A risk stratification approach is particularly suitable to settings with limited resources, where saving the greatest number of lives at lowest cost becomes imperative. However, in most guidelines, the use of risk tools has not replaced the knowledge needed to treat individual risk factors but has rather been used to augment the decision of whom to treat, so that those at the highest level of risk are easily identified for treatment. This module helps to identify those who would benefit from lifestyle changes and basic medical treatment to lower blood pressure, cholesterol, and manage diabetes mellitus in an integrated manner. Guidance on HTN and diabetes management can be found in other HEARTS Modules, as well.

Application of an approach that considers the total CVD risk will be better informed through country-level implementation research. Many areas, including the level of the facility at which CVD risk can be assessed, thresholds for treatment, adaptation of protocols, follow-up intervals and other logistics are best defined in the local context.

1 Updated WHO cardiovascular risk charts

2007 WHO/ISH cardiovascular risk prediction charts

Many risk-prediction models have been developed over the years. However, most equations are derived, recalibrated and validated in limited settings. The majority are derived from or recalibrated on populations of European descent living in high-income countries. As a result, these models might not be directly translatable or valid in low-resource and non-European settings. The WHO/ISH cardiovascular risk prediction charts were developed in 2007 to estimate 10-year risk of a fatal or non-fatal major cardiovascular event (myocardial infarction or stroke), according to age, sex, smoking status, blood pressure, total blood cholesterol and presence or absence of diabetes mellitus for 14 WHO epidemiological sub-regions (12). There are two sets of charts. One set could be used in settings where blood cholesterol can be measured, and the other in settings where blood cholesterol cannot be measured. Both sets require the status of diabetes in the individual to be known (13).

2019 Updated WHO cardiovascular disease risk charts

The WHO updated the 2007 WHO/ISH cardiovascular risk prediction charts through the formation of a cross-sectorial collaboration of academics, policy-makers and end users of CVD risk scores. The model revision took place in three steps.

First, risk prediction algorithms were developed using individual-participant data from 85 prospective cohort studies with long-term follow-up in the Emerging Risk Factors Collaboration. Endpoint definitions of fatal and non-fatal CVD outcomes used for the model are described in Annex 4.

Second, to adjust the algorithms to the contemporary circumstances of different global regions, they were recalibrated using age-specific and gender-specific incidence rates and risk-factor values obtained from the Global Burden of Disease (GBD) Study (14) and the NCD Risk Factor Collaboration (NCD-RisC) (15).

Third, performance of the algorithms was assessed by external validation, using individual-participant data from a further 19 prospective cohort studies (i.e. studies distinct from those used in the algorithm derivation) (3).

The updated WHO CVD risk prediction charts were developed and presented for 21 global regions, defined by GBD to maximize between-region variability and minimize heterogeneity in mortality and major drivers of health outcomes within each region (3) (Annex 1). The charts are intended to allow the introduction of a total risk-stratification approach for management of CVD. They are presented as laboratory-based and non-laboratory-based charts. Laboratory-based algorithms include information on age, sex, smoking status, systolic blood pressure, history or evidence of diabetes mellitus, and the total cholesterol value. In the non-laboratory-based algorithms, body mass index (BMI) is included; information on diabetes mellitus and cholesterol is not necessary for these charts.

Countries can find the charts relevant to them according to the appropriate GBD regional groups (Annex 2 and Annex 3).

WHO CVD risk (laboratory-based) charts

These are CVD risk charts that include measurements of total cholesterol and information on diabetes mellitus (Annex 2). The laboratory-based CVD risk charts should be used for treatment decisions. This is the indicated risk chart in a setting where laboratory facilities, and human and financial resources are accessible. These charts will facilitate health providers to initiate an intervention and treatment regimen, and to implement an appropriate follow-up plan based on the patient's total risk status. As an example, Figure 1 presents the WHO CVD risk charts for North Africa and Middle East Region for use when information on total cholesterol and diabetes is available.

WHO CVD risk (non-laboratory-based) charts

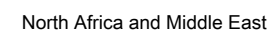
Many low-resource settings have limited testing facilities or limited financial and physical capacity for biochemical measurements (e.g. blood sugar determination and cholesterol assays). WHO CVD risk (non-laboratory-based) charts can be used to predict total CVD risk without information on total cholesterol and diabetes (Annex 3). Only age, sex, smoking status, systolic blood pressure and body mass index (BMI) will be needed to predict cardiovascular risk. These non-laboratory-based WHO CVD risk charts are aimed at stratification in low-resource communities and office settings and can be used for decisions regarding referral. In population samples, there was moderate agreement between WHO CVD risk predictions using laboratory and non-laboratory algorithms. Of those at >20% risk using the laboratory-based algorithm, >97% of men and women were also identified at >10% by using the non-laboratory-based algorithm. However, when using a 20% threshold with non-laboratory-based algorithm, only approximately 65% of men and 35% of women were identified as at the same level of risk when compared to the laboratory-based algorithm. This discrepancy is largely due to the fact that the non-laboratory-based algorithm does not allow for the extra CVD risk associated with diabetes mellitus and substantially underestimates CVD risk in individuals with diabetes mellitus. For example, among individuals with diabetes classified as being at greater than 20% risk with the laboratory-based models, about 45% of men and 25% of women were classified as being at greater than 20% risk with the non-laboratory-based models (whereas in individuals without diabetes, about 85% of men and 95% of women showed such agreement) (3). As an example, Figure 2 is the WHO CVD risk (non-laboratory-based) charts for North Africa and Middle East.

Figure 1: Example of WHO CVD risk (laboratory-based) chart



North Africa and Middle East

Risk Level  <5%  5% to <10%  10% to <20%  20% to <30%  ≥30%



Comparison between 2007 WHO/ISH risk prediction charts and 2019 WHO CVD risk charts

The updated CVD risk charts are presented in an analogous manner to the previous WHO/ISH risk charts to facilitate their continuity of use with some adaptation. However, there are several advantages of the updated WHO CVD risk charts over the previous WHO/ISH risk prediction charts and other global, regional and national CVD risk prediction models available (Table 1).

First, compared with the 2007 charts, the new risk charts use data with more scientific rigour, from diverse populations, which is much more globally relevant.

Second, the charts use more contemporary estimates of CVD incidence and risk factors. This, together with simple and generalizable methods, allows for recalibration of the models for many different populations. This means that the CVD risk estimate for an individual in a given country is likely to be much more accurate than in the older charts.

Third, the recalibration approach used allows for rapid revision of CVD models, enabling a flexible updating of models as new relevant epidemiological data for geographical areas arise.

Fourth, the new charts produce estimates for 21 Institute for Health Metrics and Evaluation (IHME) Global Burden of Disease (GBD) regions compared to the 14 WHO regions in the old charts.

The risk stratification for the updated WHO charts differs from those used in the 2007 WHO/ISH CVD risk prediction charts to recognize changes in reported GBD CVD incidence. Stratification is now as follows: <5% (green), 5% to <10% (yellow), 10% to <20% (orange), 20% to <30% (red), and ≥30% (dark red). The category >40% is not retained in the updated charts as fewer people in the new risk charts were in that group. This risk stratification is aligned to the WHO recommendations for management of CVD risk.

Annexes 5a, 5b and 6 present the comparison of different risk charts and an approach to identifying suitable risk charts.

Table 1: Comparison between 2007 WHO/ISH risk prediction charts with 2019 WHO CVD risk charts

Parameter	2007 WHO/ISH risk prediction charts			2019 WHO CVD risk charts		
Presentation	For 14 WHO epidemiological subregions			21 IHME GBD regions with more homogenous grouping of countries		
Types of charts	Two sets: One set can be used in settings where blood cholesterol can be measured. The other set is for settings in which blood cholesterol cannot be measured.			Two sets: Laboratory-based charts Non-laboratory-based charts		
Variables	1. With individual cholesterol value: • age • sex • smoking • systolic blood pressure • presence or absence of diabetes • total cholesterol 2. Without individual cholesterol value: • age • sex • smoking • systolic blood pressure • presence or absence of diabetes • national average cholesterol value			1. Laboratory-based: • age • sex • smoking • systolic blood pressure • presence or absence of diabetes • total cholesterol 2. Non-laboratory-based: • age • sex • smoking • systolic blood pressure • BMI		
Risk levels and colour code		2007			2019	
		Green	<10%		Green	<5%
		Yellow	10% to <20%		Yellow	5% to <10%
		Orange	20% to <30%		Orange	10% to <20%
		Red	30% to <40%		Red	20% to <30%
		Deep red	>40%		Deep red	≥30%
Difference in interpretation of risk levels	Green was <10%			Green is <5% and corresponding changes in other risk levels.		

2 Assessment and management of total CVD risk

This section is based on Protocol 1 in the WHO Package of Essential NCD interventions (16). This protocol remains current except for the change in the risk level in the updated WHO cardiovascular (CVD) risk charts. Assessment of total CVD risk can be used for routine management of hypertension (HTN) and diabetes mellitus (DM), and for targeting the following categories of people:

- age >40 years
- smokers
- obesity
- known to have HTN
- known to have DM
- history of premature CVD in first-degree relative
- history of DM or kidney disease in first-degree relative.

Instructions for using the WHO CVD risk (laboratory-based) charts

Table 2 and Figure 3 present a step-by-step guide to applying the WHO CVD (laboratory-based) risk charts. These charts are to be used only for individuals whose status regarding diabetes and total cholesterol is available. Tests for diabetes and cholesterol can be carried out at the time of assessment. If the information on diabetes and total cholesterol is not available, then refer to the instructions on use of non-laboratory-based risk charts.

Table 2: Using the WHO CVD risk (laboratory-based) charts

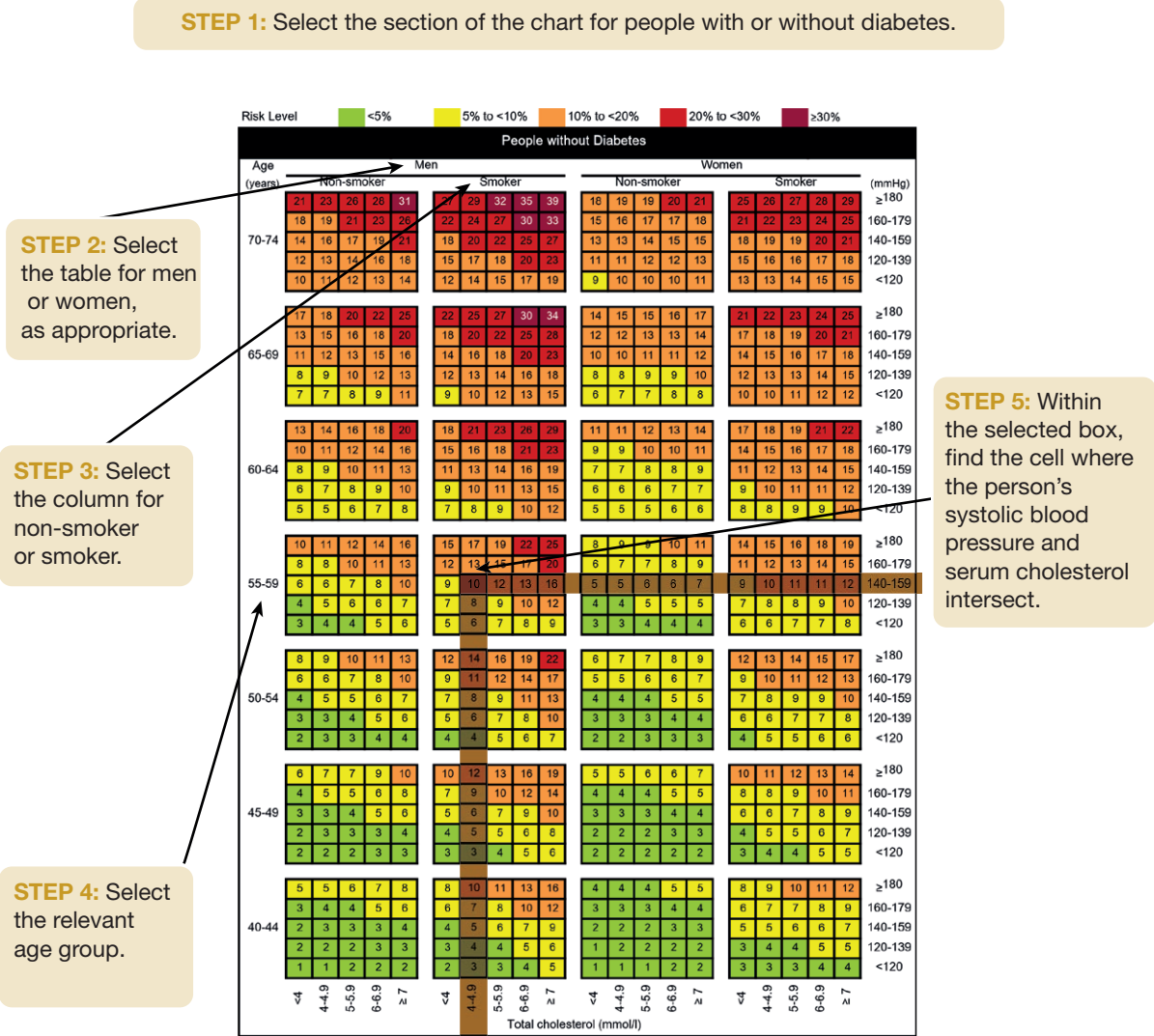
Action		
Select the regional chart covering your country: • REGION NAME is printed at the top of the charts. • Countries included in each region can be found in Annex 1.		
Have the following information ready: • age • sex • smoker* or non-smoker • presence or absence of diabetes† • systolic blood pressure • total blood cholesterol‡		
Using the charts		
STEP 1: Select the section of the chart as relevant for people with or without diabetes.		
STEP 2: Select the table for men or women, as appropriate.		
STEP 3: Select smoker or non-smoker column.		
STEP 4: Select age-group.		
STEP 5: Within the selected box find the cell where the person's systolic blood pressure and total blood cholesterol intersect.		
STEP 6: The colour of the cell indicates the 10-year risk of a fatal or non-fatal CVD event. The value within the cell is the risk percentage. Colour coding is based on the grouping.		Green <5%
		Yellow 5% to <10%
		Orange 10% to <20%
		Red 20% to <30%
		Deep red ≥30%
STEP 7: Record CVD risk percentage in person's chart.		
STEP 8: Counsel, treat and refer according to risk level		

* Current smoker

† Fasting plasma glucose ≥7.0 mmol/L (126 mg/dL), or 2-h plasma glucose ≥11.1 mmol/L (200 mg/dL), or HbA1c ≥6.5% or known diabetes

‡ Cholesterol values are to be entered in the chart as mmol/L (To convert mg/dL to mmol/L, multiply by 0.02586. e.g 200 mg/dL x 0.02586 = 5.172 mmol/L.)

Figure 3: Illustration of how to use the WHO CVD risk (laboratory-based) chart



Integrated management of total CVD risk

Management of total CVD risk in people with different risk thresholds is presented in Table 3. This is based on the WHO PEN Protocol (16) except for the change in risk levels, which are adjusted to the updated CVD risk charts. The purpose of management is to motivate and assist individuals with high risk levels to lower their total CVD risk. All individuals with blood pressure at or above 160/100 mmHg, or blood pressure raised to a lesser degree but with target organ damage should have drug treatment and specific lifestyle advice to lower their blood pressure and risk of CVD.

All patients should receive counselling on diet (which includes lipid-lowering diet and low sodium), physical activity, tobacco cessation (smoking and smokeless) and avoiding harmful use of alcohol (1). For further details on counselling please refer to the HEARTS Module “H” on Healthy-lifestyle counselling (17).

Table 3: Management guidance for total CVD risk

Management of total CVD risk (adapted from WHO PEN Protocol 1)	
Risk <10%	Counsel on diet, physical activity, smoking cessation and avoiding harmful use of alcohol. If risk <5%, follow up in 12 months. If risk 5% to <10%, follow up every 3 months until targets are met, then 6–9 months thereafter.
Risk 10% to <20%	Counsel on diet, physical activity, smoking cessation and avoiding harmful use of alcohol. Persistent BP ≥140/90 mmHg consider drugs (see below). Follow up every 3–6 months.
Risk >20%	Counsel on diet, physical activity, smoking cessation and avoiding harmful use of alcohol. Persistent BP ≥130/80, consider drugs (see below). Give a statin. Follow up every 3 months. If there is no reduction in cardiovascular risk after six months of follow-up refer to next level.
Important practical points	Management of hypertension and diabetes: - For management of hypertension refer to HEARTS technical package E module (18): https://www.who.int/cardiovascular_diseases/hearts/en/ - For management of diabetes mellitus type 2, refer to the HEARTS module Diagnosis and management of diabetes (19): https://www.who.int/publications-detail/who-ucn-ncd-20.1
	Consider drug treatment for following categories: - All patients with established DM and CVD (coronary heart disease, myocardial infarction, transient ischaemic attacks, cerebrovascular disease or peripheral vascular disease), renal disease. If stable, should continue the treatment already prescribed and be considered as having risk >20%. - People with albuminuria, retinopathy, left ventricular hypertrophy. - All individuals with persistent raised BP ≥160/100 mmHg. - All individuals with total cholesterol at or above 8 mmol/L (320 mg/dL).
Follow-up visits: - Ask about: new symptoms, adherence to advice on tobacco and alcohol use, physical activity, healthy diet, medications etc. - Assess (physical exam). - Estimate cardiovascular risk. - Refer if necessary. - Counsel all and treat as shown in protocol.	

Countries may set different thresholds for initiating treatment, based on the distribution of CVD risk in the population (Annex 7). If, after a reasonable amount of time, the non-drug treatment alone is not able to meet the hypertension (HTN) control goals, consider drug treatment.

Follow-up

Follow-up frequency will depend on the capacity of the health system and will have to be decided based on the local context. National protocols are important for the management as they will help to standardize the treatment and bring about efficiencies. A schedule of follow-up based on WHO PEN Protocol 1 is provided. If risk is <5%, follow up in 12 months. If risk is 5%–10%, follow up every 3 months, then 6–9 months thereafter (16). For patients who are not at high enough risk for medical therapy, CVD risk should be reassessed every 12 months, or earlier if clinical symptoms develop.

Instructions for using WHO CVD risk (non-laboratory-based) charts

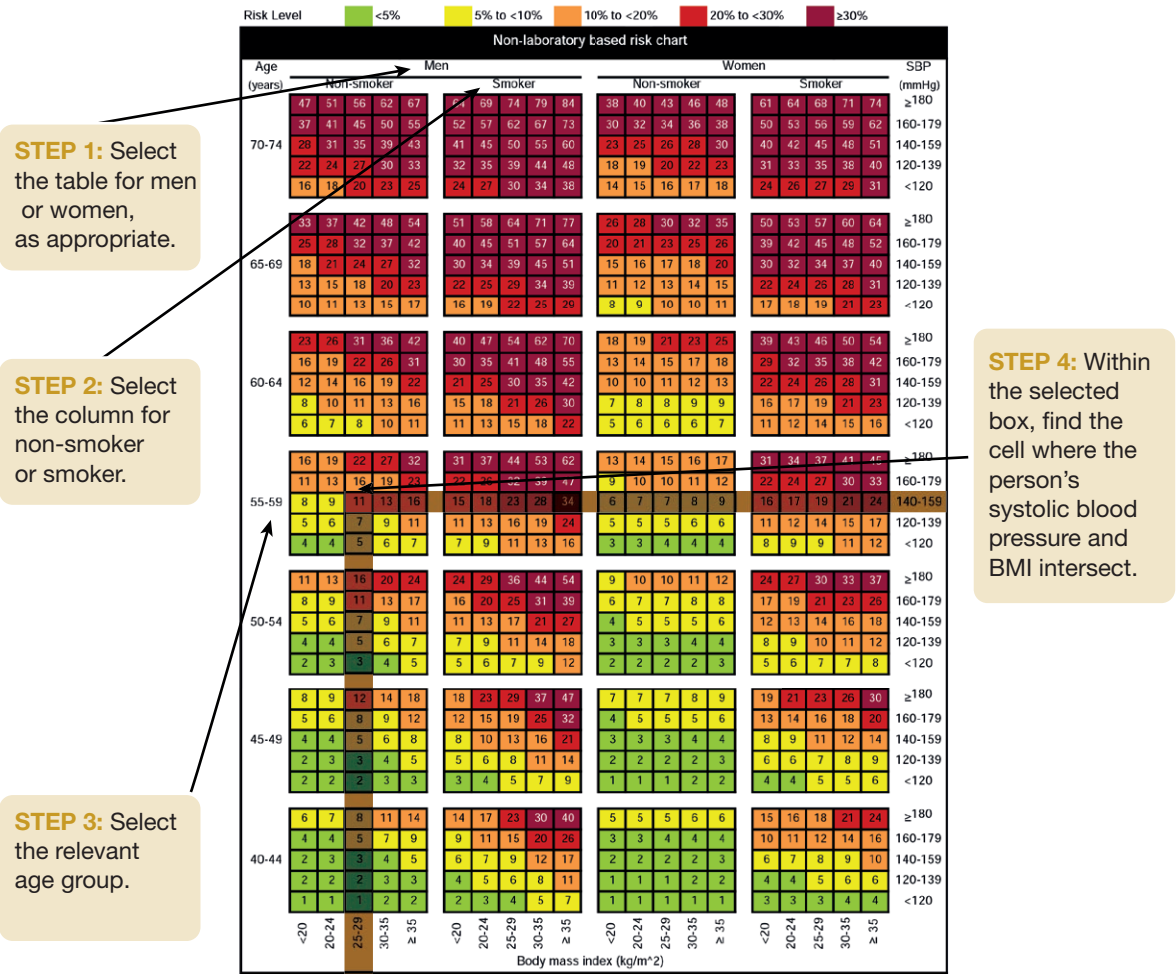
These charts are for the use of settings where diabetes and cholesterol cannot be measured. They can also be used to identify people at high risk who can be taken up for further investigations. Table 4 and Figure 4 present the steps to apply the non-laboratory WHO CVD risk charts.

Table 4: Using the WHO CVD risk (non-laboratory-based) charts

Action		
Select the regional chart covering your country: - REGION NAME is printed at the top of the charts. - Countries included in each region can be found in Annex 1.		
Have the following information ready: - age - sex - smoker* or non-smoker - systolic blood pressure - BMI (body mass index) = weight (kg) ÷ height (m)²		
Using the charts		
STEP 1: Select the table for men or women, as appropriate.		
STEP 2: Select smoker or non-smoker column.		
STEP 3: Select age-group.		
STEP 4: Within the selected box find the cell where the person's systolic blood pressure and body mass index (BMI) intersect.		
STEP 5: The colour of the cell indicates the 10-year risk of a fatal or non-fatal CVD event. The value within the cell is the risk percentage. Colour coding is based on the grouping.		Green <5%
		Yellow 5% to <10%
		Orange 10% to <20%
		Red 20% to <30%
		Deep red ≥30%
STEP 6: Record CVD risk percentage in person's chart.		
STEP 7: Counsel, treat and refer according to risk level		

* Current smoker

Figure 4: Illustration of how to use the WHO CVD risk (non-laboratory-based) chart



Use of WHO CVD risk (non-laboratory-based) charts

Currently, the WHO CVD risk (non-laboratory-based) charts can be considered for identifying a subset of the population who might benefit from laboratory-based risk assessment. Where laboratory testing may be available but extremely limited due to costs or distance, use of the non-laboratory charts could allow for a two-stage process that reduces the number of people at lower levels of risk who are subjected to unwarranted testing. Additionally, the non-laboratory risk charts can be used for education and advocacy regarding total CVD risk in areas where lab testing remains currently unavailable. In these areas, health care providers and policy-makers can use the non-laboratory risk charts to assess the general risk of the population, and to advocate for more resources if risk appears high. Lastly, some studies have shown that use of the non-laboratory-based tool for determining future risk performs as well as laboratory-based tools and may even be cost-effective for decisions regarding medical therapy (4, 5). As more data become available to confirm these findings, the WHO CVD risk (non-laboratory-based) charts may be recommended for use in treatment decisions as well. Individuals with a total CVD risk level of 10% and above should receive an assessment using laboratory-based charts after measurement for diabetes and cholesterol. Advice on lifestyle modification should be given as needed.

Annex 1: GBD regions

The first step when using the WHO CVD risk charts is to identify which Global Burden of Disease (GBD) region a country is in (Table 5).

Table 5: GBD regional groups

GBD region	Countries	Page number of charts	
		Lab-based	Non-lab-based
High-income North America	Canada, Greenland, United States of America	25	46
Caribbean	Antigua and Barbuda, Bahamas, Barbados, Belize, Bermuda, Cuba, Dominica, Dominican Republic, Grenada, Guyana, Haiti, Jamaica, Puerto Rico, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago	26	47
Central Latin America	Colombia, Costa Rica, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Venezuela (Bolivarian Republic of)	27	48
Andean Latin America	Bolivia, Ecuador, Peru	28	49
Tropical Latin America	Brazil, Paraguay	29	50
Southern Latin America	Argentina, Chile, Uruguay	30	51
Western Europe	Andorra, Austria, Belgium, Cyprus, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Israel, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom	31	52
Central Europe	Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Hungary, Montenegro, North Macedonia, Poland, Romania, Serbia, Slovakia, Slovenia	32	53
Eastern Europe	Belarus, Estonia, Latvia, Lithuania, Republic of Moldova, Russian Federation, Ukraine	33	54
North Africa and Middle East	Afghanistan, Algeria, Bahrain, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, occupied Palestinian territory, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, Tunisia, Turkey, United Arab Emirates, Yemen	34	55

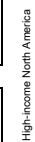
GBD region	Countries	Page number of charts	
		Lab-based	Non-lab-based
Western Sub-Saharan Africa	Benin, Burkina Faso, Cabo Verde, Cameroon, Chad, Cote d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, Togo	35	56
Central Sub-Saharan Africa	Angola, Central African Republic, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon	36	57
Eastern Sub-Saharan Africa	Burundi, Comoros, Djibouti, Eritrea, Ethiopia, Kenya, Madagascar, Malawi, Mozambique, Rwanda, Somalia, Uganda, United Republic of Tanzania, Zambia	37	58
Southern Sub-Saharan Africa	Botswana, Eswatini, Lesotho, Namibia, South Africa, Zimbabwe	38	59
Central Asia	Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, Mongolia, Tajikistan, Turkmenistan, Uzbekistan	39	60
East Asia	China, Democratic People's Republic of Korea	40	61
South Asia	Bangladesh, Bhutan, India, Nepal, Pakistan	41	62
South-East Asia	Cambodia, Indonesia, Lao People's Democratic Republic, Malaysia, Maldives, Mauritius, Myanmar, Philippines, Seychelles, Sri Lanka, Thailand, Timor-Leste, Viet Nam	42	63
High-income Asia Pacific	Brunei Darussalam, Japan, Republic of Korea, Singapore	43	64
Australasia	Australia, New Zealand	44	65
Oceania	Fiji, Kiribati, Marshall Islands, Micronesia (Federated States of), Papua New Guinea, Samoa, Solomon Islands, Tonga, Vanuatu	45	66

WHO CVD risk charts for a specific region can be downloaded from:
<https://www.who.int/news-room/detail/02-09-2019-who-updates-cardiovascular-risk-charts>

24

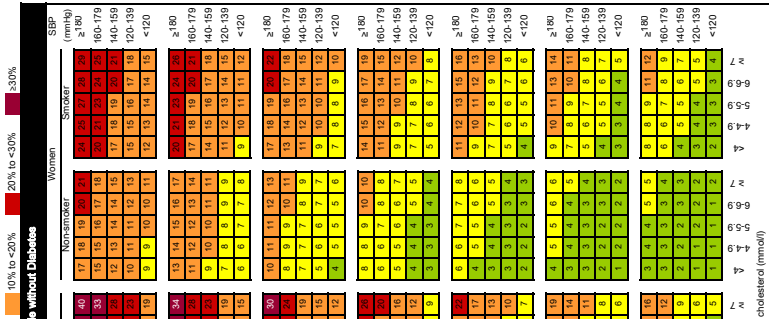
High-income North America
Canada, Greenland, United States of America

High-income North America
Canada, Greenland, United States of America

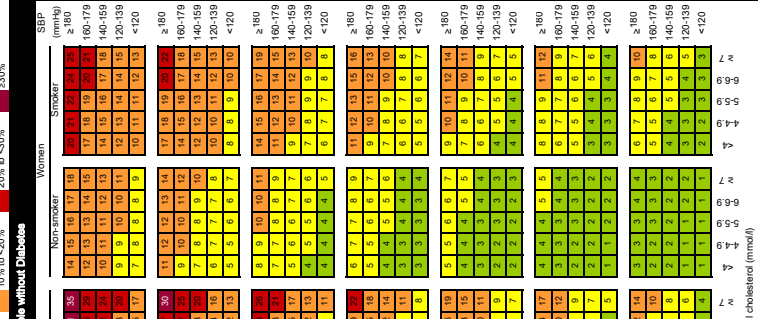


40 cardiovascular disease risk laboratory-based charts

Caribbean
Aruba, Belize, Bermuda, Cuba, Dominica, Dominican Republic, Grenada, Guyana, Haiti, Jamaica, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago



Caribbean

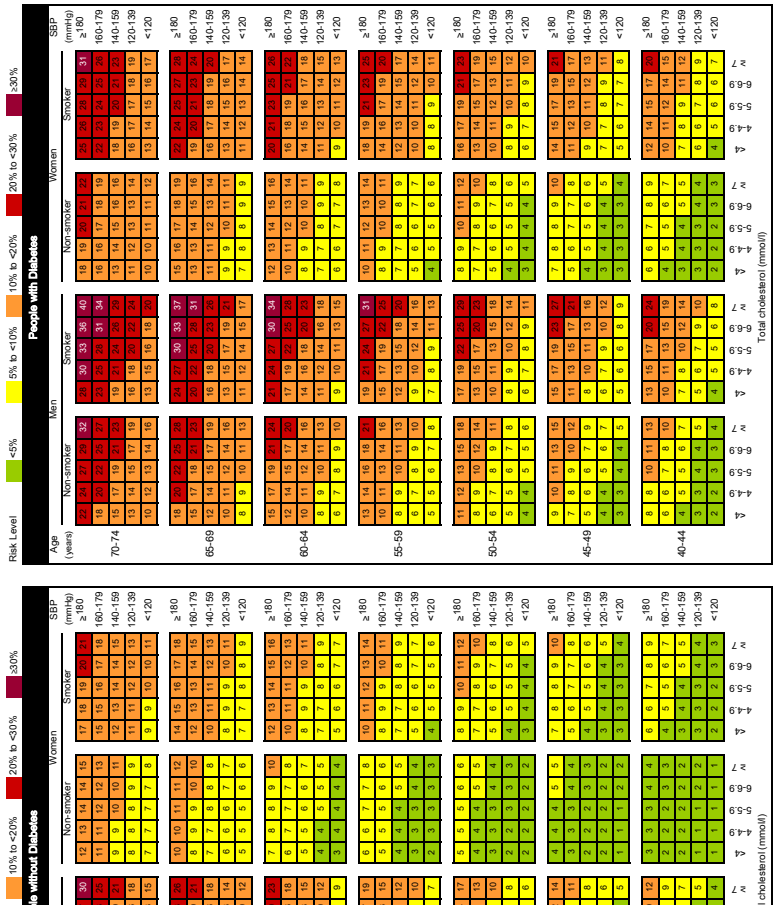


Central Latin America

10 cardiovascular disease risk laboratory-based charts

Andean Latin America

Bolivia, Ecuador, Peru

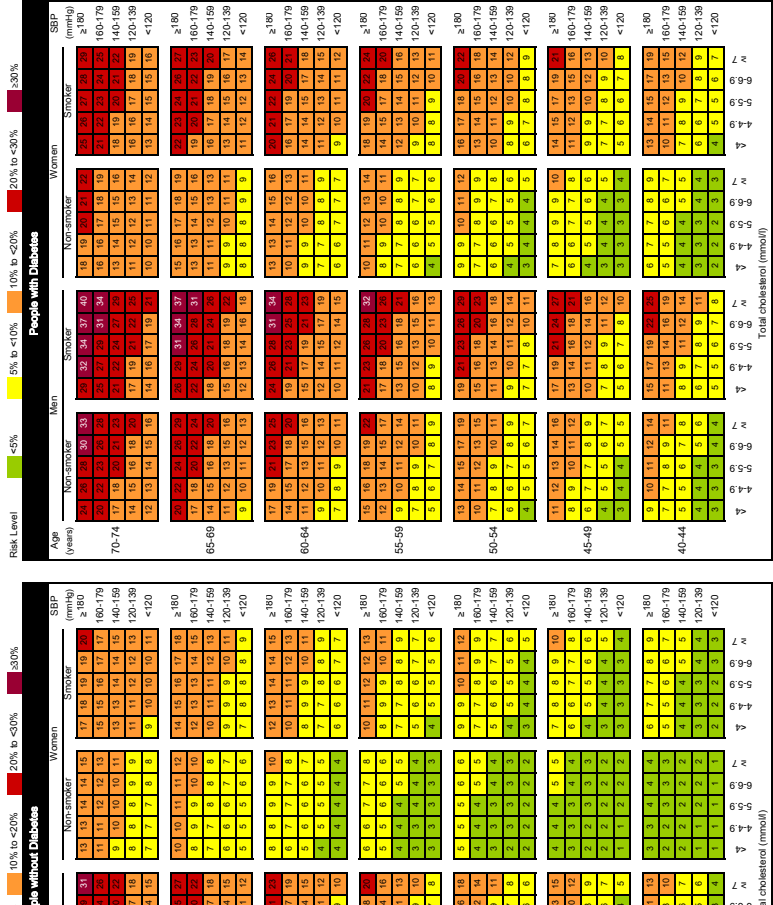


Andean Latin America

HO cardiovascular disease risk laboratory-based charts

Tropical Latin America

Brazil, Paraguay

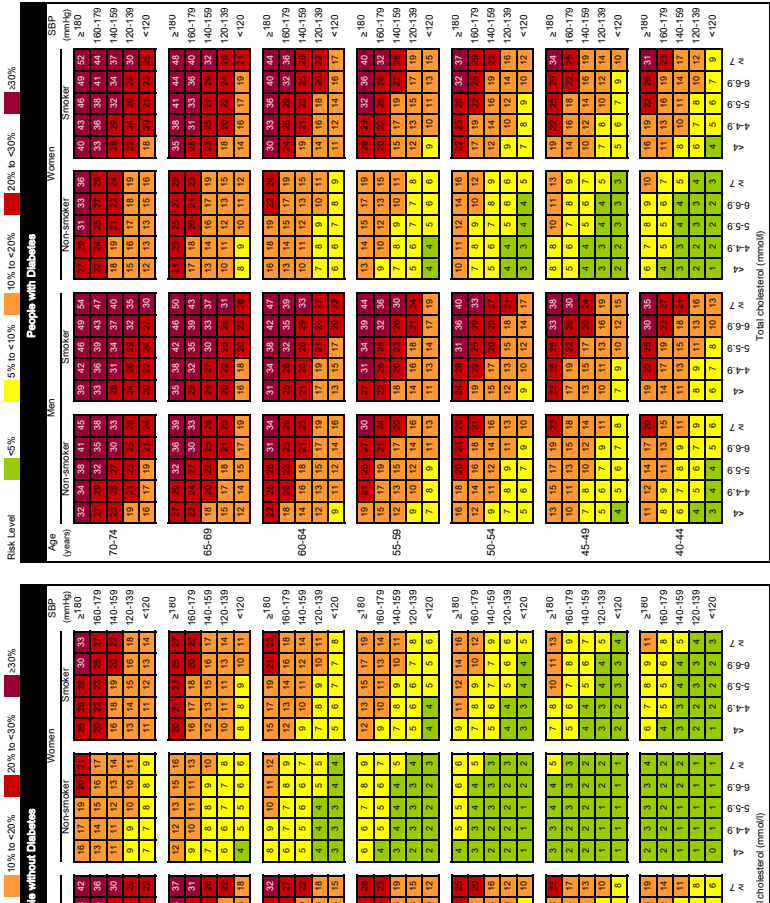


Tropical Latin America

10 cardiovascular disease risk laboratory-based charts

Southern Latin America

Argentina, Chile, Uruguay

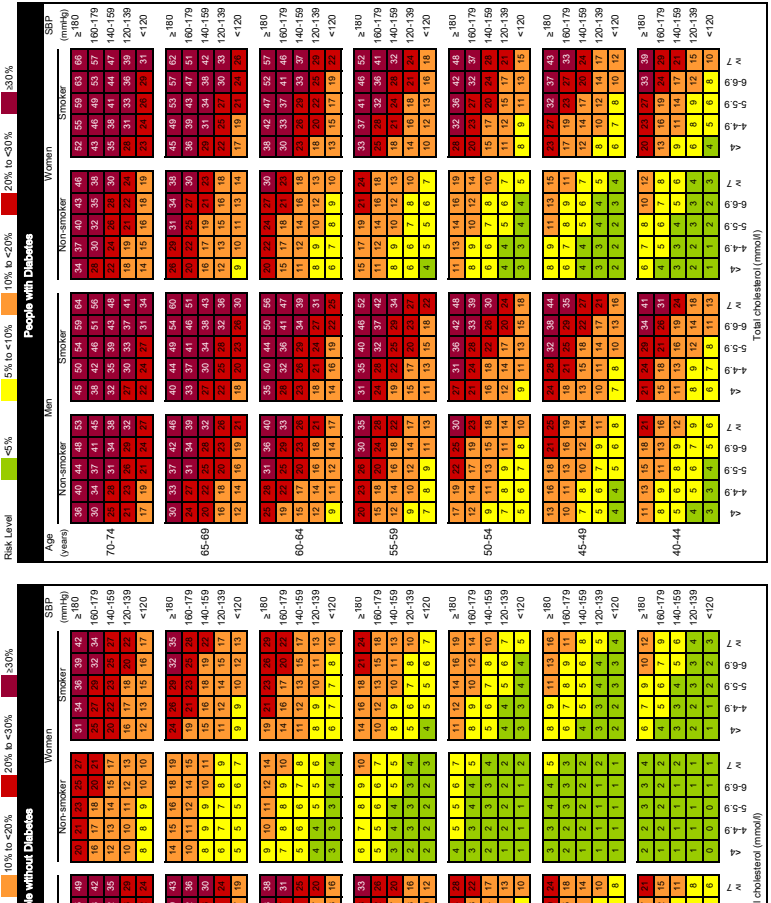


Southern Latin America

10 cardiovascular disease risk laboratory-based charts

Western Europe

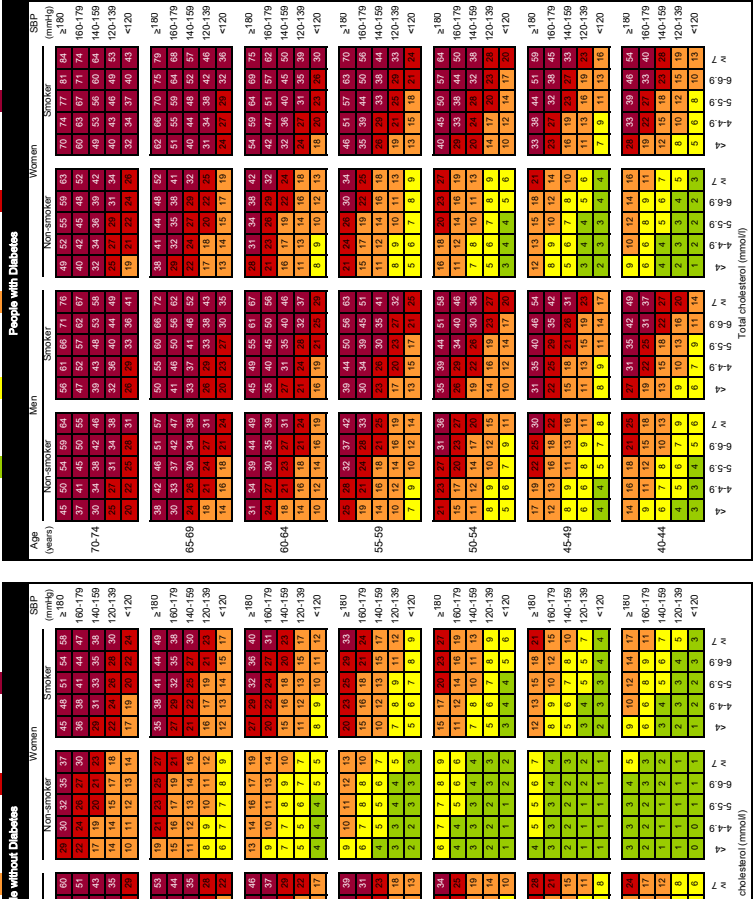
France, Germany, Greece, Iceland, Ireland, Israel, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom



Western Europe

10 cardiovascular disease risk laboratory-based charts

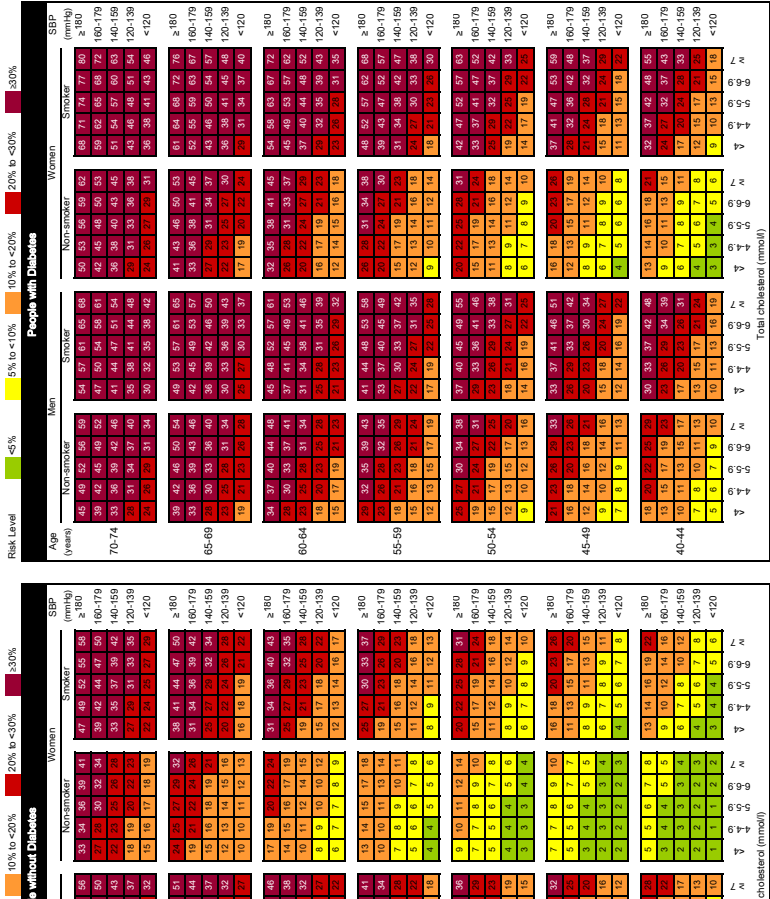
Central Europe
vina, Bulgaria, Croatia, Czech Republic, Hungary, Montenegro, North Macedonia, Poland, Romania, Serbia, Slovakia, Slovenia



Central Europe

10 cardiovascular disease risk laboratory-based charts

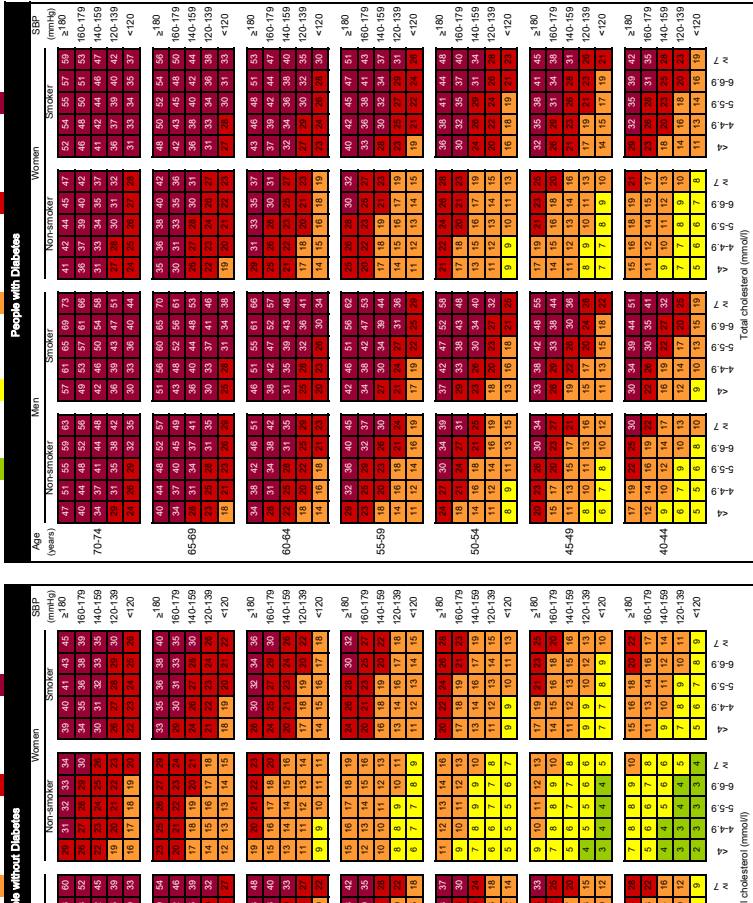
Eastern Europe
Belarus, Estonia, Latvia, Lithuania, Republic of Moldova, Russian Federation, Ukraine



Eastern Europe

10 cardiovascular disease risk laboratory-based charts

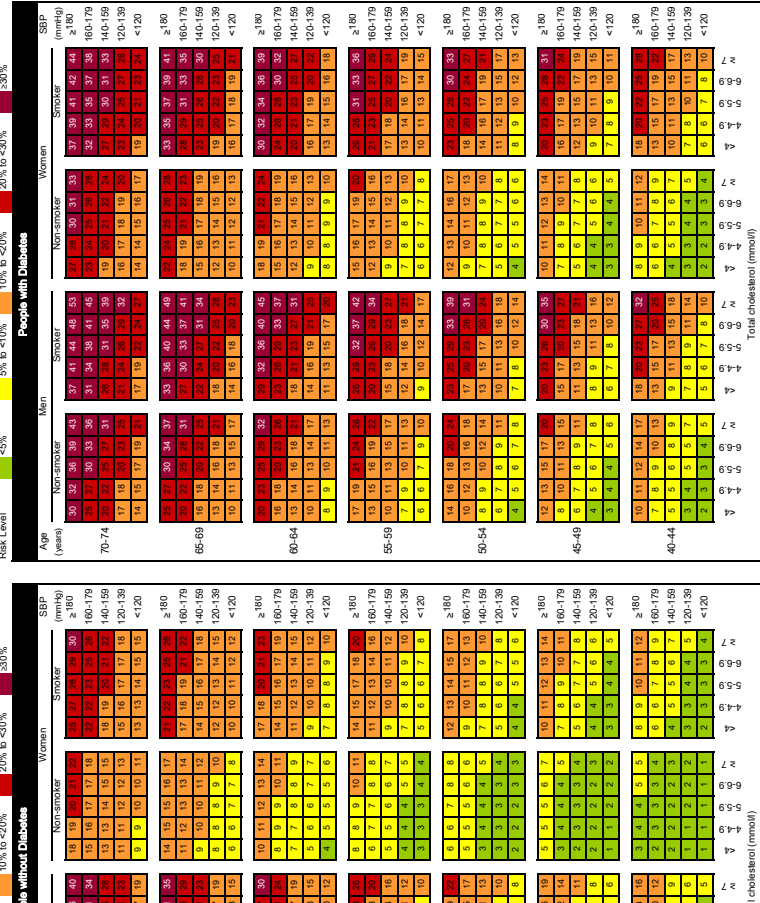
North Africa and Middle East
Republic of, Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, occupied Palestinian territory, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, Tunisia, Turkey, United Arab Emirates, Yemen



North Africa and Middle East

10 cardiovascular disease risk laboratory-based charts

Western Sub-Saharan Africa
Cote d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, Togo

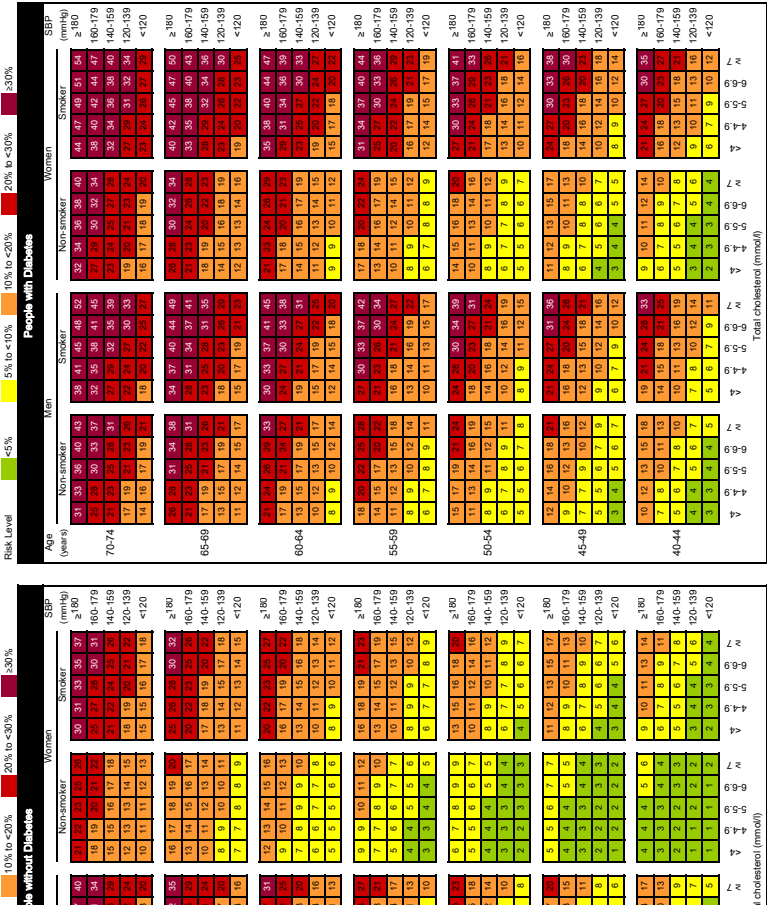


Western Sub-Saharan Africa

10 cardiovascular disease risk laboratory-based charts

Central Sub-Saharan Africa

gola, Central African Republic, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon

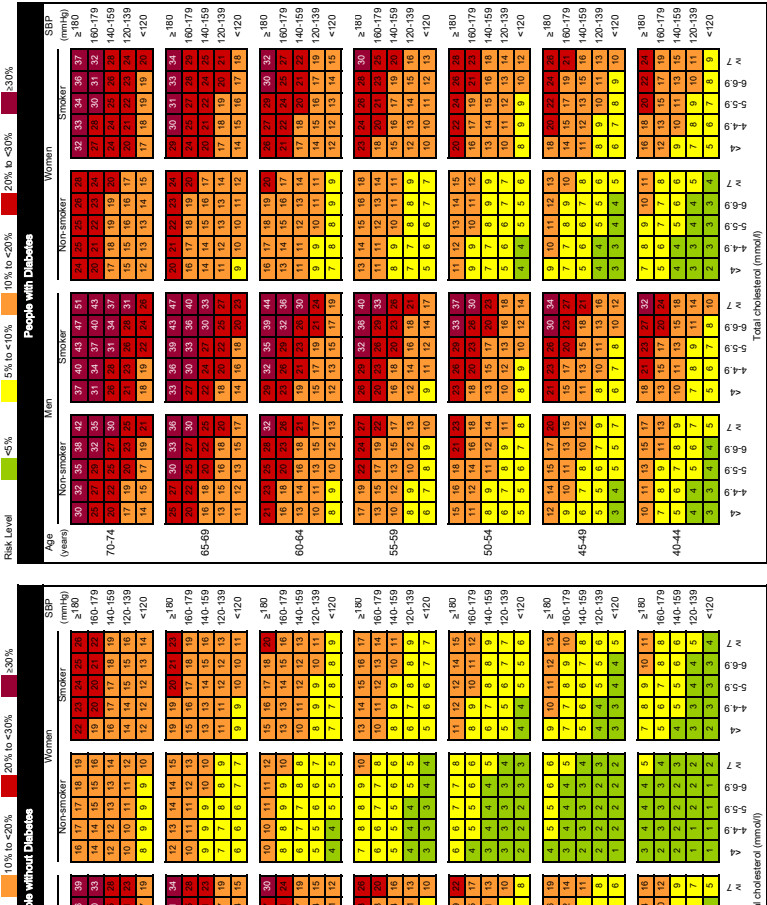


Central Sub-Saharan Africa

10 cardiovascular disease risk laboratory-based charts

Eastern Sub-Saharan Africa

Eritrea, Ethiopia, Kenya, Madagascar, Malawi, Mozambique, Rwanda, Somalia, Uganda, United Republic of Tanzania, Zambia

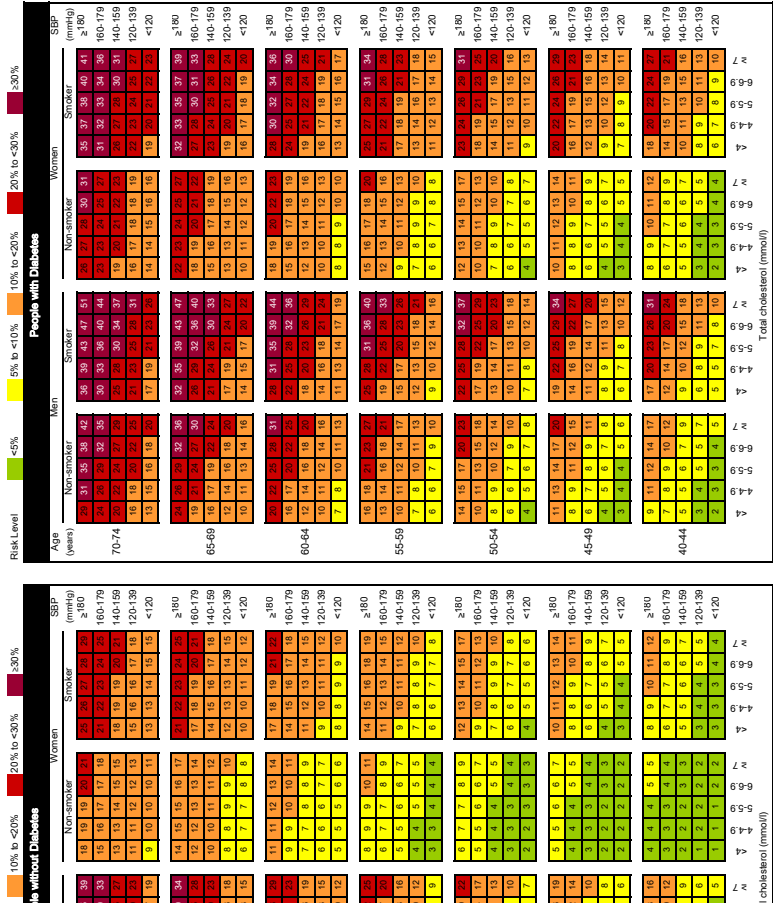


Eastern Sub-Saharan Africa

10 cardiovascular disease risk laboratory-based charts

Southern Sub-Saharan Africa

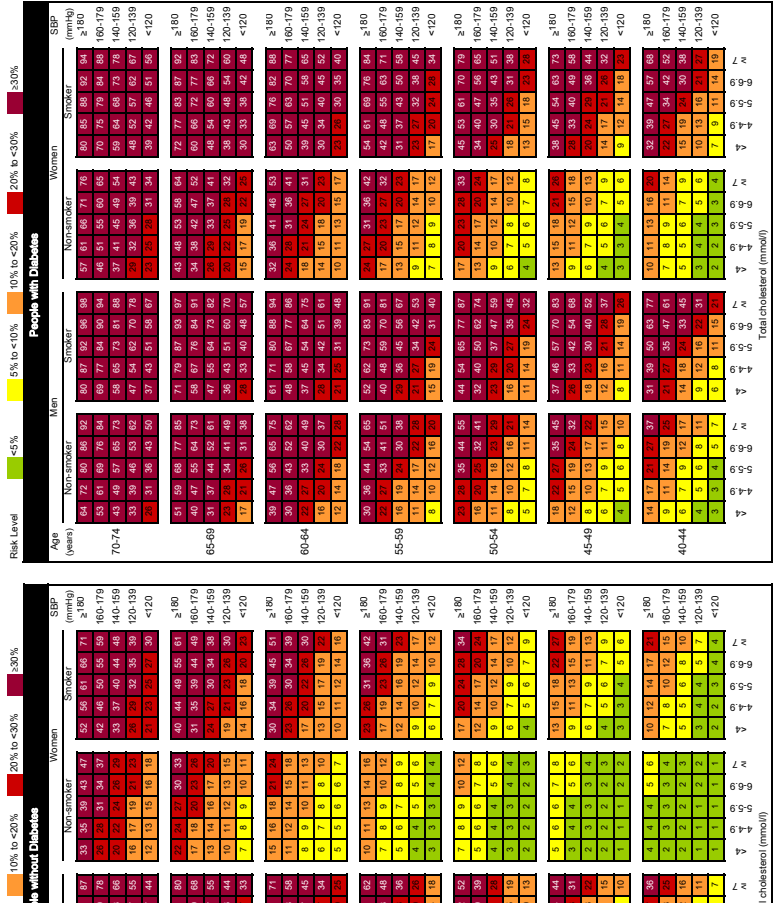
Botswana, Eswatini, Lesotho, Namibia, South Africa, Zimbabwe



10 cardiovascular disease risk laboratory-based charts

Central Asia

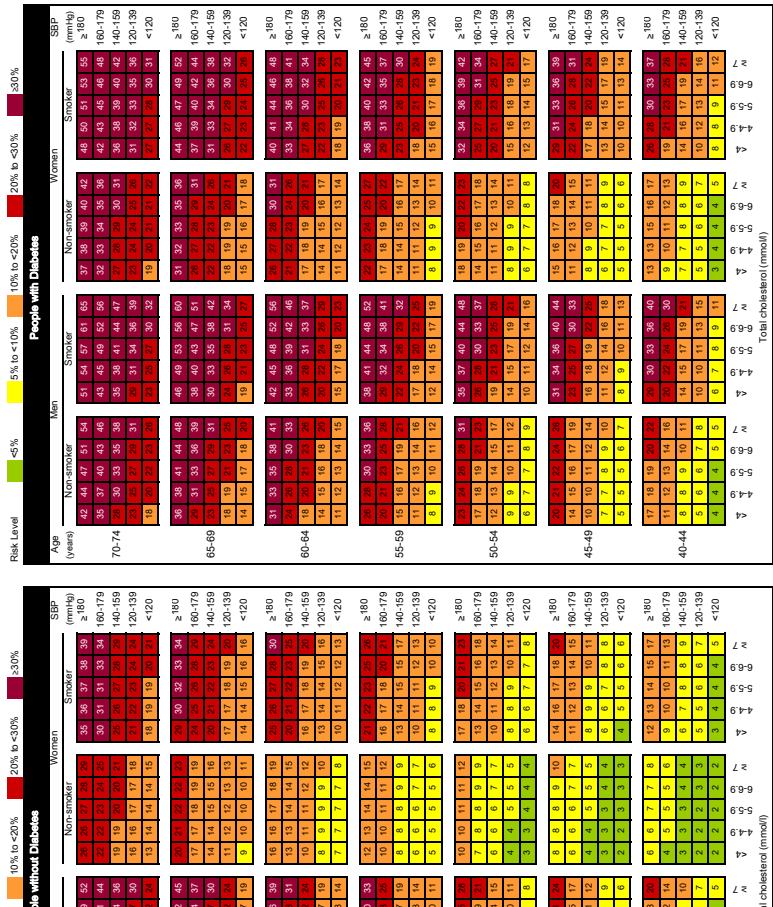
menia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, Mongolia, Tajikistan, Turkmenistan, Uzbekistan



10 cardiovascular disease risk laboratory-based charts

East Asia

China, Democratic People's Republic of Korea

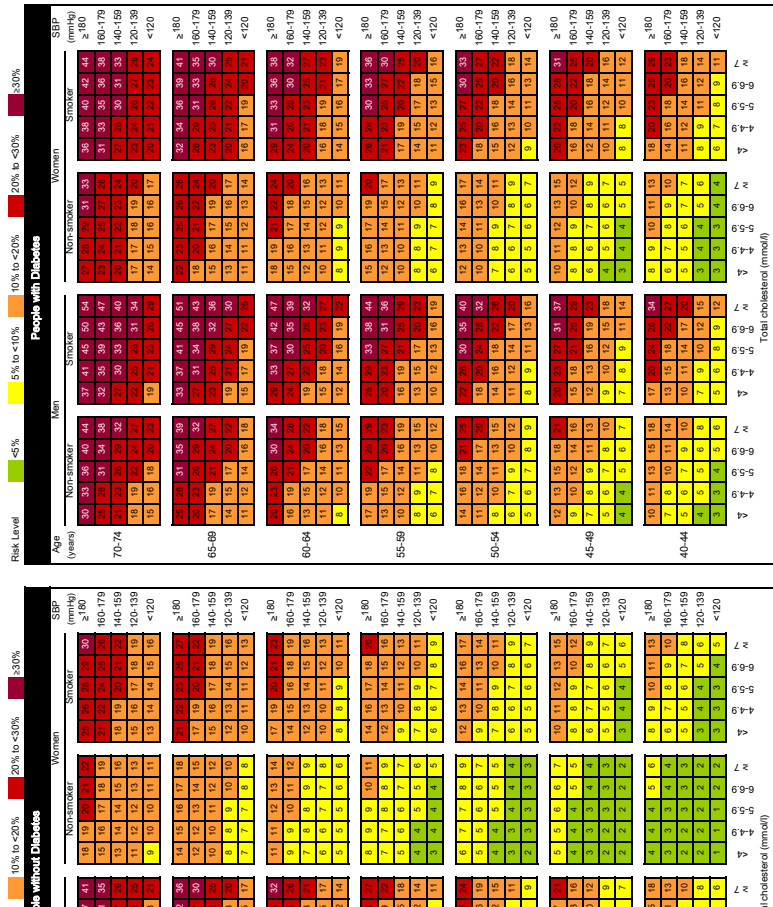


East Asia

10 cardiovascular disease risk laboratory-based charts

South Asia

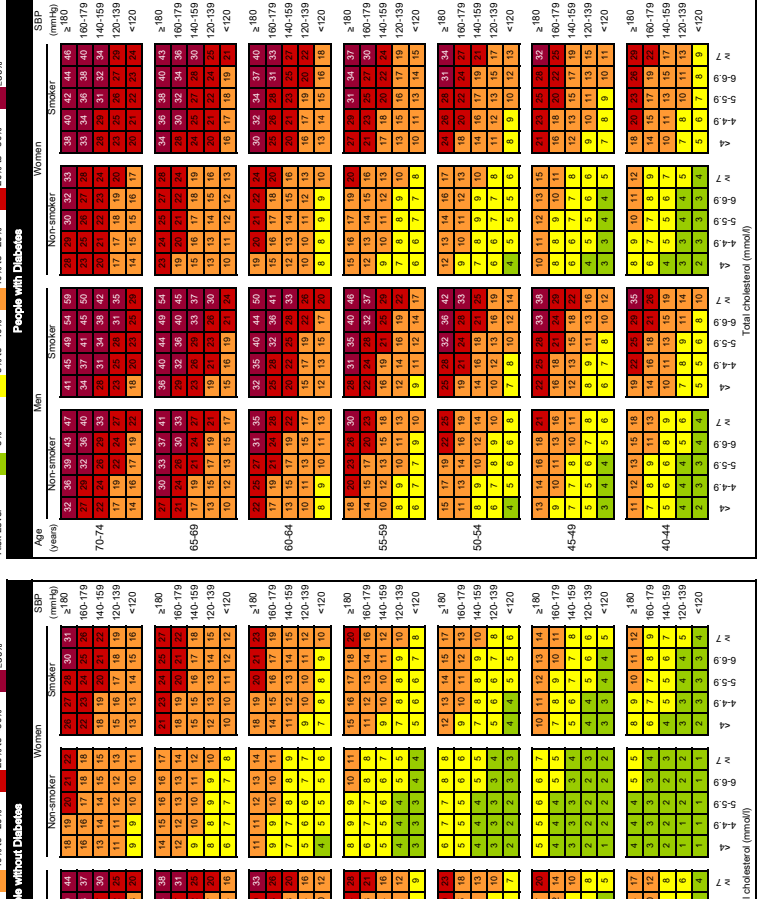
Bangladesh, Bhutan, India, Nepal, Pakistan



South Asia

10 cardiovascular disease risk laboratory-based charts

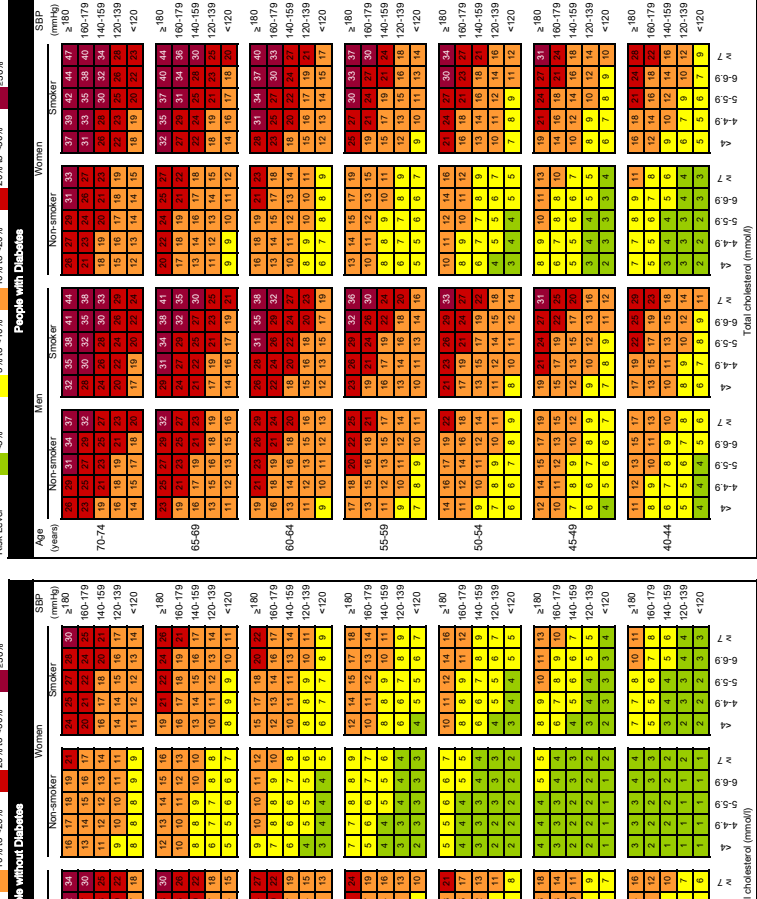
South-East Asia
Democratic Republic, Malaysia, Maldives, Mauritius, Myanmar, Philippines, Seychelles, Sri Lanka, Thailand, Timor-Leste, Viet Nam



South-East Asia

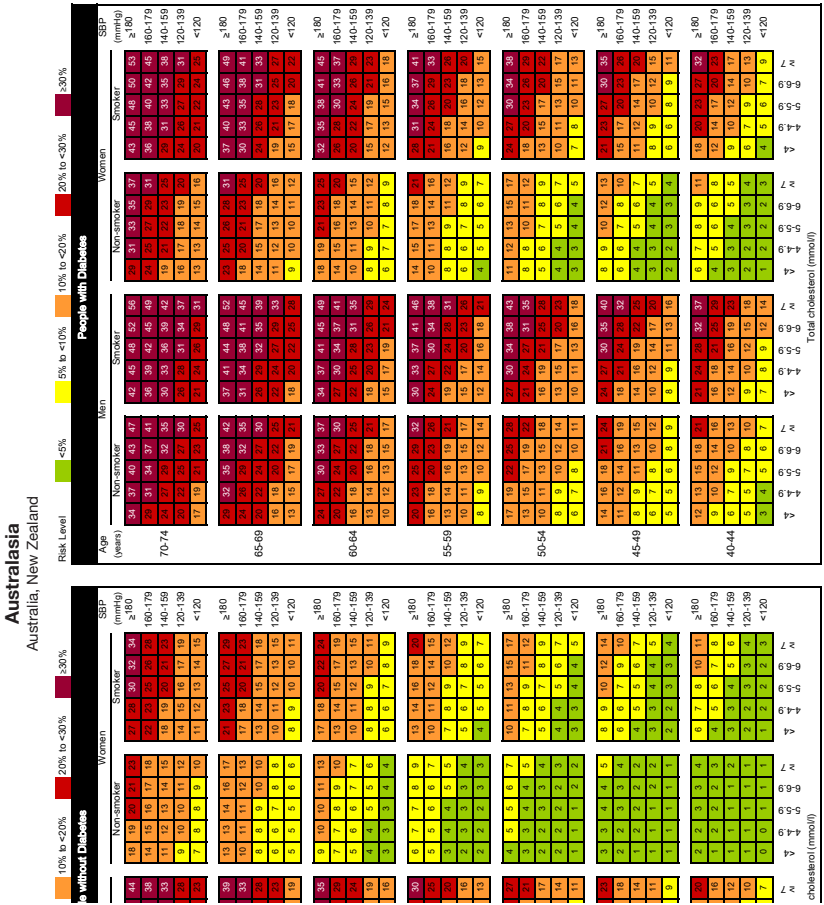
10 cardiovascular disease risk laboratory-based charts

High-income Asia Pacific
Brunei Darussalam, Japan, Republic of Korea, Singapore



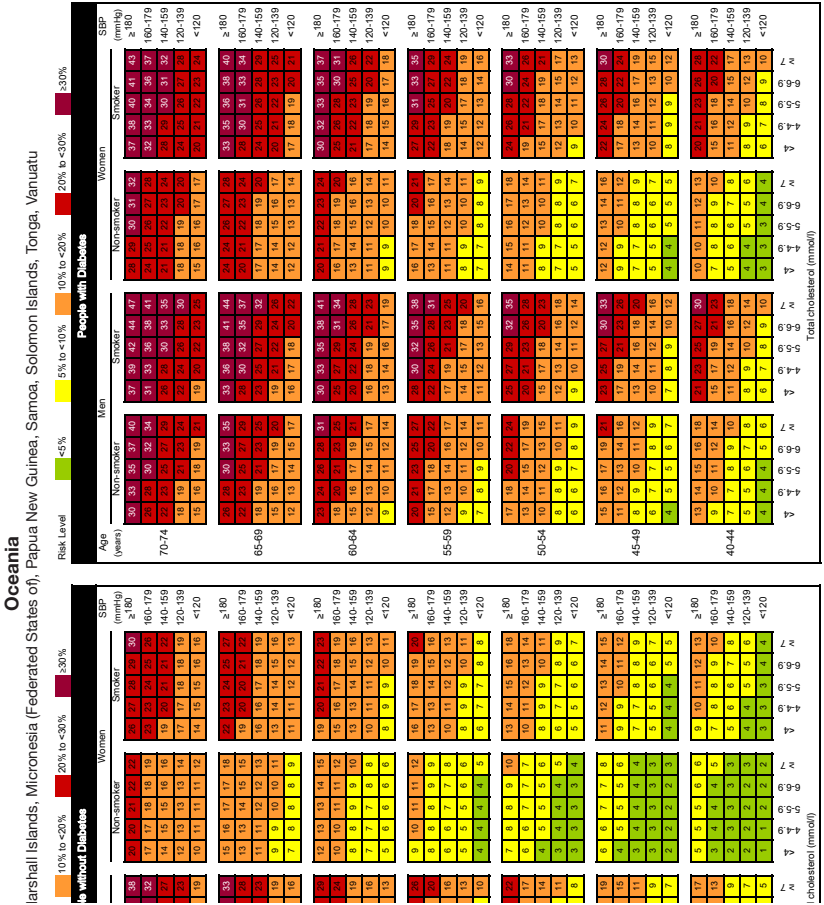
High-income Asia Pacific

10 cardiovascular disease risk laboratory-based charts



Australasia

10 cardiovascular disease risk laboratory-based charts



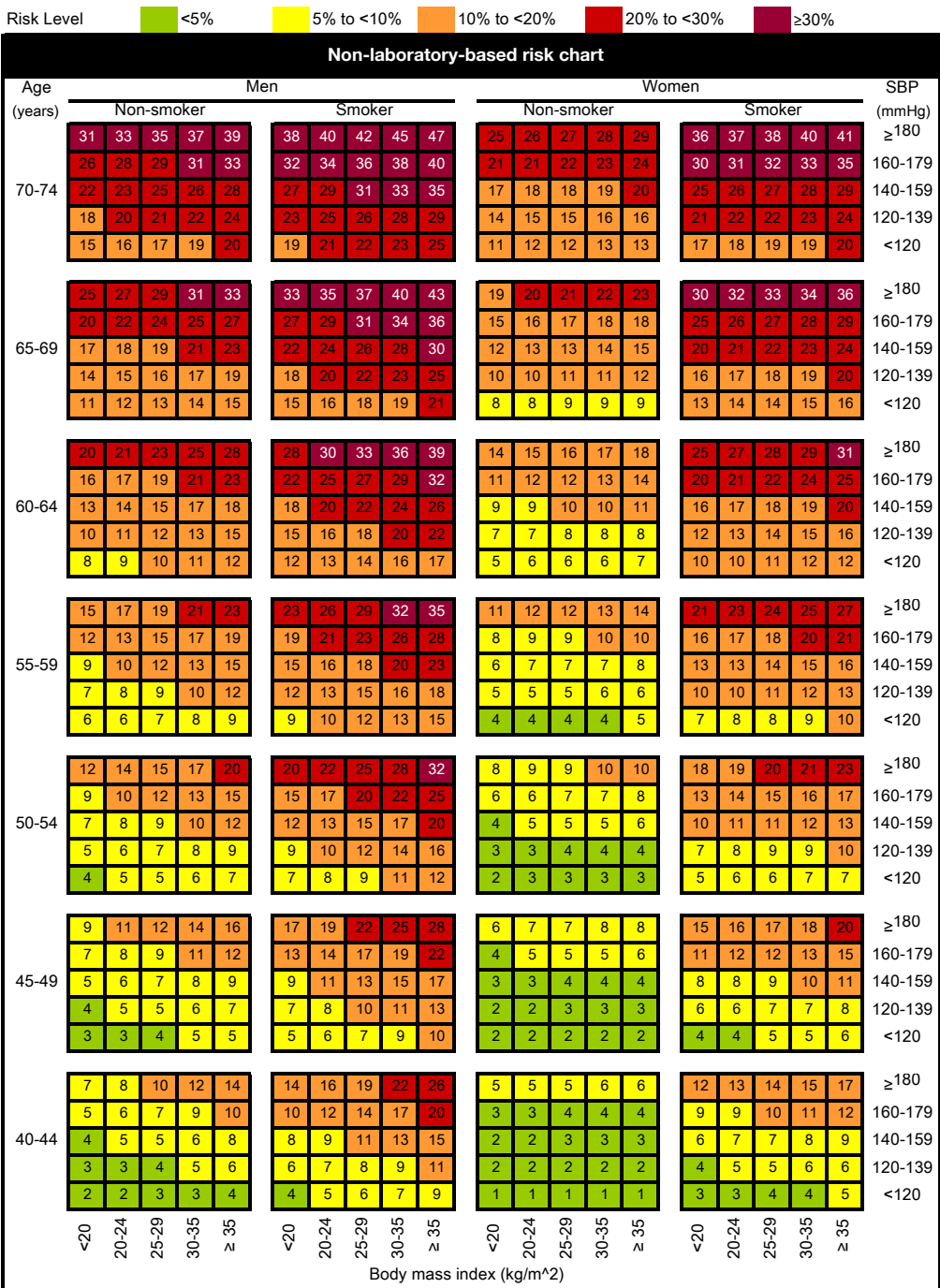
Oceania

Annex 3: WHO CVD risk (non-laboratory-based) charts

WHO cardiovascular disease risk non-laboratory-based charts

High-income North America

Canada, Greenland, United States of America

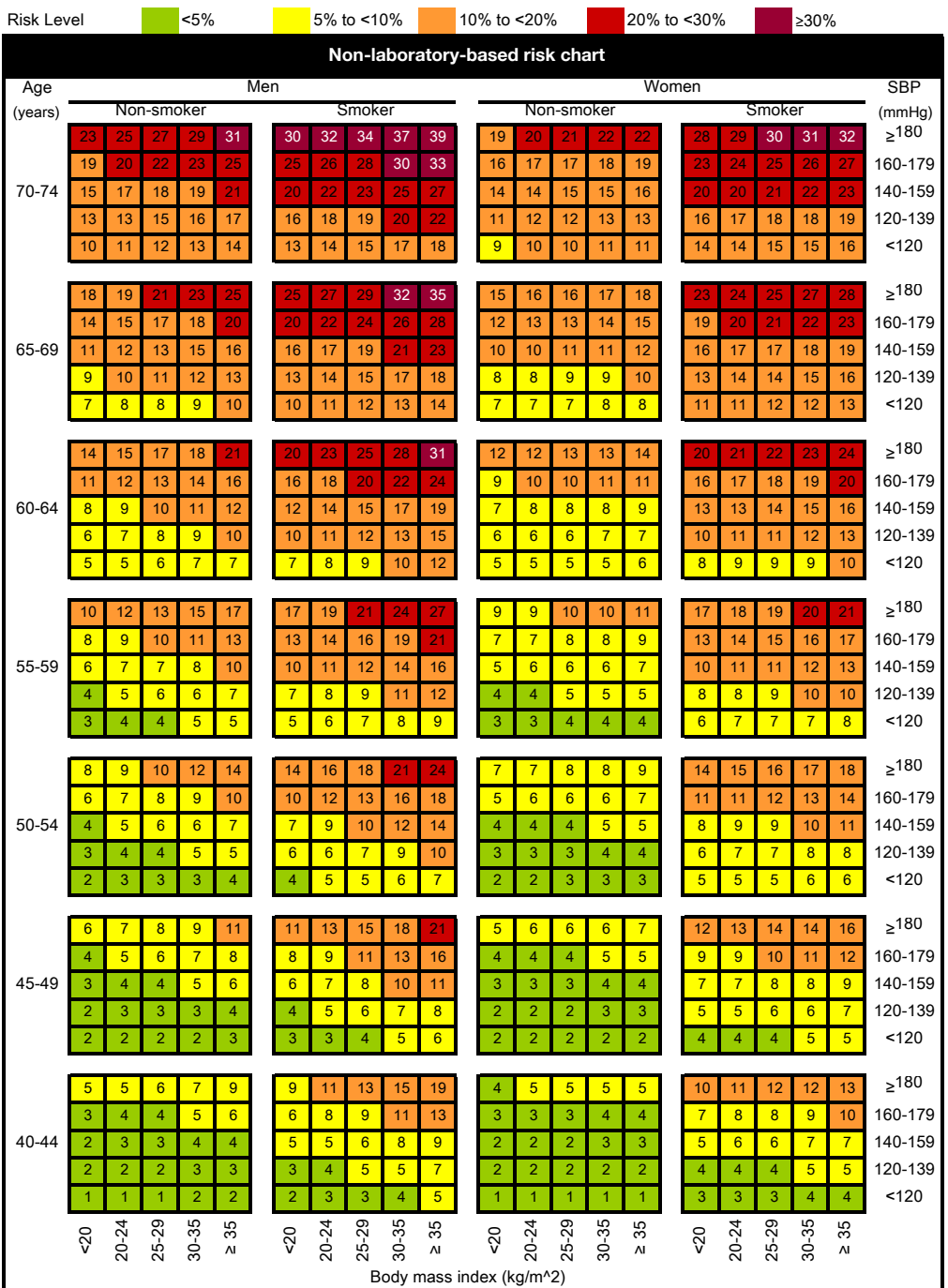


High-income North America

WHO cardiovascular disease risk non-laboratory-based charts

Caribbean

Antigua and Barbuda, Bahamas, Barbados, Belize, Bermuda, Cuba, Dominica, Dominican Republic, Grenada, Guyana, Haiti, Jamaica, Puerto Rico, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago

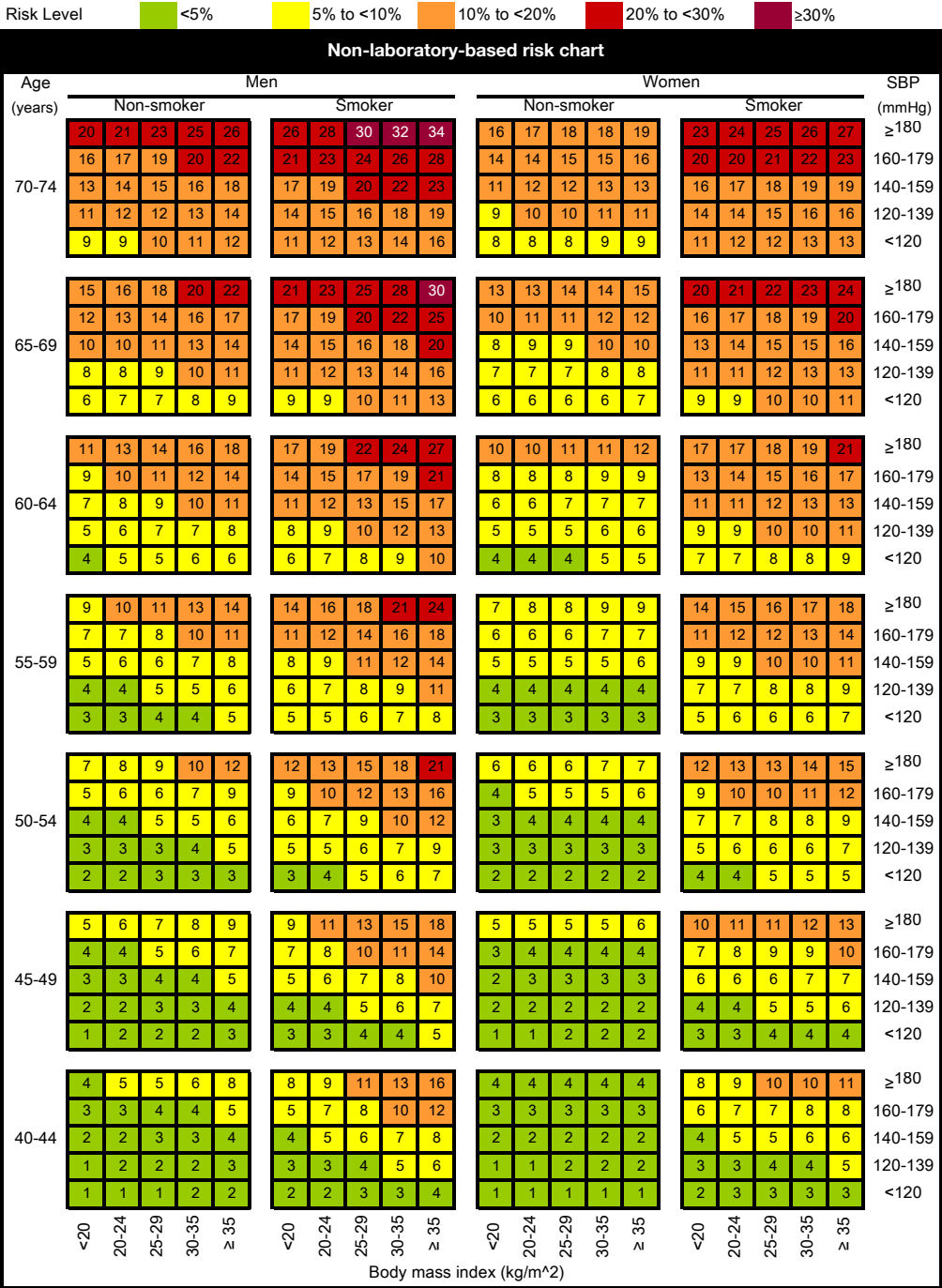


Caribbean

WHO cardiovascular disease risk non-laboratory-based charts

Central Latin America

Colombia, Costa Rica, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Venezuela (Bolivarian Republic of)

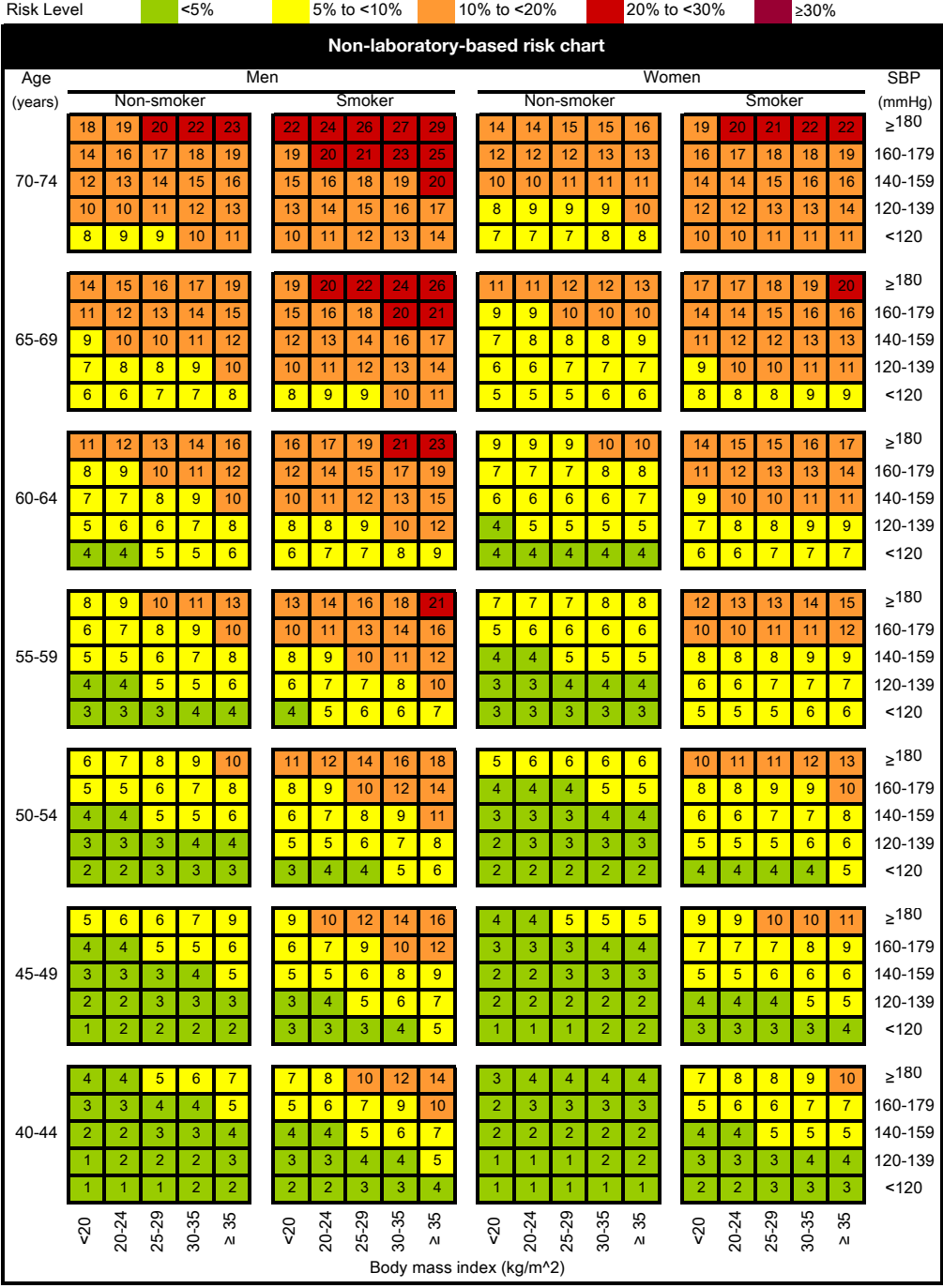


Central Latin America

WHO cardiovascular disease risk non-laboratory-based charts

Andean Latin America

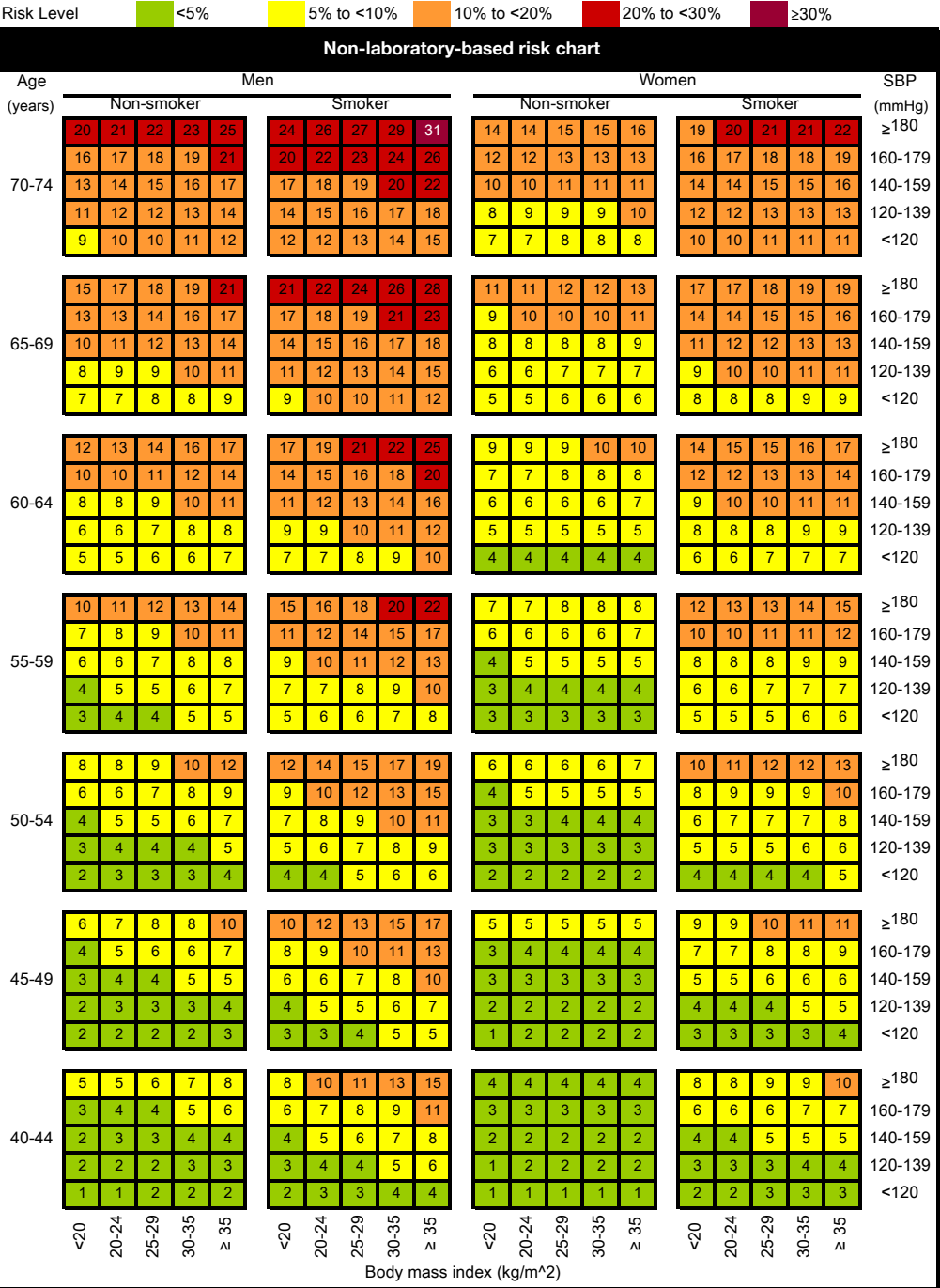
Bolivia, Ecuador, Peru



Andean Latin America

WHO cardiovascular disease risk non-laboratory-based charts

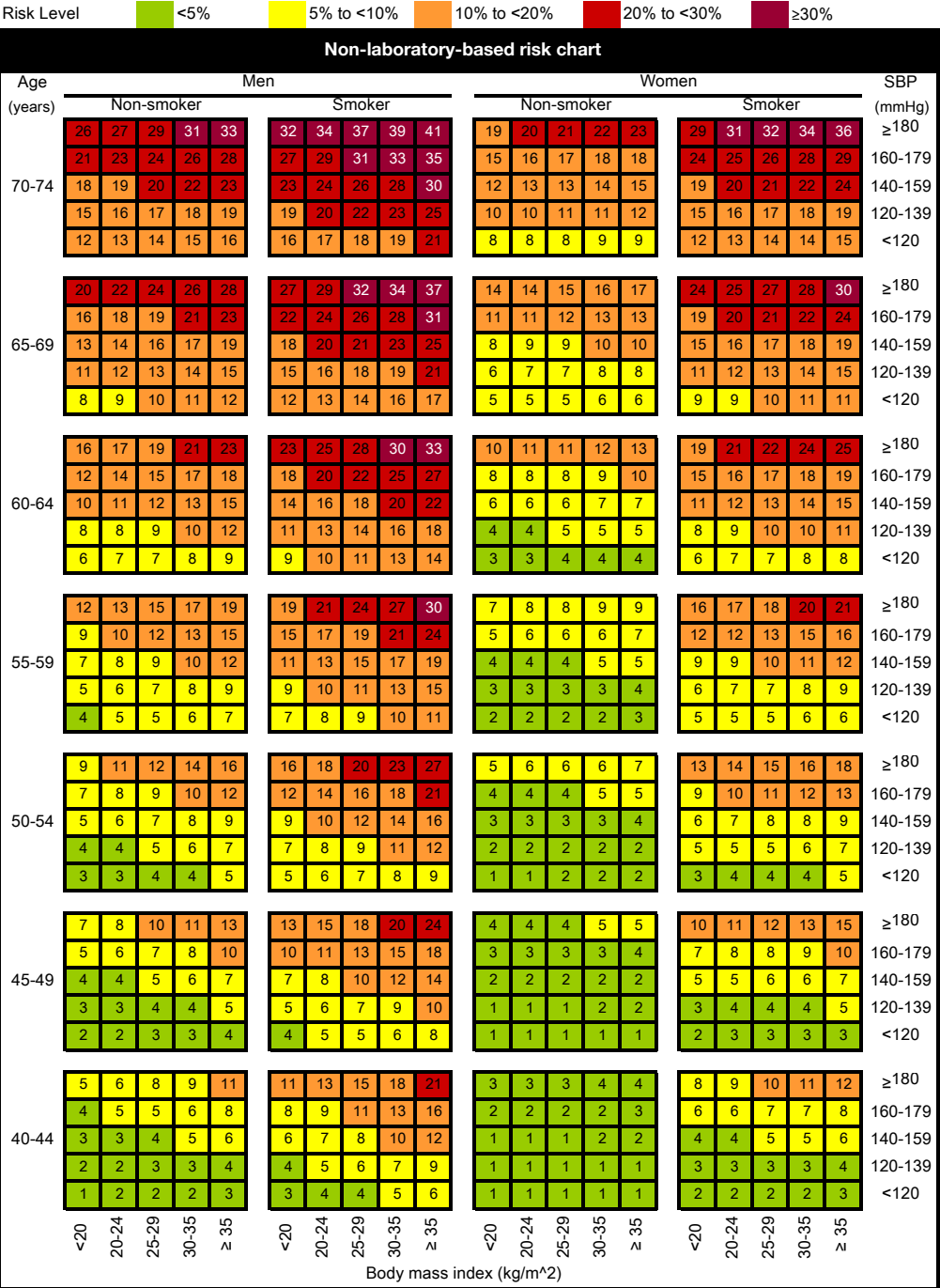
Tropical Latin America
Brazil, Paraguay



Tropical Latin America

WHO cardiovascular disease risk non-laboratory-based charts

Southern Latin America
Argentina, Chile, Uruguay

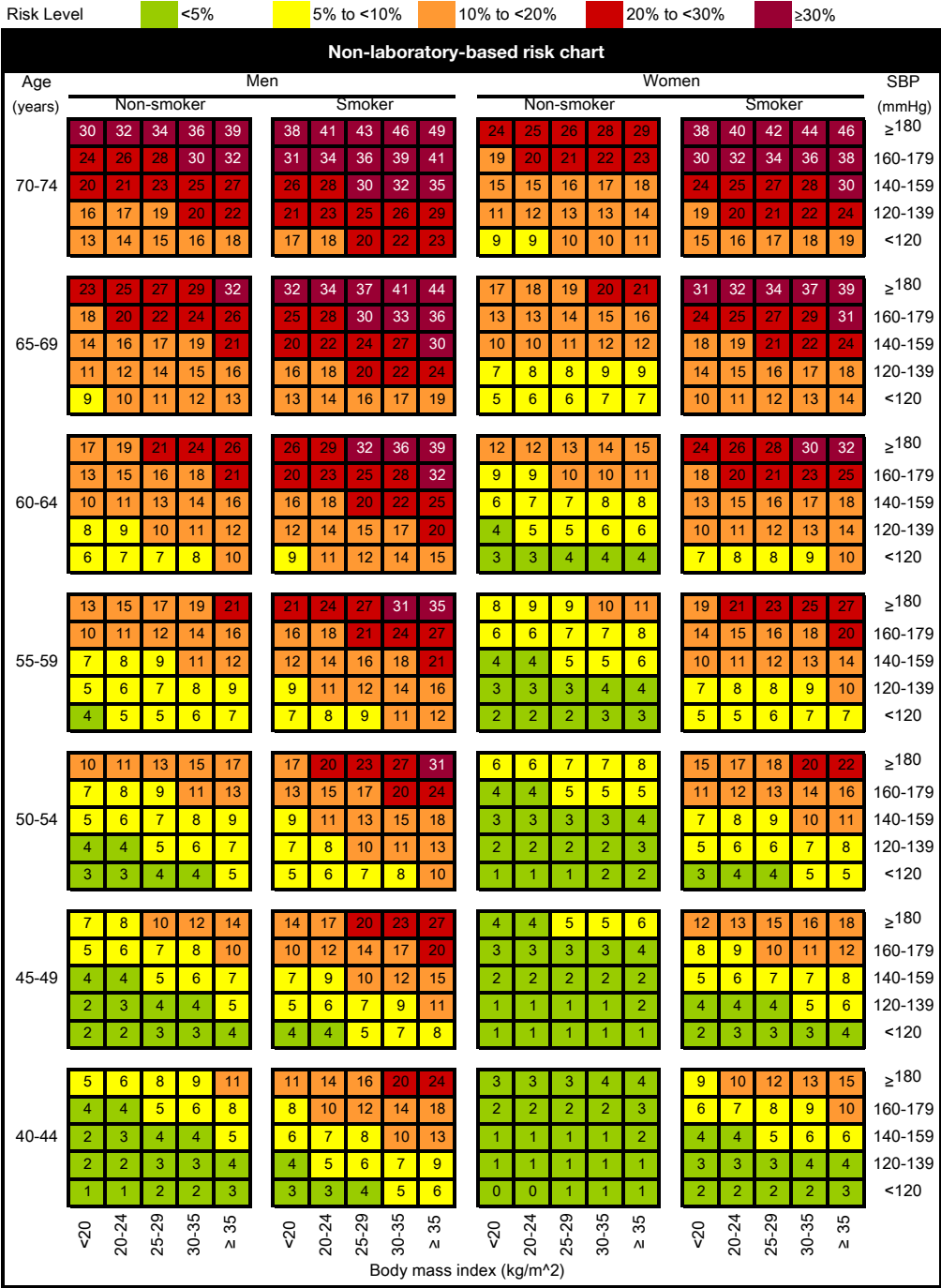


Southern Latin America

WHO cardiovascular disease risk non-laboratory-based charts

Western Europe

Andorra, Austria, Belgium, Cyprus, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Israel, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom

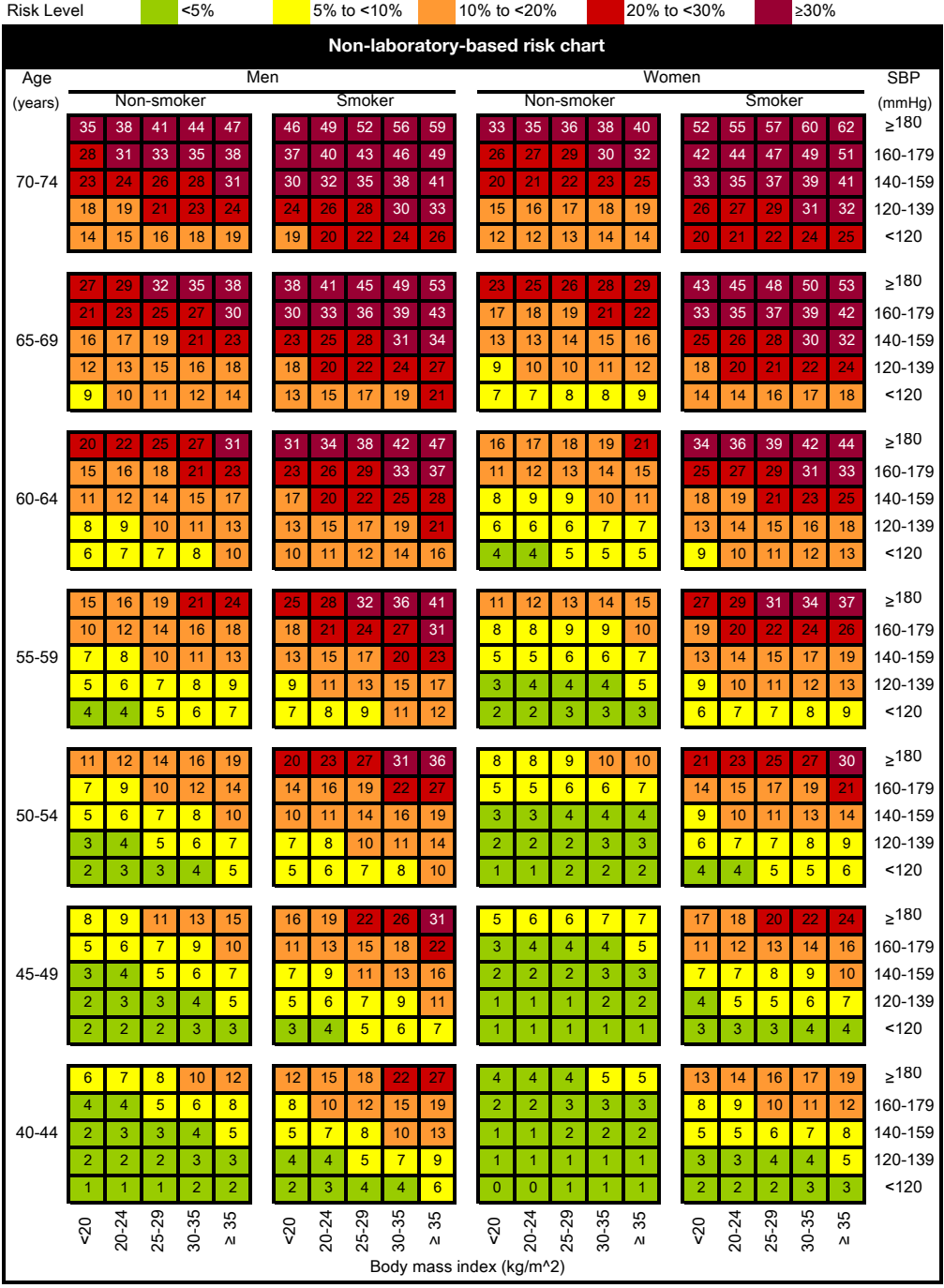


Western Europe

WHO cardiovascular disease risk non-laboratory-based charts

Central Europe

Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Hungary, Montenegro, North Macedonia, Poland, Romania, Serbia, Slovakia, Slovenia

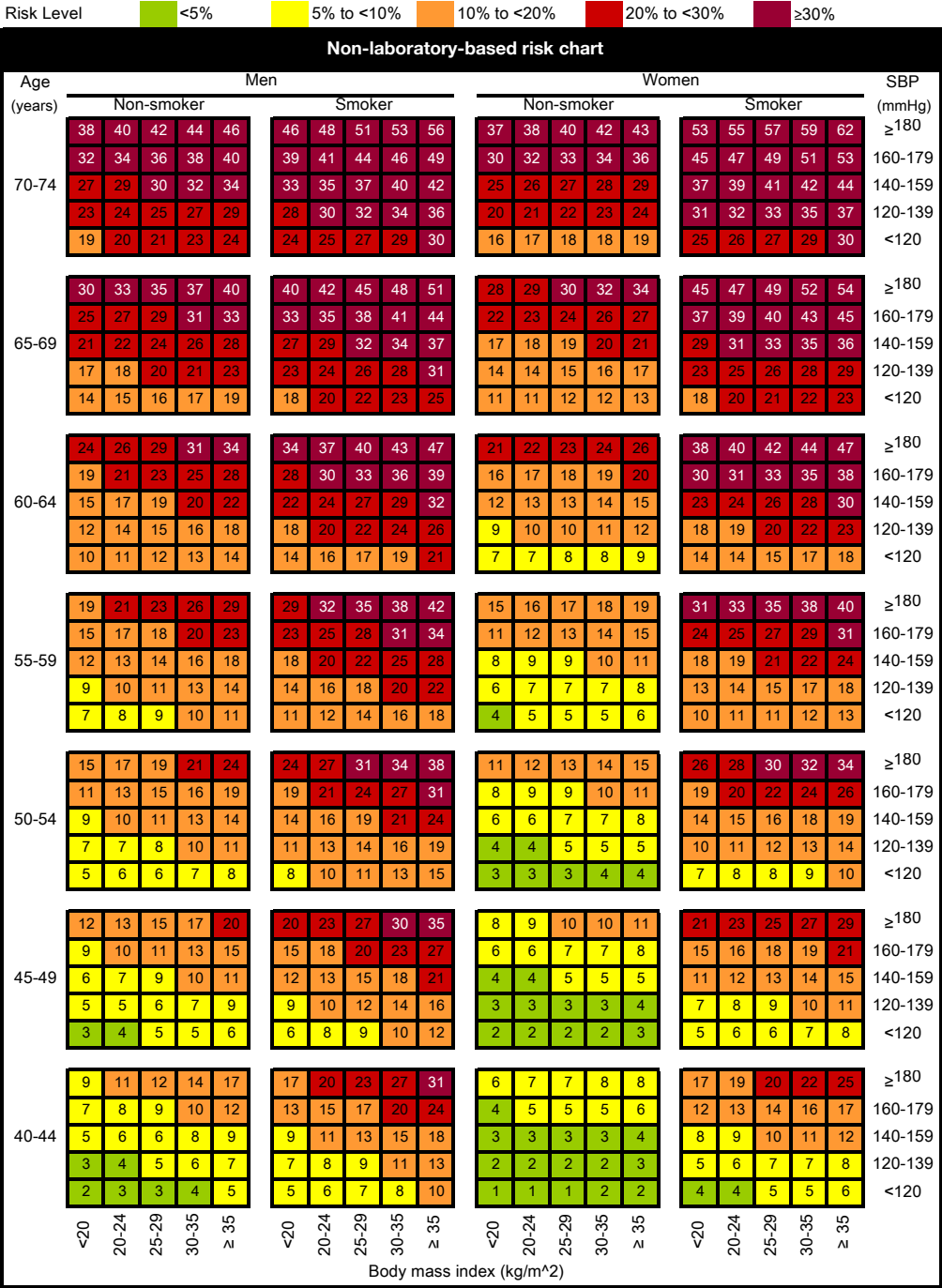


Central Europe

WHO cardiovascular disease risk non-laboratory-based charts

Eastern Europe

Belarus, Estonia, Latvia, Lithuania, Republic of Moldova, Russian Federation, Ukraine

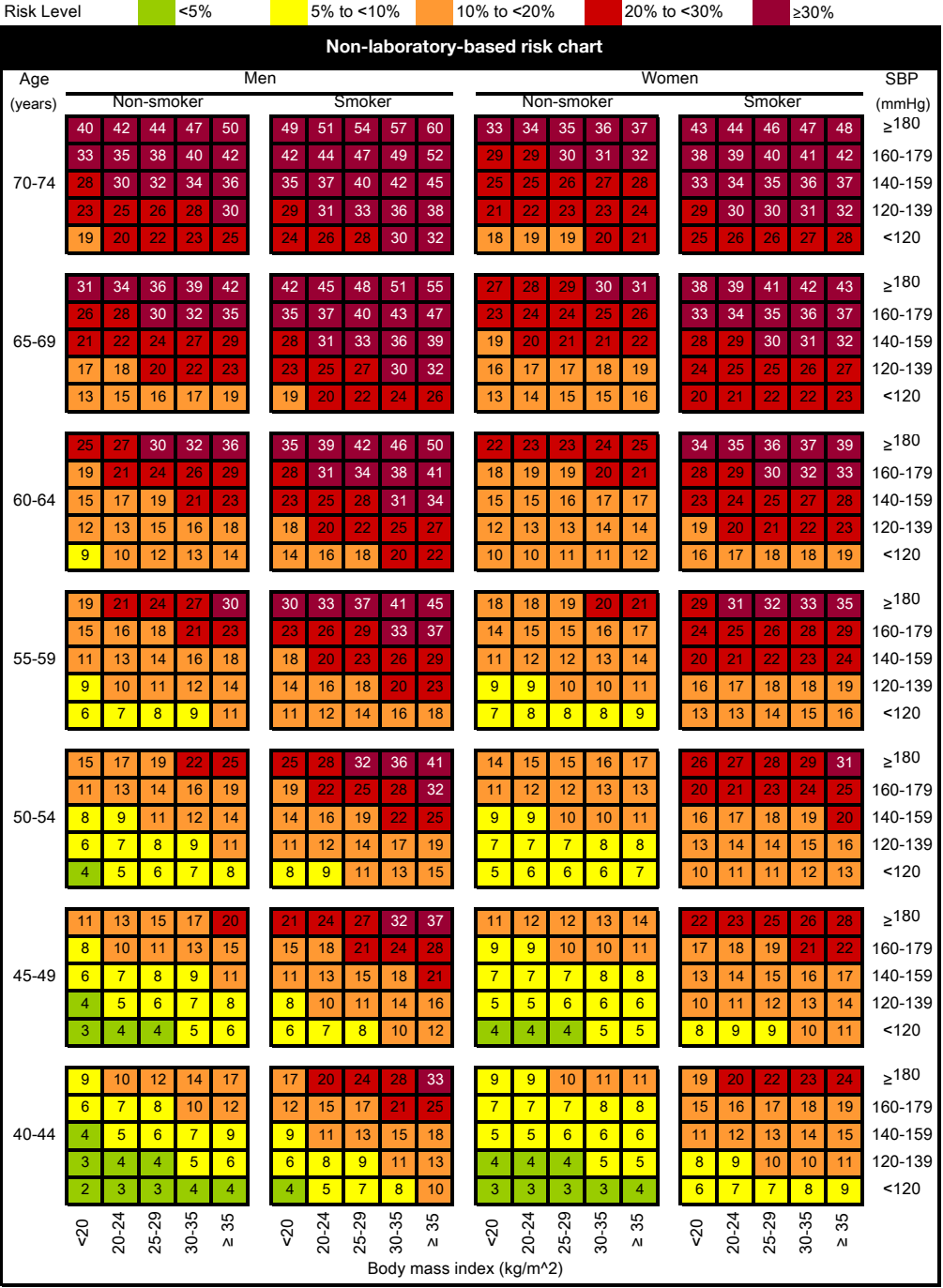


Eastern Europe

WHO cardiovascular disease risk non-laboratory-based charts

North Africa and Middle East

Afghanistan, Algeria, Bahrain, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, occupied Palestinian territory, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, Tunisia, Turkey, United Arab Emirates, Yemen

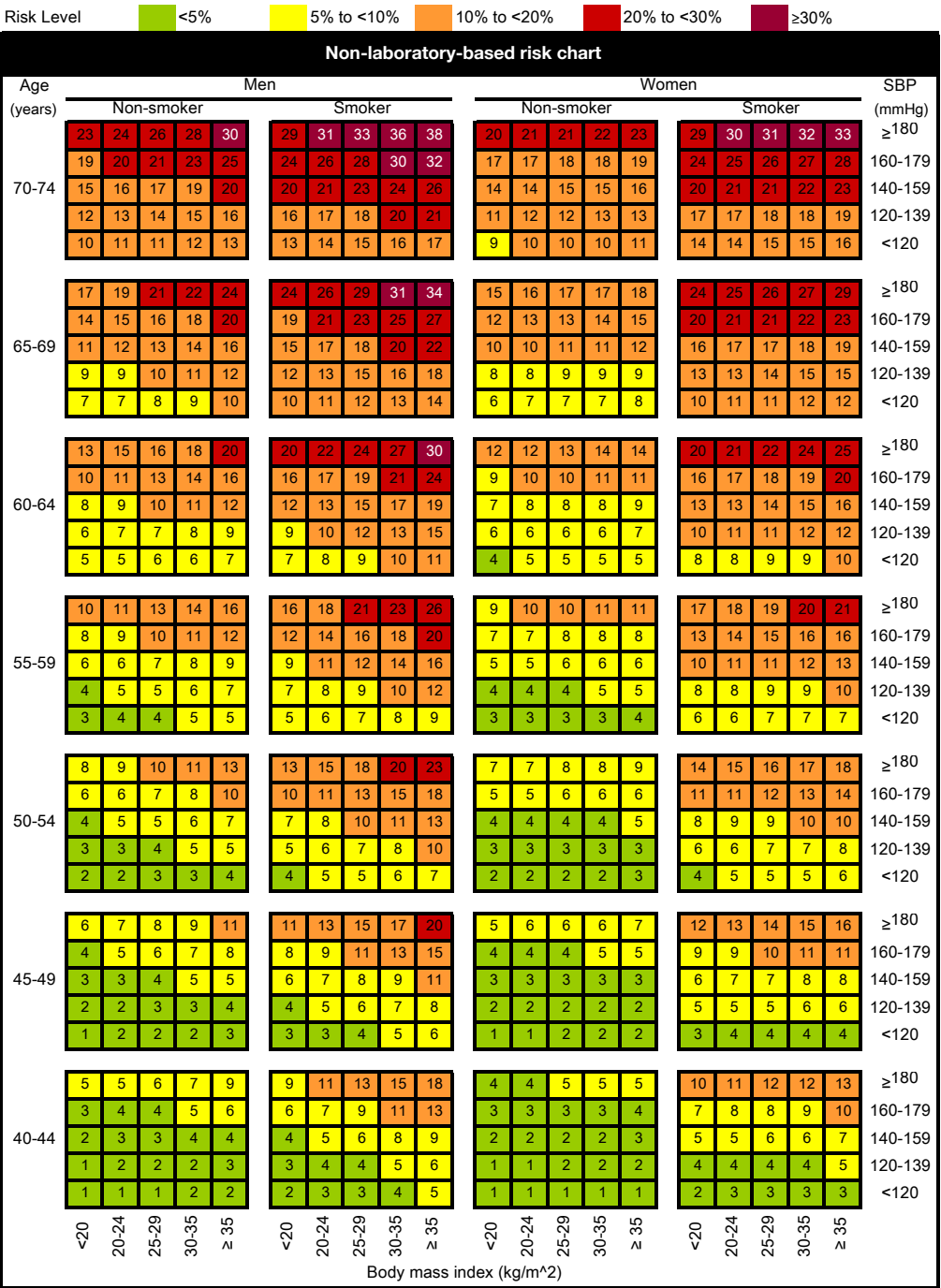


North Africa and Middle East

WHO cardiovascular disease risk non-laboratory-based charts

Western Sub-Saharan Africa

Benin, Burkina Faso, Cabo Verde, Cameroon, Chad, Cote d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, Togo

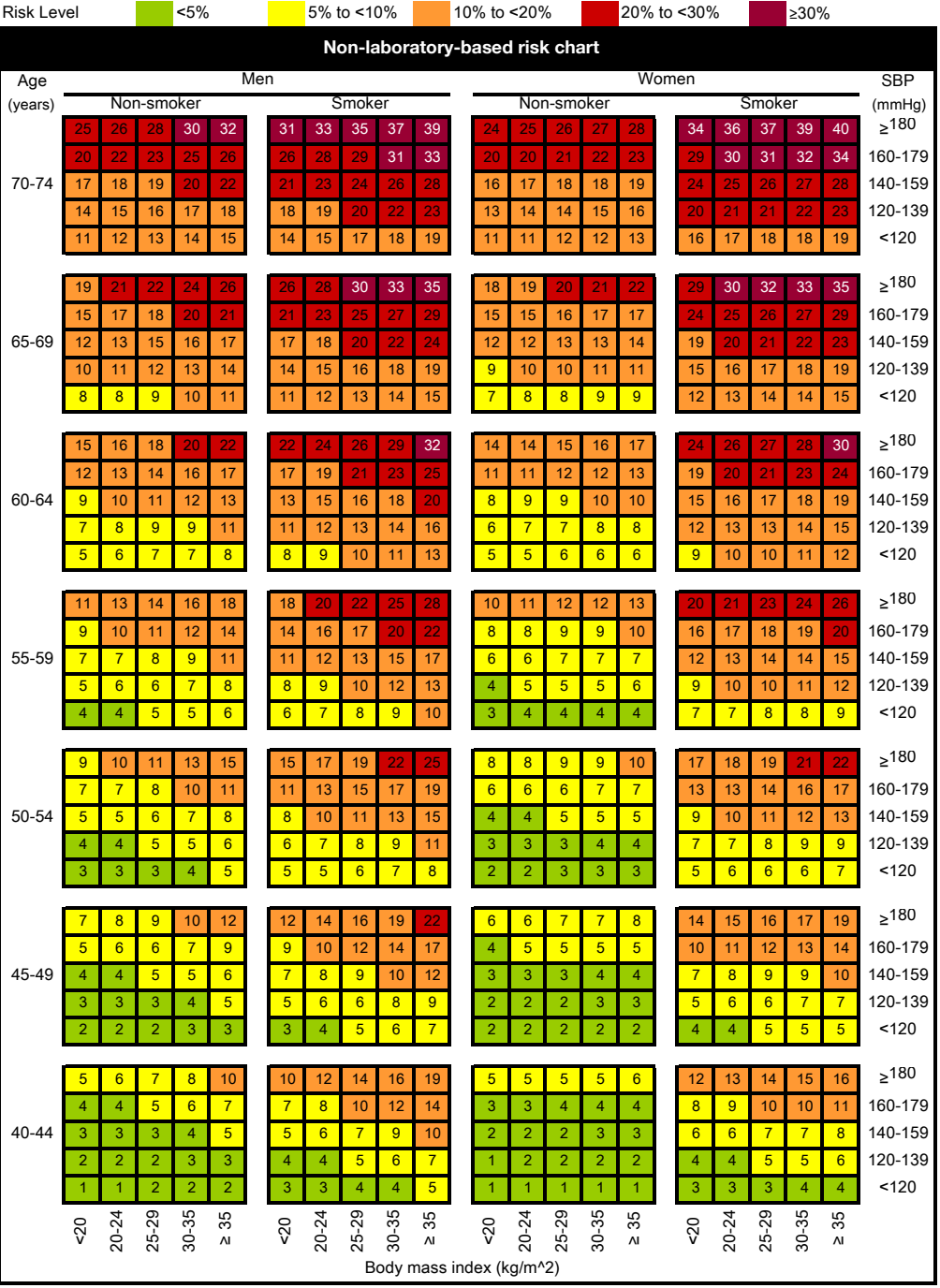


Western Sub-Saharan Africa

WHO cardiovascular disease risk non-laboratory-based charts

Central Sub-Saharan Africa

Angola, Central African Republic, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon

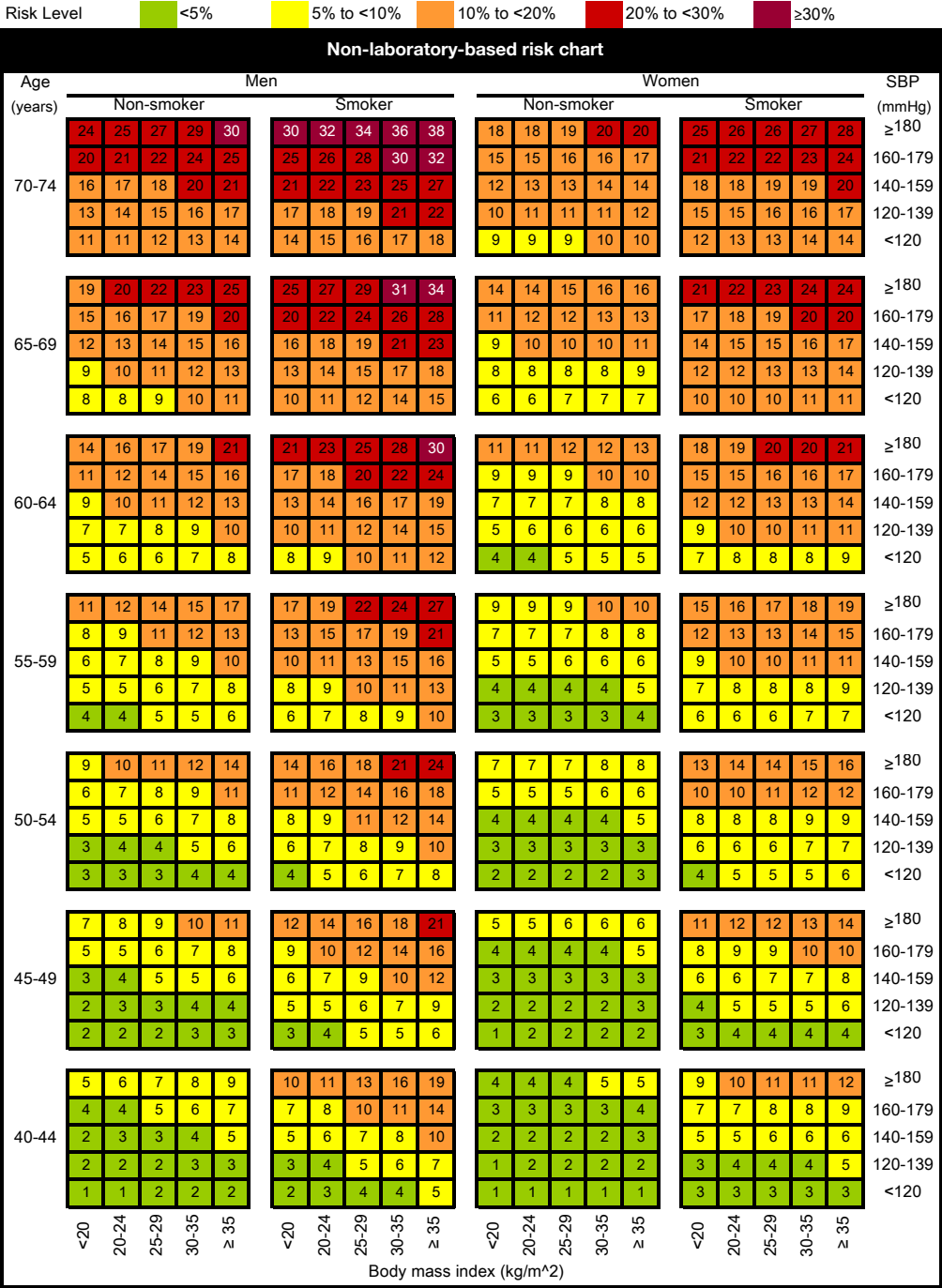


Central Sub-Saharan Africa

WHO cardiovascular disease risk non-laboratory-based charts

Eastern Sub-Saharan Africa

Burundi, Comoros, Djibouti, Eritrea, Ethiopia, Kenya, Madagascar, Malawi, Mozambique, Rwanda, Somalia, Uganda, United Republic of Tanzania, Zambia

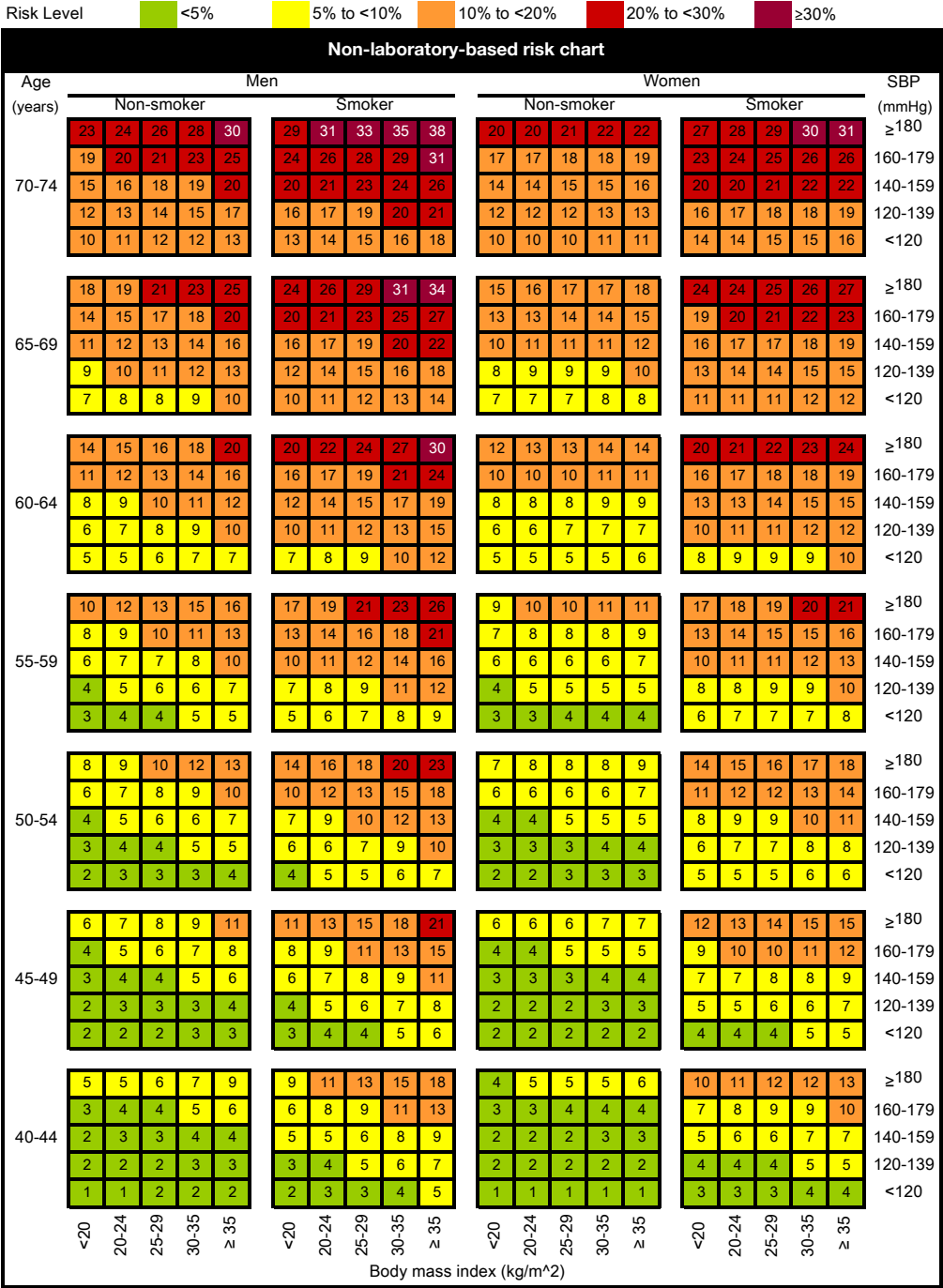


Eastern Sub-Saharan Africa

WHO cardiovascular disease risk non-laboratory-based charts

Southern Sub-Saharan Africa

Botswana, Eswatini, Lesotho, Namibia, South Africa, Zimbabwe

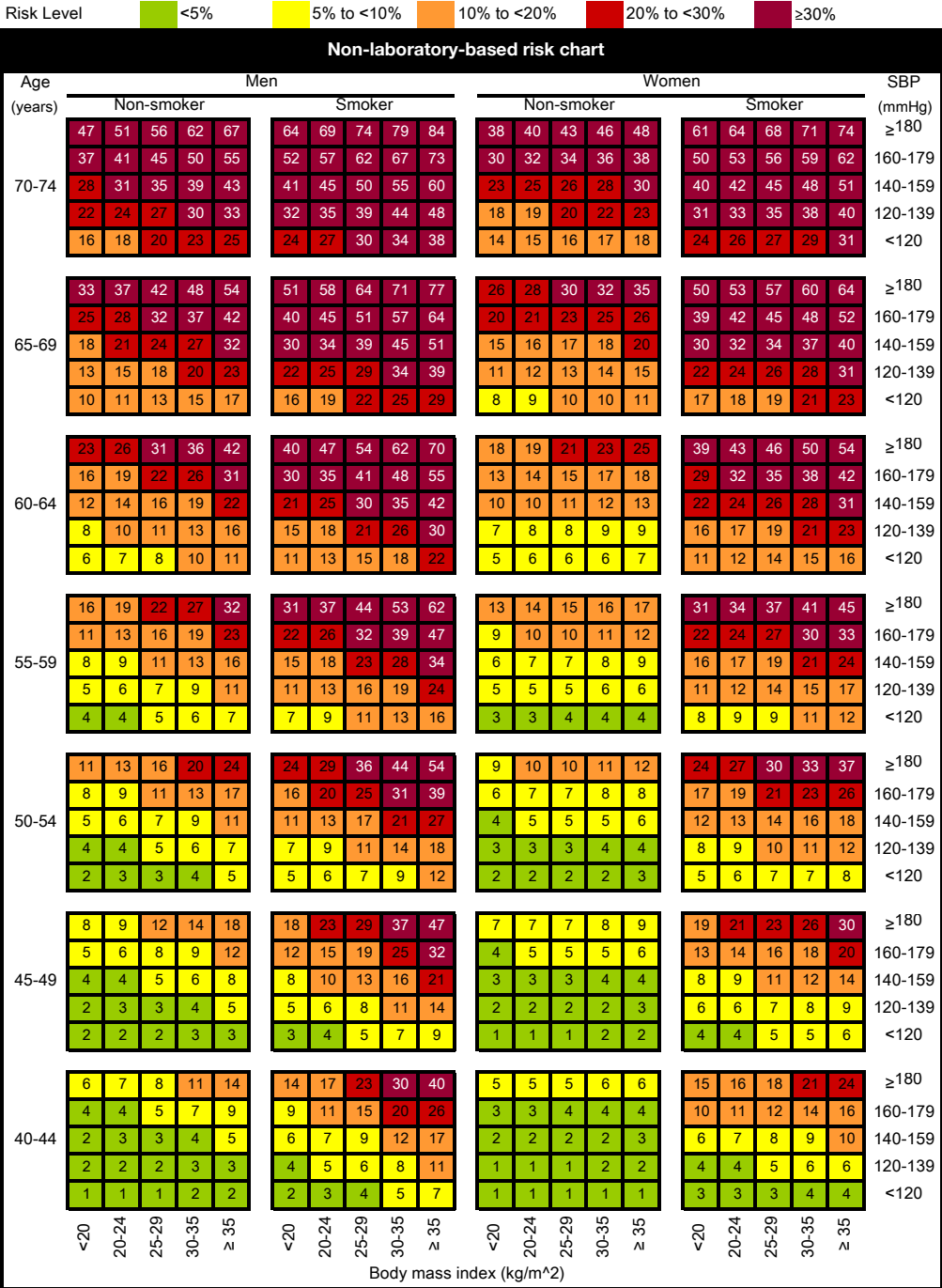


Southern Sub-Saharan Africa

WHO cardiovascular disease risk non-laboratory-based charts

Central Asia

Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, Mongolia, Tajikistan, Turkmenistan, Uzbekistan

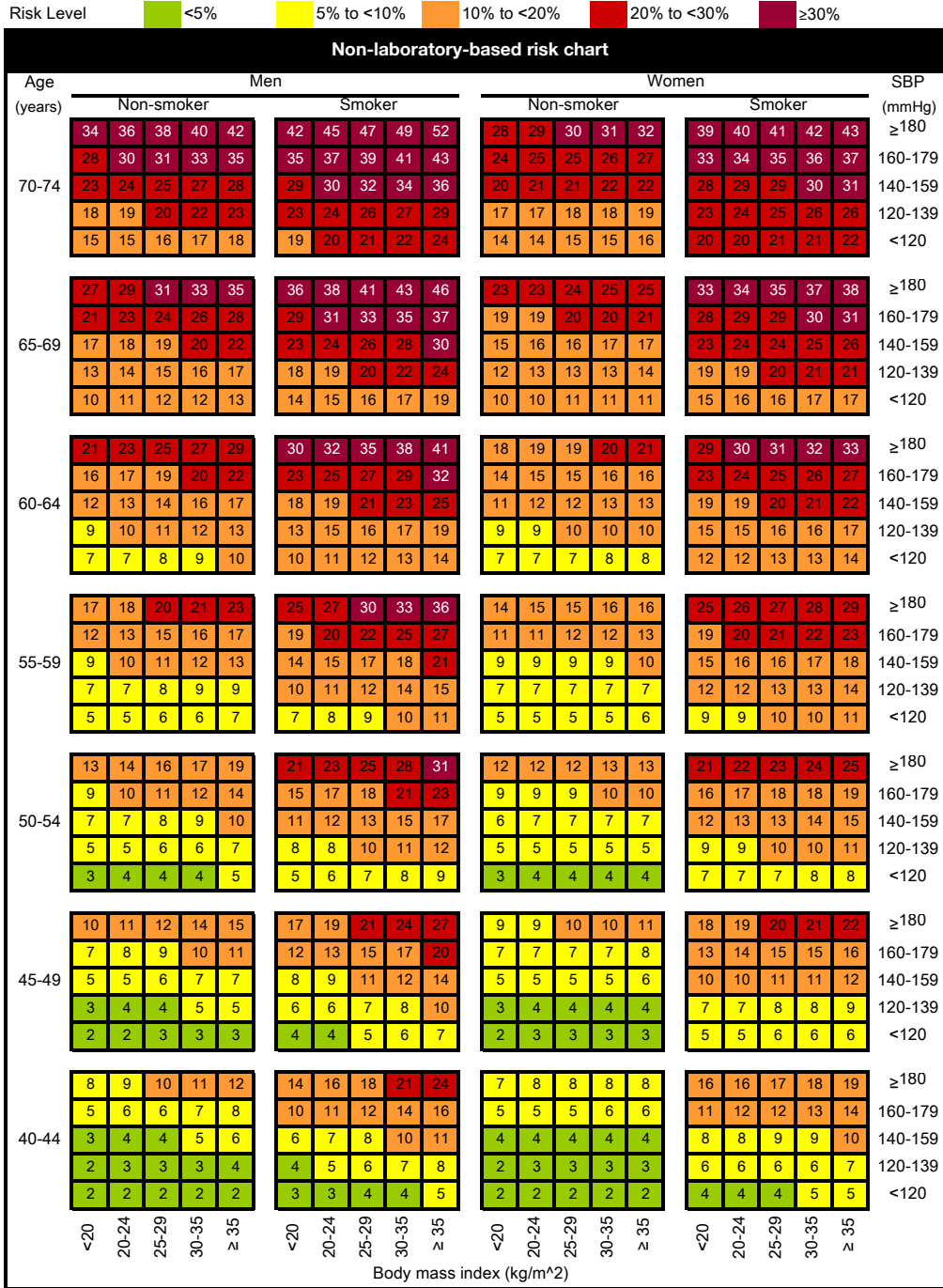


Central Asia

WHO cardiovascular disease risk non-laboratory-based charts

East Asia

China, Democratic People's Republic of Korea

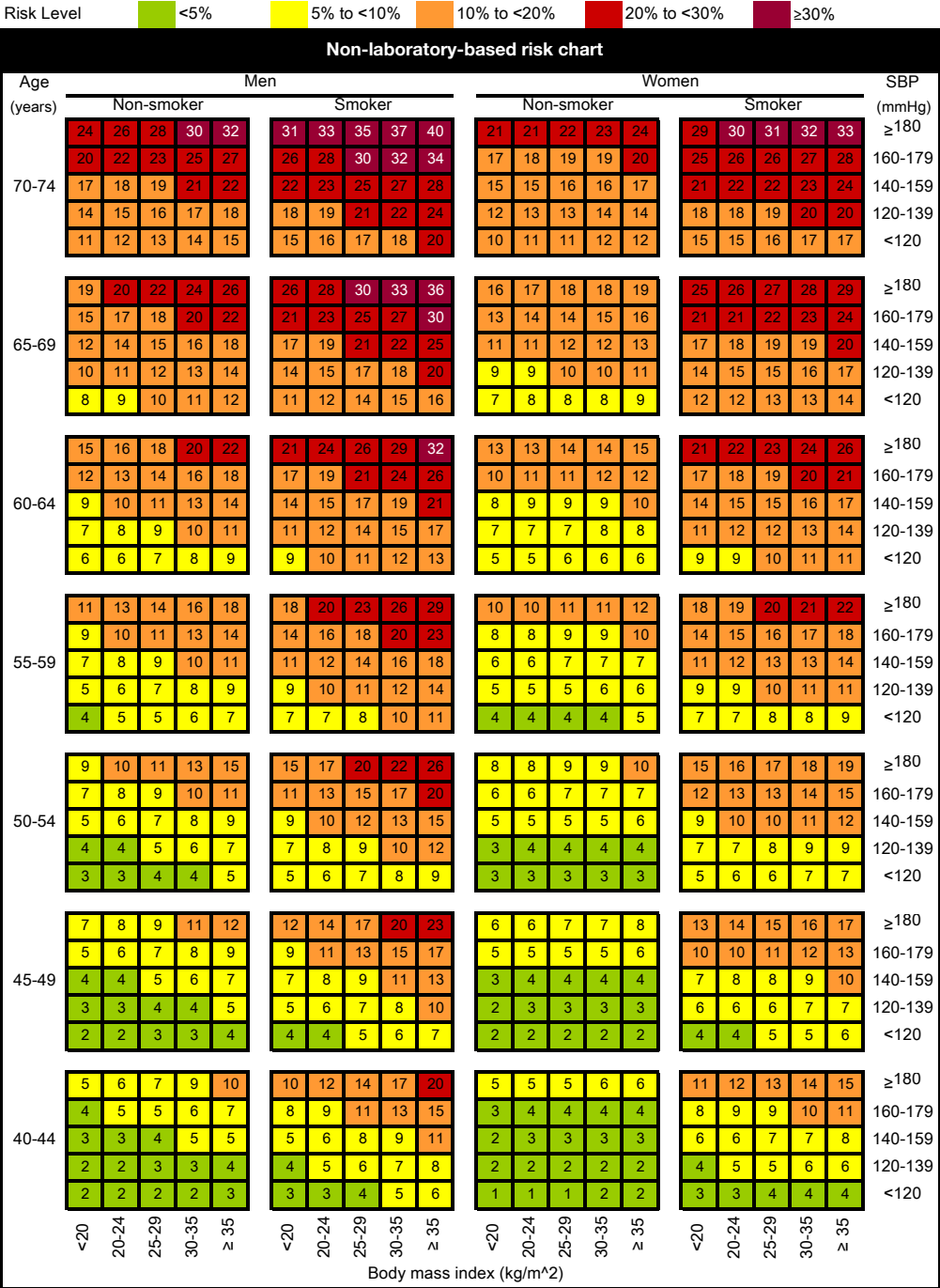


East Asia

WHO cardiovascular disease risk non-laboratory-based charts

South Asia

Bangladesh, Bhutan, India, Nepal, Pakistan

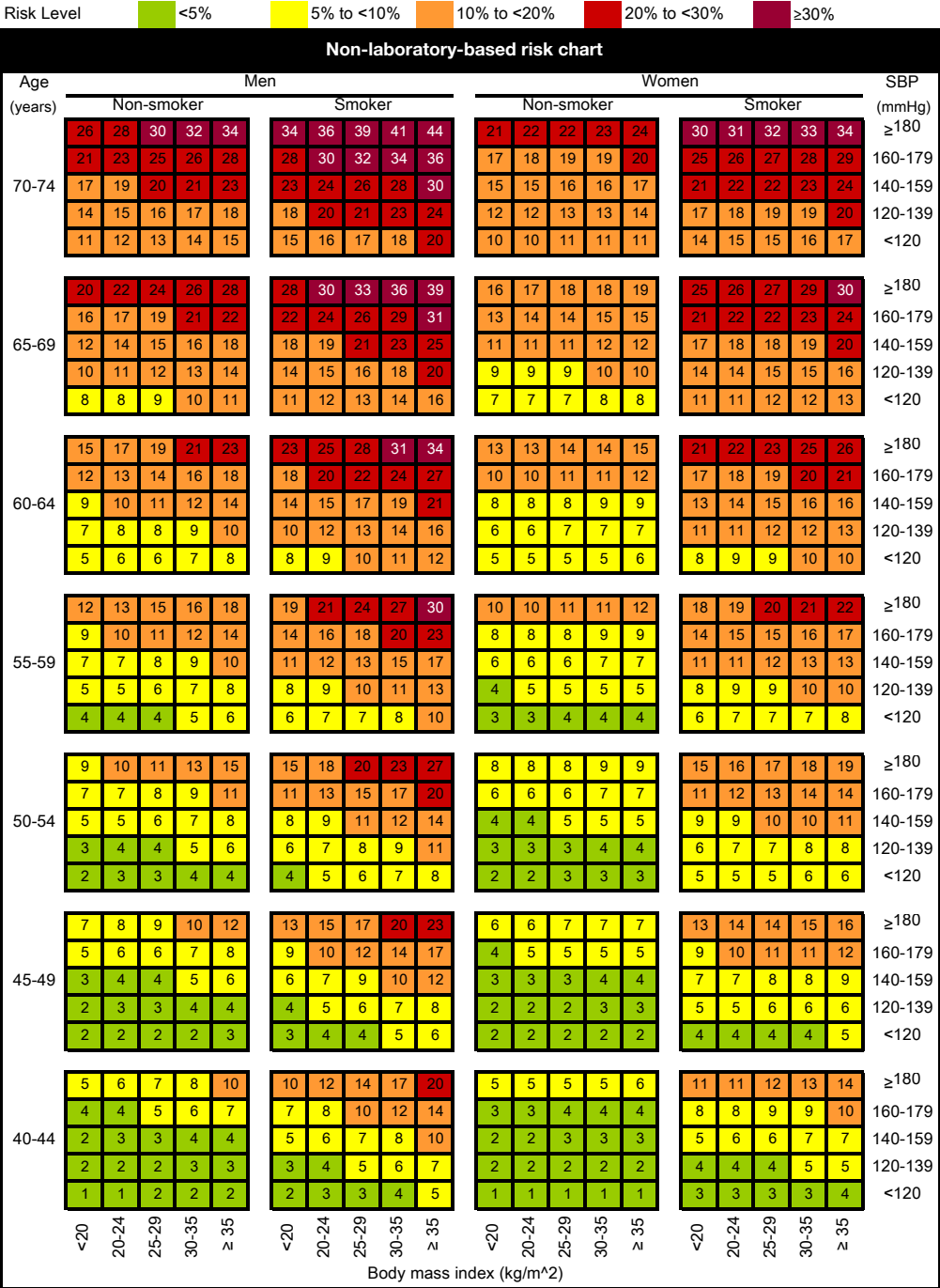


South Asia

WHO cardiovascular disease risk non-laboratory-based charts

South-East Asia

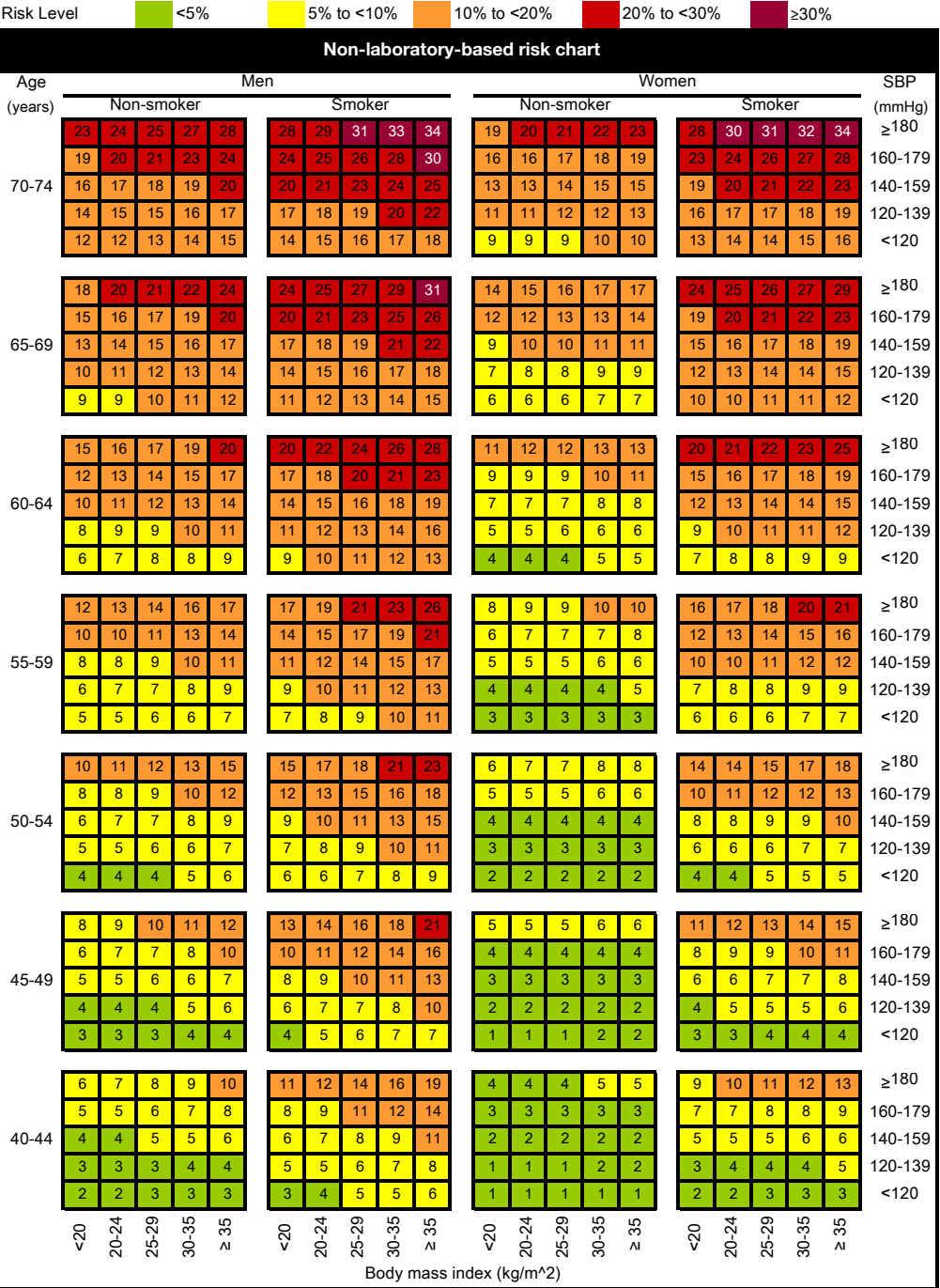
Cambodia, Indonesia, Lao People's Democratic Republic, Malaysia, Maldives, Mauritius, Myanmar, Philippines, Seychelles, Sri Lanka, Thailand, Timor-Leste, Viet Nam



South-East Asia

WHO cardiovascular disease risk non-laboratory-based charts

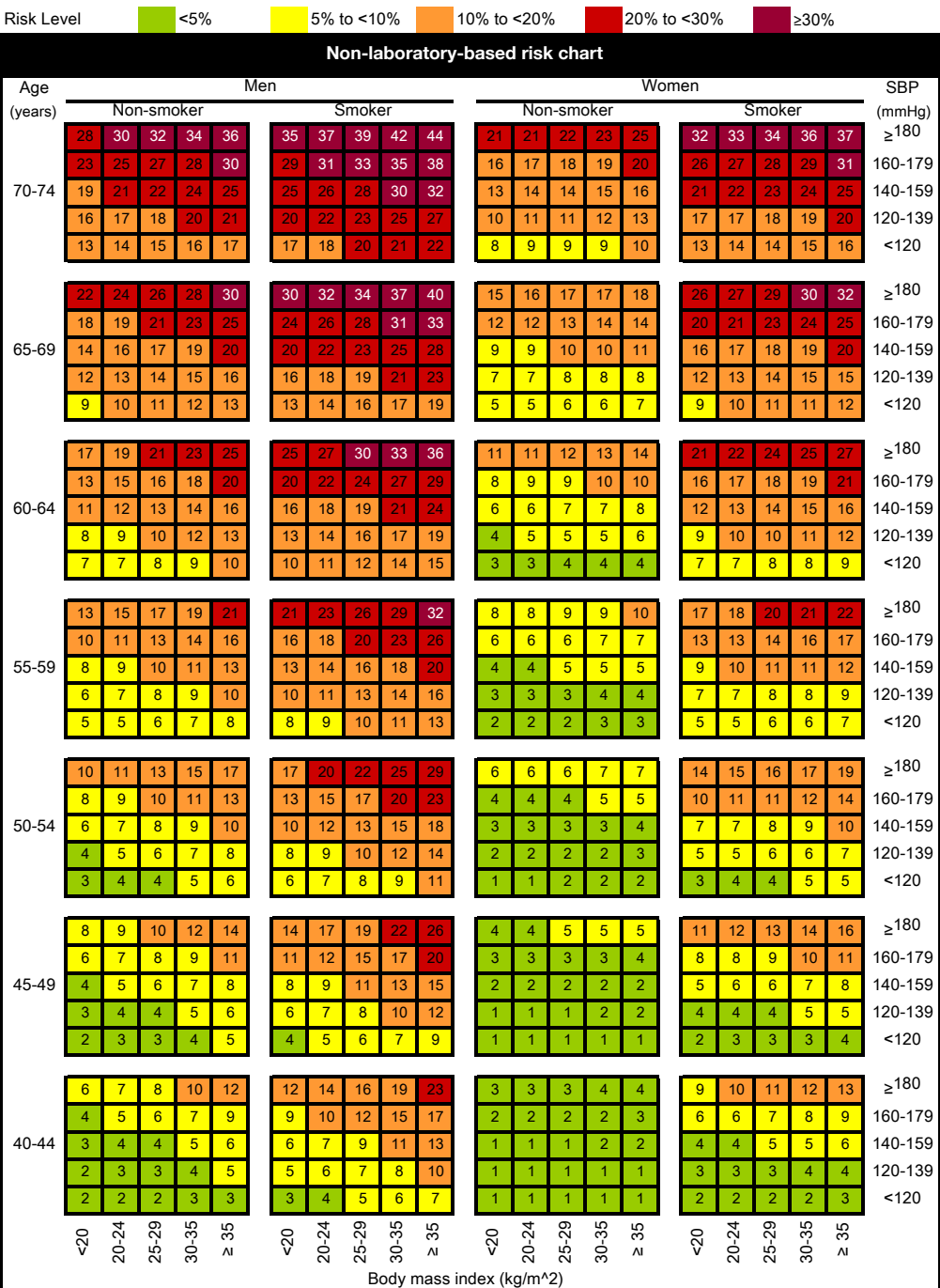
High-income Asia Pacific
Brunei Darussalam, Japan, Republic of Korea, Singapore



High-income Asia Pacific

WHO cardiovascular disease risk non-laboratory-based charts

Australasia
Australia, New Zealand

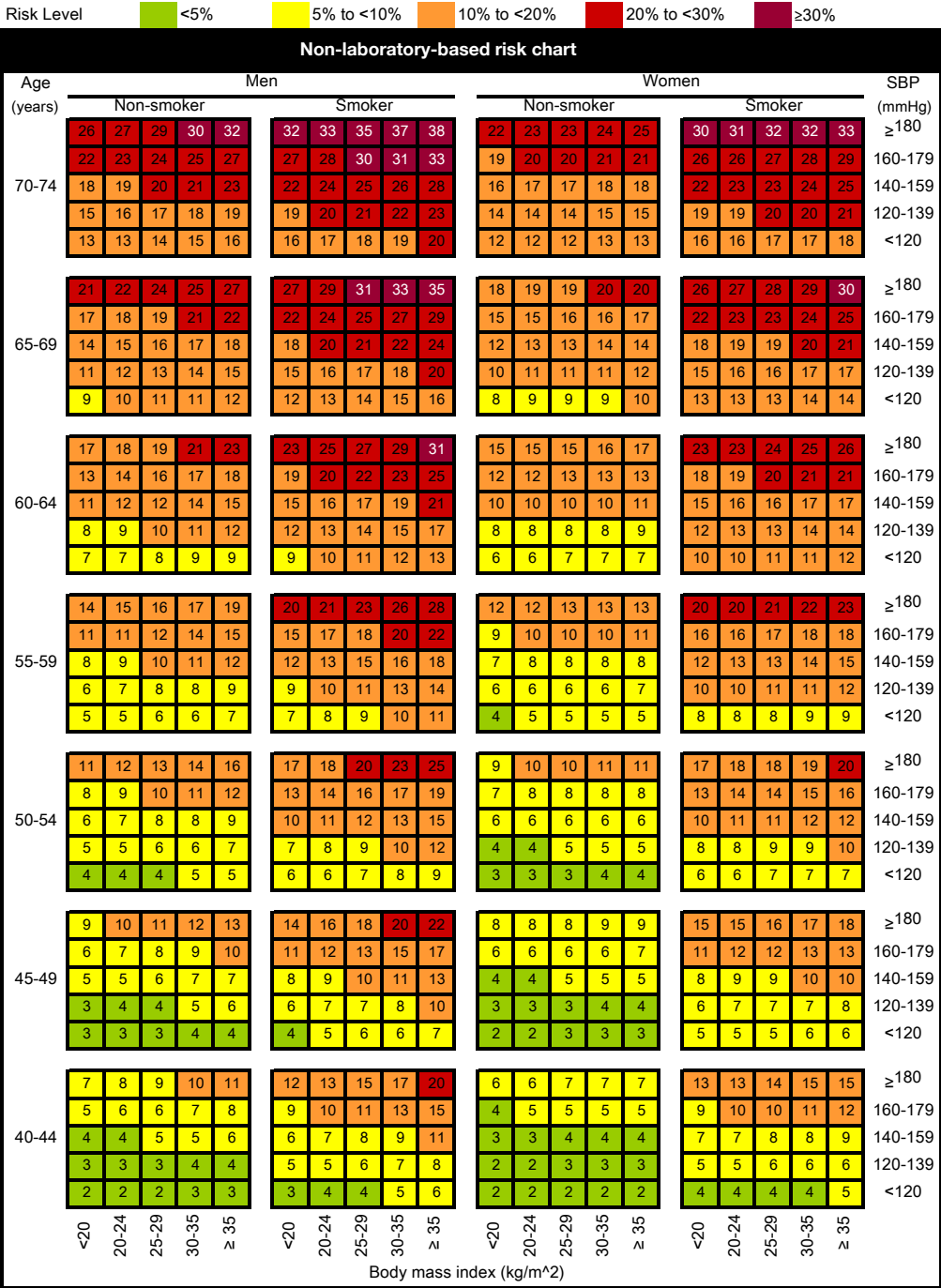


Australasia

WHO cardiovascular disease risk non-laboratory-based charts

Oceania

Fiji, Kiribati, Marshall Islands, Micronesia (Federated States of), Papua New Guinea, Samoa, Solomon Islands, Tonga, Vanuatu



Annex 4: Endpoint definitions used for incidence rates

Myocardial infarction

Case definition

Acute myocardial infarction (MI) – definite and possible MI according to the third universal definition of myocardial infarction:

- When there is clinical evidence of myocardial necrosis in a clinical setting consistent with myocardial ischaemia OR
- Detection of a rise and/or fall of cardiac biomarker values and with at least one of the following: i) symptoms of ischaemia, ii) new or presumed new ST-segment-T wave changes or new left bundle branch block, iii) development of pathological Q waves in the ECG, iv) imaging evidence of new loss of viable myocardium or new regional wall motion abnormality, or v) identification of an intracoronary thrombus by angiography or autopsy.
- Sudden (abrupt) unexplained cardiac death, involving cardiac arrest or no evidence of a noncoronary cause of death
- Prevalent MI is considered to last from the onset of the event to 28 days after the event and is divided into an acute phase (0–2 days) and subacute (3–28 days).

Stroke

Case definition

Stroke was defined according to WHO criteria – rapidly developing clinical signs of focal (at times global) disturbance of cerebral function lasting more than 24 hours or leading to death with no apparent cause other than that of vascular origin (20). Data on transient ischaemic attack (TIA) were not included.

- *Acute stroke*: stroke cases are considered acute from the day of incidence of a first-ever stroke through day 28 following the event.
- *Ischaemic stroke*: an episode of neurological dysfunction caused by focal cerebral, spinal, or retinal infarction.
- *Intracerebral haemorrhage*: a focal collection of blood within the brain parenchyma or ventricular system that is not caused by trauma.

Definitions are taken from GBD 2017 Disease and Injury Incidence and Prevalence Collaborators, Supplement 1 (20).

Annex 5a: Guide for comparison of CVD risk tools

Checklist: Cardiovascular disease risk prediction models in clinical practice		
From model derivation to implementation		Considerations
Derivation	Population	- Does the dataset used for model derivation contain information of a sufficient number of individuals (i.e. is the sample size large enough)? - Are the characteristics of individuals in the derivation sample sufficiently aligned with those of the target screening population to allow transferability of estimated risk ratios (e.g. age range, prior disease status)?
	Risk factors	- Are the relevant risk factors included? Note: main CVD risk factors (e.g. smoking, blood pressure) are key for risk prediction. - Are they accurately measured in sufficient numbers in the derivation sample? For example, use of sphygmomanometer for hypertension is more accurate than self-reported. - Can the risk factors be accurately measured in the target population?
	End points	- Does the model predict the relevant endpoint (e.g. overall CVD, stroke, CHD)? - Is endpoint collection in the derivation sample systematic and well-validated?
	Follow up	Is the follow-up time in the derivation sample sufficient to allow risk estimation over the timeframe of interest (usually 10 years)?
	Statistical Model	- Are appropriate statistical models used for the type of risk estimation? - Are relevant assumptions tested (e.g. proportional hazards assumption)?
	Internal validation	- Discrimination – is the model able to predict order of CVD event among individuals? - Calibration – is there good agreement between predicted and observed incidence (absolute risk)? - Has the model been checked for overfitting (relevant for smaller sample sizes and can be checked by cross-validation)? - Reclassification – is there appropriate movement of individuals through relevant risk categories when comparing alternative risk models?

Checklist: Cardiovascular disease risk prediction models in clinical practice		
From model derivation to implementation		Considerations
Recalibration	Recalibrated	Has the model been recalibrated for use in different populations?
	Recalibration data	Are the data used for recalibration appropriate? For example, do the recalibration data share the same characteristics as the target population?
	Recalibration methods	- Is there a methodological framework proposed / provided for future recalibration in response to changing trends with time as well as divergent CVD rates across regions / populations? For example, is a guide or statistical code provided for recalibration? - What is the ease of recalibration? - Are additional data needed for recalibration and is this available?
External validation		Is the model proven transferable to a new relevant setting / population other than used in the model derivation?
Usability	Format	Is the format appropriate to be used in the population to which the model is applied (e.g. online risk calculator, colour-coded charts)?
	Risk-factor measurement	Can all risk factors be feasibly measured and are alternative formats available for resource constraint settings (e.g. models using risk factors that don't require laboratory measurements)?
	Country settings	Is the model available for different country settings?
Implementation	Guidelines	Has the model been recommended for use by relevant guidelines?
	Health gains*	- Has the model been evaluated for health gains when used to assess total CVD risk and guide interventions (such as using statins) in high-risk populations? - Has use of the risk-prediction model resulted in significant health gains when used?
	Cost-effectiveness*	Has use of the risk-prediction model been shown to be cost-effective?

* Cost-effectiveness and health gains of (different) CVD risk models in clinical practice are highly dependent on, for example, the target population and the clinical interventions for different thresholds (e.g. statin allocation).

Table based on Cooney et al. (21), Rosello et al. (22), 2016 European Guidelines on CVD (23) and the TRIPOD guidelines (24). If of interest, the TRIPOD guidelines can be used for a more detailed checklist for statistical risk-prediction modelling (24).

mparison of CVD risk charts

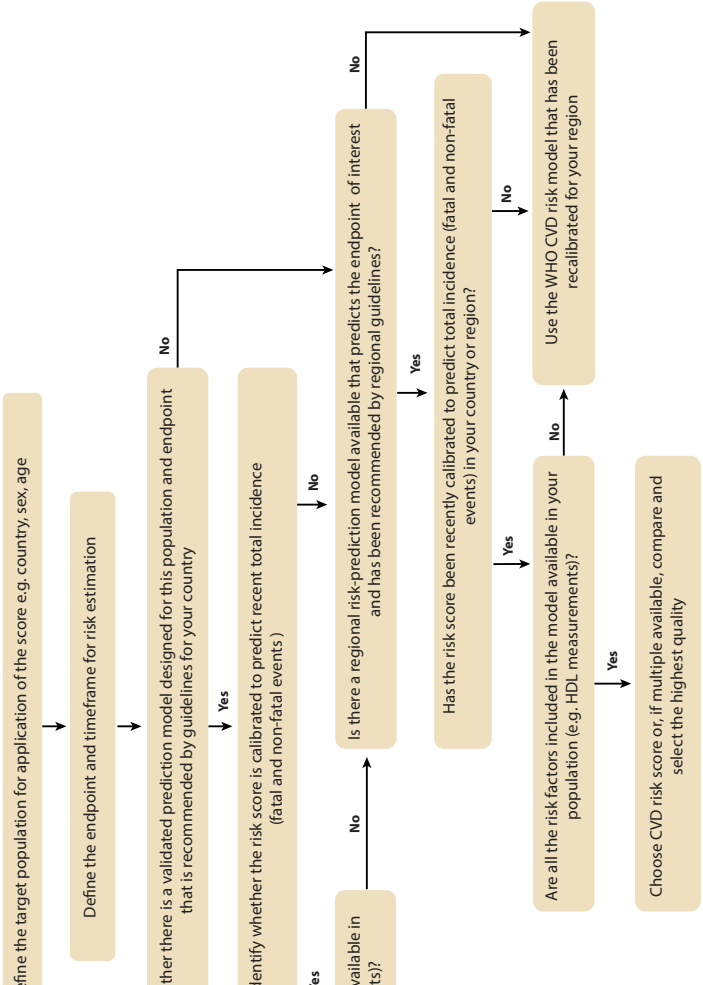
	WHO/ISH 2007	Globorisk 2015	IHMRS 2011	SCORE 2016
ign: ve cohorts (85) 177 individuals, ents ≥: 40–80 yrs aseline survey: 3 Europe, North Japan, Australia	<i>Study design:</i> No single derivation cohort. <i>Risk-factor distribution,</i> relative risks, CVD incidence from various sources combined. <i>Date of baseline survey:</i> Not applicable <i>Location:</i> Incidence, risk factors from 14 WHO regions	<i>Study design:</i> Prospective cohorts (8) with 50 129 individuals, 6042 events <i>Age range:</i> 40–84 yrs <i>Date of baseline survey:</i> 1948–1993 <i>Location:</i> North America	<i>Study design:</i> Case control study, 5349 cases 7423 controls <i>Median age:</i> 58 (49–67) yrs <i>Date of baseline survey:</i> 1999–2003 <i>Location:</i> 52 countries	<i>Study design:</i> Prospective cohorts (12) with 205 178 individuals, 7934 fatal events <i>Age range:</i> 40–65 yrs <i>Date of baseline survey:</i> 1967–1991 <i>Location:</i> Europe
smoking, SBP,	Age, sex, smoking, SBP, TC, DM	Age, sex, smoking, SBP, TC, DM	Age, sex, smoking, DM, HTN, Apolipoprotein B/A1 or TC/HDL-C ratios	Age, sex, smoking, SBP, TC or TC/HDL-C
SBP, smoking,	Sex, age, SBP, smoking, DM	Age, sex, smoking status, SBP, BMI	Age, sex, smoking, DM, HTN, family, diet, history, lifestyle, psychosocial factors	Not available
of fatal and VD troke)	10-yr risk of fatal and nonfatal CVD (CHD or stroke)	10-yr risk of fatal and nonfatal CVD (CHD or stroke)	Risk of fatal and nonfatal MI	10-yr risk of fatal CVD (CHD or stroke)
low-up in most	No actual follow-up, hypothetical 10-yr	>10-yr follow-up in 7 of 8 cohorts	No follow up, case- control design	>10-yr follow-up in all cohorts
val models	Combination of several relative and absolute risks in 'Cox model type' structure	Cox survival model	Unconditional logistic regression	Weibull survival models
ated internally	Not applicable	Well-validated internally	Well-validated internally	Well-validated internally

	WHO/ISH 2007	Globorisk 2015	IHMRS 2011	SCORE 2016
art of model ent; calibrated al regions	Integral part of construction; calibrated for 14 global regions	Integral part model development; calibrated for 187 countries	Not recalibrated for different settings	High/low risk charts provided; recalibrated post hoc in several countries
pecific s from GBD try-specific risk an NCD-RisC	Risk-factor distributions from WHO Comparative Risk Assessment study, Region-specific incidence from GBD	Used country-specific mortality rates and case-fatality to estimate incidence rates based on age trends in Swedish data; rates then modelled over past years and projected into future 10 years	Not applicable	Country-specific cohorts
amework and code provided, plied using available data	No specific method provided	Intuitive framework provided. Can be applied with routinely available data. Requires modelling if future projections of rates desired as used in original score	Not applicable since model do not predict absolute risk	No standardized approach recommended
ated in several ohorts	No adequate external validation	Well-validated in several external cohorts	Well-validated in several external cohorts	Well-validated in several external cohorts
ided charts and ode	Colour-coded charts	Colour-coded charts, online calculator	Calculation charts	Colour-coded charts and online calculator
	Yes	Yes	Yes	No
o risk	WHO/ISH	Globorisk		SCORE
harts for 21 regions.	Different charts for 14 worldwide regions	Different charts available per country.	-	Low-risk charts and high-risk charts, for grouped European countries
CVD prevention	WHO guidelines for CVD prevention 2007	-	-	ESC Guidelines 2019

	WHO/ISH 2007	Globorisk 2015	IHMRS 2011	SCORE 2016
iction charts for different ographic of recalibration with code to allow efficient oratory variant e validated in studies ibution in datasets ing 79 countries t in model were mostly ;	Risk-prediction charts provided for different ethnic-geographic regions A non-laboratory variant is available Absence of individual- level population data; models based on summary inputs No internal or external validation of the model in epidemiological cohorts	Risk-prediction charts provided for different ethnic-geographic regions Intuitive systematic recalibration approach A non-laboratory variant is available Externally validated in numerous studies	Included large number of women, youth and people from LMICs in derivation A non-laboratory variant is available Externally validated in numerous studies	Existence of low-risk and high-risk charts for European countries Externally validated in numerous studies
		Rates used in recalibration rely on many modelling steps, assumptions and projections Only USA-based data in model derivation	Case-control rather than cohort design used; may induce bias in RR estimates, prevents estimation of absolute risk Not available for different settings	Only estimates risk of fatal CVD No non-laboratory variant available Model derived only in European cohorts
: INTERHEART Modifiable Risk Score, WHO/ISH; World Health Organization / International Society of Hypertension; MI: myocardial blood pressure; HTN: hypertension; DM: diabetes mellitus; BMI: body mass index; GBD study: Global Burden of Disease Study; rtery disease; NCD Risk Factor Collaboration LMIC: Low and middle income countries				

27). Rossello et al, 2019 (22), 2016 European Guidelines on CVD (23), the TRIPOD guidelines (24), Mendis et al, 2007 (25), Hajifathalian
pean Society of Cardiology, SCORE risk charts (28).

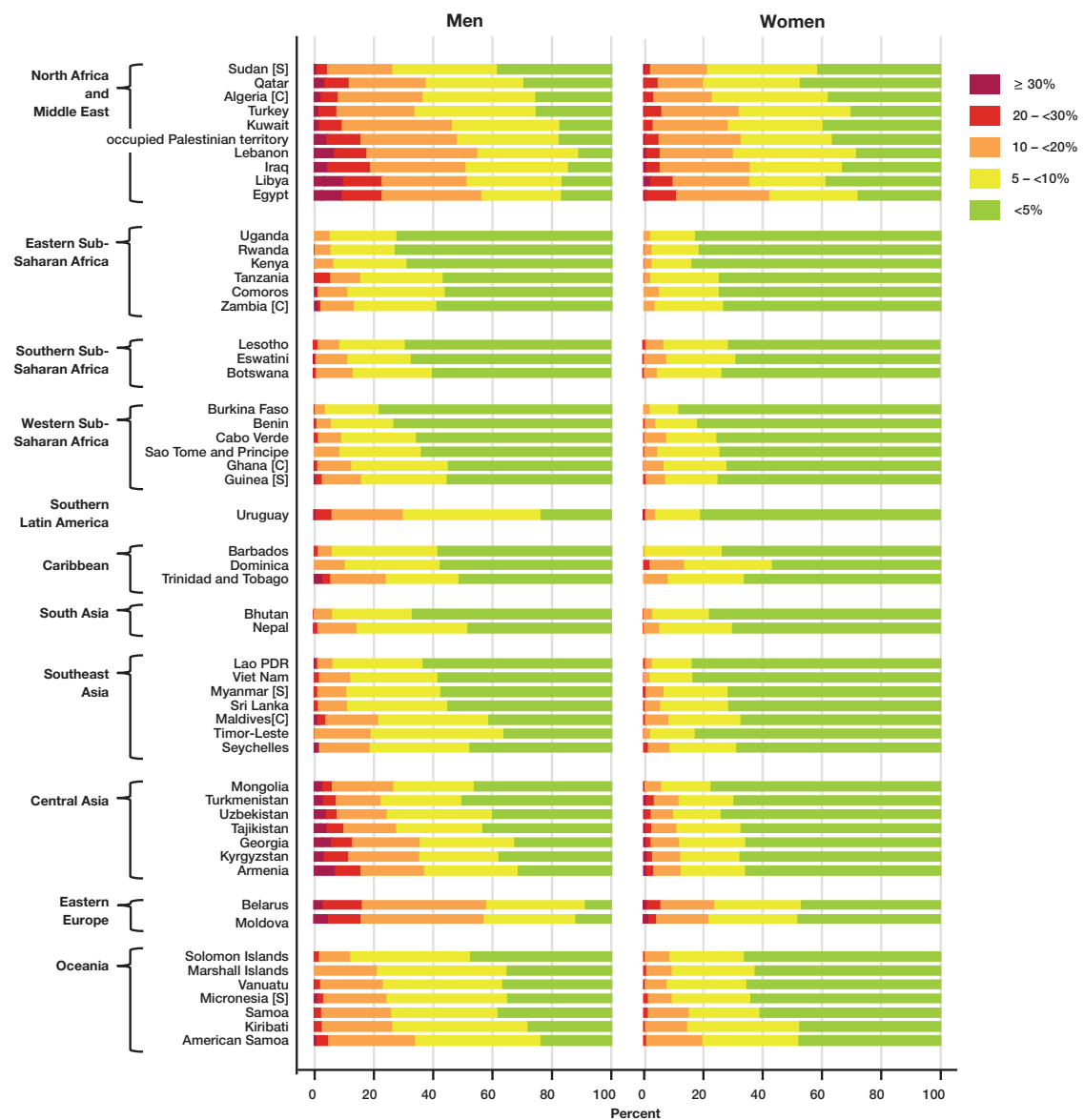
tifying a CVD risk prediction model



Annex 7: Considerations for initiation of therapy

The appropriate threshold of an individual's total risk at which intensive lifestyle interventions and drug treatment are initiated depends on the availability of resources as well as the risk level. The percentage of patients from representative countries in each region who would be eligible for treatment based on each cut-off level is presented in Figure 5 (3). As an example, <5% of the population of Uganda is at 10% or greater CVD risk, whereas approximately 50% of Egypt's population is above 10% CVD risk.

Figure 5: Distribution of 10-year CVD risk according to recalibrated laboratory-based WHO risk prediction models for individuals aged 40–64 years from example countries



Data from all countries are from adults aged 40–64 years with total cholesterol concentrations of 2.6–10.3 mmol/L and from samples representative of the national population, unless otherwise specified as subnational (S) or community based (C).

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Technical package for cardiovascular disease
management in primary health care

Tool for the development of a consensus protocol for treatment of hypertension



Contents

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Context

High blood pressure kills more people than any other condition – approximately 10 million people each year, more than all infectious diseases combined. Reducing blood pressure prevents stroke, heart attack, kidney damage, and other health problems.

An estimated 1.4 billion people worldwide have high blood pressure, but just 14% have it under control – that is roughly one person in seven. However, health providers in high- and low-income countries, urban and rural areas, and across different health systems show that a higher level of control can be achieved. Canada has reached nearly 70% control nationwide, and Barbados and Malawi have shown it is possible to increase control rates rapidly.

Health systems that are successful in supporting their patients to bring blood pressure under control do so by ensuring provision of a technical package with five key components:

1. treatment protocols that establish a standard dose- and drug-specific treatment of patients that is simple and practical yet provides sufficient detail, including specific medications and dosages and a schedule for titration or the addition of medications if blood pressure is uncontrolled
2. community-based care and task sharing so the health care workers who are most accessible to patients can provide care, including adjusting and intensifying medication regimens that follow doctor-directed protocols, allowing every member of the health care team to be optimally involved in supporting patient care
3. a regular and uninterrupted supply of quality-assured medications and equipment for accurate monitoring of blood pressure
4. patient-centered services that reduce barriers to adherence, including low-cost or free medical visits and medications; convenient medical visits and medication refills; once-daily treatment regimens with three-month refills for stable patients; the use of fewer tablets through combination medications; ready access to free blood pressure monitoring; and public education to increase awareness of the importance of controlling blood pressure
5. information systems that allow for real-time feedback on adherence and blood pressure control of individual patients, assessment of control rates by different treatment systems to strengthen tracking and accountability and facilitate continuous programme improvement.



The need for a standardized protocol

The creation and endorsement of a detailed and standardized treatment protocol is a critical first step in developing a successful large-scale hypertension programme. Standardized drug- and dose-specific treatment protocols have been shown to be superior to individualized treatment, and also facilitate the logistics of drug procurement, task sharing, staff training, data collection, and quality reporting.

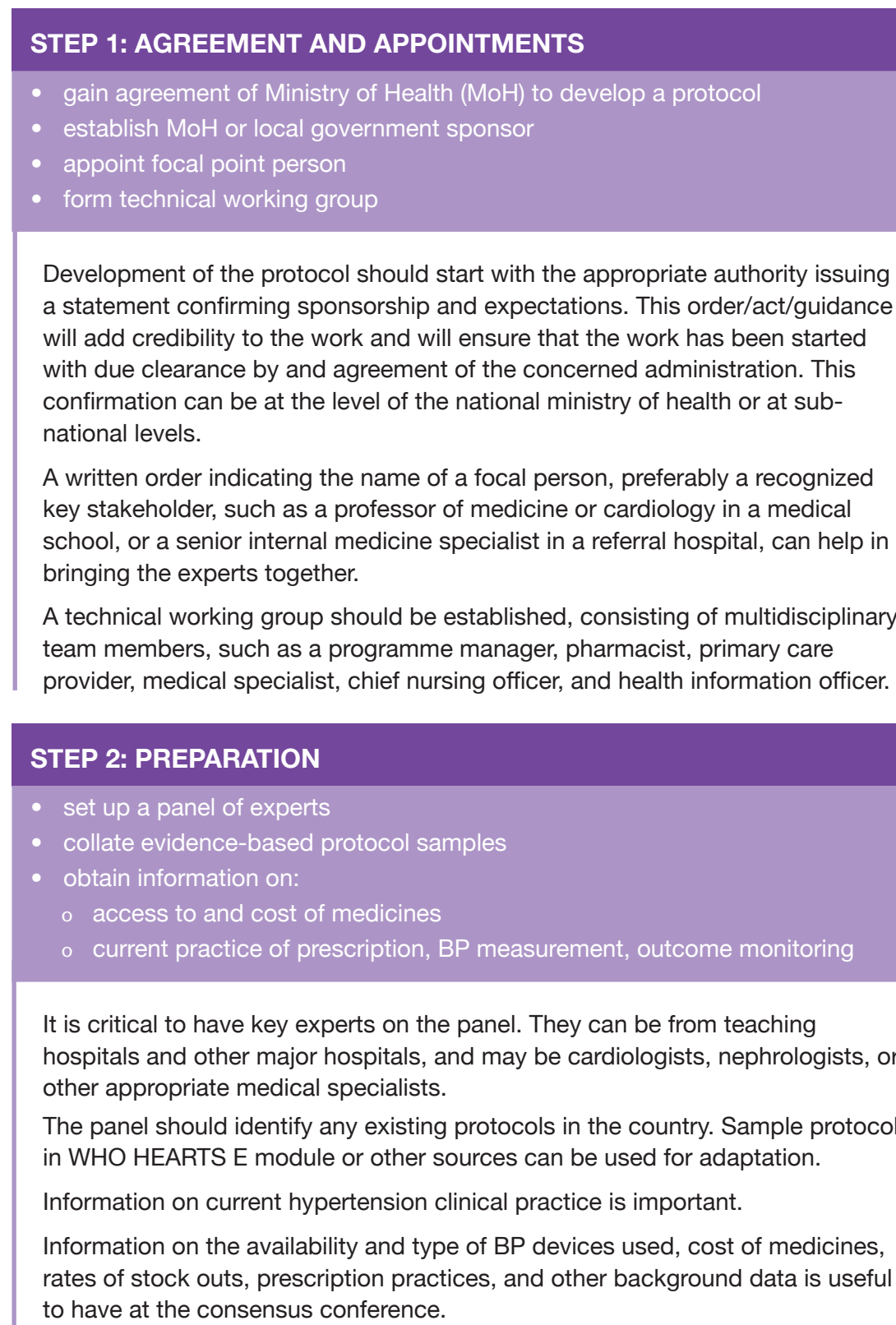
Effective protocols have:

- fewer branching and more linear pathways
- fewer options and more direction
- fewer drug classes and more specific drugs
- fewer drugs names and more drug doses
- fewer single drug pills and more fixed-dose combinations.

As more experience with treatment of hypertension is gained, it has become clear that the great majority of patients can be effectively treated with a single protocol.

There are many examples of evidence-based hypertension treatment protocols, such as those listed in the *WHO HEARTS technical package module Evidence based protocols*. Using these and other published protocols, with a focus on assessing the suitability for use in the local context, it is possible to develop a consensus protocol appropriate for their country or subnational area.

Steps for the development of a consensus protocol



STEP 3: APPROACH PAPER

- develop an approach paper and options with the technical working group and experts

The approach paper is a dossier with all the relevant materials and information. This should be developed by the technical working group. It should include:

- government order with timeline and expectations
- current status of BP measurement and management in the country/state
- national hypertension and other relevant protocols
- data describing the availability of medicines and cost
- sample protocols from E module and protocols from other relevant sources.

STEP 4: CONSENSUS MEETING

- invite participants
- facilitate discussion
- arrive at consensus

The meeting should be planned with sufficient notice to allow participation of all the relevant experts

An external facilitator can be helpful.

Consider including state/province/district health programme managers in the areas of: primary care, NCD treatment, and access to medications; medication procurement agency personnel working in stock control and distribution; academics and leading physicians; and healthcare providers from the facility.

Sessions that can be considered during the meeting are:

- current status of hypertension treatment including protocols
- presentation on algorithms for hypertension management
- group work – discussion of proposed protocols
- finalizing the protocol and reaching consensus.

If possible, it is preferable to have a meeting for two days, although if preparation is good a single day may suffice. Start with government stating the need for the protocols and the current status, on the basis of the approach paper. Consider:

- the benefits of a standard protocol
- current practice for hypertensive patients
- availability and costs of hypertensive medications
- exploration of current supply-chain mechanism for anti-hypertensives
- special considerations for the population of the country.

Consider establishing sub-groups on diagnosis, management of and access to medicines and technology.

Plan to agree on the protocol during the second day.

Considerations should include whether it is evidence based, any local issues such as availability and cost.

Arrive at agreement on a specific drug- and dose-specific protocol that is easy to follow.

STEP 5: DESIGN, ENDORSEMENT AND DISTRIBUTION OF PROTOCOL

- design a drug- and dose-specific protocol, taking into account:
 - the delivery platform
 - access to medicines
 - prescription rights
- obtain endorsement from the appropriate authority
- distribute to all relevant parties

Design and share the endorsed protocol. Ensure that appropriate government endorsement and other details are provided to ensure that it has been officially endorsed. Examples of consensus protocols developed in India are included in the Annex.

Develop a guide for implementation, with training needs, estimated quantity of required medicines, agreement on prescription privileges based on position for initiation and for refills, and definition of referral criteria.

Ensure wide dissemination such that the protocol reaches all appropriate staff, including central supply-store personnel, all doctors who are expected to provide services, nurses, and all health care providers who care for people with hypertension.

Provide as single-page, easy-to-use protocols.

Develop wall charts where needed and communicate widely.

Annex: Consensus protocols developed by states in India

Maharashtra

Hypertension Protocol

Measure blood pressure of **all adults** over 18 years

High BP: **SBP $\geq$ 140 or DBP $\geq$ 90 mmHg**

Check for compliance at each visit before titration of dose or addition of drugs



Step 1

If BP is high*

Prescribe Amlodipine 5 mg + adherence counseling

Step 2

After 30 days measure BP again. If still high:
Add Telmisartan 40mg**

Step 3

After 30 days measure BP again. If still high:
Increase Telmisartan to 80mg

Step 4

After 30 days measure BP again. If still high:
Increase Amlodipine to 10mg

Step 5

After 30 days measure BP again. If still high:
Add Chlorthalidone 6.25mg

Step 6

After 30 days measure BP again. If still high:
Increase Chlorthalidone to 12.5mg

...

After 30 days measure BP again. If still high:

Check that patient has been taking drugs regularly and correctly. If so, refer patient to a specialist.

Women who are or could become pregnant

▲ DO NOT give Telmisartan or Chlorthalidone.

- ACE inhibitors, angiotensin receptor blockers (ARBs), thiazide/thiazide-like diuretics and statins should not be given to pregnant women or to women of childbearing age not on highly effective contraception.

- Calcium channel blocker (CCB) can be used. If not controlled with intensification dose, refer to specialist.

Diabetic patients

- Treat diabetes according to protocol.
- Aim for BP target of <140/90.

Heart attack in last 3 years

- Add beta blocker to Amlodipine at initial treatment.

Heart attack or stroke ever

- Begin low-dose aspirin (75 mg) and statin.

Chronic kidney disease

- ACE inhibitor or ARB preferred if close clinical and biochemical monitoring possible after specialist opinion.

* If SBP 140-159 and/or DBP 90-99, start on lifestyle management for one month prior to initiation of medications.

If SBP $\geq$ 180 and/or DBP $\geq$ 110 start treatment and refer to specialist immediately.

Recommended investigations at initiation of therapy: CBC, blood sugar, serum creatinine, electrolytes (optional). If S creatinine $>$ 1.5 mg, refer to specialist.

** If Telmisartan not available: replace with Enalapril 5 mg (initiation dose) and 10 mg (intensification dose).

Lifestyle advice for all patients



Eat less than 1 tsp of salt per day: avoid papads, chips, chutneys, dips, pickles, etc.



If overweight, lose weight.

- Limit intake of fried foods.

- Avoid foods with high amounts of saturated fats (e.g. cheese, ice cream, fatty meat).

- Avoid processed foods containing trans fats.

- Avoid added sugar.

- Eat 5 servings of fruits and vegetables per day.

- Use healthy oils: polyunsaturated and monounsaturated oils.

- Reduce fat intake by changing how you cook: remove the fatty part of meat; use vegetable oil; boil, steam or bake rather than fry; limit reuse of oil for frying.



Exercise regularly: 2.5 hours per week



Avoid alcohol and tobacco



India Hypertension Management Initiative: Maharashtra 1.00-9-18

Punjab

Hypertension Protocol

Measure blood pressure of **all adults** over 18 years

High BP: **SBP $\geq$ 140 or DBP $\geq$ 90 mmHg**



Step 1

If BP is high:*

Prescribe Amlodipine 5mg

Step 2

After 30 days[#] measure BP again. If still high:
Increase to Amlodipine 10mg

Step 3

After 30 days[#] measure BP again. If still high:
Add Telmisartan 40mg

Step 4

After 30 days[#] measure BP again. If still high:
Increase to Telmisartan 80mg

Step 5

After 30 days[#] measure BP again. If still high:
Add Chlorthalidone 12.5mg**

Step 6

After 30 days[#] measure BP again. If still high:
Increase to Chlorthalidone 25mg**

...

After 30 days measure BP again. If still high:

Check if the patient has been taking medications regularly and correctly. If yes, refer to a specialist.

Pregnant women and women who may become pregnant

▲ DO NOT give Telmisartan or Chlorthalidone.

- Statins, ACE inhibitors, angiotensin receptor blockers (ARBs), and thiazide/thiazide-like diuretics should not be given to pregnant women or to women of childbearing age not on effective contraception.
- Calcium channel blocker (CCB) can be used. If not controlled with intensification dose, refer to a specialist.

Diabetic patients

- Treat diabetes according to protocol.
- Aim for a BP target of < 140/90 mmHg.

Heart attack in last 3 years

- Add beta blocker to Amlodipine with initial treatment.

Heart attack or stroke, ever

- Begin low-dose aspirin (75mg) and statin.

People with high CVD risk

- Consider aspirin and statin.

Chronic kidney disease

- ACEI or ARB preferred if close clinical and biochemical monitoring is possible.

* If SBP $\geq$ 180 or DBP $\geq$ 110, refer patient to a specialist after starting treatment.

If SBP 160-179 or DBP 100-109, start treatment on the same day.

If SBP 140-159 or DBP 90-99, check on a different day and if still elevated, start treatment.

Dose of anti-hypertension medications can be titrated at 15 days frequency if required.

** Hydrochlorothiazide can be used if Chlorthalidone is not available (25 mg starting dose, 50 mg intensification dose).

Lifestyle advice for all patients



Avoid tobacco and alcohol



Exercise 2.5 hours/week



Reduce weight, if overweight



Reduce salt, under 1 tsp/day



Eat less fried foods

- Eat 5 servings of fruits and vegetables per day.

- Avoid papads, chips, chutneys, dips, and pickles.

- Use healthy oils like sunflower, mustard, or groundnut.

- Limit consumption of foods containing high amounts of saturated fats.

- Reduce fat intake by changing how you cook:

- Remove the fatty part of meat

- Use vegetable oil

- Boil, steam, or bake instead of fry

- Limit reuse of oil for frying

- Avoid processed foods containing trans fats.

- Avoid added sugar.

India Hypertension Management Initiative: Punjab 1.00-5-15

Resources

Technical package for cardiovascular disease management in primary health care. Evidence-based treatment protocols <https://apps.who.int/iris/bitstream/handle/10665/260421/WHO-NMH-NVI-18.2-eng.pdf;jsessionid=1668E30A538A5066985C863BE6CB5973?sequence=1>

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Guideline for the pharmacological treatment of hypertension in adults



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Acronyms and abbreviations

ACE1	angiotensin-converting enzyme 1
ACE2	angiotensin-converting enzyme 2
ACEi	angiotensin-converting enzyme inhibitor
ARB	angiotensin-II-receptor blocker
BB	beta-blocker
BP	blood pressure
CAD	coronary artery disease
CCB	calcium channel blocker
CKD	chronic kidney disease
CRE	WHO Office of Compliance, Risk Management and Ethics
CV	cardiovascular
CVD	cardiovascular disease
DBP	diastolic blood pressure
DM	diabetes mellitus
DOI	declaration of interest
ECG	electrocardiogram
EML	Essential Medicines List
ERG	External Review Group
ESRD	end-stage renal disease
GDG	Guideline Development Group
GRADE	Grading of Recommendations Assessment, Development and Evaluation
HCW	health care worker (nonphysician)
HIC	high-income country
HTN	hypertension
LIC	low-income country
LMIC	low- and middle-income country
LVH	left ventricular hypertrophy
MACE	major adverse cardiovascular event
MI	myocardial infarction
MIC	middle-income country
NCD	noncommunicable disease
PEN	WHO package of essential NCD interventions
PICO	population intervention comparator outcome
QALY	quality-adjusted life year
RAAS	renin-angiotensin-aldosterone system
SBP	systolic blood pressure

Executive summary

More people die each year from cardiovascular diseases than from any other cause. Over three quarters of heart disease and stroke-related deaths occur in low-income and middle-income countries. Hypertension – or elevated blood pressure – is a serious medical condition that significantly increases the risk of heart, brain, kidney and other diseases. Hypertension can be defined using specific systolic and diastolic blood pressure levels or reported use of antihypertensive medications. An estimated 1.4 billion people worldwide have high blood pressure, but just 14% have it under control. However, cost-effective treatment options do exist.

In this guideline, the World Health Organization (WHO) provides the most current and relevant evidence-based global public health guidance on the initiation of treatment with pharmacological agents for hypertension in adults. The recommendations target adult, non-pregnant patients who were appropriately diagnosed with hypertension and counselled about life-style modifications.

The guideline provides new recommendations on the threshold for the initiation of pharmacological treatment for hypertension, as well as recommendations on intervals for follow up, target blood pressure to be achieved for control, and the cadre of health care workers who may initiate treatment. The guideline provides the basis for deciding whether to initiate treatment with monotherapy, dual therapy or single-pill combinations, as well as guidance for countries selecting medicines and algorithms for hypertension control for their national guidelines for hypertension management.

The guideline was developed in accordance with the WHO Handbook for Guideline Development. In brief, the WHO Steering Group, in collaboration with the Guideline Development Group, developed key questions and rated outcomes to identify those critical for the development of the guideline. Conflicts of interest were handled in line with the current Compliance, Risk Management and Ethics (CRE) policy and all members of the GDG were asked to fill in the standard WHO Declaration of Interest (DOI) forms, which were reviewed. An overview of systematic reviews of the evidence was used to build summary of findings tables according to the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) approach. The Guideline Development Group developed recommendations, considering the certainty of the evidence; the balance between desirable and undesirable effects; resource requirements and cost-effectiveness; health equity; acceptability, patient values and preferences, and feasibility.

Recommendations

1. RECOMMENDATION ON BLOOD PRESSURE THRESHOLD FOR INITIATION OF PHARMACOLOGICAL TREATMENT
WHO recommends initiation of pharmacological antihypertensive treatment of individuals with a confirmed diagnosis of hypertension and systolic blood pressure of ≥140 mmHg or diastolic blood pressure of ≥90 mmHg.
<i>Strong recommendation, moderate- to high-certainty evidence</i>
WHO recommends pharmacological antihypertensive treatment of individuals with existing cardiovascular disease and systolic blood pressure of 130–139 mmHg.
<i>Strong recommendation, moderate- to high-certainty evidence</i>

WHO suggests pharmacological antihypertensive treatment of individuals without cardiovascular disease but with high cardiovascular risk, diabetes mellitus, or chronic kidney disease, and systolic blood pressure of 130–139 mmHg.
<i>Conditional recommendation, moderate- to high-certainty evidence</i>
2. RECOMMENDATION ON LABORATORY TESTING
When starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment.
<i>Conditional recommendation, low-certainty evidence</i>
3. RECOMMENDATION ON CARDIOVASCULAR DISEASE RISK ASSESSMENT
WHO suggests cardiovascular disease risk assessment at or after the initiation of pharmacological treatment for hypertension, but only where this is feasible and does not delay treatment.
<i>Conditional recommendation, low-certainty evidence</i>
4. RECOMMENDATION ON DRUG CLASSES TO BE USED AS FIRST-LINE AGENTS
For adults with hypertension requiring pharmacological treatment, WHO recommends the use of drugs from any of the following three classes of pharmacological antihypertensive medications as an initial treatment: 1. thiazide and thiazide-like agents 2. angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs) 3. long-acting dihydropyridine calcium channel blockers (CCBs).
<i>Strong recommendation, high-certainty evidence</i>
5. RECOMMENDATION ON COMBINATION THERAPY
For adults with hypertension requiring pharmacological treatment, WHO suggests combination therapy, preferably with a single-pill combination (to improve adherence and persistence), as an initial treatment. Antihypertensive medications used in combination therapy should be chosen from the following three drug classes: diuretics (thiazide or thiazide-like), angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs), and long-acting dihydropyridine calcium channel blockers (CCBs).
<i>Conditional recommendation, moderate-certainty evidence</i>
6. RECOMMENDATIONS ON TARGET BLOOD PRESSURE
WHO recommends a target blood pressure treatment goal of <140/90 mmHg in all patients with hypertension without comorbidities.
<i>Strong recommendation, moderate-certainty evidence</i>

WHO recommends a target systolic blood pressure treatment goal of <130 mmHg in patients with hypertension and known cardiovascular disease (CVD).
<i>Strong recommendation, moderate-certainty evidence</i>
WHO suggests a target systolic blood pressure treatment goal of <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease).
<i>Conditional recommendation, moderate-certainty evidence</i>
7. RECOMMENDATIONS ON FREQUENCY OF ASSESSMENT
WHO suggests a monthly follow up after initiation or a change in antihypertensive medications until patients reach target.
<i>Conditional recommendation, low-certainty evidence</i>
WHO suggests a follow up every 3–6 months for patients whose blood pressure is under control.
<i>Conditional recommendation, low-certainty evidence</i>
8. RECOMMENDATION ON TREATMENT BY NONPHYSICIAN PROFESSIONALS
WHO suggests that pharmacological treatment of hypertension can be provided by nonphysician professionals such as pharmacists and nurses, as long as the following conditions are met: proper training, prescribing authority, specific management protocols and physician oversight.
<i>Conditional recommendation, low-certainty evidence</i>

1 Introduction

More people die each year from cardiovascular diseases than from any other cause. Over three quarters of heart disease and stroke-related deaths occur in low-income and middle-income countries (1). Blood pressure is the force exerted by circulating blood against the walls of the body's arteries, the major blood vessels in the body. Blood pressure is written as two numbers. The first number (systolic) represents the pressure in blood vessels when the heart contracts or beats. The second number (diastolic) represents the pressure in the vessels when the heart rests between beats. Hypertension – or elevated blood pressure – is a serious medical condition that significantly increases the risk of diseases of the heart, brain, kidneys and other organs (2). Hypertension can be defined using specific systolic and diastolic blood pressure levels or reported use of antihypertensive medications. An estimated 1.4 billion people worldwide have high blood pressure, but just 14% have it under control (2). However, cost-effective treatment options do exist.

A World Health Organization (WHO) guideline dealing specifically with raised blood pressure was last published more than 20 years ago – in 1999 – and is now outdated. In 2007, a comprehensive guideline on cardiovascular risk included some recommendations on hypertension but this also needs revision and updating in the light of new evidence and practice (3). Guidance is particularly needed on some controversial issues, such as when to start treatment and whether laboratory testing and cardiovascular risk assessment are needed prior to starting treatment. In the past decade, WHO has included diagnosis and management of hypertension in a total cardiovascular risk approach as part of the WHO package of essential noncommunicable disease (NCD) Interventions (WHO PEN) 2007, 2010 and 2013. However, this approach has not included the most recent advances in pharmacological treatment.

The WHO Essential Medicines List (EML) identifies all classes of antihypertensive drugs – angiotensin-converting enzyme inhibitors (ACEi), calcium channel blockers (CCB), angiotensin receptor blockers (ARB) and thiazide diuretics – as essential. In June 2019, the EML included single-pill combination medications for hypertension. This further supports the evaluation of all classes of antihypertensive drugs as well as single-pill combinations in this current guideline.

Scope and objectives of the hypertension guideline

The 2021 WHO hypertension guideline aims to provide the most current and relevant evidence-based global public health guidance on the initiation of treatment (with pharmacological agents) for hypertension in adults. The recommendations target the general adult, non-pregnant, hypertensive population.

Although several countries and professional societies have guidelines on the topic of hypertension, these are specific to the population of that particular country or the specific setting or constituency of the professional society. Recent shifts in hypertension management, such as moving away from using beta-blockers as a first-line agent or the increased research and adoption of combination therapies and single-pill combinations, are all additional reasons for new guidance. The *Guideline for the pharmacological treatment of hypertension in adults* will be the first global guideline in the past two decades on the topic and will have specific relevance to low- and middle-income countries (LMICs).

The guideline provides new recommendations on the threshold for the initiation of pharmacological treatment for hypertension, recommendations on intervals for follow up, target blood pressure to be achieved for control, and the cadre of health care workers who may initiate treatment. It provides the basis for deciding whether to initiate treatment with monotherapy, dual therapy, or single-pill combination, as well as guidance for countries on selecting medicines for hypertension control in their national guidelines for hypertension management.

This guideline will replace the guidance in the modules *Evidence-based protocols* and *Risk-based CVD management* of the HEARTS technical package, as well as the guidance in the WHO PEN package regarding the threshold for initiation of treatment, and for the preferred pharmacological treatment for hypertension.

The guideline does not address measurement of blood pressure or diagnosis of hypertension. It addresses pharmacotherapy in individuals with a “confirmed” diagnosis of hypertension, such as hypertension diagnosed when blood pressure is found to be elevated on two different days.

Although this guideline does not address modifiable risk factors for hypertension such as unhealthy diet, physical inactivity, consumption of tobacco and alcohol, and being overweight or obese, a comprehensive treatment plan for hypertension must include addressing these risk factors through life-style modifications and other interventions (2).

Nonpharmacological approaches to treatment or prevention of hypertension include:

- reducing salt intake (to less than 5g daily)
- eating more fruit and vegetables
- being physically active on a regular basis
- avoiding use of tobacco
- reducing alcohol consumption
- limiting the intake of foods high in saturated fats
- eliminating/reducing trans fats in diet (2).

The guideline does not address hypertensive crisis as it is focused on chronic blood pressure management in regular care settings.

The objectives of the hypertension guideline are to:

- provide a blood pressure threshold for the initiation of treatment for hypertension;
- determine if laboratory tests or cardiovascular risk assessment are required prior to initiation of treatment for hypertension;
- determine the pharmacological agents with which to initiate treatment;
- determine the need to initiate treatment with monotherapy, dual therapy, or single-pill combinations;
- provide targets for blood pressure control in hypertension;
- provide intervals for follow up for patients with hypertension; and
- determine how nonphysician health care workers can participate in the management of hypertension.

Target audience

Primary audience

Clinicians/health care providers at all levels of health care.

Secondary audience

National NCD/CVD programme managers, health care academics, policy-makers setting practice recommendations, students, and hypertension medicine manufacturers.

2 Method for developing the guideline

2.1 Guideline contributors

In order to develop the hypertension guideline, WHO established three groups:

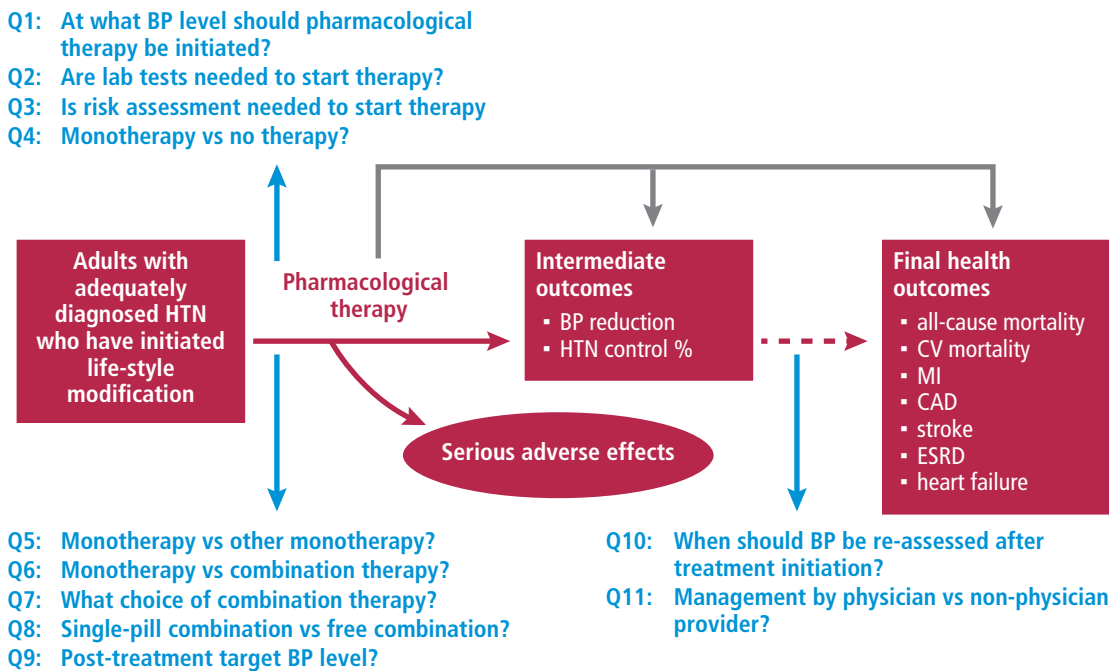
- 1. An internal WHO Steering Group to coordinate the guideline development process.
- 2. A Guideline Development Group (GDG), composed of hypertension (HTN) physicians, nephrologists, cardiologists, pharmacists, nurses, researchers, academics, policy-makers and representatives of patient groups, to review the evidence and develop recommendations. WHO selected the members of the GDG based on relevant expertise but it also considered appropriate representation by region and sex.
- 3. An External Review Group (ERG), composed of technical experts, representatives of HTN patient groups and ministries of health from low-resource countries, to provide peer review of the guideline and ensure recommendations are aligned with current global needs.

Annex 1 lists the contributors in each group. Annex 2 describes the process for declaring and managing conflicts of interest.

2.2 Analytical framework and PICO

An initial GDG meeting was held in Geneva in July 2019 to determine the scope and PICO (population, intervention, comparison, outcome) questions of the guideline. The GDG first developed an analytical framework (Fig. 1) that demonstrates the impact of interventions on intermediate and final outcomes and displays the order of the key questions to better visualize and place them along the patient-flow pathway. Following this, and a preliminary scoping review and discussion between the steering group and methodologist, PICO questions were developed. These were considered, deliberated on, further refined and voted on during the meeting.

Fig. 1 Analytic framework for antihypertensive medication treatment



METHOD FOR DEVELOPING THE GUIDELINE

2.3 Outcome importance rating

Members of the WHO Steering Group, in consultation with the GDG and methodologist, developed a list of treatment outcomes most relevant to the care of individuals with HTN. The GDG then rated each outcome on a scale from 1 to 9 and indicated whether it considered each outcome critical (rated 7–9), important (rated 4–6) or not important (rated 1–3) for decision-making. The mean scores are provided in Annex 3.

2.4 Reviews of evidence

The WHO Steering Group, with the participation of the GDG, determined the scope of the guideline and identified eleven questions in the format of population, intervention, comparison, and outcomes (PICO) to guide the search for systematic reviews (Annex 4). Eleven overviews of reviews informed the guideline development process. A systematic search was carried out in PubMed, Embase, The Cochrane Library, and Epistemonikos to identify existing systematic reviews that answered the PICO questions published in 2015 or after. Suitable systematic reviews were then evaluated based on the following criteria:

- their methodology as appraised by the AMSTAR (Assessing the Methodological Quality of Systematic Reviews) tool;
- how directly they matched the PICO questions;
- whether they reported sufficient information to allow for an assessment of the certainty of the evidence (e.g. tables with characteristics of included studies, risk of bias assessments at the study level, results of meta analyses in forest plots);
- whether they reported evidence in subgroups of interest (e.g. patients with diabetes mellitus (DM), cardiovascular disease (CVD), chronic kidney disease (CKD) etc); and
- the date of the most recent review to ensure the most updated evidence was used.

The Systematic Review Team prioritized the most useful reviews for each question, comparison, outcome and subpopulation within questions, and included as many reviews as necessary to address each question comprehensively. (Web Annex A provides details of the search terms and strategy used.) No reviews were updated.

Two questions (PICO 2 and 10, see Annex 4) were not addressed in an existing systematic review, and evidence from primary studies was therefore used. In this case, the Systematic Review Team reviewed the list of studies used in existing guidelines, queried the GDG and content experts, and searched for primary studies.

A total of 159 systematic reviews and 17 additional primary studies were included. Most of these were traditional systematic reviews in which the authors conducted pairwise meta-analysis, whereas nine analysed individual patient data. The Systematic Review Team also identified several published network meta-analyses. (See Web Annex A for full details.)

Most of the included reviews were found to be of high certainty when assessed using the AMSTAR tool. Details of the selection process, and the reviews and studies included for each PICO question, are presented in Web Annex A.

Another overview of reviews was conducted to identify systematic reviews to inform other decision criteria in the evidence-to-decision framework, including people's values, resources, acceptability, feasibility and equity, presented in Web Annex A. This review was enriched by primary studies identified by GDG members. This review focused on evidence informing HTN management in low- and middle-income settings.

2.5 Certainty of evidence and strength of recommendations

The GDG rated the certainty of evidence and developed the recommendations using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach (4). When making recommendations, GRADE defines the certainty of a body of evidence as “the extent of our confidence that the estimates of an effect are adequate to support a particular decision or recommendation” (5).

Members of the GDG, with the help of the methodologist, developed evidence profiles to summarize relative and absolute estimates of effects, and an assessment of the certainty of the evidence. One evidence profile for each comparison within a PICO question was constructed, using the online Guideline Development Tool GRADEpro (<https://gradepr.org>). When there were systematic reviews addressing the relative effects in specific subpopulations, separate evidence profiles were constructed for each subpopulation.

According to the GRADE approach, the certainty of the evidence can be high, moderate, low, or very low. Bodies of evidence from randomized controlled trials – which comprised almost the totality of those included in this guideline – start the assessment as high-certainty evidence but can be down-rated due to considerations of risk of bias, inconsistency, imprecision, indirectness, and publication bias. Table 1 presents an explanation of the different levels of certainty of the evidence (5).

Table 1 Certainty of evidence and its implications

Certainty level	Definition
High	We are very confident that the true effect lies close to that of the estimate of effect.
Moderate	We are moderately confident in the effect estimate. (The true effect is likely to be close to the estimate of effect, but there is a possibility that it is substantially different.)
Low	Our confidence in the effect estimate is limited. (The true effect may be substantially different from the estimate of the effect.)
Very low	We have very little confidence in the effect estimate. (The true effect is likely to be substantially different from the estimate of effect.)

The strength of the recommendations reflects the degree of confidence of the GDG that the desirable effects (e.g. beneficial health outcomes) of the recommendations outweigh the undesirable effects (e.g. adverse effects). The strength of recommendations in this guideline was graded into two categories:

A strong recommendation is one for which the GDG was confident that the desirable effects of adhering to the recommendation outweigh the undesirable effects.

A weak or conditional recommendation is one for which the GDG concluded that the desirable effects of adhering to the recommendation probably outweigh the undesirable effects, but was not confident about these trade-offs.

2.6 Deciding upon recommendations

The GDG met virtually for four sessions in February 2021. Systematic reviews and GRADE tables were presented at the meeting. Formulation of recommendations and their strength were facilitated by the chair and supported by the methodologist, with the aim of achieving unanimous consensus. While the plan was to use a simple majority vote, full consensus was reached on all recommendations.

The GDG used evidence-to-decision tables to guide the process of developing recommendations. These tables addressed the certainty of evidence, the balance between desirable and undesirable effects, values, resource use and cost-effectiveness, equity, acceptability and feasibility. These tables are available in Web Annex B, and the full evidence profiles in Web Annex A. The WHO Steering Group noted remarks made by members of the External Review Group and considered incorporating these into the final guideline. Some evidence-to-decision frameworks were consolidated to provide recommendations that are more practical and implementable from an end-user perspective; hence, the 11 questions led to eight recommendations.

2.7 Funding

The development of this guideline was financially supported by the US Centers for Disease Control and Prevention and the World Health Organization.

3 Recommendations

3.1 Blood pressure threshold for initiation of pharmacological treatment

1. RECOMMENDATION ON BLOOD PRESSURE THRESHOLD FOR INITIATION OF PHARMACOLOGICAL TREATMENT
WHO recommends initiation of pharmacological antihypertensive treatment of individuals with a confirmed diagnosis of hypertension and systolic blood pressure of ≥140 mmHg or diastolic blood pressure of ≥90 mmHg.
Strong recommendation, moderate- to high-certainty evidence
WHO recommends pharmacological antihypertensive treatment of individuals with existing cardiovascular disease and systolic blood pressure of 130–139 mmHg.
Strong recommendation, moderate- to high-certainty evidence
WHO suggests pharmacological antihypertensive treatment of individuals without cardiovascular disease but with high cardiovascular risk, diabetes mellitus, or chronic kidney disease, and systolic blood pressure of 130–139 mmHg.
Conditional recommendation, moderate- to high-certainty evidence
Implementation remarks: Initiation of pharmacological hypertension (HTN) treatment should start no later than four weeks following diagnosis of HTN. If blood pressure level is high (e.g. systolic ≥160 mmHg or diastolic ≥100 mmHg) or there is accompanying evidence of end organ damage, initiation of treatment should be started without delay.

Evidence and rationale

The GDG reviewed evidence from 14 relevant systematic reviews that summarized data from a large number of randomized trials involving over 120 000 adult participants (Web Annex A). Evidence summaries are presented for the general population and for higher-risk populations (with diabetes (DM), coronary artery disease (CAD), prior stroke) and were presented for various systolic blood pressure (SBP) thresholds (Web Annex A).

The anticipated benefits of a lower blood pressure (BP) target (140 SBP in the general population and 130 SBP in a high-risk population) were reduction in mortality, cardiovascular mortality, stroke, myocardial infarction (MI) and heart failure events. The anticipated harms were mostly not serious side-effects, and some were a surrogate outcome, such as rise in creatinine that may not be clinically relevant. On average, treatment was associated with a reduction in deaths and cardiovascular events that ranged from 5 to 10/1000 and harms that ranged from 20 to 30/1000. The benefits were a reduction in severe events with significant morbidity and mortality whereas the harms were mostly not clinically significant.

In summary, the anticipated benefits were large and clearly outweighed the harms. The overall certainty varied from moderate to high, depending on the BP level and agent used.

Evidence-to-decision considerations

The value of antihypertensive therapy is well accepted by most patients, health care providers, health systems, professional societies and government agencies. From a patient’s perspective, preventing cardiovascular events is highly valued. However, some individuals who are eligible for antihypertensive treatment may not present to care, may be lost to follow up, or are prescribed a treatment but fail to take/adhere to the treatment. Treatment may be perceived as low value from the perspective of an asymptomatic patient unless the person is convinced of a trade-off between immediate inconvenience/ side-effects and potential long-term health gains (6). The patient perception of an unfavourable cost-benefit may be further exacerbated by the requirement for out-of-pocket payment for appointments or medications. Therefore, the GDG considered that although there is important variability in stakeholder values, overall initiation of hypertension (HTN) medications is likely to be feasible and acceptable overall. Given that the barriers to accessing HTN care in low-income settings include low patient health literacy, lack of financial protections, and limited resources (7), the GDG felt that health inequalities would probably be reduced by HTN treatment.

In terms of costs and resource requirements, the GDG acknowledged variability, based on the structure of a country’s public health system and its economic status. Other costs, including human resources and medications, were considered moderate, given the benefits. Multiple sources of cost effectiveness are available from various countries, such as Argentina, Nigeria, the USA and UK (8, 9, 10, 11, 12, 13), and for lower thresholds and higher-risk individuals (14, 15, 16). Most cost-effectiveness estimates were clustered below USD 1000 per averted disability-adjusted life year (DALY) – well below the average 2017 GDP per capita for lower-middle income countries of USD 2188 (17), suggesting that pharmacological treatment could be very cost-effective for this group of countries. HTN treatment (treating all with BP ≥140/90 mmHg) has been shown to be cost-effective and a “best buy” intervention by the Kostova study (8). Treating high-risk/CVD patients with baseline 130–139 mmHg has been shown to be cost effective, but not cost saving (SPRINT cost-effective analyses) (15); value depends on maintaining the intervention effect for more than five years.

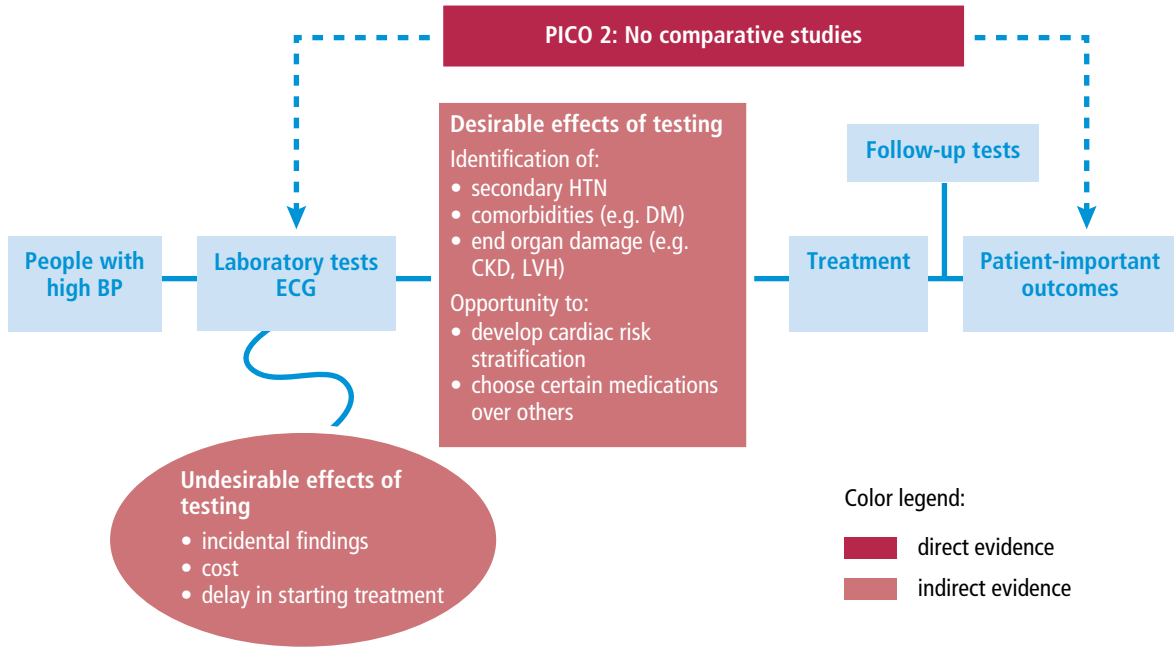
3.2 Laboratory testing before and during pharmacological treatment

2. RECOMMENDATION ON LABORATORY TESTING
When starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment.
Conditional recommendation, low-certainty evidence
Implementation remarks: - Suggested tests include serum electrolytes and creatinine, lipid panel, HbA1C or fasting glucose, urine dipstick, and electrocardiogram (ECG). - In low-resourced areas or non-clinical settings, where testing may not be possible because of additional costs, and lack of access to laboratories and ECG, treatment should not be delayed, and testing can be done subsequently. - Some medicines, such as long-acting dihydropyridine calcium-channel blockers (CCBs) are more suitable for initiation without testing, compared to diuretics or angiotensin-converting enzyme inhibitors (ACEi)/angiotensin-II receptor blockers (ARBs).

Evidence and rationale

Comparative studies that evaluate various testing strategies prior to the initiation of antihypertensive treatment were not identified, despite a search of the literature. Therefore, indirect evidence was sought to evaluate this question. An analytic framework was developed to conceptualize the rationale for obtaining diagnostic testing, such as laboratory tests and electrocardiogram (ECG), in this context (see Fig. 2). This framework identified the four most important reasons for obtaining testing, which are to diagnose secondary HTN, identify comorbidities (e.g. DM), identify end organ damage (e.g. chronic kidney disease (CKD) or left ventricular hypertrophy (LVH)), and for cardiac risk stratification.

Fig. 2 Framework for analysis



In terms of secondary HTN, various studies suggest a prevalence of 5–10% among patients with a diagnosis of HTN and a higher prevalence of 10–30% among patients with particularly high BP (e.g. over 175/115 mmHg) or BP that is resistant to treatment (18, 19, 20). Morbidities and end organ damage that could be identified by testing patients with HTN are also common. One estimation indicates that 23%, 24% and 39% of patients with a diagnosis of HTN have one, two, three or more comorbidities respectively. Common comorbidities among patients with HTN that can be discovered via laboratory testing are hyperlipidaemia and diabetes, which have a prevalence of 56% and 27% respectively (21). Testing at the point of starting HTN medications or for subsequent monitoring can also identify patients who develop certain adverse events after treatment (e.g. hyperkalaemia and acute kidney injury), thus providing a rationale for testing. Testing also had the additional advantage of identifying compelling indications for choosing certain medications over others. For example, identifying diabetes would favour the use of ACEis/ARBs, and identifying hyponatraemia would lead to not starting diuretics. Overall, the desirable effects of testing were judged to be at least moderate.

The framework also identified the most undesirable effects of obtaining testing, which were delay of starting treatment, cost and incidental findings. Delay of treatment was judged to be the most important concern since it can lead to losing the patient for follow up and the potential for adverse cardiovascular (CV) outcomes. Incidental findings on testing were thought to be less important. The undesirable effects of testing were judged to have smaller magnitude. On balance, desirable effects are likely to outweigh undesirable effects. The certainty of evidence across outcome was judged to be very low due to serious concerns about the indirectness of evidence.

RECOMMENDATIONS

Evidence-to-decision considerations

There is uncertainty about patients’ values and preferences regarding the issue of testing before starting treatment for HTN. The cost of tests such as electrolytes, creatinine, lipid panel, glucose, HbA1C, urine dipstick, and ECG relative to overall costs of treatment and complications of HTN are small (22). However, in less well-resourced settings, this cost can have a large impact. Furthermore, if additional tests like ECG or 24-hour ambulatory BP monitoring were added, the cost can become a barrier (23). It is unknown whether testing would lead to cost saving or be cost-effective. Testing was judged to be acceptable to most stakeholders, particularly patients and health care providers, and to a lesser extent to those governing health systems. Requiring testing before starting HTN medications can exacerbate health inequities and may not be feasible in low-resource settings.

3.3 Cardiovascular disease risk assessment as guide to initiation of antihypertensive medications

3. RECOMMENDATION ON CARDIOVASCULAR DISEASE RISK ASSESSMENT
WHO suggests cardiovascular risk assessment at or after the initiation of pharmacological treatment for hypertension, but only where this is feasible and does not delay treatment.
Conditional recommendation, low-certainty evidence
Implementation remarks: - Most patients with SBP ≥140 or DBP ≥90 mmHg are high risk and indicated for pharmacological treatment; they do not require cardiovascular (CVD) risk assessment prior to initiating treatment. CVD risk assessment is most important for guiding decisions about initiating pharmacological treatment for hypertension (HTN) in those with lower SBP (130–139 mmHg). It is critical in those with HTN that other risk factors must be identified and treated appropriately to lower total cardiovascular risk. - Many CVD risk-assessment systems are available. In the absence of a calibrated equation for the local population, the choice should depend on resources available, acceptability and feasibility of application. - Whenever risk assessment may threaten timely initiation of HTN treatment and/or patient follow up, it should be postponed and included in the follow-up strategy, rather than taken as a first step to indicate treatment.

Evidence and rationale

The most direct evidence is derived from an individual patient data meta-analysis by Karmali that compared the number of major adverse cardiovascular events (MACE) at five years when using a CVD risk assessment strategy (based on age, sex, body mass index, blood pressure, previous antihypertensive treatment, smoking, diabetes mellitus (DM), and history of CVD) vs BP levels alone for determining which patients receive treatment (24). This analysis suggested that risk assessment can potentially prevent 310 MACE events in 1000 people over five years, which the GDG considered to be a moderate-to-large benefit. However, this evidence was indirect for many reasons, including the effect being dependent on the BP at presentation (graphs diverge at higher level of BP, compared with starting medications without risk assessment) and the fact that these trials did not actually randomize patients to the two strategies sought in PICO question 3 (see Annex 4). Furthermore this evidence should not suggest that people with intermediate risk would not have important treatment benefit.

There was no evidence of the undesirable anticipated effects of starting treatment based on cardiovascular risk assessment. However, delay in initiating care for HTN management and loss to follow up are important considerations, especially in low-resource settings.

The GDG deduced that benefits of risk assessment may not all be attributable to risk assessment per se, but rather to the various treatments provided for risk factors identified during risk assessment. The certainty of evidence across outcome was judged to be very low due to serious concerns about the indirectness of evidence. Overall, the desirable effects of risk assessment at or after initiating HTN medications outweighed the plausible undesirable effects, particularly when risk assessment was deemed not to delay the initiation of treatment.

Evidence-to-decision considerations

There is important uncertainty over the value stakeholders place on conducting a CVD risk assessment prior to starting pharmacological treatment, and it was noted that patient perspectives may vary, depending on the setting. In low-resource settings, patients may focus more on immediate treatment without having to bear additional costs for screening for other risk factors and treating them. Studies have also shown that in high-income countries (HICs) such as the United States, people of a lower socioeconomic status have lower control of blood pressure and higher CVD risk over the years (25). Thus, in low-resource settings, adding one more step before initiating treatment may increase inequities, as those patients who have limited access to health care services may suffer delays in treatment or even end up not receiving HTN treatment at all.

In terms of costs, there is no direct evidence of whether treatment of HTN with or without risk stratification is more cost effective. The cost of implementation of CVD risk assessment should also account for capacity building of health care providers and the time taken to do so for each patient.

Cost of testing and delay in initiating care can be significant following a CVD risk stratification strategy in low-resource settings. Modelling by Gaziano et al. showed significant cost reduction using CVD risk-stratification before initiation of treatment in low-resource settings. However, screening costs, including the cost of obtaining risk-factor information, productivity costs due to work loss, care costs and travel time were not included in the analysis (26).

A meta-analysis showed that proportional relative risk reduction in major CVD events from BP lowering did not differ substantially with the presence or absence of previous CVD events, coronary heart disease, or cerebrovascular disease. Hence, the absolute benefit of blood pressure lowering would be greatest in those with the highest absolute risk of CVD (27).

3.4 Drug classes to be used as first-line agents

To develop a recommendation that is practical and implementable by end-users, the evidence-to-decision frameworks of PICO questions 4 and 5 (see Annex 4) were used to develop one recommendation.

4. RECOMMENDATION ON DRUG CLASSES TO BE USED AS FIRST-LINE AGENTS

For adults with hypertension requiring pharmacological treatment, WHO recommends the use of drugs from any of the following three classes of pharmacological antihypertensive medications as an initial treatment:

1. thiazide and thiazide-like agents
2. angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs)
3. long-acting dihydropyridine calcium channel blockers (CCBs).

Strong recommendation, high-certainty evidence

Implementation remarks:

- Long-acting antihypertensives are preferred.
- Examples of indications to consider specific agents include diuretics or CCBs in patients over 65 years or those of African descent, beta-blockers in ischaemic heart disease, ACEis/ARBs in patients with severe proteinuria, diabetes mellitus, heart failure or kidney disease.

RECOMMENDATIONS

Evidence and rationale

Data from 32 systematic reviews were used to derive evidence on benefits and harms of various medication classes (19 for comparisons against placebo and 13 for head-to-head comparisons). These reviews summarized the results of a many large randomized trials (Web Annex A). The anticipated benefits were considered to be large. Mortality and major adverse cardiac events (MACE) reduction per 1000 treated people for the various classes were 3 and 14 (low-dose thiazide), 12 and 39 (high-dose thiazide), 23 and 48 (ACEi), 8 and 23 (CCB), 2 and 8 (beta-blockers) and 14 and no data for MACE (ARBs) respectively. The anticipated adverse events were judged to be moderate. Compared to placebo, 60 and 100 additional adverse events per 1000 treated people were observed for thiazides and beta-blockers respectively. Withdrawal from ACEi treatment and cough per 1000 treated people were 12 and 26 respectively. A systematic review of studies of pharmacotherapy for HTN in sub-Saharan Africa showed a rate of side-effects of CCB of 6% (headache), 2% (dizziness) 2% (ankle oedema) (28).

In terms of the head-to-head comparisons among various classes, there was a smaller body of evidence, with less data available on hard endpoints and patient-important outcomes. Comparisons showed overall minimal differences in SBP or DBP. For example, ACEis/ARBs vs CCBs differed by less than 2 mmHg, and so did comparisons between ACEis/ARBs vs thiazide or ACEis vs ARB. There were more stroke events with beta-blockers than CCBs or ACEis/ARBs.

The anticipated benefits clearly outweighed the potential harms for three classes of medications: thiazide and thiazide-like agents, ACEis/ARBs, and long-acting dihydropyridine CCBs. The adverse events of these three classes were infrequent, usually mild, and can be managed or another agent can be substituted. The amount of BP reduction appeared to be a more major determinant of reduction in CV events than the choice between these three classes of antihypertensive medications, as was shown in several landmark trials (ALLHAT, VALUE, CAMELOT trials) (29, 30, 31). This balance of benefits and harms was not as clear for beta-blockers as a first choice for HTN management.

In terms of potential subgroups of patients that may benefit more from specific medication classes, ALLHAT (Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial) suggested greater BP reduction in individuals of African descent with chlorthalidone than lisinopril, and that stroke was significantly less likely with the diuretic than with the lisinopril in this group of patients than in those of Caucasian ethnicity (32). Other studies suggested benefit of diuretics or CCBs in patients over 65 years of age or of African descent, beta-blockers in patients with HTN who are post-myocardial infarction, ACEis/ARBs in diabetes mellitus, heart failure or kidney disease (33, 34). Diuretics were likely to be the most efficacious medications, and CCBs the least efficacious medications for the prevention of heart failure.

The overall certainty of evidence varied from high to moderate for these three classes of medications when compared against placebo. It was noted that diuretic trials were older and practice patterns may have changed over time, and that the severity and stage spectrum of diabetes mellitus and chronic kidney disease varied in the available trials. In addition, evidence supporting the efficacy of antihypertensive drug therapy is derived from trials conducted in adults at high risk for CVD/atherosclerotic CVD. Since CVD risk increases with higher levels of BP, and given that risk factors for CVD tend to track together, the assumption of greater benefits using CVD risk could be attributed to this.

Evidence-to-decision considerations

The value of antihypertensive therapy is well accepted by most patients, health care providers, health systems, professional societies and government agencies. From a patient's perspective, preventing cardiovascular events is highly valued. However, some individuals who are eligible for antihypertensive treatment may evade efforts aimed at treatment or are prescribed a treatment but fail to take/adhere to the treatment. The asymptomatic nature of the disease and concern about adverse events are the likely driver for this perspective. Interviews of patients in England suggested greater acceptance of antihypertensive drug therapy with higher socioeconomic status.

In one study, as many as 35% of the Caucasians and 20% of the South Asians in the two lowest socioeconomic categories told their interviewer that they would not accept antihypertensive drug therapy (35). Shahaj et al. (6) synthesized six qualitative and 29 quantitative reviews and identified a range of individual and social factors that affect treatment adherence, including familial (lack of support, need for separate meals), and environmental (sense of security, local amenities, healthy food availability). A review by Fragasso et al. (36) suggested that quality of life on antihypertensive therapy is an important issue because clinicians are asked to initiate drug therapy in mostly asymptomatic patients who are never happy to become, instead, symptomatic because of side-effects. Therefore, the GDG considered that there is important variability in stakeholder values, but overall initiation of HTN medications is likely to be feasible and acceptable overall. Considering the ample literature on disparities in adherence to BP medication regimes and cardiovascular outcomes based on race or socioeconomic status, treatment was judged to reduce health inequities.

In terms of costs and resource requirements, thiazide-like agents, ACEis/ARBs and long-acting dihydropyridine CCBs are available as generic drugs, are simple to manufacture, and should be available at low cost globally. Other costs related to workforce requirements, provision of infrastructure, laboratory testing, lost work time etc. are real but modest. Numerous modelling studies demonstrate cost effectiveness of antihypertensive therapy, which is especially beneficial in LMICs where large numbers of adults have untreated HTN, as long as medications are available at low cost. Models were available from many countries, including Bangladesh, Ghana and Nigeria (37, 38, 39, 40).

3.5 Combination therapy

To develop a recommendation that is practical and implementable by end-users, the evidence-to-decision frameworks of PICO questions 6, 7 and 8 (see Annex 4) were used to develop one recommendation.

5. RECOMMENDATION ON COMBINATION THERAPY

For adults with hypertension requiring pharmacological treatment, WHO suggests combination therapy, preferably with a single-pill combination (to improve adherence and persistence), as an initial treatment. Antihypertensive medications used in combination therapy should be chosen from the following three drug classes: diuretics (thiazide or thiazide-like), angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs), and long-acting dihydropyridine calcium channel blockers (CCBs).

Conditional recommendation, moderate-certainty evidence

Implementation remarks:

- Combination medication therapy may be especially valuable when the baseline BP is ≥20/10 mmHg higher than the target blood pressure.
- Single-pill combination therapy improves medication-taking adherence and persistence and BP control.

RECOMMENDATIONS

Evidence and rationale

The GDG developed three PICO questions to address: monotherapy vs combination therapy as a first-line treatment for HTN, a comparison of the various combination therapies, and a comparison of single-pill combinations vs multiple-pill combinations. These three questions were addressed separately in the evidence profiles and evidence-to-decision framework, but eventually led to one recommendation. The evidence base consisted of six, seven and eight systematic reviews respectively (Web Annex A).

Evidence summaries demonstrate several comparisons of combination therapy to monotherapy. Data on mortality, MACE and other hard endpoints were imprecise. Combination therapy lowered SBP more than monotherapy did (e.g. standard dose CCB combined with ARB vs high dose CCB; or ACEi and ARB combination vs either drug class alone) and had fewer adverse events (standard dose CCB combined with ARB vs high dose CCB). Data on cardiovascular outcomes are limited from randomized trials. A large nonrandomized study from Italy (125 635 patients, age 40–85 years) evaluated those who started antihypertensive treatment with one drug vs a two-drug single-pill or free combination. Propensity score adjusted analysis suggests that an initial two-drug single-pill or free combination was associated with significant reductions in the risk of death (20%, 11–28%) and hospitalization for cardiovascular events (16%, 10–21%) compared with initial monotherapy (41). Combination antihypertensive therapy may be associated with fewer side-effects due to use of lower doses of each drug.

A comparison of the various combination therapies suggested overall effectiveness of combination therapies that contained the three drug classes of diuretic, ACEi/ARB and CCB. Other desirable effects of a combination therapy are improved treatment adherence and persistence. However, many of these studies used a single-pill combination, thereby confounding the question of monotherapy vs combination therapy. A meta-analysis compared adherence and persistence between groups of patients taking antihypertensives as single-pill combinations vs free-equivalent components based on 12 retrospective database studies. Adherence, measured as the mean difference in medication possession ratio, was 8–14% higher with a single-pill combination. Persistence was also twice as likely (42). A second systematic review demonstrated that simplifying dosing regimens results in significant improvements in medication adherence, ranging from 6% to 20% (43).

The desirable effects of greater adherence/persistence, improved BP control, and potentially improved clinical outcomes of combinations of the three classes of antihypertensive therapy compared outweigh the undesirable effects such as side-effects, particularly when provided as a single-pill combination. The overall certainty in evidence was low across the outcomes of interest, noting that evidence was limited in terms of hard endpoints.

Evidence-to-decision considerations

In terms of stakeholder values and preferences about monotherapy vs combination therapy or the various combination therapies, data were minimal. No important variability in values was expected with regard to the critical outcomes. A systematic review demonstrated that simplifying dosing regimens results in significant improvements in medication adherence, ranging from 6% to 20% (43). Considering the comparative ease of using a single-pill combination over multiple-pill combinations, and the anticipated impact on adherence and persistence, the GDG judged that from a patient perspective the single-pill option will be favoured by most.

Combination therapy is accompanied initially by a moderate increase in resource requirements, such as procurement, supply chain, and direct medication costs. Some combinations may be expensive, or not allow for exact dosing of both agents. However, the net benefit of improved BP control and reduction of major events associated with the hypertensive process compared to the increase in cost is large. BP control is also likely to be achieved sooner with combination therapy. Many modelling studies that evaluated combination vs monotherapy used a fixed dose (thereby not truly addressing the question). One model from Japan used data from randomized trials and compared low-dose combination therapy of controlled-release nifedipine (20 mg/day) plus candesartan (8 mg/day) vs titrated monotherapy of

candesartan. In the combination therapy group, higher efficacy and lower incremental treatment cost (dominance) were observed when compared to the monotherapy group (44). A retrospective cohort study that used the 2008–2012 BlueCross BlueShield of Texas claims suggests that mean annual drug utilization costs were highest for a single-pill combination strategy. However, disease-related inpatient services utilization costs were lower compared with the up-titration strategy, which may offset initial costs (45). In one model from China, olmesartan/amlodipine as a single pill was dominant, compared with olmesartan and amlodipine free combination and valsartan/amlodipine single-pill combination (46). In a second study, there was a reduction in the cost of therapy of 33%, with a saving of USD 19 per patient/month after switching from free combination to the single-pill combination (47).

Since single-pill combination therapy increases medication adherence and persistence, which could improve HTN control rates and decrease major clinical events, the impact on health equities is expected to be favourable. In terms of acceptability, combination therapy, including in a single-pill form, can initially be met with scepticism among stakeholders, including health care providers. However, this initial scepticism may improve once BP control improves. Despite effective, safe, affordable, and available pharmacological antihypertensive agents, the control rates of HTN are dismal worldwide, and over the last 5 to 10 years have been decreasing in some HICs, and in LMICS, in tandem with increasing major cardiovascular events. Over 30% of the world population has HTN and only 13.8% of cases are considered controlled (48). One major reason for this poor level of control (one in seven) is that most patients only receive monotherapy, whereas empirical evidence demonstrates that most patients require two drugs or more to achieve optimal and sustained control (44, 46, 49, 50, 51, 52). The rationale for recommending a combination therapy, particularly in a single-pill approach, is based on the following considerations:

- most individuals with HTN will eventually require two or more antihypertensive agents to achieve BP control;
- the combination of two agents from complementary classes yields greater BP-reduction efficacy (at least equal to the sum of the efficacy of each chosen agent);
- lower doses of each agent are needed, which results in a reduction of side-effects and the fact that use of complementary classes of antihypertensive agents may mitigate the side-effects of each agent;
- adherence and persistence are increased; and
- simplified logistics can lead to fewer stock-outs and a reduced pharmacy inventory (53, 54).

In terms of feasibility, a study from India compared prices of antihypertensive single-pill combinations and equivalent single-agent pills in the private health care sector. The results suggested that manufacturers have priced the combination higher than the price of its components. These data demonstrate that the price of combination pills could be lowered to match the combined price of the component, and that manufacturing costs and market forces do not present a barrier to the implementation of antihypertensive combination pills (55). Thus, the intervention is likely feasible to implement. The GDG acknowledged some challenges to single-pill combinations, such as limited flexibility in modifying the doses of individual components, and difficulty in attributing side-effects to one of its components (56).

Although randomized trials addressing this issue are not abundant, and those available are not sufficiently large or conducted for a long enough period to clearly address differences in major clinical events, the initial combination treatment approach has been in place for over 15 years in large health systems, such as the Kaiser Permanente system in the United States (57) and is a major component of the WHO Global HEARTS Programme and the PAHO HEARTS in the Americas Initiative (53). Recently, combination antihypertensive medications in a single pill have been added to the WHO Essential Medicines List (49). This approach has demonstrated general acceptance by government, public, and private stakeholders and is demonstrating success in increasing HTN control rates worldwide.

3.6 Target blood pressure

6. RECOMMENDATION ON TARGET BLOOD PRESSURES
WHO recommends a target blood pressure treatment goal of <140/90 mmHg in all patients with hypertension without comorbidities.
Strong recommendation, moderate-certainty evidence
WHO recommends a target systolic blood pressure treatment goal of <130 mmHg in patients with hypertension and known cardiovascular disease (CVD).
Strong recommendation, moderate-certainty evidence
WHO suggests a target systolic blood pressure treatment goal of <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease).
Conditional recommendation, moderate-certainty evidence

Evidence and rationale

The evidence base consisted of five systematic reviews as well as a review of the SPRINT trial (58). Evidence profiles were constructed for various BP treatment targets, based on age and comorbidities (Web Annex A).

The desirable effects of lower target BP (per 1000 treated patients) were: a reduction in mortality of 27 (for SBP <120 vs <130–139) and of 7 (for SBP 140/90 vs 150–160/95–105); a reduction in cardiovascular mortality of 40 (for SBP <130 vs <130–149) and 6 (for SBP 140/90 vs 150–160/95–105); and a reduction in stroke of 17 (for SBP <130 vs <140). The increase in serious adverse events per 1000 treated patients was 20 (for SBP <130 vs <140) and 5 (for SBP 140/90 vs 150–160/95–105).

Summary results from a systematic review focusing on adults 65 years and older by Murad et al (59). suggests that treatment to a lower BP target in individuals 65 years or older leads to a significant reduction in all-cause and CVD mortality, chronic kidney disease, myocardial infarction, or stroke outcomes. Similar conclusions were provided by another systematic review by Reboussin et al (60). Neither of these meta-analyses was able to account for the high risk of patients enrolled in the available trials – at least in SPRINT and ACCORD (11, 61). Therefore, the GDG cautions against applying this evidence to lower-risk patients with raised BP or HTN – specifically, those not meeting trial eligibility criteria for SPRINT, ACCORD or SPS3 (62). Network meta-analyses found a similar direction of effect but more optimistic beneficial effect sizes with intensive treatment (63, 64).

In patients with comorbidity (CAD, DM, CKD) there is consistent benefit with lower targets (variable thresholds); however, data in these subgroups were imprecise and the evidence was less certain. Adverse events such as dizziness in intensive control group and ischaemia in patients with coronary artery disease can shift the balance of benefits and harms in those aged 65 years or older. Concern about lower adherence due to the need for extra patient and provider effort to reach lower targets should also be balanced against intensive control. The overall certainty of the evidence was judged to be moderate, with large benefits and moderate harms. The GDG made a judgement that the desirable effects outweigh the undesirable effects at a treatment goal of <140/90 mmHg in all patients with HTN without comorbidities and of <130 mmHg in high-risk patients with HTN – those with high CVD risk, diabetes, chronic kidney disease.

Evidence-to-decision considerations

From a patient perspective, HTN is often a silent disease and patients may not take antihypertensive medications as directed because the positive effects of these medications are not as obvious as potential side-effects (61). Society and patients want to avoid premature mortality or disability. Serious adverse events are also feared, but their duration and severity are often not well characterized in trials. Lower targets are likely acceptable to other stakeholders, such as governments and providers, though there are usually several competing priorities and interests – especially the more acute demands of, and a higher priority placed on, acute conditions and health emergencies. Many well-known barriers to access to HTN care in low-income settings exist (6). Investment in the primary health care platform required for effective HTN management is often a challenge. Countries with low rates of HTN control using more conservative BP thresholds may feel burdened by any request to set more ambitious BP treatment goals, even if only in selected high-risk patients.

Intensive treatment for selected patients adds complexity for health workers; emphasis on team-based care in low-resource settings means that simple, protocolized care is needed. Intensive treatment for some patients complicates treatment protocols and may lead to decisional overload, especially for health workers with more limited training and/or autonomy.

On the other hand, strict BP targets in the general population with HTN are likely to be less acceptable to stakeholders. Most available evidence is derived from high-risk patients receiving intensive treatment and not the general population living with HTN. Treating HTN will reduce health inequity because preventing CV events reduces mortality across the population. Uncontrolled HTN might be over-represented in vulnerable populations. Therefore, improvement of HTN treatment and control through better treatment and a lower BP target could reduce long-standing inequality.

Regarding costs, intensive treatment in the SPRINT trial meant one additional medication, one additional office visit, and one additional laboratory test evaluation on average, and additional titration visits per participant over 3.25 years, compared with standard treatment. In the United States, this translates to about USD 23 000 more per patient per year over their remaining lifetime (14, 15). Health care costs are much less in countries other than the United States. Treating to lower BP targets will have diminishing returns in progressively lower-risk patients as the magnitude of benefit becomes smaller. A cost-effectiveness study of screening and optimal management of HTN, diabetes mellitus and chronic kidney disease in an Australian setting found that an intensive management of previously uncontrolled HTN compared with usual care resulted in an incremental cost-effectiveness ratio of AUD 2588. The study does not specify the target BP for the comparisons (65). A SPRINT trial health economic analysis provided similar inferences (48, 50).

3.7 Frequency of re-assessment

7. RECOMMENDATION ON FREQUENCY OF ASSESSMENT
WHO suggests a monthly follow up after initiation or a change in antihypertensive medications until patients reach target.
Conditional recommendation, low-certainty evidence
WHO suggests a follow up every 3–6 months for patients whose blood pressure is under control.
Conditional recommendation, low-certainty evidence

RECOMMENDATIONS

There was a minimal number of comparative studies that evaluated different follow-up lengths after initiation of HTN medications. One randomized controlled trial compared a follow-up interval of three months to an interval of six months in family practice clinics in Canada. Participants (age 30–74 years) had essential HTN that was controlled for at least three months before entry into the study. Mean BP, control of HTN, patient satisfaction and adherence to treatment were similar between the two groups (66). A retrospective population-based cohort study of family-practice clinics in the UK (67) studied 88 756 adults with HTN (1986–2010). This study showed that in patients newly diagnosed with HTN, those with >1.4 months prior to initiation of treatment had a hazard ratio of 1.12 (1.05–1.20) for a major adverse cardiovascular event (MACE) compared to those who started treatment at <1.4 months. For patients who were initiated on treatment, those who waited >2.7 months before re-evaluation had a hazard ratio of 1.18 (1.11–1.25) for MACE compared to those reassessed at <2.7 months. In addition, when reviewing protocols of large HTN trials that demonstrated important improvement in cardiovascular events with BP control, such as ACCORD and SPRINT, the initial length of time to follow up was one month (68, 69). This evidence indirectly suggests the appropriateness of this initial follow up in settings that conferred important benefit.

The anticipated desirable consequences of shorter follow up are better BP control and monitoring of side-effects, and perhaps improved adherence. Longer follow-up times are expected to lead to loss to follow up. A systematic review of the impact of interventions to improve medication adherence in adults prescribed antihypertensive medications suggested a decrease in adherence with an increase in time between intervention and follow up (70). The undesirable consequences of shorter follow up are the burden on patients and the health system. Certainty relating to these effects is very low.

The GDG found no evidence related to the question of the optimal follow-up time after the point that the treated HTN patient achieves stable blood pressure control.

Evidence-to-decision considerations

Data on what patients consider significant in terms of the length of follow up after initiation of HTN medications are lacking. Many patients, particularly those aged 65 years or older, or who live alone, are likely to be reassured by more frequent monitoring of BP, which can identify early signs of clinical deterioration and provide a sense of security (71). However, younger asymptomatic patients may have the opposite perspective and find that frequent monitoring interferes with work and family responsibilities. Telemonitoring may reduce the need for follow up, especially for patients living in areas remote from health care facilities. However, despite existing evidence on the effectiveness of telemonitoring for patients with HTN, there is no empirical evidence of its long-term outcomes or generalizability to patients with various backgrounds and educational levels (72).

Data on costs, resources and cost-effectiveness were unavailable. Frequent follow up is anticipated to be associated with additional resource requirements, which may be offset by improved adherence, BP control and improved outcomes that are important to patients. Burden on the health system may be reduced by involving nonphysician providers in follow up.

The WHO GDG considered a one-month follow up after initiation of medications for HTN to be a reasonable approach, whereas intervals of 3 to 6 months can be applied when patients' BP is close to the target and stable. Due to the lack of comparative data, these recommended intervals should be viewed as suggestions and may be modified, based on feasibility and other contextual factors. Such intervals were judged to be feasible and acceptable to key stakeholders. The impact of such follow-up lengths on health equity is unclear.

3.8 Administration of treatment by nonphysician professionals

8. RECOMMENDATION ON TREATMENT BY NONPHYSICIAN PROFESSIONALS
WHO suggests that pharmacological treatment of hypertension can be provided by nonphysician professionals such as pharmacists and nurses, as long as the following conditions are met: proper training, prescribing authority, specific management protocols and physician oversight.
Conditional recommendation, low-certainty evidence
Implementation remarks: - Community health care workers (HCWs) may assist in tasks such as education, delivery of medications, blood pressure (BP) measurement and monitoring through an established collaborative care model. The scope of hypertension care practised by community HCWs depends on local regulations and currently varies by country. - Telemonitoring and community or home-based self-care are encouraged to enhance the control of BP as a part of an integrated management system, when deemed appropriate by the treating medical team and found feasible and affordable by patients. - Physician oversight can be done through innovative methods such as telemonitoring or similar to ensure access to treatment is not delayed.

Evidence and rationale

PICO question 11 (see Annex 4) addressed BP management by nonphysician health care workers (HCWs) as well as self-management by patients. The evidence base for this question consisted of 11 systematic reviews (Web Annex A). The available evidence focused on evaluating care models in which BP control was managed by pharmacists, nurses, dietitians, and community HCWs. The outcome assessed in these studies were BP level and control. There was no data on cardiovascular events. Although the certainty of evidence was in general low, the magnitude of effect showed better control in 91–264 more patients per 1000 (pharmacist studies) and an SBP/DBP reduction of 1–8 mmHg (nurse/HCW/dietitian studies). No study showed that nonphysician management was inferior to physician management.

A systematic review by Greer et al., showed that pharmacist-managed care led to better BP control (relative risk 1.44 or 170 more controlled per 1000) with no obviously reported difference in adherence, clinical events or quality of life (73). A systematic review by Anand has shown that in LICs and MICs, task sharing with pharmacists led to reductions of 8 mmHg SBP and 3.74 mmHg DBP. Similar results were yielded by task sharing with nurses (5.34 mmHg lower), dieticians (4.67 mmHg lower), and community HCWs (3.67 mmHg lower) (74). Data on undesirable effects (harms) were unavailable, which may be due to publication bias, or reflect minimal harms.

In terms of self-management, a systematic review by Tucker (75) shows that self-monitoring by patients led to a 3.24 mmHg lower level SBP and 1.5 DBP, both statistically significant, and better BP control, as long as self-monitoring was remotely managed by a HCW. However, the study limitation was the inability to adequately blind participants to the intervention. There was minimal evidence about self-titration on BP medications.

The GDG also concluded that since the evidence was from HICs, it may be less applicable to other settings and that training of nonphysician HCWs varies considerably between countries. Overall the certainty of evidence was low, with large anticipated desirable effects and a small magnitude of undesirable effects.

Evidence-to-decision considerations

There is significant variability in patient and provider perspectives. Overall, society and patients want to reduce the risk of premature mortality or morbidity. Most of the available quantitative data were focused on remote monitoring and not specifically on whether patients preferred BP being managed by physicians vs other providers, which was the primary question. Limited information provided mixed results, with some patients appreciating some applications of self-care while others were concerned that being managed by others could harm the patient–doctor relationship, but these comments were related to use of home-monitoring devices. In some studies in which BP was managed by nonphysicians, there was good patient satisfaction and high retention, suggesting at least willingness, if not a preference, to have BP managed by nonphysicians (76, 77). Conversely, in-depth interviews with a sample of patients in the UK explored nurse and pharmacist prescribing and demonstrated that patients had concerns about clinical governance, privacy and whether sufficient space was available to provide the service in community pharmacies. Participants had less concern about nursing management (78). Another study from Scotland explored patients’ perspectives on pharmacist prescribing and reported high patient satisfaction, but 65% stated that they would prefer to consult a doctor (79). Presumably, health inequities are reduced, since task shifting in the public sector increases access to those using public health vs private health. Increasing access in underserved areas can improve inequities.

Regarding costs, Jacob et al. (80) synthesized data from 31 studies (24 from the US) and suggested that studies that use community team approaches project a cost around USD 200/person/year to implement, but with cost-savings for prevention of CVD outcomes such that net costs had a median cost of USD 65/person/year, with 10 studies indicating negative or cost-savings overall. Most cost/quality-adjusted life year (QALY) estimates were between USD 3888 and USD 24 000/QALY, with pharmacist-led programmes being more cost-effective than nurse-led ones. Only two were >USD 50 000/QALY out of 28 studies.

Most of the remaining cost data presented were related to self-monitoring and not to the question of physician-led vs nonphysician-led care. However, if it is assumed that nonphysician salaries are lower, then potentially costs will be lower, but that assumes that only limited effort by physicians is involved in any oversight of nonphysicians. Kulchaitanaroai et al. found similar results with a physician–pharmacist collaborative system (81).

The two available analyses, by Jacob et al. and Kulchaitanaroai et al., focused on team-based interventions as opposed to specifically physician vs other provider, and it is not clear if incremental cost-effectiveness ratios fit countries in all economic categories, nor whether the countries’ willingness- to-pay thresholds were analysed. All values appear to be below USD 50 000/QALY. For the US, the results were highly cost-effective, with most estimates well under USD 50 000/QALY but it remains unclear exactly how these may be translated to countries in lower economic categories. Even at USD 10 000/QALY, however, this would be acceptable for most MICs, though perhaps not for all LICs. However, if the costs were the same or lower in programmes led by nurses or pharmacists compared to those led by physicians then cost-saving was likely.

The GDG proposed four conditions that must be met for nonphysician-prescribing of antihypertensives. These focused on the prescribers having proper training, prescribing authority in their locale, working within specific management protocols and having physician oversight. Community HCWs were suggested as personnel who could assist with tasks such as education, delivery of medications, BP measurement and monitoring through an established collaborative care model.

Telemonitoring supervised by HCWs, and community- or home-based self-care, were considered as tools to enhance BP control as part of an integrated management system.

RECOMMENDATIONS

4 Special settings

4.1 Hypertension in disaster, humanitarian and emergency settings

Hypertension (HTN) is seen in a range of humanitarian crises and disaster settings (natural or humanmade). This includes, but is not limited to, the wars in Syria and Iraq, the impact of the Great East Japan Earthquake and Hurricane Katrina, and the living conditions of Palestinian refugees. The burden of HTN on those populations can be considerable (82). There are very little data on HTN control, access to care and treatment, and patient understanding of HTN from Africa and Asia (except Japan), despite protracted refugee situations on these continents. Violent and protracted conflicts are disastrous to civilian populations and their health care systems, and result in interruptions to treatment and care (83, 84). Armed conflicts are associated with increased short-term and long-term cardiac morbidity and mortality and increases in blood pressure (BP) (85). Following exposure to conflict, research in military populations shows that post-traumatic stress disorder and severe injury are independent risk factors for the development of HTN (86). The rates of treatment ranged from 53.4% to 98.1% of patients with HTN in this population (87, 88).

There are currently no data regarding target BP or the best antihypertensive agent to treat disaster-related HTN. Opinion-based recommendation is that the target BP control level should be less than 140 mmHg for SBP and less than 90 mmHg for DBP. According to Kario et al, long-acting CCBs are preferred because they are metabolically neutral, and best at reducing BP variability, which is an independent predictor of clinical outcomes, especially stroke. In addition, the BP-lowering effect of long-acting CCBs is dose-dependent, and the degree of BP reduction that can be anticipated from these agents is known (89). Despite the challenges of working in humanitarian settings, several agencies have produced guidelines for the identification and management of HTN. The WHO's Interagency Emergency Health Kit has included a supplementary module with antihypertensive medications since 2017, but it is unclear how widely these are being used (90, 91). According to a personal communication from a physician who treated HTN in Syrian refugees, the treatment was variable, and dependent on whatever drug samples were available in the clinic. They had limited choices, including atenolol, lisinopril, and verapamil. Treatment was tailored to the patient's history. For example, patients with a history of coronary artery disease received atenolol and lisinopril, patients with diabetes received lisinopril, and patients with migraine received verapamil.

Assessment of HTN and appropriate resourcing to treat it should be a priority for agencies providing emergency and longer-term care for patients after or during humanitarian crises to prevent significant mortality and morbidity. Further studies are needed to accurately estimate prevalence of HTN in crisis-affected populations throughout the world and to evaluate the best treatment approach for this population.

Humanitarian crises and disaster settings (natural or humanmade) can affect health care and services in many different ways. A list of potential barriers that can affect the management of individuals with hypertension is as follows:

- significant decline of living standards
- loss/destruction of health care facilities
- flight of medical personnel causing shortage of medical care providers
- severe shortage of medicines
- lack or absence of essential supplies, equipment and materials
- compromise of the provision of primary and secondary health care
- interruption of water, food and electricity
- lack of morbidity and mortality data due to destruction of information systems and data collection
- high psychological stress burden on both general population and health care personnel.

4.2 COVID-19 and hypertension

Almost all available evidence suggests that hypertension increases the risk of severe COVID-19, defined as admission to intensive care, clinically defined severity or a combination of these; or mortality. It was sometimes unclear, however, whether this risk was independent of other risk factors (92). Initial reports have identified higher rates of HTN among severely ill, hospitalized COVID-19 patients, with overall HTN rates of 50–56% (93, 94). It had been unclear if this relationship was causal or confounded by age and other comorbidities associated with HTN, including obesity, diabetes and chronic kidney disease. Concerns regarding use of angiotensin-converting enzyme inhibitors (ACEis) in these patients were raised due to identification of angiotensin-converting enzyme 2 (ACE2), the monocarboxypeptidase that inactivates angiotensin II and thereby counters the activation of the classic renin–angiotensin–aldosterone system (RAAS), as the functional receptor for the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) (95, 96). The WHO conducted a rapid review of evidence related to the use ACEis or ARBs in COVID patients which identified 11 observational studies. No studies were found that were designed to directly assess whether ACEis or ARBs increase the risk of acquiring COVID-19. After adjustment for confounders, history of ACEi or ARB use was not found to be associated with increased severity of COVID-19 illness. There were no studies that addressed the potential benefits and harms of initiating ACEis or ARBs as treatment for patients with COVID-19 (97). Accordingly, discontinuation of ACEis or ARBs may yield worse outcomes than continuation of their use in patients with a diagnosis of COVID-19. In contrast to the uncertainty about the potential benefit of initiating RAAS blocker use in patients with COVID-19, there is a clear potential for harm in withdrawing these agents in high-risk COVID-19 patients with established myocardial injury, HTN or heart failure (96). Most of the world's professional societies either recommend or strongly encourage continuing ACEis/ARBs in COVID-19-infected patients (98). Further research that will address key unanswered questions about the role of the RAAS in the pathogenesis and possible treatment of COVID-19 and other coronavirus-based diseases is urgently needed. Prospective studies – in particular, ongoing randomized, placebo-controlled trials such as the Ramipril for the Treatment of COVID-19 (RAMIC) trial (ClinicalTrials.gov number, NCT04366050) may provide clearer insight regarding the effect of ACEis or ARBs in patients with COVID-19.

4.3 Pregnancy and hypertension

Hypertension, including chronic HTN, gestational HTN, pre-eclampsia, and eclampsia, is a very common medical condition in pregnancy (99). Due to the adverse consequences of increased morbidity and mortality to both the women and fetus, HTN in pregnancy must be diagnosed, treated (when appropriate), and followed up diligently. It is important to note that up to 10% of pregnancy-related deaths are attributed to HTN, and its presence in pregnancy leads to long-term adverse cardiovascular consequences. Unfortunately, hypertensive disorders of pregnancy are markedly increasing (100, 101). For instance, in the United States between 1998 and 2006, hypertensive disorders in pregnancy increased from 6.7% to 8.3%, chronic HTN in pregnancy increased from 1.1% to 1.7%, and pre-eclampsia/eclampsia from 0.9% to 1.2%. The potential serious consequences of HTN and pregnancy and the contraindication in pregnancy of some of the commonly prescribed pharmacological antihypertensive medications discussed below should be discussed with women who are or could become pregnant.

The normal haemodynamic state of pregnancy is one of systemic vasodilation accompanied by an increase in cardiac output and decrease in total peripheral resistance. This results in a normal decrease in BP in the second trimester. HTN in pregnancy is generally diagnosed when BP is ≥ 140 mmHg and/or ≥ 90 mmHg on at least two occasions, at least six hours apart. Chronic HTN is defined as a diagnosis of HTN before 20 weeks gestation, while gestational HTN is defined as a diagnosis of HTN at 20 weeks or later. Pre-eclampsia and eclampsia are pregnancy-specific medical conditions requiring immediate and specific medical management.

While BP treatment thresholds for HTN in pregnancy continue to change, it is generally recommended for both chronic and gestational HTN that pharmacologic treatment be initiated when the SBP is ≥ 160 mmHg and/or the DBP is ≥ 105 mmHg. In chronic HTN, frequently the woman has already been diagnosed with HTN prior to the pregnancy and thus may already be on chronic antihypertensive pharmacological therapy. In this case, the current regimen may be continued, with the caveat that the medication regimen may have to be changed to preferred medications, and certain antihypertensive medications that are contraindicated in pregnancy must be discontinued. The recommended treatment BP goal/target also has been subject to debate and is changing. For instance, achieving a lower BP target (DBP of 85 mmHg vs 100 mmHg) has recently been shown to decrease the maternal development of severe HTN while not increasing maternal or fetal risk. If target organ damage is present, initiating antihypertensive pharmacological treatment at a DBP of ≥ 90 mmHg should be considered.

As with most, if not all, other medical conditions requiring pharmacological treatment during pregnancy, the treatment considerations in HTN are no different from those of non-pregnant adults. Thus, since medications are not studied specifically for efficacy and safety in pregnancy, medication selection is usually based on long-term clinical use and experience. This usually means older medications that have had a substantial long-term track record of efficacy and safety are to be considered. For the pharmacological treatment of HTN in pregnancy, preferred medications include methyldopa, beta-blockers (particularly labetalol), CCBs (particularly nifedipine and, as an alternative, verapamil), and the direct-acting vasodilators (particularly hydralazine). There is evidence to suggest that among these agents, beta-blockers and CCBs appear to be more effective than methyldopa in decreasing the development of severe HTN later in the pregnancy. The use of thiazide diuretics has been debated, particularly if the individual is already chronically on a thiazide prior to the pregnancy. In this situation the thiazide diuretic may be continued during the pregnancy.

There are clear contraindications to the use of some antihypertensive medications during pregnancy. These include all the renin–angiotensin system inhibitors, such as the ACEis, the ARBs and, although not used any more, the direct-acting renin inhibitors, due to direct adverse effects on the fetus, and the mineralocorticoid receptor antagonist spironolactone due to fetal anti-androgen effects. The use of the beta-blocker atenolol is also contraindicated due to the observation of intrauterine fetal growth inhibition (102).

In summary, HTN in pregnancy, manifested by the various hypertensive pregnancy disorders, is a very common medical condition. HTN pregnancy disorders have serious maternal and fetal consequences. There are currently several preferred oral antihypertensive pharmacological agents available to treat chronic HTN and gestational HTN during pregnancy. In addition, there are antihypertensive pharmacological agents that are contraindicated in pregnancy. There is evidence to support the pharmacological treatment of HTN in pregnancy at given BP thresholds without and with the presence of end organ damage to decrease the likelihood of the development of severe HTN later in the pregnancy. Even with the effective lowering of BP during the pregnancy and in the immediate post-partum period, the presence of hypertensive disorders of pregnancy significantly increases long-term CV risk, including future HTN, coronary disease, and stroke.

5 Publication, implementation, evaluation and research gaps

5.1 Publication

This guideline is available to download from the WHO website. Given that an overview of published systematic reviews was used for the development of the guideline, all reviews are already published and available online.

5.2 Implementation and dissemination

WHO regional and country offices, through their contacts with ministries of health, will encourage implementation at country level. WHO will provide technical assistance if substantial country adaptation is required. The HEARTS technical package, which is currently being implemented in 18 countries and has significant partner endorsement, membership and engagement, will be the platform used to implement and disseminate this guideline. The package will be revised to include the implementation tools from this guideline. Separate implementation aspects are being considered as implementation or derivative tools for nonphysician treatment and the treatment of hypertension in areas of humanitarian crises, following publication of the guideline. Implementation support will be extended to countries through all three levels of WHO.

5.3 Evaluation

WHO will monitor uptake and implementation of the guideline in national policies and programmes by reviewing the number of countries that have adapted or endorsed the guideline nationally.

5.4 Future updating of the guideline

The guideline is expected to be valid for a period of five years. This period reflects the fact that new research findings are likely to become available in the meantime but also represents a feasible time frame, considering the costs, time and other resources that are needed for the updating process. If the evidence base or user needs change before the five-year mark, consideration will be given to producing updates sooner.

5.5 Research gaps

Several research gaps were identified by the GDG according to the theme of the PICOs.

Thresholds to determine initiation of therapy and targets to achieve for control

- More evidence is required regarding treatment of those in the SBP 130–139 range who fall into one or more of the following subgroups: diabetes, chronic kidney disease, heart failure, 65 years or older.
- There is a need for better outcomes data from, for example, trials that include heart failure and cognitive impairment among outcomes.
- Clinical significance of adverse events registered in clinical trials needs greater clarity.
- There is a need to quantify the difference in estimates between blinded, placebo-controlled trials and unblinded, active control trials using a standard framework.

- There is a need for periodic analysis of trials in order to capture effects of changes over time in background epidemiology of CVD, non-BP treatments, competing risks, etc.
- More evidence is needed in LICs, MICs and other non-North American/European countries.
- An assessment of the feasibility, resource needs, and costs of intensive treatment in real clinical practice is needed. The resource commitment required for more intensive treatment in LMICs needs to be quantified.
- The opportunity cost of directing resources towards achieving SBP <130 in high-risk individuals needs to be established.
- Research is needed on the feasibility, acceptability, and efficacy of intensive treatment, especially in high-risk populations in LICs and MICs.

Laboratory tests to determine initiation of treatment

- A greater understanding of the essential tests to be performed in all patients to reduce costs and improve outcomes is required.

Role of cardiovascular risk in hypertension treatment

- An exploration is needed of key operational aspects of the implementation of a risk-based approach to CVD prevention and BP-lowering pharmacological treatment in primary health care settings.

Monotherapy versus combination therapy

- A comparison is required of long-term data about hard clinical endpoints between monotherapy and combination therapy.
- There is a need for research studies on real-world experiences, designed and statistically powered, to determine if there is a difference in clinical outcomes, such as reduction in MACE, mortality, and serious adverse events, between single-pill combinations vs multiple-pill combinations.
- Health economic analyses are needed to quantify cost-effectiveness and budget implications of implementing incremental initial combination therapy compared with initial monotherapy.

Frequency of re-assessment

- Criteria establishing the clinical definition of stable BP control will be needed to guide the selection of patients for less frequent follow-up visits.
- Research is needed for early and accurate identification of patients less likely to achieve BP control and less likely to follow up as requested by their health care provider.
- Better evidence is needed on the timing, frequency, and intensity of interventions that improve treatment adherence.

Team-based care for hypertension

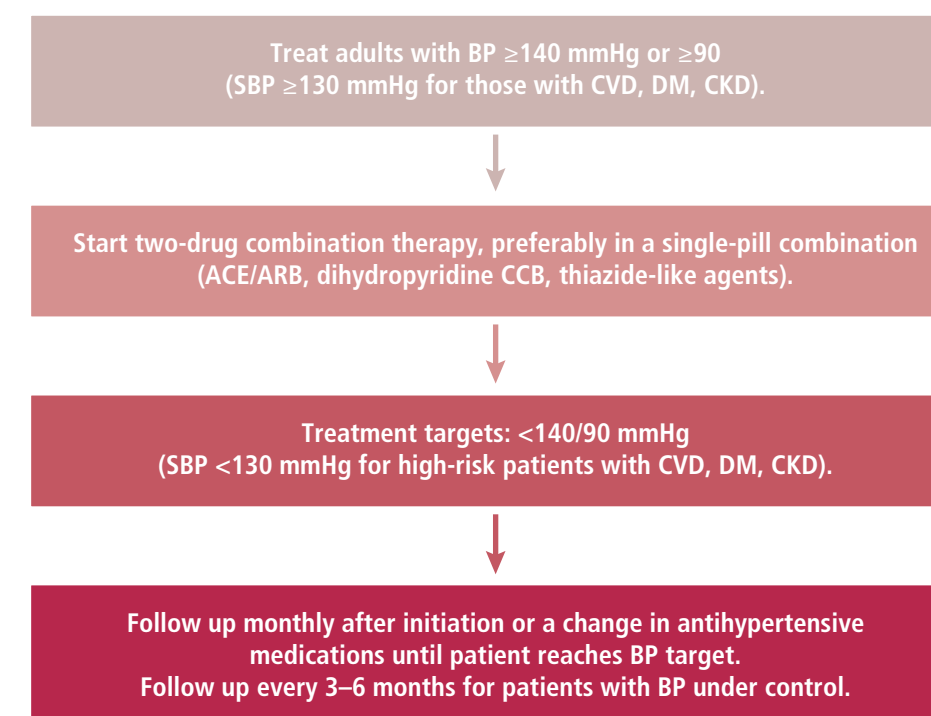
- Evidence is needed that remote monitoring and use of community HCWs/navigators can assist in the management of BP.
- Evidence of the feasibility, costs, and effectiveness of community/home-based monitoring of BP is needed.

6 Implementation tools

6.1 Guideline recommendations

Graphic summaries of the guideline recommendations are presented below in an algorithmic approach (Figs 3 and 4). This maps the recommendations to a patient-care pathway.

Fig. 3 An approach for starting treatment with a single-pill combination



Pharmacological treatment to be initiated under the following circumstances:

- A diagnosis of HTN has already been made.
- Initiation of pharmacological HTN treatment should start no later than four weeks after diagnosis of HTN.
- If BP level is high or there is accompanying evidence of end organ damage, initiation of treatment should be started without delay.
- Patient should be counselled about starting medication therapy.
- Basic laboratory testing (electrolytes, creatinine, lipogram, glucose, HbA1C, urine dipstick, and ECG) to occur as long as it does not delay treatment.
- A CV risk assessment can be conducted immediately (as long as it does not delay initiation of treatment) or at a later visit.
- Consider using diuretics or CCB in patients 65 years or older, or those of African or Afro-Caribbean descent, beta-blockers (BBs) post MI, ACEis/ARBs in those with DM, heart failure or CKD.

Fig. 4 An approach for starting treatment not using a single-pill combination (i.e. with monotherapy or free combination therapy)



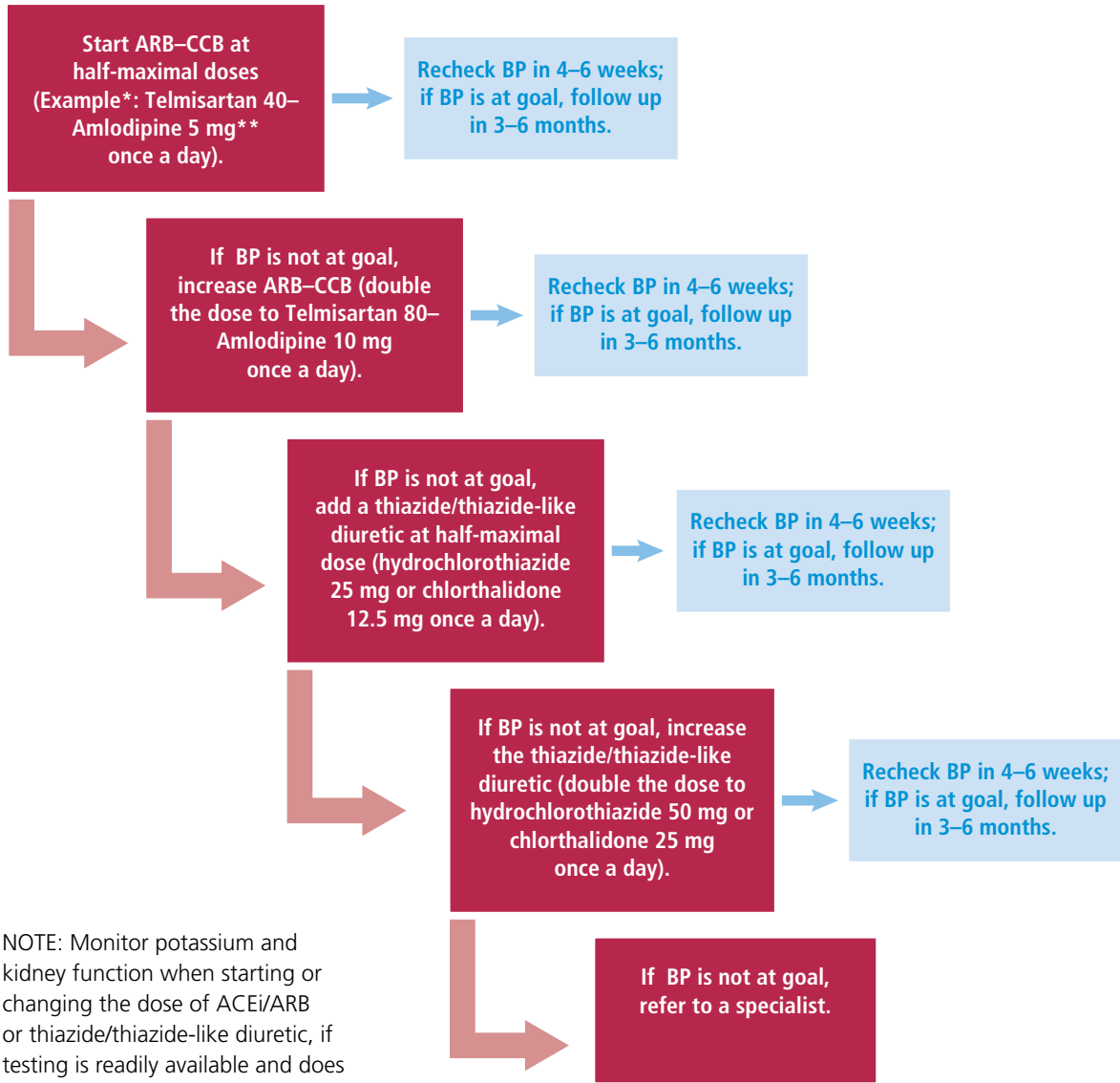
6.2 Drug- and dose-specific protocols

Two examples of suggested drug and dose-specific protocols are presented below (Figs 5 and 6). These should be viewed as examples and other approaches are possible.

Algorithm 1: Initiation of treatment with a single-pill combination

- Beginning treatment with two antihypertensive drugs from different classes is recommended when baseline BP is $\geq 20/10$ mmHg above goal, and should be considered when baseline BP is $\geq 140/90$ mmHg.
- Drugs affecting the renin–angiotensin system (ACEis, ARBs, and aliskiren) have been associated with serious fetal toxicity, including renal and cardiac abnormalities and death; they are contraindicated for use during pregnancy.

Fig. 5 Algorithm 1



NOTE: Monitor potassium and kidney function when starting or changing the dose of ACEi/ARB or thiazide/thiazide-like diuretic, if testing is readily available and does not delay treatment.

This protocol is contraindicated for women who are or could become pregnant. Neither an ACEi or ARB should be given to pregnant women.

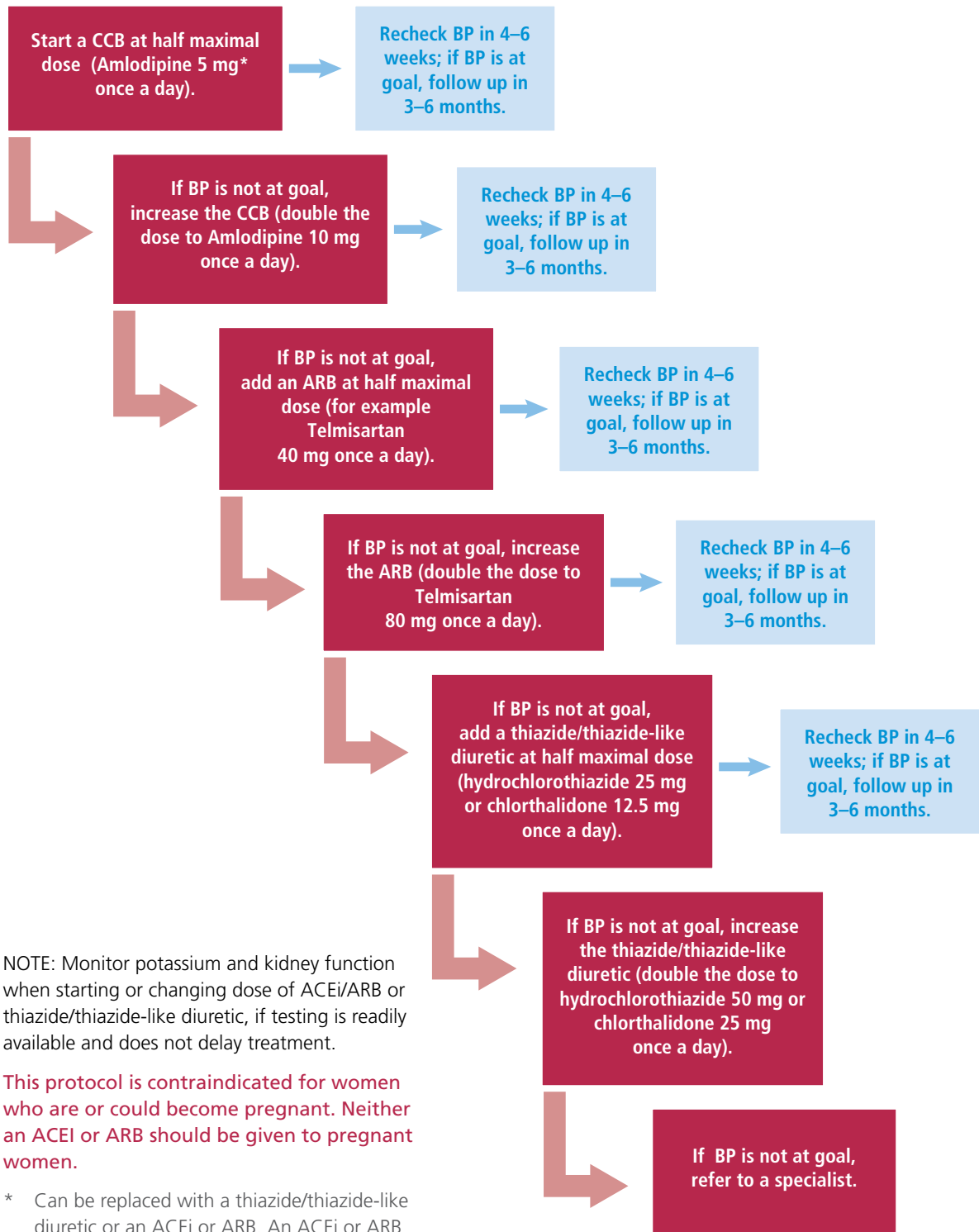
* The medications mentioned serve as examples and can be replaced with any two medications from any of the three drug classes (ACEis/ARBs, CCBs or thiazide/thiazide-like diuretics). Start two individual pills or, if available, both in a single-pill combination (fixed-dose combination).

** Can be replaced with other individual pills or, if available, other single-pill combinations (fixed-dose combinations).

Algorithm 2: Initiation of treatment not using a single-pill combination (i.e. with monotherapy or free combination therapy)

- A CCB, rather than a thiazide-type diuretic or ACEi/ARB, was selected as first-line medication if one agent is used, to avoid the need for electrolyte measurements or to alleviate concerns regarding potential change in GFR.
- Drugs affecting the renin-angiotensin-aldosterone system (ACEis, ARBs, and aliskiren) have been associated with serious fetal toxicity, including renal and cardiac abnormalities and death; they are contraindicated for use during pregnancy.

Fig. 6 Algorithm 2



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Annex 1: List of contributors

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Overall coordination and writing of the guideline

The guideline process was coordinated by the WHO Department of Noncommunicable Diseases. The first draft was written by Taskeen Khan. Drafts were reviewed by the Guideline Development Group and External Review Group, and subsequently revised by Taskeen Khan.

Annex 2. Managing declarations of interest and conflicts of interest

The steering group followed the current Compliance, Risk Management and Ethics (CRE) policy.

All members of the Guideline Development Group (GDG) were asked to fill in the standard WHO Declaration of Interest (DOI) forms, which were reviewed. The WHO Secretariat reviewed the curriculum vitae of each potential participant and conducted internet searches (PubMed, Open Payments Data, Google Scholar) for information on potential financial and academic conflicts of interest related to the subject of the meeting. All DOIs are on file at the WHO Department of Noncommunicable Diseases.

The WHO Steering Group published the names and brief biographies of potential GDG members on the WHO website for more than two weeks, together with a description of the objective of the meeting, for public review and comment. There were no concerns raised about any members.

None of the declared interests was judged sufficient to affect any of the experts’ objective judgement during the guidelines development process or on the recommendations, or therefore to preclude their full participation in the development of the guidelines.

During the course of the development of the guideline, one GDG member’s status of conflict of interest changed as she accepted a temporary appointment as a staff member at WHO. She was removed from further GDG engagements immediately upon accepting appointment.

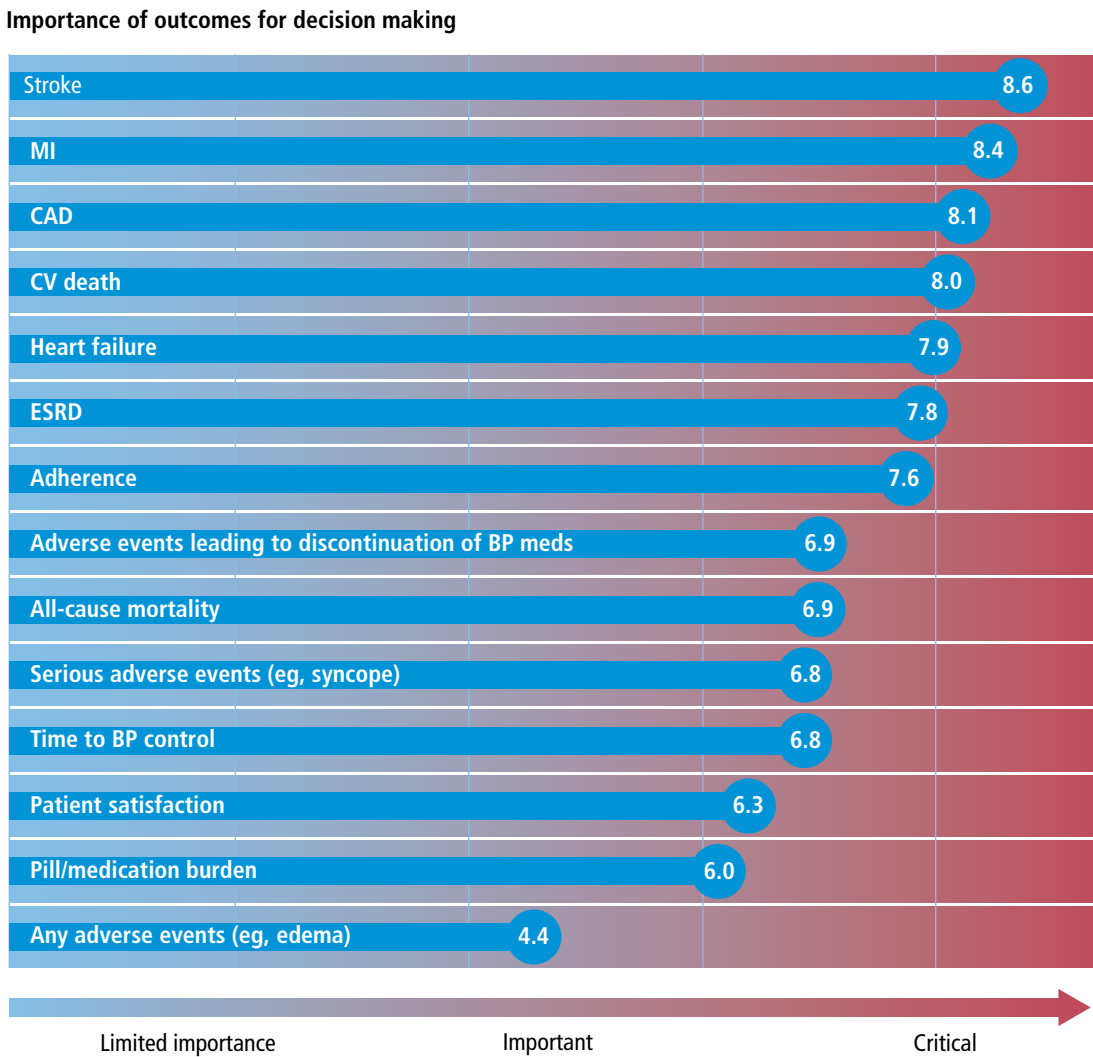
All members of the GDG and all meeting observers were required to sign a confidentiality agreement before participating in the meeting.

All members of the External Review Group (ERG) were asked to fill in the standard WHO DOI forms, which were reviewed. None of the declared interests was judged sufficient to affect any of the judgement during the review process, or therefore to preclude their participation as expert reviewers.

Annex 3: Treatment outcomes relevant to hypertension

Members of the WHO Steering Group, in consultation with the GDG and methodologist, developed a list of treatment outcomes most relevant to the care of individuals with hypertension. The GDG then rated each outcome on a scale from 1 to 9 and indicated whether it considered each outcome critical (rated 7–9), important (rated 4–6) or not important (rated 1–3) for decision-making (Fig. A3.1).

Fig. A3.1 Rating of outcomes



Annex 4: PICO questions

The eleven questions in population, intervention, comparison, and outcomes (PICO) format used to guide the systematic reviews. In addition, relevant subgroups (s) were identified.

1 At what level of blood pressure should pharmacological therapy be started to prevent cardiovascular events?

P	Adult men and women
I	Specific systolic and diastolic blood pressure thresholds:*
C	Placebo or systolic or diastolic blood pressure threshold that is higher than intervention thresholds
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease, cognitive impairment/dementia, heart failure events and adverse events
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity

* Each BP threshold in the intervention category (I) will be compared with a higher threshold. For example, I (<140) will be compared to C (≥140)

2 Is any laboratory testing necessary prior to initiation or during titration of pharmacological treatments

P	Adult men and women requiring antihypertensive treatment
I	Initiation or titration of antihypertensives without lab tests
C	Initiation or titration of antihypertensives with lab tests
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease and heart failure events, cognitive impairment/decline BP control Time to control BP Adherence Adverse effects Patient satisfaction
s	Individual drugs and doses Patients with no comorbidities Baseline blood pressure Type of lab test (ECG, blood, etc)

3 Should cardiovascular risk assessment be used to guide initiation of antihypertensive medications?	
P	Adult men and women without pre-identified CVD
I	Initiating antihypertensives drug therapy based on a formal CVD risk estimation
C	Initiating antihypertensives drug therapy without formal CVD risk assessment (i.e. using only BP threshold)
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease heart failure events, cognitive impairment/dementia, and adverse events Proportion of people prescribed with antihypertensives BP levels
s	BP levels

4 In adults with hypertension requiring pharmacological treatment, which drugs should be used as first-line agents?	
P	Adult men and women with hypertension requiring pharmacological treatment
I	BB, CCB, diuretics, ACEi, or ARB
C	Placebo
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, cognitive impairment/dementia, myocardial infarction, end-stage renal disease and heart failure events Adverse effects such as bradycardia, acute kidney injury, angioedema, asthma, electrolyte abnormalities or hypotension BP reduction and control (if data on CVD events are absent)
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline BP

5 In adults with hypertension requiring pharmacological treatment, which drugs (BB, CCB, diuretics, ACEi, or ARB vs BB, CCB, diuretics, ACEi, or ARB in head-to-head studies) should be used as first-line agents?	
P	Adult men and women with HTN requiring pharmacological treatment
I	BB, CCB, diuretics, ACEi, or ARB
C	BB, CCB, diuretics, ACEi, or ARB (head-to-head studies)
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, cognitive impairment/dementia, myocardial infarction, end-stage renal disease and heart failure events Adverse effects such as bradycardia, acute kidney injury, angioedema, asthma, electrolyte abnormalities or hypotension Blood pressure reduction and control (if data on CVD events are absent)
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline blood pressure

6 In adults with hypertension requiring pharmacological treatment, which drugs (monotherapy using BB, CCB, diuretics, ACEi or ARB vs combination therapy using BB, CCB, diuretics, ACEi or ARB) should be used as first-line agents?	
P	Adult men and women with HTN requiring pharmacological treatment
I	BB, CCB, diuretics, ACEi, or ARB
C	Combination therapy
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, cognitive impairment/dementia, myocardial infarction, end-stage renal disease and heart failure events Adverse effects such as bradycardia, acute kidney injury, angioedema, asthma, electrolyte abnormalities or hypotension BP reduction and control (if data on CVD events are absent)
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline BP

7 In adults with hypertension requiring pharmacological treatment, which combination therapy of two or more drugs (BB, CCB, diuretics, ACEi, or ARB) vs different combination therapy of two or more drugs (BB, CCB, diuretics, ACEi, or ARB) should be used as first-line agents?	
P	Adult men and women with HTN requiring pharmacological treatment
I	Combination therapy of two or more drugs (BB, CCB, diuretics, ACEi, or ARB)
C	Different combination therapy of two or more drugs (BB, CCB, diuretics, ACEi, or ARB)
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, cognitive impairment/dementia, myocardial infarction, end-stage renal disease and heart failure events Adverse effects such as bradycardia, acute kidney injury, angioedema, asthma, electrolyte abnormalities or hypotension BP reduction and control (if data on CVD events are absent)
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline BP

8 In adults with hypertension requiring pharmacological intervention, is use of a single-pill combination of antihypertensive drugs associated with improved outcomes?

P	Adult men and women with HTN requiring pharmacological intervention
I	Single-pill combination (FDC) of antihypertensive drugs – five classes (any two or more from the five)
C	Pharmacological interventions that do not involve use of single-pill combinations
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease and heart failure events. Adverse effects Patient satisfaction Adherence BP level/change Number of antihypertensive medications
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline BP

9 What target blood pressure should pharmacological treatment aim to achieve?

P	Adult men and women
I	Specific systolic and diastolic blood pressure targets: systolic (mmHg): <120, <130, <140, <150 diastolic (mmHg): <70, <80, <90
C	Systolic or diastolic blood pressure targets that are higher than the intervention targets
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease heart failure events, cognitive impairment/dementia, and adverse events
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity

10 In adults with hypertension given pharmacological treatment, when should blood pressure be reassessed?

P	Adult men and women with HTN receiving a pharmacological intervention
I	Specific interval
C	Alternative interval
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end stage renal disease and heart failure events Adverse effects BP control Adherence Patient satisfaction
s	Titration phase vs controlled HTN follow up, level of initial BP, other conditions, remote monitoring vs clinical visit

11 Can pharmacological management of hypertension be provided by nonphysician care providers?

P	Adult men and women
I	Pharmacological management by nonphysician care providers
C	Pharmacological management by medically qualified practitioners (doctors)
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease and heart failure events BP control Adherence Serious adverse effects Patient satisfaction
s	Initiation vs follow up Self-care vs CHW vs nurse vs pharmacist vs physician assistant vs in or out of clinic Levels of care Rural vs urban settings Ethnicity

HEARTS

Compendium of essential clinical terminology for hypertension-control programmes



Compendium of essential clinical terminology for hypertension-control programmes

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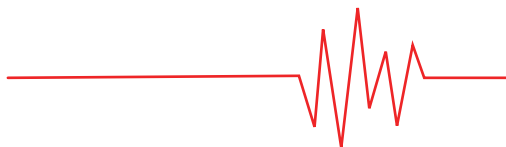
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Abbreviations

ABPM	ambulatory blood pressure monitoring
AOBP	automated office blood pressure
ACE inhibitor (ACEi)	angiotensin-converting enzyme inhibitor
ARB	angiotensin receptor blocker
BP	blood pressure
BPMD	blood pressure measuring device
CDS	clinical decision support
CVD	cardiovascular disease
DASH	dietary approaches to stop hypertension
DBP	diastolic blood pressure
EOD	end-organ damage
HBPM	home blood pressure monitoring
HEARTS	healthy-lifestyle counselling; evidence-based treatment protocols; access to essential medicines and technology; risk-based CVD management; team-based care; systems for monitoring
HTN	hypertension
LTFU	loss to follow-up
NCD	noncommunicable disease
SBP	systolic blood pressure
SPC	single pill combination
TLC	therapeutic lifestyle changes
WHO	World Health Organization

1. Introduction

Cardiovascular diseases (CVDs) are the leading cause of death globally, and hypertension is the most important modifiable risk factor contributing to premature morbidity and mortality (1). According to the WHO *Global report on hypertension 2025*, about 1.4 billion adults live with hypertension (2). Effective detection, treatment, and long-term control of elevated blood pressure are essential components of population-level CVD prevention.

The WHO HEARTS technical package provides a comprehensive, evidence-based framework for improving CVD prevention and hypertension control in primary health care (3). It does this by promoting eight interrelated technical components (3):

- healthy-lifestyle counselling
- evidence-based treatment protocols
- access to essential medicines and technology
- risk-based CVD management
- team-based care
- systems for monitoring
- implementation guide
- tool for the development of a consensus protocol for treatment of hypertension



Successful implementation of the HEARTS framework requires not only standardized clinical protocols, but also harmonized terminology across service delivery, monitoring systems, quality improvement processes, and programme reporting. Inconsistent use or interpretation of key terms such as “patients under care”, “blood pressure control”, “loss to follow-up”, “cohort control”, or “standardized treatment protocol” can undermine indicator comparability, impede performance assessment, and weaken accountability mechanisms (1, 3). This compendium responds to the need for a structured and standardized set of definitions to support consistent implementation of hypertension-control programmes within the HEARTS framework.

This compendium provides a consolidated reference for essential clinical, operational, and programme-related terminology used in hypertension-control programmes, aligned with the WHO HEARTS technical package and WHO guidelines on the pharmacological treatment of hypertension.

Specifically, this compendium aims to:

- support consistent interpretation and application of HEARTS-related terminology across countries and health-system levels;
- strengthen the “Systems for monitoring” component by clarifying operational definitions of core programme indicators;
- facilitate implementation of “Evidence-based treatment protocols” through standardized clinical terminology;
- enhance programme performance assessment, cohort monitoring, and quality-improvement processes;
- promote harmonization of reporting across national and subnational hypertension programmes.

The definitions are intended for adult hypertension-control programmes implemented as part of the HEARTS framework and may be adapted to national contexts while maintaining conceptual integrity.

This document is intended for:

- national and subnational programme managers implementing HEARTS;
- primary health-care clinicians and multidisciplinary teams;
- monitoring and evaluation officers;
- health-information system specialists;
- policy-makers and technical partners supporting CVD prevention.

By standardizing terminology, this compendium aims to strengthen implementation fidelity, improve indicator comparability, and support continuous quality improvement within HEARTS-based hypertension-control programmes.

2. How to use this compendium



This compendium is organized into three categories:

- **Clinical terminology:** terms used in diagnosis, risk assessment, treatment and patient follow-up. These support consistent clinical decision-making and documentation.
- **Operational terminology:** terms describing service-delivery processes, screening, registration, continuity of care, and quality improvement. These support programme implementation and management.
- **Monitoring and evaluation terminology:** indicator definitions used for reporting and performance assessment. Where applicable, these specify numerator, denominator and time reference to promote consistent measurement.

Users should first identify whether a term is clinical terminology, operational terminology or indicator-related to ensure appropriate application. Countries may adapt terminology to national contexts, but conceptual definitions should be preserved to maintain comparability across settings. Regular review is recommended to ensure continued alignment with updated WHO guidance and HEARTS implementation materials.

3. Methodology



The methodology used for this compendium aimed to ensure conceptual consistency, technical accuracy, and alignment with existing WHO recommendations and monitoring frameworks. The development process consisted of four main stages: (1) identifying relevant source documents; (2) selection and prioritization of terms; (3) drafting and harmonization of definitions; and (4) technical review and refinement.

Terminology was identified by reviewing key WHO normative and implementation documents related to CVD prevention and hypertension control, including but not limited to:

- WHO guidelines for the pharmacological treatment of hypertension in adults (4);
- The WHO HEARTS technical package and associated implementation guides (3);
- WHO cardiovascular risk-prediction charts and related materials;
- WHO noncommunicable disease (NCD) global monitoring frameworks and indicator metadata (5);
- Relevant WHO publications on adherence (6), essential medicines (7), and primary health care (8).

Where appropriate, definitions were aligned with established WHO guidance. In cases where no formal WHO definition was available, terminology commonly used in programme implementation was defined operationally to promote clarity and consistency.

Draft definitions were reviewed internally by technical staff with expertise in hypertension management, primary health-care implementation, and monitoring and evaluation. The review process focused on consistency with WHO guidance; internal coherence across sections; clarity of language; and applicability across different health-system contexts, particularly in low- and middle-income countries. Feedback from the review process was incorporated to refine definitions, resolve inconsistencies, and eliminate duplication.

This compendium is intended as a reference document to support standardization and does not replace WHO clinical guidelines or national regulatory frameworks.

4. Clinical terminology ^(9–11)



4.1 Diagnosis and classification

Hypertension (HTN) (high blood pressure):

Condition in which the pressure in the blood vessels is too high (140/90 mmHg or higher). This includes having a systolic blood pressure (SBP) $\geq$ 140 mmHg, or diastolic blood pressure (DBP) $\geq$ 90 mmHg, or taking blood pressure-lowering medication.

Elevated blood pressure

A SBP between 130–139 mmHg, or DBP between 80–89 mmHg.

Hypertensive emergency

Severe elevation in blood pressure, typically exceeding 180/120 mmHg, accompanied by signs of target-organ damage, such as injury to the brain, cardiovascular system, or kidneys.

White-coat hypertension

White-coat hypertension, also known as white coat syndrome, is a condition where a patient's blood pressure is elevated in a clinical setting but normal at home or during ambulatory blood pressure monitoring (ABPM). This phenomenon is often attributed to anxiety experienced during a medical visit, leading to higher readings in the presence of health-care professionals.

Masked hypertension

Masked hypertension, also known as white coat normotension, is defined as a condition where a patient's blood pressure readings appear normal in a clinical setting (e.g. doctor's office) but are elevated outside of that setting, such as at home or during daily activities. This phenomenon occurs when a person has normotensive office blood pressure (OBP) while experiencing hypertensive readings outside of it, often due to stress or other factors.

Resistant hypertension

Resistant hypertension is defined as blood pressure that remains above the therapeutic goal despite concurrent use of at least three antihypertensive agents of different classes, commonly including a long-acting calcium channel blocker, a blocker of the renin-angiotensin system (angiotensin-converting enzyme inhibitor or angiotensin receptor blocker), and a diuretic. Resistant hypertension is also diagnosed if blood pressure control requires four or more antihypertensive drugs.

4.2 Blood pressure measurement and monitoring

Systolic Blood Pressure (SBP)

SBP is the pressure in the blood vessels when the heart contracts or beats.

Diastolic Blood Pressure (DBP)

DBP is the pressure in the blood vessels when the heart relaxes between beats.

Blood pressure variability

This refers to the fluctuations in blood pressure readings over time, which can be influenced by factors such as heart rate, respiration, and sympathetic nervous system responses.

Ambulatory blood pressure monitoring (ABPM)

This is a non-invasive method used to measure blood pressure over a 24-hour period while the patient is in their own environment. This technique provides a true reflection of blood pressure throughout the day and night, making it a valuable tool for diagnosing and managing hypertension. ABPM involves taking blood pressure readings every 15–30 minutes during the day and every hour at night, allowing for a comprehensive assessment of blood pressure variability.

Home blood pressure monitoring (HBPM)

Consists of the measurement of blood pressure by a person outside of the clinical setting, at home, and is a validated approach for out-of-office blood pressure (BP) measurement. HBPM provides valuable data for diagnosing and managing hypertension.

Automated office blood pressure (AOBP)

This is a method of measuring blood pressure that utilizes automated devices to provide accurate and reliable readings, minimizing human error and the “white coat” effect.

Target blood pressure

This is the recommended blood pressure level to be achieved, which may vary depending on patient characteristics or comorbidities. WHO recommends a target blood pressure treatment goal of:

- < 140/90 mmHg in all patients with hypertension without comorbidities;
- <130 mmHg in patients with hypertension and known CVD;
- <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease) (4).

4.3 Risk stratification and pathophysiology

Primary (essential) hypertension

Primary hypertension, also called essential or idiopathic hypertension, is high blood pressure without a specific identifiable cause, defined by WHO as a SBP consistently ≥ 140 mmHg and/or a diastolic blood pressure consistently ≥ 90 mmHg, measured on two separate days.

This form of hypertension is termed primary or essential because there is no underlying identifiable medical condition causing it, unlike secondary hypertension, which has specific causes such as kidney disease or hormonal disorders.

Secondary hypertension

Secondary hypertension, as defined by WHO, refers to arterial hypertension due to an identifiable cause. It affects approximately 5–10% of the general hypertensive population and can be caused by various conditions such as kidney disease, endocrine disorders, and obstructive sleep apnoea. Unlike essential hypertension, which has no identifiable cause, secondary hypertension often responds well when the underlying condition is addressed.

End-organ damage (EOD)

This is a structural and/or functional injury in organs commonly affected by hypertension and vascular disease (e.g., heart, brain, kidneys, retina, and large arteries), attributable to chronic elevated blood pressure and related risk factors.

Atherosclerosis

Atherosclerosis is a common disease that develops when a sticky plaque builds up in the arteries. Diseases linked to atherosclerosis are the leading cause of death worldwide.

Sympathetic overactivity

This is a sustained increase in activity of the sympathetic nervous system (e.g., increased sympathetic nerve firing and/or circulating catecholamines) that can contribute to higher blood pressure and cardiovascular strain.

Cardiovascular-risk score

The cardiovascular-risk score is a calculated estimate of an individual’s absolute probability of experiencing a major cardiovascular event over a defined time horizon, typically expressed as a percentage risk over 10 years. WHO has developed 10-year CVD-risk charts to assess CVD risk among people aged 40 years and over (lab-based and non-lab-based).

4.4 Clinical assessment and management

Baseline evaluation

This is defined as the initial clinical assessment at the start of care, used to establish diagnosis, identify comorbidities and complications, and set a reference point for follow-up and treatment planning.

Risk-factor profiling

This refers to the systematic identification and documentation of cardiovascular-risk factors (e.g., smoking, diabetes, lipids, obesity) to guide total cardiovascular-risk reduction alongside blood pressure management.

Treatment adherence

This is the extent to which a person's behaviour (taking medication, following dietary advice, and/or making lifestyle changes) corresponds with agreed recommendations from a health-care provider (6).

Effective antihypertensive treatment

Effective antihypertensive treatment refers to the combination of evidence-based pharmacological and non-pharmacological interventions (based on WHO or national recommendations, protocols or guidelines) that achieve and maintain blood pressure control to recommended targets, reduce cardiovascular risk, and prevent hypertension-related morbidity and mortality.

Lifestyle assessment

This is a structured assessment of behavioural and environmental factors that influence cardiovascular risk and blood pressure, e.g., diet, including salt intake, physical activity, tobacco, alcohol, weight etc.

Therapeutic inertia

This can be defined as a failure of clinicians to intensify antihypertensive therapy when blood pressure treatment goals are unmet ($\geq 140/90$ mmHg), even when the patient has been seen two or more occasions. This contributes significantly to uncontrolled hypertension and often occurs because of concerns about side effects such as hypotension; assumptions that a slightly lower reading represents improvement; patient reports of missed doses; or competing priorities during the clinical encounter.

Diagnostic inertia

Diagnostic inertia is the failure of health workers to diagnose hypertension in adults who consistently present with elevated blood pressure ($\geq 140/90$ mmHg), even when they have been seen twice (or more) in health facilities. It represents a critical lapse in care in which clinicians miss the opportunity to identify and initiate treatment. This failure contributes significantly to poor population-level blood pressure control and can have life-threatening consequences, as missing a diagnosis of hypertension can be as dangerous as missing other serious medical conditions.

Titration

Titration refers to stepwise dose adjustment and/or addition of medicines over time to achieve target blood pressure while monitoring tolerance and safety.

Side-effect profile

This is the expected pattern and frequency of adverse effects associated with a medicine or regimen, used to guide selection and patient counselling.

Polypharmacy

Polypharmacy is the use of multiple medicines by one person (often to manage multimorbidity), which can increase regimen complexity and the risk of adverse events and non-adherence if not regularly reviewed.

Contraindications

Contraindications are the specific clinical circumstances in which a medicine should not be used because the risk of harm outweighs potential benefit.

First-line antihypertensives

These are the classes of drug recommended for initial pharmacological treatment of hypertension. WHO's 2021 guideline for the pharmacological treatment of hypertension in adults recommends (as first-line antihypertensive drugs): thiazide/thiazide-like diuretics, ACE inhibitors or ARBs, and long-acting dihydropyridine calcium-channel blockers.

Combination therapy

Combination therapy refers to the use of two (or more) antihypertensive medicines (drugs) together to achieve blood pressure control, preferably initiated as a combination when pharmacological treatment is required.

Single-pill combination

This is a fixed-dose combination of two or more antihypertensive medicines in a single pill, used to simplify treatment and improve adherence and persistence.

4.5 Healthy-lifestyle and non-pharmacological interventions

Dietary approaches to stop hypertension (DASH)

DASH is a flexible, balanced eating plan designed to support a heart-healthy dietary pattern and help lower blood pressure, emphasizing fruits/vegetables/whole grains and less sodium.

Sodium restriction

This involves reducing dietary sodium intake to recommended levels to lower blood pressure and cardiovascular risk; for adults, WHO recommends < 2000 mg/day sodium ($\approx < 5$ g/day salt).

Therapeutic lifestyle changes (TLC)

Refers to a structured lifestyle programme combining diet, physical activity, and weight management to improve cardiovascular risk.

Behavioural counselling

Behavioural counselling is a patient-centred approach that supports individuals to adopt and sustain healthier behaviours (e.g., diet, physical activity, tobacco cessation), typically using structured brief interventions and follow-up.

5. Programme operations terminology (12–14)



5.1 Service delivery and health systems capacity

Guideline

This broadly refers to comprehensive, evidence-based documents developed by expert bodies (WHO/others) that contain recommendations for clinical practice or public health policy. They describe recommended drug classes, multiple treatment options, and general principles for dose titration. These guidelines are designed to help end-users make informed decisions on specific actions, such as clinical interventions or public health measures, aiming to achieve the best possible health outcomes. They are often lengthy and best suited for specialist interpretation and clinical decision-making.

Standardized treatment protocol

Typically simple, one-page documents, these protocols are short, clear, and uniform sets of clinical instructions that specify the exact medicines, doses, and step-by-step titration needed to manage a condition for primary care, making them easy to use and suitable for national or subnational scale-up. They represent an adaptation of guideline evidence to a specific health system by providing short, clear, standardized instructions.

Availability of hypertension core medicines

This sets out the extent to which essential antihypertensive medicines (as defined by a programme's "core" list) are in stock and available at the point of care on the day of assessment/survey. Availability of hypertension medicines ensures an uninterrupted supply of essential medicines, which in turn supports patient adherence to care.

Validated blood pressure devices

A validated device is one that has undergone independent accuracy testing by an organization such as an accredited testing laboratory or academic institution, using publicly available, internationally accepted protocols (e.g., ISO 81060-2). This process involves rigorous clinical comparison of the device model against a reference standard, among real patients. Validation is performed once per device model (not for each individual unit), and demonstrates that the model can reliably measure blood pressure. Proof of validation is required before procurement.

Accurate blood pressure measuring devices

Accurate blood pressure measuring devices (BPMs) are those that have successfully passed rigorous, independent validation testing against a recognized reference standard and are available with a range of appropriate cuff sizes to ensure correct measurement across different arm circumferences.

Calibrated blood pressure devices

Calibration is a routine maintenance procedure performed on each individual device to verify and adjust its readings against a known standard over time. For manual (aneroid) devices, this typically involves adjusting the pressure gauge so that the needle accurately reflects true cuff pressure when compared with a mercury column or calibrated digital reference. Calibration must be repeated periodically (e.g., every 6–12 months) to ensure continued measurement accuracy. Automated BPMs do not require user calibration, as they use built-in sensors and algorithms. Instead, periodic inspection is recommended to ensure that the cuffs and tubes are intact, and sensors are functioning properly. Only authorized agencies should be contracted for repairs or servicing.

Recommended blood pressure devices

These are automated, upper-arm cuff blood pressure measuring devices that meet quality and accuracy requirements (including clinical validation) for use in screening, diagnosis and management.

Availability of a functional blood pressure measuring device

This means the presence of at least one recommended blood pressure measuring device that is operational and can be used to measure blood pressure at the time of assessment (including necessary cuffs/power/working condition).

Supportive supervisory visit

This is a planned, collaborative, on-site visit where supervisors work with facility teams to guide, mentor, and solve problems together to improve the quality of hypertension care. Instead of fault-finding, these visits focus on correct BP measurement, adherence to treatment protocols, medicine availability, accurate data, and effective patient follow-up. Using clear objectives and standardized tools, supportive supervision helps strengthen performance and improve blood pressure control.

Clinical decision support (CDS)

This includes tools (paper-based or digital) that provide person-specific, evidence-based recommendations to health workers at the point of care (e.g., treatment algorithms, prompts, risk tools) to support clinical decisions.

Care pathways

Care pathways represent a structured, protocol-based sequence-of-care processes that standardize diagnosis, treatment, follow-up, and referral for hypertension/CVD risk management to improve quality and outcomes. They are multidisciplinary tools designed to optimize care delivery by standardizing the sequence and timing of key interventions. They aim to improve service efficiency, enhance coordination among care teams, and promote consistent, high-quality care. Care pathways reduce unnecessary practice variation, address therapeutic inertia, support timely and appropriate clinical decisions, optimize resource use, and strengthen health-system performance.

5.2 Screening and case finding

Opportunistic screening

This refers to screening for hypertension as part of routine blood pressure measurement among all adults who visit health facilities for routine health service encounters, enabling the detection of elevated blood pressure/hypertension and a pathway towards diagnosis and care.

Case finding

This involves provider-initiated activities (e.g. mandatory, accurate BP measurement) to identify people with a condition (or at high risk of one) such as hypertension, who may not yet be diagnosed, followed by confirmatory diagnosis and linkage to care.

Assessment of CVD risk

This is an estimation of the 10-year absolute CVD risk among adults ≥ 40 years using a validated CVD risk-assessment tool such as the WHO CVD risk prediction charts/models to guide risk-based management.

5.3 Registration and continuity of care

Registrations

This is the number of patients with hypertension enrolled under the programme in a reporting period after diagnosis; includes those already on treatment and those newly diagnosed through screening.

Cumulative number of patients enrolled in care

This is the total number of patients registered/enrolled since programme initiation up to the reporting date (sum of registrations over time).

Patients under care

These are the patients with hypertension who had at least one visit (new or follow-up) to a health-care facility in the 12 months before the reporting date.

Missed visit

These are the registered patients with hypertension who missed their last appointment in the past 3 months.

Loss to follow-up (LTFU)

These are registered patients with hypertension who did not have a single follow-up visit in the 12 months before the reporting date.

6. Monitoring and evaluation definition and indicators



Indicator definitions are clearly presented in WHO's **Noncommunicable disease facility-based monitoring guidance (15)**.

For each indicator, the document explicitly specifies the numerator and denominator, ensuring consistent interpretation and facilitating standardized measurement across facilities



Hypertension new patients

Number of people with hypertension who were newly registered during the past 3 months.

Hypertension treated patients

Number of people with diagnosed hypertension who were treated with antihypertensive medication during the reporting period.

Hypertension patient retention

Proportion of established patients with hypertension with at least one visit in the past 12 months.

Blood pressure under control

This is defined as SBP below 140 mmHg (< 140 mmHg) and DBP below 90 mmHg (< 90 mmHg) at 3–6 months after treatment initiation.

Blood pressure control among people with hypertension

This refers to achieving a defined blood pressure threshold for “controlled” blood pressure based on WHO or national treatment protocol/guidelines, among the total number of people registered for hypertension treatment in the facility, excluding those who were newly diagnosed within the past 3 months.

Based on WHO guidelines, BP is considered controlled when:

- SBP < 140 mmHg and DBP < 90 mmHg
- SBP < 130 mmHg among people with history of CVD
- SBP < 130 mmHg among high-risk people with hypertension, i.e., those with high CVD risk, diabetes mellitus, chronic kidney disease

Facility cross-sectional blood pressure control

This indicator refers to the proportion of patients under care at a facility whose blood pressure was controlled at their most recent recorded visit within a defined period (cross-sectional snapshot). The indicator covers the total number of people registered for hypertension treatment in the facilities whose BP was controlled at the last clinical visit in the reporting period.

Blood pressure not under control

This indicator is defined as a systolic BP equal to or above 140 mmHg (≥ 140 mmHg) or DBP equal to or above 90 mmHg (≥ 90 mmHg) at 3–6 months after treatment initiation.

Quarterly blood pressure control

This indicator is the proportion of patients registered for hypertensive treatment during a quarter (cohort) whose blood pressure is controlled 3–6 months after registration (measured quarterly).

Community-level blood pressure control

This is the proportion of the estimated population with hypertension in a defined geographic area (e.g., district/state) with controlled blood pressure (measured annually).

Cohort blood pressure control

This is the proportion of a defined registration cohort whose blood pressure is controlled at a specified follow-up time (“window”) after registration.

Community cross-sectional blood pressure control

This is the proportion of the estimated population with hypertension in a geographic area whose blood pressure is controlled, as measured at a defined time point/period (cross-sectional).

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Global report on hypertension

The race against
a silent killer

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Foreword

High blood pressure kills. In fact, cardiovascular diseases are the world’s leading cause of death. Globally, an estimated 1.3 billion adults were affected by hypertension in 2019. I know the dangers of hypertension from personal experience, because it is a condition I live with. I am fortunate that I was diagnosed, have access to good medical care and medicines, and that I understand my condition and can make sure it is controlled. Unfortunately, the same is not true for the majority of those with hypertension, a disease that is often dubbed the “silent killer”.

This report compiles data for the first time on the far-reaching consequences of uncontrolled hypertension, including heart attacks, strokes and premature death, along with substantial economic losses for communities and countries. It also contains information on the global, regional and country-level burden of hypertension and progress of control efforts.

Perhaps most importantly, the report shows a way forward, by highlighting key aspects of guidance from the World Health Organization’s *HEARTS: technical package for cardiovascular disease management in primary health care* and the *Guideline for the pharmacological treatment of hypertension in adults*. These tools were developed to provide guidance for countries in establishing early detection, management, control and monitoring of hypertension as part of primary health care services.

In fact, hypertension can be controlled effectively with simple, low-cost medication regimens, and yet only about one in five people with hypertension has controlled it. Hypertension control programmes remain neglected, under-prioritized and vastly underfunded. Strengthening hypertension control must be part of every country’s journey towards universal health coverage, based on well-functioning, equitable and resilient health systems, built on a foundation of primary health care.

There are many challenges to tackling hypertension but, as this report shows, there are also many reasons for hope. By implementing programmes we can improve hypertension control rates and prevent millions of deaths from heart attack and stroke.

Dr Tedros Adhanom Ghebreyesus
Director-General
World Health Organization (WHO)



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Acronyms and abbreviations

BP	blood pressure
BPMD	blood pressure monitoring device
CKD	chronic kidney disease
CVD	cardiovascular disease
DALYs	disability-adjusted life years
DBP	diastolic blood pressure
IHCI	India Hypertension Control Initiative
LMICs	low- and middle-income countries
PAHO	Pan American Health Organization
PHC	primary health care
PM _{2.5}	particulate matter of a diameter ≤2.5 µm
SBP	systolic blood pressure
SDG	Sustainable Development Goal
SMASH	Shandong Ministry of Health Action on Salt and Hypertension
NCD	noncommunicable disease
WHO	World Health Organization

Executive summary

High blood pressure is one of the world’s leading risk factors for death and disability. The number of people living with hypertension (blood pressure of ≥ 140 mmHg systolic or ≥ 90 mmHg diastolic or on medication) doubled between 1990 and 2019, from 650 million to 1.3 billion (1).

This common, deadly condition is an important public health problem that leads to stroke, heart attack, heart failure, kidney damage and many other health problems (2, 3). A study of 87 behavioural, environmental, occupational and metabolic risk factors found that high systolic blood pressure (≥ 110 –115 mmHg) was the single most important risk factor for early death worldwide, leading to an estimated 10.8 million avoidable deaths every year, and a burden of 235 million years of life lost or lived with a disability (disability-adjusted life years, DALYs) annually (3). High blood pressure causes more deaths than other leading risk factors, including tobacco use and high blood sugar.

Hypertension and its associated complications also have enormous economic costs – for patients and their families, health systems and national economies. People living with the condition incur direct medical costs and lose wages, often in their prime working years, which can be impoverishing for entire families. Hospital and outpatient care for heart attacks and strokes caused by uncontrolled hypertension are expensive for health systems. National economies lose tax income, have decreased productivity, increased health care costs, and increased societal needs for the support of adults surviving heart attack and stroke, and of children whose parents have died or become disabled. By one estimate, the economic benefits of improved hypertension treatment programmes outweigh the costs by about 18 to one.

Addressing risk factors and managing hypertension

The good news is that hypertension and its associated complications can be addressed. Risk-factor strategies include ensuring that people eat a healthy diet low in sodium, maintain a healthy weight, avoid alcohol and tobacco and take regular physical activity (all of which also contribute to good health more generally). Policies and practices to support these strategies are most effective when implemented across entire populations or within specific settings, such as in schools and at workplaces.

“Hypertension and its associated complications have enormous economic costs for patients and their families, health systems and national economies.”

“The WHO HEARTS technical package provides cost-effective strategies to control blood pressure and prevent heart attack, stroke, and other complications.”

For people who have hypertension, there are ways to minimize its impact on health and wellbeing. The starting point for living well with hypertension and preventing complications is early diagnosis and early, effective treatment – the longer a person lives with undiagnosed and inadequately treated hypertension, the worse their health outcomes are likely to be. Currently, among adults aged 30–79 years with hypertension, only 54% have been diagnosed with the condition, 42% are being treated for their hypertension, and 21% are considered to have their hypertension controlled (4).

In some countries and health care systems, large-scale hypertension control programmes have achieved hypertension control for many patients, giving them longer, healthier lives. The World Health Organization (WHO) HEARTS technical package provides cost-effective strategies that can be implemented at the primary health care level to control blood pressure and prevent heart attack, stroke and other complications. Components of the package include the use of a standardized drug- and dose-specific treatment protocol, uninterrupted access to quality-assured medications, team-based care, patient-centred services, and a robust monitoring system to track patient progress and health system performance. Follow-up and referral systems are also important to improve treatment outcomes. Implementation of the HEARTS package and the system improvements this requires in terms of integrated health care services, demonstrates that many of its components support, and can also be applied to, other health problems requiring consistent, long-term care, including diabetes, chronic kidney disease and HIV/AIDs.

Although hypertension can be prevented and treated, few countries currently do so effectively. Better hypertension management will save lives. Increasing the percentage of people whose hypertension is under control globally to 50% would prevent 76 million deaths between 2023 and 2050. Treating hypertension is one of the most important interventions to meet the Sustainable Development Goal (SDG) target 3.4 of a one third reduction in premature mortality from the leading noncommunicable diseases.

Conclusions and key messages

Improving the prevention and treatment of hypertension is feasible, low-cost, safe and necessary to achieve global and national targets, including those of the SDGs. Effective hypertension management will bring health, wellbeing, and economic benefits. It will reduce burdens on acute-care services, increase integration of health care systems and, most importantly, reduce the deaths, suffering and costs arising from complications such as heart attack, stroke and kidney failure.

To improve the prevention and treatment of hypertension, countries should:

- Provide national leadership accountable for steady progress in preventing heart attack and stroke, especially through improved control of hypertension by:
 - establishing national mechanisms such as a national steering committee, coordinated by strategic leadership from the ministry of health, to monitor and improve resource allocation and implementation of an integrated response to noncommunicable diseases, with specific attention to hypertension control rates;
 - ensuring representation on such a steering committee comes from inside and outside government and includes clinicians committed to improving the management and control of hypertension through primary health care services;
 - setting and monitoring, at least annually and ideally quarterly, national targets and indicators to foster accountability, which should include the number of patients with hypertension whose treatment has brought their blood pressure to the target level of control selected by the country, and this number as a proportion of the estimated total number of people living with hypertension in the country;
 - ensuring that national policies and plans addressing hypertension are fully costed, funded and implemented.
- Develop comprehensive programmes that address risk factors for hypertension and other noncommunicable diseases (e.g. encouraging healthy diets low in sodium, reducing tobacco and alcohol use, and making regular physical activity a normal part of daily life).
- Implement the WHO HEARTS package by:
 - selecting a drug- and dose-specific treatment protocol at the national, subnational or health system level;
 - ensuring an uninterrupted supply of quality-assured drugs and of externally validated automated digital blood pressure monitors;
 - establishing team-based care for patients with hypertension and other chronic conditions, with nurses, pharmacists, outreach workers and others collaborating with clinicians to support patient care;
 - making health care services patient-friendly, reducing barriers to care through improved patient education and treatment at convenient times and locations close to the patient's home or work. Care should include medications that are free of cost to the patient and, for stable patients, multi-month (e.g. 3- and 6-month) refills, reducing the effort needed for patients to continue to take medications and the burden on health care systems as treatment coverage increases;
 - establishing an accurate and practical information system that supports

“ Better hypertension management will save lives. ”

frontline health care workers and enables longitudinal tracking of the WHO data indicators for hypertension care, with particular attention to facility-based control rate and estimated community-wide control rates.

- Address broader factors to enable health care systems to support patient care by:
 - ensuring there are enough trained and supported health care staff at all levels;
 - including hypertension management in essential benefit packages as part of universal health coverage;
 - providing sustainable finance for the delivery of health care programmes so that treatment is not interrupted in times of financial or other difficulties;
 - improving information systems to empower providers, programme managers and patients;
 - progressively adding services, as permitted by human and financial resources, to provide integrated, comprehensive primary care to all people; and
 - strengthening the national capacity to collect, analyse and use representative data on the burden and trends of hypertension in order to continuously improve and optimize the programme.

Hypertension and the global public health agenda

1

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, and many of these deaths can be prevented by treating hypertension.

The United Nations General Assembly political declaration of 2011 on the prevention and control of noncommunicable diseases (NCDs) (5) recognized that the incidence and impacts of CVDs and other NCDs can be largely prevented or reduced with an approach that incorporates evidence-based, affordable, cost-effective, population-wide and multisectoral interventions. To catalyse national action, the World Health Assembly adopted a comprehensive Global Monitoring Framework in 2013, comprised of nine voluntary global targets to be reached by 2025 (Fig. 1), which included a “25% relative reduction in the prevalence of raised blood pressure”.

These targets were included in Appendix 2 of the *Global action plan for the prevention and control of noncommunicable diseases 2013–2020* (6), which provides a roadmap and policy options to attain the nine voluntary global targets. More recently, the Seventy-sixth World Health Assembly in 2023 (7) approved an updated Appendix 3 Menu of policy options and cost-effective interventions. This reflects the most recent scientific evidence, and includes, as a recommended intervention to address cardiovascular diseases, the pharmacological treatment of hypertension in adults in line with the 2021 *Guideline for the pharmacological treatment of hypertension in adults* (8).

Commitments to reducing hypertension were deepened in 2015 following the United Nations General Assembly’s adoption of the 2030 Agenda for Sustainable Development and, specifically, Sustainable Development Goal (SDG) target 3.4: reduce by one third premature mortality from noncommunicable diseases through prevention and treatment (9). Actions to control the burden of hypertension – including population-level interventions to prevent hypertension and scale up treatment – are key steps to achieving this, and also make a contribution to other SDG targets, including those relating to universal health coverage and access to affordable essential medicines.

At present, approximately four out of every five people with hypertension are not adequately treated (4). At the same time, high systolic blood pressure – a silent killer – is responsible for more than 10 million deaths every year – more deaths than from any other risk to health. Many of these deaths are

“High blood pressure – the silent killer – is responsible for more than 10 million deaths every year.”

preventable through policies that reduce the risk of hypertension, including a healthy diet with less sodium and healthy levels of potassium, and improved treatment. Improving treatment of hypertension can save millions of lives and also strengthen primary health care systems.

Effective management of hypertension in primary health care requires accurate diagnosis, with screening of every adult patient, immediate treatment initiation for patients with confirmed very high (>160/100) blood pressure and prompt rechecking of all other patients with elevated blood pressure within one to four weeks, accessible standardized treatment free to the patient, and rigorous monitoring of progress at each facility, each sub-national area and nationally.

Every national health leader should regularly assess progress towards improving control of blood pressure to meet their country’s targets.

Fig. 1. Voluntary global targets for prevention and control of noncommunicable diseases (as of August 2023)

1.		A one third relative reduction in the overall mortality from cardiovascular diseases, cancer, diabetes, or chronic respiratory diseases by 2030 ^a
2.		A 20% relative reduction in the harmful use of alcohol by 2030 ^b
3.		A 15% relative reduction in prevalence of insufficient physical activity by 2030 ^a
4.		A 30% relative reduction in mean population intake of salt/sodium ^c
5.		A 30% relative reduction in prevalence of current tobacco use ^c
6.		A 25% relative reduction in the prevalence of raised blood pressure or to contain the prevalence of raised blood pressure ^c
7.		Halt the rise in diabetes and obesity ^c
8.		At least 50% of eligible people (aged 40 years and older with a 10-year cardiovascular risk ≥20%) including those with CVD to receive drug therapy and counselling (including glycaemic control) to prevent heart attacks and strokes ^c
9.		An 80% availability of the affordable basic technologies and essential medicines, including generics, required to treat major noncommunicable diseases in both public and private facilities ^c

Note: The target date for targets 4–9 is 2025.

Source: a. WHO NCD Accountability Framework, including Global Monitoring Framework for NCD prevention and control (2021 update) in alignment with the extension of the NCD Global Action Plan to 2030 (10); b. Decision WHA75(11). 2022 (11); c. *Global action plan for the prevention and control of noncommunicable diseases 2013–2020*, Appendix 2 (6)

1.1 Purpose of the report

This report advances global work to address noncommunicable diseases. It aims to draw focused attention to the priority noncommunicable diseases targeted by world leaders, in particular the public health challenge of hypertension, and to generate momentum for national, regional and global action. The world has made substantial progress by focusing on specific communicable diseases such as HIV, tuberculosis, malaria, vaccine-preventable diseases and childhood infections. Progress in reducing today’s leading causes of death, of which hypertension is the most deadly, will similarly require focused attention along with coordination to strengthen primary health care.

Section 1 has introduced hypertension as a global health issue and has outlined the goals of this report.

Section 2 presents an overview of the global prevalence, treatment and control of, and risk factors for, hypertension, the burden of mortality related to high systolic blood pressure, and the progress of countries towards the voluntary global target to reduce raised blood pressure by 25% (6). It discusses the importance of strong health information systems to enable the collection, analysis, interpretation and dissemination of data, and for robust systems for monitoring and recording blood pressure.

Section 3 focuses on the health and cost impacts of hypertension, and provides scenarios for how we can save lives by scaling up hypertension treatment.

Section 4 presents some of the risk factors for hypertension: unhealthy diet containing excess sodium, use of tobacco and alcohol, physical inactivity and air pollution.

Section 5, which includes case studies from the WHO regions, discusses diagnosis and early detection of hypertension, and models of care, including those for co-morbidities, and describes the HEARTS approach to managing hypertension.

Section 6 presents a short conclusion.

Annex 1 discusses the methodology used in the analysis of the impact and cost of the treatment of hypertension.

Annex 2 provides country-by-country profiles of the status of hypertension.

“Every national health leader should regularly assess progress toward their targets for control of blood pressure.”

The global burden of hypertension

2

Hypertension is a serious, chronic medical condition that increases mortality from cardiovascular and kidney disease. It occurs when the pressure in blood vessels is too high (12). Blood pressure is created by the force of blood pushing against the walls of blood vessels (arteries) as it is pumped by the heart. The higher the pressure, the harder the heart must pump and the more damage this excess pressure causes to many parts of the body, especially the brain, heart and kidneys (13).

Consideration of the burden of high blood pressure, including hypertension, involves looking at it from different angles, using different types of data and statistical methods. Table 1 lists key terms and their definitions used in this section to build up as complete a picture as possible. There is further discussion of the source of the data used in this section, and suggestions for ways in which Member States can improve data collection on hypertension, in Section 2.7.

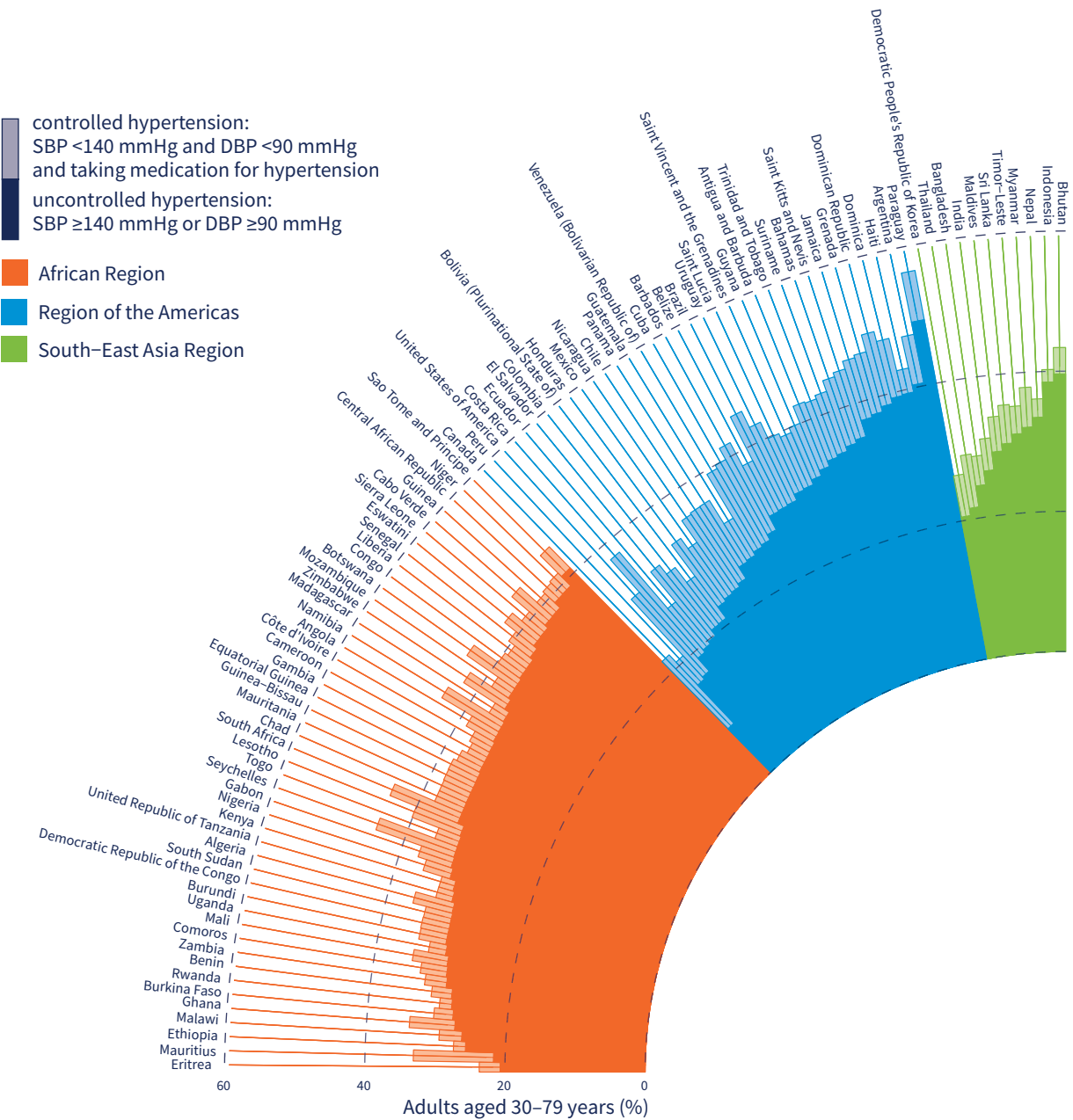
Table 1. Key terms and their definitions used for monitoring blood pressure and hypertension in populations

Hypertension prevalence	
Numerator: systolic blood pressure (SBP) ≥140 mmHg or diastolic blood pressure (DBP) ≥90 mmHg or taking medication for hypertension	Denominator: adults aged 30–79 years
Uncontrolled hypertension prevalence	
Numerator: SBP ≥140 mmHg or DBP ≥90 mmHg	Denominator: adults aged 30–79 years
Diagnosis coverage	
Numerator: has hypertension and has received previous hypertension diagnosis	Denominator: adults aged 30–79 with hypertension, as defined above
Treatment coverage	
Numerator: has hypertension and is taking medication for hypertension	Denominator: adults aged 30–79 with hypertension, as defined above
Effective treatment coverage	
Numerator: has controlled hypertension: taking medication for hypertension and SBP <140 mmHg and DBP <90 mmHg	Denominator: adults aged 30–79 with hypertension, as defined above
High systolic blood pressure	The concept of high SBP is used for comparative risk assessment (3). It refers to any SBP above the theoretical minimum risk level, estimated to be at 110–115 mmHg.

2.1 The global prevalence of hypertension

For the purpose of monitoring population health, hypertension is defined as having systolic blood pressure (SPB) ≥140 mmHg or diastolic blood pressure (DPB) ≥90 mmHg or taking medication for hypertension (Table 1). Hypertension is estimated to affect 33% of adults aged 30–79 worldwide (age-standardized estimate) (4).^{1, 2}

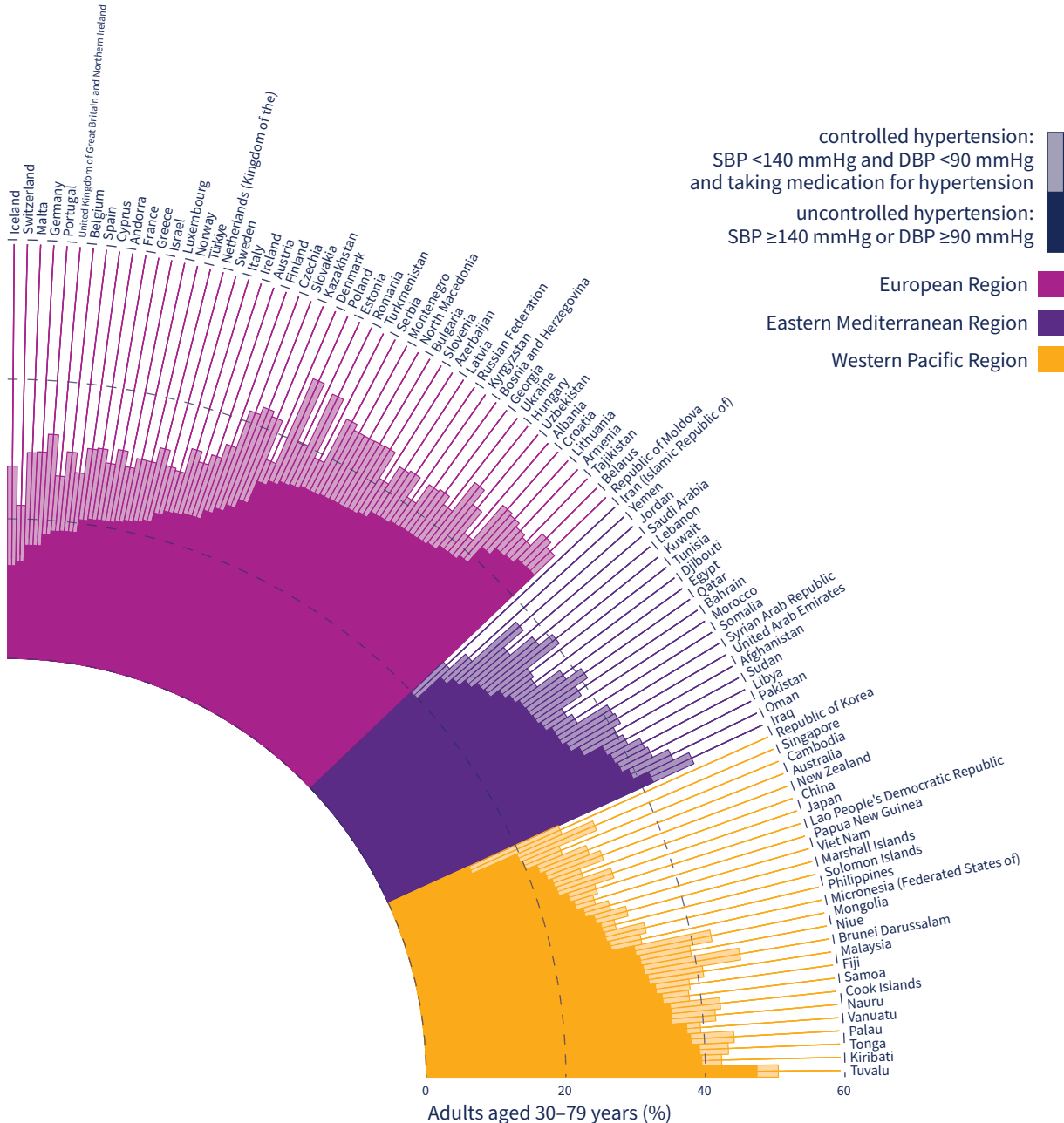
Fig. 2. Prevalence of hypertension, by control status, among adults aged 30–79, by country, in 2019



¹ Blood pressure and hypertension prevalence increase with age. In this report, prevalence estimates are age-standardized to ensure that comparisons between populations with different proportions of younger and older adults are not affected by their respective age structures.

² All hypertension, hypertension coverage and uncontrolled hypertension estimates in this section are from, or are calculated from, the estimates produced by the NCD Risk Factor Collaboration (NCD-RisC) (1), which are also available from the Global Health Observatory (GHO) (4).

Prevalence of hypertension is similar across groups of countries defined by income level (14), with only a slight difference from 32% of adults aged 30–79 years in high-income countries to 34% in low-income countries. Regional and country variability is more notable. Regional variation ranges from 28% in the WHO Western Pacific Region to 38% in the WHO Eastern Mediterranean Region (Table 2). At the country level, 29 countries (out of 192 with data) distributed across all WHO regions and country income groups have a prevalence below 30%, and 27 countries in all income groups and all WHO regions except the WHO South-East Asia Region have a prevalence above 45% (Fig. 2).



Note: for further explanation of controlled and uncontrolled hypertension see Table 1. The sum of controlled hypertension and uncontrolled hypertension is hypertension.

Source: Global Health Observatory (GHO). Noncommunicable diseases: risk factors [online database] (4).

“The number of adults with hypertension doubled from 650 million in 1990 to 1.3 billion in 2019.”

Globally, the prevalence of hypertension is slightly higher among males (34%) than females (32%). This female advantage is age-related: the global age-standardized prevalence of hypertension among people aged 30–49 years is 19% for women versus 24% for men. This pattern of lower hypertension prevalence among women aged under 50 years holds in most countries worldwide (15). However, for people aged 50–79 years, both men and women globally are estimated to have equivalent hypertension prevalence of 49%.

Globally, the estimated age-standardized prevalence of hypertension changed little from 1990 to 2019, increasing from 32% to 33%. Hypertension prevalence declined in high-income countries (from 38% in 1990 to 32% in 2019) and in the WHO European Region (from 45% in 1990 to 37% in 2019). These declines were offset by slight to moderate increases in other regions, including the WHO Western Pacific Region (24% to 28%) and the WHO South-East Asia Region (29% to 32%).

However, the total number of adults in the world is increasing as a result of population growth and because the age structure of the population is shifting, with older groups accounting for a larger proportion of the total population. Since the rate of hypertension increases with age, these two trends are increasing the number of adults aged 30–79 years with hypertension. The number of adults with hypertension doubled from 650 million in 1990 to 1.3 billion in 2019 (Fig. 3).

About 78% of adults with hypertension live in low- and middle-income countries (LMICs). The greatest number of people with hypertension live in the most populous WHO regions: Western Pacific Region and South-East Asia Region.

Table 2. Age-standardized prevalence of hypertension among adults aged 30–79 years, and among those with hypertension, diagnosis, treatment and effective treatment coverage in 2019, by WHO region

Region	Hypertension (%)	Diagnosis coverage (%)	Treatment coverage (%)	Effective treatment coverage ^a (%)
African	36 (38, 33)	43 (46, 39)	27 (30, 24)	12 (14, 9)
The Americas	35 (38, 33)	70 (73, 67)	60 (64, 57)	36 (41, 32)
South-East Asia	32 (36, 29)	39 (44, 34)	30 (34, 25)	14 (18, 10)
European	37 (39, 35)	66 (69, 63)	53 (56, 50)	26 (29, 23)
Eastern Mediterranean	38 (41, 35)	49 (53, 45)	39 (43, 34)	15 (19, 13)
Western Pacific	28 (32, 25)	54 (59, 48)	41 (47, 35)	18 (23, 14)
Global	33 (35, 32)	54 (56, 51)	42 (45, 40)	21 (23, 19)

a. Controlled hypertension among all hypertension. Controlled hypertension is defined as blood pressure <140 mmHg systolic and <90 mmHg diastolic and taking medication for hypertension.

Note: Data in parentheses are 95% uncertainty intervals.

Source: Global Health Observatory (GHO). Noncommunicable diseases: risk factors [online database] (4).

2.2 Risk factors for age-related increase in blood pressure

Many factors contribute to the age-related increase in blood pressure that is common in most countries, although the exact causes remain unknown. Genetics undoubtedly play a role, but their specific contribution remains unclear. Several general environmental exposures can also contribute to increases in blood pressure, including pollution (16, 17), very cold temperatures and extreme elevations in altitude (18). However, the main contributors seem to be personal environmental exposures such as poor-quality diet high in sodium and low in potassium, overweight and obesity, consumption of alcohol, use of tobacco and physical inactivity (19).

“More than three-quarters of adults with hypertension live in LMICs.”

Fig. 3. Number of adults aged 30–79 years with hypertension, by region and year, 1990 and 2019



Notes: The area of each circle is proportional to the number of adults with hypertension in the region. m: millions.

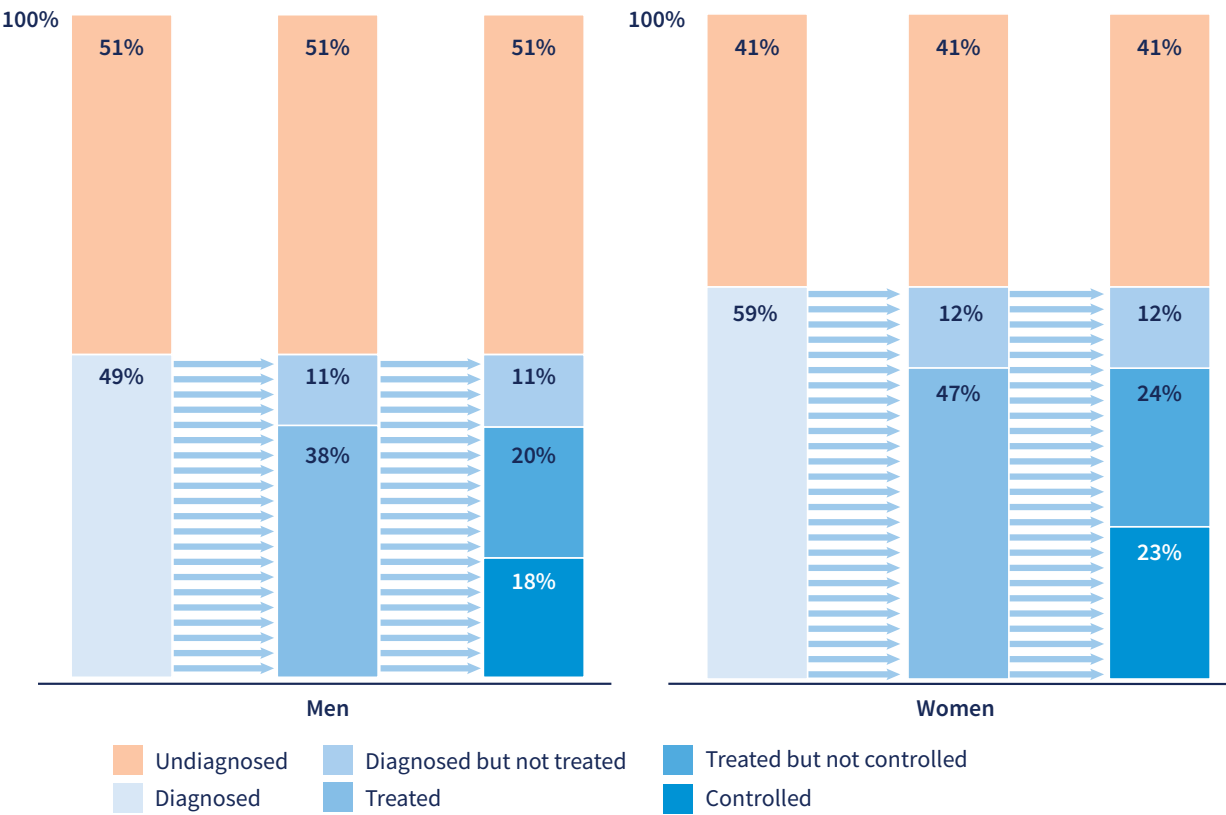
Source: NCD Risk Factor Collaboration (NCD-RisC) (1) and additional calculations.

2.3 Status of hypertension care

Among adults aged 30–79 years with hypertension, an estimated 54% have been diagnosed with hypertension, 42% are being treated for their hypertension, and 21% have had their hypertension controlled (see Table 2 for confidence intervals). Diagnosis, treatment, and effective treatment coverage varies by sex, region and country income level. In every region and in all but four countries, women with hypertension were more likely to be treated than men (Fig. 4). Globally, the percentage of women with hypertension who are currently being treated is estimated to be 47%, compared with only 38% of men. The same gender pattern holds for the likelihood of being diagnosed and the likelihood of having controlled hypertension.

Treatment coverage varies by WHO region and country income level. Treatment coverage is highest in the Region of the Americas (60%) and lowest in the African Region (27%). Treatment coverage tends to be higher in wealthier countries. In low-income countries, only 26% of adults with hypertension were receiving treatment vs 58% in high-income countries. Nevertheless, some middle-income countries have achieved treatment coverage as high as the best-performing high-income countries (Fig.5).

Fig. 4. Hypertension treatment cascade in 2019, for adults 30–79 years of age globally, by sex. Age-standardized rates



Source: NCD Risk Factor Collaboration (NCD-RisC) (1).

The rate of effective treatment coverage, or hypertension control (out of those with hypertension), follows similar patterns to treatment coverage: rates are highest in high-income countries (36%) and the Region of the Americas (36%) and lowest in low-income countries (11%) and the African Region (12%).

Among those with uncontrolled hypertension who had not previously been diagnosed, globally nearly 30% have systolic blood pressure measurements ≥ 160 mmHg or diastolic blood pressure measurements ≥ 100 mmHg. This is significantly higher than the threshold for hypertension (SPB ≥ 140 mmHg, DPB ≥ 90 mmHg), and indicates an urgent need for treatment.

Fig. 5. Percentage of adults aged 30–79 years currently being treated (taking medication) for their hypertension in 2019, out of those with hypertension, by country, sex, and country income group



Source: NCD Risk Factor Collaboration (NCD-RisC) (1).

“ The decline in uncontrolled hypertension from 29% in 2010 to 26% in 2019 is insufficient to achieve the voluntary global target of a 25% relative decline by 2025. ”

2.4 Uncontrolled hypertension and progress toward the 2025 target

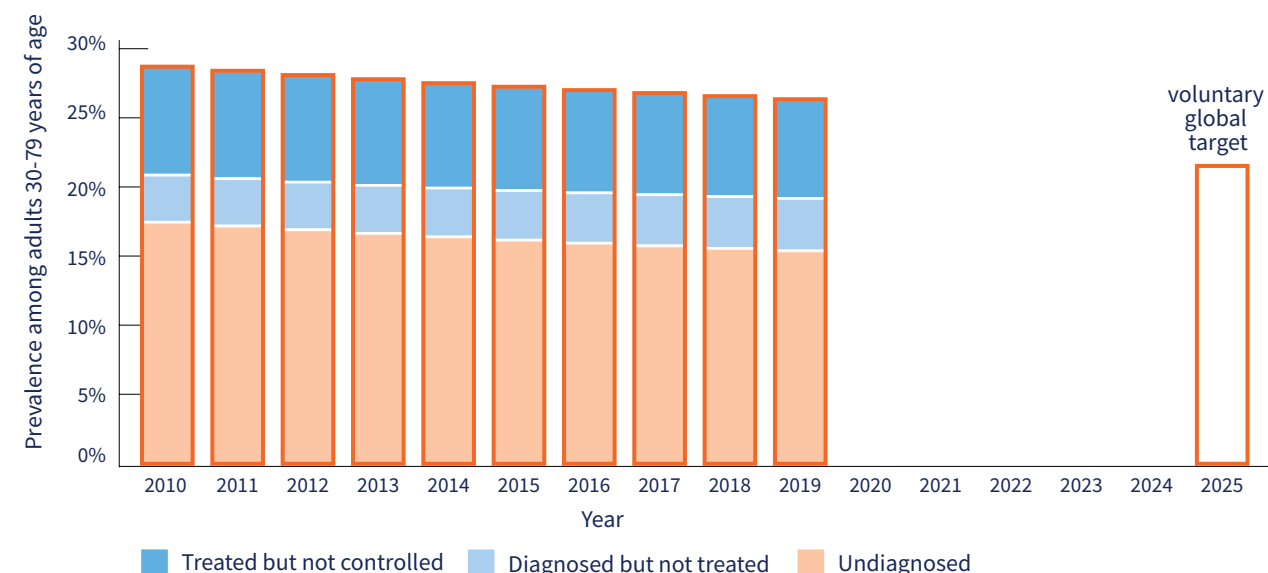
The voluntary global target on blood pressure, endorsed by the Sixty-sixth World Health Assembly in 2013, envisages a 25% reduction in raised blood pressure (uncontrolled hypertension) by 2025 against a 2010 baseline (20). Uncontrolled hypertension is defined as systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg. It includes individuals who have not been diagnosed with hypertension and those who have been diagnosed but are not on treatment or whose blood pressure is not controlled despite treatment (Fig. 6).

Globally, the prevalence of uncontrolled hypertension declined slightly from 29% in 2010 to 26% in 2019, but this rate of decline is insufficient to achieve the voluntary global target of 21% (a 25% relative decline 2010–2025).¹ Improving systems in order to provide effective treatment to about half of the adults who have already been diagnosed with hypertension would be sufficient to meet the target. Although no WHO region or country income group is on track to meet the target, a handful of countries are on track.²

¹ Target prevalence was computed on unrounded prevalence estimates, and then rounded.

² WHO/NCD-RisC estimates (1, 4) were projected to 2025 as described in WHO NCD Accountability Framework, including Global Monitoring Framework for NCD prevention and control (2021 update), in alignment with the extension of the NCD Global Action Plan to 2030.

Fig. 6. Global prevalence of uncontrolled hypertension, by category among adults aged 30–79 years of age, and the voluntary global target for uncontrolled hypertension (raised blood pressure) for 2025



Note: The 2025 voluntary global target is to reduce prevalence of raised blood pressure (uncontrolled hypertension) from 2010 levels by 25% by 2025. Estimates are age-standardized values for adults aged 30–79 years.

Source: NCD Risk Factor Collaboration (NCD-RisC) (1) and additional calculations.

2.5 Trends in hypertension care

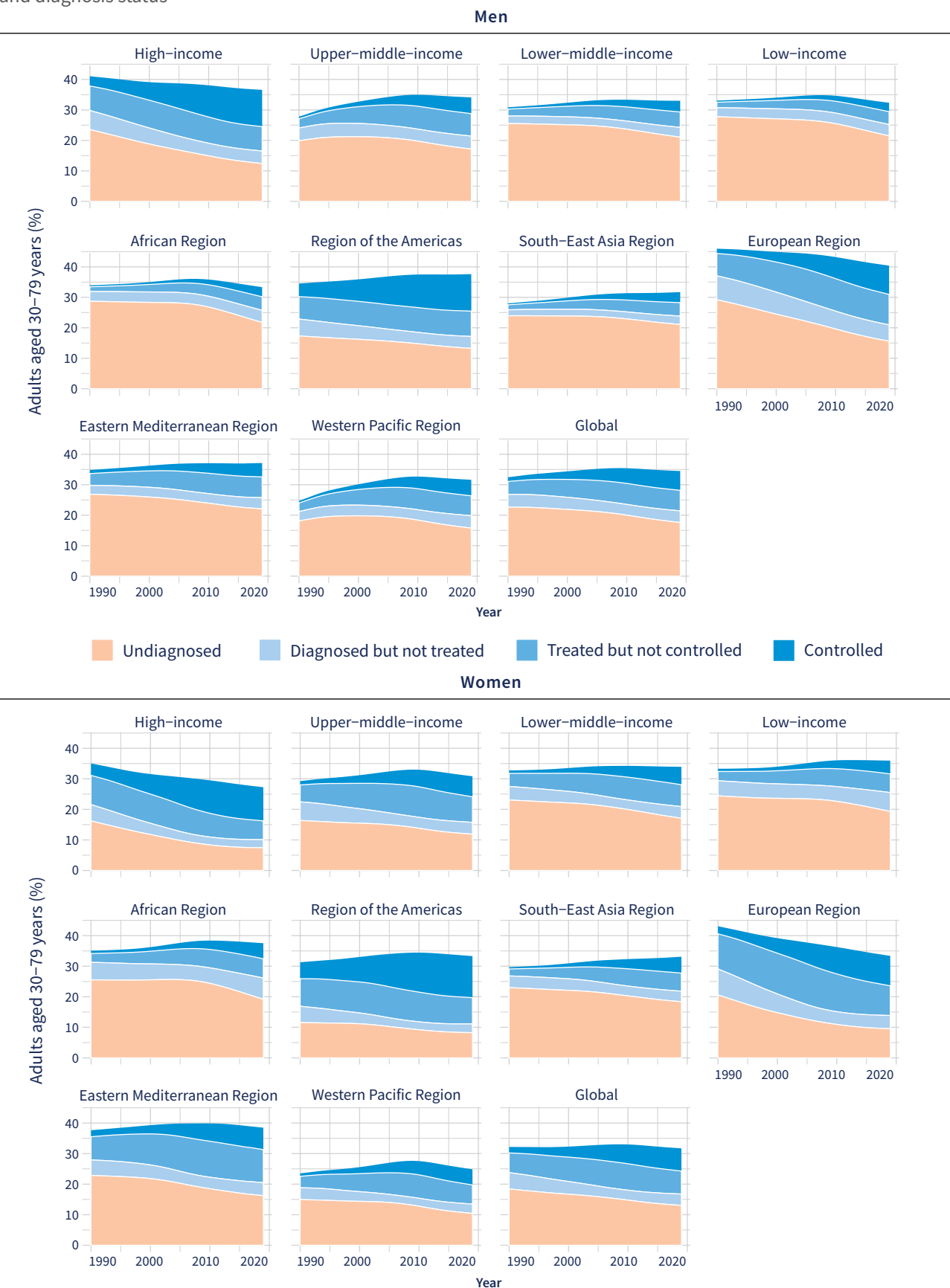
As described above, global hypertension prevalence changed little from 1990 to 2019. Although declines in the prevalence of hypertension were observed in high-income countries and in the WHO European Region, these were offset by small increases in other regions. Nevertheless, detection, treatment and control of hypertension improved globally in all WHO regions and country income levels between 1990 and 2019 (Fig. 7).

Globally, treatment coverage – the percentage of adults aged 30–70 years with hypertension who were taking medication for their hypertension – increased from 22% in 1990 to 42% in 2019 (age-standardized estimates). Effective treatment coverage is estimated to have quadrupled from 5% to 21% in three decades.

A pharmacist preparing medicine at a hospital in Ethiopia.



Fig. 7. Trends in age-standardized prevalence of hypertension, 1999–2019, disaggregated by control, treatment, and diagnosis status



Source: NCD Risk Factor Collaboration (NCD-RisC) (1) and additional calculations.

2.6 Deaths due to high systolic blood pressure

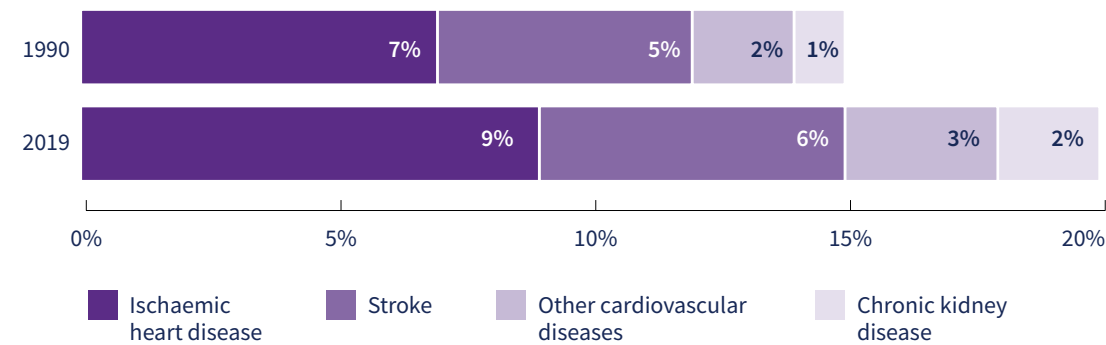
Hypertension increases mortality from cardiovascular diseases and kidney disease. In fact, high blood pressure increases risk of death even when systolic blood pressure is in the 115 to 130 mmHg range, which is below the threshold for treatment of hypertension in most guidelines (21, 22, 23). Calculating additional mortality due to high blood pressure compared to a theoretical level which results in the lowest mortality risk – called the “theoretical minimum risk level” – can give an idea of the impact of high blood pressure on population mortality (24). While the theoretical minimum risk level may not be achievable given current interventions, comparing multiple risk factors to their theoretical minimum risk levels allows a comparison of how these risks affect populations.

When considered in this framework, high systolic blood pressure (i.e. ≥ 110 –115 mmHg) is the world’s leading risk factor for mortality, causing more deaths than any other behavioural, metabolic or environmental risk factor assessed (3). If all adults in the world had a theoretical minimum risk-level systolic blood pressure in 2019 (estimated as 110–115 mmHg), about 19% of all deaths worldwide (10.8 million) would have been averted in that year (Fig. 8, Fig. 9).

“ If all adults had 110–115 mmHg SBP in 2019, about 19% of deaths would have been averted in that year. ”

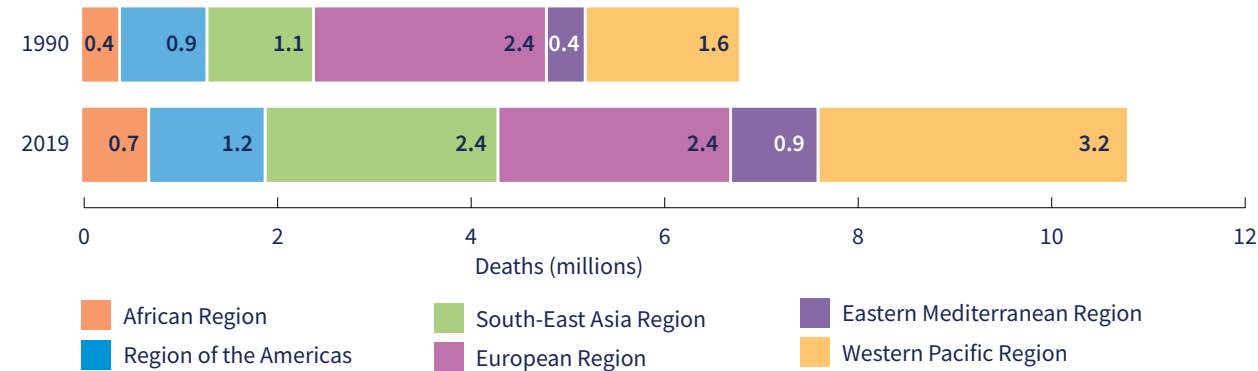
“ High systolic blood pressure is responsible for one in every five deaths. ”

Fig. 8. Percentage of global deaths attributable to high systolic blood pressure (1990 and 2019), by cause of death



Source: Global Burden of Disease Collaborative Network (25) and additional calculations.

Fig. 9. Deaths attributable to high systolic blood pressure (1990 and 2019), by WHO region



Source: Global Burden of Disease Collaborative Network (25) and additional calculations.

“ 38% of deaths attributed to high systolic blood pressure occur in adults under 70 years of age. ”

Globally, in 2019, more than half of all cardiovascular deaths were attributable to high systolic blood pressure – including all deaths from hypertensive heart disease, 53% of ischaemic heart disease deaths, and 53% of stroke deaths (25). In addition, 62% of deaths from chronic kidney disease are attributed to high systolic blood pressure. Together, these deaths from high systolic blood pressure comprise 18% of male deaths and 20% of female deaths. Although 62% of deaths attributed to high systolic blood pressure occur among adults aged 70 and over, 38% – 4 million deaths – occur in adults under the age of 70 years. This proportion is even higher in LMICs, in which 42% of deaths attributable to high blood pressure are among people younger than age 70.

High systolic blood pressure affects people in all geographic regions and at all country income levels. Even in low-income countries, 11% of deaths are caused by higher-than-ideal systolic blood pressure. This increases to 24% of deaths in upper-middle-income countries. Altogether, more than 80% of deaths attributed to high systolic blood pressure occur in low- and middle-income countries (14).

High systolic blood pressure is also a leading risk factor for disease burden when considering years of life lived in ill health and age at time of death, causing 9% of disability-adjusted life years (DALYs; 235 million) (25).

2.7 Data used to monitor hypertension and hypertension treatment

Strong health information systems that enable the continuous and systematic collection of data on hypertension – and the analysis, interpretation and dissemination of that data – are crucial in the race against hypertension. Only with such comprehensive information on the burden of hypertension – including its underlying determinants and risk factors – can health ministries devise effective prevention and treatment strategies and monitor their progress. And only with such information can an accurate assessment be made of the problem presented by hypertension at the regional and global level.

Effective surveillance of hypertension relies on multiple sources of data: death registration data that include cause of death, population-based surveys, and health-facility data.

Recording the cause of every death is the most fundamental source of information for tracking the health of populations. Calculating the burden of mortality due to hypertension requires information on deaths from cardiovascular diseases and kidney diseases, together with health examination survey data on the distribution of blood pressure levels in the population.

Although hypertension is diagnosed at a primary health care facility, in most cases, health facility data cannot be used for determining hypertension

coverage and control in the total population. Instead, data on hypertension prevalence, diagnosis, treatment and control needs to be captured in population-based surveys.

Systematic population surveys of blood pressure status, previous diagnosis and treatment is essential if Member States are to address the prevalence and distribution of hypertension, and design targeted and effective prevention and control strategies to mitigate the impact of hypertension on individuals and communities. In these surveys, such as the WHO STEPwise approach surveys (STEPS) (26), blood pressure is measured, and diagnosis and treatment are reported by individuals. Respondents are considered to be receiving treatment if they report taking medication for hypertension, and to have their blood pressure controlled if, in addition, their systolic blood pressure is <140 mmHg and diastolic is <90 mmHg.

However, survey data are not available for every Member State, or for every year. A median of only one nationally representative data source per Member State (since 2010) was identified for the estimates presented in this report. To make country estimates, national data were used together with subnational data. These data were then combined in a statistical model that borrows strength from other time periods and countries to make estimates for every country and year.

Survey data are limited, as hypertension diagnosis is typically confirmed between one and three weeks after an initial measurement; identification of undiagnosed hypertension within a survey carried out in one day may misclassify some individuals.

Hypertension monitoring systems in primary health care are vital, as these take place during the patient’s first point of contact with integrated, comprehensive and continuous care. Tracking data in these services, ideally electronic, can play a pivotal role in monitoring and guiding improvements in service quality and coverage, patient outcomes and overall public health. WHO has strengthened guidance on facility-based patient monitoring through the *Noncommunicable disease facility-based monitoring guidance*, which includes a comprehensive framework and list of indicators, including the indicators for hypertension (27).

Regular monitoring of the health governance and infrastructure within a Member State is also of paramount importance in ensuring a strong and efficient response to hypertension in that it allows policy-makers and stakeholders to assess the capacity of the system to address hypertension.

Monitoring the health infrastructure ensures that facilities are adequately equipped, maintained, and staffed to meet the hypertension-related needs of the population. This process also assists in addressing disparities in access to

“ Strong health information systems that enable the continuous and systematic collection, analysis, interpretation and dissemination of data on hypertension are crucial in the race against hypertension. ”

health care and in strengthening health care systems’ resilience during emergencies. Ultimately, continuous monitoring of health governance and infrastructure plays a vital role in fostering a robust health care system, capable of providing optimal health outcomes and well-being for the entire population.

2.8 Summary

- High systolic blood pressure is the world’s leading risk factor for mortality.
- Hypertension affects one in three adults and presents with no symptoms – the “silent killer”. Almost half of people with hypertension are unaware of their condition.
- Over the past three decades, the number of adults aged 30–79 years living with hypertension has doubled to an estimated 1.3 billion. More than three-quarters of adults with hypertension live in LMICs.
- Only about one in five people with hypertension have controlled their hypertension.
- The world is not on track to meet the voluntary global target of a 25% reduction in the prevalence of raised blood pressure (uncontrolled hypertension) by 2025.
- Comprehensive data, including population-based surveys and death registration data, are essential to monitor hypertension in countries.
- Monitoring the performance of health governance and infrastructure assists in the creation of a robust health care system, capable of addressing pervasive health problems such as hypertension.

Benefits of hypertension interventions

Pharmacological treatment of hypertension is not new; drugs that treat hypertension have been around since the 1950s (28), and improved population blood pressure has been responsible for at least 20% of the decline in cardiovascular disease mortality in high-income countries over the past four decades (29).

Section 5 will demonstrate that hypertension management is a core element of primary health care, and that a practical, simple hypertension treatment approach quickly leads to improved hypertension control. Unfortunately, most health systems in LMICs were designed to provide episodic care for acute problems or routine services such as immunization, and not to provide care for lifelong, chronic conditions such as hypertension (30). As a result, effective coverage of hypertension care globally is low, with four out of five people with hypertension not having controlled blood pressure (1).

Providing drugs to adults with asymptomatic elevation in blood pressure rarely captures hearts and minds in the same way as some health problems do, but ignoring a public health problem as enormous as hypertension can cause huge strains on health systems over the long term because of costly downstream cardiovascular care demands that could have been avoided with preventive care. Section 5 highlights the WHO HEARTS technical package (31), which provides specific tools for planners and providers, including simple protocols for screening and treatment. We now have evidence that implementation of HEARTS in several demonstration sites across Latin America and the Caribbean, South and East Asia, Europe and sub-Saharan Africa raises hypertension control substantially, and that programmes can be scaled up relatively quickly if there is political and financial commitment. However, hypertension programme coverage and blood pressure control are only steps along the way to the ultimate goal of saving lives by preventing and treating hypertension. This section of the report makes the case for investing in hypertension control by exploring the potential health and economic benefits of implementing HEARTS recommendations worldwide.

“ Ignoring hypertension as a public health programme can cause huge strains on health systems over the long term. ”

3.1 Health benefits of hypertension prevention and control programmes

Decades of robust clinical trials have established the efficacy and safety of hypertension treatment. Numerous systematic reviews have summarized this evidence in guidelines (32). This section focuses on the impact of hypertension treatment on preventing future incident cardiovascular disease. A 2016 meta-analysis looked at this issue, and the evidence for risk reduction at different levels of blood pressure elevation (i.e. stages of hypertension). Overall, cardiovascular risk declines by 20% for every 10-mmHg reduction in systolic blood pressure, though the reduction is somewhat larger for stroke and hypertensive heart failure as compared to ischaemic heart disease, and there is a trend towards greater reductions among those with more severe hypertension. Table 3 summarizes the pooled effect sizes for these subgroups (33).

Using this evidence and recent experience with the HEARTS country demonstration programmes, a modelling study in 2022 assessed the potential impact of scaling up hypertension treatment coverage (34). For this section of the report the findings of that study were updated in several important ways:

- Cardiovascular mortality inputs were updated using the 2019 Global Health Estimates (35).

Table 3. Impact of a 10-mmHg reduction in blood pressure on risk of different cardiovascular outcomes and by level of blood pressure elevation

Outcome	Relative risk per 10 mmHg reduction in blood pressure (95% confidence interval)
<i>By type of cardiovascular outcome</i>	
Ischaemic heart disease	0.83 (0.78 - 0.88)
Stroke	0.73 (0.68 - 0.77)
Heart failure	0.72 (0.67 - 0.78)
<i>By level of blood pressure elevation for the composite outcome of major cardiovascular events^a</i>	
130–139mmHg	0.87 (0.82 - 0.92)
140–149mmHg	0.79 (0.72 - 0.87)
150–159mmHg	0.80 (0.71 - 0.91)
160mmHg	0.74 (0.69 - 0.79)

a. Fatal and non-fatal myocardial infarction, sudden cardiac death, revascularization, fatal and non-fatal stroke, and fatal and non-fatal heart failure

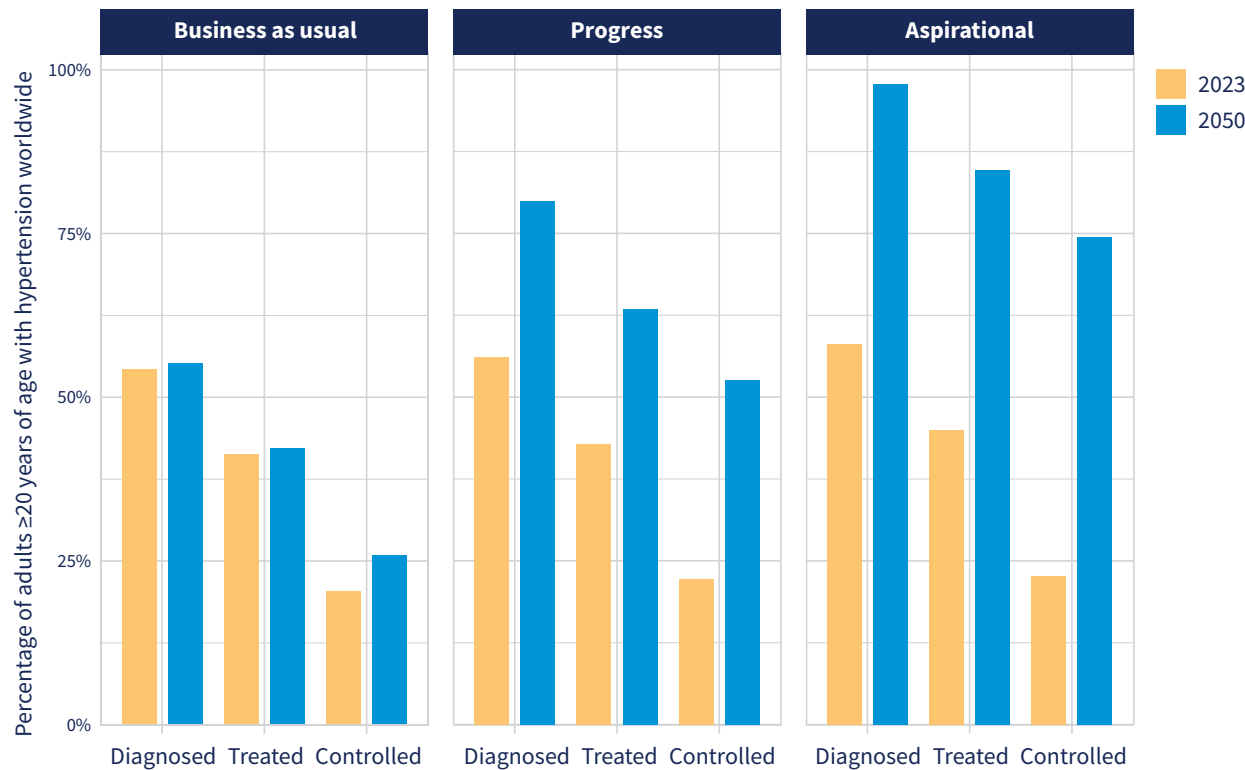
Note: data are from a meta-analysis by Ettehad and colleagues of large-scale blood-pressure-lowering trials (33)

- The nonfatal outcomes of myocardial infarction, stroke, and heart failure were added.
- Population health impact was summarized in disability-adjusted life years (DALYs).
- The cost of hypertension treatment, and the economic impact of hypertension treatment, were added.

In quantifying the health and economic gains from hypertension control, this report follows the assumptions of three different scenarios: the “business as usual”, “progress” and “aspirational” scenarios (34), (for details see Annex 1).

- **Business as usual:** The hypertension cascade of care (including trends in treatment and control) continues to improve gradually into the future, according to rates of change observed in the recent past (1). This scenario shows what would happen if Member States continued to scale up hypertension treatment at their current rates with no additional effort to scale up interventions, and it serves as a comparator for the progress and aspirational scenarios.
- **Progress:** The hypertension cascade of care improves at rates observed historically in high-performing countries (e.g. South Korea, Canada, and

Fig. 10. Required improvements in hypertension cascade of care to achieve 50% (progress scenario) or 75% (aspirational scenario) blood pressure control by 2050



Note: Business-as-usual scenario data are based on NCD Risk Factor Collaboration (NCD-RisC) estimates (1). Given uncertainty about scale-up during the COVID-19 pandemic, the values for 2023 are assumed to be the same as for the NCD-RisC estimates for 2019. Improvements in the cascade by 2050 are extrapolated from annual rates of change in hypertension treatment and control, by country, over 2010–2019.

“Improved global blood pressure control to $\geq 50\%$ would avert 76 million cardiovascular deaths, 120 million strokes, 79 million myocardial infarctions, and 17 million cases of heart failure between 2023 and 2050.”

others) (34). This scenario implies an acceleration of trends in improved access to quality hypertension care.

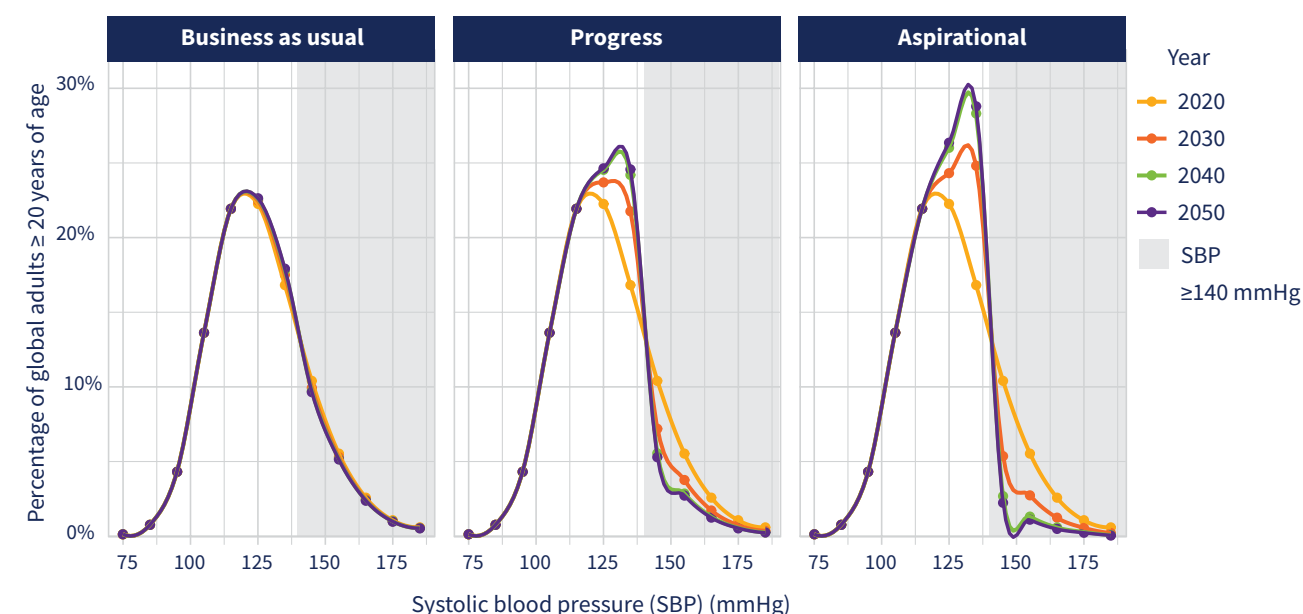
- **Aspirational:** Hypertension control accelerates to rates of improvement that are faster than observed historically in high-performing countries (for example, about 4% per year at intermediate levels of intervention coverage) but slower than the most successful global health programme: anti-retroviral drug therapy for HIV (about 5% per year).

The difference in projected health outcomes between the two “intervention scenarios” as compared to business as usual gives the deaths and nonfatal events averted. Fig. 10 illustrates how the cascade of care would evolve between 2023 and 2050 in the average Member State included in this analysis.

These improvements translate into a leftward shift in the population blood pressure distribution for the average Member State, illustrated in Fig. 11, so that most individuals with hypertension would have systolic blood pressure levels below 140 mmHg.

The consequences of these shifts on projected cardiovascular disease outcomes are shown in Table 4. The estimates are aggregated from 182 Member States that cover 98% of the world population. The figures are cumulative, beginning in 2023, when accelerations in hypertension care are assumed to start, and ending in 2050, the year in which nearly all Member States could feasibly achieve the targets specified in the progress and aspirational scenarios by replicating the successes of high-performing countries (34). Improved blood pressure

Fig. 11. Implications of improved hypertension care for global blood pressure distribution



Note: See notes for Fig. 10. In the progress scenario, improvements in hypertension treatment and control lead to a leftward shift in the population blood pressure distribution over time, which results in fewer incident and fatal cases of CVD. It was conservatively estimated that individuals whose blood pressure is newly controlled would have an SBP of 130–139 mmHg; some individuals will achieve greater reductions, so future population blood pressure levels will probably be more normally distributed.

control would avert 76 million cardiovascular deaths, 120 million strokes, 79 million myocardial infarctions, and 17 million cases of heart failure between 2023 and 2050.

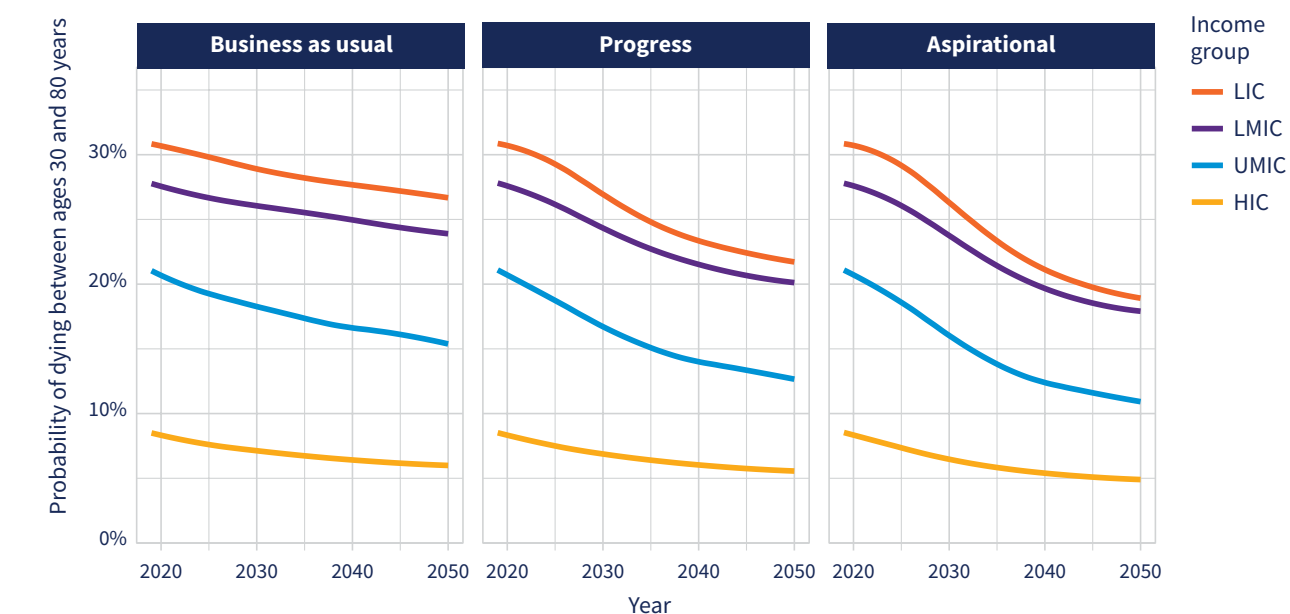
Better hypertension care would not only improve cardiovascular outcomes in the aggregate, but would also improve the distribution of these outcomes across Member States. Fig. 12 illustrates how the largest absolute reductions in cardiovascular mortality would be achieved in the least-developed countries, where mortality rates are the highest because of low access to care (36).

Table 4. Cumulative health gains (2023–2050) from improved population blood pressure control

Outcome	Projected number (progress scenario)	Projected number (aspirational scenario)
Nonfatal myocardial infarctions averted	79 million	140 million
Nonfatal strokes averted	120 million	200 million
Nonfatal cases of heart failure averted	17 million	31 million
Cardiovascular deaths averted	76 million	130 million

Note: Projections represent the difference in outcomes between the progress and aspirational scenarios as compared to the business-as-usual scenario, with results presented as the aggregate of 182 Member States. See Pickersgill et al. (34) for list of Member States included in the analysis.

Fig. 12. Reductions in cardiovascular mortality inequalities resulting from improved blood pressure control



Note: The plots show the changes in probability of death from cardiovascular disease between ages 30 and 80 (i.e. CVD-specific 50q30), a modification of the Sustainable Development Goal 3.4 indicator (37) in the business-as-usual and two intervention scenarios, disaggregated into the four World Bank income groups. LIC: low-income countries; LMIC: lower-middle-income countries; UMIC: upper-middle-income countries; HIC: high-income countries.

3.2 Costs and benefits of hypertension prevention and control interventions

Better access to hypertension care can provide obvious health benefits. It is also important to consider the costs and economic benefits of accelerated progress towards prevention and control of hypertension.

A recent systematic review summarized the findings of 71 studies on the cost and/or cost-effectiveness of hypertension treatment (38). While these studies employed a range of designs (including trials, cross-sectional analyses and modelling), together they provide a reasonably clear picture of the overall value for money in hypertension care.

There are large variations in the cost of hypertension treatment across countries. Differences in costing methods explain some of this variation, but a lot of the variation is due to different medication costs (38). The largest variations in prices among countries appear to be for ACE inhibitors and angiotensin II receptor blockers, especially in studies that looked at monotherapy with these agents.

A systematic review (38) identified 42 cost-effectiveness analyses conducted in 15 countries. Again, the studies reveal considerable variation in cost-effectiveness ratios, but most studies found that hypertension treatment cost less than US\$ 1000 per DALY averted. In studies that stratified cost-effectiveness ratios by individual patients' baseline cardiovascular risk, there was a trend towards better value for money in treating higher-risk subgroups. While this concept is not new (39), selectively treating high-risk hypertension might be impractical in real-world settings and raises important ethical problems in not giving treatment to patients with a life-threatening condition.

The incremental cost of scaling up hypertension care according to the assumptions of the progress scenario reported in Section 3.1 can be estimated using a range of studies. The cost of scaling up hypertension care would include the health system costs of implementing hypertension control programmes, including outpatient visit costs, costs of drugs and, the costs of delivering drugs. First, the cost of the drugs comes from the updated Appendix 3 of the Global action plan for the prevention and control of noncommunicable diseases (40, 41), specifically the intervention "Pharmacological treatment of hypertension in adults using either of the following: thiazide and thiazide-like agents; angiotensin-converting enzyme inhibitors (ACE-Is)/angiotensin-receptor blocker (ARBs); calcium channel blockers (CCBs)". Including health care delivery costs, the average unit cost of hypertension care (drugs and delivery costs) across all 62 countries included in the analysis is Int\$ 43 per person-year. With increased volumes and more efficient procurement, drug costs may be able to be reduced substantially.

The additional costs required to do population-based hypertension screening were calculated using default data on typical screening programme costs from the HEARTS costing tool (42). However, implementing a hypertension programme involves more than just the costs incurred at primary health care facilities. Health systems also need to be strengthened to provide screening and care. Estimates by a 2009 WHO task force were used to generate programme costs such as health management information systems, administration and financing (43). CHOICE estimates of the cost of strengthening supply chains were used to generate a markup for logistics (44). In total, the health system strengthening costs add 26% to the direct costs of service delivery.

The economic benefits from the progress scenario compared to the business-as-usual scenario were estimated by calculating the difference in the number of healthy life years remaining, based on estimates of fatal and nonfatal events averted, age-specific remaining life expectancy, and disability weights from the Global Burden of Disease Study. Following the human capital approach, the difference in DALYs averted were valued at the world average gross domestic product per capita, expressed in International Dollars.

Table 5 shows the results of the economic analysis. Achieving 50% population blood pressure control by 2050 (compared with the business-as-usual scenario) would cost about Int\$ 660 per DALY averted, on average, which is comparable to some of the best investments in global health – including interventions for the "unfinished agenda" of communicable, perinatal and nutritional diseases (45). Overall, the economic benefits of hypertension control could outweigh these costs by nearly 18 to one.

“The economic benefits of hypertension control could outweigh the costs by nearly 18 to one.”

Table 5. Total incremental costs and cumulative economic gains (2023–2050) from scenarios involving improved population blood pressure control compared to business-as-usual scenario

Outcome	Projected number (progress scenario)	Projected number (aspirational scenario)
Incremental costs (compared to current level of blood pressure control globally)	Int\$ 300 billion	Int\$ 630 billion
DALYs averted	450 million	720 million
Cost per DALY averted	Int\$ 660	Int\$ 870
Economic benefits	Int\$ 5.3 trillion	Int\$ 8.5 trillion
Benefit–cost ratio	18	14

Source: DALYs are calculated from World Population Prospects 2022 life tables (46) and average disability weights for different cardiovascular outcomes as reported in the Global Burden of Disease 2019 Study (47). Costs are taken from the Draft updated Appendix 3 to the Global action plan for the prevention and control of noncommunicable diseases (40), and are expressed in 2020 International Dollars. A DALY averted is valued at average global GDP per capita, also in International Dollars. The International Dollar is a hypothetical unit of currency that has the same purchasing power parity that the US dollar had in the United States of America at a given point in time. Costs and economic benefits are discounted at 3%.

The additional resources required to achieve these two scenarios would be considerable. The progress scenario would require an average of US\$ 6.7 billion annually worldwide.¹ The aspirational scenario would require an average of US\$ 14 billion annually worldwide. Achieving the greater population health benefits of the aspirational scenario would cost more, but could generate significant returns (14 to one).

3.3 Summary

- If countries mobilize to achieve the goal of 50% population hypertension control by the year 2050, 76 million cardiovascular deaths and 450 million DALYs would be avoided.
- Though the costs of achieving 50% blood pressure control are substantial, the economic benefits generated are estimated to be 18 times greater than business as usual.

¹ These costs are presented in US dollars rather than International Dollars since they represent the value of the monetary transactions that would be required to finance hypertension programmes worldwide. They are different from the incremental cost estimates in International Dollars in Table 6, which are discounted at 3% to construct cost-effectiveness and benefit-cost ratios.



Man reading leaflet promoting healthy diet at celebration of 2013 World Health Day in Timor-Leste.

Risk factors for hypertension

4.1 Population-based hypertension risk factors

The WHO *Guideline for the pharmacological treatment of hypertension in adults* notes the importance of addressing noncommunicable disease risk factors, such as a diet high in sodium and low in potassium, the consumption of alcohol and physical inactivity, in both the treatment and the prevention of hypertension (8).

This section presents some risk factors for hypertension, with the limitation that systematic reviews may not have been conducted for all of the approaches presented below.

4.1.1 Dietary sodium

Consuming a healthy diet throughout the life-course helps to prevent and control hypertension. The exact make-up of a diversified, balanced and healthy diet will vary, depending on individual characteristics, cultural context, locally available foods and dietary customs. However, the basic principles of what constitutes a healthy diet remain the same (48).

For hypertension specifically, reducing sodium and eating a diet that includes lots of fibre, such as wholegrain rice, bread and pasta, and plenty of fruit and vegetables high in potassium and nitrates also helps lower blood pressure (49).

Excess sodium intake increases blood pressure. Of all the deaths from cardiovascular causes that occurred in 2019, almost 2 million were attributed to sodium consumption that results in 24-hour urinary sodium excretion above the reference level of 1 to 5 grams per day (3). This is a larger number than from any other dietary factor (50).

Fortunately, blood pressure is reduced when dietary sodium intake is reduced and when potassium intake is increased. Reducing sodium intake is one of the most cost-effective ways to improve health, as it can avert many cardiovascular events and deaths at very low total programme costs (51).

In 2011, the United Nations General Assembly adopted a political declaration that committed Member States to the prevention and control of noncommunicable diseases. A reduction in sodium intake was noted as one of the top priority actions, and a 30% reduction in sodium intake by 2025, with an eventual target of <2000 mg/day worldwide, was adopted in 2013.

“Of all the deaths from cardiovascular causes that occurred in 2019, almost 2 million were attributed to daily sodium consumption above optimal levels.”

“Reducing sodium intake is one of the most cost-effective ways to improve health.”

The global average sodium intake is estimated to be 4310 mg/day (10.78 g of salt per day)(50), which far exceeds the physiological requirement and is more than double the WHO recommendation of <2000 mg/day of sodium (equivalent to <5 g of salt) in adults. WHO regional sodium estimates range from 2687 mg/day (6.7 g/day salt) in the WHO African Region to 6247 mg/day (15.6 g/day salt) in the WHO Western Pacific Region. In many high-income countries, and increasingly in LMICs, a significant proportion of sodium in the diet comes from processed foods such as bread, cereal and grains, processed meats and dairy products (50).

Sodium reduction policies and other measures

WHO provides several sodium-related best-buy policies as practical actions that should be undertaken immediately, to prevent cardiovascular disease and its associated costs (40).

These include:

- lowering sodium content in food products;
- implementing front-of-pack labelling to help consumers select food products with lower sodium content;
- conducting mass-media campaigns to alter consumer behaviour around sodium; and
- implementing public food procurement and service policies to reduce sodium content in food served or sold.

Box 1: Population sodium reduction programme

Shandong Province, China

The Shandong Ministry of Health Action on Salt and Hypertension (SMASH) programme was a government-led initiative aimed at curbing salt intake and reducing the prevalence of hypertension among adults. The programme employed various strategies, including a media campaign, distribution of scaled salt spoons and public education activities.

The results of the programme were remarkable, with a significant 24.8% reduction in sodium intake observed over a five-year period. This decrease was measured through 24-hour urine sodium excretion, which dropped from 5338 mg/day to 4013 mg/day. Additionally, potassium excretion increased by 15.1%, and the sodium to potassium ratio decreased by 37.7%, indicating positive dietary

changes. Beyond decreased sodium consumption, the SMASH programme demonstrated downstream improvement in blood pressure levels. The adjusted mean SBP decreased by 1.8 mmHg, and the adjusted mean DBP decreased by 3.1 mmHg, most likely due to decreased sodium intake.

Alongside these physiological changes, the programme positively impacted systematically assessed knowledge, attitudes and behaviours related to sodium reduction and hypertension.

Participants showed increased awareness of the recommended salt intake, paid more attention to the labelling on processed foods, and took action to reduce sodium in their diets.

A moderate reduction in sodium intake and increase in potassium intake causes a significant reduction in blood pressure and can prevent millions of premature deaths. Potassium-enriched salt substitutes is an affordable strategy to reduce sodium and increase potassium intake simultaneously, with proven benefit to reduce blood pressure and prevent cardiovascular events (52).

WHO monitors country progress in making national commitments and taking a multifaceted approach to implementing policies to reduce sodium intake. The result is published as the Sodium Country Score Card (51). In 2023, just over a quarter (27.6%) of the world's population lived in countries with any mandatory measures towards sodium reduction.

4.1.2 Alcohol consumption

Alcohol consumption ranked eighth among the mortality risks worldwide in 2019 (25, 3). The risks associated with alcohol use manifest in various ways, including injuries, liver toxicity and an increased risk of cancer.

The relationship between alcohol consumption and the incidence of coronary heart disease in different racial groups (53), along with other pieces of evidence, implies that the reduced risk observed among moderate drinkers of wine or other beverages could be attributed to a health cohort effect (54).

Although alcohol acutely lowers blood pressure, this is followed by a rebound increase a few hours after ingestion (55, 56). Furthermore, chronic alcohol consumption is linked to a high incidence of hypertension (57), even when consumed at low to moderate levels (58).

Due to the absence of clinical trials demonstrating alcohol's beneficial impact on major cardiovascular events, and considering the risks of hypertension and other diseases, alcohol intake to promote cardiovascular disease prevention is not supported.

WHO introduced the SAFER initiative in 2018, in collaboration with partners. "SAFER" (59) is an acronym for the five most cost-effective interventions to reduce alcohol-related harm, and stands for the following interventions:

- **Strengthen restrictions on alcohol availability**
- **Advance and enforce drink driving counter measures**
- **Facilitate access to screening, brief interventions and treatment**
- **Enforce bans or comprehensive restrictions on alcohol advertising, sponsorship, and promotion**
- **Raise prices on alcohol through excise taxes and pricing policies.**

“Chronic alcohol consumption, even at low levels, is linked to a high incidence of hypertension.”



Campaigners in Indonesia raising awareness of the harmful effects of tobacco use and exposure to secondhand smoke.

4.1.3 Tobacco use

Tobacco caused 8.7 million deaths globally in 2019, including 3.2 million deaths from cardiovascular diseases (60). The relationship between tobacco uses and blood pressure is complex. While scientific evidence on the impact of tobacco use on chronic high blood pressure levels is inconclusive (61), tobacco smoking and exposure to secondhand smoke acutely exert a hypertensive effect, mainly through the stimulation of the sympathetic nervous system (62, 63, 64). Similarly, research has shown that smokeless tobacco use may acutely elevate blood pressure (65, 66, 67, 68). These and other effects of tobacco use not only reduce life expectancy amongst those who consume tobacco and are exposed to tobacco smoke but also adversely affect quality of life.

Cessation of tobacco use is the most effective lifestyle measure for the prevention of many cardiovascular diseases and may have significant benefits for hypertensive patients (8). The WHO Framework Convention on Tobacco Control (FCTC) (69) is an evidence-based treaty that reaffirms the right of all people to the highest standard of health. Based on a review of the evidence, the MPOWER package was introduced in 2008 by WHO as a technical package to help countries implement the cost-effective demand-reduction measures of the WHO FCTC. The MPOWER technical package is made up of the following measures.

- **Monitoring tobacco use and prevention policies**
- **Protecting people from secondhand smoke**
- **Offering people help to quit**
- **Warning about the dangers of tobacco**
- **Enforcing tobacco advertising, promotion and sponsorship bans**
- **Raising taxes.**

All MPOWER measures are part of the NCD “best-buy” interventions. The WHO best-buys are considered affordable, cost-effective and evidence-based interventions that can help tackle the global burden caused by noncommunicable diseases.

4.1.4 Physical inactivity

Physical activity promotes many physiological responses that result in lowered risk of hypertension due to beneficial short- and long-term autonomic and haemodynamic adaptations. There is strong evidence of an inverse relationship between physical activity and incidence of hypertension among adults with normal blood pressure, and that physical activity reduces blood pressure among adults with prehypertension and normal blood pressure (70).

Physical activity is associated with hypertension based on evidence that increased physical activity improves physical function and reduces



cardiovascular disease progression and mortality in people living with hypertension (71). For example, compared with control groups of people who do not exercise, people with hypertension who are physically active can reduce systolic blood pressure by approximately 12 mmHg and diastolic blood pressure by approximately 6mmHg (72).

In addition, emerging evidence shows that people with hypertension who are physically active can significantly improve their health-related quality of life compared with those who are inactive (73). For adults living with hypertension, evidence supports aerobic activity, muscle-strengthening activity, and combinations of the two for improving cardiovascular disease progression (74).

Sedentary behaviour, on the other hand, and particularly for extended periods of time, is associated with higher all-cause mortality, cardiovascular mortality and cardiovascular disease incidence (74).

If the current prevalence of physical inactivity does not change, almost 240 million new, preventable cases of hypertension will occur globally from 2020 to 2030, which will place a burden of more than Int\$ 115 billion on the public health care system (75).

The WHO toolkit ACTIVE, which was launched in 2019, proposes policy options that can be adapted and tailored to local cultures and contexts to help increase levels of physical activity globally (76). These include:

- the development and implementation of national guidelines for physical activity for all age groups;

A “health ball” in Guarulhos, Brazil, arranged by the city council’s physical exercise team.

- establishing national coordinating mechanisms involving all relevant government departments and key non-government stakeholders to develop and implement coherent and sustainable policy and actions plans;
- implementing community-wide communication campaigns to raise awareness and knowledge of the multiple health, economic and social benefits of being physically active;
- investing in new technologies, innovation and research to develop cost-effective approaches to increasing physical activity, particularly in low-resource contexts;
- ensuring regular surveillance and monitoring of physical activity and policy implementation.

Further WHO publications offering guidance for policy-makers and programme managers include:

- the *Global action plan on physical activity (GAPPA) 2018–2030* (77), which sets out multiple policy actions to support people in being regularly active as part of work, transport (walking and cycling), recreation or play; and
- *Promoting physical activity through primary health care: a toolkit*, which promotes physical activity through brief counselling interventions in primary health care (78).

Many other interventions support and encourage physical activity, whether through promotion of walking and cycling, provision of public open space, or affordable and accessible digital programmes such as *Be he@lthy, be mobile: a handbook on how to implement mobile health for physical activity* (79).

4.1.5 Air pollution

Air pollution is one of the major risk factors for noncommunicable diseases (80); it accounts for an estimated 6.7 million deaths every year worldwide (81). Around the globe, 99% of the population is exposed to air quality that does not meet the levels recommended in the WHO global air quality guidelines (81) and 2.3 billion people globally relied primarily on polluting fuels and devices for cooking in 2021, which exposed them to dangerous levels of air pollutants, especially in LMICs where access to clean energy is still challenging (80, 82). The best indicator for estimating the health impacts of air pollution, fine particulate matter (PM_{2.5}), ranked fourth among mortality risks worldwide in 2019 (25). Cardiovascular diseases such as ischaemic heart disease and stroke are among the main diseases attributable to air pollution exposure (83). These are health outcomes for which hypertension is a well-known etiological factor. In addition, consolidated evidence shows an increased risk of high blood pressure in adults with each 10-µg/m³ increase in long-term exposure to PM_{2.5} (84, 85), while the evidence is limited and inconsistent for short-term exposure (86). Pre-existing conditions such as cardiovascular and respiratory diseases,

obesity, and low socioeconomic status increase the risk of experiencing the negative health effects of air pollution, particularly those related to PM_{2.5}.

While struggling to find consideration in most clinical guidelines, the burden of disease that can be attributed to air pollution should not be neglected by the public health community and it needs to be incorporated in guideline development, among other interlinked environmental and planetary health risk factors (87).

Improving air quality would constitute an opportunity for primary prevention of hypertension as well as other noncommunicable diseases. WHO has provided guidance on personal interventions and risk communication on air pollution (88), and training opportunities for the education of health professionals (89). Yet the major efforts needed to improve air quality remain to be addressed by countries and regions in reducing emissions of health-damaging air pollutants. WHO promotes interventions and initiatives for healthy sectoral policies (including energy, transport, housing, urban development and electrification of health care facilities), addressing key risks to health from air pollution indoors and outdoors, and contributing to achieving health co-benefits from climate change mitigation policies (90).

“The burden of disease that can be attributed to air pollution should not be neglected.”

4.2 Summary

- About one-third of the world’s adults live with hypertension and are at risk of heart disease, stroke, or death.
- WHO supports the reduction of dietary sodium (and increase in potassium), the reduction of alcohol consumption, the control of tobacco use, the promotion of physical activity, and a reduction in air pollution to prevent hypertension.
- The most effective ways to control these hypertension risk factors is via large-scale, population-level public health programmes.

Diagnosing and managing hypertension

5.1 Clinical presentation of high blood pressure and hypertension

Because it is usually asymptomatic, hypertension has appropriately been labelled the “silent killer”; unless their blood pressure is measured, most people will be unaware of their condition until they have a clinical complication such as a heart attack, stroke or kidney failure.

Cohort studies and reports using electronic record linkage have both documented a strong, continuous and progressive association between systolic blood pressure and the risk of CVD (91, 92, 93). The association has been documented at all ages for both systolic blood pressure (SBP) and diastolic blood pressure (DBP), for all major forms of CVD, and throughout the entire range of blood pressure (from SBP as low as 115 mmHg) (94). Classic cohort studies have shown that, starting at a blood pressure of 115/75 mmHg, the risk of death from a heart attack or stroke at ages 35–69 doubles with every 20-point increase in systolic blood pressure (95). For most patients, lower blood pressure is better, down to the 115–120 mmHg range, although for clinical practice considerations, a target of <140/90 mmHg is most common.

The significance of high blood pressure as a risk factor for CVD is modified by the presence or absence of other CVD risk factors (96). As a result, the absolute risk associated with elevated blood pressure is much higher at an older age, when it is common to find high blood pressure accompanied by several other CVD risk factors. However, high blood pressure at younger ages is also harmful, since long-term studies show that damage to blood vessels accumulates over a lifetime (97). In addition to CVD, high blood pressure is associated with many other adverse outcomes, including kidney disease, cognitive impairment and dementia (94).

5.2 Diagnosis and early detection

The starting point for living well with hypertension is early diagnosis. The longer a person lives with undiagnosed and inadequately treated hypertension, the worse the health outcomes are likely to be. Easy and sustained access to basic validated automated blood pressure measuring devices in primary health care

“Starting at a blood pressure of 115/75 mmHg, the risk of death from a heart attack or stroke at ages 35–69 doubles with every 20-point increase in systolic blood pressure.”

“The longer a person lives with undiagnosed and inadequately treated hypertension, the worse the health outcomes are likely to be.”

facilities is therefore essential, and diagnosis should be available in primary health care settings. Blood pressure measurements should be conducted by a trained staff member in a standardized way, with an appropriate cuff and the patient comfortably seated with their back supported, an empty bladder and legs uncrossed. It is efficient and acceptable to conduct two blood pressure readings at first and use results of the second reading to guide decisions about the need to schedule a follow-up visit to complete the diagnostic work-up.

If blood pressure is confirmed to be either ≥ 160 mmHg systolic or ≥ 100 mmHg diastolic on two readings on a single day, medications to reduce blood pressure should begin on that day. If blood pressure is in the 140–159 mmHg/90–99 mmHg range, the diagnosis of hypertension should be confirmed at an additional patient follow-up visit, usually 1 to 4 weeks after the first set of measurements.

In general, hypertension is diagnosed if, on the two visits:

- systolic blood pressure on both days is ≥ 140 mmHg and/or
- diastolic blood pressure on both days is ≥ 90 mmHg (98).

Counselling to improve diet, physical activity and other risk factors should be provided, but should not delay initiation of pharmacological treatment. Note that for people a pre-existing CVD and those without CVD but living with co-morbid diabetes, chronic kidney disease, or considered at high cardiovascular risk, the WHO *Guideline for the pharmacological treatment of hypertension in adults* (8) recommends also considering blood-pressure-lowering treatment if SBP is ≥ 130 –139 mmHg. It is now clear that reducing blood pressure to lower levels provides additional benefits to essentially all patients.

5.2.1 Early detection of hypertension

Evidence supports the benefits of early diagnosis and effective treatment of hypertension, both because vascular disease results from cumulative exposure to high blood pressure, and because earlier screening reduces the lead time between diagnosis and treatment (99). The asymptomatic nature of high blood pressure also means that active clinical screening in the health care system is essential if the necessary treatment for hypertension is to be provided.

Screening for hypertension is only useful when it is closely linked to follow up at a health care facility able to measure blood pressure accurately and treat high blood pressure effectively. Many individuals with high blood pressure in a population screening setting do not meet the criteria for hypertension when they are more carefully evaluated on follow up. Also, many individuals are lost to follow up; in population screening programmes in countries around the world, effective follow up of patients with abnormal blood pressure is the exception rather than the norm, which is why detection in health care facilities is the priority to improve hypertension control.

Population screening programmes increase the number of clinically diagnosed cases of hypertension and, as such, will increase the workload in the health care system, not only in terms of the process of early detection but, more importantly, in dealing with the increased number of clinically diagnosed cases who will be identified and will require long-term, usually life-long, care (100). With prevalence rates close to a quarter of the adult population globally (1), no population screening system should be established without consideration of whether local health care resources are sufficient to cope with delivering standard hypertension treatment services to the large number of extra patients identified. Simply adding new cases to a health care system without expanding the capacity for treatment will, in the absence of compensating efficiencies, result in low-quality care unlikely to result consistently in adequate control of blood pressure (101).

It is recommended that blood pressure measurements be done “opportunistically”: that is, on adults during routine visits to primary health care facilities, including all adults at first presentation to the facility, and, if normal, periodically thereafter (e.g. annually). Every patient with elevated blood pressure readings requires immediate follow up for treatment. Population-based screening is not recommended as the primary method for detection and early diagnosis of hypertension (102).

5.3 Management of hypertension

5.3.1 Guideline for the pharmacological treatment of hypertension

In 2021, the World Health Organization launched its *Guideline for the pharmacological treatment of hypertension in adults*, which provides recommendations on the threshold for the initiation of pharmacological treatment for hypertension, recommendations for the basis on which to decide whether to initiate treatment with monotherapy, dual therapy or single-pill combinations, as well as guidance for countries selecting medicines and algorithms for hypertension control for their national guideline for hypertension management.

The guideline provides implementation targets, including those on intervals for follow up, target blood pressure to be achieved for control, and the cadre of health care workers who may initiate treatment. Lastly, the guideline provides examples of a drug- and dose-specific protocol for managing hypertension which can be adapted based on country context (8). A summary of the 2021 WHO guideline recommendations is provided in Table 6.

5.4 HEARTS for hypertension control

WHO and the United States Centers for Disease Control (CDC) launched the Global Hearts Initiative in 2016 (103), including the *HEARTS: technical package for cardiovascular disease management in primary health care* (104) to support

“In population screening programmes, effective follow up of patients with abnormal blood pressure is the exception, which is why detection in health care facilities is the priority to improve hypertension control.”

Table 6. WHO guideline recommendations on the pharmacological treatment of hypertension in adults

1. Recommendation on blood pressure threshold for initiation of pharmacological treatment
WHO recommends initiation of pharmacological antihypertensive treatment of individuals with a confirmed diagnosis of hypertension and systolic blood pressure of ≥ 140 mmHg or diastolic blood pressure of ≥ 90 mmHg.
WHO recommends pharmacological antihypertensive treatment of individuals with existing cardiovascular disease and systolic blood pressure of 130–139 mmHg.
WHO suggests pharmacological antihypertensive treatment of individuals without cardiovascular disease but with high cardiovascular risk, diabetes mellitus, or chronic kidney disease, and systolic blood pressure of 130–139 mmHg.
2. Recommendation on laboratory testing
When starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment.
3. Recommendation on cardiovascular disease risk assessment
WHO suggests cardiovascular disease risk assessment at or after the initiation of pharmacological treatment for hypertension, but only where this is feasible and does not delay treatment.
4. Recommendation on drug classes to be used as first-line agents
For adults with hypertension requiring pharmacological treatment, WHO recommends the use of drugs from any of the following three classes of pharmacological antihypertensive medications as an initial treatment: 1. thiazide and thiazide-like agents 2. angiotensin converting-enzyme inhibitors (ACEis)/angiotensin receptor blockers (ARBs) 3. long-acting dihydropyridine calcium channel blockers (CCBs).
5. Recommendation on combination therapy
For adults with hypertension requiring pharmacological treatment, WHO suggests combination therapy, preferably with a single-pill combination (to improve adherence and persistence), as an initial treatment. Antihypertensive medications used in combination therapy should be chosen from the following three drug classes: diuretics (thiazide or thiazide-like), angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs), and long-acting dihydropyridine calcium channel blockers (CCBs).
6. Recommendations on target blood pressure
WHO recommends a target blood pressure treatment goal of $<140/90$ mmHg in all patients with hypertension without comorbidities.
WHO recommends a target systolic blood pressure treatment goal of <130 mmHg in patients with hypertension and known cardiovascular disease (CVD).
WHO suggests a target systolic blood pressure treatment goal of <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease).
7. Recommendations on frequency of assessment
WHO suggests a monthly follow up after initiation or a change in antihypertensive medications until patients reach target.
WHO suggests a follow up every 3–6 months for patients whose blood pressure is under control.
8. Recommendation on treatment by nonphysician professionals
WHO suggests that pharmacological treatment of hypertension can be provided by nonphysician professionals such as pharmacists and nurses, as long as the following conditions are met: proper training, prescribing authority, specific management protocols and physician oversight.

governments to strengthen the prevention and control of CVD and to facilitate achievements of the UN health-related SDG goals. The package is endorsed by 11 partner organizations: American Heart Association (AHA), Centre for Chronic Disease Control (CCDC), International Society of Hypertension (ISH), International Society of Nephrology (ISN), Pan American Health Organization, Resolve to Save Lives (RTSL), US CDC, World Heart Federation (WHF), World Hypertension League (WHL), WHO and World Stroke Organization (WSO). Hypertension control is a pathfinder for improved primary health care and universal health coverage.

The HEARTS technical package was launched in 2018 as the core technical package of the Global Hearts Initiative. The package is the public health approach to managing hypertension and other CVD risk factors, such as dyslipidaemia, at the primary health care level, with the aim of incorporating it into universal health coverage benefit packages. Six modules and an implementation guide make up the package: *Healthy-lifestyle counselling, Evidence-based treatment protocols, Access to essential medicines and technology, Risk-based CVD management, Team-based care and Systems for monitoring* (31).

The sections below will expand on each of these modules, highlighting relevant and country examples of contextualization and implementation.

5.5 Healthy lifestyle counselling

The WHO *Guideline for the pharmacological treatment of hypertension in adults* emphasises that non-pharmacological treatment is an important component of hypertension management. Smoking cessation, limiting use of alcohol, healthy diet and sufficient physical activity are advised for all newly diagnosed patients. An evidence-based diet to lower blood pressure in populations includes a lower-sodium diet and potassium-enriched foods. Recent evidence from China demonstrates that low-sodium, potassium-enriched cooking salts lower blood pressure and prevent stroke in people living with hypertension (52). However, the impact of diet and lifestyle changes is modest, as evidenced by primary-care-based trials that have demonstrated only a modest blood pressure-lowering effect (overall 1.3 mmHg lower systolic and 0.5 mmHg lower diastolic blood pressure) (105). Thus, nearly all people diagnosed with hypertension will continue to have uncontrolled blood pressure, despite making diet and exercise changes to lower blood pressure, and will have to take medications to keep their blood pressure controlled whilst still maintaining healthy lifestyles (106). Lifestyle changes may be considered as adjunctive therapy to antihypertensive medications, which will be essential in the care of nearly all patients diagnosed with hypertension.

Self-care may include making lifestyle changes, such as adopting a healthy diet and engaging in regular physical activity, as well as ensuring proper medication adherence and monitoring blood pressure.

“Lifestyle changes may be considered as adjunctive therapy to antihypertensive medications, which will be essential in the care of nearly all patients diagnosed with hypertension.”

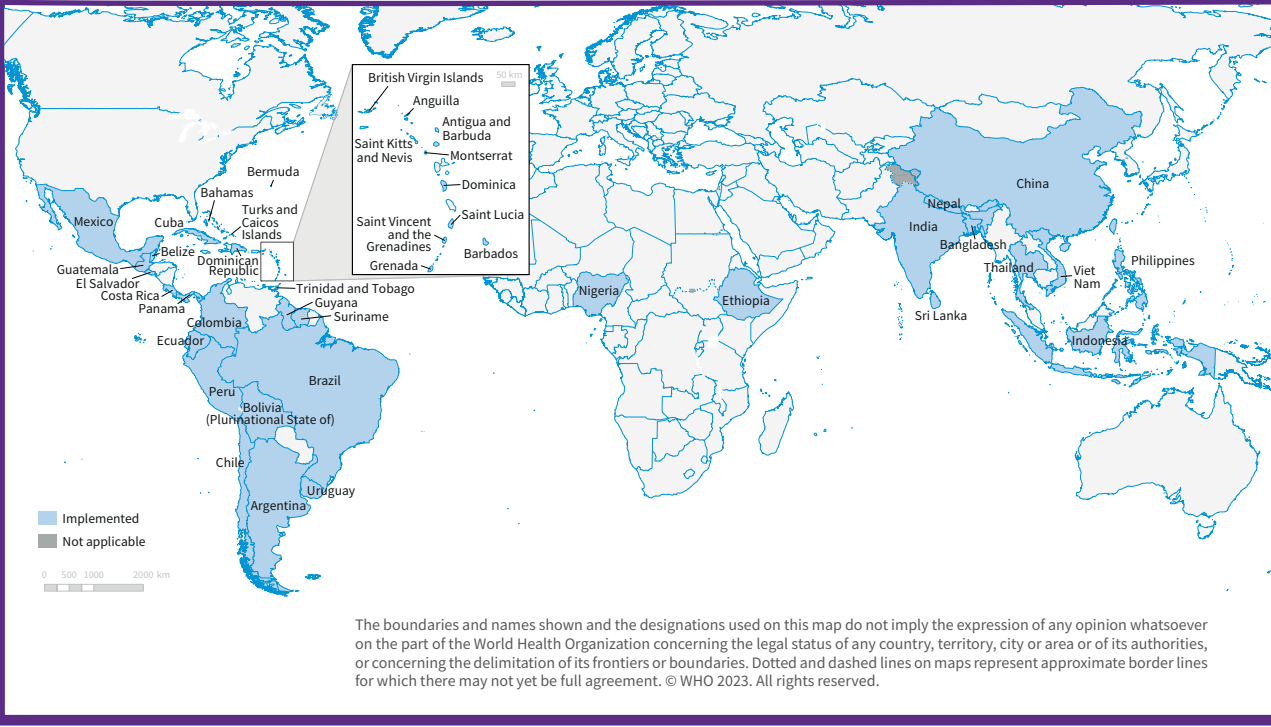
Box 2: Implementation of HEARTS

The roll-out of HEARTS across the world

The HEARTS technical package is being implemented in nearly 40 Member States. Approximately 35 000 facilities are implementing HEARTS. Regional programmes facilitate dissemination of best practices across countries within the region. Additionally, three regional programmes, HEARTS in the Americas (Pan American Health Organization/WHO), SEA-HEARTS (WHO South-East Asia Region) initiative and

the Signature Initiative (WHO European Region), demonstrate the accelerated commitments and movements to scaling up HEARTS.

Over 17.4 million patients are now enrolled in a contextualized HEARTS hypertension management programme and are being treated according to standardized treatment.



5.5.1 Evidence-based protocols

Simple evidence-based treatment protocols for the treatment of individuals with hypertension increase treatment effectiveness and efficiency (107, 108). Recent guidelines (8) and successful hypertension control programmes (109) improved on early, complicated treatment guidelines, and advise treatment using a simple evidence-based protocol with a linear progression of treatment intensification steps and specific medicines and doses. Medications recommended in protocols are selected based on value (low cost and high efficacy), availability, infrequent adverse event rates, reduced need for laboratory monitoring, and once-daily dosing. An additional benefit of a simple protocol is that it also delineates the return visit time (usually four weeks or less) to allow for rapid up-titration of doses and medications until hypertension control is achieved. It is becoming increasingly clear that, in addition to the overarching goal of improved hypertension control rate within a population, goals such as

“A simple evidence-based treatment protocol for hypertension clarifies the treatment steps for both patient and clinician, and helps reduce therapeutic inertia.”

reduced time to control and longer time kept under control also result in prevented events related to CVDs. A simple evidence-based treatment protocol for hypertension clarifies the treatment steps for patient and clinician, and helps reduce therapeutic inertia (a tendency not to act to intensify treatment despite blood pressure remaining uncontrolled).

Developing and adopting a standardized and evidence-based simple treatment protocol has additional programme implementation benefits. Training activities will be more efficient and simplified, as treatment becomes more standardized using a shared protocol. Because protocols identify a short-list of between two and three preferred medicines, medicine procurement,

inventory, forecasting, and supply chain can all be simplified. Because most patients should receive similar, standardized care for hypertension, attention to known health disparities can be assessed and addressed.

The process of generating wide consensus to develop, select or adapt a hypertension treatment protocol can catalyse launching, or recommitting to, hypertension control in the community adopting the protocol. Participants often include clinical leaders and multidisciplinary teams, consisting of academic leaders, medical system administrators, pharmacy and equipment procurement experts, clinicians (such as primary care clinicians, family physicians, cardiologists, internal medicine physicians and nephrologists), ministers of health or government representatives and other key stakeholders. In addition, selecting just two or three medications can result in substantial reductions in the unit cost of each, which can facilitate extending treatment and improving adherence as more patients are initiated into, and continue in, care.

“There is no component of a hypertension control programme more important than the adequate supply of good-quality, essential antihypertensive medicines.”

5.5.2 Access to essential medicines and health technologies

Improving access to medicines and health technologies for the diagnosis, management and treatment of hypertension is essential for ensuring universal access to quality health care. Ensuring an adequate supply of good-quality medicines and health technologies requires a robust health system, which includes good leadership and governance, adequate financing, access to information and evidence, quality service delivery, a strong health workforce, and equitable access to essential medicines and health products of assured quality, safety, efficacy and cost-effectiveness. When hypertension control programmes are expanding, good procurement practices play a key role in securing quality products at affordable prices and ensuring adequate and timely supply, while good supply-chain management ensures that quality products are available at all levels of the health system to meet the growing needs for medicines and health technologies (111).

“Prices for the same essential antihypertensive medicines vary by more than ten-fold between countries.”

Although about 80% of hypertensive patients reside in LMICs (112), the demand in LMICs for antihypertensive medicines is undermined by significant under-diagnosis, weak quantification, and under-developed treatment and follow-up programmes, along with at times weak procurement and supply-chain systems. Furthermore, in most LMICs, social security systems, when they exist, tend not to cover all citizens and/or not to reimburse all essential medicines and medical devices for noncommunicable diseases, including those for treating hypertension. As a result, in many countries, most of the costs of treatment, including medications, end up being paid out-of-pocket by people living with hypertension, who additionally may not recognize the need for lifetime medication when the disease is asymptomatic (113, 114).

There is no component of a hypertension control programme more important than the adequate supply of good-quality, essential antihypertensive medicines. Currently, prices for the same essential antihypertensive medicines vary

Box 3: Incorporating WHO guideline into simple treatment protocol

Sri Lanka’s adaptation of the WHO guideline

In Sri Lanka, 35.6% of adults have hypertension and 15.5% of them have their blood pressure under control (110). To tackle this burden, the Sri Lanka Ministry of Health began implementing the WHO HEARTS technical package for hypertension control in 2021. Along with key stakeholders, it developed a simple hypertension treatment protocol based on recommendations from the 2021 WHO *Guideline for the pharmacological treatment of hypertension in adults*. This was adapted to the Sri Lankan context with the introduction of a two-part protocol: for all adults aged 18–79 years old the five-step protocol follows the WHO recommendation to initiate therapy with a dual drug combination (Fig. 13), whereas for patients over 80 years old, the seven-step protocol recommends initiating treatment with an initial amlodipine monotherapy to avoid potential side-effects in elderly patients.

By 2023 the treatment protocol covered a population of 96 900 adults in Kalutara district and had registered 8600 patients in the programme; the cross-sectional facility-based blood pressure control rate was 57%.

The programme has a high retention rate – 86% of adults in the programme are returning for care every three months, which contributes in part to the high programme control. Lessons learned from the demonstration project include the ease of training on, and implementation of, a simple treatment protocol, the feasibility of treating hypertension at

primary care level, success with referring to higher levels of care when needed, and the straightforward monitoring of patient progress to blood pressure control due to clear blood pressure control goals.

“Due to the success of the demonstration project, the programme will be expanded to a new district and then scaled up nationally.”

Fig. 13. Simple hypertension treatment protocol, Sri Lanka, 2021–present



Box 4: Strengthening access to medicines

The India Hypertension Control Initiative (IHCI)

The IHCI is a collaborative initiative of the Ministry of Health and Family Welfare, Government of India, state governments, Indian Council of Medical Research (ICMR), WHO-India and Resolve to Save Lives as a technical partner. The IHCI was started in 2018 to improve hypertension control in the community using the WHO HEARTS strategy of evidence-based treatment protocol, uninterrupted drug supply, patient-centred care, task sharing and digital information system. IHCI launched in selected districts of five states and, as of June 2023, had scaled up to 155 districts of 27 states with enrolment of 5.8 million hypertension patients.

Although protocol-based treatment and minimum 30 days' prescription were key agreed and accepted strategies, availability of medicines emerged as the biggest challenge soon after initiation of the programme. This situation initially discouraged a patient from returning to facilities for follow-up visits in some areas. The programme was at risk of losing credibility.

In addition to the challenges of public procurement experienced in most countries, the conventional method of forecasting and procurement – adding a percentage to the prior year's use – was not adequate to meet the medication requirements of a rapidly scaled-up hypertension control programme. Also, there was no mechanism to facilitate linking patient load to drug stock in order to ensure a continuous supply at health facilities. The WHO-recruited consultants, in conjunction with the Ministry of Health and Family Welfare and partners, provided technical and training support on strengthening the medicine supply chain, including facilitating development of state-specific hypertension treatment protocols, inclusion of protocol drugs in states' list

of essential medicines, forecasting requirements of drugs, support for budgeting, facilitating last-mile distribution and developing a ready-reckoner tool linking drug requirements to the patient load (Fig. 14. Supply-chain strengthening interventions). These proved helpful in ensuring adequate stock and equitable distribution to all service delivery points. With an adequate and uninterrupted availability of drugs, it was possible to decentralize the programme to 18 000+ Ayushman Bharat Health and Wellness Centres for improved access to care.

Fig. 14. Supply-chain strengthening interventions

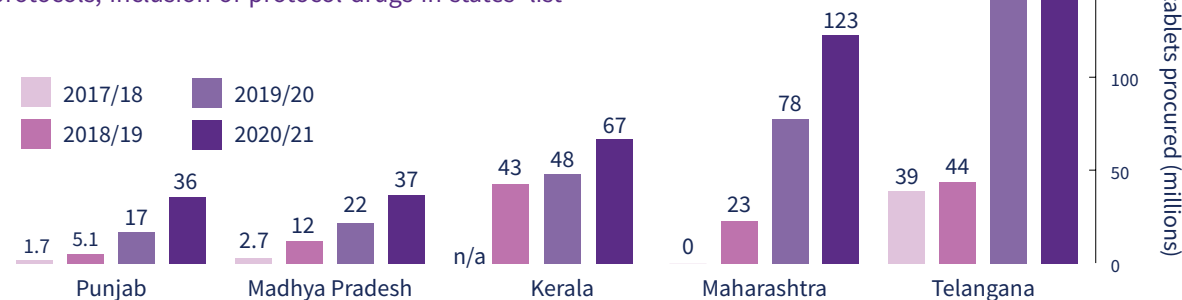


Source: The Indian Hypertension Control Initiative

Together, these efforts resulted in increased procurement of protocol medicines by the participating states (Fig. 15. Increment in procurement of amlodipine over four years in four states of India).

By 2020, the IHCI had ensured that more than 70% of health care facilities had ensured one month's stock of the protocol medicines, and fewer than 10% had experienced stock-outs.

Fig. 15. Increment in procurement of amlodipine over four years in four states of India



“Manual blood pressure measurement devices need to be replaced by independently validated electronic automated devices.”

by more than ten-fold between countries, and many country governments and hypertension patients are paying prices that are higher than necessary (115). Medicines for hypertension are too often not reliably available in primary health care facility pharmacies. A lack of standardized national protocols and treatment algorithms for hypertension that name preferred, evidence-based medicines incentivize a market fragmented into a vast number of formulations prescribed and sold within the same therapeutic classes, especially in the private sector. This leads to poor visibility of market opportunities for manufacturers, potentially resulting in irregular or no supply. Other potential factors include challenges to timely registration of medications in all countries (particularly in the case of single-pill combinations) or establishing a reliable supply chain and distribution in-country.

Components of the medicine supply chain system, including forecasting, procurement, storage, stock-management and distribution, need to be strengthened and maintained to avoid stock-outs and disruption of treatment. Additionally, there needs to be a robust monitoring system in place to ensure timely reporting, and so that these challenges can be identified early and mitigated promptly. Lastly, medication must be dispensed correctly and used rationally by the health care provider and patient alike (116).

Although there are differences in countries and regions, two of the basic anti-hypertensive medications, amlodipine and hydrochlorothiazide, are generally available at a price of US\$ 0.01 per tablet, or less. Losartan, telmisartan and enalapril, the other commonly used class of medications, can cost only slightly more. Therefore, a full year of treatment with quality-assured, state-of-the-art medications in most countries need not cost more than between US\$ 5 and US\$ 10 (114).

Finally, in markets where there are multiple manufacturers of varying quality, manufacturers of high-quality products may find it challenging to offer competitive prices, particularly where patients are purchasing medicines out-of-pocket. Quality health products are imperative and such sustainable production comes at a cost (117). At the same time, some quality products that are sold by the manufacturer at affordable prices may be subsequently rendered unaffordable by multiple markups in the supply chain (118). In the absence of consolidated procurement, manufacturers sell to distributors, who sell to wholesalers, who sell to dispensing points, often with substantial markups along the way. For example, one study in Kenya found highly variable pricing throughout the public and private health sector, with markups exceeding 340% on amlodipine and 280% on hydrochlorothiazide (119).

Box 5: HEARTS in the Americas

Using clinically validated automated devices in primary health care settings

HEARTS in the Americas is an initiative being implemented in 32 countries of the Region of the Americas with technical cooperation by PAHO. It includes approximately 3000 health centres and covers approximately 30 million adults (120). The goal is to improve population hypertension control and promote the secondary prevention of CVDs, with the emphasis on primary health care settings. Measurement of blood pressure is an important component of continuous quality improvement of the initiative, which includes clinical training, use of a protocol and clinically validated devices.

When the initiative was first launched, a small study showed that not only did countries not have the regulations necessary to ensure that only validated blood pressure monitoring devices (BPMs) were registered to be sold, but ministries of health and other health authorities had no record of the types of devices, brands and models that were being used in primary health care facilities (121). As in other parts of the world (122), the market in Latin America is dominated by non-validated BPMs.

To address that issue, HEARTS started working with both health authorities and national regulatory authorities, and this work evolved into a two-fold approach:

- efforts to introduce regulation of marketing authorization; and
- measures to regulate purchases of BPMs using public funds

A guidance document was published by PAHO to be used as a tool in this process (123) and the manifold strategy to improve the regulatory environment has been described elsewhere (124).

The initiative promoted the use of validated BPMs by increasing awareness of the importance of validation and accuracy, knowledge of what validation is, and mechanisms to verify whether a device has been validated or not.

While working towards implementation of regulatory measures, health authorities of several

countries began procuring validated BPMs with technical support from PAHO. Ministries of health that are not able to regulate purchasing of BPMs at the national level have issued recommendations for local health authorities to purchase validated BPMs exclusively.

Because of the increase in demand and the observed market limitations, the PAHO Strategic Fund has included selected validated BPMs in its list of products, and countries in the Region of the Americas can use its pooled procurement and financing mechanisms.

HEARTS is advocating for government transparency, which could be achieved by the creation or improvement of publicly available databases documenting brands and models granted marketing authorization and bought with public funds.

The challenges have been numerous, and include lack of knowledge and political will, market limitations, resistance by the medical profession, budgetary constraints, and the complexity of institutional and legislative environment, among others.

Results have been promising, as many countries are in the process of developing regulatory measures, while others have already changed their purchase profile, or are promoting validation through social media and training health personnel (125).

The importance of having exclusive use of clinically validated BPMs in primary health care has become part of the public health debate in the Region of the Americas.



5.5.3 Blood pressure monitoring devices

Although a deceptively simple aspect, major barriers in obtaining quality blood pressure measurements have been reported across low- and high-resource settings. These barriers refer to the type of devices used, the accuracy of the devices and a lack of knowledge in purchasing preferred devices.

Regarding the type of devices used, WHO published, in 2020 (126), clear technical guidance on the specifications of blood pressure measuring devices to use. Due to environmental concerns about mercury, key issues with poor calibration of aneroid devices, and improper manual blood pressure measurement techniques, manual blood pressure measurement devices need to be replaced by independently validated electronic automated devices.

Across many recent guideline documents (32, 127), the consensus is that due to these limitations of manual devices, automated upper-arm cuff monitors are strongly recommended, as they eliminate human error (e.g. caused by hearing impairment and digit preference) and require less training. In recent years, numerous new types of cuffless devices have entered the global market, including popular wrist-band devices. Despite the considerable potential of these devices to take many readings day and night without any user awareness, the use of these devices is not currently recommended for the management of hypertension due to remaining questions regarding their accuracy, performance and ability to track changes in blood pressure (128).

While there are many benefits of automated cuff-based electronic monitors, the most significant drawback is that only one in five of the available automated upper-arm and wrist-based cuff devices has evidence of validation testing for accuracy (122) and, in general, validated devices are more costly than those without validation. New technologies may make blood pressure measurement easier, but as of 2023 there is no alternative to upper-arm validated blood pressure cuffs that has sufficient consistency and accuracy to be recommended for diagnosis and management of hypertension in health care facilities.

The response to the challenge of access to medicine and technology requires a multi-faceted approach. Bulk procurement resulting from a protocol specifying the medicines required can drive down costs, even within a country, especially as most health care systems tend to become decentralized to state, district or local levels of government.

5.5.4 Risk-based management

The WHO *Guideline for the pharmacological treatment of hypertension in adults* strongly recommends that all hypertension patients should have blood pressure controlled to <140 mmHg systolic and <90 mmHg diastolic. The guideline also recommends further treatment to <130 mmHg systolic in patients with hypertension complicated by existing CVD, diabetes, chronic kidney disease, or

at high risk of developing CVD (8). Complete assessment of CVD risk in patients without a CVD diagnosis requires laboratory testing, which many low-resource settings do not have ready access to, therefore there is a potential to delay or miss the opportunity for treatment. The guideline therefore recommends initiating immediate treatment for hypertension on the day of hypertension diagnosis, based on raised blood pressure alone, followed by a CVD risk assessment (8). This new guidance is clear in that the approach to initiating treatment for hypertension does not have to be limited by a CVD risk approach and that dose- and drug-specific protocols with clear thresholds should be used for implementation in countries.

Box 6: Scaling up hypertension care in India

The role of specialist health consultants

The scale-up of the India Hypertension Control Initiative (IHCI) (see Box 4) was implemented at state and district level by government officials and staff, with technical support from specially trained consultants, recruited by WHO.

WHO consultants supported more than 20 states in the implementation of the IHCI and, as of June 2023, 27 states or union territories in India had developed a standard treatment protocol for hypertension, based on the WHO HEARTS technical package. In addition, a few states were also working on adding fixed-dose combinations to the protocol in order to improve patient adherence. WHO and other partners support states in calculating the annual drug demand for antihypertensives so that they can plan effectively for the implementation of the treatment protocols. Regular feedback given by WHO consultants on drug availability at facility and district level helps districts to maintain optimum drugs at every level.

The technical support provided by these consultants has played a pivotal role in training more than 8000 medical officers, 4500 pharmacists, 9000 staff nurses and more than 20 000 other health care workers on IHCI strategies, blood pressure measurement, use of standard protocol, follow-up care, drug stock management, team-based care, patient-centred care and counselling on lifestyle management.

WHO is assisting states to implement a national NCD portal, a uniform digital information system for

noncommunicable diseases such as hypertension, diabetes and three common cancers. Feedback about the NCD Portal, and expertise in this domain, is helping to improve uptake. As of June 2023, there were 5.8 million hypertension patients under treatment in IHCI implementation districts in 27 states.

WHO assisted states to plan decentralization of hypertension follow-up care to Ayushman Bharat Health and Wellness Centres – primary health care health units under national health mission. Drug refills to ensure continuity of care at these centres helped more than 38% of patients in the first wave of the COVID-19 pandemic (129). These centres are providing follow-up care to hypertension patients near to their village.

WHO consultants and government staff monitor the quality of programme implementation by supportive supervision to identify gaps and share good practices. Regular data analysis and feedback is crucial in data-driven policy level decisions.

With special focus on hypertension and diabetes care, the Ministry of Health and Family Welfare, Government of India has decided to put 75 million patients with hypertension or diabetes on standard care by 2025.

Consultants from WHO and the Government of India, as well as state and district governments, will play important role in achieving this ambitious target.

“Team-based care is effective in improving hypertension control among patients in a cost-effective way.”

The HEARTS module on risk-based management of CVD was launched in 2020 after updated risk-prediction charts were developed for 21 global regions. The charts are divided into non-laboratory-based and laboratory-based charts. The threshold for the level of CVD risk when treatment is required remains something for countries to decide upon using the laboratory-based charts (130).

Application of the total CVD risk approach will be better informed through country-level implementation research. Many areas, including the level of the facility at which CVD risk can be assessed (primary, secondary or tertiary), screening factors, thresholds for treatment, adaptation of protocols, follow-up intervals and other logistics are best defined in the local context (130).

5.5.5 Team-based care

The world is facing a substantial and ongoing shortage of trained health workers. WHO projects a global shortfall of 10 million health workers by 2030, mostly in LMICs (131). In order to provide patient-centred continuous care more effectively, primary care systems can include team-based care strategies in their clinic workflows and protocols.

Team-based care uses multidisciplinary teams (which may involve new staff or the shifting of tasks among existing staff). Teams can include patients themselves, primary care physicians, and other allied health professionals such as nurses, pharmacists, counsellors, social workers, nutritionists, community health workers and others. Teams reduce the burden on physicians by utilizing the skills of trained health workers. Strong evidence shows that team-based care is effective in improving hypertension control among patients in a cost-effective way. A degree of task shifting/team-based care is already taking place in many settings; WHO provides further guidance on how to maximize this approach for greater impact (132).

The context in which teams operate varies significantly between and within countries, as does the task allocation. Recommendation 8 in the *Guideline for the pharmacological treatment of hypertension in adults* encourages team-based care: “that pharmacological treatment of hypertension can be provided by nonphysician professionals such as pharmacists and nurses, as long as the following conditions are met: proper training, prescribing authority, specific management protocols and physician oversight (8). However, local implementation factors are fundamental when designing team-based care for hypertension management:

- Community health care workers may assist in tasks such as education, delivery of medications, blood pressure measurement and monitoring through an established collaborative care model. The scope of hypertension care practised by community health care workers depends on local regulations and currently varies by country.
- Telemonitoring, community- or home-based blood pressure monitoring, and health care worker support are encouraged to enhance hypertension

Box 7: Team-based care by local health workers

The barangay health workers' role in hypertension control in the Philippines

The Republic of the Philippines, a country composed of more than 42 000 barangays or villages, has long been implementing the delivery of team-based health care services, cascaded by the Department of Health. At the community level, barangay health workers serve as the first point of contact between the community and the health care system under the primary health care framework. They are community volunteers who provide primary health care services and cover at least 20 households at the minimum. They report to the barangay health station under the supervision of a nurse or a midwife. After complying with training requirements and being accredited by the local health board in their village, barangay health workers become official members of the health service delivery team.

Barangay health workers have significant roles in realizing universal health coverage in the country, where health services are provided to all, especially the poor and disadvantaged. The Republic Act 7883, otherwise known as the Barangay Health Workers' Benefits and Incentives Act of 1995, outlines their critical role in providing a voluntary health service in the community, where they are considered community organizers, educators, and frontline providers of primary health care at the grassroots level.

Cognizant of the roles of the barangay health workers in the community, the Philippines Healthy Hearts Programme (HHP) leveraged their support toward reaching its initial objectives – implementing the standard WHO HEARTS package in selected facilities within Iloilo Province, thereby improving population hypertension control in the catchment area. They were engaged as part of the health care team

implementing the following community-level interventions across the hypertension care cascade:

- Measuring and recording blood pressure during risk assessment of patients at the barangay health station or at home during house visits, after which blood pressure readings are reported to the nurse or midwife supervisor for collation and submission to the rural health unit or urban health centre;
- Providing education on the prevention and control of noncommunicable diseases, particularly hypertension, during patient visits;
- Informing the community on the availability of hypertension care services and medicines;
- Delivering medications to far-flung rural areas and to households where patients are unable to visit to regularly visit the health facility; and
- Tracking patients who have missed their quarterly clinic visits for hypertension treatment follow up.

With the expansion of the Healthy Hearts programme, the barangay health workers remain a driving force in reaching out to patients in the community.

A barangay health worker with patient.



“User-centred, simple digital information systems facilitate rapid capture of essential patient-level data, reduce health care worker data entry burden, and support rapid scale-up.”

management as part of an integrated chronic disease management system, if community-based care is deemed appropriate by the treating medical team and found feasible and affordable by patients.

- Physician oversight can be done through innovative methods such as tele-monitoring of patients and coordination with team members to ensure access to treatment is not delayed.
- When team-based hypertension care is delivered in the community, this allows hypertension patients with stable, controlled blood pressure to visit the clinic less frequently. Corresponding changes must be made to ensure access to essential antihypertensive medicines, namely multi-month refills (ideally, 90-days) and community or home delivery of medication refills.

5.5.6 Systems for monitoring

Effective hypertension control programmes require a robust information system to enable the monitoring of blood pressure control at the individual patient level and to assess progress on improving outcomes at the facility and health system levels (133). In many LMIC primary health care facilities, staff record and report data on inefficient paper-based systems. The creation of a secure, accurate, non-duplicated longitudinal patient record, readily available during clinical visits, is essential for the care of patients with hypertension, diabetes and other conditions requiring long-term care. In many LMIC primary care contexts, patient encounters last less than five minutes; digital information systems therefore need to be simple and quick to use by front-line health care workers.

User-centred, simple digital information systems facilitate rapid capture of essential patient-level data, reduce health care worker data entry burden, and support rapid scale-up. An optimal system replaces paper, so that health care workers only need to record the information once, reducing the risk of loss of data and of inaccurate data. One example of a user-friendly mobile application for hypertension management is the DHIS-2 Hypertension Tracker app used to monitor patients and programme performance (134).

Whether digital or paper-based, strong information systems provide rapid data feedback loops on key indicators to enable supportive supervision and continuous quality improvement. Focus on a list of core HEARTS indicators directs attention to the most important outcomes:

- number of hypertension patients enrolled from the catchment area (to estimate programme coverage);
- proportion of enrolled patients with controlled blood pressure (<140/90 mmHg in the past three months);
- proportion of patients not retained in care (with no visit in the past three and twelve months); and

- availability of core hypertension medications and blood pressure monitoring devices (135).

When programme managers receive monthly data on this limited set of core indicators, they can identify lagging facilities and implement programme improvements. HEARTS global and country leaders have identified that improving hypertension control requires formalized quality improvement teams, including supervisory staff and facility managers, who set annual outcome goals and plan interventions to address root causes and improve outcomes. The quality improvement strategy relies upon accurate, timely, actionable data to address core areas of hypertension quality gaps: a reliable supply of antihypertensive medications, retention in care, decentralization and therapeutic inertia.

Improved hypertension control demands retaining patients in care and maintaining them on effective treatment. Digital information systems can be leveraged as a powerful mechanism to support retention in care and facilitate team-based care. Loss to follow up is a leading barrier to hypertension control, often occurring in 50% or more of patients in low-resource settings (136). Digital tools can generate lists of patients overdue for care, send automated SMS text and telephone reminders, and prompt community outreach, thereby reducing loss to follow up. Evidence shows that digital interventions can also contribute to improving patients’ adherence to medication and their engagement with their treatment, can provide background information and increase patients’ adherence to their health provider’s guidelines, thereby increasing blood pressure control (137, 138).

The primary goal of a robust information system for large-scale hypertension programmes is ultimately to support clinicians and programme managers by providing data feedback loops to improve quality of care and thereby increase the proportion of patients whose blood pressure is controlled.

5.6 National scale-up of HEARTS for hypertension

As of 2023, multiple countries have successfully implemented the WHO HEARTS package, demonstrating progressively improved population coverage, diagnosis rates, linkage to care, and blood pressure control. However, to maximize the population health impact of the HEARTS hypertension control programmes and reduce deaths and disability, it is crucial to expand these successful demonstration programmes from limited areas to full national scale. HEARTS programmes in India and Cuba (see Boxes 6 and 8) have made significant progress towards achieving national scale, and other countries are poised to reach national scale.

When is a HEARTS hypertension control programme ready for national scale-up? Determining the readiness involves assessing key scale-up factors, which can be undertaken once the programme’s demonstration phase is completed.

Box 8: National scale-up of HEARTS

Cuba’s hypertension control programme

HEARTS in the Region of the Americas is a regional programme involving 32 countries in Latin America and the Caribbean. Cuba joined HEARTS in the Americas as one of four founder countries in 2016. The country developed a simple hypertension treatment protocol, starting with dual drug treatment, and leveraged a strong primary health care infrastructure centred on a polyclinic facility that offers both treatment and preventative care, and from which doctor and nurse teams deploy to serve the wider local community. Unique features of the Cuba HEARTS programme include health care team home visits to patients to follow up on hypertension care, local manufacturing of the protocol medicines and the design, validation, and manufacture of the automated blood pressure monitoring device used for the programme.

In 2016 the HEARTS package technical elements were introduced at a demonstration site and, in 2018, Cuba HEARTS expanded from one to three community primary health care facilities; the programme expanded to 22 facilities by 2021. By that time, 78% of the catchment area hypertension population were enrolled in the programme, and the programme achieved 59% facility-based blood pressure control.

Inspired by the success of the first expansion phase, the Ministry of Public

Health of Cuba declared 2022–2023 the “Year of Hypertension Control”, and HEARTS hypertension control programme expanded nationally to implement the same standard package in all 451 primary care polyclinics in the country. A national database for hypertension was established between September 2022 and January 2023, facilitating data audits in each area and identifying individuals with uncontrolled hypertension as baseline data for the new areas. National scale-up was supported by a strong public health infrastructure, team-based care, locally manufactured medicines and blood pressure monitoring devices, and a national campaign to inform and inspire health workers and the general population about the importance of controlling hypertension in communities.

By the beginning of 2023, out of an estimated 3 million individuals with hypertension in Cuba, over 2 million were currently registered under the programme.



A patient discussing treatment for her hypertension.

WHO provides general guidance on how to systematically analyse and support necessary actions for sustainable national scale-up using a systems approach (139, 140, 141, 142). For hypertension control programmes, the key components of readiness-for-scale include clinical capacity, delivery of a standard service package based on HEARTS, clinical quality of care, a reliable supply of protocol medicines sufficient to meet the demand from the anticipated influx of new patients, a robust health information system for tracking patients and monitoring the programme, and government commitment in terms of political will and financing (Table 7). At the same time, pragmatism is important. When political will and commitments align, HEARTS programmes can transition to a national scale-up process with most scale-up factors addressed, and minor gaps able to be filled in along the way.

5.7 Co-morbidities and hypertension

5.7.1 Diabetes

Diabetes is not only a risk factor for CVD; people with diabetes are particularly vulnerable to health damage if they also have untreated hypertension, which is common in people with diabetes in many countries (143). To support this evidence, WHO Member States have adopted global targets on diabetes including a target focusing on blood pressure: 80% of people with diagnosed diabetes have good control of blood pressure (144).

Table 7. Examples of specific factors influencing HEARTS hypertension control programme readiness-to-scale domains

Scale-up domain	Scale factor 1	Scale factor 2	Scale factor 3
Clinical capacity	PHC facilities	PHC staffing	Decentralization
Service delivery	Simple protocol	90-day Rx	Team-based care
Clinical quality	Blood pressure control >50%	Missed visit rate <25%*	–
Medication supply	Protocol medicine patient-days ≥30 in 100% of pharmacies	Facility digital drug inventory of protocol medicines	Morbidity-based medicine supply forecasting
Information system	Digital information system for managing hypertension	HEARTS indicators integrated into information system	Hypertension programme outcomes synthesized in programme-wide dashboard
Political will	Hypertension services included in national universal health coverage package	Medicines free for patients or minimal out-of-pocket cost	No out-of-pocket fees for hypertension follow-up visits

*Defined as no return for scheduled clinic visit in the prior three or six months

Given the overlap of diabetes and CVD risk factors, combined management for risk of both conditions is a rational approach. The decision to systematically look for people at high risk of diabetes and CVD is a strategic one specific to each health care setting, and will depend, at least in part, on the number likely to be identified and the resources available to adequately care for them.

Uncontrolled hypertension and high cholesterol – not hyperglycaemia – are the risk factors most strongly associated with premature death and disability among people living with diabetes. Treatment for hypertension is among the best-value treatments for people living with diabetes, as the treatment is inexpensive and treatment lowers the risk of CVD, the biggest killer for people with diabetes (143). Treatment of hypertension can also reduce the risk of kidney failure, an important and common complication of diabetes. Primary health care services need to be equipped to manage people with type 2 diabetes.

5.7.2 Chronic kidney disease hypertension

The bidirectional relationship between hypertension and chronic kidney disease (CKD) is an interplay involving intricate pathophysiological mechanisms and mutual exacerbation of each condition. Hypertension is a well-established risk factor for development and progression of CKD and is responsible for the second-highest global age-standardized rates of DALYs related to CKD (145). Optimal treatment and control of hypertension with commonly available therapies are critical to reducing the risk of CKD progression.

On the other hand, the appearance of hypertension can often be the first indicator of CKD, as renal parenchymal disease is the most common cause of secondary hypertension. Therefore, people with newly diagnosed hypertension should be screened for the presence of kidney disease whenever possible. The interplay between hypertension and CKD creates a vicious cycle, with each condition exacerbating the other. This bidirectional association underscores the imperative for early CKD and hypertension detection, rigorous blood pressure control, and optimal management of CKD to prevent disease progression and thereby avoid or delay downstream cardiovascular events and renal complications. Guideline-recommended blood pressure targets are more stringent in people with CKD to optimise the preservation of kidney function over time and protect from adverse cardiovascular outcomes. For patients with heavy proteinuria and renal damage, drugs that block the renin-angiotensin system are particularly valuable in this regard (146).

5.8 HIV and hypertension

Ageing among people living with HIV, coupled with the epidemiological transition underway in many LMICs, has led to an increased burden of CVD (147, 148, 149). Hypertension is the most prevalent CVD risk factor among people living with HIV; globally, an estimated 27% of people living with HIV on antiretroviral

therapy (ART) also have hypertension (150), and it is estimated that only one quarter of these individuals are receiving treatment. People living with HIV may be at higher risk of CVD than HIV-negative adults owing to the higher prevalence of traditional CVD risk factors, adverse effects of some antiretroviral therapy medications, and HIV-related chronic inflammation (147, 151).

National HIV programmes, in collaboration with partners such as PEPFAR and the Global Fund, have already successfully strengthened health systems in LMICs for HIV care and treatment, and these can be leveraged to improve care for hypertension among people living with HIV via integration (148, 149, 152): the coordination, location or simultaneous delivery of HIV and hypertension services to patients who need it, when they need it (149). This makes sense because of the marked similarities in the care cascade for both conditions. Furthermore, integrated treatment models reduce the number of clinic appointments people living with HIV with comorbid disease must keep, promote early hypertension diagnosis and treatment, reduce service duplication which may lead to cost savings, allow providers to have a comprehensive view of their patient's history, and promote patient-centred care (153, 154). HIV-hypertension integration is also highly effective for blood pressure control without negatively impacting HIV disease outcomes (155, 156, 157, 158).

Once hypertension has been diagnosed, the provision of organized and evidence-based health services can ensure that people with this chronic condition receive key interventions and regular follow up necessary to minimize complications and help them live long and healthy lives.

Box 9: HIV and hypertension

The integration of HIV and hypertension care, based on the WHO HEARTS technical package, has demonstrated that high levels of HIV viral suppression can be maintained while simultaneously achieving high blood pressure control.

A HIV-hypertension integration pilot in Uganda resulted in control of 73% at 24 months, up from 5.1% at baseline, while maintaining an HIV viral load suppression rate of 98%. Ninety-six percent of the patients were also retained in care while receiving integrated multi-month dispensing for both hypertension and HIV medications within a differentiated service delivery model (155).

This arose from a two-year grant awarded in 2019 by Resolve to Save Lives (RTSL) to Makerere University Joint AIDS Program (MJAP) to pilot HIV-hypertension care at Mulago ISS, the largest HIV clinic in Uganda. Uganda is home to about 1.4 million people living with HIV with an HIV prevalence of 5.1% among adults aged 15–49 years (159; 160). Prior to the pilot, while more than 95% of the clinic's HIV patients were virally suppressed, of the 24.4% of patients diagnosed with hypertension, only 1% were initiated on treatment. Of those initiated on treatment, only 5.1% had achieved blood pressure control (155; 156).

5.9 Summary

- When hypertension is diagnosed, simple, low-cost treatments are available to control blood pressure and prevent heart disease, stroke or death.
- The WHO-HEARTS technical package has been implemented in nearly 40 Member States and demonstrates that hypertension can be controlled in large-scale national programmes based in primary health care facilities and communities.
- Little more than five years after the HEARTS package was introduced, country governments are mobilizing the resources to grow HEARTS-based hypertension control programmes to national scale.

Conclusion

This first WHO *Global report on hypertension* underscores the enormous scale of the hypertension problem, which affects populations across the world, and the potential to reverse current trends. The political basis for concerted action to address hypertension is there, woven into targets and actions on cardiovascular diseases in the Sustainable Development Goals, the 2011 Political declaration of the high-level meeting of the General Assembly on the prevention and control of noncommunicable diseases, the WHO *Global action plan for the prevention and control of noncommunicable diseases* and the WHO Implementation roadmap 2013–2030. Where they are built upon, these foundations will catalyse action by all.

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Annex 1. Methodology for analysis of impact and cost

The methods used in the analysis of hypertension treatment impact and cost have been detailed in previous publications (1, 2). This annex provides a brief overview, including adaptations made for Section 3 of this report.

The hypertension outcomes model used a combination of state-transition modelling and demographic projections methods. We modelled the evolution of population demography, blood pressure distributions, and fatal and nonfatal cardiovascular disease (CVD) by age and sex. We looked at mortality from four specific CVD causes, ischaemic heart disease (IHD), hypertensive heart disease (HHD), ischaemic stroke, and haemorrhagic stroke. Additionally, we looked at nonfatal myocardial infarction (MI), stroke, and heart failure (HF). One hundred and eighty-two Member States comprising 98% of the world population were included in our analysis.

As described by Pickersgill and colleagues (1), we used four state-transition (Markov) models to represent the risk of incident and fatal CVD from the four causes listed above. Each model includes four mutually exclusive states: “well”, “sick”, “dead from cause X” and “dead from any other cause”. The transition probabilities within these models were specific to each age, sex, and location. We calculated the transition probabilities using data from the Global Burden of Disease (GBD) 2019 study (3), which captures the relationship between new CVD, prevalent CVD, and fatal CVD risk over time within 5-year age cohorts. Transition probabilities were calibrated by running our model over 2010–2019 and making post hoc adjustments to the transition probabilities to maximize the fit to “observed” GBD 2019 values. Pickersgill and colleagues provide more details on the models and relevant formulae (1). Nonfatal event rates were modelled as a function of fatal event rates using data from the Prospective Urban-Rural Epidemiological (PURE) Study (4).

The state-transition models were then linked to a demographic projection model that tracked survival in age cohorts over time as they age. Population and all-cause mortality data were taken from the World Population Prospects (WPP) 2022 Revision, and the base-case (i.e. no additional scale-up) demographic projections in the model were calibrated to WPP medium variant projections over the period 2020–2050. We incorporated excess mortality from COVID-19 into our model as a temporary “shock” in all-cause mortality, as described by Watkins and colleagues (2).

Mortality levels for each CVD cause were re-scaled from GBD 2019 values to

values reported in the Global Health Estimates 2019 (5). We assumed that the age-, sex-, and location-specific incidence of each CVD cause would be constant over 2020–2050 (at 2019 levels). We assumed that case-fatality from each CVD cause, as well as background mortality, would continue to (modestly) decline over 2020–2050, following average annual rates of change observed over 2010–2019, reflecting continued advances in CVD care and in health care more generally.

To generate population blood pressure distributions by age, sex, and location, we used hypertension “cascade of care” data and mean systolic blood pressure estimates from the NCD Risk Factors Collaboration (NCD-RisC) (6) along with estimates of age-specific and sex-specific 5th-percentile blood pressure levels in populations free of CVD and anti-hypertensive drug use. These parameters, and an assumed Gaussian distribution, allowed us to compute the prevalence of raised blood pressure corresponding to increasing levels of risk (e.g., 130–139mmHg, 140–149mmHg, and so on). We combined these distributions with estimates of blood-pressure-specific reductions in the incidence of CVD, as described in a recent meta-analysis of hypertension treatment trials (7). This allowed us to model the reduction in CVD incidence that would result from a greater proportion of the population with hypertension having controlled blood pressure. Pickersgill and colleagues provide more details on this procedure and relevant formulae (1).

In addition to modelling a reduction in CVD incidence, we modelled a reduction in CVD case-fatality from greater use of antihypertensive drug therapy, i.e. secondary prevention. We assumed a 26% relative reduction in case-fatality among persons with IHD without heart failure (8) (70% of IHD cases) and a 20% relative reduction among persons with heart failure (9). We assumed 36% and 76% relative reductions in case-fatality from ischaemic and haemorrhagic stroke respectively (10). Finally, we assumed the same effect size for HHD as for IHD with heart failure.

Pickersgill and colleagues explore historical trends in the rate of improvement in the hypertension cascade of care and describe a series of assumptions around realistic scale-up patterns for hypertension care (1). In brief, they looked at the rates of change in blood pressure treatment and control observed in high-performing countries (e.g. Canada) in the NCD-RisC data, as well as recent experiences implementing the HEARTS technical package in several countries. They modelled annual increases in hypertension coverage using a parabolic function, which implies that improvements in hypertension coverage occur faster at intermediate coverage levels as compared to low or high coverage levels. Practically, this approach is intended to represent the slowing of progress that countries inevitably experience as they expand health care to harder-to-reach populations, meaning that our projections are conservative.

Finally, we estimated the cost of hypertension care using our model and cost data from the Updated Appendix 3 of the Global action plan for the prevention and control of noncommunicable diseases (11). Specifically, we obtained estimates of the “unit cost” of hypertension treatment per person per year; these were multiplied by hypertension prevalence counts and treatment coverage rates to compute total costs. These estimates assume all persons with hypertension would be on a calcium-channel blocker and 60% would require a second medication, assumed to be an angiotensin II receptor blocker.

The additional costs required to do hypertension screening were calculated using default data on typical screening programme costs from the HEARTS costing tool (12). However, implementing a hypertension programme involves more than just the costs incurred at primary health care facilities. Health systems also need to be strengthened to provide screening and care. Estimates by a 2009 WHO task force were used to generate programme costs such as health management information systems, administration and financing. CHOICE estimates of the cost of strengthening supply chains were used to generate a markup for logistics (13). In total, the health system strengthening costs add 26% to the direct costs of service delivery.

The difference in costs between scenarios (progress or aspirational vs. business as usual) gives the incremental cost of accelerated progress.

As noted in the report, incremental health gains were converted into disability adjusted life-years, which were multiplied by average gross domestic product per capita to give the incremental economic benefits and benefit–cost ratio, calculated as DALYs using established procedures for cost-effectiveness analysis, but without an age-weighting function (14). (Of note, the DALYs estimated for economic evaluation are not directly comparable with the DALYs estimated for burden of disease analysis since they use different reference life-expectancy values, and disease-burden DALYs are not discounted.) We discounted both costs and benefits at 3% when computing cost-effectiveness and benefit–cost ratios.

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Annex 2. Hypertension profiles and explanatory notes

Background

Hypertension – also known as high blood pressure – is one of the world’s leading risk factors for death and disability. The aim of the hypertension profiles is to highlight the current status of hypertension for each WHO Member State (194 total) curated from multiple data sources. These profiles help to establish a baseline for the country and support monitoring trends toward the prevention and control of hypertension as a public health problem.

The profiles include the following information:

- **Morbidity and mortality:** prevalence of hypertension; coverage of hypertension diagnosis, treatment and control; trends in the prevalence of uncontrolled hypertension; hypertension control rate scenarios; probability of premature mortality from NCDs; cardiovascular disease (CVD) deaths; and the percentage of CVD deaths attributable to high systolic blood pressure;
- **Risk factors:** related risk factors including dietary salt intake, tobacco use, obesity, alcohol consumption and physical inactivity prevalence; and
- **National response:** national targets for blood pressure (BP) and salt consumption; national survey on BP/hypertension and sodium intake; functioning system for mortality data; guidelines for hypertension management.

All data are rounded to the nearest whole number; additional rounding was carried out on some statistics as described in the explanatory notes.

For further details see the Explanatory notes at the end of the document.

Afghanistan

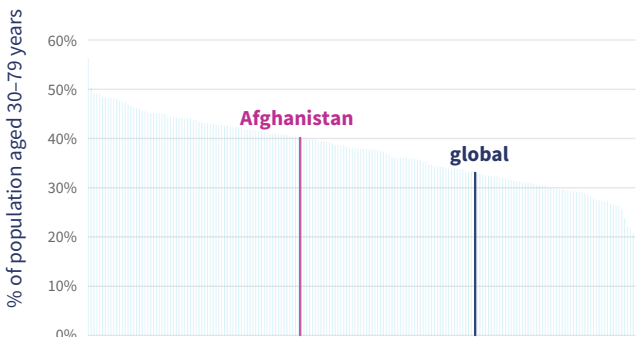
Hypertension profile

Total population (2019): 37 769 000

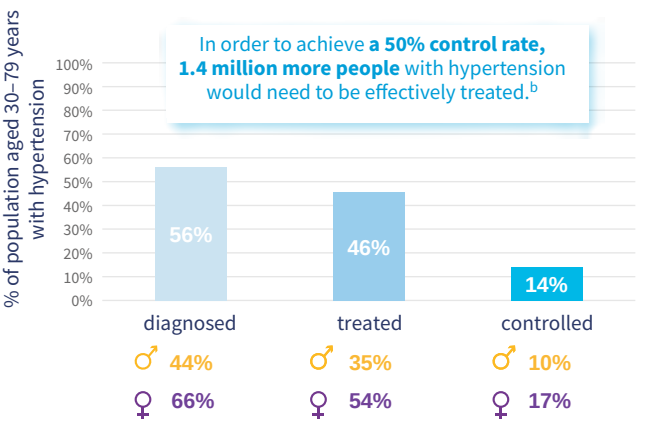
Total deaths (2019): 255 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 40% ♂ 35% ♀ 45%

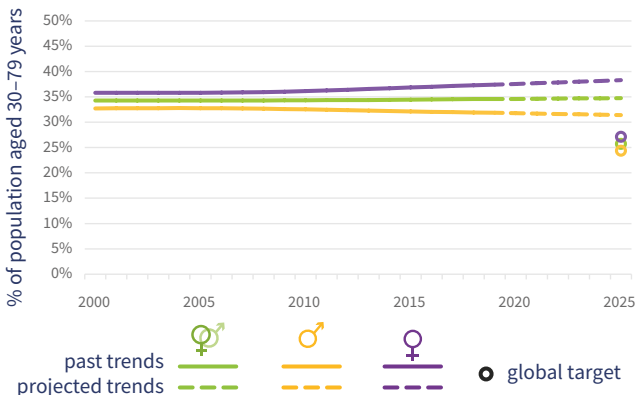
Prevalence of hypertension – global comparison (both sexes)^a



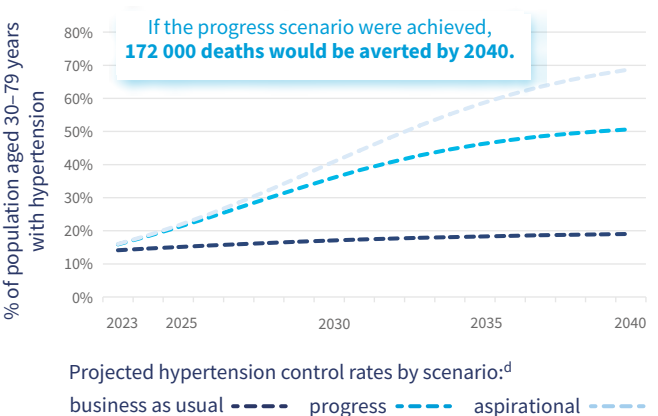
Of the 3.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality				
	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	35	34	36	2019
Cardiovascular disease deaths	71 300	33 800	37 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	52	55	2019
Risk factors ^e				
	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	24	40	8	2019
Obesity, adults aged 18+ years (%)	6	3	8	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016
National response				
Targets		Treatment		
National target for blood pressure	✓	Guidelines for management of hypertension		
National target for salt consumption	✓			
Surveillance				
Conducted recent, national survey measuring raised blood pressure/hypertension	✓			
Conducted recent, national survey on salt/sodium intake	✓			
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗			

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Albania

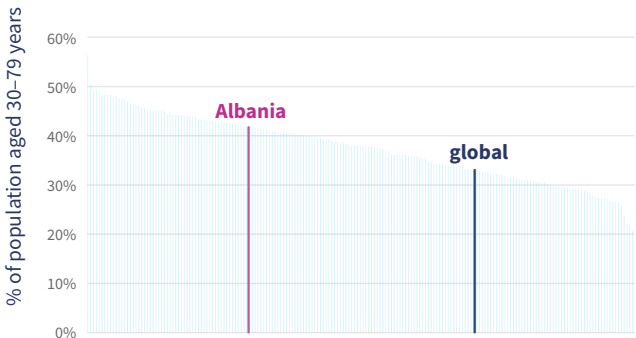
Hypertension profile

Total population (2019): 2 874 000

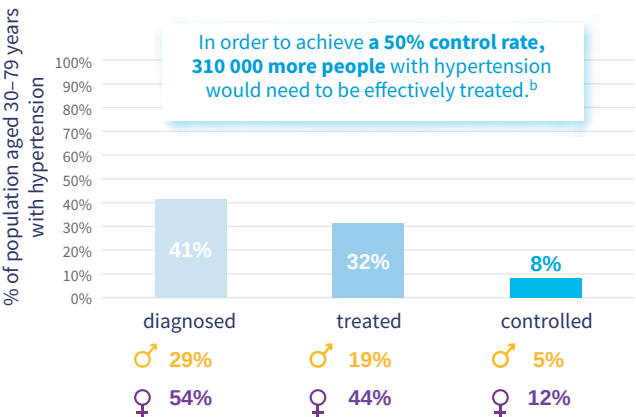
Total deaths (2019): 31 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 42% ♂ 43% ♀ 41%

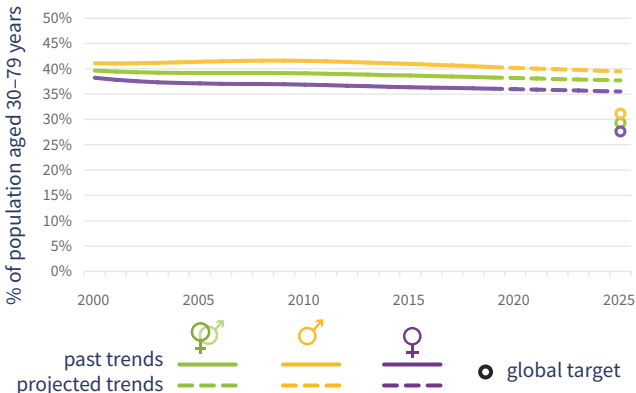
Prevalence of hypertension – global comparison (both sexes)^a



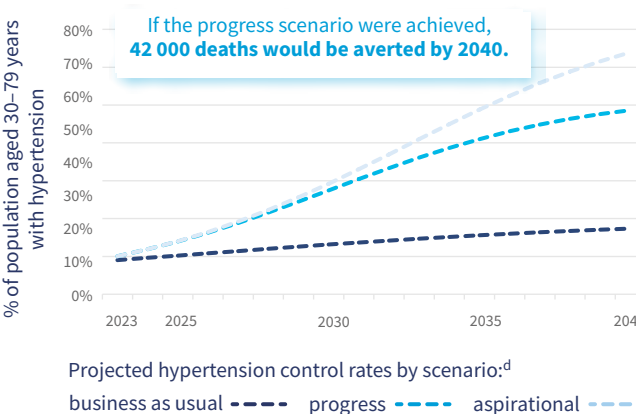
Of the 757 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	17	6	2019
Cardiovascular disease deaths	19 500	8300	11 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	57	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%) ^f	23	40	6	2019
Obesity, adults aged 18+ years (%)	22	22	22	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	8	2	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✓
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Algeria

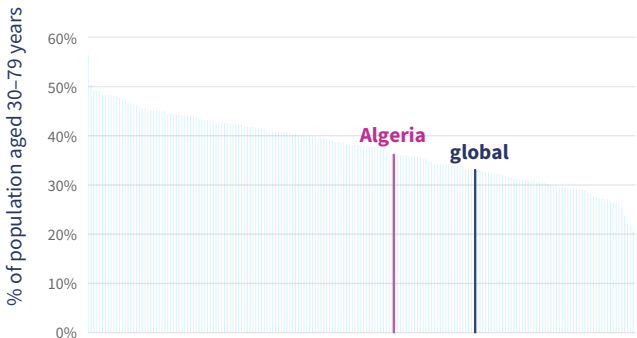
Hypertension profile

Total population (2019): 42 705 000

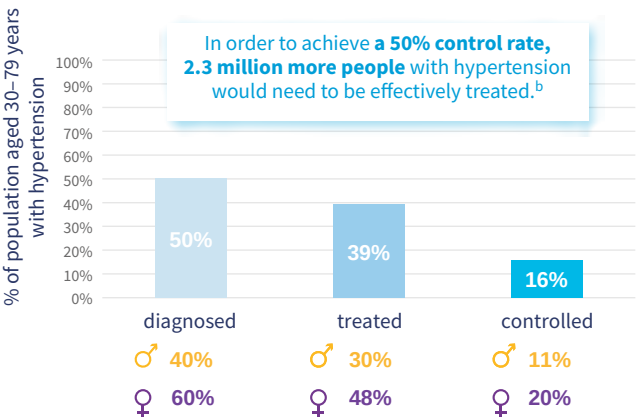
Total deaths (2019): 203 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 36% ♂ 35% ♀ 37%

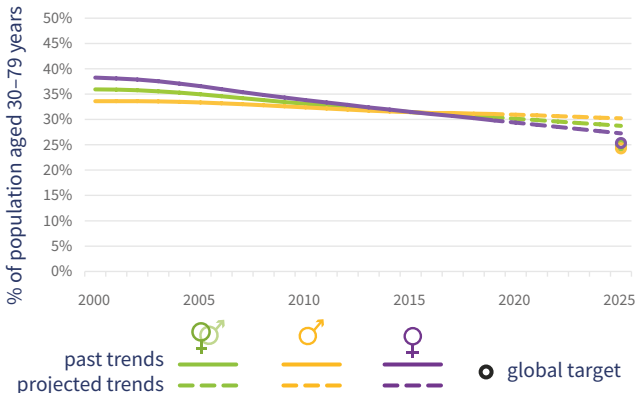
Prevalence of hypertension – global comparison (both sexes)^a



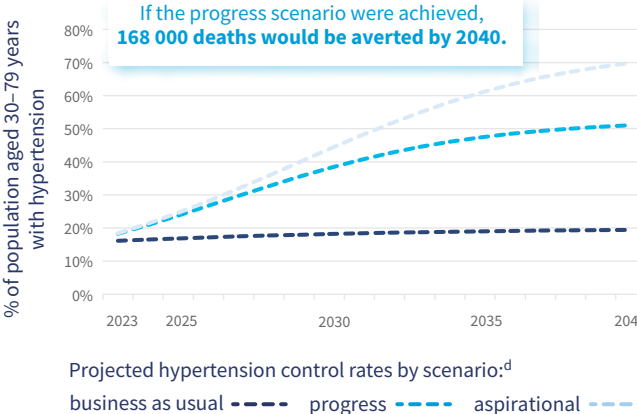
Of the 6.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	14	15	13	2019
Cardiovascular disease deaths	91 500	46 500	45 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	53	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	21	41	1	2019
Obesity, adults aged 18+ years (%)	27	20	35	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	1	0	2019
Physical inactivity, adults aged 18+ years (%)	34	27	40	2016

National response

Targets	Treatment
National target for blood pressure	✓
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Andorra

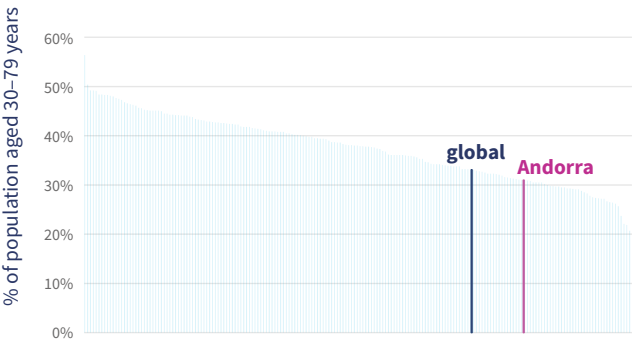
Hypertension profile

Total population (2019): 76 300

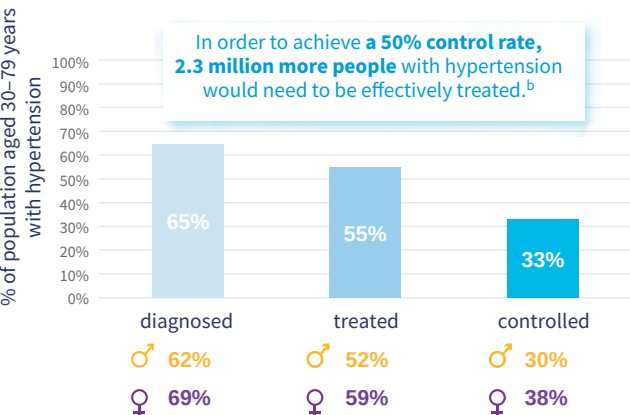
Total deaths (2019): no data

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 36% ♀ 26%

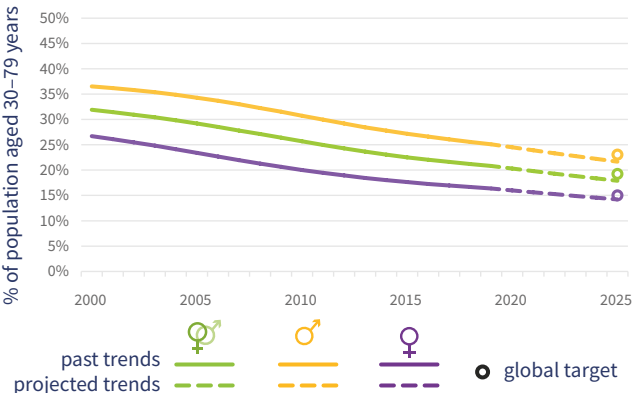
Prevalence of hypertension – global comparison (both sexes)^a



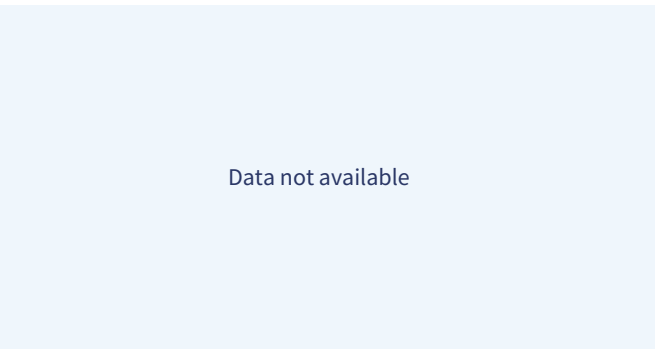
Of the 6.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	53	53	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	7	2019
Current tobacco use, adults aged 15+ years (%) ^e	32	36	28	2019
Obesity, adults aged 18+ years (%)	26	26	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	17	5	2019
Physical inactivity, adults aged 18+ years (%)	38	37	40	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake. e. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Angola

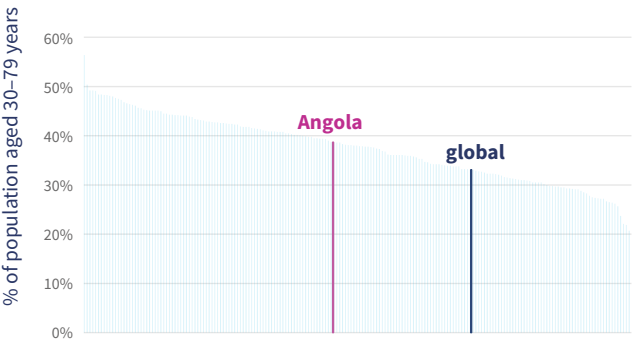
Hypertension profile

Total population (2019): 32 354 000

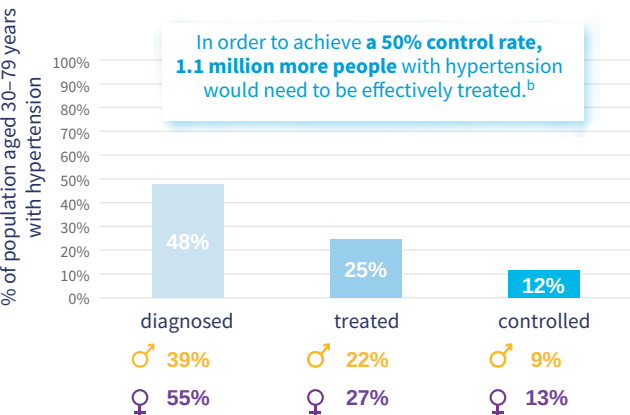
Total deaths (2019): 229 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 39% ♂ 37% ♀ 41%

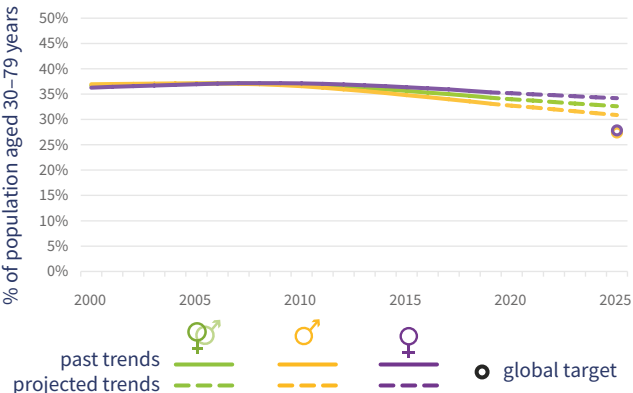
Prevalence of hypertension – global comparison (both sexes)^a



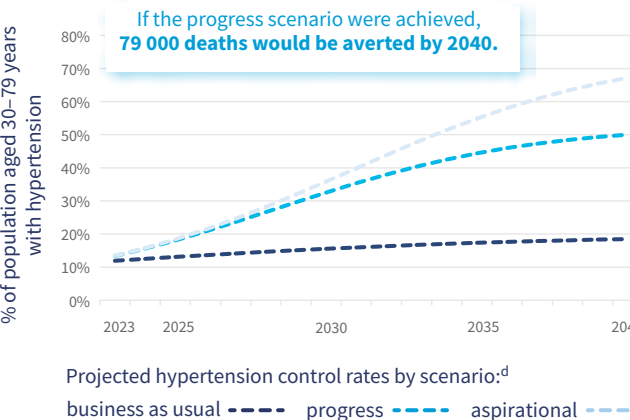
Of the 2.9 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	25	19	2019
Cardiovascular disease deaths	27 800	14 500	13 300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	61	66	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	8	4	12	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	10	3	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Antigua and Barbuda

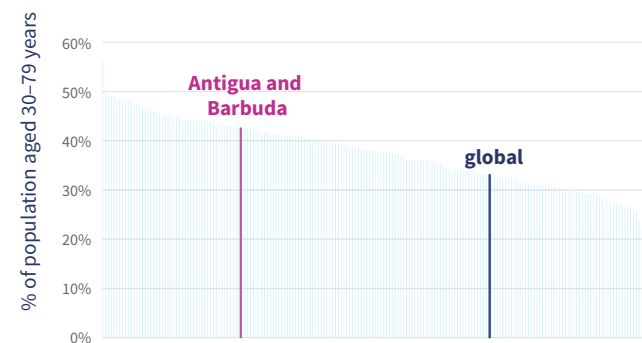
Hypertension profile

Total population (2019): 92 100

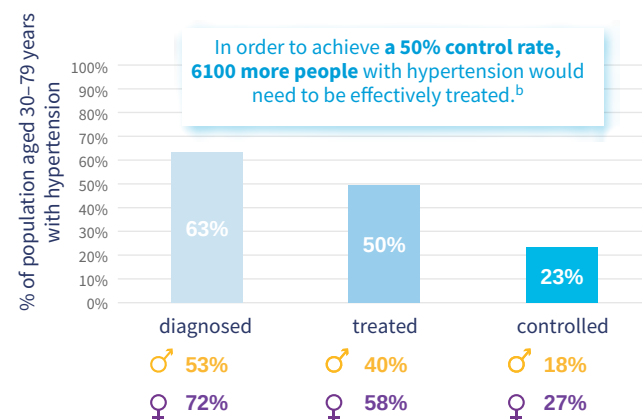
Total deaths (2019): 650

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 43% ♂ 42% ♀ 43%

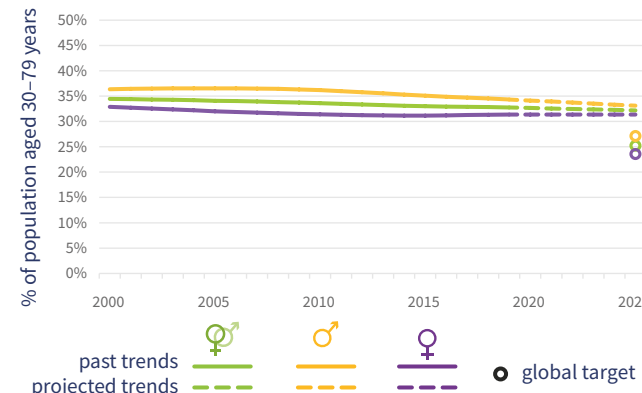
Prevalence of hypertension – global comparison (both sexes)^a



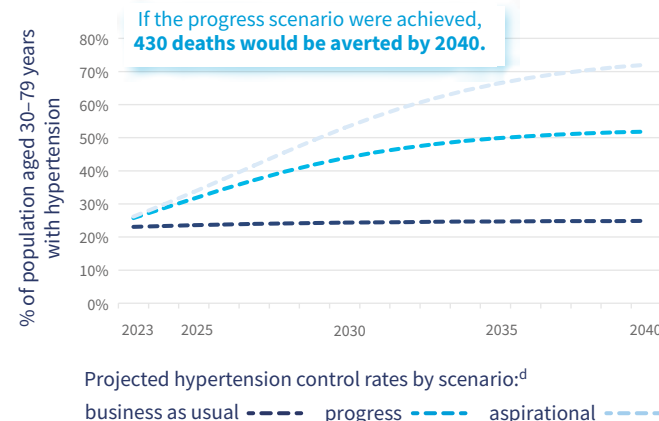
Of the 23 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	18	18	18	2019
Cardiovascular disease deaths	220	110	110	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	55	53	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	19	12	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	14	4	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of	
National target for salt consumption	✗	hypertension	✗
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Argentina

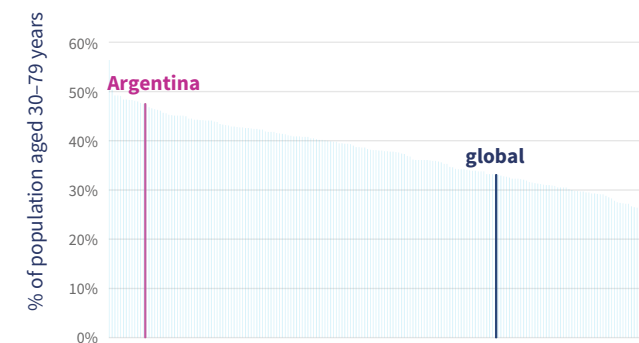
Hypertension profile

Total population (2019): 44 746 000

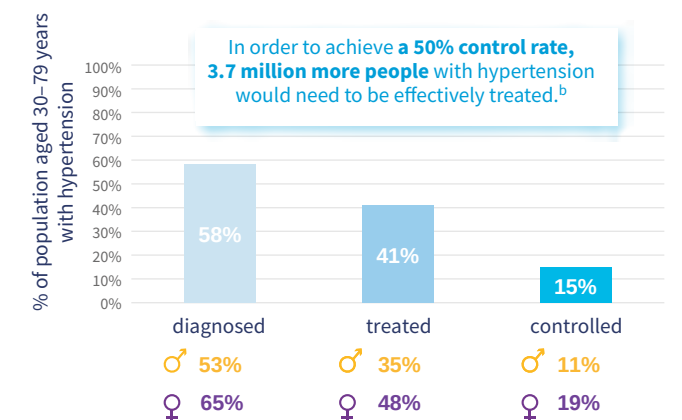
Total deaths (2019): 349 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 48% ♂ 54% ♀ 41%

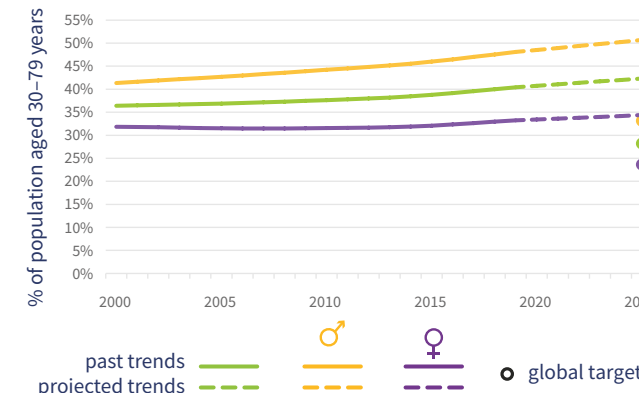
Prevalence of hypertension – global comparison (both sexes)^a



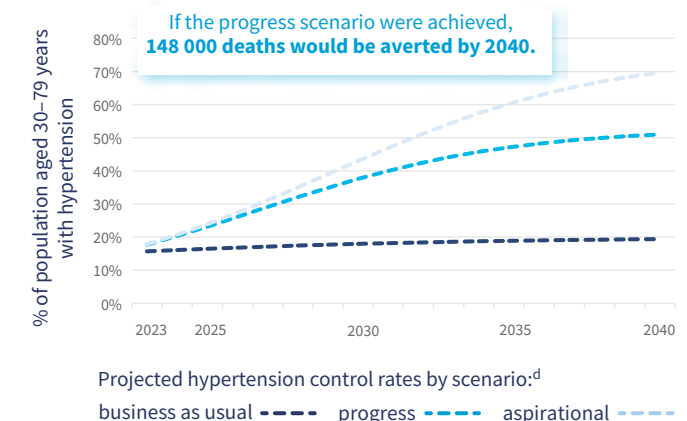
Of the 10.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	20	12	2019
Cardiovascular disease deaths	98 800	49 200	49 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	48	48	48	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%)	25	30	20	2019
Obesity, adults aged 18+ years (%)	28	27	29	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	13	4	2019
Physical inactivity, adults aged 18+ years (%)	42	38	45	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Armenia

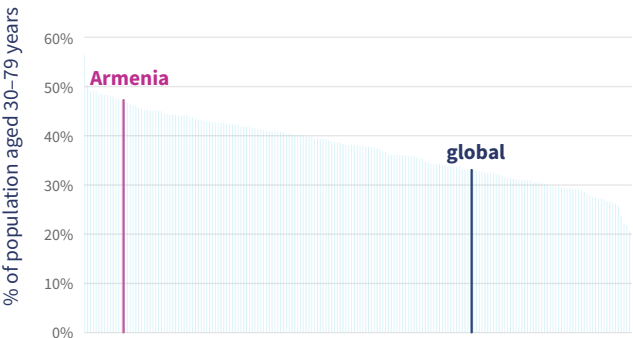
Hypertension profile

Total population (2019): 2 821 000

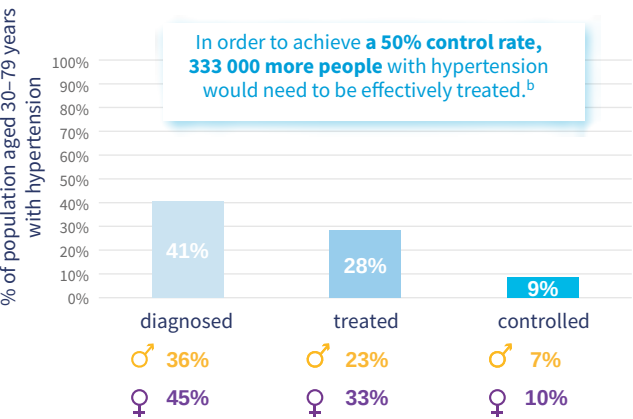
Total deaths (2019): 25 400

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 47% ♂ 49% ♀ 46%

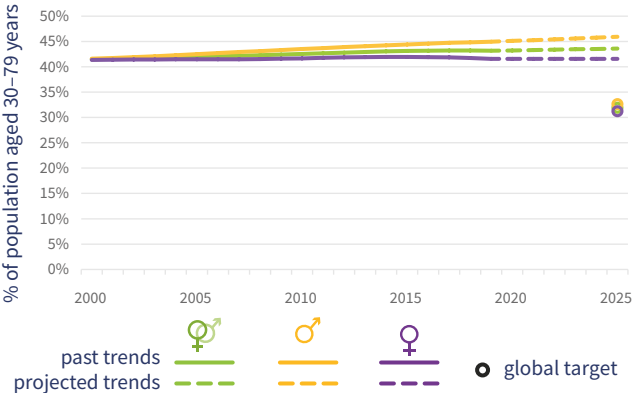
Prevalence of hypertension – global comparison (both sexes)^a



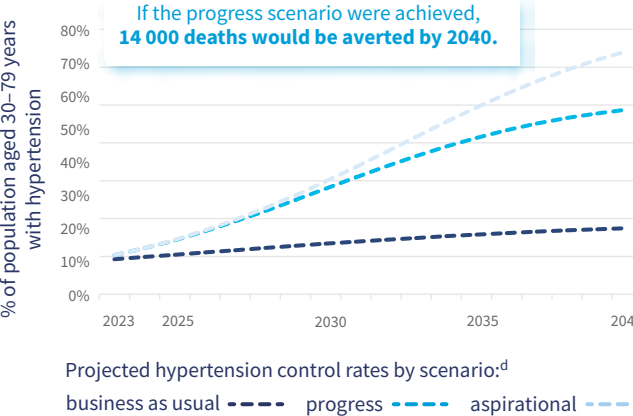
Of the 813 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	20	28	13	2019
Cardiovascular disease deaths	13 100	6300	6800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	56	57	54	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%)	26	50	2	2019
Obesity, adults aged 18+ years (%)	20	17	23	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	9	2	2019
Physical inactivity, adults aged 18+ years (%)	23	23	22	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Australia

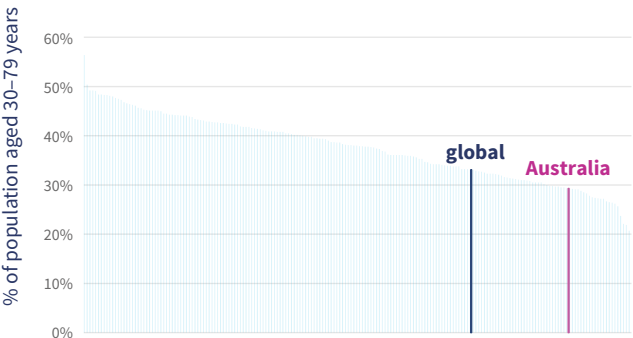
Hypertension profile

Total population (2019): 25 357 000

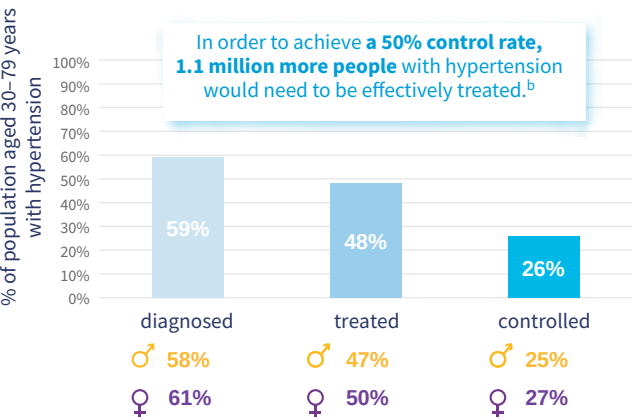
Total deaths (2019): 165 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 29% ♂ 32% ♀ 26%

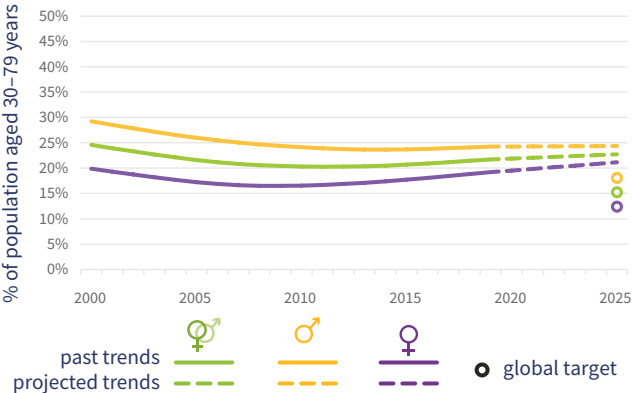
Prevalence of hypertension – global comparison (both sexes)^a



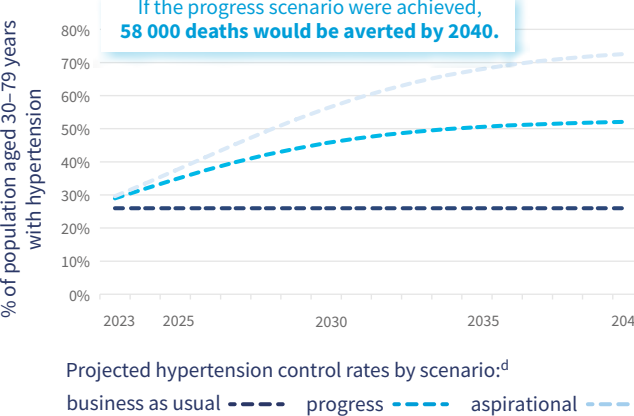
Of the 4.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	9	10	7	2019
Cardiovascular disease deaths	42 200	21 200	21 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	43	43	43	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%) ^f	14	16	12	2019
Obesity, adults aged 18+ years (%)	29	30	28	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	16	5	2019
Physical inactivity, adults aged 18+ years (%)	30	27	34	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Austria

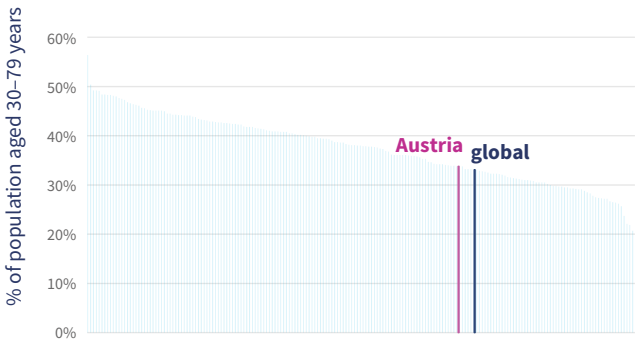
Hypertension profile

Total population (2019): 8 880 000

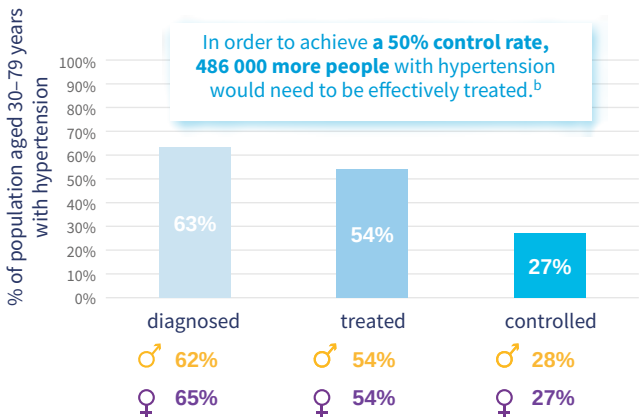
Total deaths (2019): 85 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 38% ♀ 30%

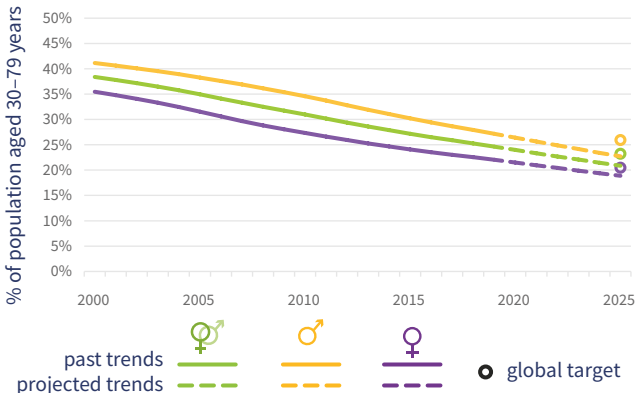
Prevalence of hypertension – global comparison (both sexes)^a



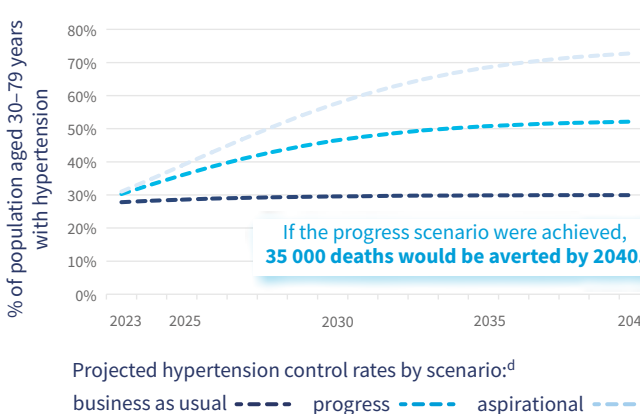
Of the 2.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	13	8	2019
Cardiovascular disease deaths	33 100	14 800	18 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	52	53	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	8	2019
Current tobacco use, adults aged 15+ years (%) ^f	27	29	26	2019
Obesity, adults aged 18+ years (%)	20	22	18	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	19	5	2019
Physical inactivity, adults aged 18+ years (%)	30	26	34	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Azerbaijan

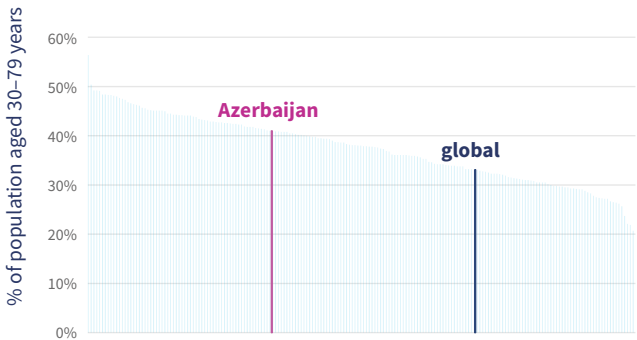
Hypertension profile

Total population (2019): 10 233 000

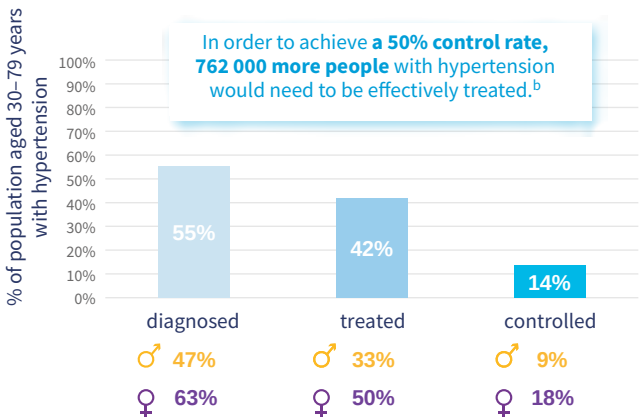
Total deaths (2019): 82 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 40% ♀ 42%

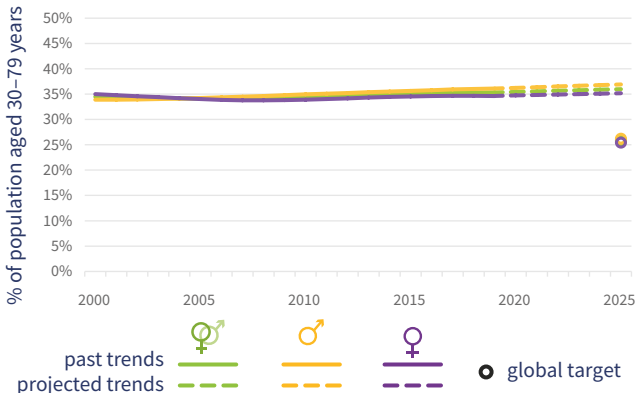
Prevalence of hypertension – global comparison (both sexes)^a



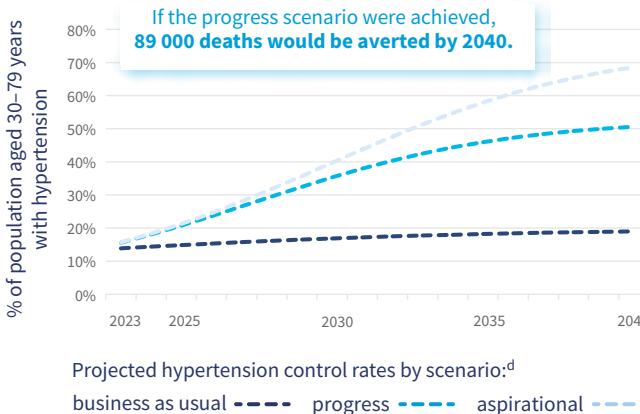
Of the 2.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	27	34	20	2019
Cardiovascular disease deaths	51 300	25 200	26 100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	56	54	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%)	24	48	0	2019
Obesity, adults aged 18+ years (%)	20	16	24	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	4	1	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	no data Guidelines for management of hypertension
National target for salt consumption	no data ✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Bahamas

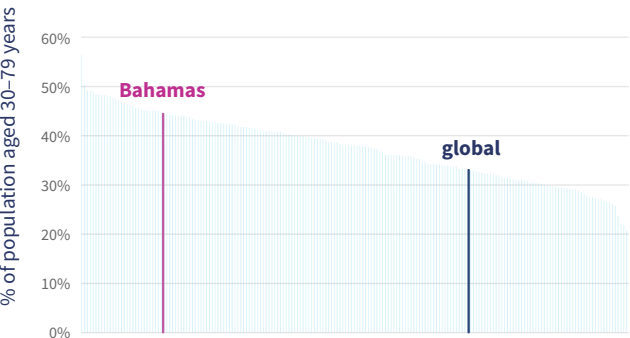
Hypertension profile

Total population (2019): 405 000

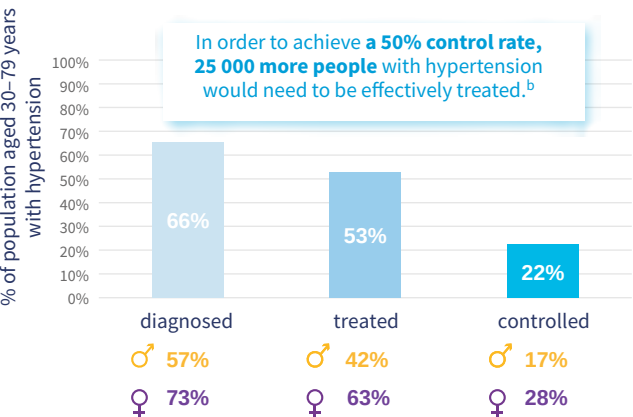
Total deaths (2019): 2670

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 45% ♂ 46% ♀ 44%

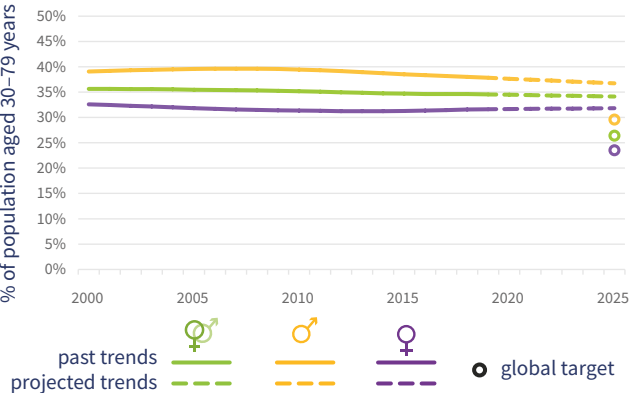
Prevalence of hypertension – global comparison (both sexes)^a



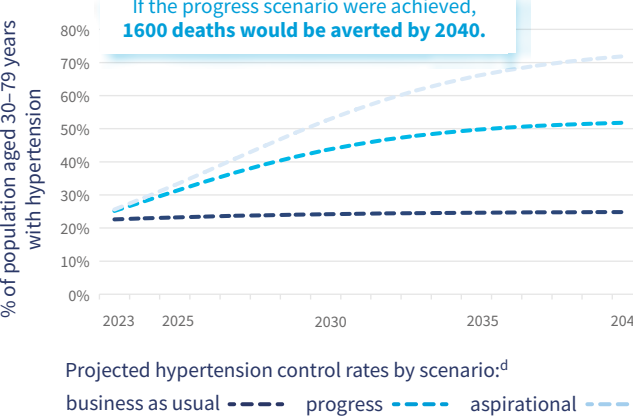
Of the 90 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	20	24	17	2019
Cardiovascular disease deaths	890	490	400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	61	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	11	19	3	2019
Obesity, adults aged 18+ years (%)	32	24	38	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	7	2	2019
Physical inactivity, adults aged 18+ years (%)	43	30	56	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Bahrain

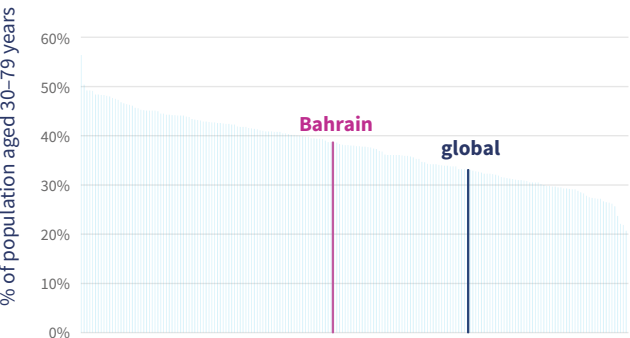
Hypertension profile

Total population (2019): 1 494 000

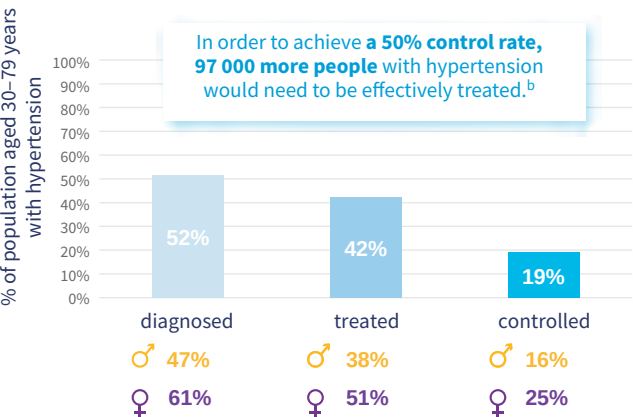
Total deaths (2019): 4620

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 39% ♂ 40% ♀ 35%

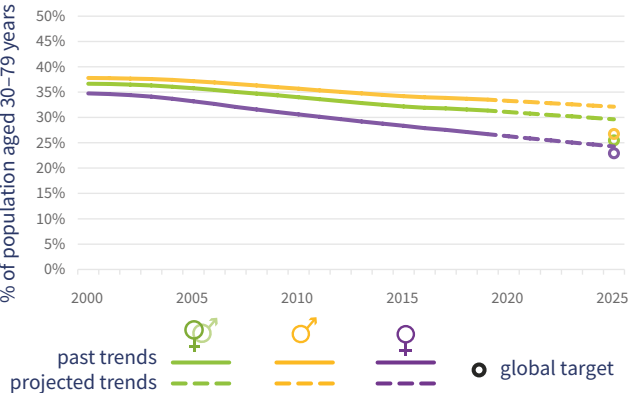
Prevalence of hypertension – global comparison (both sexes)^a



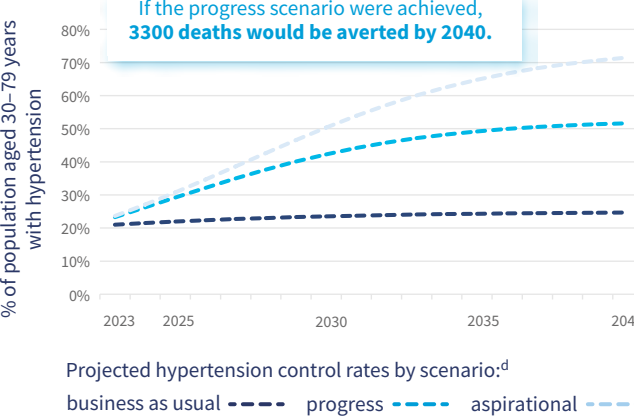
Of the 282 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	16	15	2019
Cardiovascular disease deaths	1400	910	540	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	56	56	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%) ^f	15	26	5	2019
Obesity, adults aged 18+ years (%)	30	26	37	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	2	0	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Bangladesh

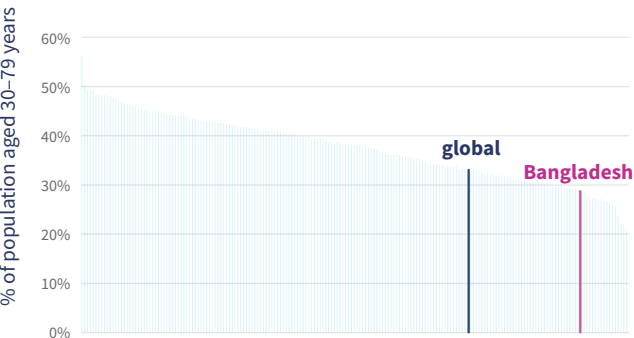
Hypertension profile

Total population (2019): 165 500 000

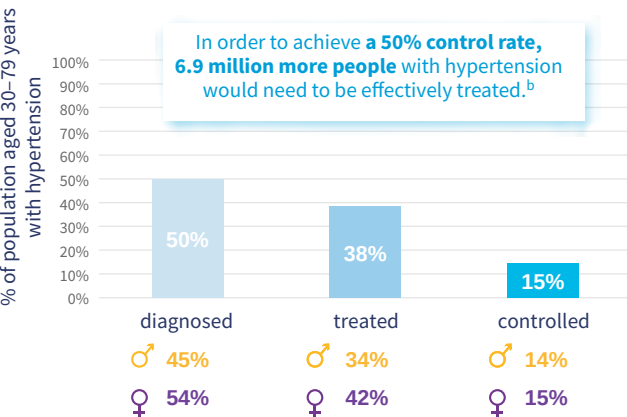
Total deaths (2019): 793 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 29% ♂ 24% ♀ 34%

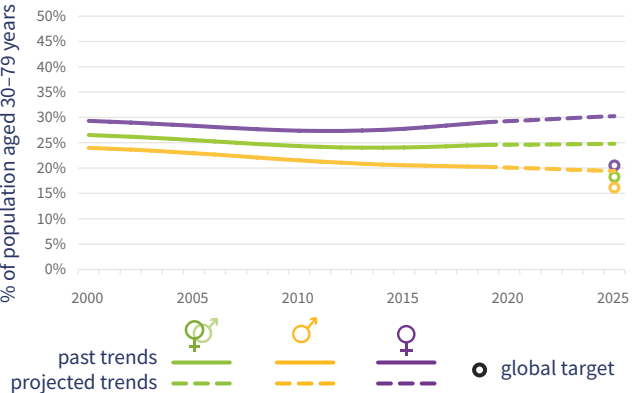
Prevalence of hypertension – global comparison (both sexes)^a



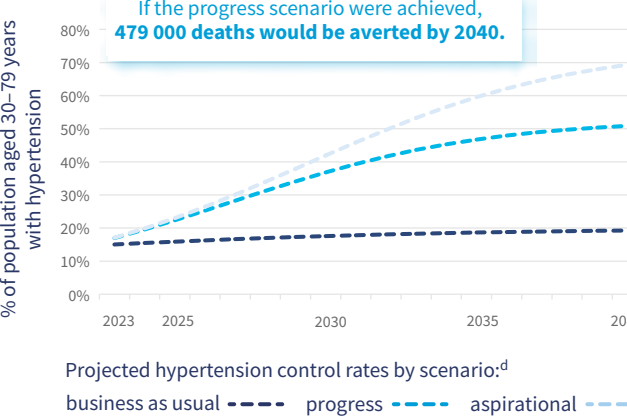
Of the 19.4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	19	21	16	2019
Cardiovascular disease deaths	273 000	144 000	129 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	51	58	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	9	8	2019
Current tobacco use, adults aged 15+ years (%)	35	53	18	2019
Obesity, adults aged 18+ years (%)	4	2	5	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	28	16	39	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Barbados

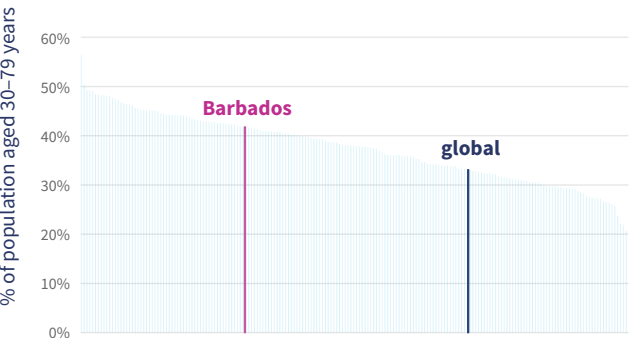
Hypertension profile

Total population (2019): 280 000

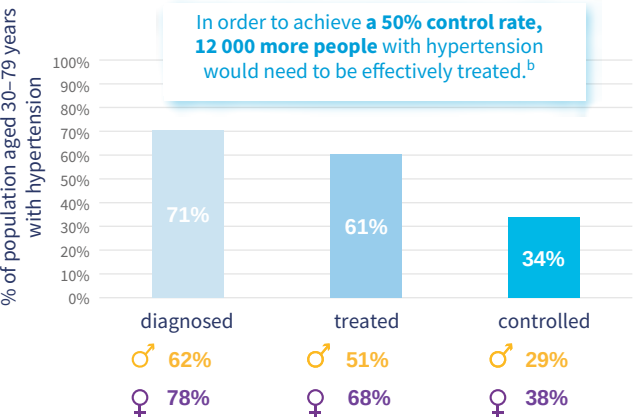
Total deaths (2019): 3420

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 42% ♂ 40% ♀ 43%

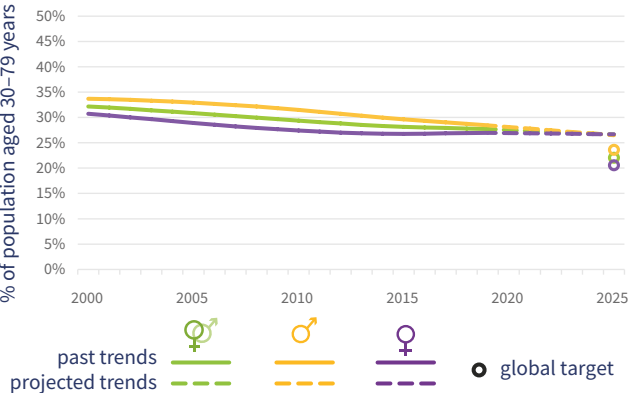
Prevalence of hypertension – global comparison (both sexes)^a



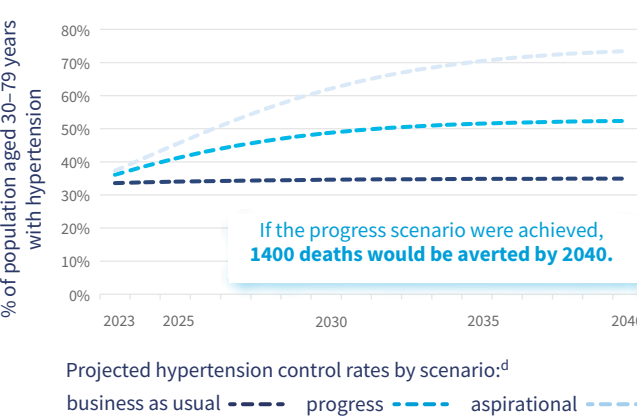
Of the 80 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	17	15	2019
Cardiovascular disease deaths	1000	460	550	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	54	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	6	2019
Current tobacco use, adults aged 15+ years (%)	9	15	2	2019
Obesity, adults aged 18+ years (%)	23	15	31	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	16	4	2019
Physical inactivity, adults aged 18+ years (%)	43	29	55	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Belarus

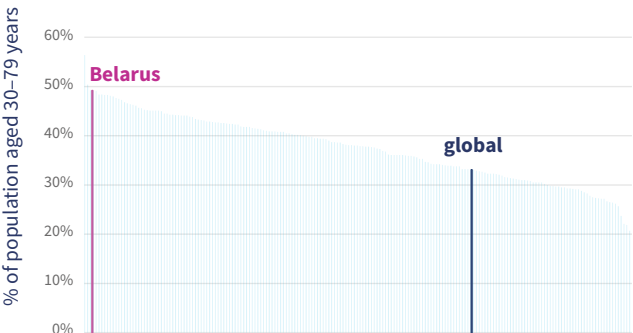
Hypertension profile

Total population (2019): 9 674 000

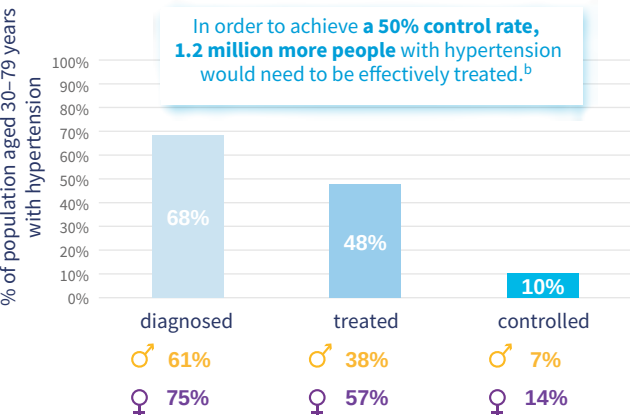
Total deaths (2019): 116 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 49% ♂ 52% ♀ 47%

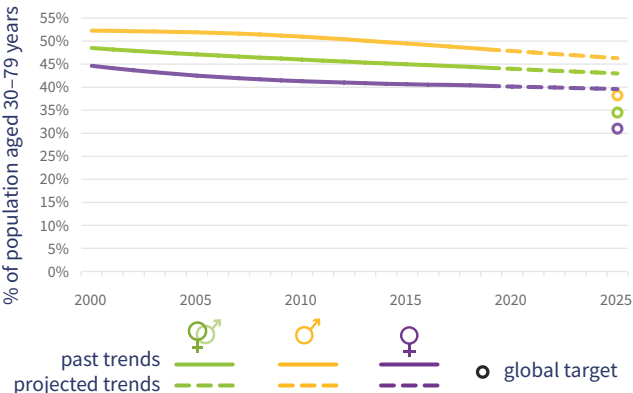
Prevalence of hypertension – global comparison (both sexes)^a



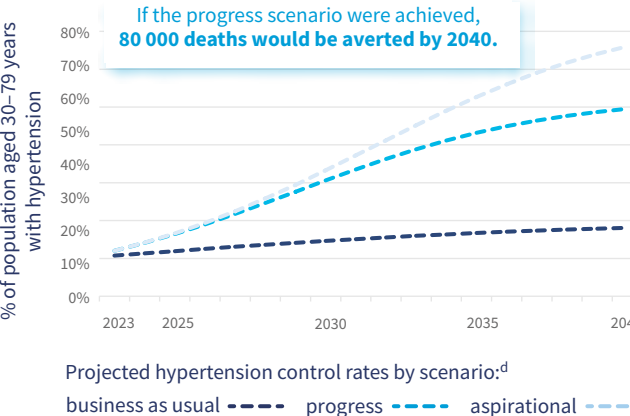
Of the 3.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	36	14	2019
Cardiovascular disease deaths	73 200	33 000	40 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	61	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%) ^f	31	48	14	2019
Obesity, adults aged 18+ years (%)	25	22	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	18	5	2019
Physical inactivity, adults aged 18+ years (%)	14	14	14	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Belgium

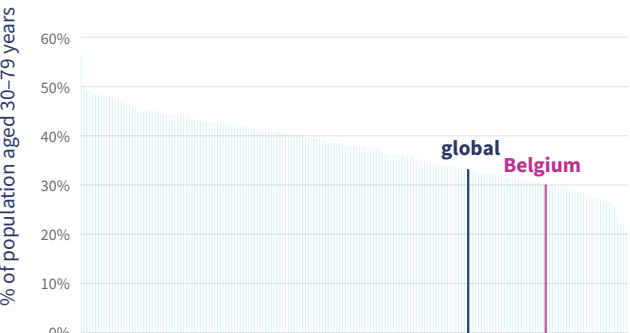
Hypertension profile

Total population (2019): 11 511 000

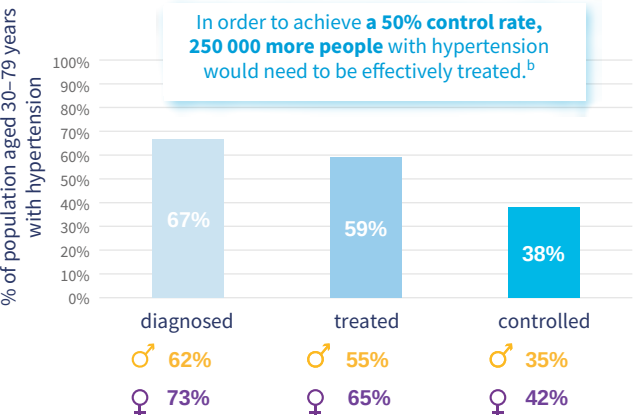
Total deaths (2019): 113 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 30% ♂ 34% ♀ 26%

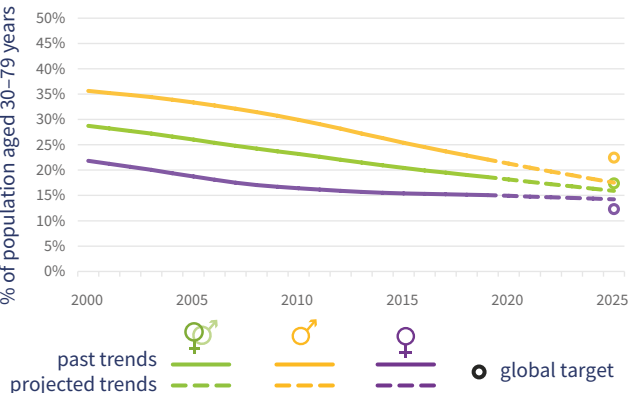
Prevalence of hypertension – global comparison (both sexes)^a



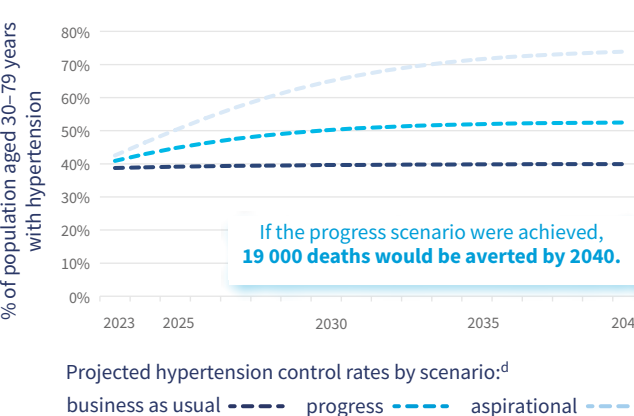
Of the 2.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	13	8	2019
Cardiovascular disease deaths	30 800	14 000	16 800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	49	49	49	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%)	24	26	21	2019
Obesity, adults aged 18+ years (%)	22	23	21	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	16	5	2019
Physical inactivity, adults aged 18+ years (%)	36	31	41	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Belize

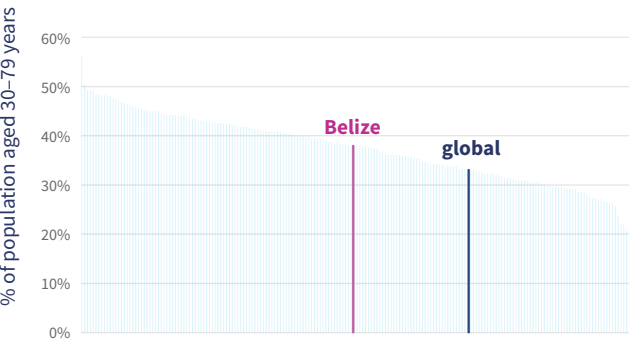
Hypertension profile

Total population (2019): 389 000

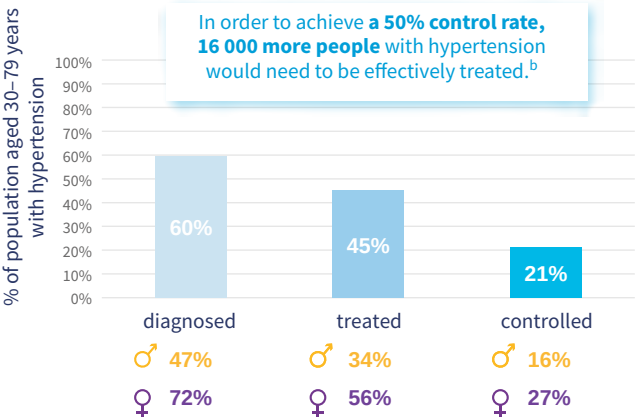
Total deaths (2019): 1880

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 38% ♀ 38%

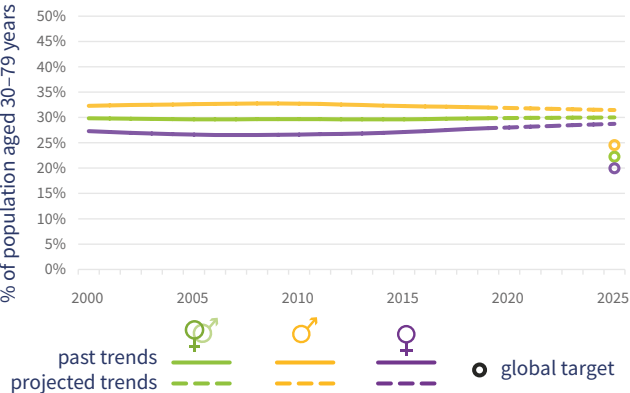
Prevalence of hypertension – global comparison (both sexes)^a



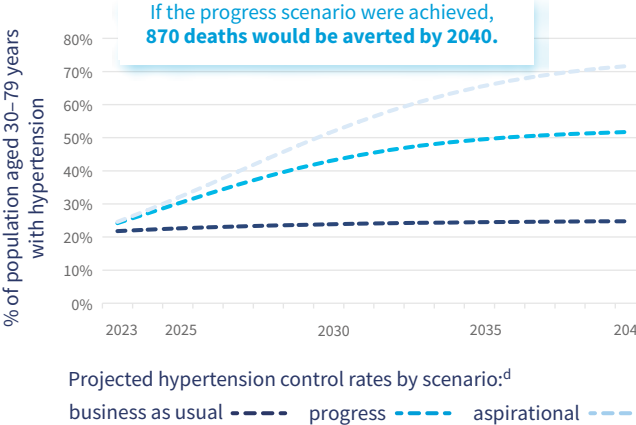
Of the 55 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	19	14	2019
Cardiovascular disease deaths	420	240	180	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	54	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%) ^f	9	16	2	2019
Obesity, adults aged 18+ years (%)	24	17	32	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	9	2	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Benin

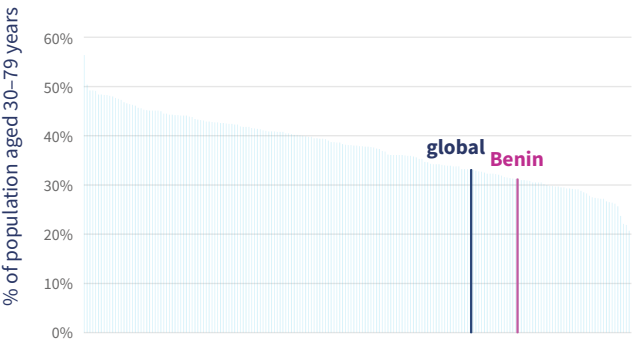
Hypertension profile

Total population (2019): 12 290 000

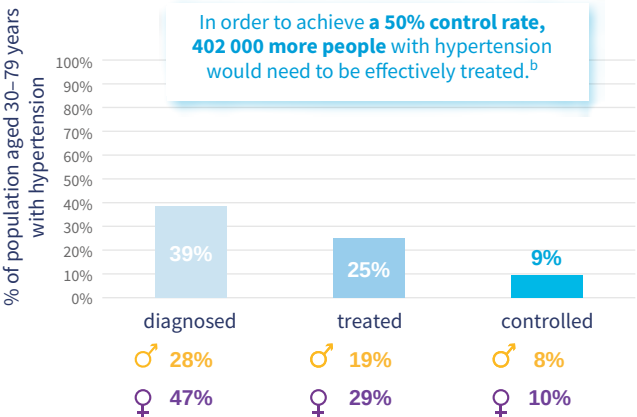
Total deaths (2019): 93 400

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 29% ♀ 33%

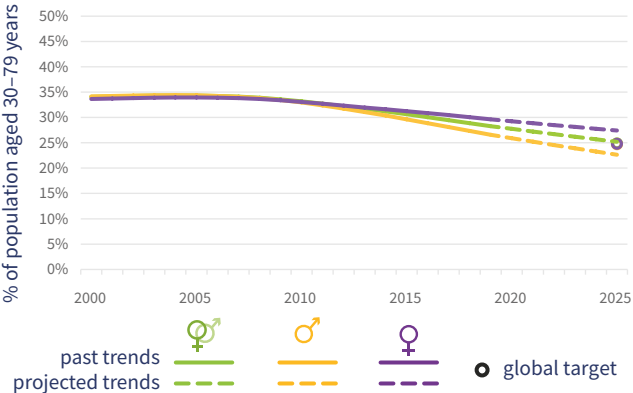
Prevalence of hypertension – global comparison (both sexes)^a



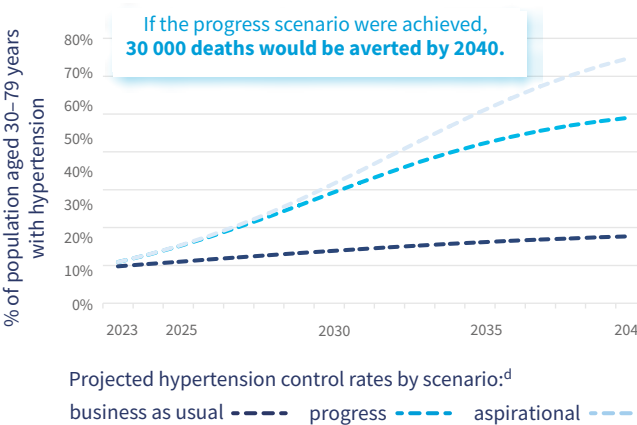
Of the 979 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	24	21	2019
Cardiovascular disease deaths	13 900	6700	7200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	56	54	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	8	2019
Current tobacco use, adults aged 15+ years (%)	7	12	2	2019
Obesity, adults aged 18+ years (%)	10	5	14	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	14	3	2019
Physical inactivity, adults aged 18+ years (%)	16	14	18	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Bhutan

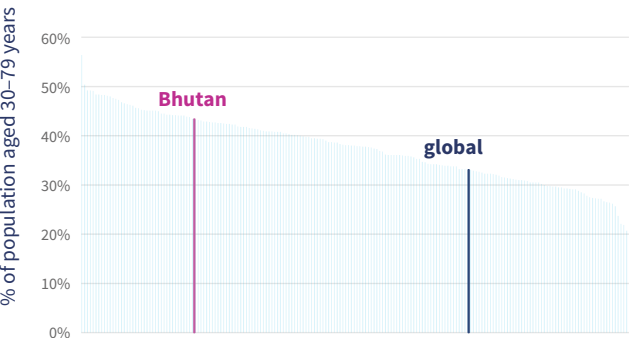
Hypertension profile

Total population (2019): 767 000

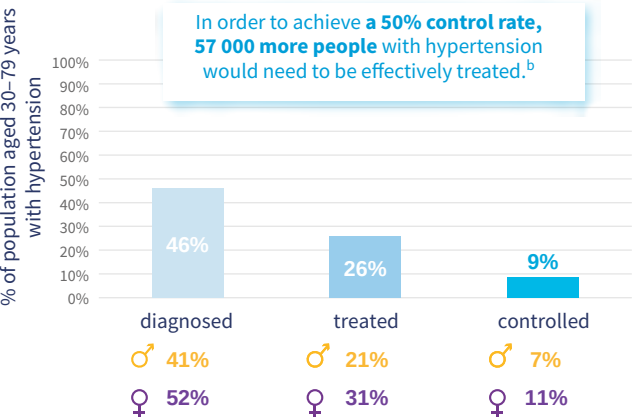
Total deaths (2019): 4530

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 43% ♂ 44% ♀ 43%

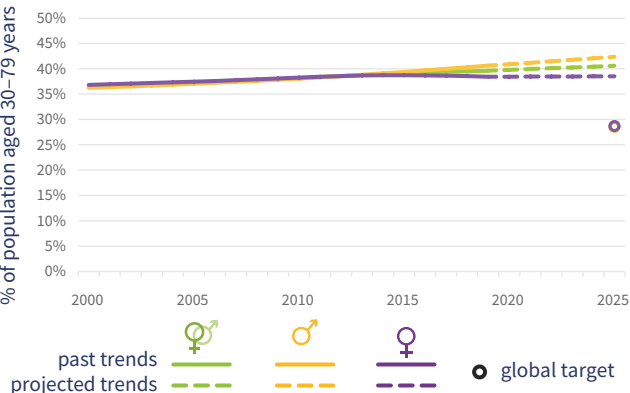
Prevalence of hypertension – global comparison (both sexes)^a



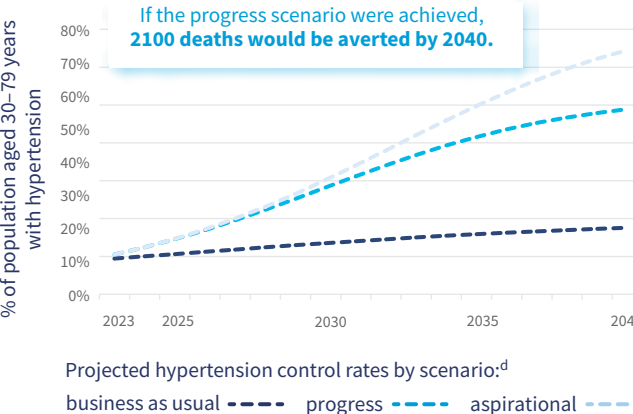
Of the 137 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	18	20	17	2019
Cardiovascular disease deaths	1300	750	530	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	51	51	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	9	8	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	6	5	9	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	23	18	29	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Bolivia (Plurinational State of)

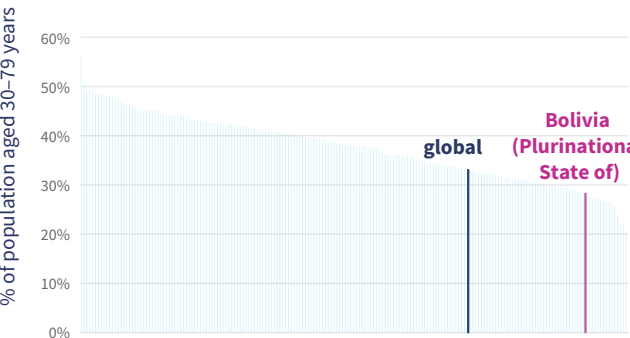
Hypertension profile

Total population (2019): 11 777 000

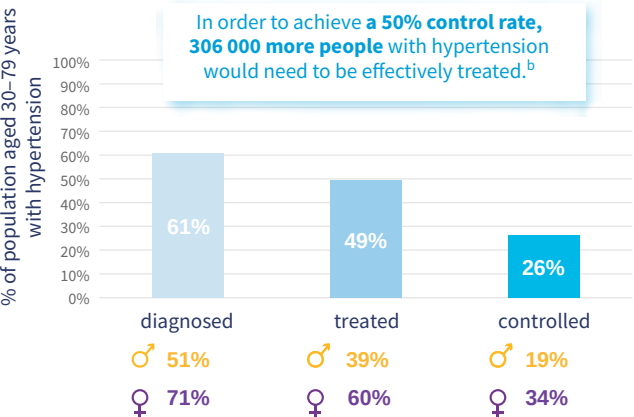
Total deaths (2019): 86 400

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 28% ♂ 29% ♀ 27%

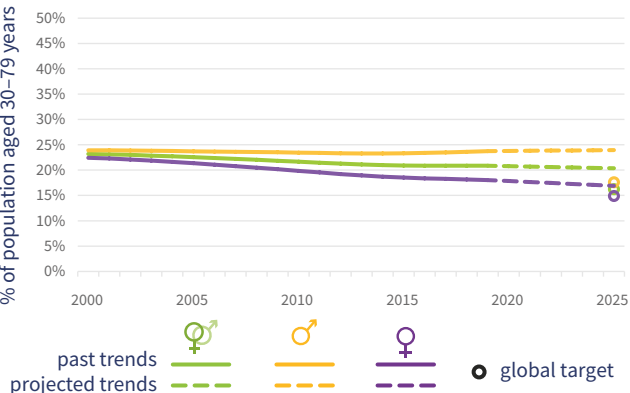
Prevalence of hypertension – global comparison (both sexes)^a



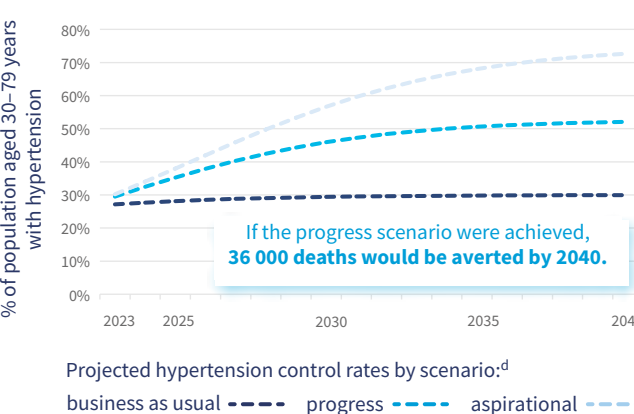
Of the 1.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	18	17	18	2019
Cardiovascular disease deaths	22 400	10 400	12 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	43	42	44	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%)	13	21	5	2019
Obesity, adults aged 18+ years (%)	20	15	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	6	2	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Bosnia and Herzegovina

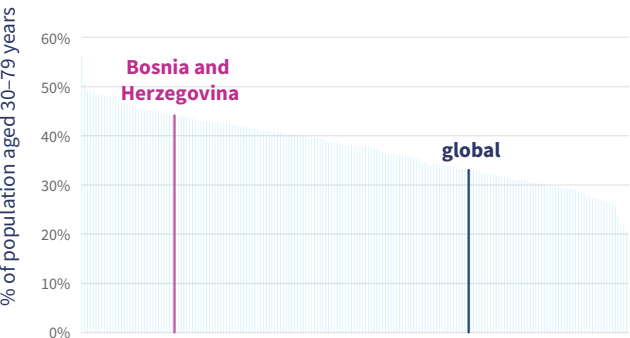
Hypertension profile

Total population (2019): 3 361 000

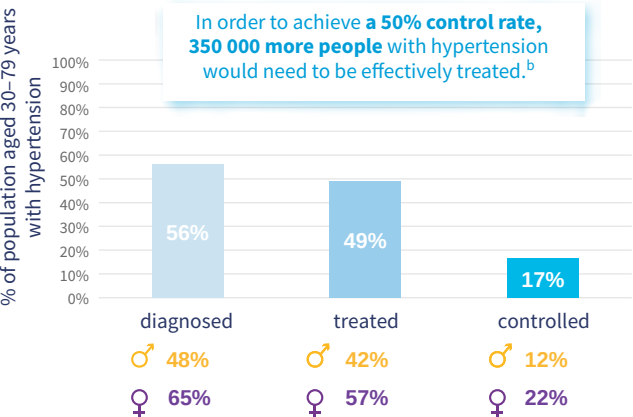
Total deaths (2019): 37 800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 44% ♂ 47% ♀ 41%

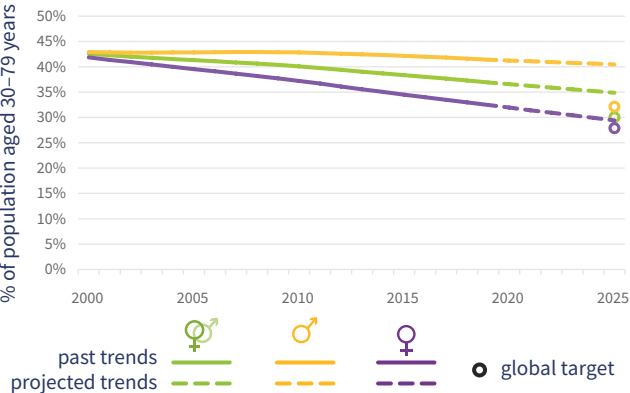
Prevalence of hypertension – global comparison (both sexes)^a



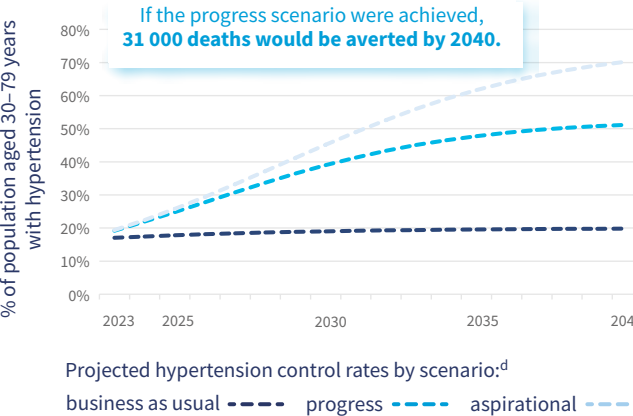
Of the 1.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	19	24	14	2019
Cardiovascular disease deaths	19 700	9000	10 700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	54	55	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%) ^f	36	43	28	2019
Obesity, adults aged 18+ years (%)	18	17	18	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	11	2	2019
Physical inactivity, adults aged 18+ years (%)	25	23	28	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Botswana

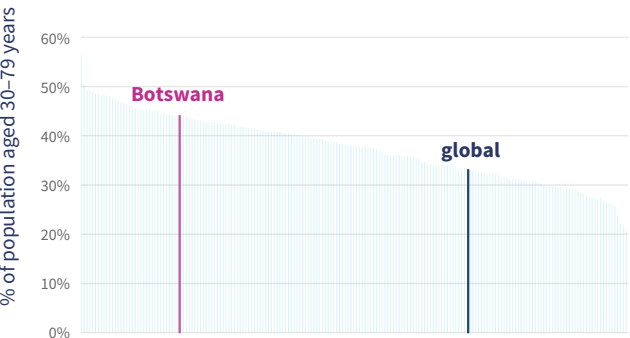
Hypertension profile

Total population (2019): 2 500 000

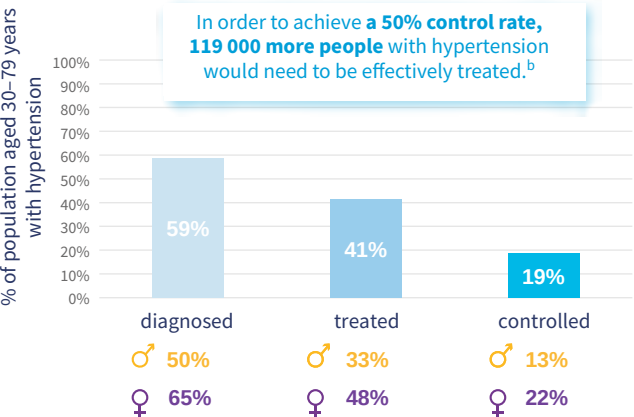
Total deaths (2019): 20 700

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 44% ♂ 40% ♀ 47%

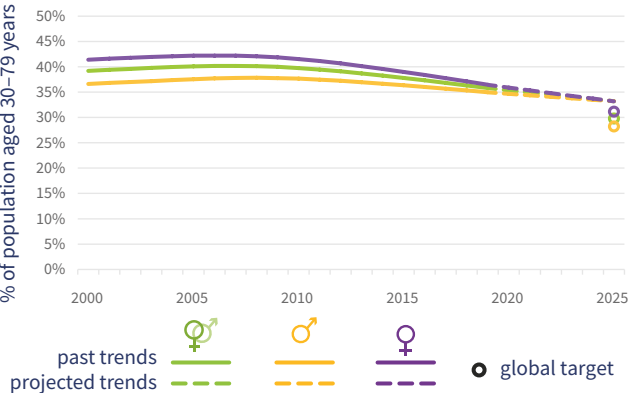
Prevalence of hypertension – global comparison (both sexes)^a



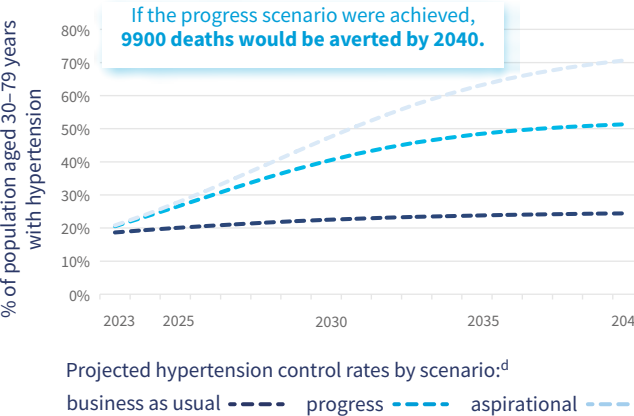
Of the 364 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	27	33	23	2019
Cardiovascular disease deaths	4100	2000	2100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	64	61	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	7	2019
Current tobacco use, adults aged 15+ years (%)	20	31	9	2019
Obesity, adults aged 18+ years (%)	19	8	29	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	14	3	2019
Physical inactivity, adults aged 18+ years (%)	22	17	26	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Brazil

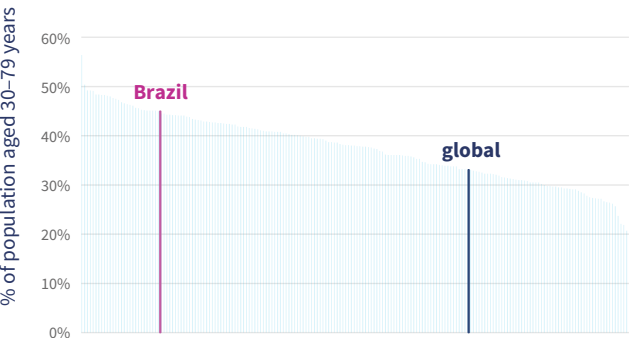
Hypertension profile

Total population (2019): 211 800 000

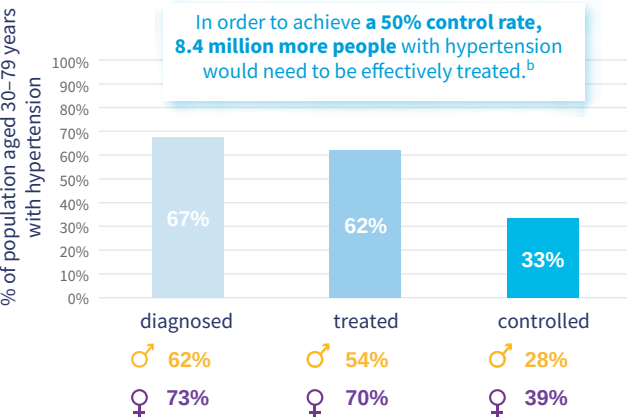
Total deaths (2019): 1 372 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 45% ♂ 48% ♀ 42%

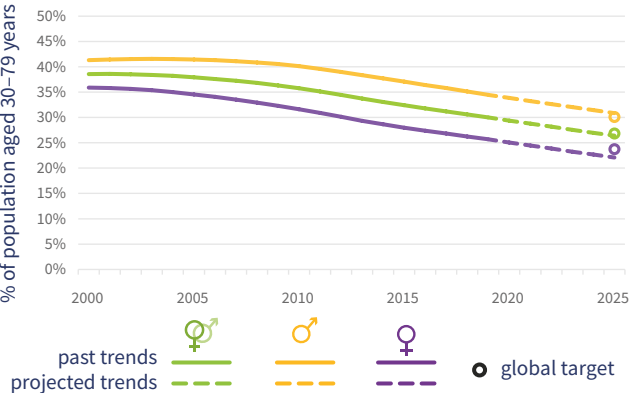
Prevalence of hypertension – global comparison (both sexes)^a



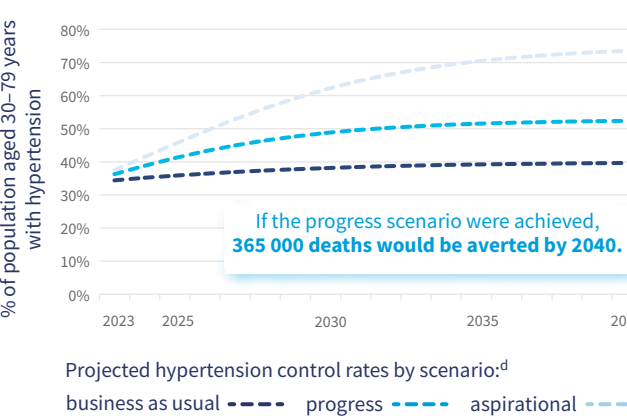
Of the 50.7 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	15	19	13	2019
Cardiovascular disease deaths	381 000	202 000	179 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	55	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%)	13	17	10	2019
Obesity, adults aged 18+ years (%)	22	19	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	12	3	2019
Physical inactivity, adults aged 18+ years (%)	47	40	53	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Brunei Darussalam

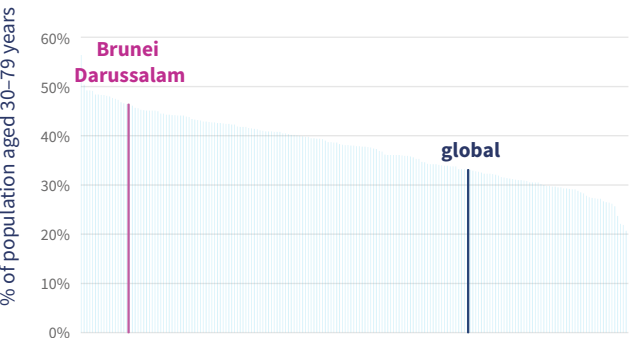
Hypertension profile

Total population (2019): 438 000

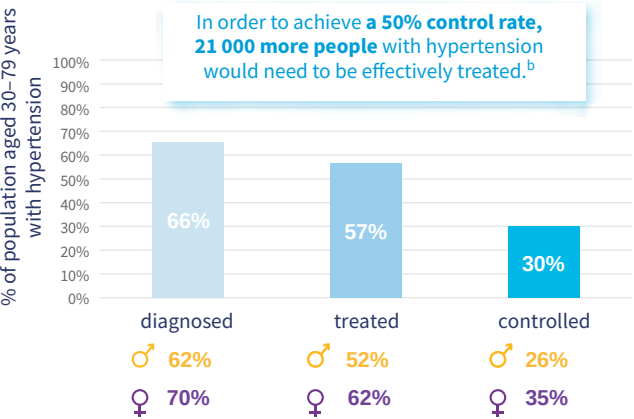
Total deaths (2019): 2 230

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 46% ♂ 47% ♀ 46%

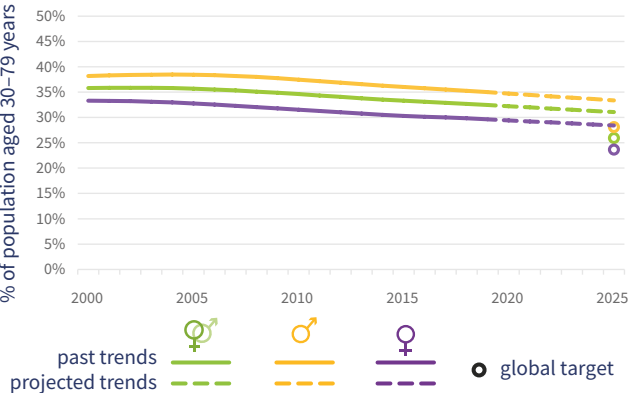
Prevalence of hypertension – global comparison (both sexes)^a



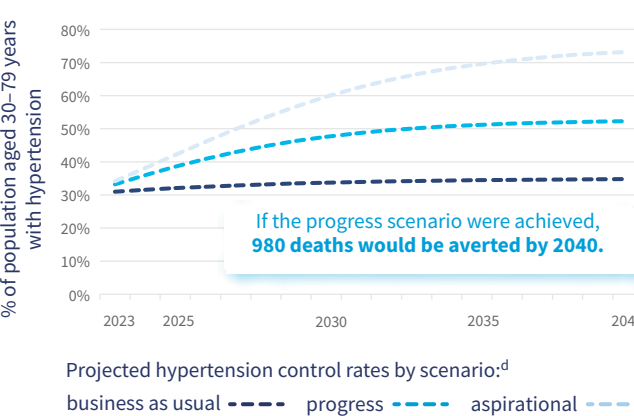
Of the 100 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	19	20	17	2019
Cardiovascular disease deaths	670	400	270	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	55	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	11	12	10	2019
Current tobacco use, adults aged 15+ years (%) ^f	16	30	2	2019
Obesity, adults aged 18+ years (%)	14	13	16	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	1	0	2019
Physical inactivity, adults aged 18+ years (%)	27	21	34	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Bulgaria

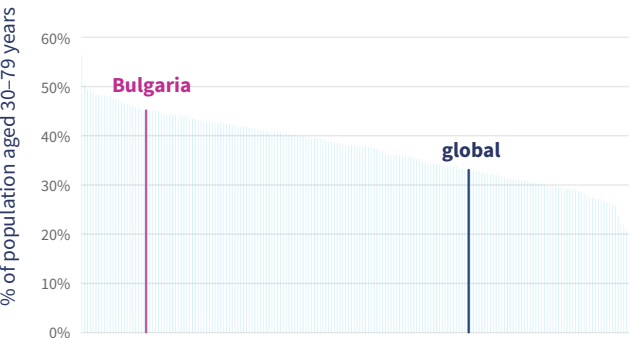
Hypertension profile

Total population (2019): 7 053 000

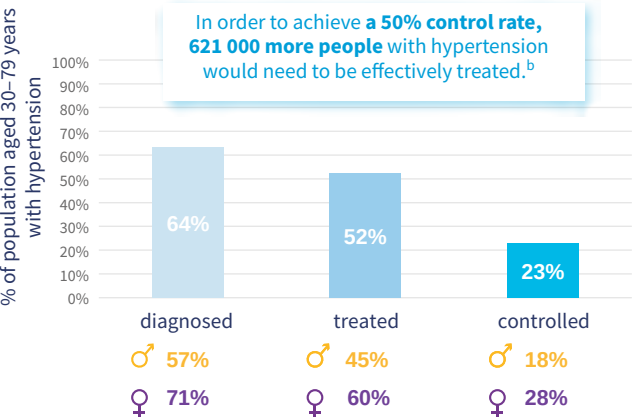
Total deaths (2019): 106 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 45% ♂ 49% ♀ 41%

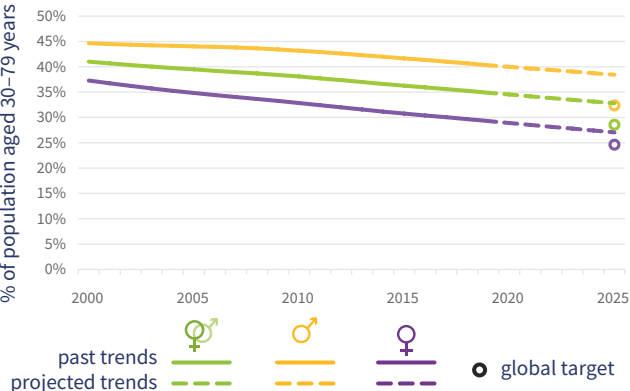
Prevalence of hypertension – global comparison (both sexes)^a



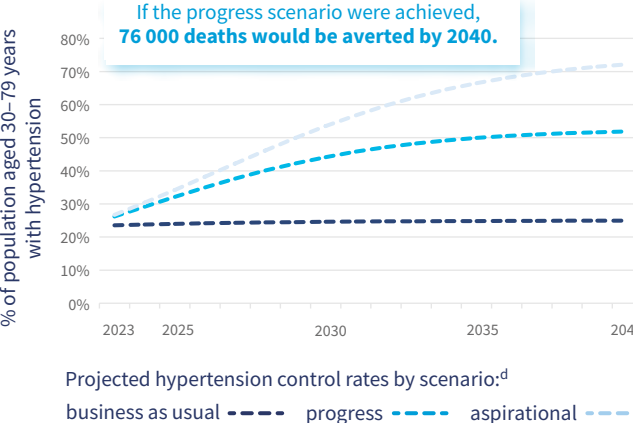
Of the 2.4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	32	16	2019
Cardiovascular disease deaths	68 900	33 200	35 700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	61	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%) ^f	39	42	37	2019
Obesity, adults aged 18+ years (%)	25	26	24	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	20	5	2019
Physical inactivity, adults aged 18+ years (%)	39	36	41	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Burkina Faso

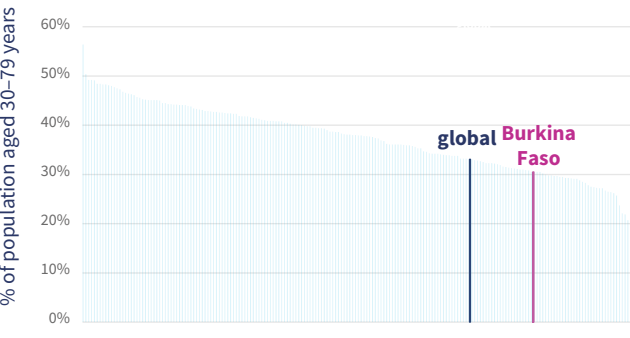
Hypertension profile

Total population (2019): 20 952 000

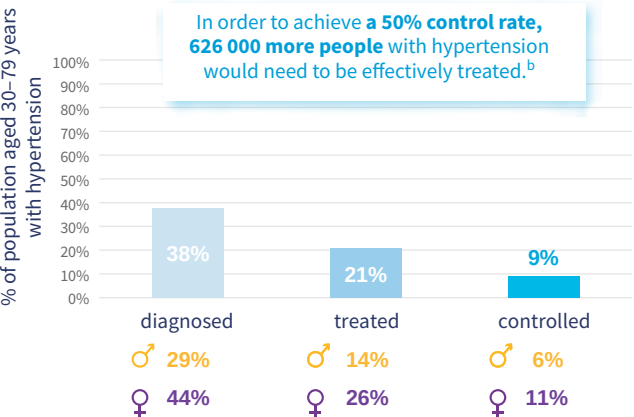
Total deaths (2019): 152 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 29% ♀ 32%

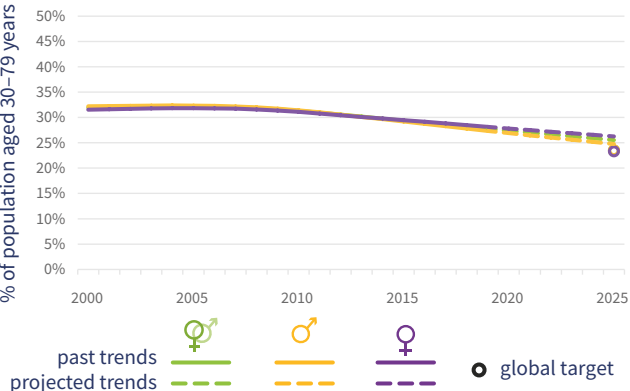
Prevalence of hypertension – global comparison (both sexes)^a



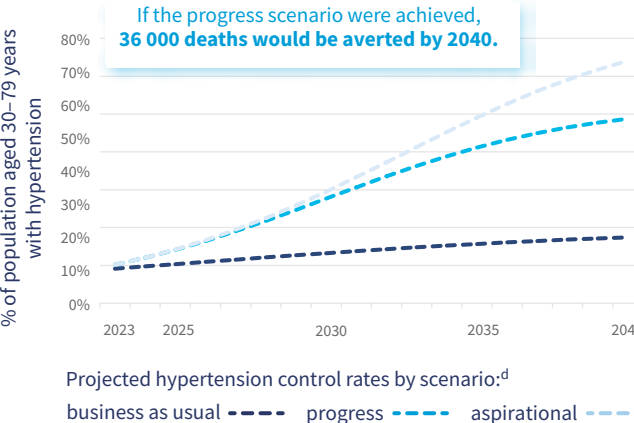
Of the 1.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	26	22	2019
Cardiovascular disease deaths	19 000	9500	9500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	51	55	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	15	23	7	2019
Obesity, adults aged 18+ years (%)	6	3	8	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	17	3	2019
Physical inactivity, adults aged 18+ years (%)	20	18	23	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Burundi

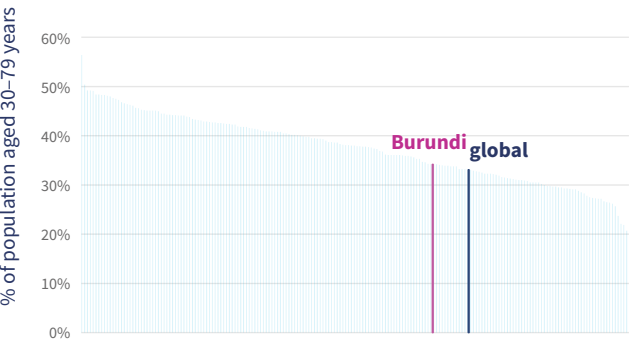
Hypertension profile

Total population (2019): 11 875 000

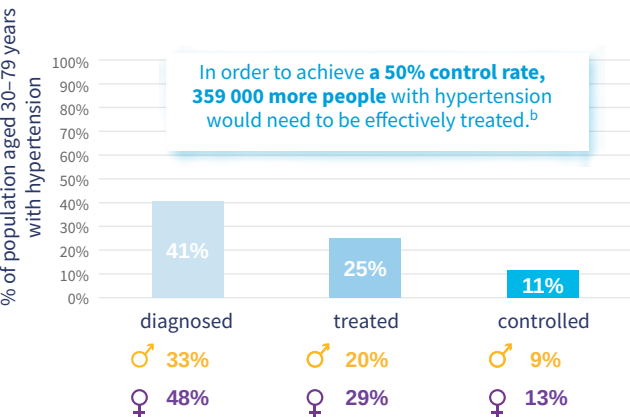
Total deaths (2019): 76 800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 33% ♀ 36%

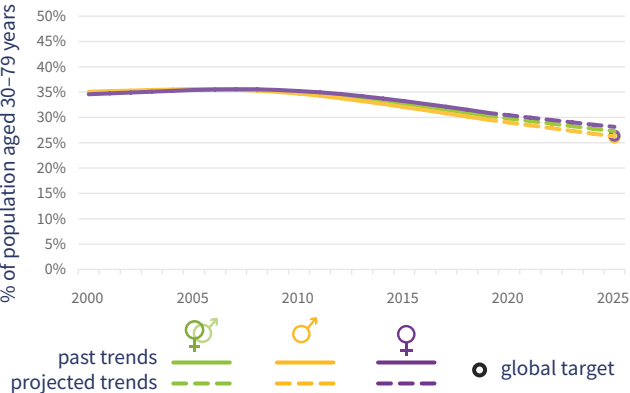
Prevalence of hypertension – global comparison (both sexes)^a



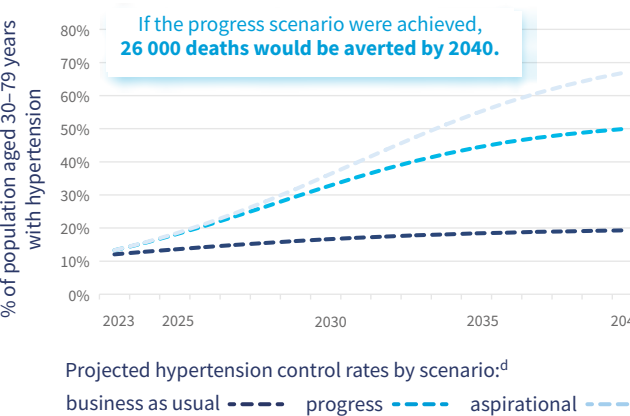
Of the 907 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	27	23	2019
Cardiovascular disease deaths	10 200	5200	5000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	57	58	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	12	18	6	2019
Obesity, adults aged 18+ years (%)	5	2	9	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	7	2	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Cabo Verde

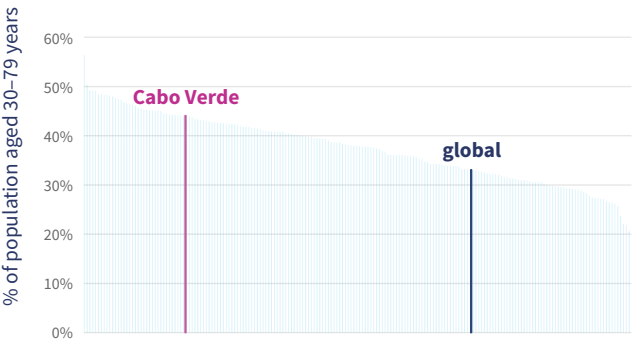
Hypertension profile

Total population (2019): 577 000

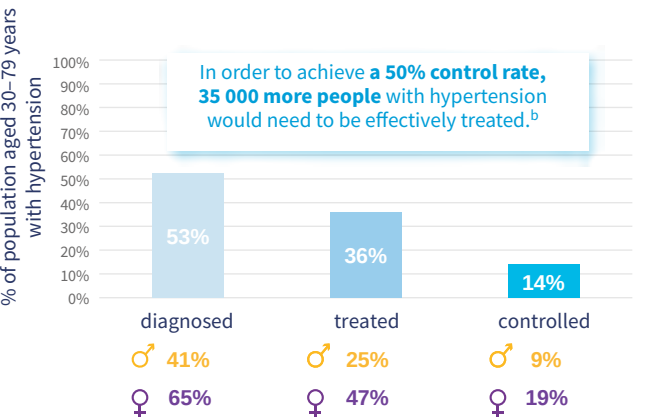
Total deaths (2019): 2850

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 44% ♂ 46% ♀ 42%

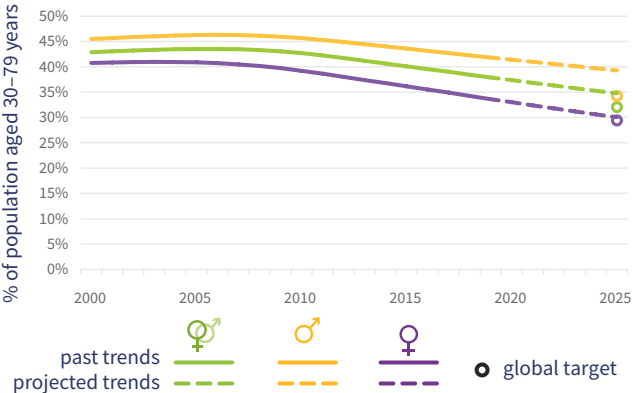
Prevalence of hypertension – global comparison (both sexes)^a



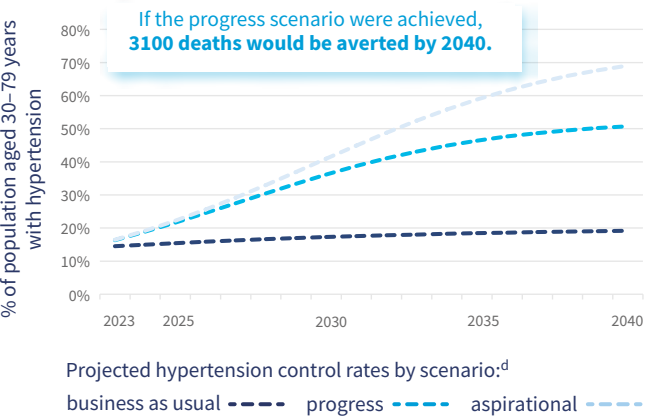
Of the 95 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	17	23	13	2019
Cardiovascular disease deaths	870	410	450	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	59	60	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	12	18	6	2019
Obesity, adults aged 18+ years (%)	12	7	16	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	10	3	2019
Physical inactivity, adults aged 18+ years (%)	20	14	25	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Cambodia

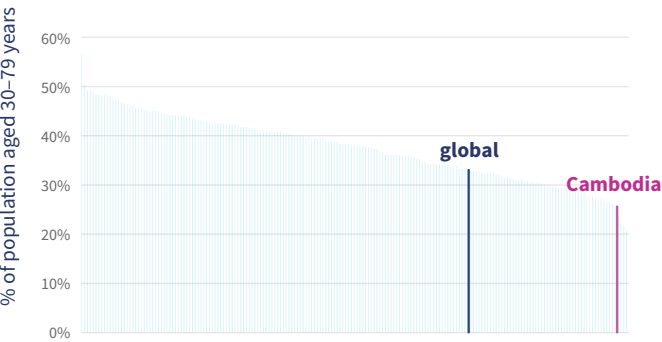
Hypertension profile

Total population (2019): 16 208 000

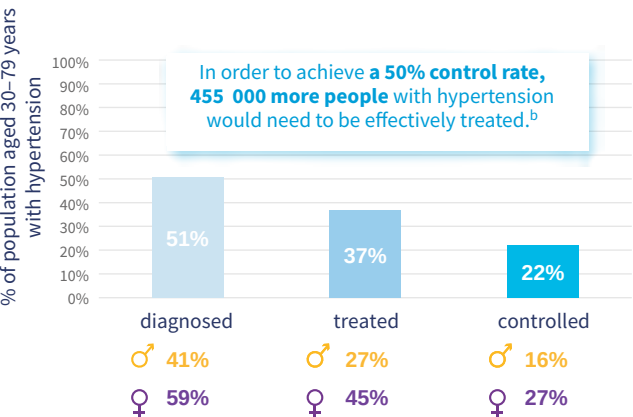
Total deaths (2019): 96 600

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 26% ♂ 26% ♀ 25%

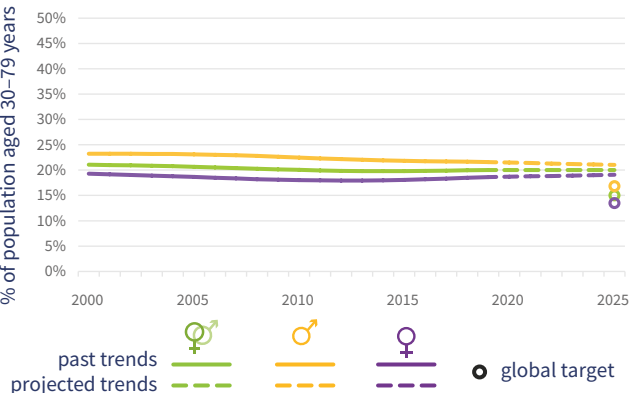
Prevalence of hypertension – global comparison (both sexes)^a



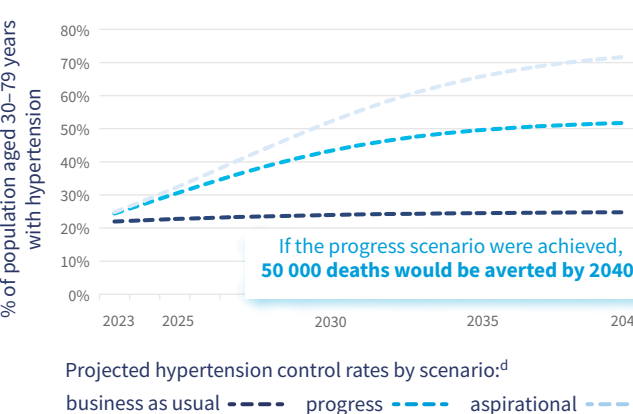
Of the 1.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	27	19	2019
Cardiovascular disease deaths	27 100	13 400	13 700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	44	45	44	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	10	2019
Current tobacco use, adults aged 15+ years (%)	22	37	6	2019
Obesity, adults aged 18+ years (%)	4	3	5	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	14	3	2019
Physical inactivity, adults aged 18+ years (%)	11	10	11	2016

National response

Targets	Treatment
National target for blood pressure	✓
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Cameroon

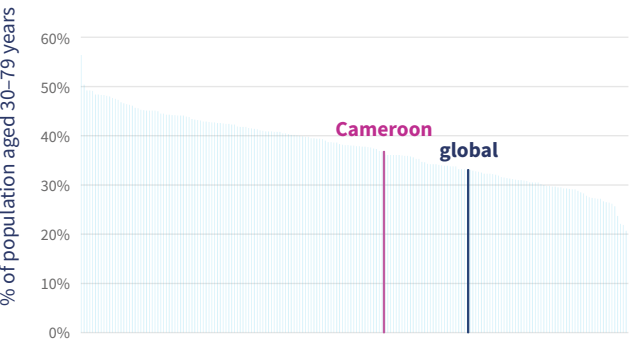
Hypertension profile

Total population (2019): 25 782 000

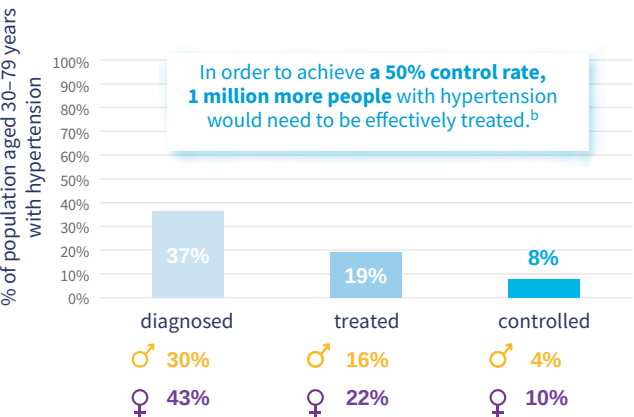
Total deaths (2019): 196 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 37% ♂ 35% ♀ 39%

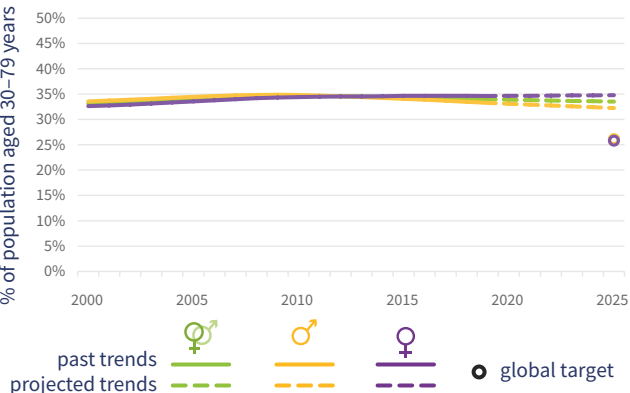
Prevalence of hypertension – global comparison (both sexes)^a



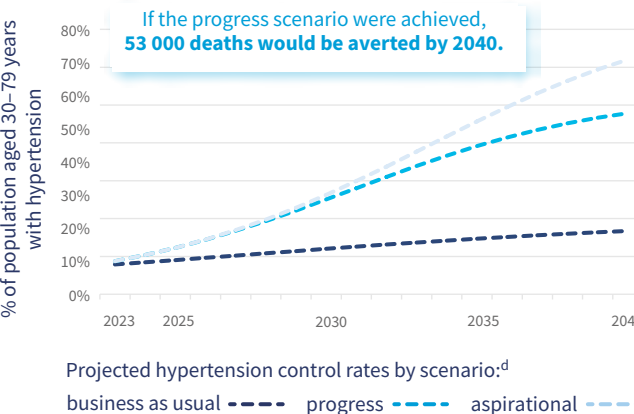
Of the 2.4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	26	22	2019
Cardiovascular disease deaths	27 200	14 100	13 100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	58	61	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	8	14	2	2019
Obesity, adults aged 18+ years (%)	11	6	16	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	16	4	2019
Physical inactivity, adults aged 18+ years (%)	29	22	35	2016

National response

Targets	Treatment
National target for blood pressure	✓
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Canada

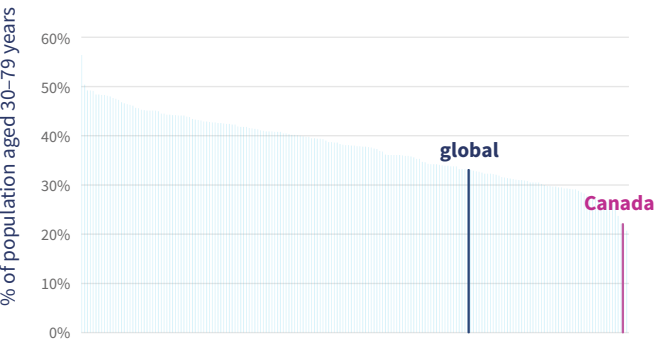
Hypertension profile

Total population (2019): 37 523 000

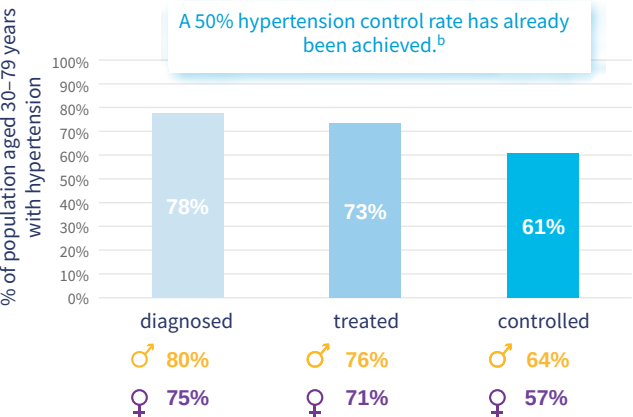
Total deaths (2019): 278 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 22% ♂ 24% ♀ 20%

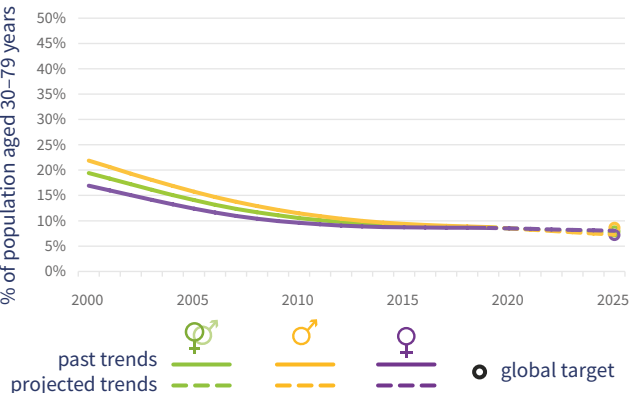
Prevalence of hypertension – global comparison (both sexes)^a



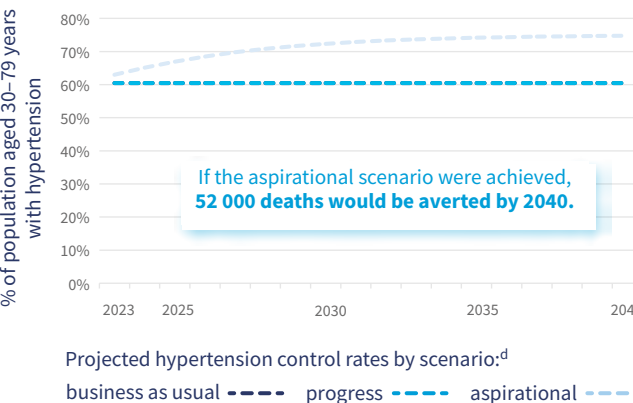
Of the 6.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	11	8	2019
Cardiovascular disease deaths	70 200	36 200	34 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	44	45	43	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%)	14	16	11	2019
Obesity, adults aged 18+ years (%)	29	30	29	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	16	4	2019
Physical inactivity, adults aged 18+ years (%)	29	26	31	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Central African Republic

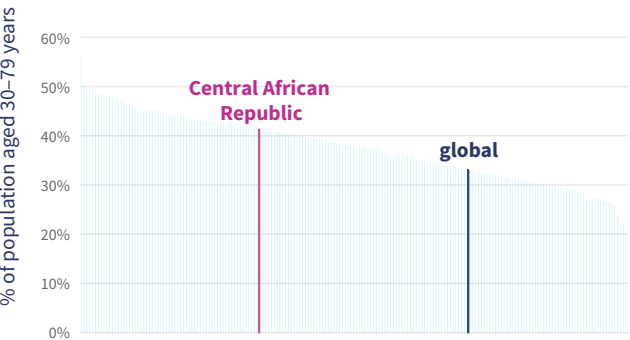
Hypertension profile

Total population (2019): 5 209 000

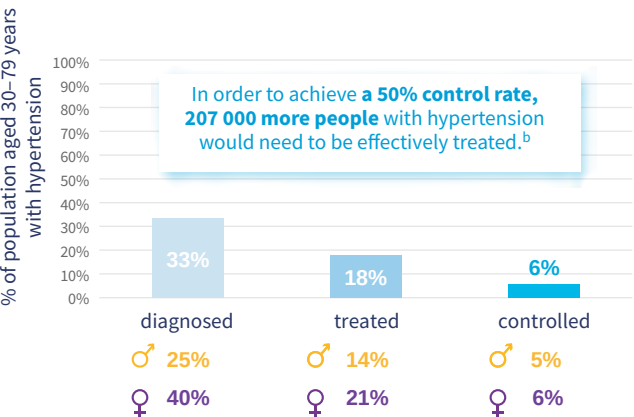
Total deaths (2019): 56 700

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 40% ♀ 43%

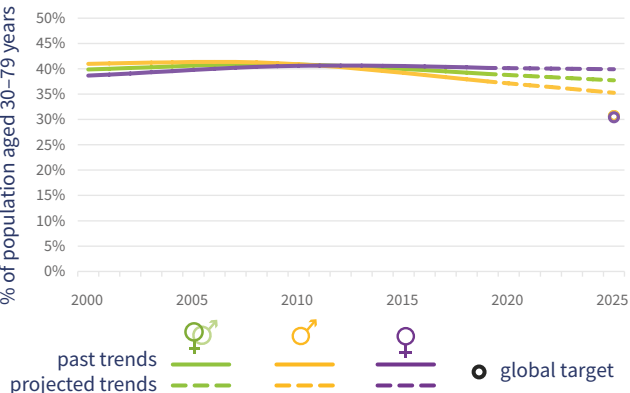
Prevalence of hypertension – global comparison (both sexes)^a



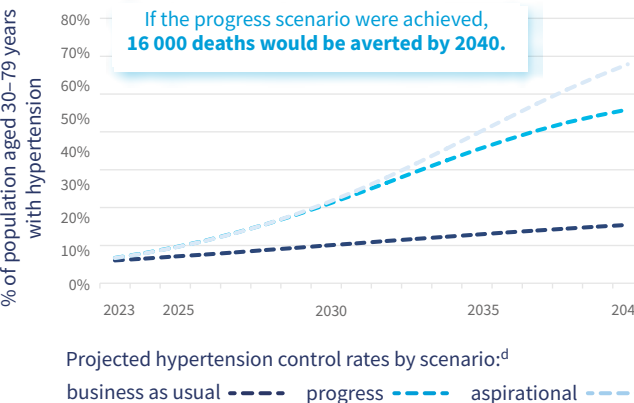
Of the 467 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	36	41	32	2019
Cardiovascular disease deaths	8300	4000	4300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	61	65	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	8	4	11	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	1	2019
Physical inactivity, adults aged 18+ years (%)	14	13	16	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Chad

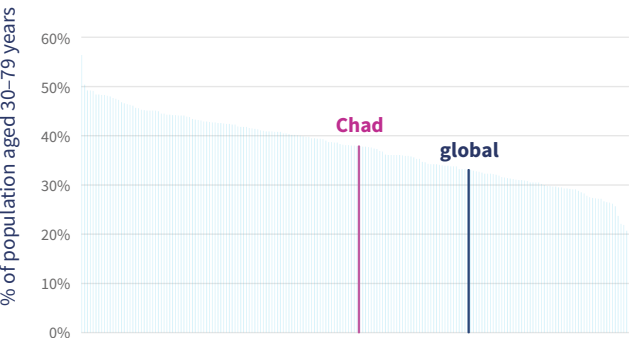
Hypertension profile

Total population (2019): 16 127 000

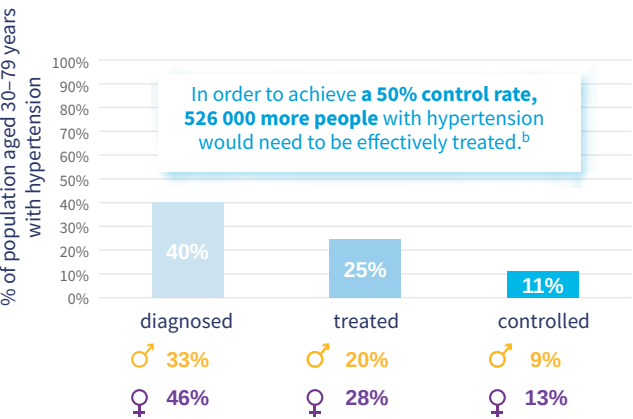
Total deaths (2019): 149 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 35% ♀ 41%

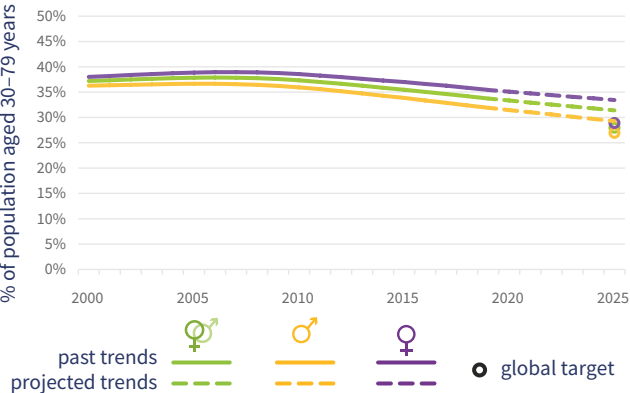
Prevalence of hypertension – global comparison (both sexes)^a



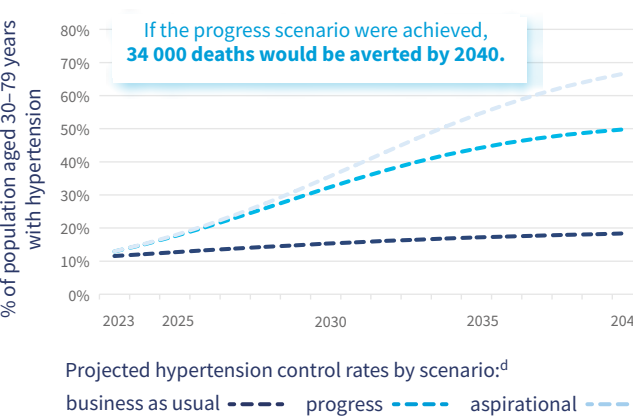
Of the 1.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	23	23	2019
Cardiovascular disease deaths	14 700	7200	7 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	51	54	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	9	14	3	2019
Obesity, adults aged 18+ years (%)	6	3	9	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	6	1	2019
Physical inactivity, adults aged 18+ years (%)	23	20	27	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Chile

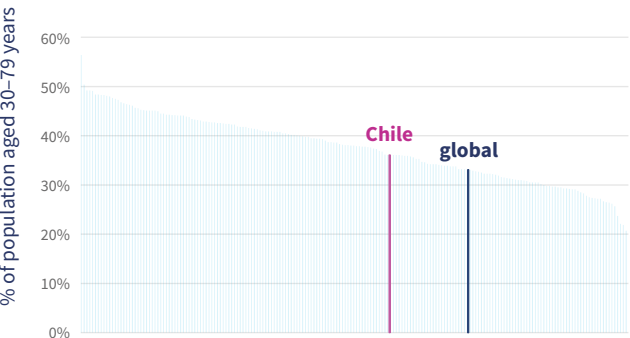
Hypertension profile

Total population (2019): 19 039 000

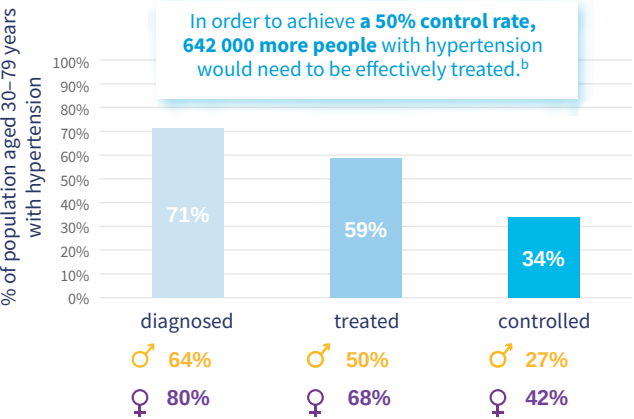
Total deaths (2019): 110 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 36% ♂ 39% ♀ 33%

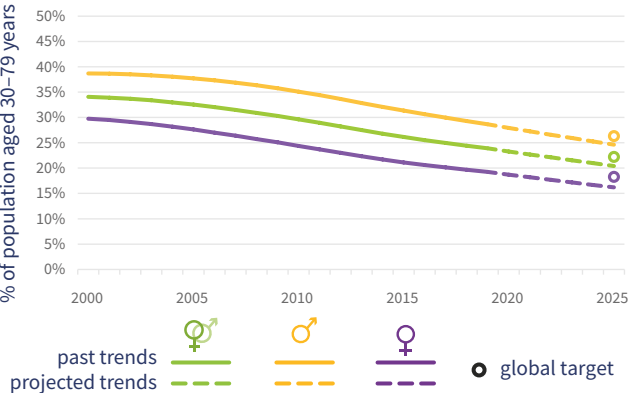
Prevalence of hypertension – global comparison (both sexes)^a



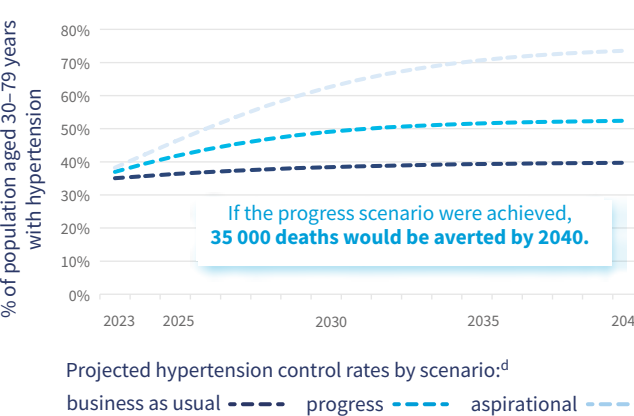
Of the 4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	12	8	2019
Cardiovascular disease deaths	28 500	14 400	14 100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	61	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%) ^f	30	32	28	2019
Obesity, adults aged 18+ years (%)	28	25	31	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	11	3	2019
Physical inactivity, adults aged 18+ years (%)	27	24	29	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

China

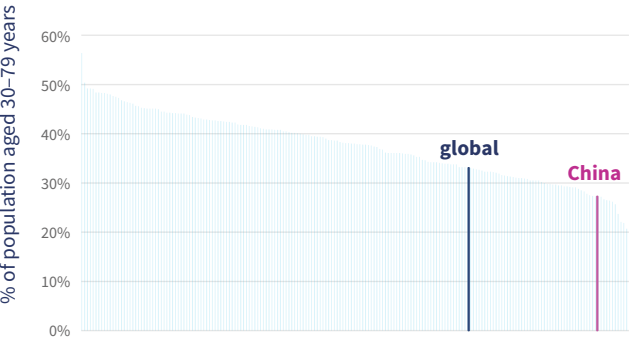
Hypertension profile

Total population (2019): 1 422 000 000

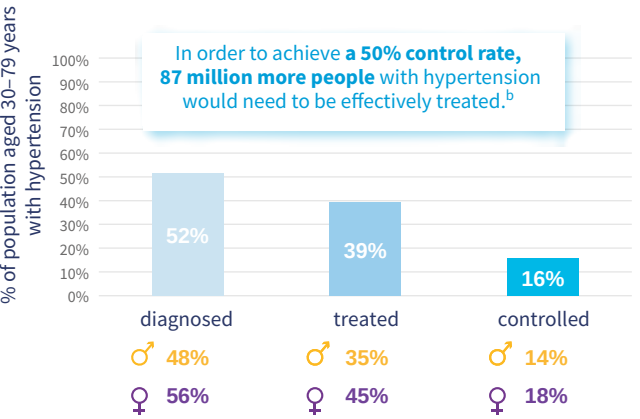
Total deaths (2019): 10 106 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 27% ♂ 30% ♀ 24%

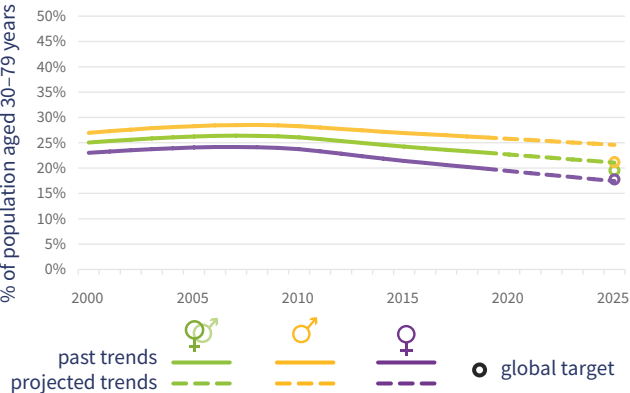
Prevalence of hypertension – global comparison (both sexes)^a



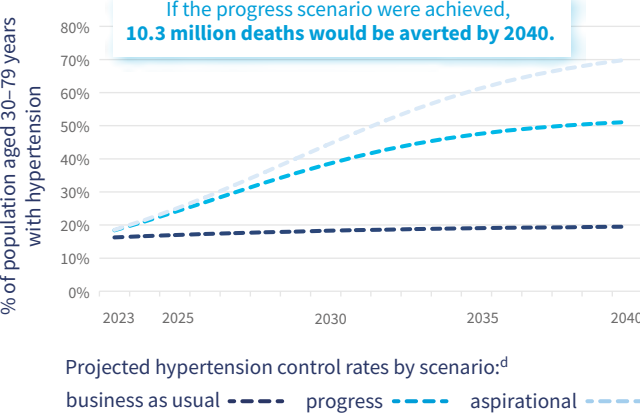
Of the 256.7 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	20	11	2019
Cardiovascular disease deaths	4 307 000	2 429 000	1 877 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	54	54	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	17	19	16	2019
Current tobacco use, adults aged 15+ years (%) ^f	26	50	2	2019
Obesity, adults aged 18+ years (%)	6	6	7	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	9	2	2019
Physical inactivity, adults aged 18+ years (%)	14	16	12	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Colombia

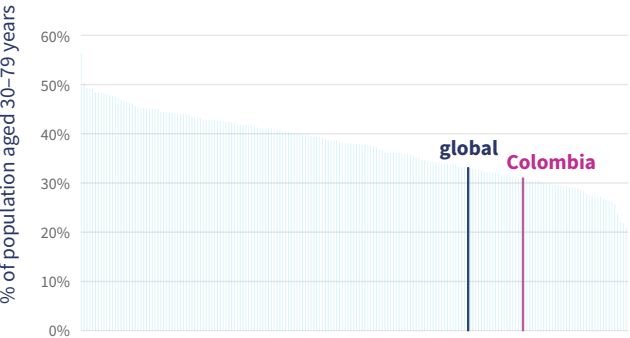
Hypertension profile

Total population (2019): 50 187 000

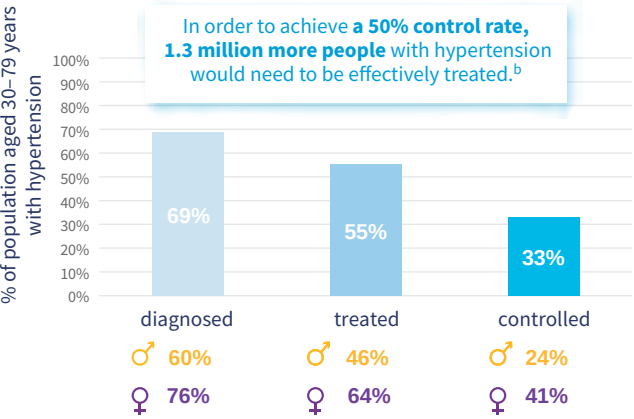
Total deaths (2019): 239 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 31% ♀ 31%

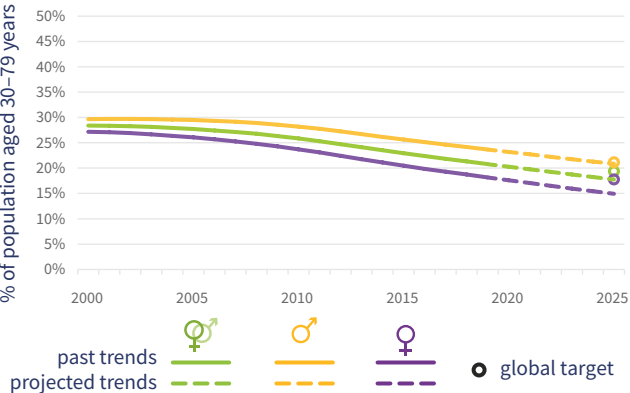
Prevalence of hypertension – global comparison (both sexes)^a



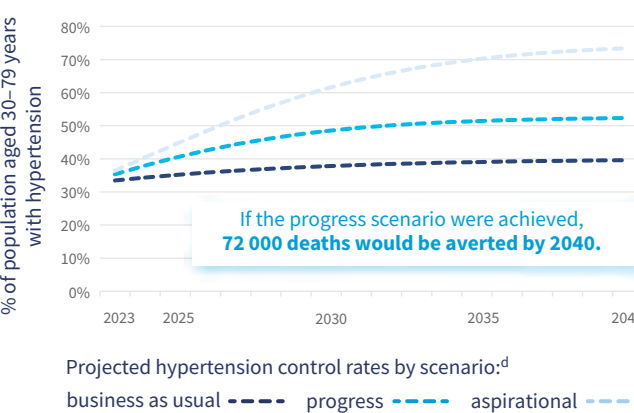
Of the 7.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	11	9	2019
Cardiovascular disease deaths	73 500	37 000	36 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	48	50	46	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	12	14	10	2019
Current tobacco use, adults aged 15+ years (%)	9	13	5	2019
Obesity, adults aged 18+ years (%)	22	18	27	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	8	2	2019
Physical inactivity, adults aged 18+ years (%)	44	39	49	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Comoros

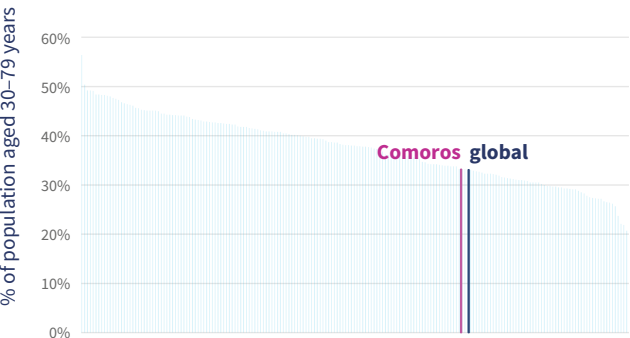
Hypertension profile

Total population (2019): 791 000

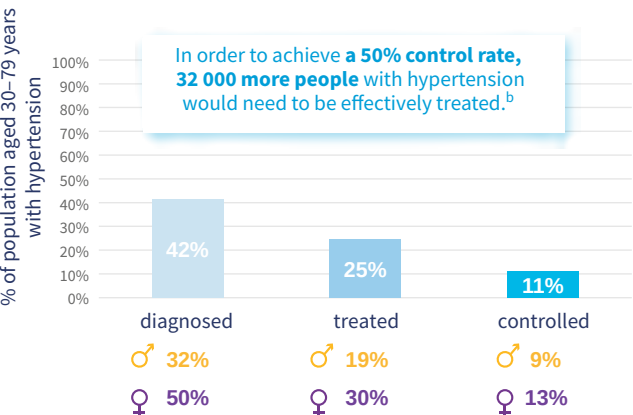
Total deaths (2019): 5030

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 33% ♂ 30% ♀ 36%

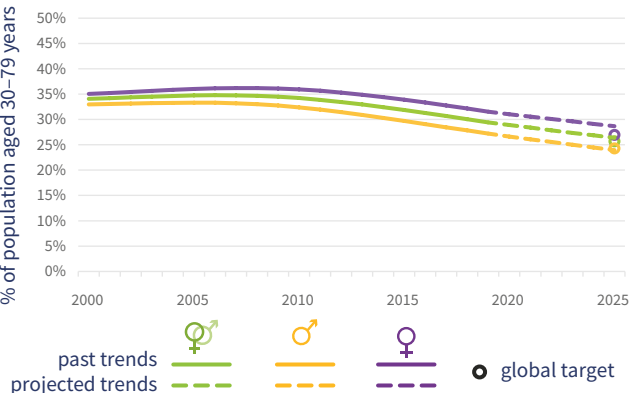
Prevalence of hypertension – global comparison (both sexes)^a



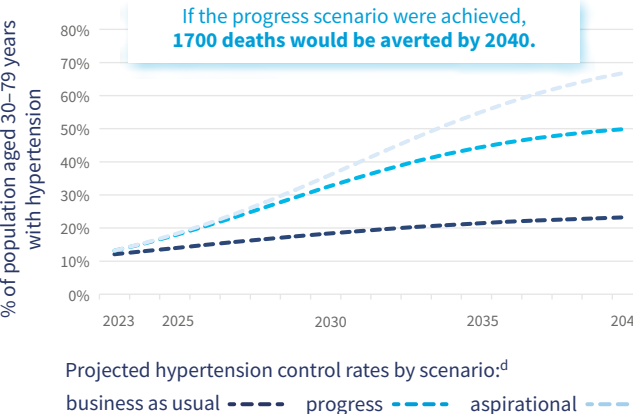
Of the 82 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	21	21	2019
Cardiovascular disease deaths	910	430	480	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	56	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	8	2019
Current tobacco use, adults aged 15+ years (%)	21	30	12	2019
Obesity, adults aged 18+ years (%)	8	3	12	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	14	10	19	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Congo

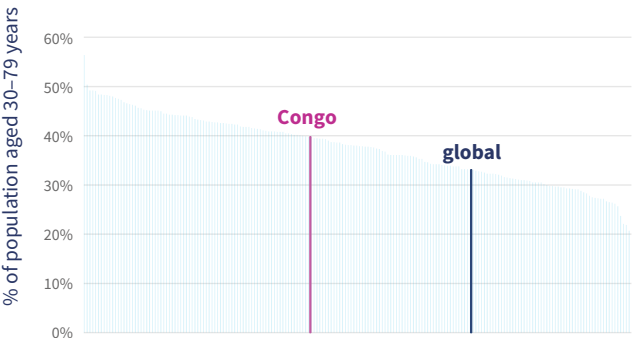
Hypertension profile

Total population (2019): 5 571 000

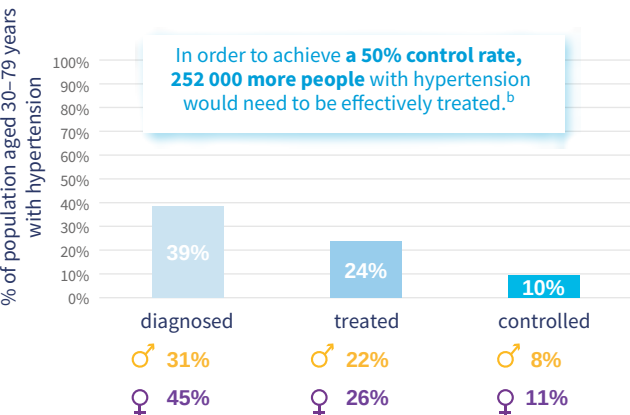
Total deaths (2019): 35 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 40% ♂ 38% ♀ 42%

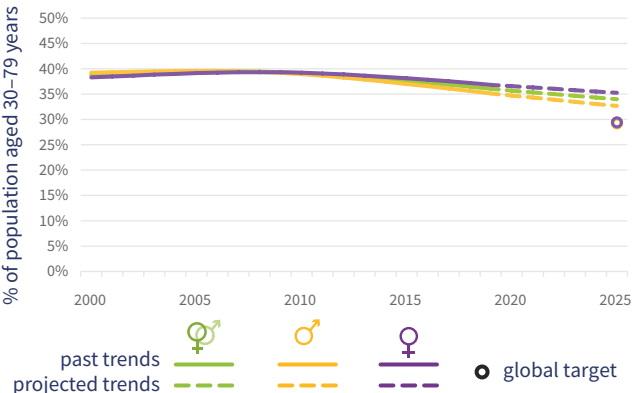
Prevalence of hypertension – global comparison (both sexes)^a



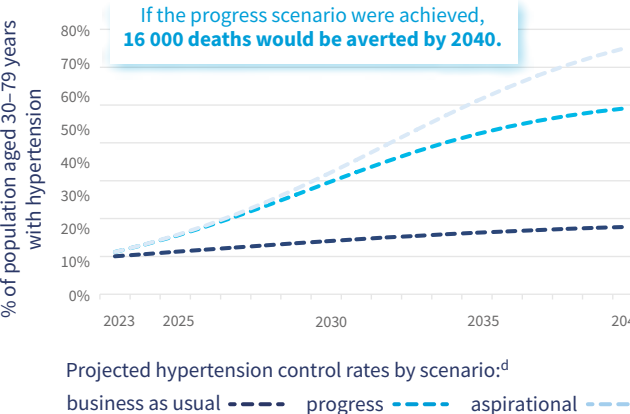
Of the 617 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	23	22	2019
Cardiovascular disease deaths	6300	2700	3600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	66	63	68	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	14	26	2	2019
Obesity, adults aged 18+ years (%)	10	6	14	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	10	3	2019
Physical inactivity, adults aged 18+ years (%)	28	25	31	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Cook Islands

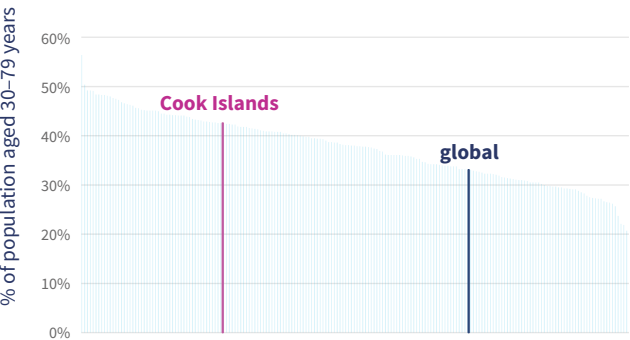
Hypertension profile

Total population (2019): 17 100

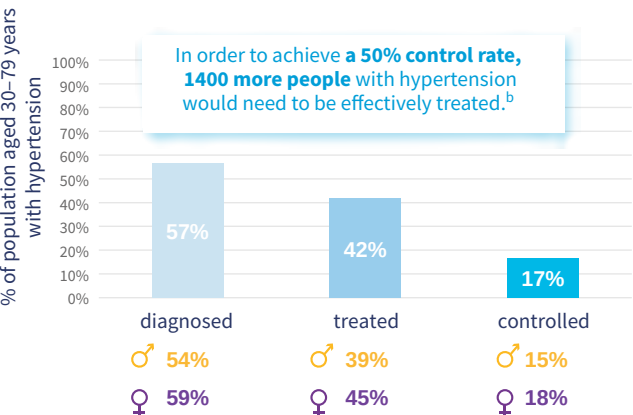
Total deaths (2019): no data

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 43% ♂ 45% ♀ 41%

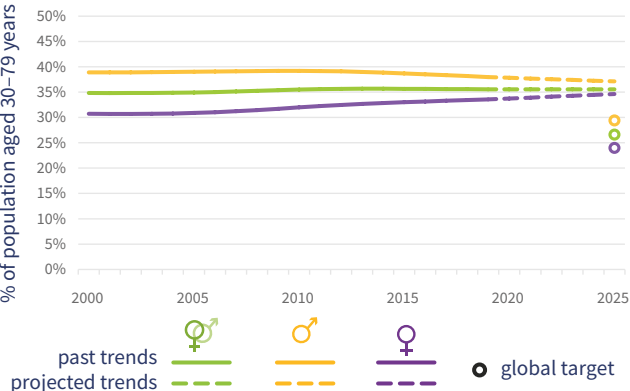
Prevalence of hypertension – global comparison (both sexes)^a



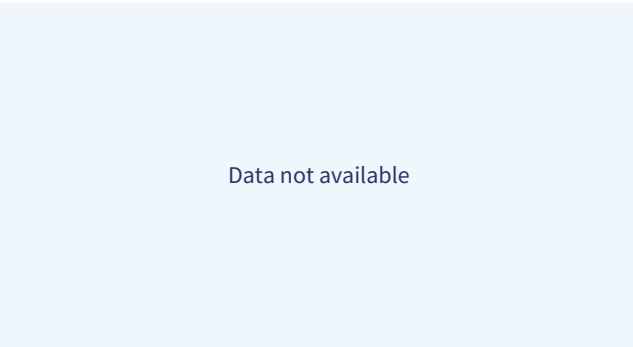
Of the 4500 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	64	63	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	8	2019
Current tobacco use, adults aged 15+ years (%) ^e	25	28	21	2019
Obesity, adults aged 18+ years (%)	56	53	59	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	18	4	2019
Physical inactivity, adults aged 18+ years (%)	18	10	27	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake. e. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Costa Rica

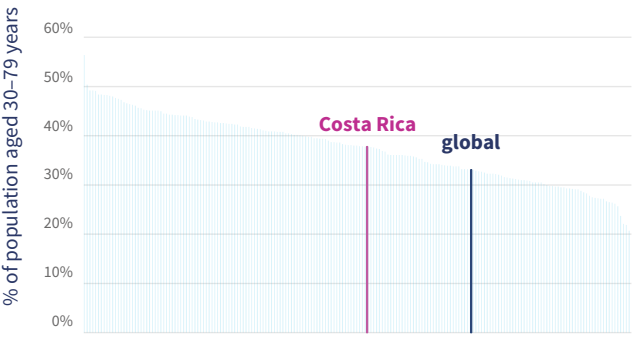
Hypertension profile

Total population (2019): 5 085 000

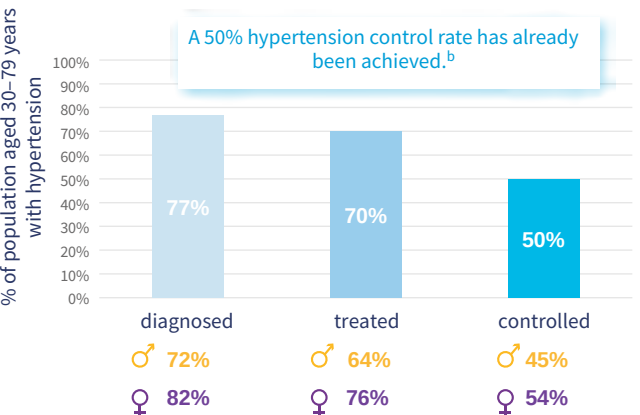
Total deaths (2019): 23 900

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 36% ♀ 39%

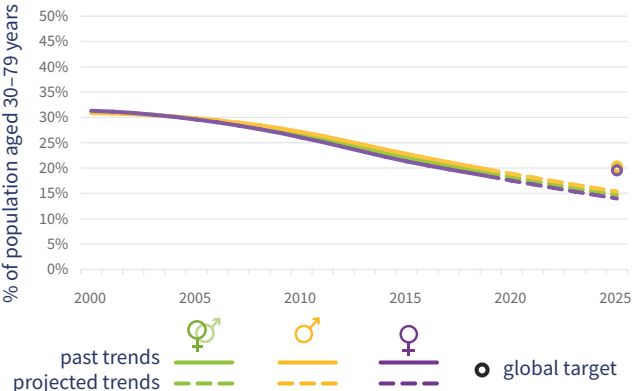
Prevalence of hypertension – global comparison (both sexes)^a



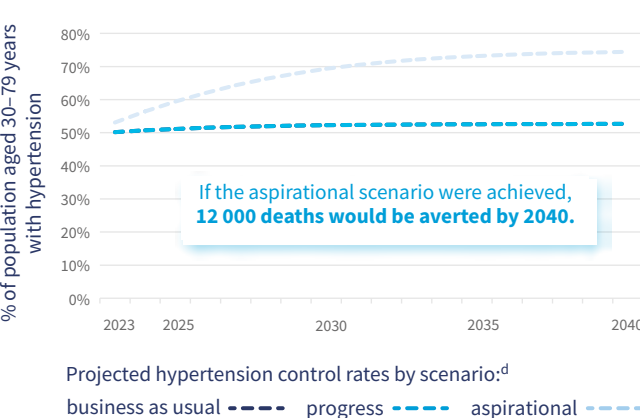
Of the 1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	11	8	2019
Cardiovascular disease deaths	6100	3300	2800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	59	59	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	9	2019
Current tobacco use, adults aged 15+ years (%)	9	13	5	2019
Obesity, adults aged 18+ years (%)	26	21	30	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	6	1	2019
Physical inactivity, adults aged 18+ years (%)	46	38	54	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Côte d'Ivoire

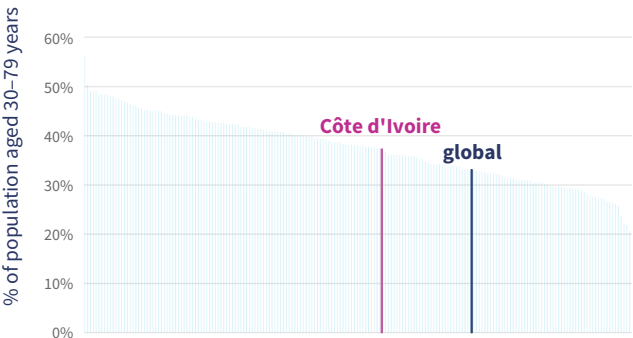
Hypertension profile

Total population (2019): 26 148 000

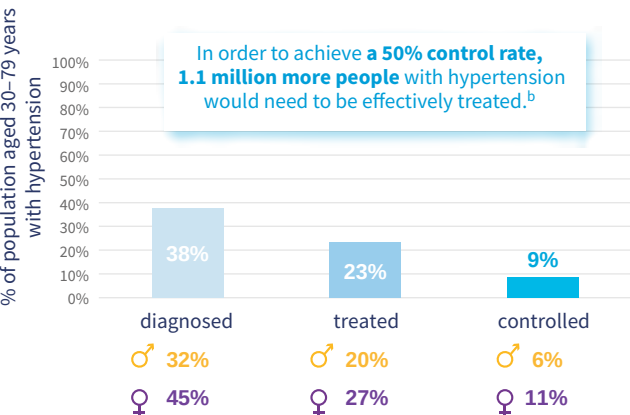
Total deaths (2019): 195 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 37% ♂ 37% ♀ 38%

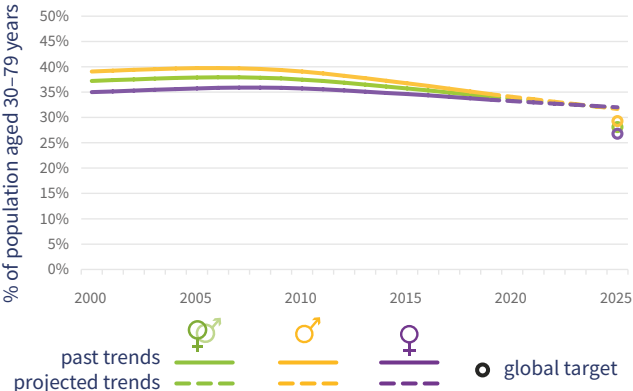
Prevalence of hypertension – global comparison (both sexes)^a



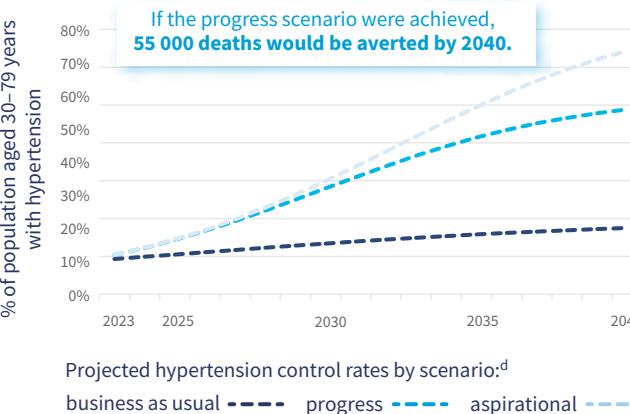
Of the 2.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	24	19	2019
Cardiovascular disease deaths	25 900	15 800	10 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	63	63	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	10	19	1	2019
Obesity, adults aged 18+ years (%)	10	6	15	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	33	29	37	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Croatia

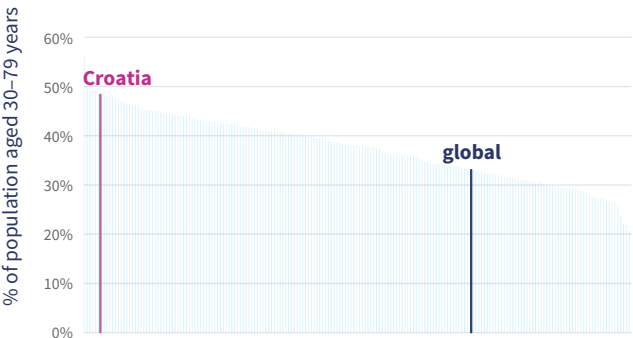
Hypertension profile

Total population (2019): 4 130 000

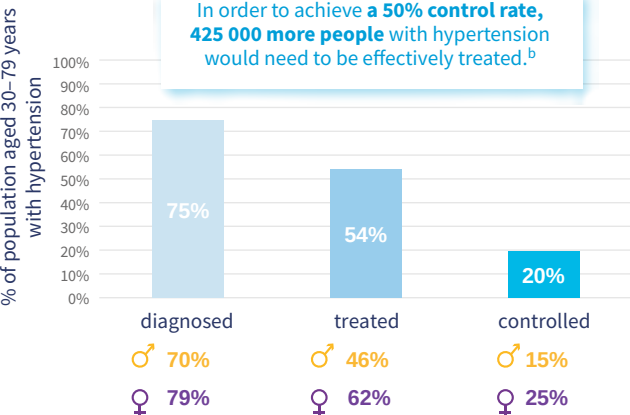
Total deaths (2019): 51 800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 48% ♂ 51% ♀ 45%

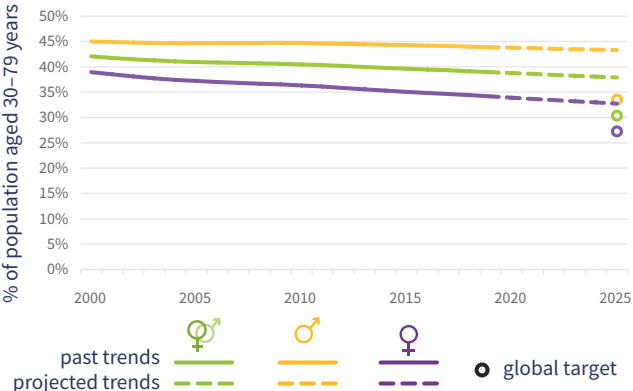
Prevalence of hypertension – global comparison (both sexes)^a



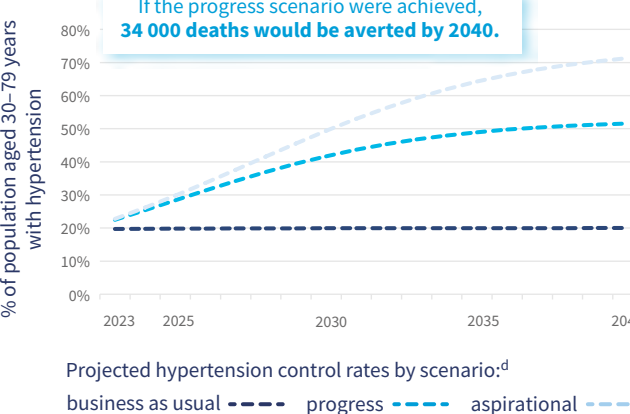
Of the 1.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	22	11	2019
Cardiovascular disease deaths	22 100	9700	12 400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	57	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%) ^f	37	38	36	2019
Obesity, adults aged 18+ years (%)	24	24	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	14	4	2019
Physical inactivity, adults aged 18+ years (%)	31	26	36	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Cuba

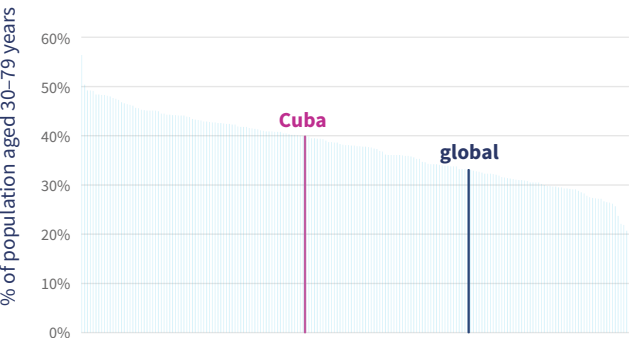
Hypertension profile

Total population (2019): 11 317 000

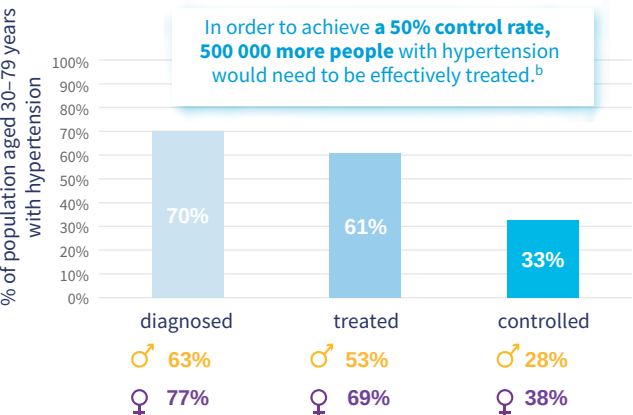
Total deaths (2019): 111 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 40% ♂ 40% ♀ 40%

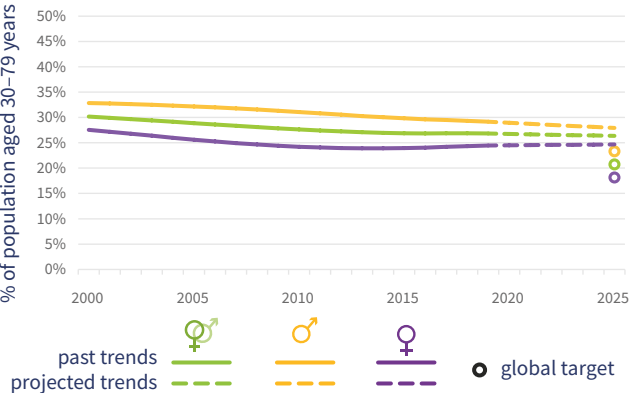
Prevalence of hypertension – global comparison (both sexes)^a



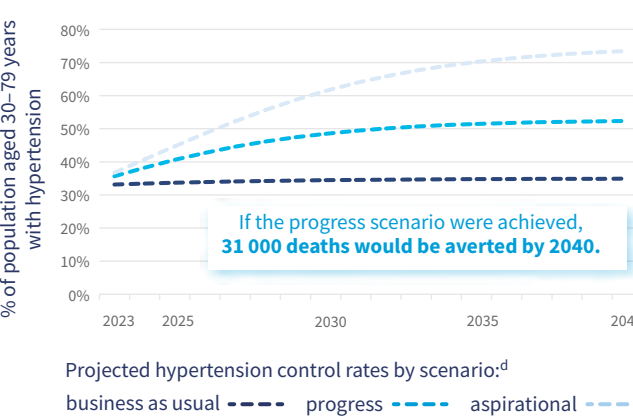
Of the 3.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	17	20	13	2019
Cardiovascular disease deaths	40 600	21 400	19 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	45	46	44	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	19	26	11	2019
Obesity, adults aged 18+ years (%)	25	19	30	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	10	2	2019
Physical inactivity, adults aged 18+ years (%)	37	31	43	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Cyprus

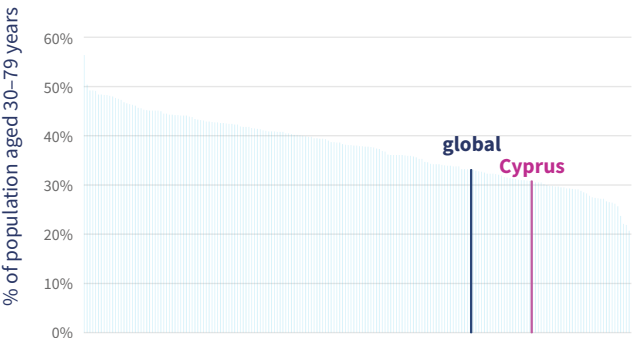
Hypertension profile

Total population (2019): 1 229 000

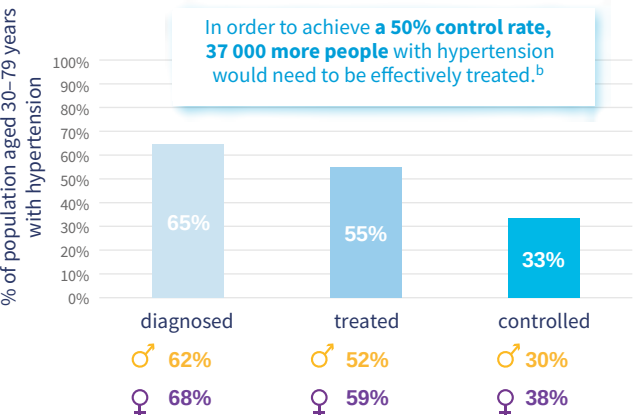
Total deaths (2019): 6760

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 36% ♀ 26%

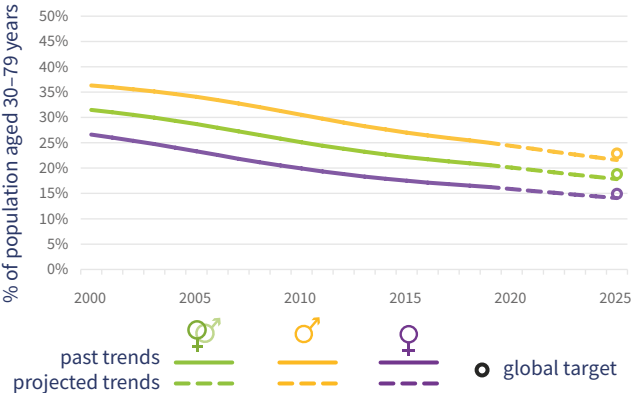
Prevalence of hypertension – global comparison (both sexes)^a



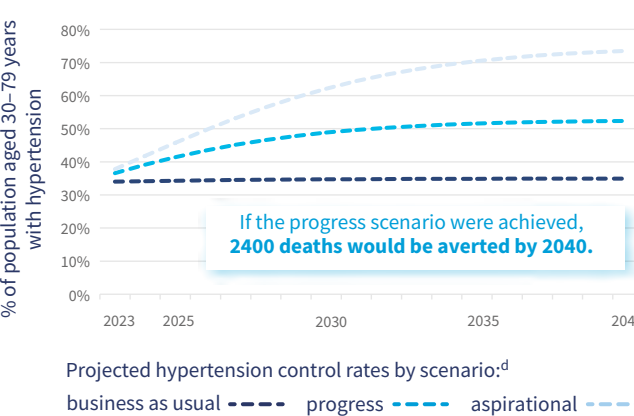
Of the 235 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	8	11	6	2019
Cardiovascular disease deaths	2100	1100	1000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	53	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	10	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	36	48	23	2019
Obesity, adults aged 18+ years (%)	22	22	22	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	13	3	2019
Physical inactivity, adults aged 18+ years (%)	44	38	51	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	no data		
Conducted recent, national survey on salt/sodium intake	no data		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Czechia

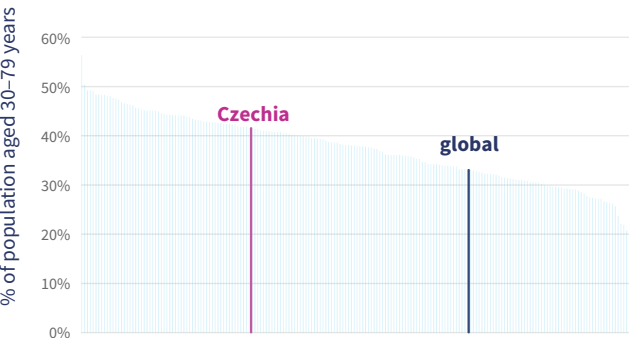
Hypertension profile

Total population (2019): 10 537 000

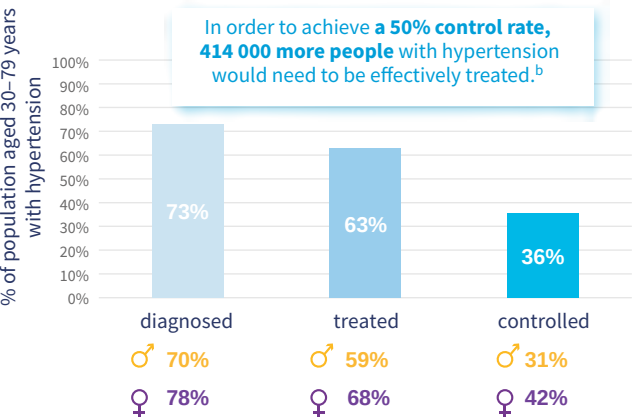
Total deaths (2019): 113 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 42% ♂ 49% ♀ 34%

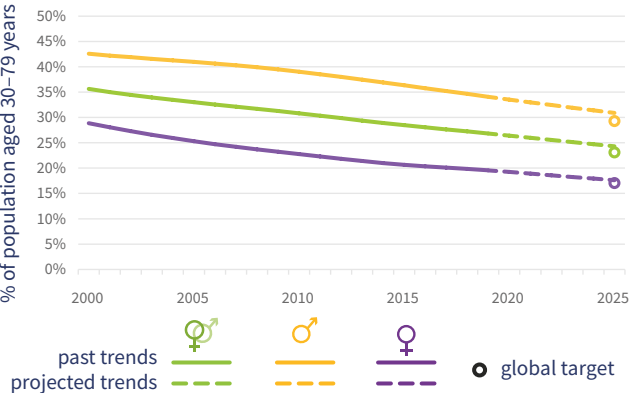
Prevalence of hypertension – global comparison (both sexes)^a



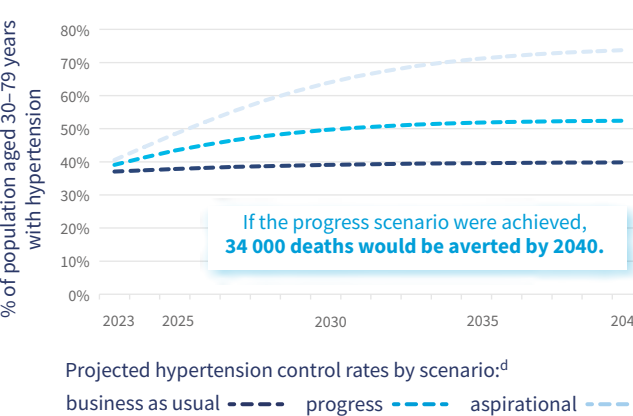
Of the 3.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	14	19	10	2019
Cardiovascular disease deaths	47 900	22 600	25 400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	47	51	43	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%) ^f	31	35	27	2019
Obesity, adults aged 18+ years (%)	26	26	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	13	21	6	2019
Physical inactivity, adults aged 18+ years (%)	31	28	34	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Democratic People's Republic of Korea

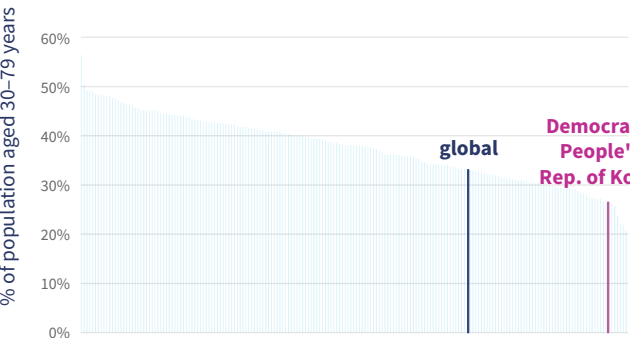
Hypertension profile

Total population (2019): 25 755 000

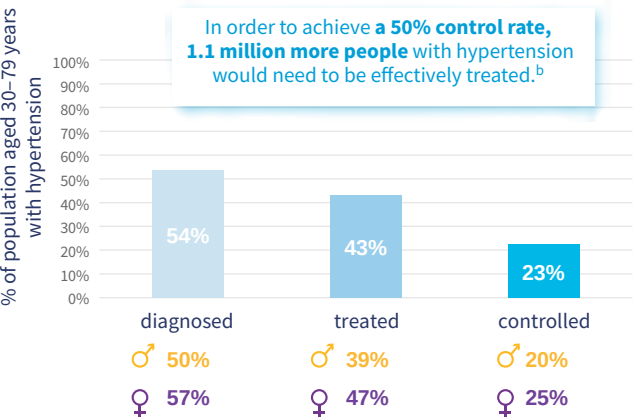
Total deaths (2019): 225 000[!]

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 27% ♂ 28% ♀ 25%

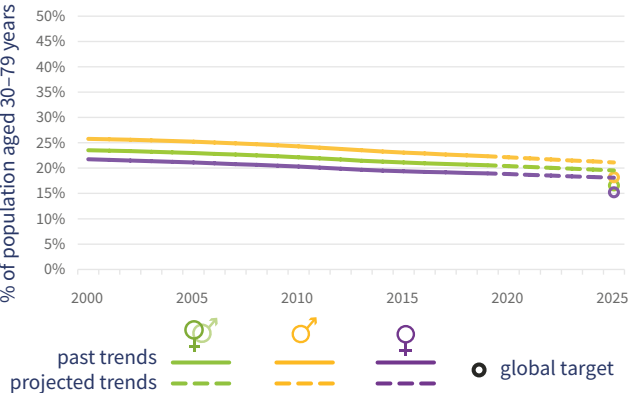
Prevalence of hypertension – global comparison (both sexes)^a



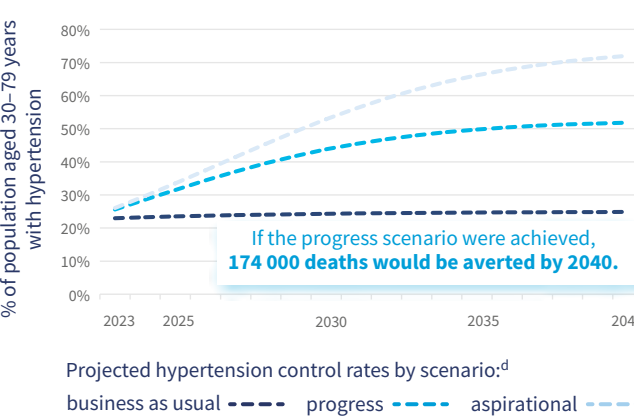
Of the 4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	30	19	2019
Cardiovascular disease deaths	90 900	41 400	49 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	47	50	44	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	12	13	12	2019
Current tobacco use, adults aged 15+ years (%) ^f	18	36	0	2019
Obesity, adults aged 18+ years (%)	7	6	7	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	7	2	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Democratic Republic of the Congo

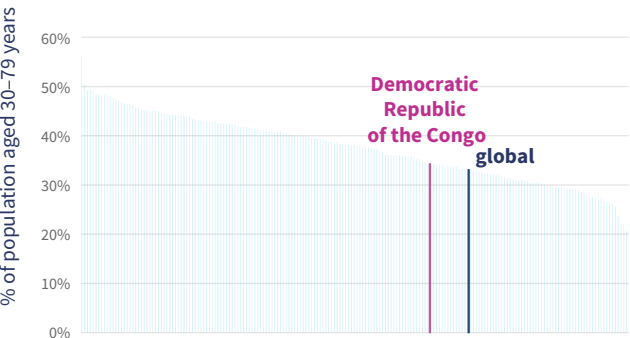
Hypertension profile

Total population (2019): 89 907 000

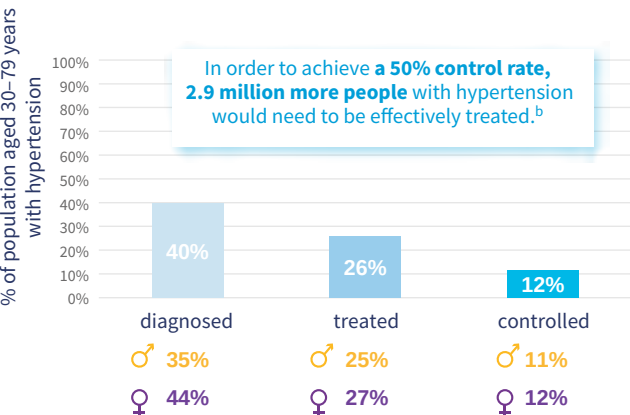
Total deaths (2019): 722 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 33% ♀ 36%

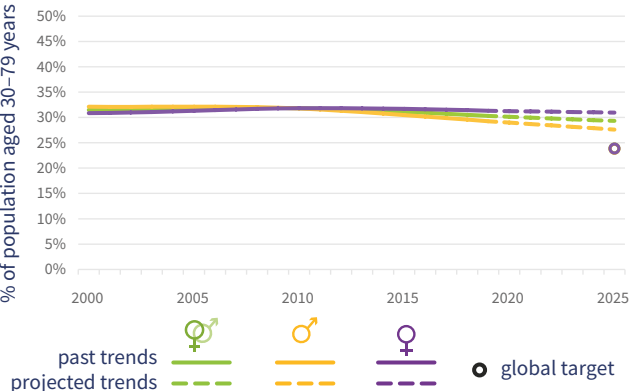
Prevalence of hypertension – global comparison (both sexes)^a



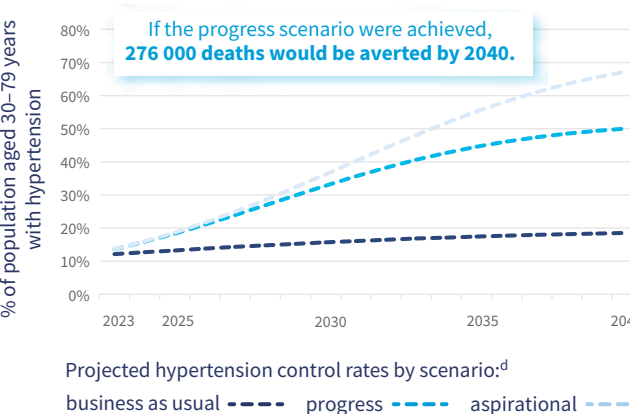
Of the 7.4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	26	22	2019
Cardiovascular disease deaths	98 600	47 700	50 900	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	61	59	62	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	6	5	2019
Current tobacco use, adults aged 15+ years (%)	13	23	3	2019
Obesity, adults aged 18+ years (%)	7	4	10	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	1	2019
Physical inactivity, adults aged 18+ years (%)	24	21	27	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Denmark

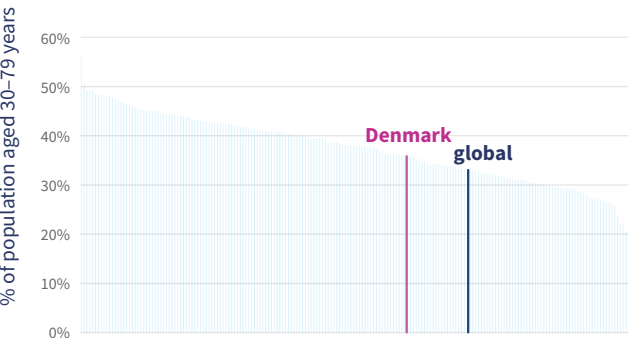
Hypertension profile

Total population (2019): 5 796 000

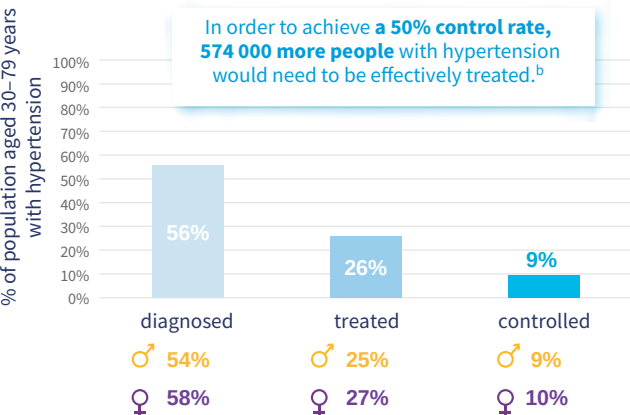
Total deaths (2019): 54 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 36% ♂ 43% ♀ 29%

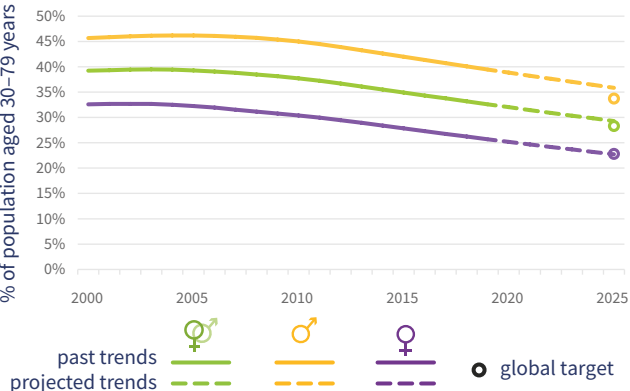
Prevalence of hypertension – global comparison (both sexes)^a



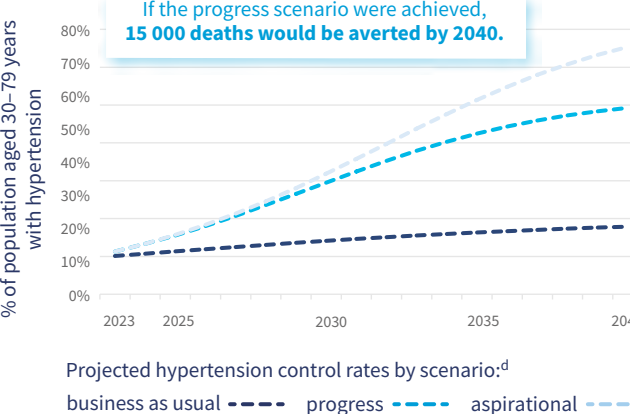
Of the 1.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	13	9	2019
Cardiovascular disease deaths	12 800	6600	6200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	53	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	18	18	18	2019
Obesity, adults aged 18+ years (%)	20	22	17	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	15	4	2019
Physical inactivity, adults aged 18+ years (%)	29	26	31	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Djibouti

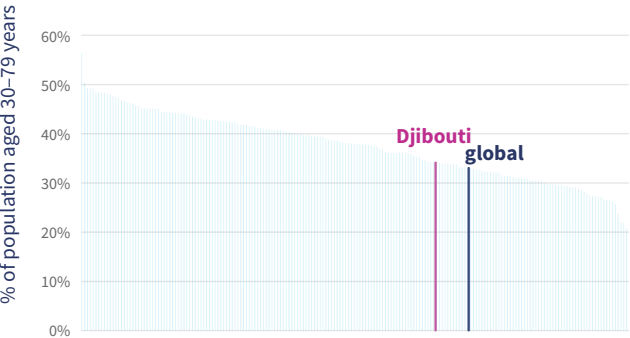
Hypertension profile

Total population (2019): 1 074 000

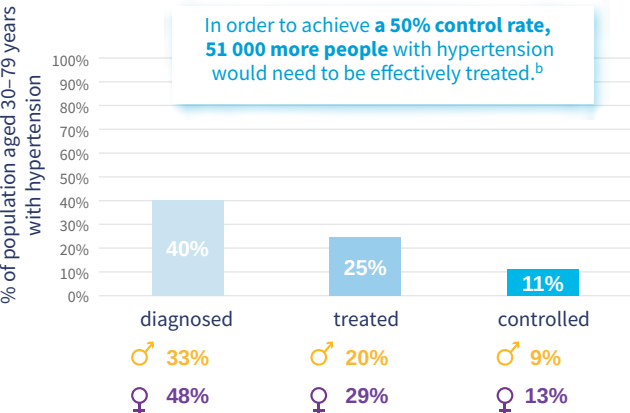
Total deaths (2019): 7270

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 33% ♀ 36%

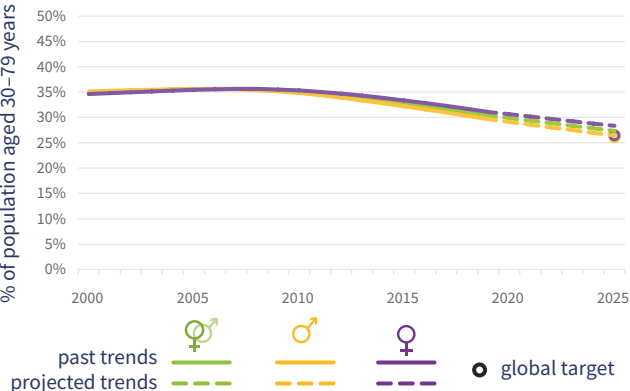
Prevalence of hypertension – global comparison (both sexes)^a



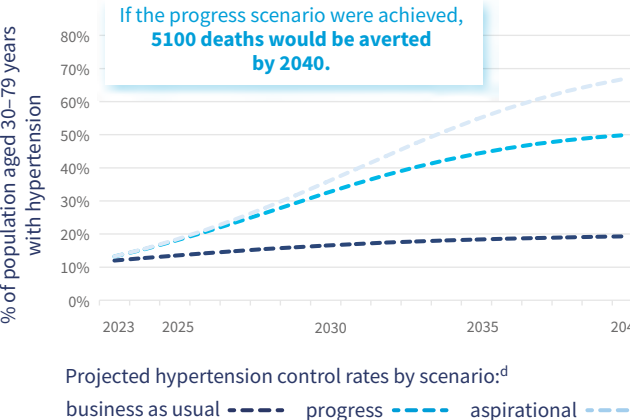
Of the 130 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	23	21	2019
Cardiovascular disease deaths	1700	910	820	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	60	61	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	14	9	18	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	1	0	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Dominica

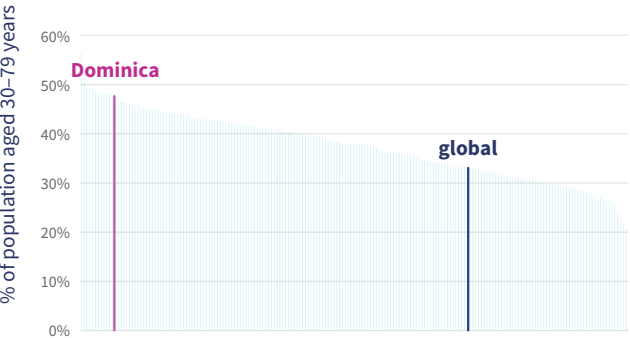
Hypertension profile

Total population (2019): 71 400

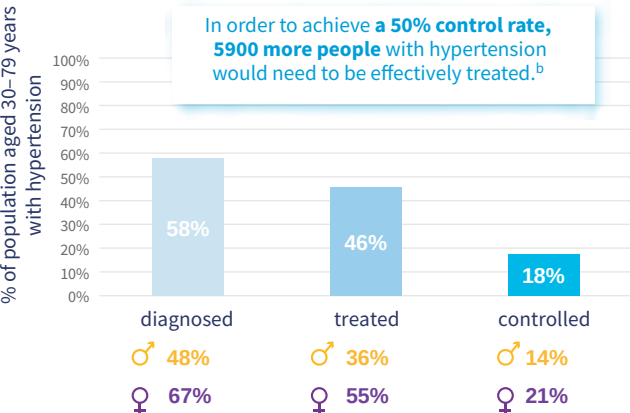
Total deaths (2019): no data

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 48% ♂ 46% ♀ 50%

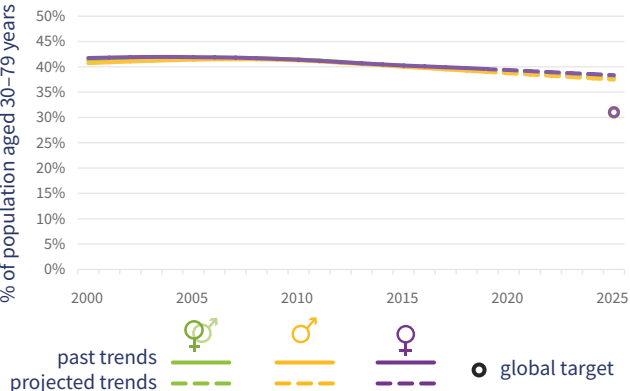
Prevalence of hypertension – global comparison (both sexes)^a



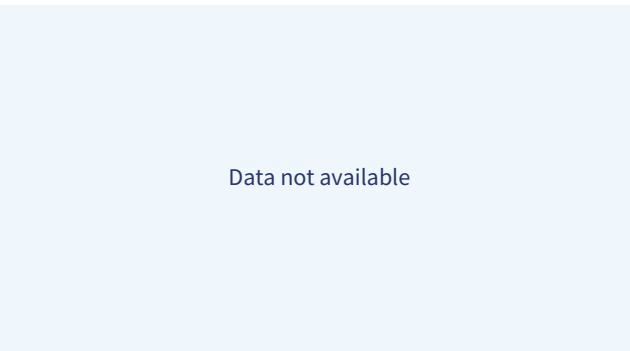
Of the 18 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	56	50	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	28	20	36	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	10	2	2019
Physical inactivity, adults aged 18+ years (%)	22	13	30	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		✓
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake.

Dominican Republic

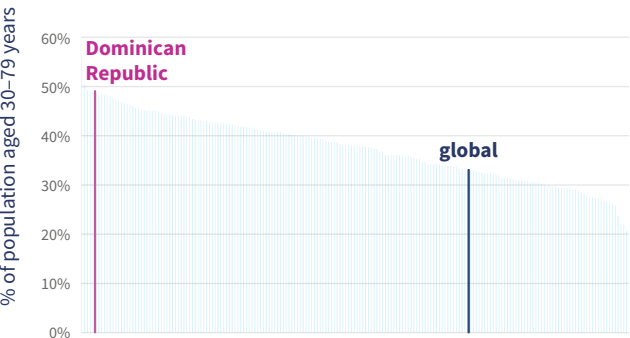
Hypertension profile

Total population (2019): 10 882 000

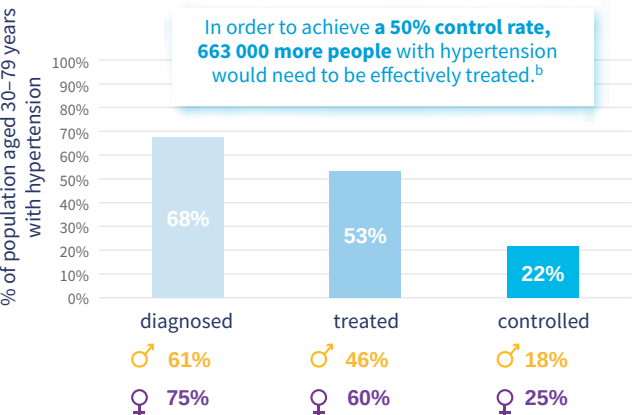
Total deaths (2019): 72 900

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 49% ♂ 49% ♀ 49%

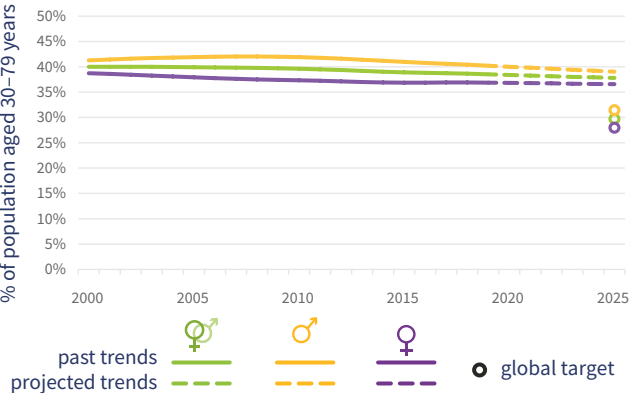
Prevalence of hypertension – global comparison (both sexes)^a



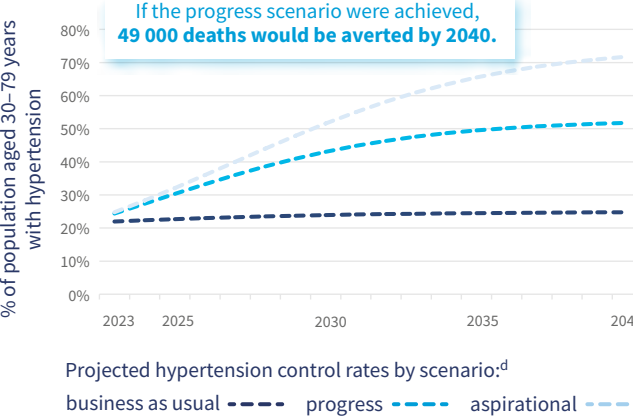
Of the 2.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	19	22	16	2019
Cardiovascular disease deaths	26 500	14 100	12 400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	54	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	11	15	7	2019
Obesity, adults aged 18+ years (%)	28	21	34	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	11	3	2019
Physical inactivity, adults aged 18+ years (%)	39	34	43	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Ecuador

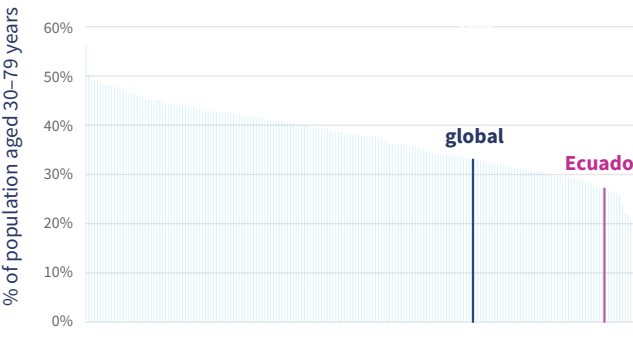
Hypertension profile

Total population (2019): 17 344 000

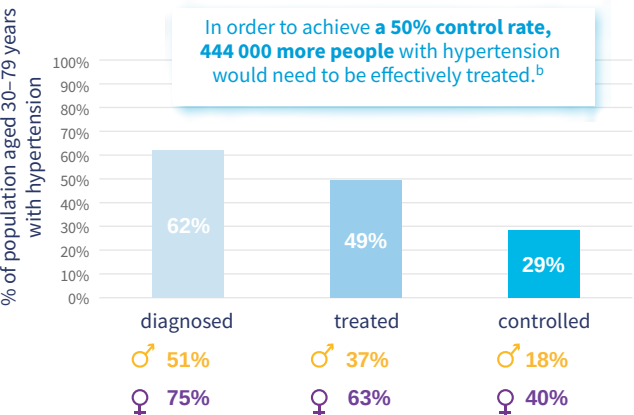
Total deaths (2019): 80 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 27% ♂ 29% ♀ 25%

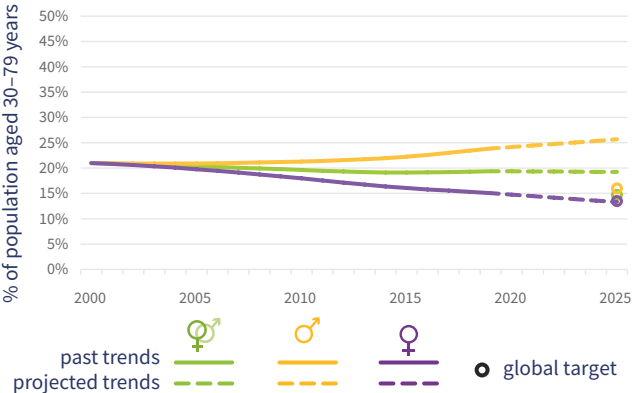
Prevalence of hypertension – global comparison (both sexes)^a



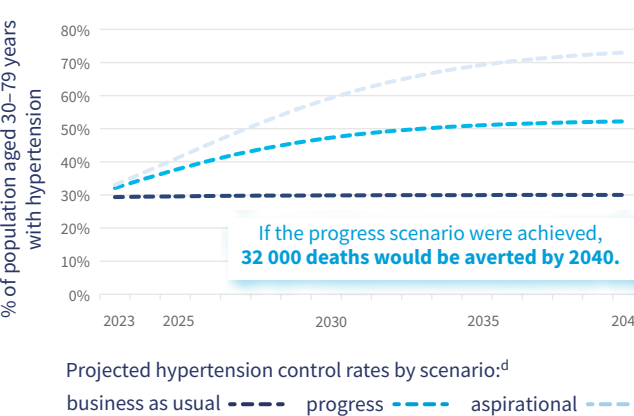
Of the 2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	12	10	2019
Cardiovascular disease deaths	19 600	10 200	9400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	43	43	44	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%)	12	19	4	2019
Obesity, adults aged 18+ years (%)	20	15	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	27	25	30	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Egypt

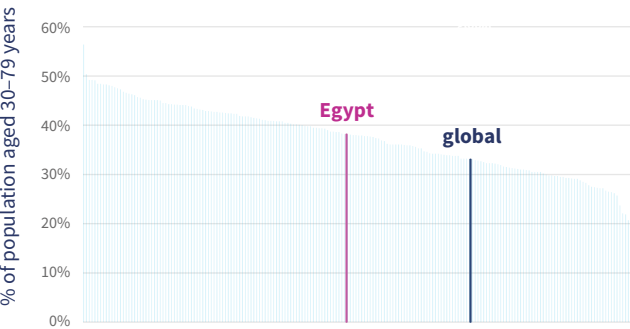
Hypertension profile

Total population (2019): 105 600 000

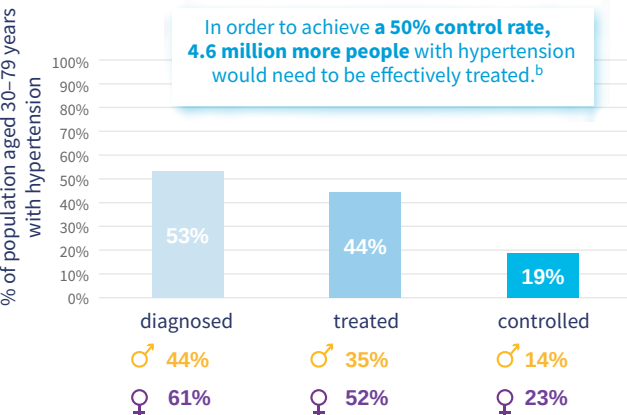
Total deaths (2019): 575 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 36% ♀ 41%

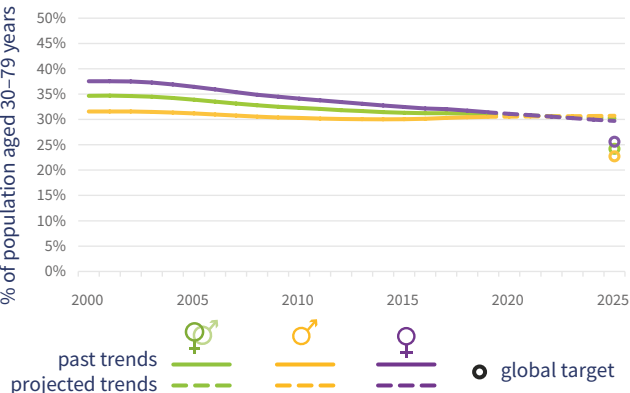
Prevalence of hypertension – global comparison (both sexes)^a



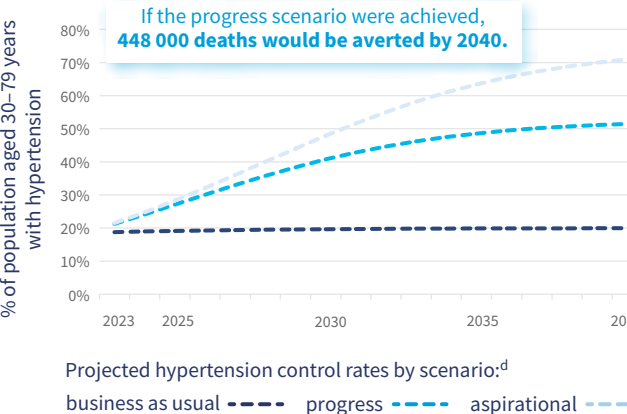
Of the 14.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	28	33	23	2019
Cardiovascular disease deaths	253 000	136 000	117 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	55	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	24	48	0	2019
Obesity, adults aged 18+ years (%)	32	23	41	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	31	23	39	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

El Salvador

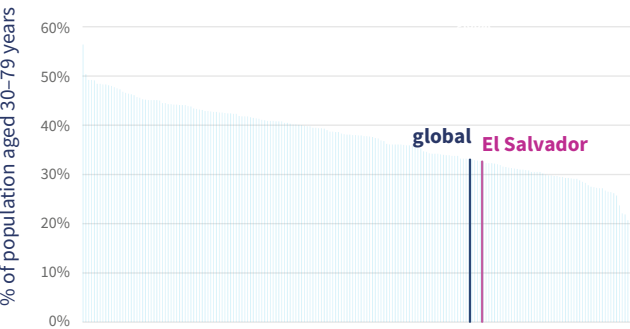
Hypertension profile

Total population (2019): 6 280 000

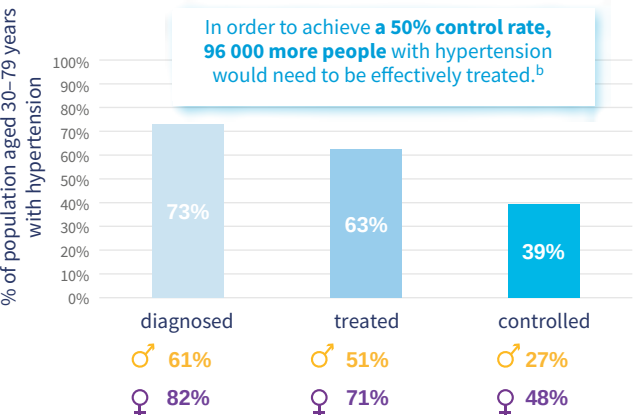
Total deaths (2019): 40 700

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 33% ♂ 31% ♀ 34%

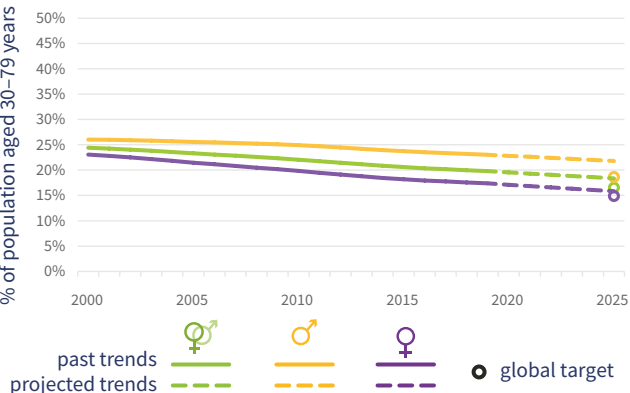
Prevalence of hypertension – global comparison (both sexes)^a



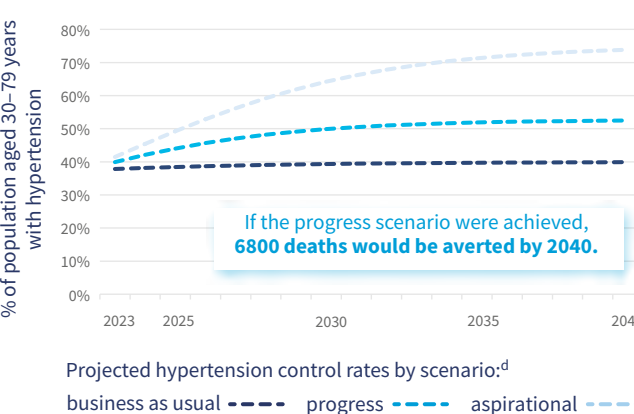
Of the 918 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	12	10	2019
Cardiovascular disease deaths	8200	4200	4100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	52	51	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	9	2019
Current tobacco use, adults aged 15+ years (%) ^f	8	15	2	2019
Obesity, adults aged 18+ years (%)	25	19	29	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	6	1	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Equatorial Guinea

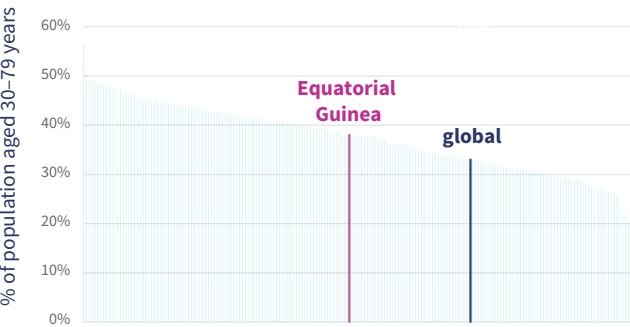
Hypertension profile

Total population (2019): 1 553 000

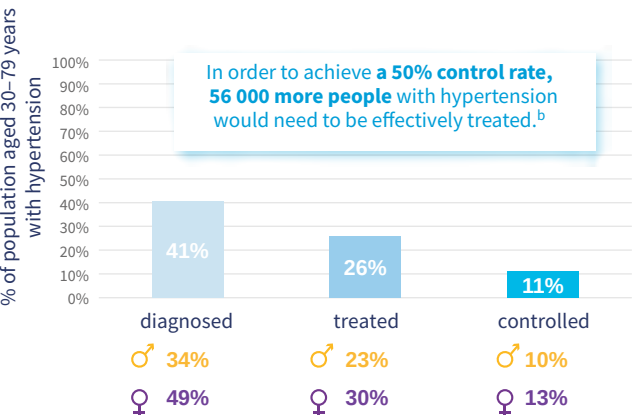
Total deaths (2019): 10 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 37% ♀ 40%

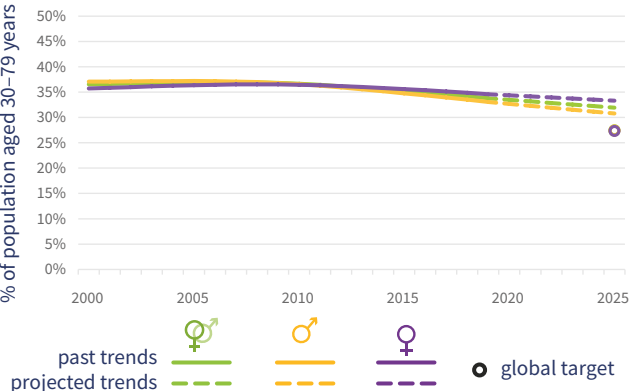
Prevalence of hypertension – global comparison (both sexes)^a



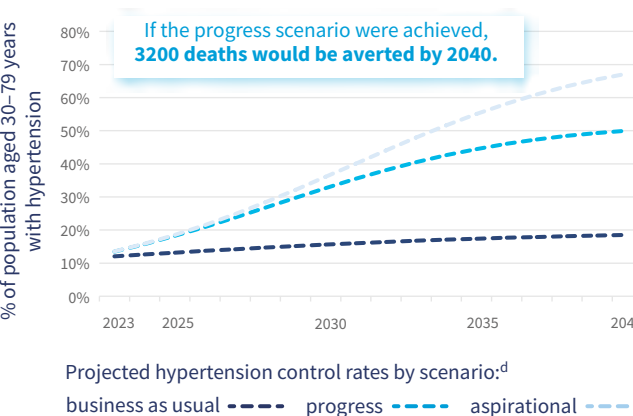
Of the 141 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	22	23	2019
Cardiovascular disease deaths	1200	530	660	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	65	61	68	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	8	4	13	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	10	3	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Eritrea

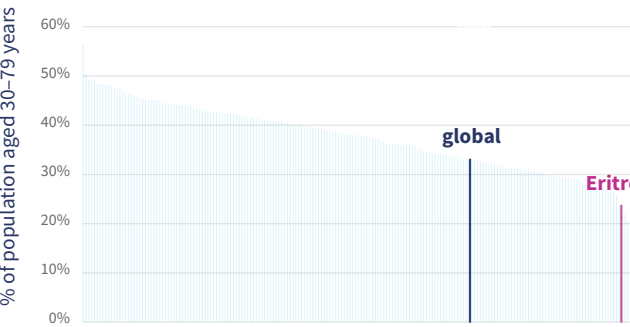
Hypertension profile

Total population (2019): 3 499 000

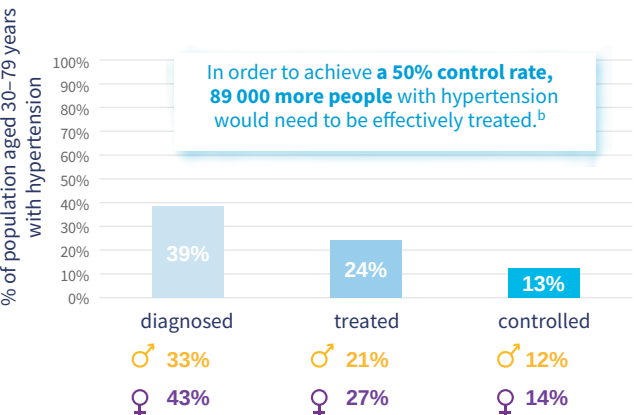
Total deaths (2019): 27 800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 24% ♂ 23% ♀ 25%

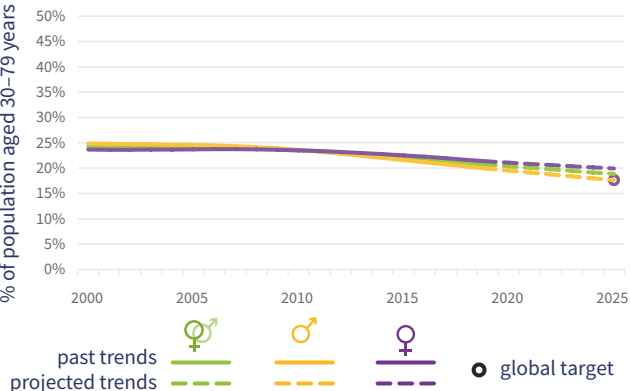
Prevalence of hypertension – global comparison (both sexes)^a



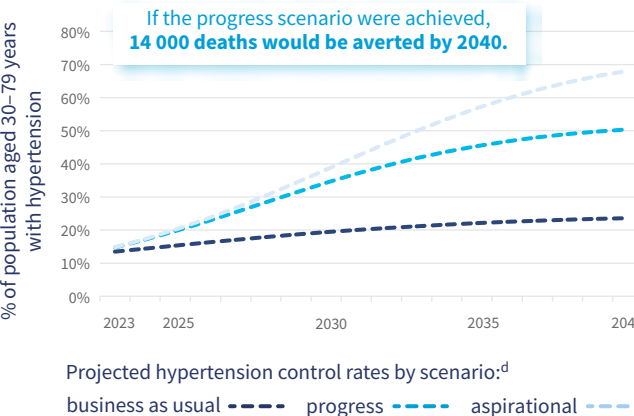
Of the 236 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	27	29	24	2019
Cardiovascular disease deaths	6200	3100	3100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	53	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	8	15	0	2019
Obesity, adults aged 18+ years (%)	5	2	8	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	2	0	2019
Physical inactivity, adults aged 18+ years (%)	22	14	31	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Estonia

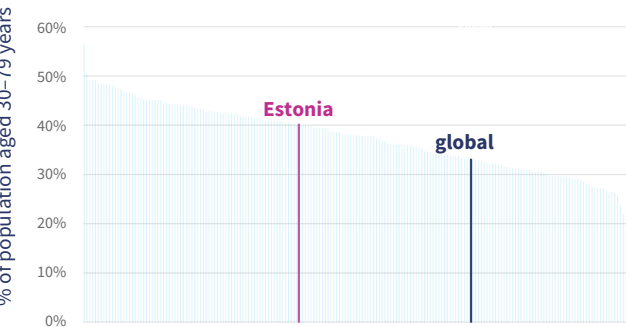
Hypertension profile

Total population (2019): 1 327 000

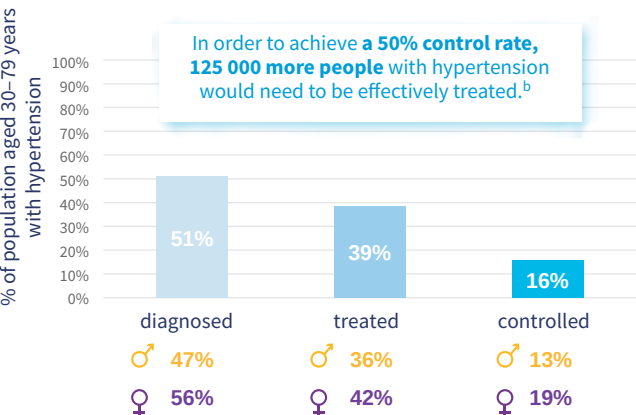
Total deaths (2019): 15 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 40% ♂ 46% ♀ 34%

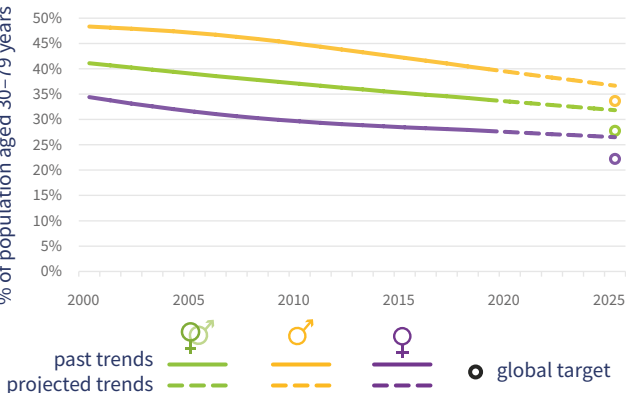
Prevalence of hypertension – global comparison (both sexes)^a



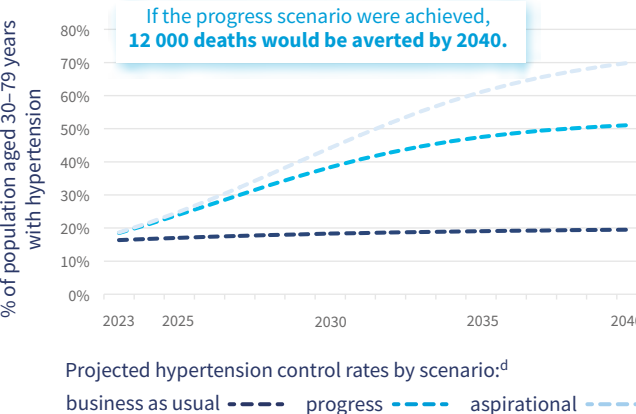
Of the 368 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	15	22	9	2019
Cardiovascular disease deaths	7700	3000	4700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	63	63	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	31	37	24	2019
Obesity, adults aged 18+ years (%)	21	20	22	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	18	5	2019
Physical inactivity, adults aged 18+ years (%)	32	29	35	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Eswatini

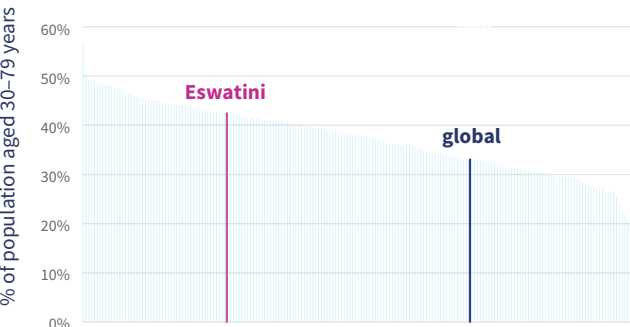
Hypertension profile

Total population (2019): 1 170 000

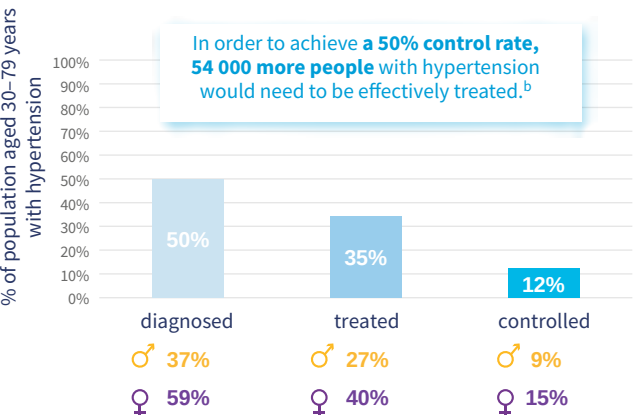
Total deaths (2019): 11 800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 43% ♂ 37% ♀ 47%

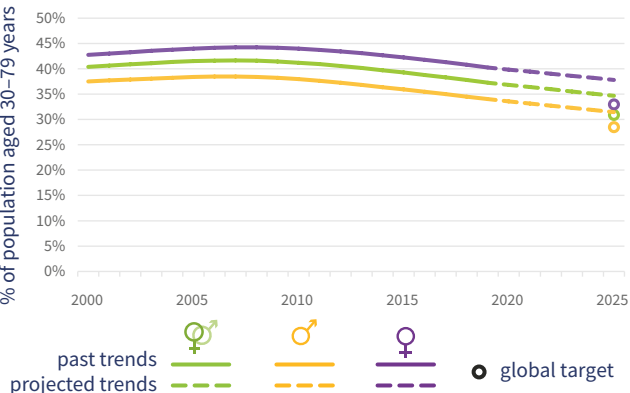
Prevalence of hypertension – global comparison (both sexes)^a



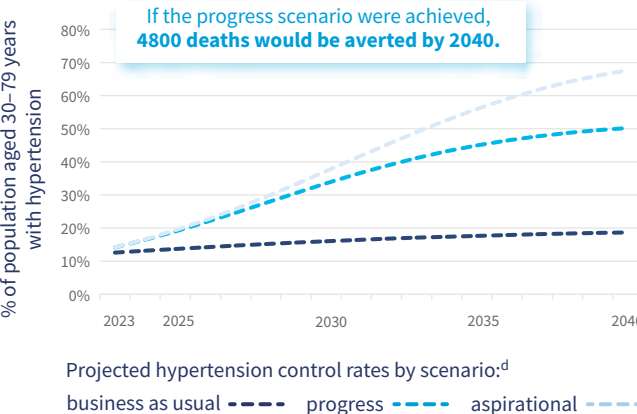
Of the 140 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	35	45	24	2019
Cardiovascular disease deaths	2100	1100	950	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	61	65	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	7	2019
Current tobacco use, adults aged 15+ years (%)	9	17	2	2019
Obesity, adults aged 18+ years (%)	17	5	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	14	3	2019
Physical inactivity, adults aged 18+ years (%)	28	24	32	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Ethiopia

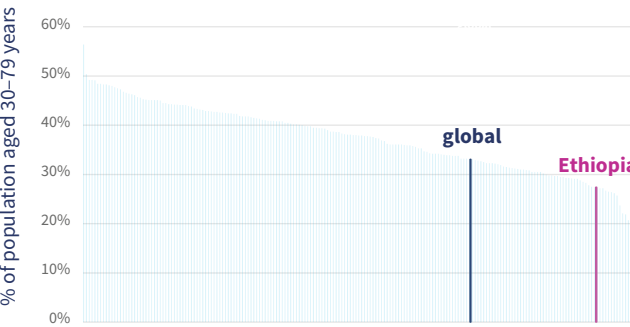
Hypertension profile

Total population (2019): 114 100 000

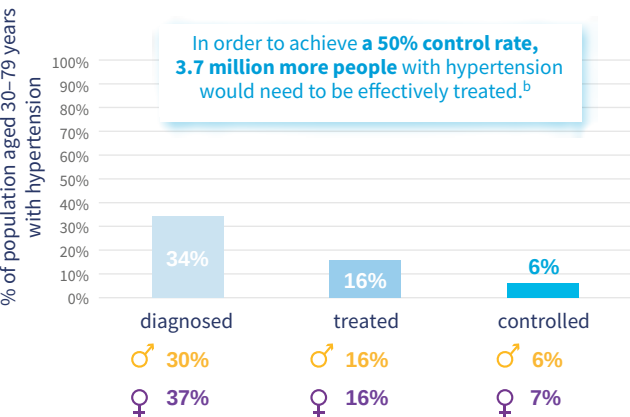
Total deaths (2019): 627 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 27% ♂ 25% ♀ 30%

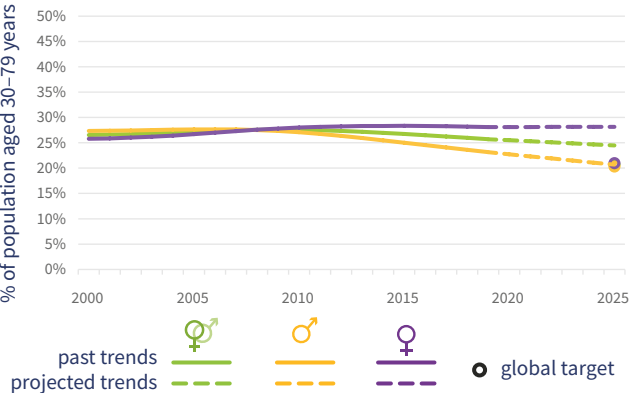
Prevalence of hypertension – global comparison (both sexes)^a



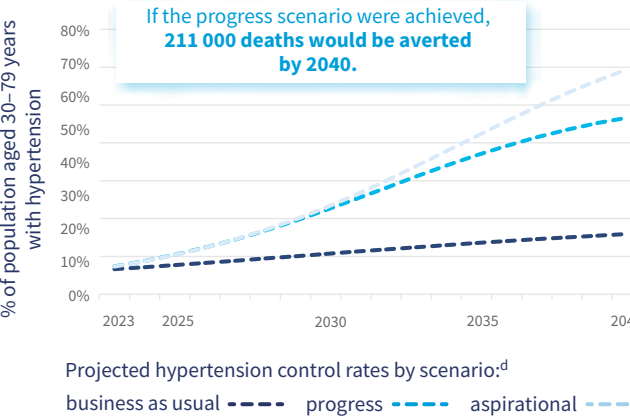
Of the 8.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	17	17	17	2019
Cardiovascular disease deaths	98 400	48 100	50 300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	48	45	51	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	5	9	1	2019
Obesity, adults aged 18+ years (%)	5	2	7	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	6	1	2019
Physical inactivity, adults aged 18+ years (%)	15	11	18	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Fiji

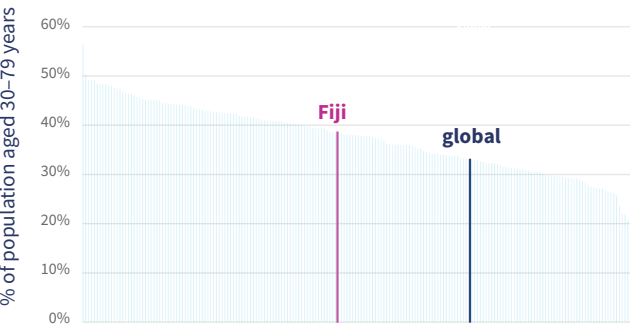
Hypertension profile

Total population (2019): 918 000

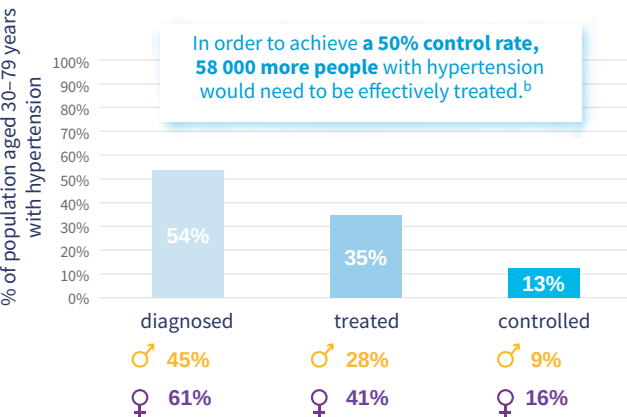
Total deaths (2019): 7130

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 39% ♂ 37% ♀ 41%

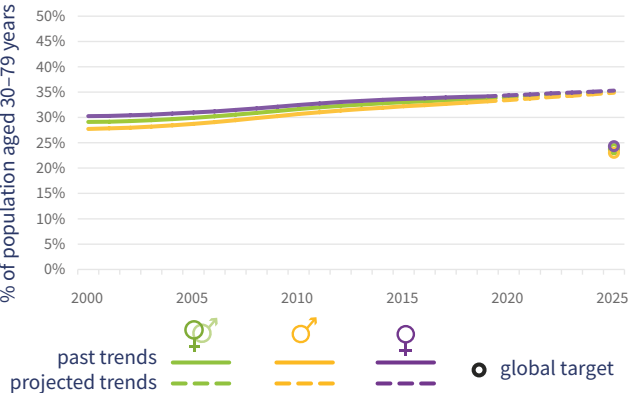
Prevalence of hypertension – global comparison (both sexes)^a



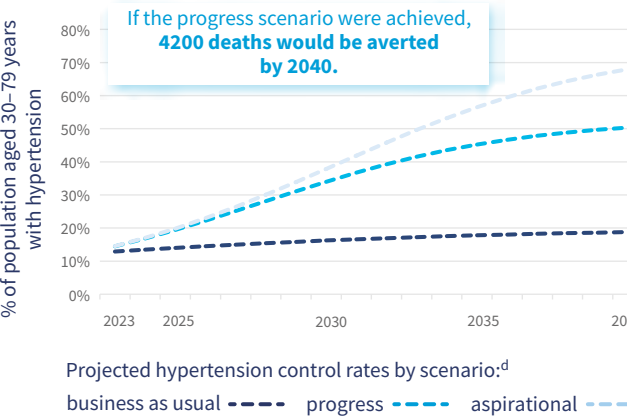
Of the 153 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	38	42	33	2019
Cardiovascular disease deaths	2500	1500	970	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	62	61	65	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	23	36	11	2019
Obesity, adults aged 18+ years (%)	30	25	35	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	6	1	2019
Physical inactivity, adults aged 18+ years (%)	17	11	24	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Finland

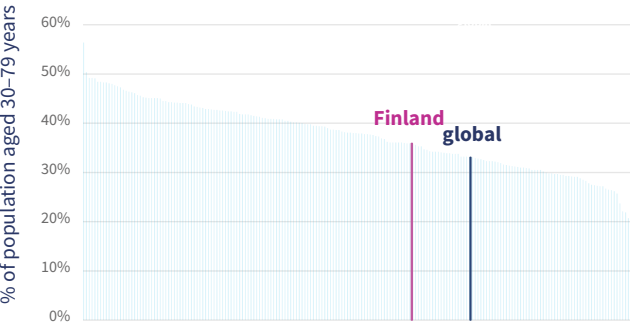
Hypertension profile

Total population (2019): 5 522 000

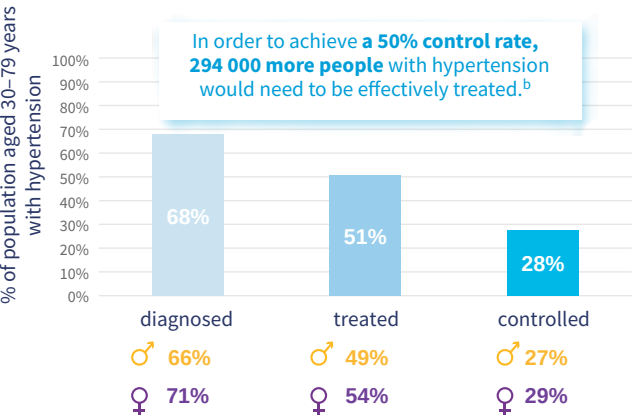
Total deaths (2019): 55 300

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 36% ♂ 41% ♀ 31%

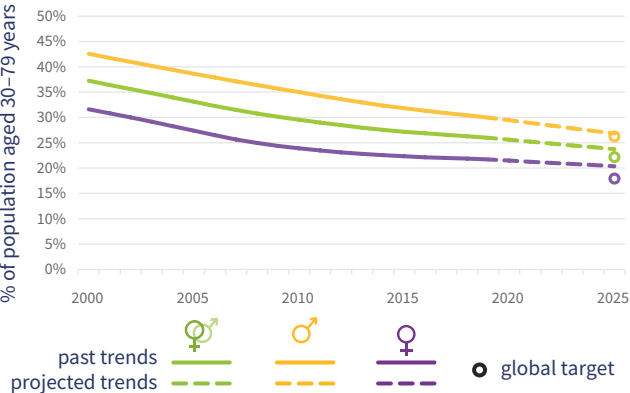
Prevalence of hypertension – global comparison (both sexes)^a



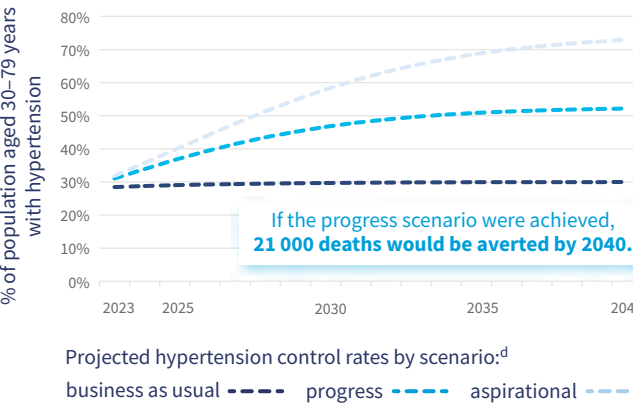
Of the 1.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	12	7	2019
Cardiovascular disease deaths	18 900	9600	9300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	54	51	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	10	7	2019
Current tobacco use, adults aged 15+ years (%)	22	28	17	2019
Obesity, adults aged 18+ years (%)	22	24	21	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	14	4	2019
Physical inactivity, adults aged 18+ years (%)	17	17	16	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

France

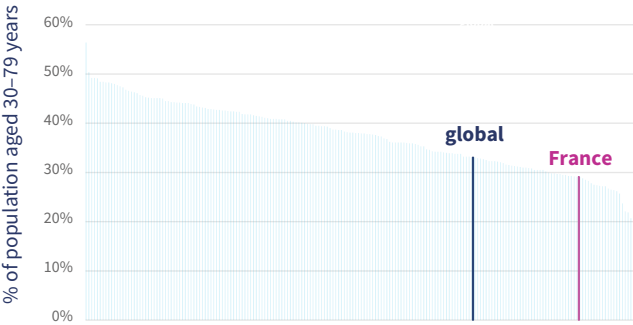
Hypertension profile

Total population (2019): 64 400 000

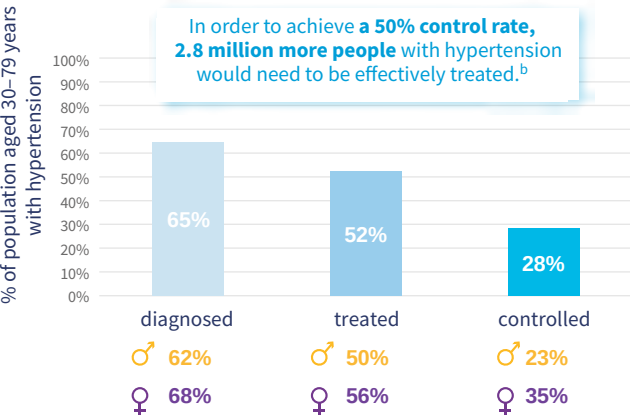
Total deaths (2019): 598 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 29% ♂ 34% ♀ 24%

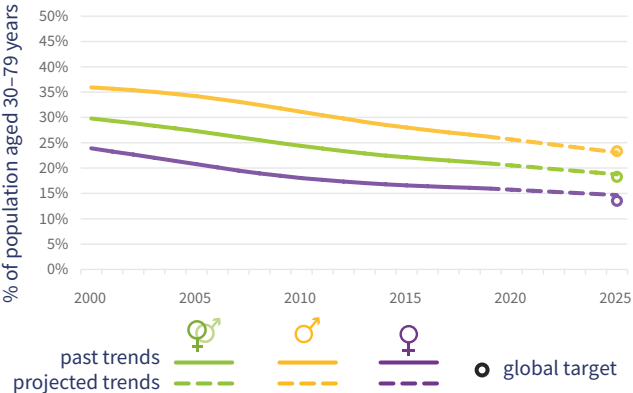
Prevalence of hypertension – global comparison (both sexes)^a



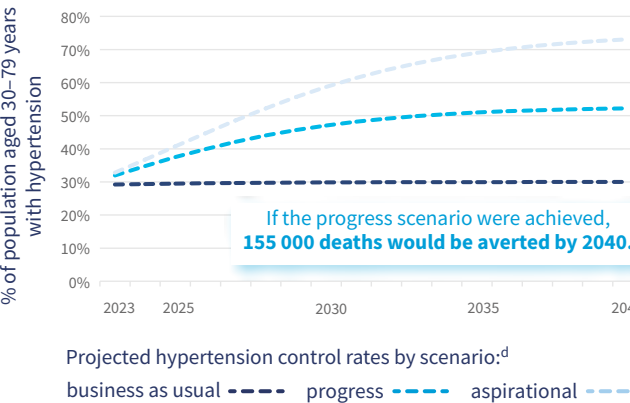
Of the 14.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	14	7	2019
Cardiovascular disease deaths	149 000	67 800	80 800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	48	50	47	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	34	35	32	2019
Obesity, adults aged 18+ years (%)	22	22	21	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	18	5	2019
Physical inactivity, adults aged 18+ years (%)	29	24	34	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Gabon

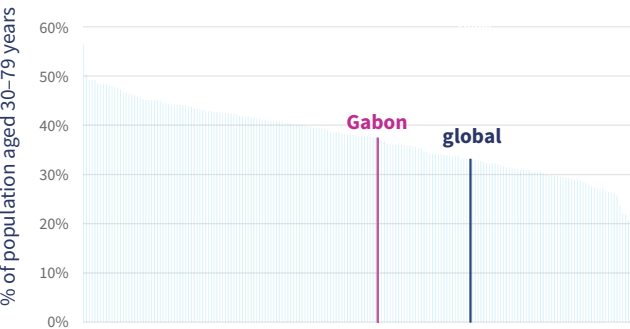
Hypertension profile

Total population (2019): 2 243 000

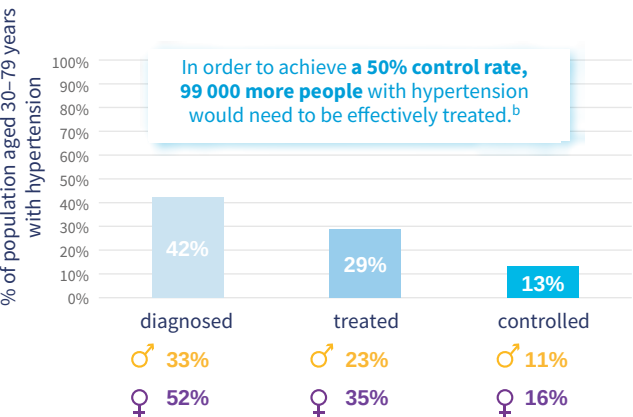
Total deaths (2019): 14 300

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 37% ♂ 36% ♀ 39%

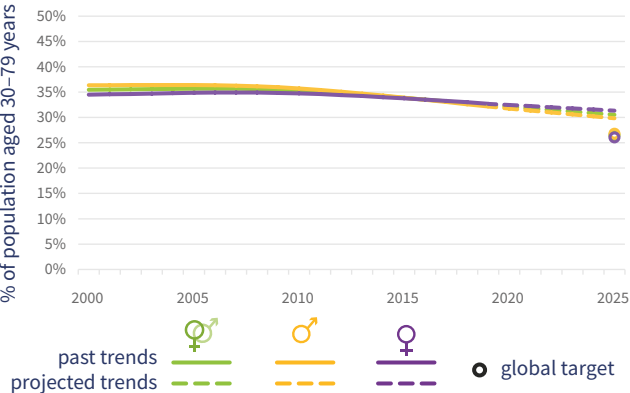
Prevalence of hypertension – global comparison (both sexes)^a



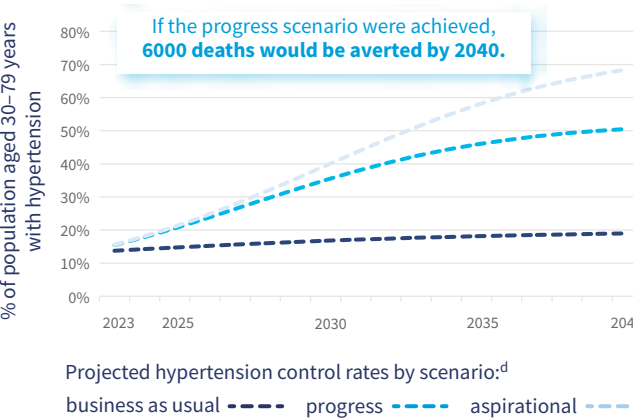
Of the 260 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	26	17	2019
Cardiovascular disease deaths	2600	1400	1200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	62	61	63	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	15	10	20	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	11	3	2019
Physical inactivity, adults aged 18+ years (%)	25	18	33	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Gambia

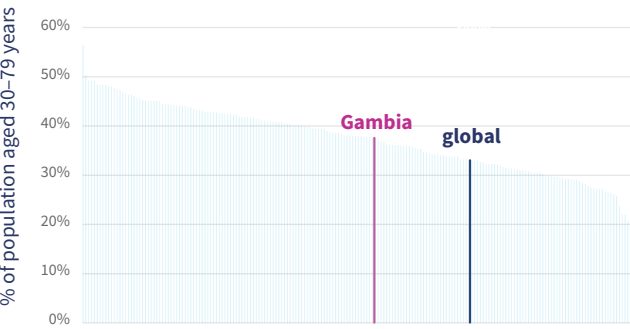
Hypertension profile

Total population (2019): 2 509 000

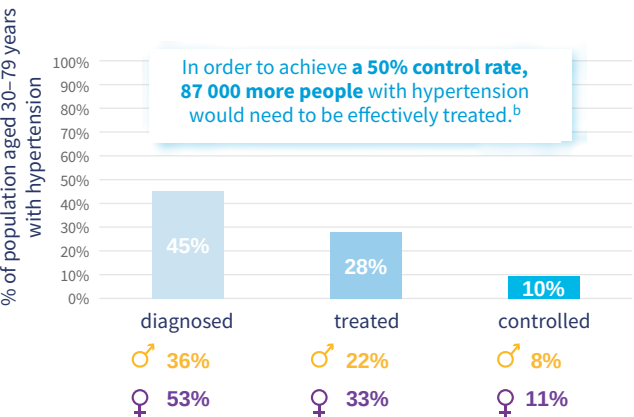
Total deaths (2019): 14 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 35% ♀ 40%

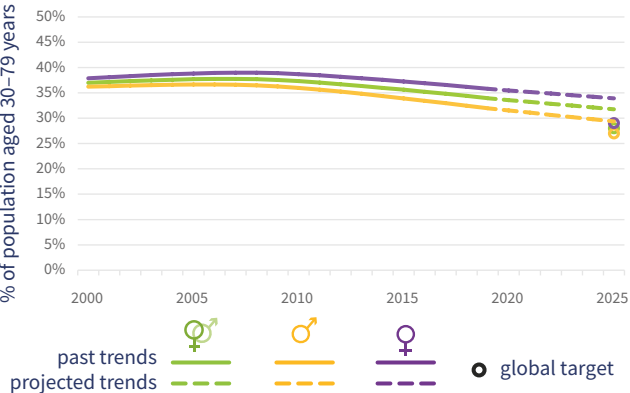
Prevalence of hypertension – global comparison (both sexes)^a



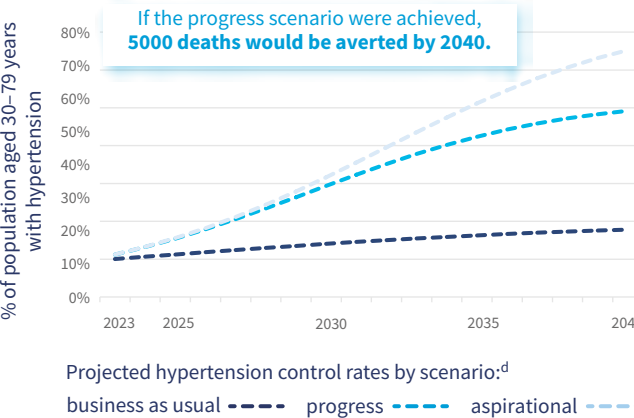
Of the 211 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	22	20	2019
Cardiovascular disease deaths	2200	1100	1100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	61	60	63	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	12	22	1	2019
Obesity, adults aged 18+ years (%)	10	6	15	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	2	0	2019
Physical inactivity, adults aged 18+ years (%)	21	16	26	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Georgia

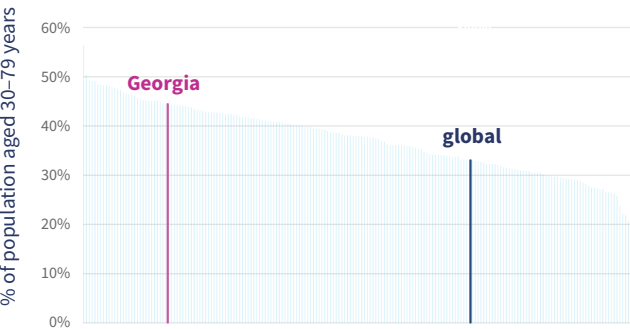
Hypertension profile

Total population (2019): 3 771 000

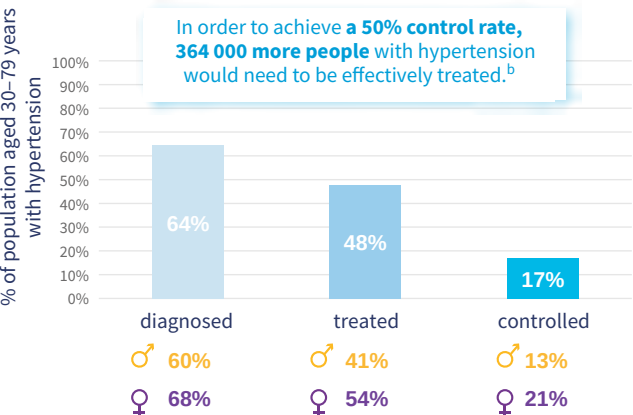
Total deaths (2019): 53 400

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 45% ♂ 47% ♀ 42%

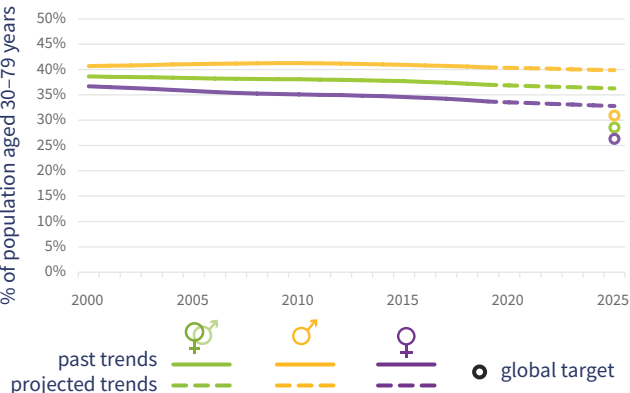
Prevalence of hypertension – global comparison (both sexes)^a



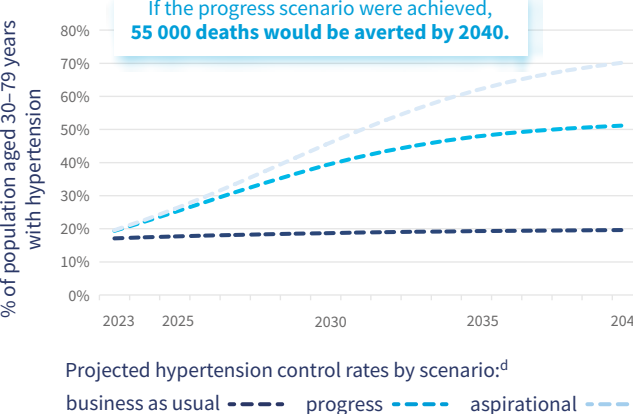
Of the 1.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	36	15	2019
Cardiovascular disease deaths	34 200	16 500	17 700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	62	64	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%)	32	56	7	2019
Obesity, adults aged 18+ years (%)	22	19	24	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	14	24	6	2019
Physical inactivity, adults aged 18+ years (%)	18	17	19	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Germany

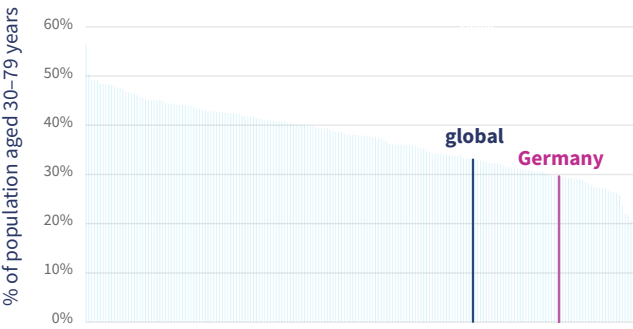
Hypertension profile

Total population (2019): 83 148 000

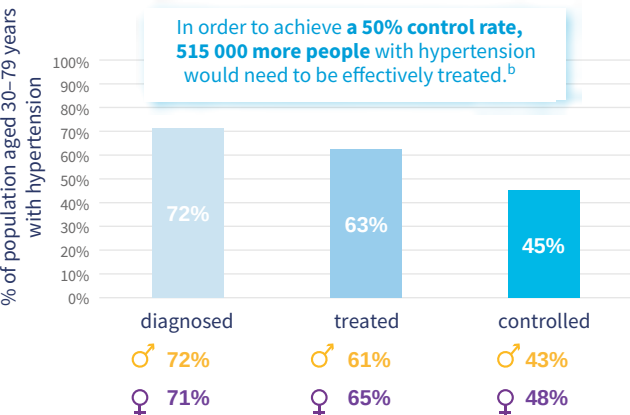
Total deaths (2019): 845 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 30% ♂ 34% ♀ 25%

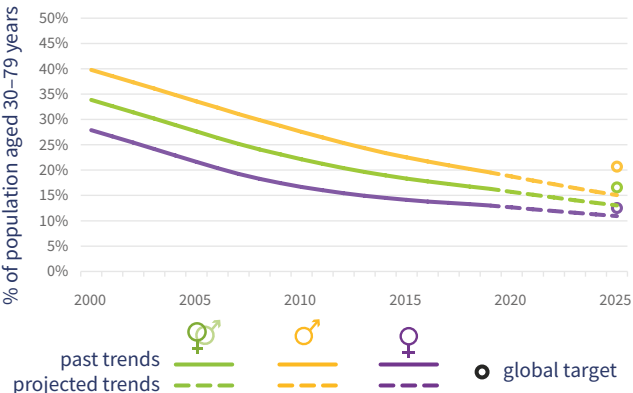
Prevalence of hypertension – global comparison (both sexes)^a



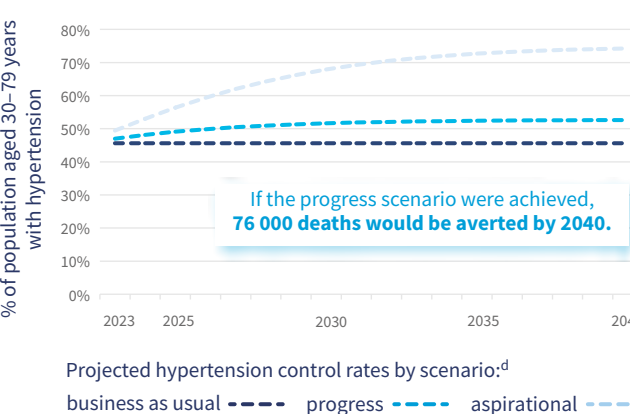
Of the 19.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	12	15	9	2019
Cardiovascular disease deaths	285 000	158 000	127 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	52	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	23	25	20	2019
Obesity, adults aged 18+ years (%)	22	24	20	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	19	5	2019
Physical inactivity, adults aged 18+ years (%)	42	40	44	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Ghana

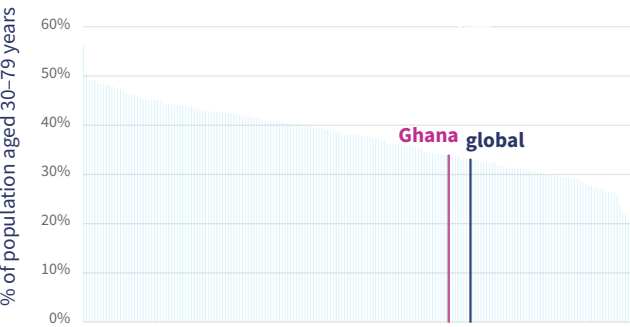
Hypertension profile

Total population (2019): 31 522 000

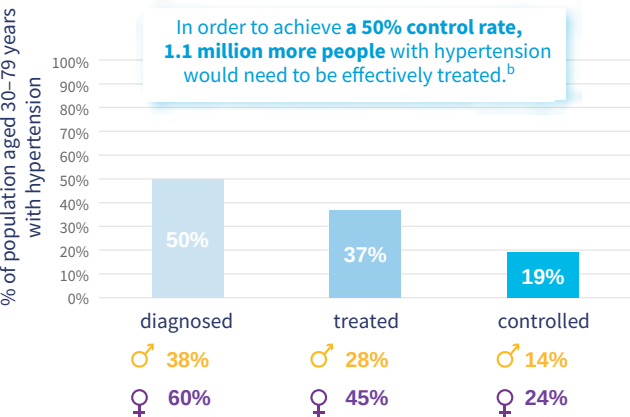
Total deaths (2019): 190 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 31% ♀ 36%

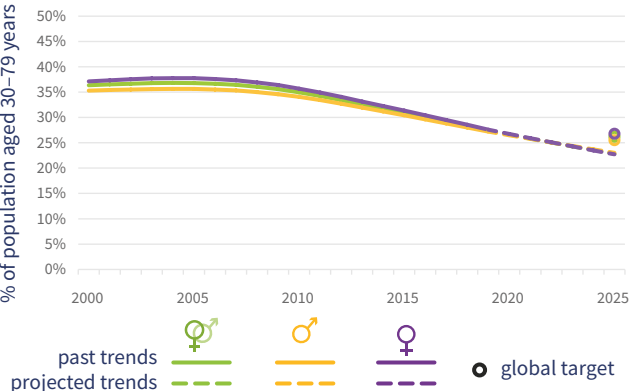
Prevalence of hypertension – global comparison (both sexes)^a



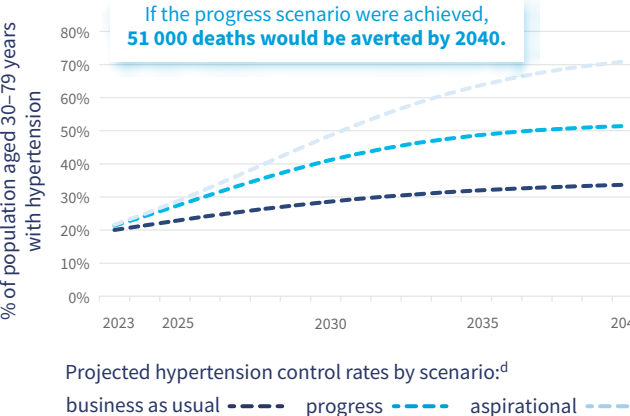
Of the 3.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	24	21	2019
Cardiovascular disease deaths	35 100	16 500	18 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	61	56	64	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	8	2019
Current tobacco use, adults aged 15+ years (%)	4	7	0	2019
Obesity, adults aged 18+ years (%)	11	5	17	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	7	2	2019
Physical inactivity, adults aged 18+ years (%)	22	19	25	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Greece

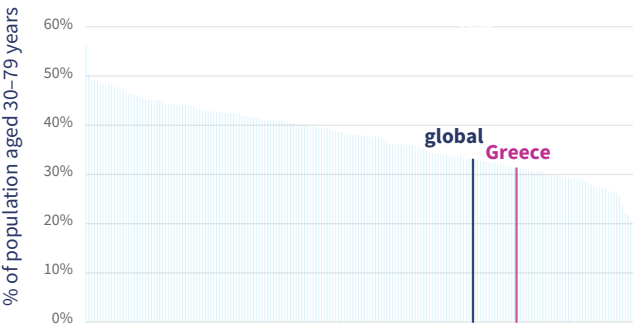
Hypertension profile

Total population (2019): 10 574 000

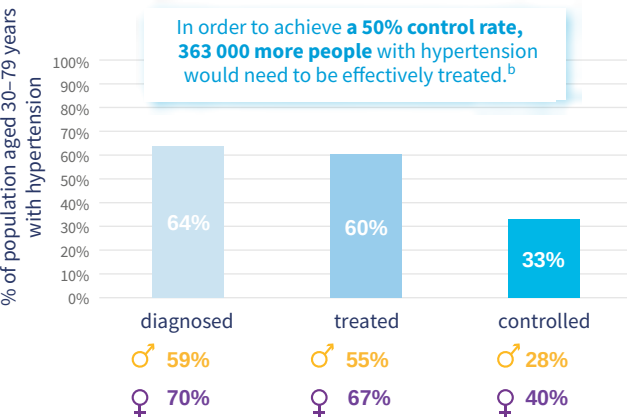
Total deaths (2019): 126 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 37% ♀ 26%

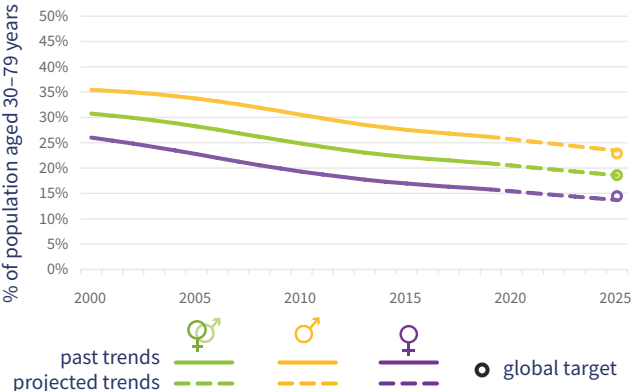
Prevalence of hypertension – global comparison (both sexes)^a



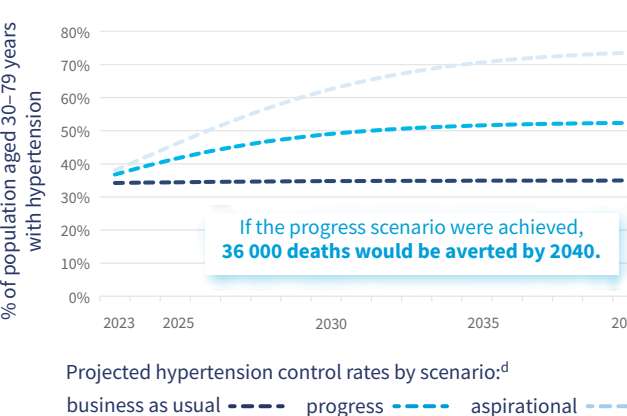
Of the 2.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	12	17	8	2019
Cardiovascular disease deaths	42 300	20 400	21 900	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	47	46	47	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%)	35	38	31	2019
Obesity, adults aged 18+ years (%)	25	24	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	11	3	2019
Physical inactivity, adults aged 18+ years (%)	38	34	41	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Grenada

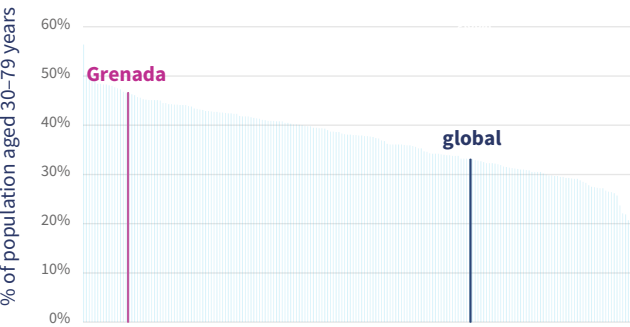
Hypertension profile

Total population (2019): 123 000

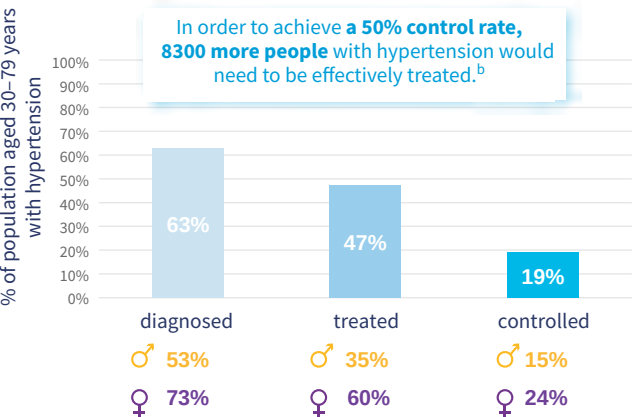
Total deaths (2019): 1010

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 47% ♂ 48% ♀ 46%

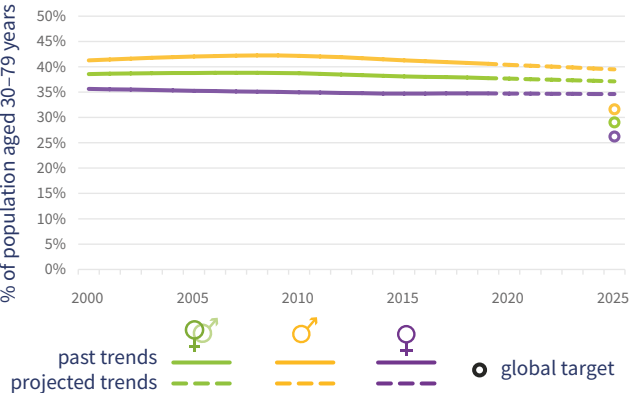
Prevalence of hypertension – global comparison (both sexes)^a



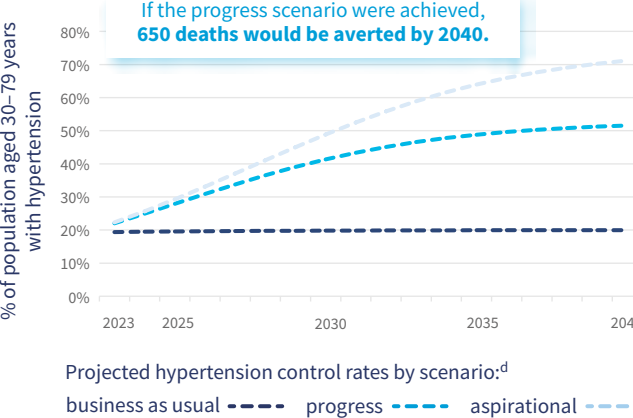
Of the 27 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	26	20	2019
Cardiovascular disease deaths	330	170	160	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	53	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	21	13	29	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	13	3	2019
Physical inactivity, adults aged 18+ years (%)	29	22	35	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Guatemala

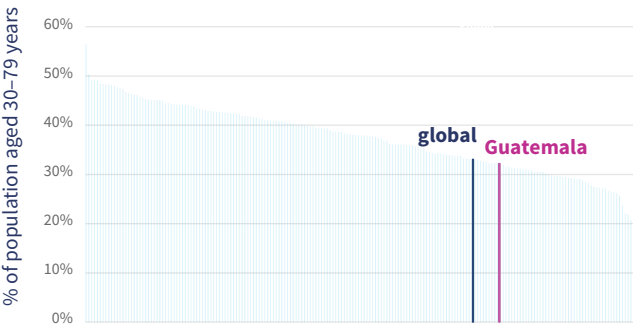
Hypertension profile

Total population (2019): 17 106 000

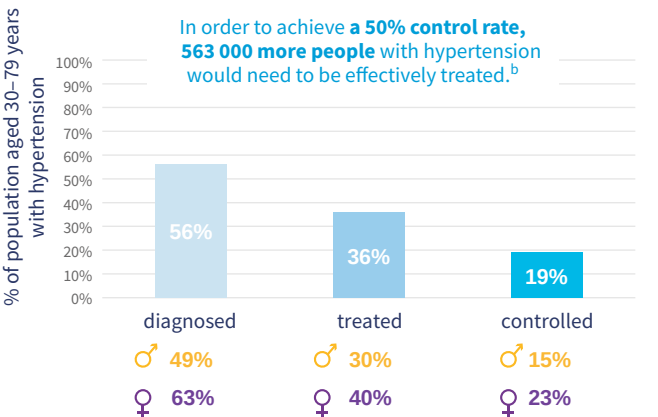
Total deaths (2019): 95 100

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 32% ♂ 32% ♀ 33%

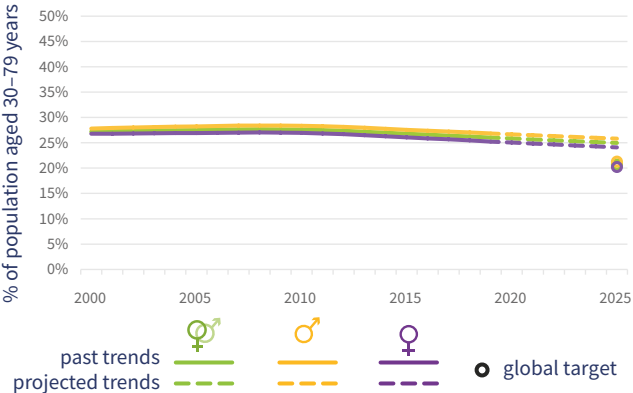
Prevalence of hypertension – global comparison (both sexes)^a



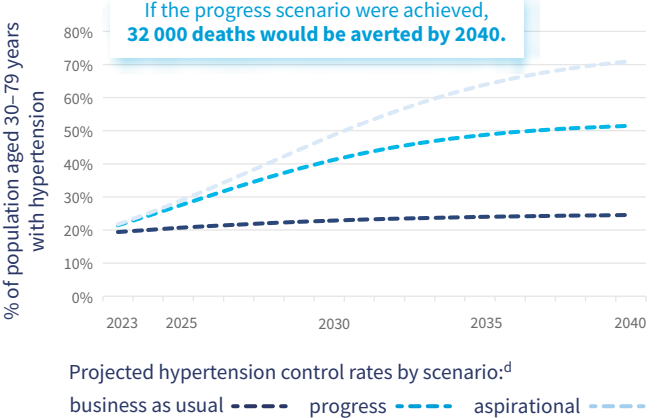
Of the 1.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	17	16	2019
Cardiovascular disease deaths	16 500	8100	8400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	50	49	51	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	9	2019
Current tobacco use, adults aged 15+ years (%)	11	20	2	2019
Obesity, adults aged 18+ years (%)	21	15	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	1	2019
Physical inactivity, adults aged 18+ years (%)	37	37	37	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Guinea

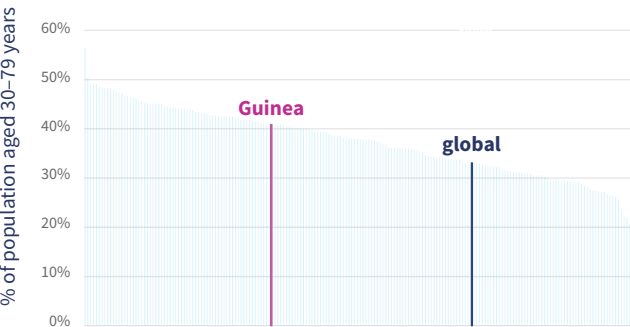
Hypertension profile

Total population (2019): 12 878 000

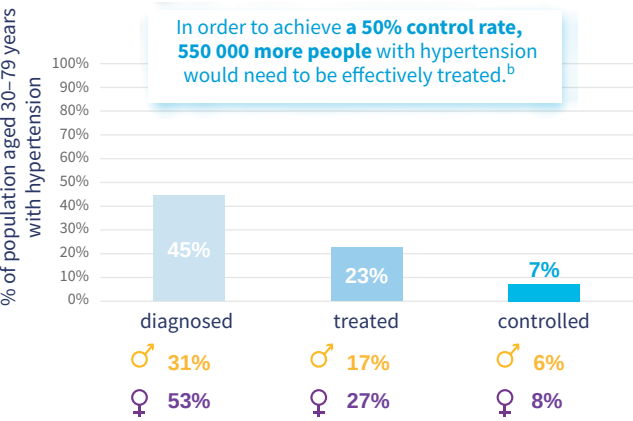
Total deaths (2019): 108 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 38% ♀ 43%

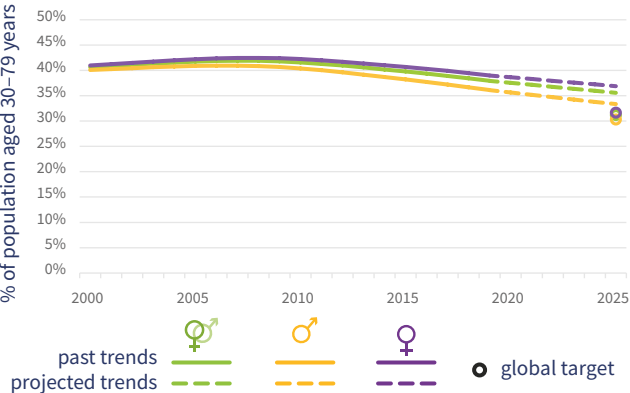
Prevalence of hypertension – global comparison (both sexes)^a



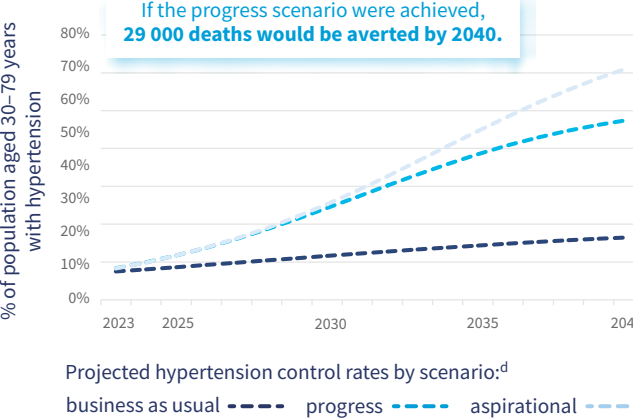
Of the 1.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	25	25	2019
Cardiovascular disease deaths	14 400	5800	8600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	52	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	8	4	12	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	1	0	2019
Physical inactivity, adults aged 18+ years (%)	15	11	18	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Guinea-Bissau

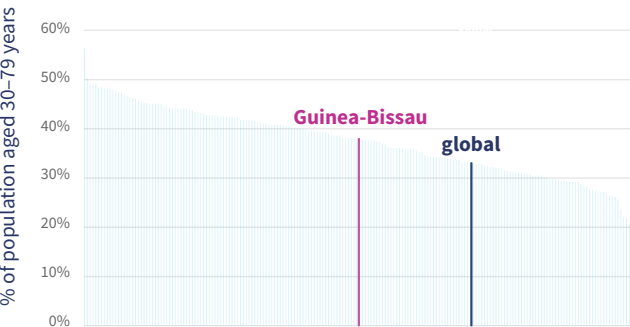
Hypertension profile

Total population (2019): 1 970 000

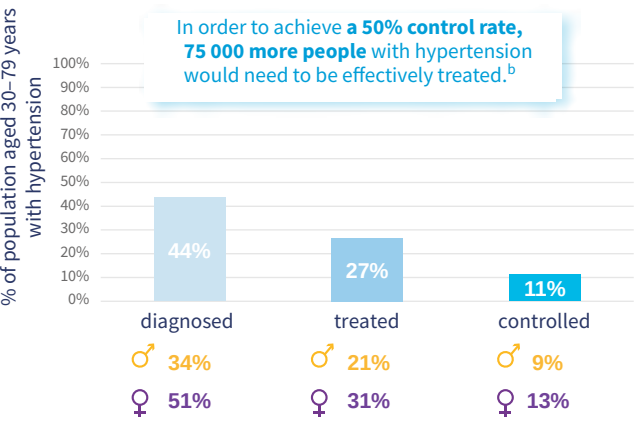
Total deaths (2019): 16 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 36% ♀ 40%

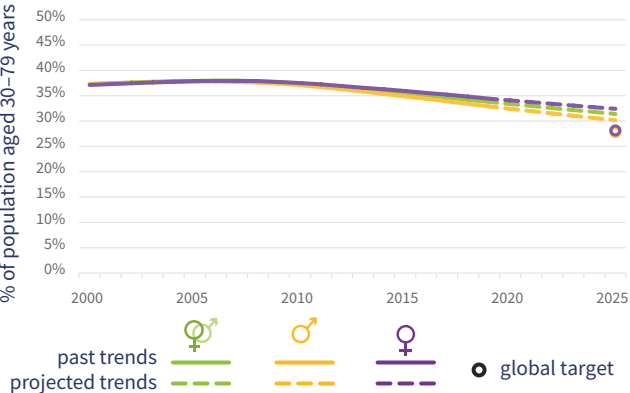
Prevalence of hypertension – global comparison (both sexes)^a



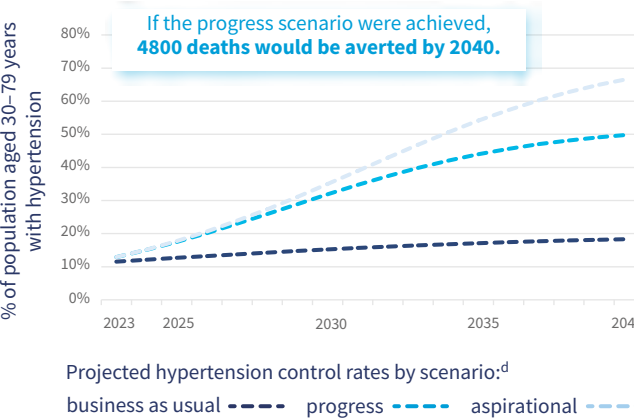
Of the 191 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	27	23	2019
Cardiovascular disease deaths	2100	1000	1100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	58	62	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	9	18	1	2019
Obesity, adults aged 18+ years (%)	10	5	14	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	7	1	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Guyana

Hypertension profile

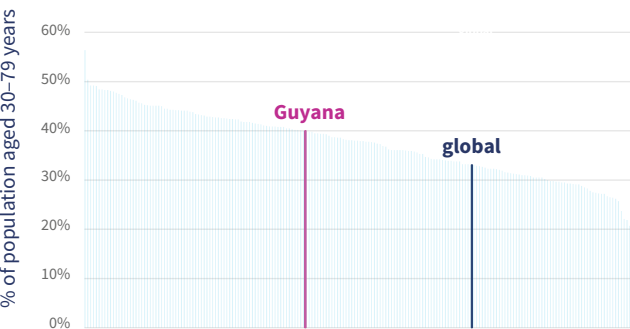
Total population (2019): 799 000

Total deaths (2019): 8050¹

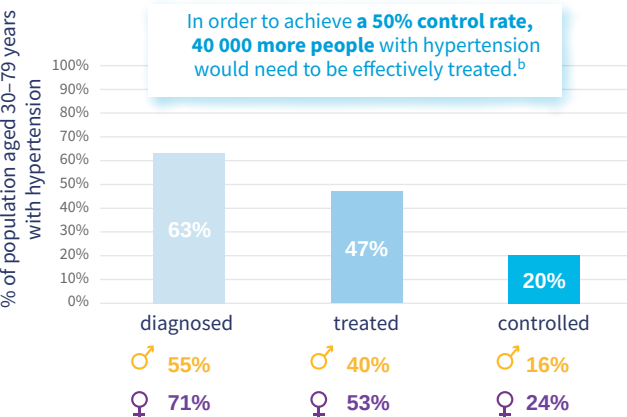
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

♀ 40% ♂ 38% ♀ 42%

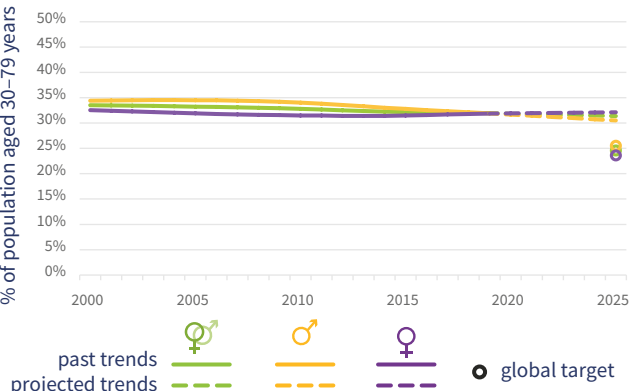
Prevalence of hypertension – global comparison (both sexes)^a



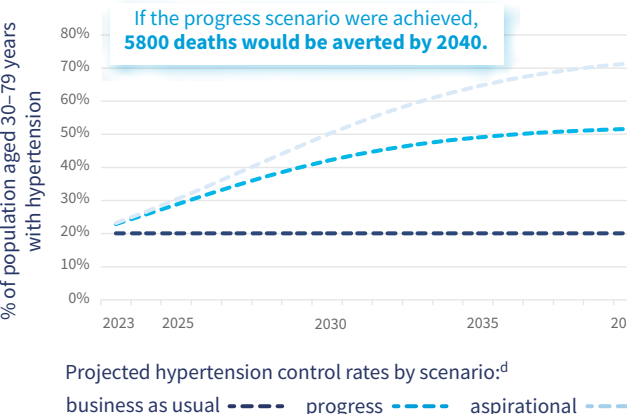
Of the 136 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	29	32	26	2019
Cardiovascular disease deaths	3000	1600	1400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	56	53	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	13	23	3	2019
Obesity, adults aged 18+ years (%)	20	13	27	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	9	2	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Haiti

Hypertension profile

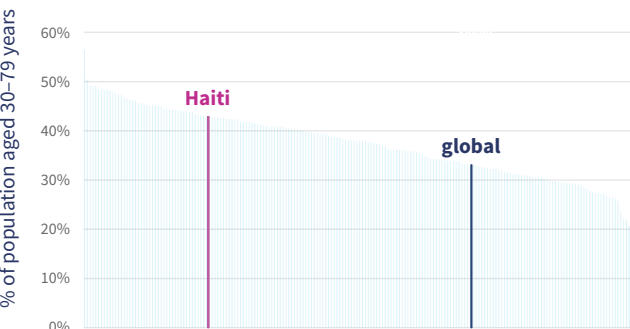
Total population (2019): 11 160 000

Total deaths (2019): 97 200

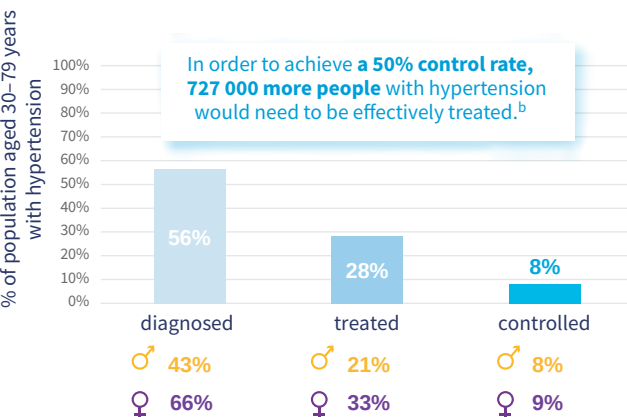
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

♀ 43% ♂ 38% ♀ 48%

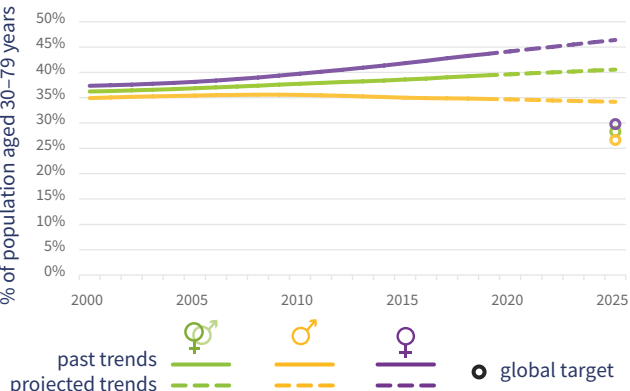
Prevalence of hypertension – global comparison (both sexes)^a



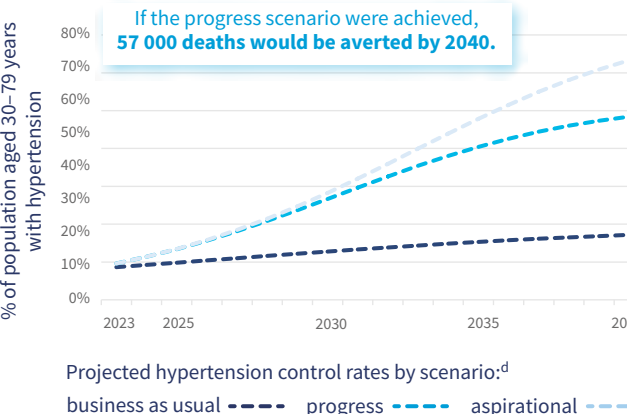
Of the 1.7 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	31	30	33	2019
Cardiovascular disease deaths	31 100	13 000	18 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	57	62	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	8	13	3	2019
Obesity, adults aged 18+ years (%)	23	18	27	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	6	1	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	
National target for salt consumption	✗		✗
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Honduras

Hypertension profile

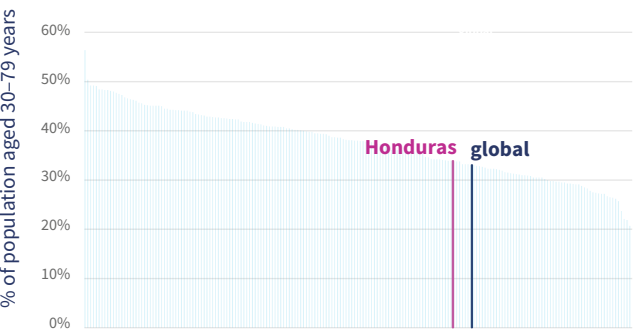
Total population (2019): 9 959 000

Total deaths (2019): 55 000¹

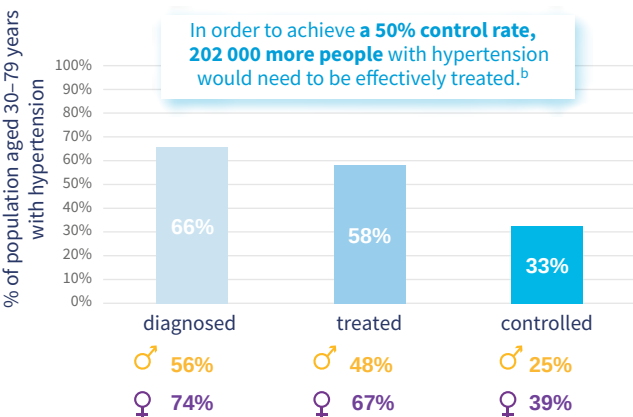
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

♀ 34% ♂ 33% ♀ 34%

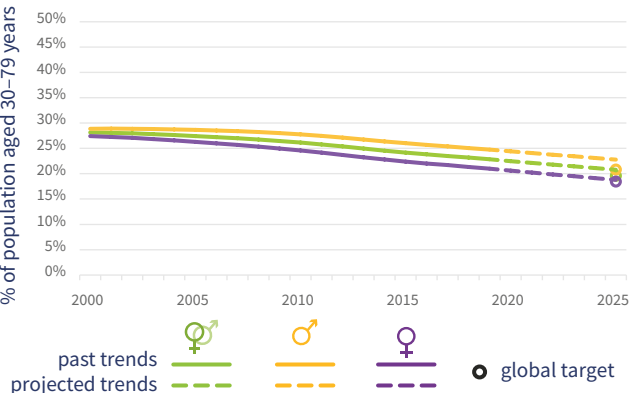
Prevalence of hypertension – global comparison (both sexes)^a



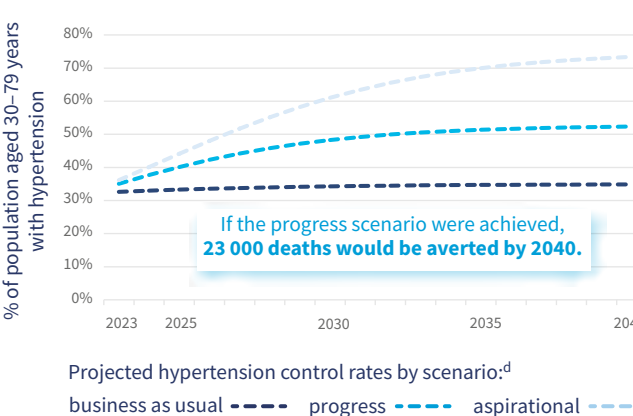
Of the 1.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	19	18	19	2019
Cardiovascular disease deaths	15 800	7000	8800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	53	54	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	9	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	21	16	27	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✓
National target for salt consumption	✗
Guidelines for management of hypertension	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Hungary

Hypertension profile

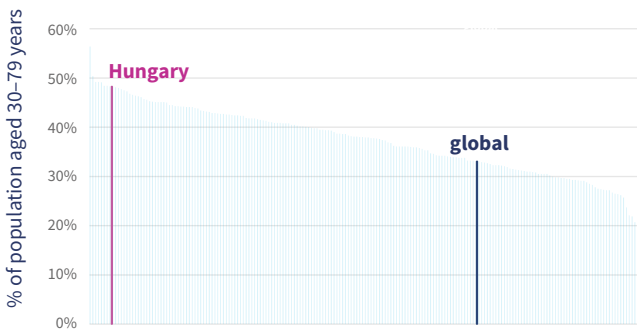
Total population (2019): 9 772 000

Total deaths (2019): 127 000

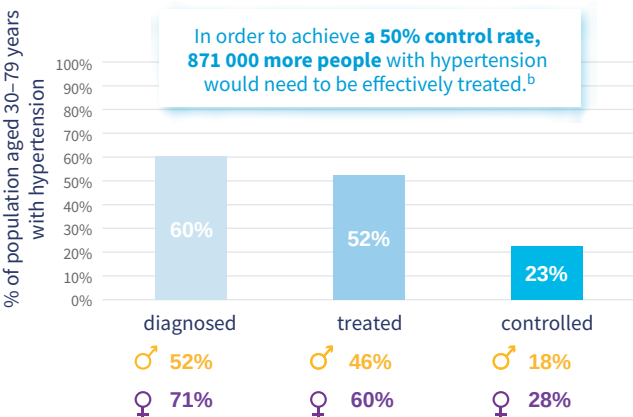
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

♀ 48% ♂ 56% ♀ 41%

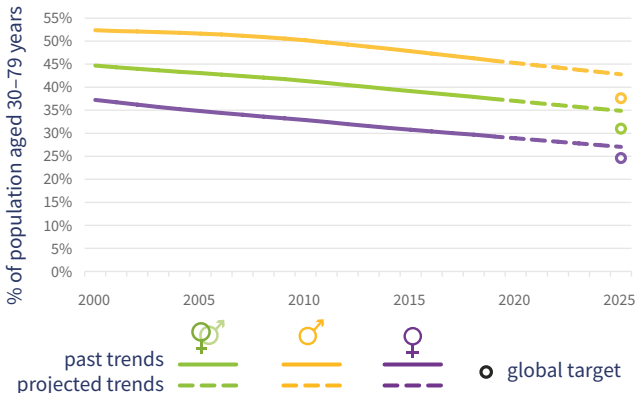
Prevalence of hypertension – global comparison (both sexes)^a



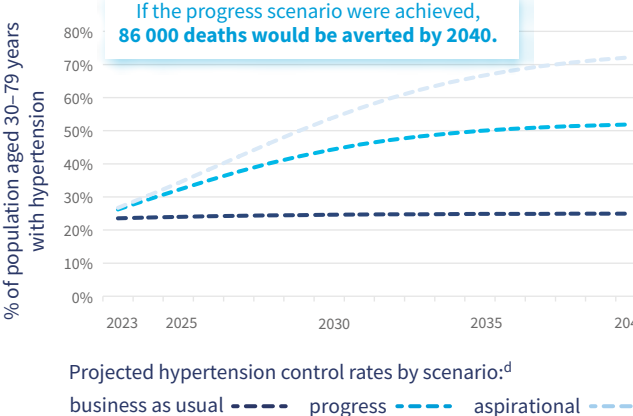
Of the 3.4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	30	15	2019
Cardiovascular disease deaths	62 100	27 300	34 800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	58	51	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	14	17	12	2019
Current tobacco use, adults aged 15+ years (%) ^f	32	36	28	2019
Obesity, adults aged 18+ years (%)	26	28	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	17	5	2019
Physical inactivity, adults aged 18+ years (%)	39	33	43	2016

National response

Targets	Treatment
National target for blood pressure	✗
National target for salt consumption	✗
Guidelines for management of hypertension	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Iceland

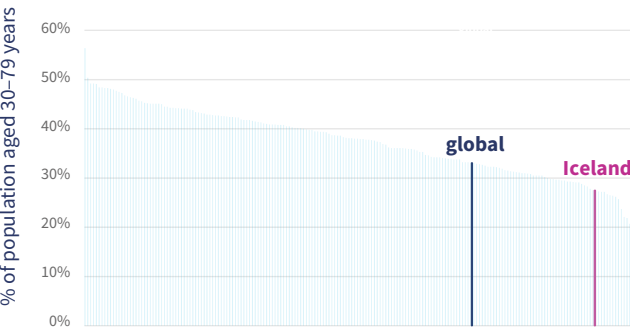
Hypertension profile

Total population (2019): 361 000

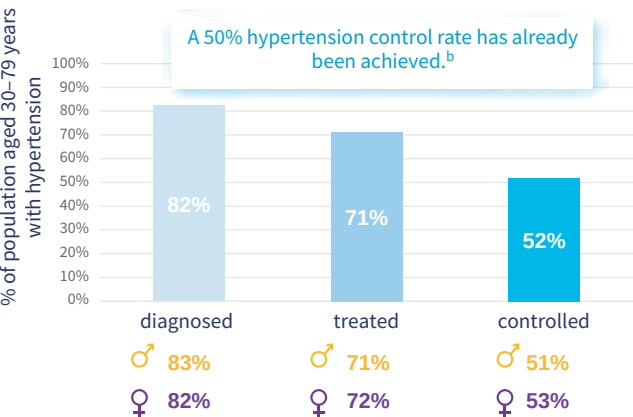
Total deaths (2019): 2340

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 28% ♂ 31% ♀ 24%

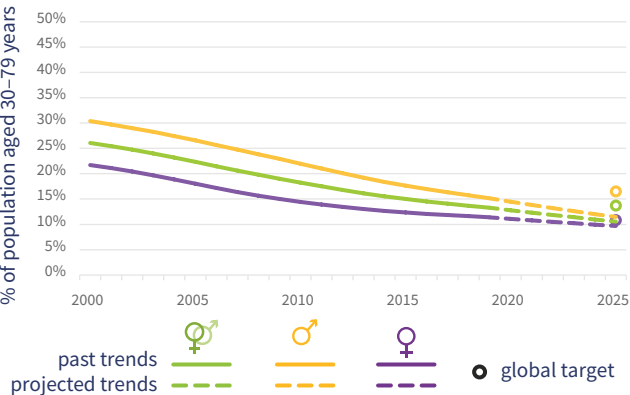
Prevalence of hypertension – global comparison (both sexes)^a



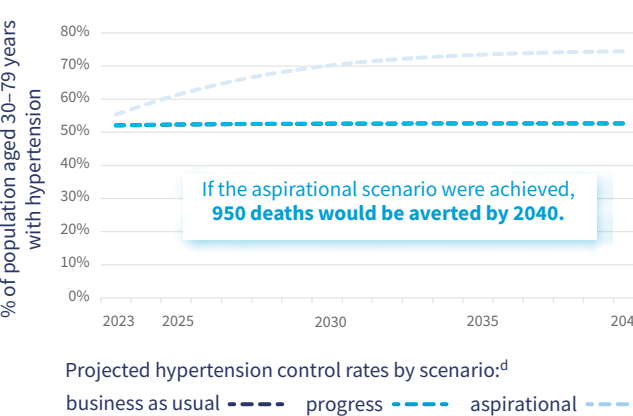
Of the 63 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	9	10	7	2019
Cardiovascular disease deaths	700	370	330	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	48	50	47	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	10	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	13	13	13	2019
Obesity, adults aged 18+ years (%)	22	24	19	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	12	4	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

India

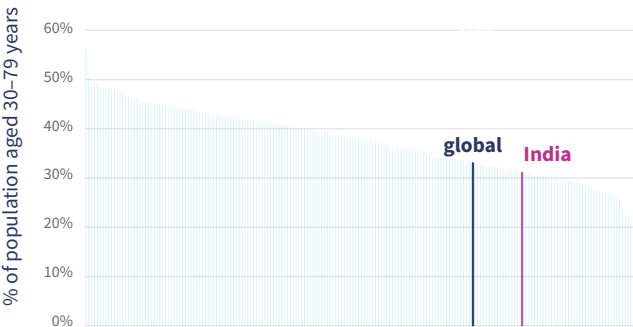
Hypertension profile

Total population (2019): 1 383 000 000

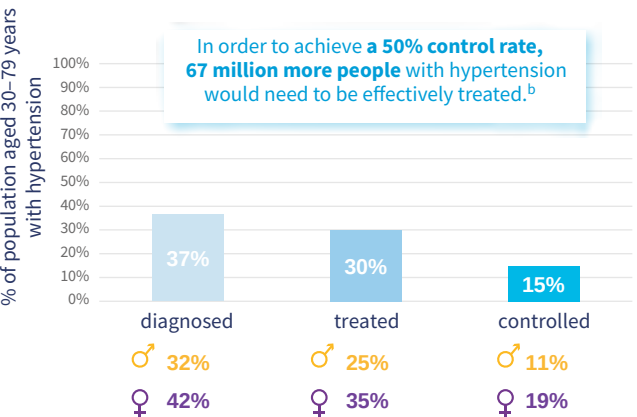
Total deaths (2019): 9 171 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 32% ♀ 31%

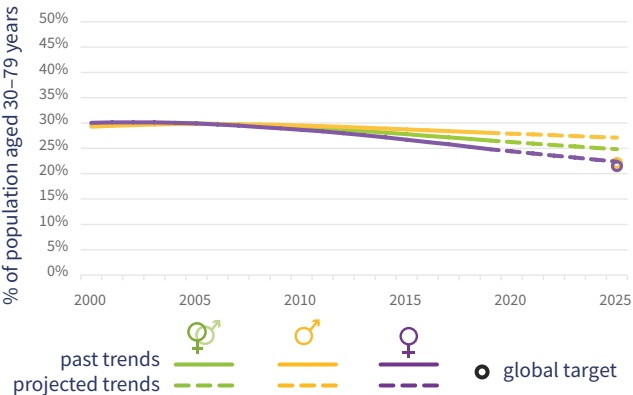
Prevalence of hypertension – global comparison (both sexes)^a



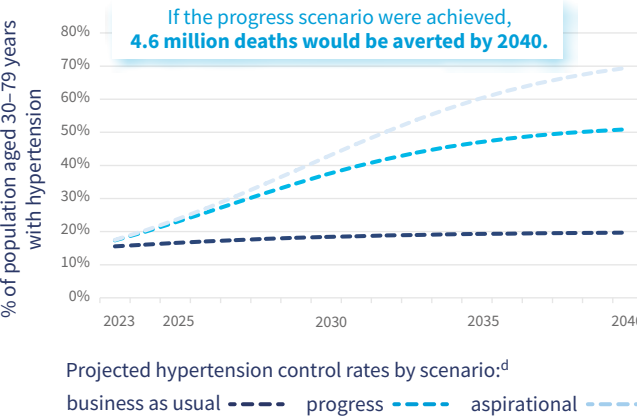
Of the 188.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	25	19	2019
Cardiovascular disease deaths	2 566 000	1 451 000	1 116 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	51	54	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	9	2019
Current tobacco use, adults aged 15+ years (%)	28	42	14	2019
Obesity, adults aged 18+ years (%)	4	3	5	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	8	2	2019
Physical inactivity, adults aged 18+ years (%)	34	25	44	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Indonesia

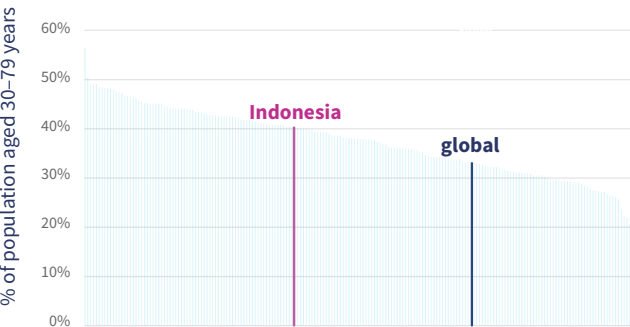
Hypertension profile

Total population (2019): 269 600 000

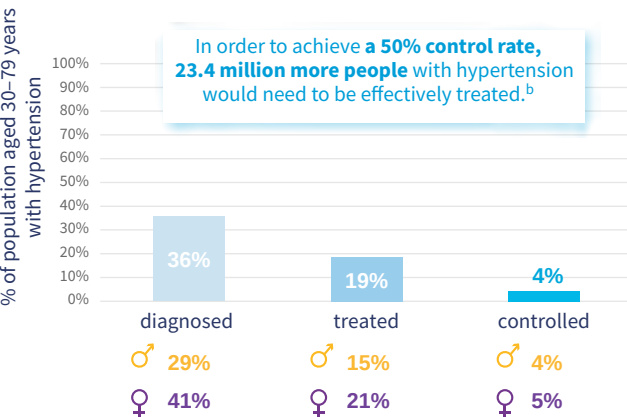
Total deaths (2019): 1 816 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 40% ♂ 36% ♀ 45%

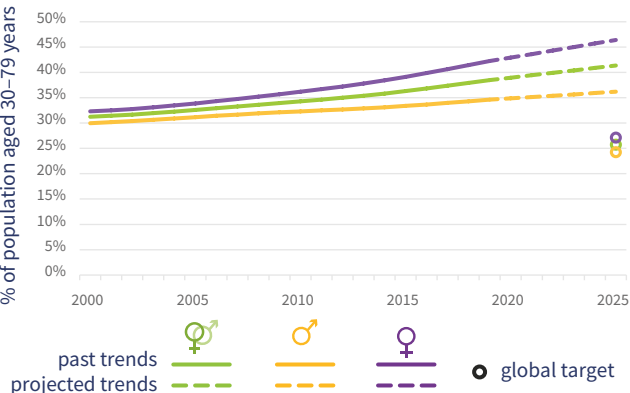
Prevalence of hypertension – global comparison (both sexes)^a



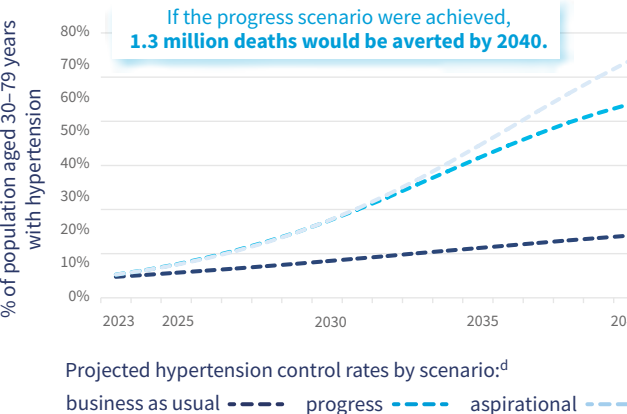
Of the 51.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	28	21	2019
Cardiovascular disease deaths	697 000	355 000	342 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	69	67	71	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	10	2019
Current tobacco use, adults aged 15+ years (%)	37	71	4	2019
Obesity, adults aged 18+ years (%)	7	5	9	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	23	23	22	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Iran (Islamic Republic of)

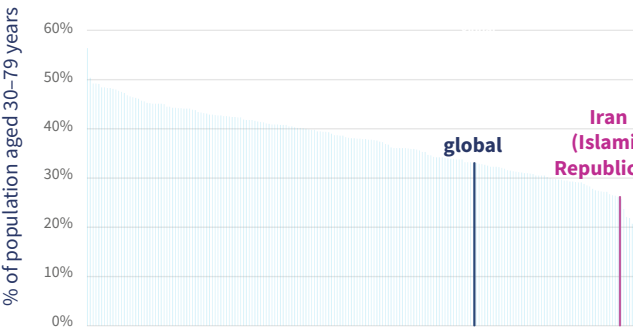
Hypertension profile

Total population (2019): 86 564 000

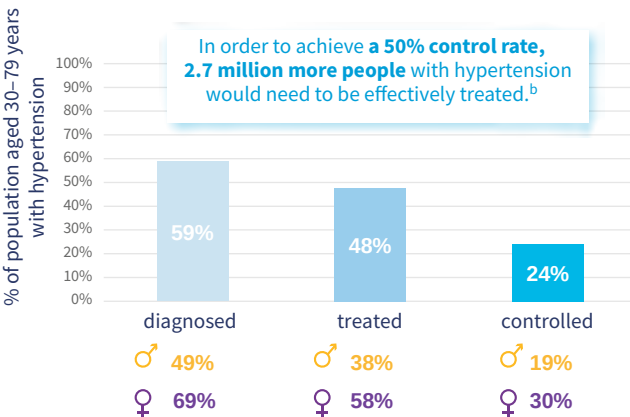
Total deaths (2019): 365 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 26% ♂ 27% ♀ 26%

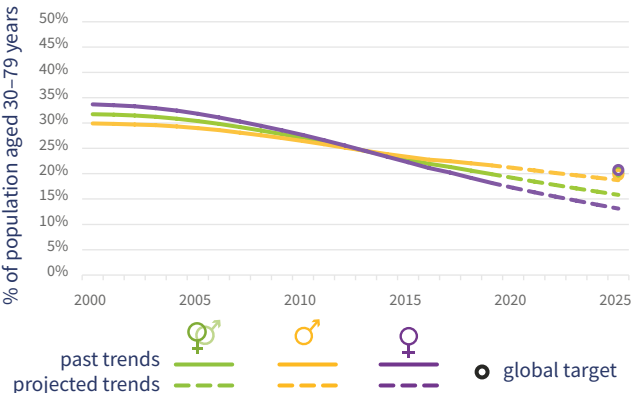
Prevalence of hypertension – global comparison (both sexes)^a



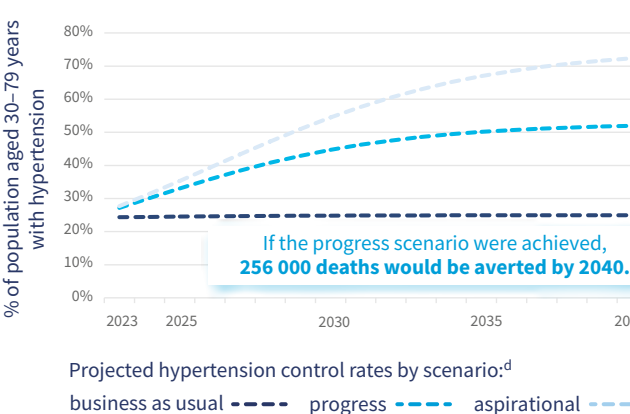
Of the 10 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	15	18	12	2019
Cardiovascular disease deaths	157 000	90 200	66 800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	52	54	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	14	25	3	2019
Obesity, adults aged 18+ years (%)	26	19	32	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	1	0	2019
Physical inactivity, adults aged 18+ years (%)	33	23	43	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Iraq

Hypertension profile

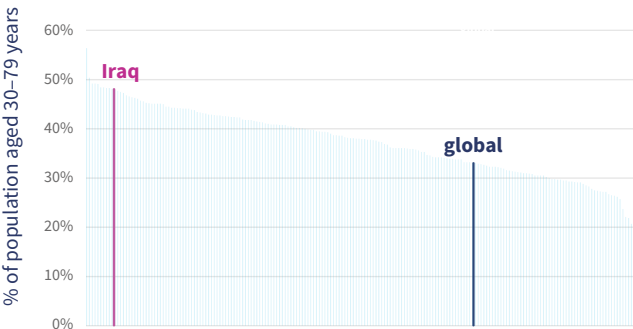
Total population (2019): 41 564 000

Total deaths (2019): 160 000

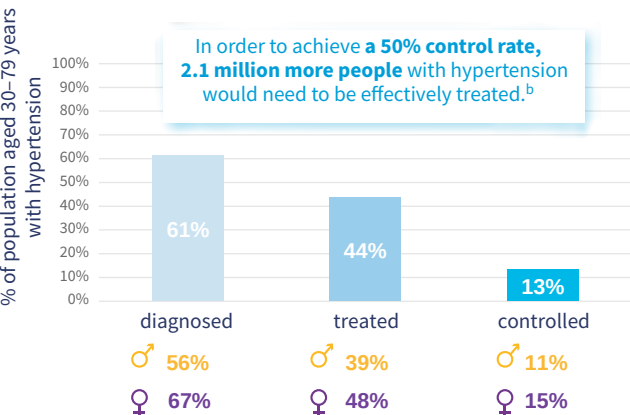
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

48% 48% 48%

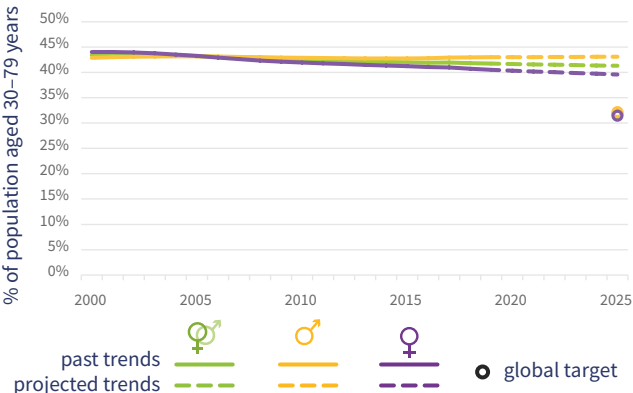
Prevalence of hypertension – global comparison (both sexes)^a



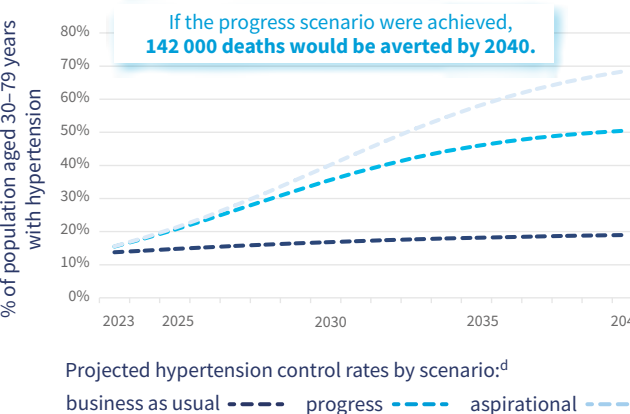
Of the 5.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	28	20	2019
Cardiovascular disease deaths	62 900	33 300	29 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	64	62	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	19	36	2	2019
Obesity, adults aged 18+ years (%)	30	23	37	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	52	39	65	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Ireland

Hypertension profile

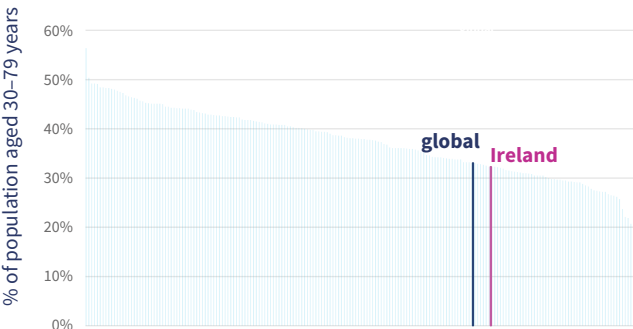
Total population (2019): 4 896 000

Total deaths (2019): 30 800

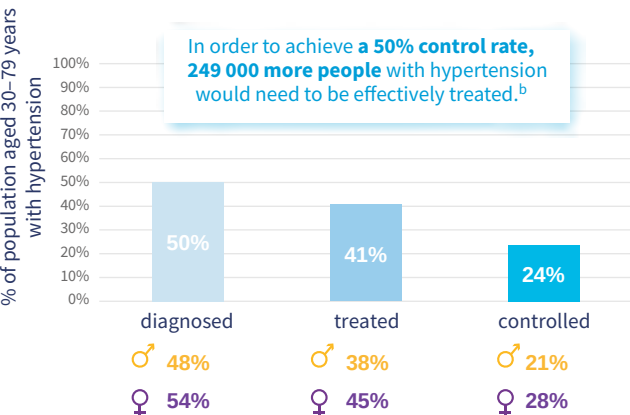
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

32% 38% 27%

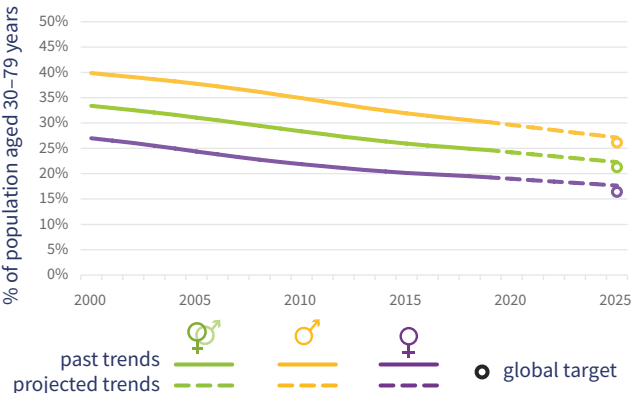
Prevalence of hypertension – global comparison (both sexes)^a



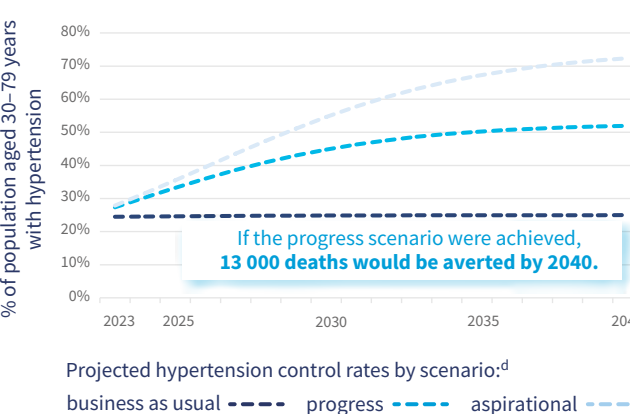
Of the 1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	11	8	2019
Cardiovascular disease deaths	8300	4300	4000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	51	53	49	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%) ^f	21	23	20	2019
Obesity, adults aged 18+ years (%)	25	25	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	18	6	2019
Physical inactivity, adults aged 18+ years (%)	33	28	37	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	no data		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Israel

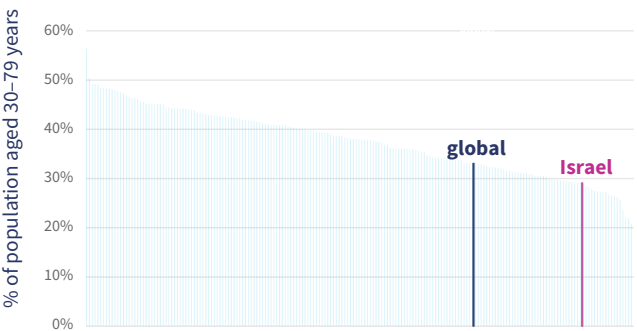
Hypertension profile

Total population (2019): 8 608 000

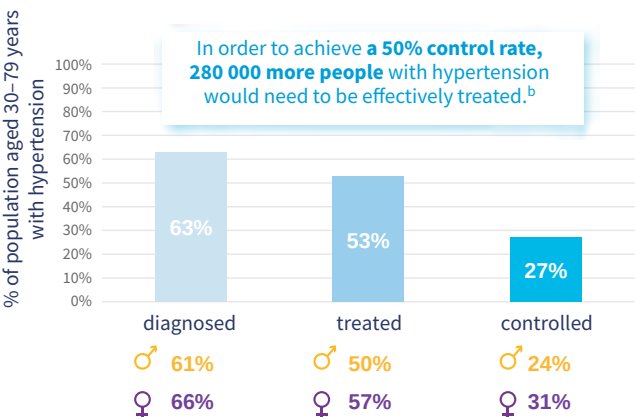
Total deaths (2019): 45 400

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 29% ♂ 33% ♀ 25%

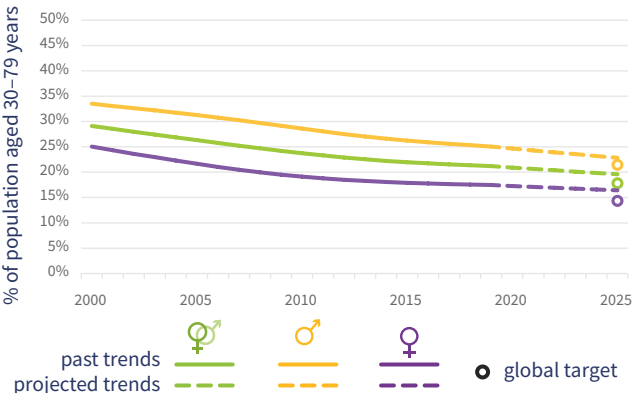
Prevalence of hypertension – global comparison (both sexes)^a



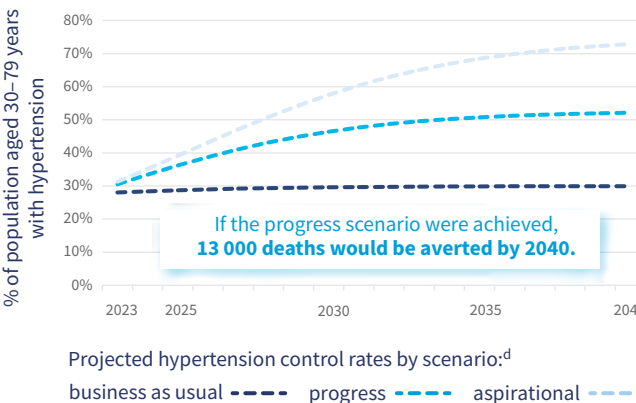
Of the 1.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	9	11	7	2019
Cardiovascular disease deaths	10 200	5100	5100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	51	51	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	10	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	22	29	14	2019
Obesity, adults aged 18+ years (%)	26	26	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Italy

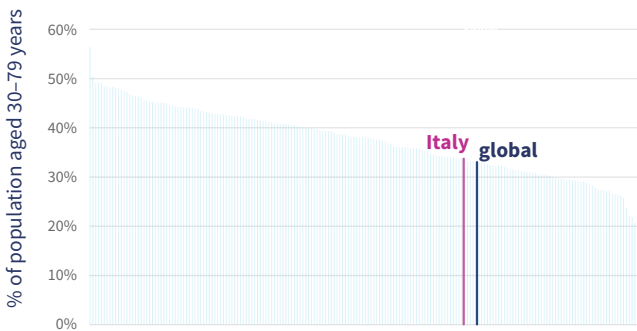
Hypertension profile

Total population (2019): 59 728 000

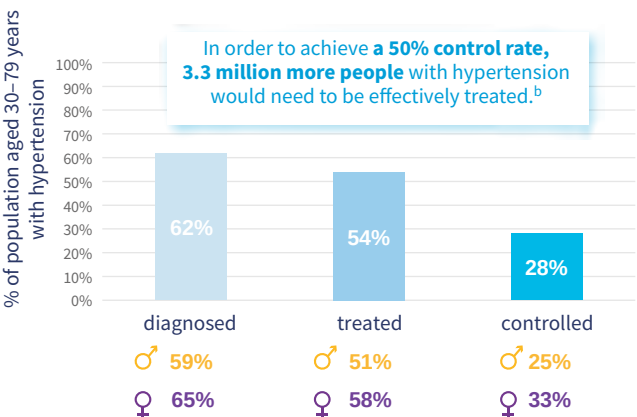
Total deaths (2019): 651 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 39% ♀ 29%

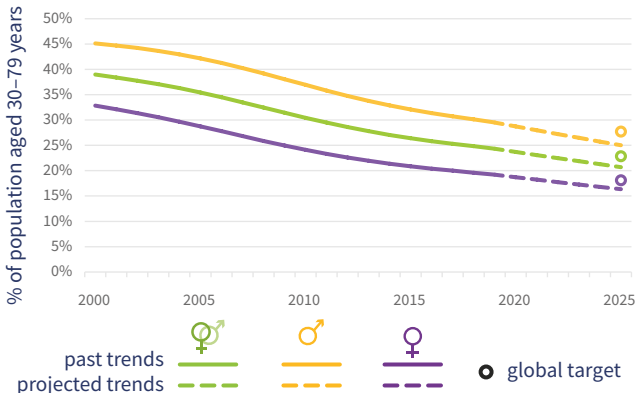
Prevalence of hypertension – global comparison (both sexes)^a



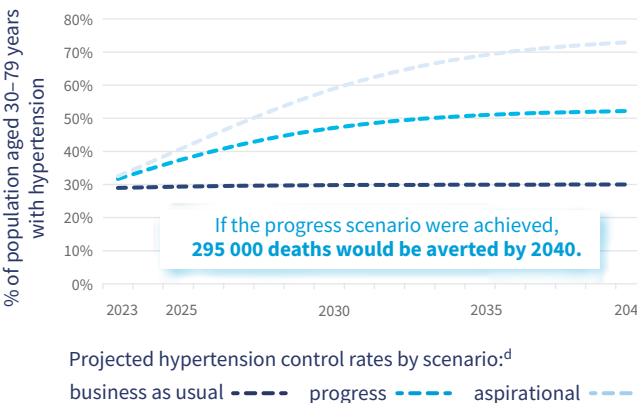
Of the 16.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	9	11	7	2019
Cardiovascular disease deaths	227 000	98 800	129 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	47	47	48	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	8	2019
Current tobacco use, adults aged 15+ years (%) ^f	23	27	20	2019
Obesity, adults aged 18+ years (%)	20	20	20	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	13	4	2019
Physical inactivity, adults aged 18+ years (%)	41	36	46	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Jamaica

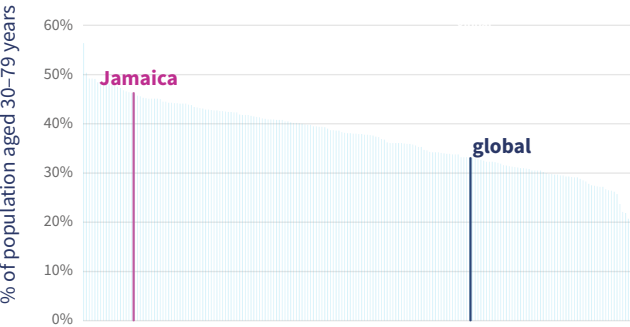
Hypertension profile

Total population (2019): 2 814 000

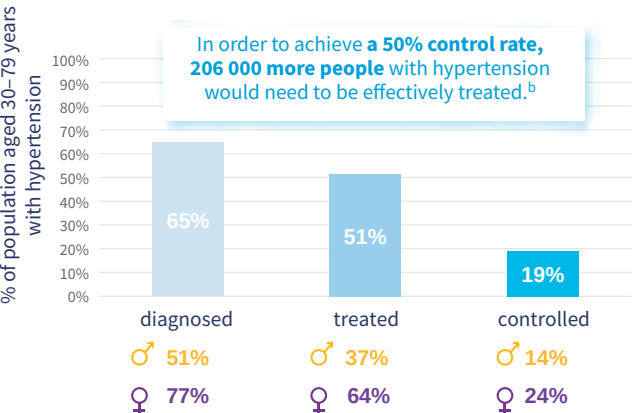
Total deaths (2019): 19 300

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 46% ♂ 45% ♀ 48%

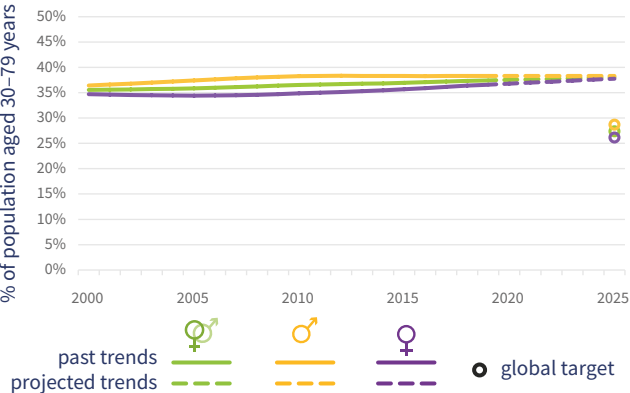
Prevalence of hypertension – global comparison (both sexes)^a



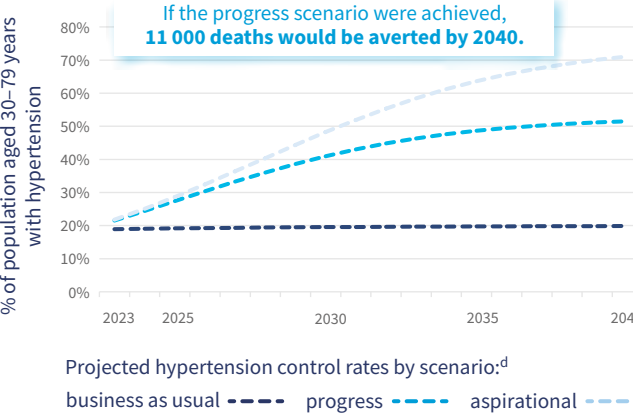
Of the 670 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	17	16	17	2019
Cardiovascular disease deaths	5400	2600	2800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	51	51	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%) ^f	10	16	4	2019
Obesity, adults aged 18+ years (%)	25	15	33	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	6	1	2019
Physical inactivity, adults aged 18+ years (%)	33	28	37	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Japan

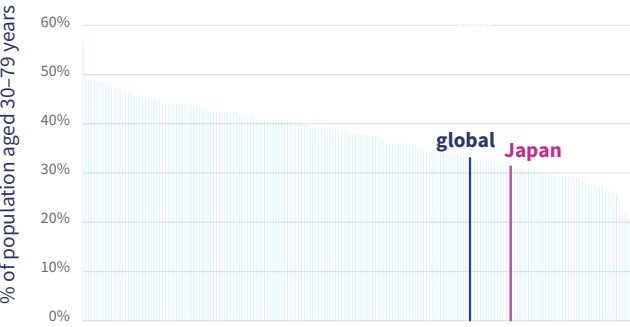
Hypertension profile

Total population (2019): 125 800 000

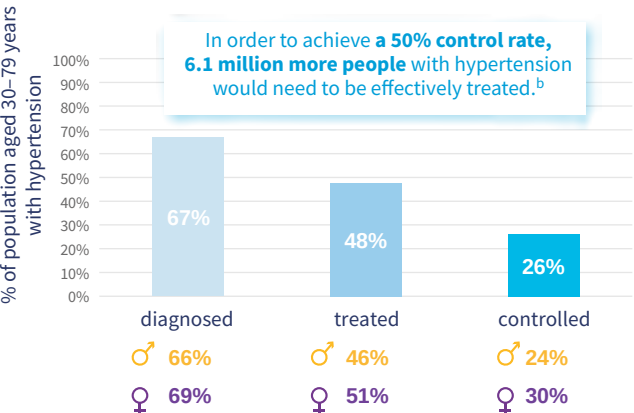
Total deaths (2019): 1 352 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 40% ♀ 23%

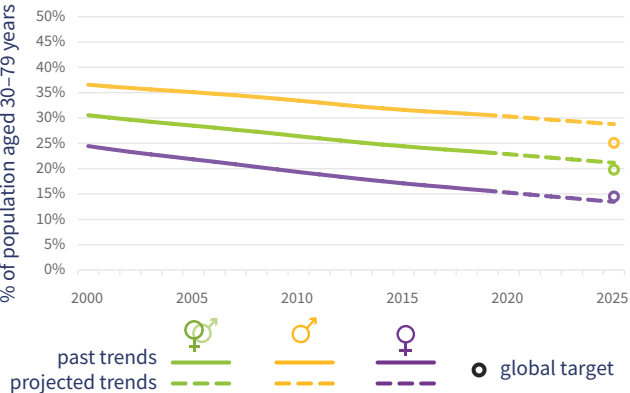
Prevalence of hypertension – global comparison (both sexes)^a



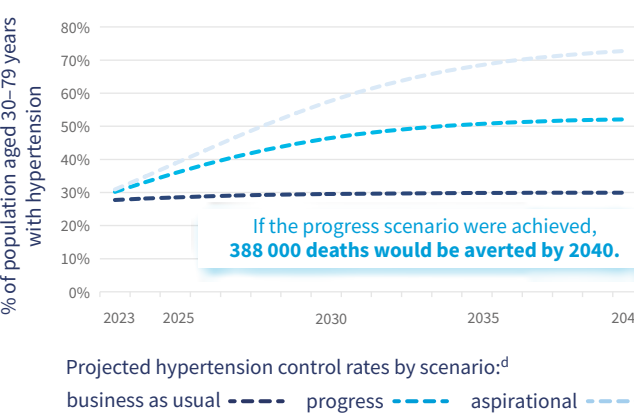
Of the 33.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	8	11	6	2019
Cardiovascular disease deaths	364 000	165 000	199 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	45	48	42	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	9	2019
Current tobacco use, adults aged 15+ years (%) ^f	21	31	10	2019
Obesity, adults aged 18+ years (%)	4	5	4	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	11	3	2019
Physical inactivity, adults aged 18+ years (%)	35	34	37	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Jordan

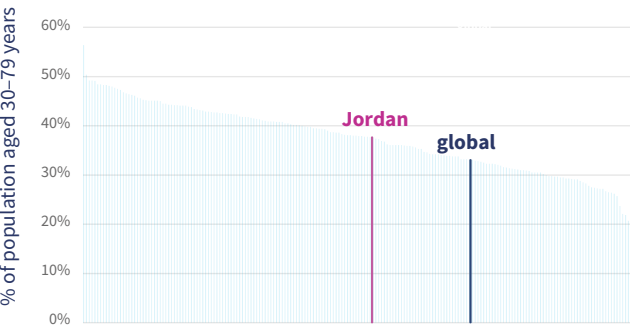
Hypertension profile

Total population (2019): 10 699 000

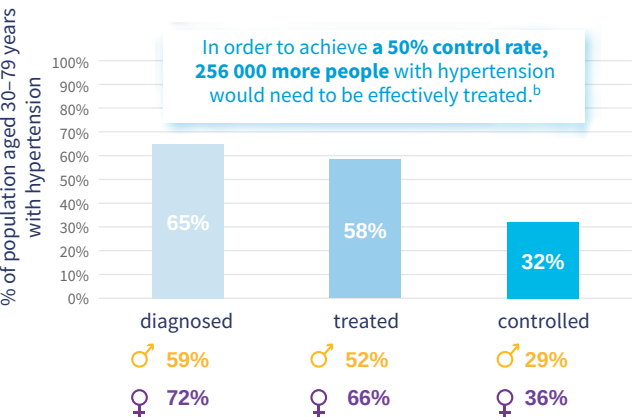
Total deaths (2019): 28 900

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 40% ♀ 36%

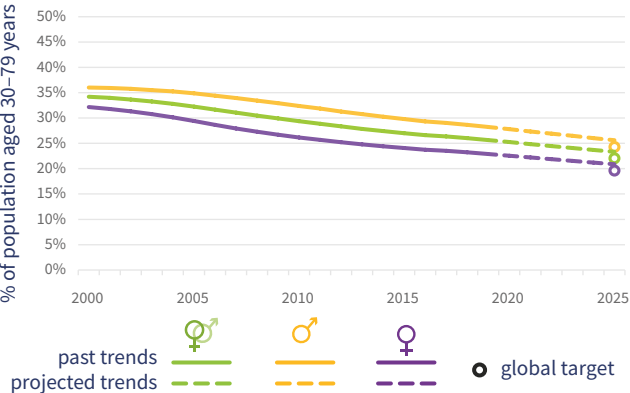
Prevalence of hypertension – global comparison (both sexes)^a



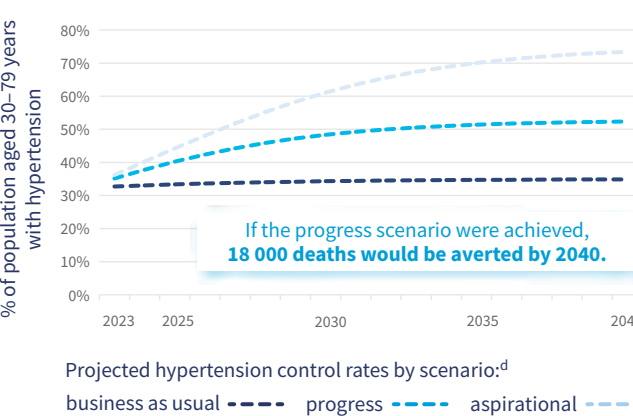
Of the 1.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	15	17	14	2019
Cardiovascular disease deaths	9700	4700	5000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	58	62	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%) ^f	35	57	13	2019
Obesity, adults aged 18+ years (%)	36	28	43	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	1	0	2019
Physical inactivity, adults aged 18+ years (%)	12	10	13	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Kazakhstan

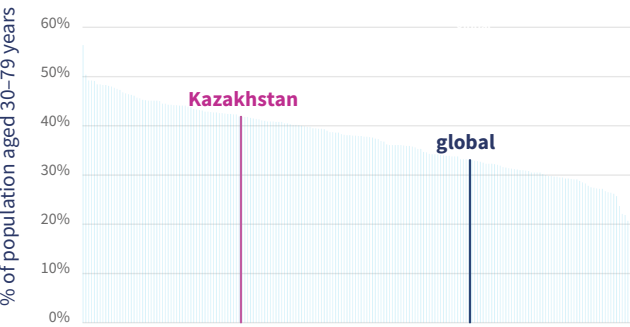
Hypertension profile

Total population (2019): 18 754 000

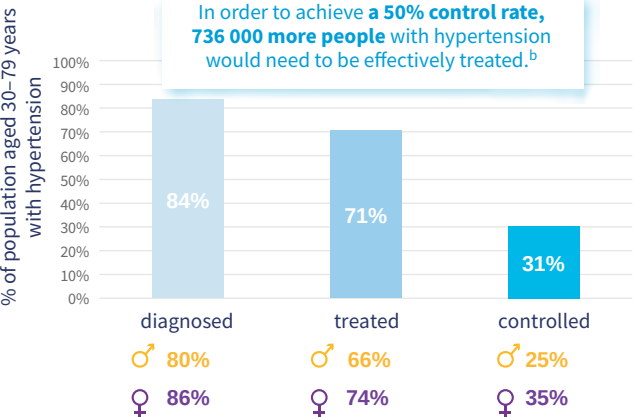
Total deaths (2019): 127 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 42% ♂ 41% ♀ 43%

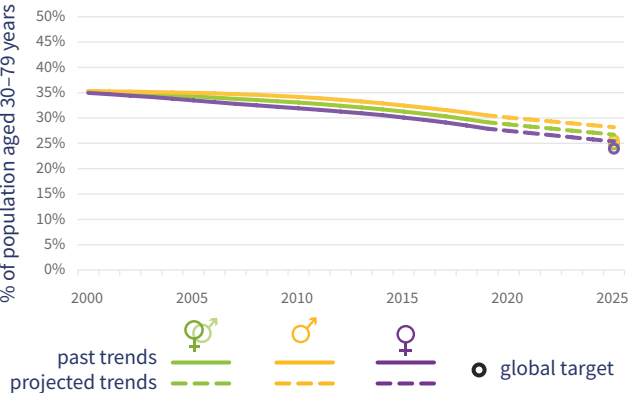
Prevalence of hypertension – global comparison (both sexes)^a



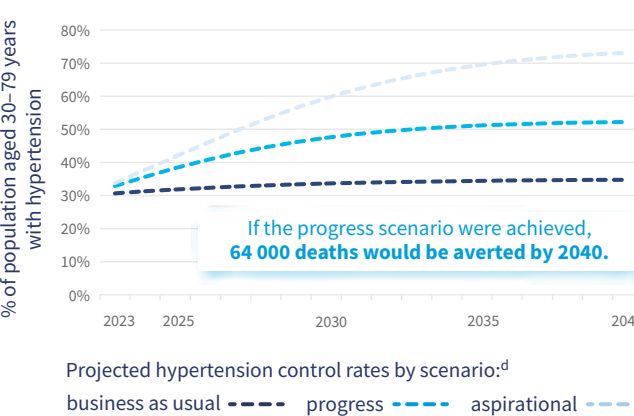
Of the 3.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	31	15	2019
Cardiovascular disease deaths	60 300	27 500	32 800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	64	63	65	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%)	24	40	7	2019
Obesity, adults aged 18+ years (%)	21	19	23	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	8	1	2019
Physical inactivity, adults aged 18+ years (%)	28	26	29	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Kenya

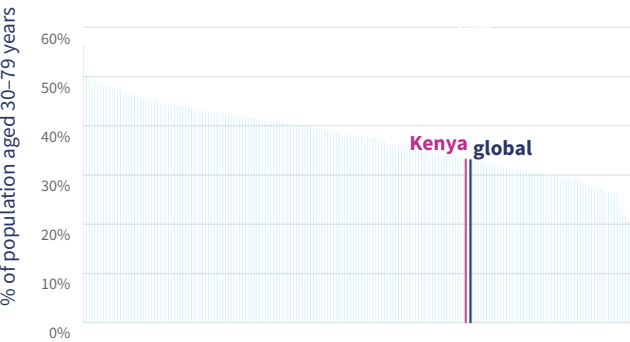
Hypertension profile

Total population (2019): 50 951 000

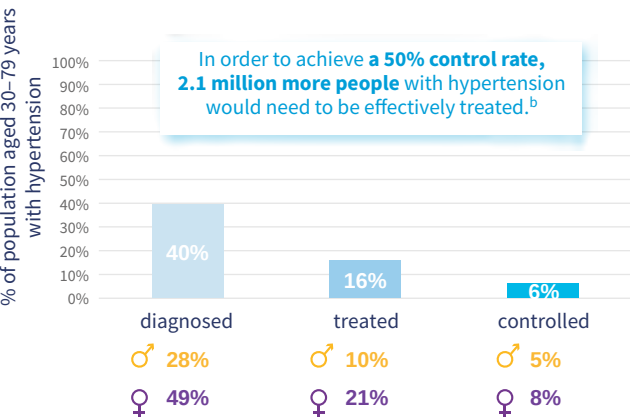
Total deaths (2019): 294 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 33% ♂ 31% ♀ 35%

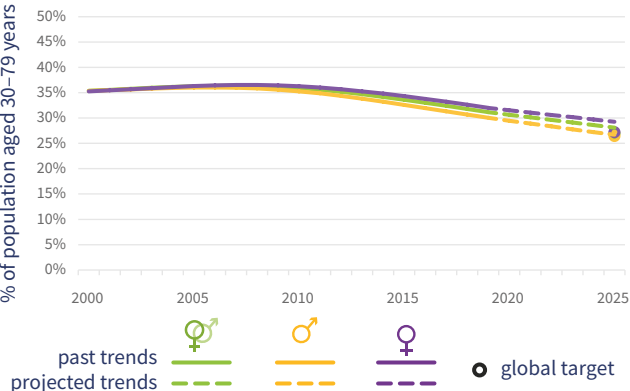
Prevalence of hypertension – global comparison (both sexes)^a



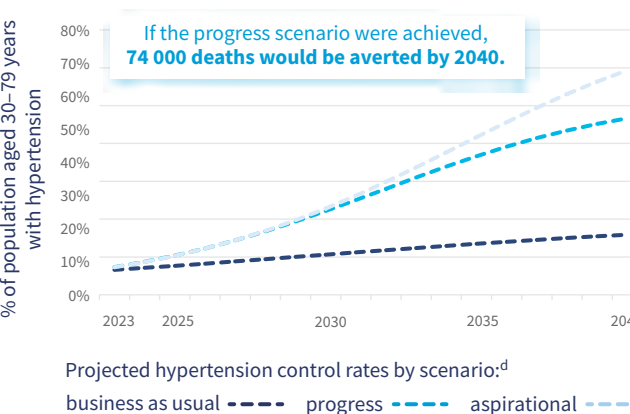
Of the 4.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	23	20	2019
Cardiovascular disease deaths	36 000	18 000	18 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	57	64	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	4	8	2019
Current tobacco use, adults aged 15+ years (%)	12	20	3	2019
Obesity, adults aged 18+ years (%)	7	3	11	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	15	14	17	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Kiribati

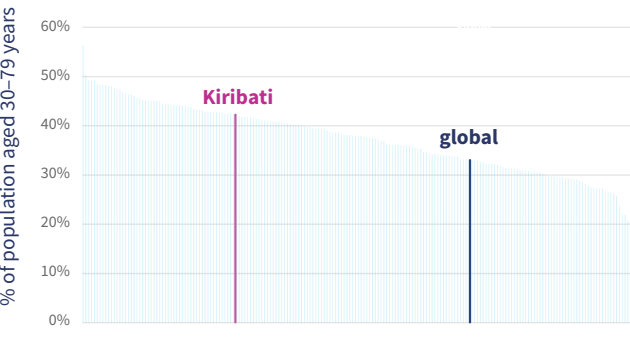
Hypertension profile

Total population (2019): 124 000

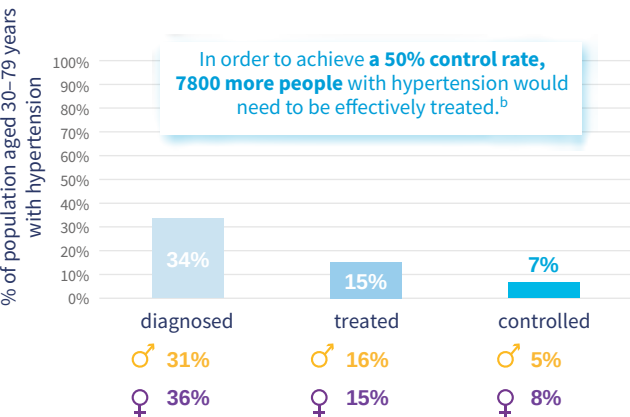
Total deaths (2019): 1280

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 42% ♂ 40% ♀ 44%

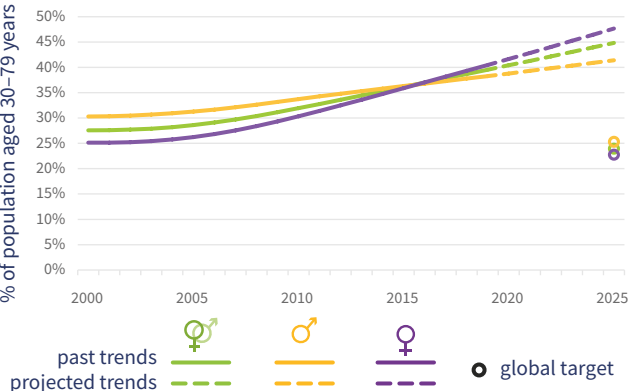
Prevalence of hypertension – global comparison (both sexes)^a



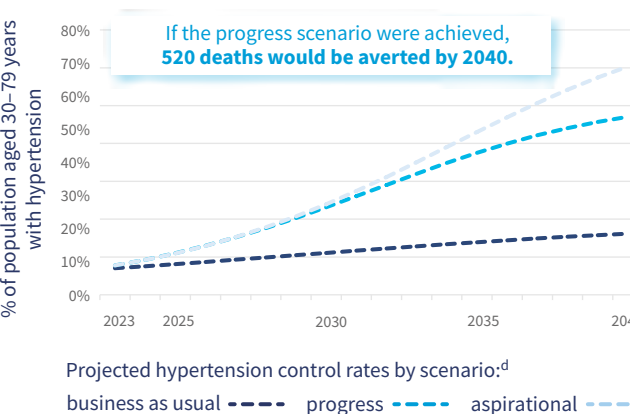
Of the 18 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	51	58	44	2019
Cardiovascular disease deaths	390	220	170	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	46	50	39	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	42	55	28	2019
Obesity, adults aged 18+ years (%)	46	42	50	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	1	0	2019
Physical inactivity, adults aged 18+ years (%)	40	34	46	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Kuwait

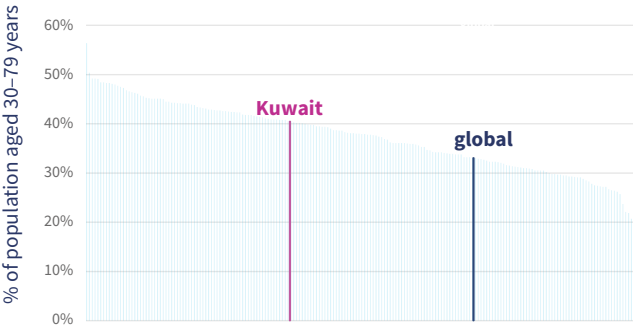
Hypertension profile

Total population (2019): 4 441 000

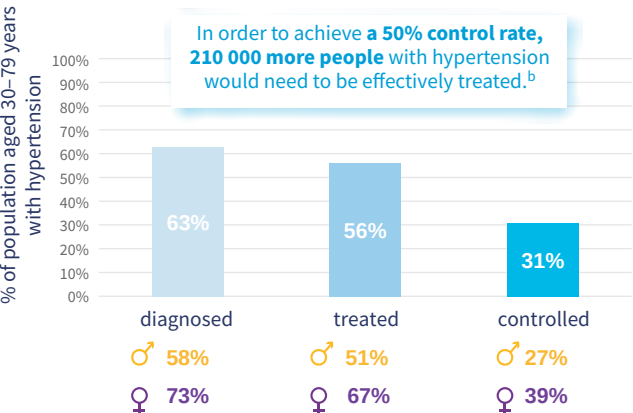
Total deaths (2019): 8240

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 44% ♀ 35%

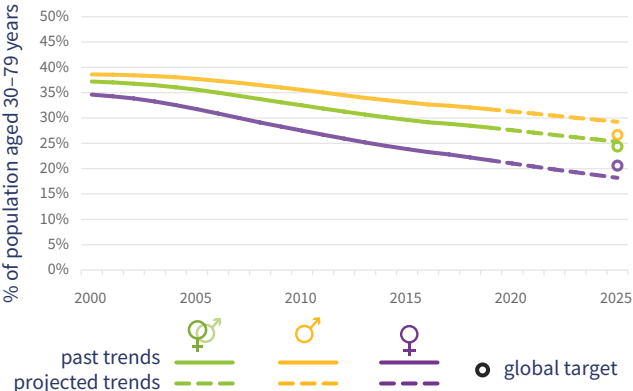
Prevalence of hypertension – global comparison (both sexes)^a



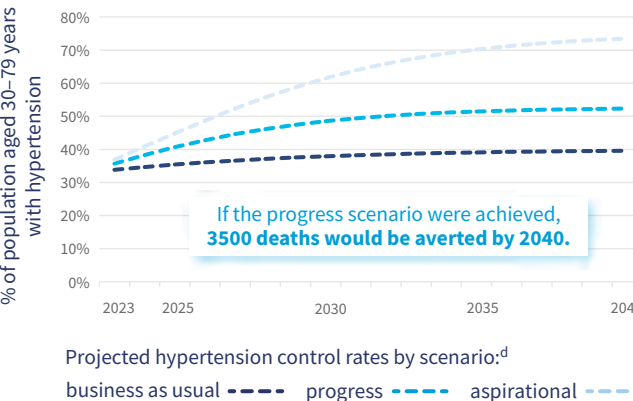
Of the 921 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	12	14	8	2019
Cardiovascular disease deaths	3300	2900	460	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	55	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%)	18	34	2	2019
Obesity, adults aged 18+ years (%)	38	33	46	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	67	61	75	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Kyrgyzstan

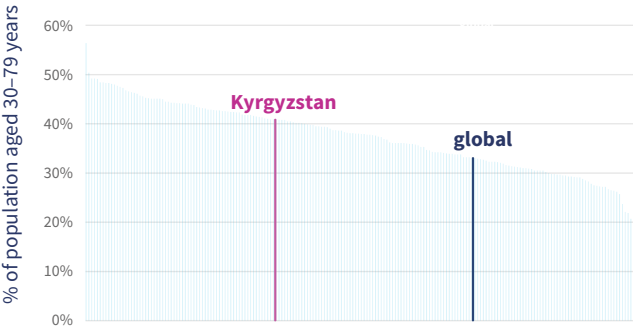
Hypertension profile

Total population (2019): 6 324 000

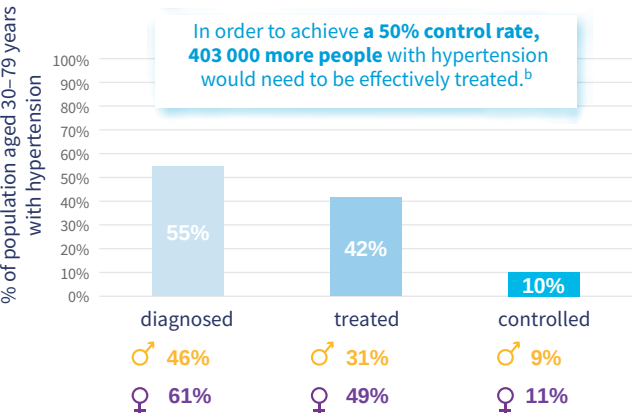
Total deaths (2019): 29 900

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 38% ♀ 43%

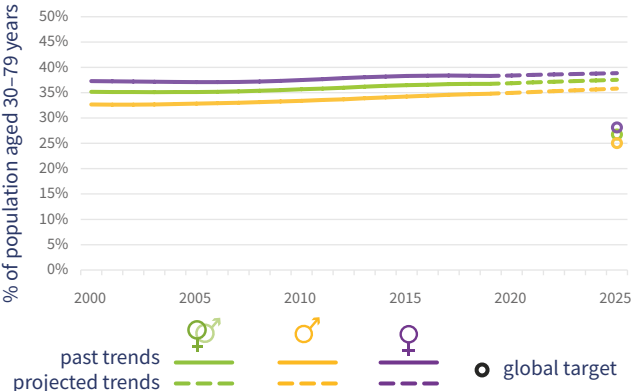
Prevalence of hypertension – global comparison (both sexes)^a



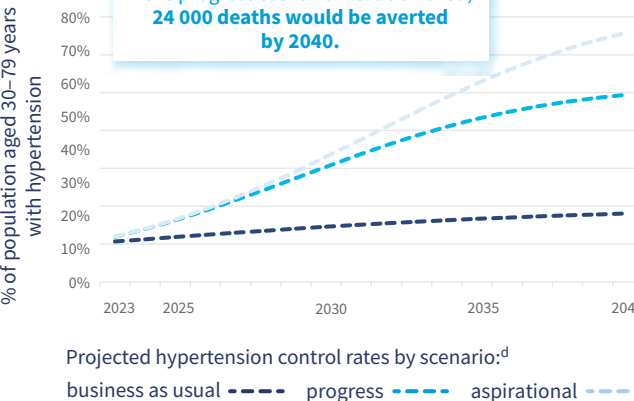
Of the 1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	20	27	15	2019
Cardiovascular disease deaths	14 000	6800	7100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	44	44	44	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%)	28	52	3	2019
Obesity, adults aged 18+ years (%)	17	14	19	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	9	1	2019
Physical inactivity, adults aged 18+ years (%)	14	11	17	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Lao People's Democratic Republic

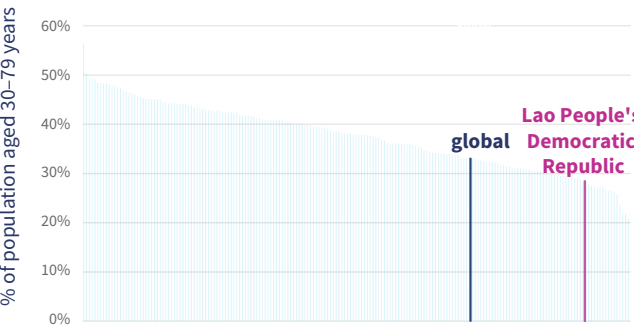
Hypertension profile

Total population (2019): 7 212 000

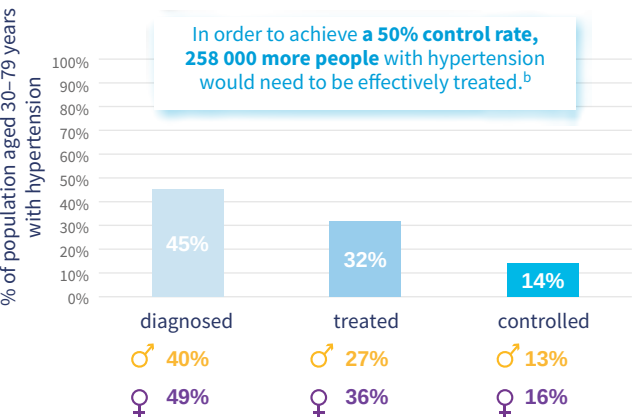
Total deaths (2019): 43 500

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 29% ♂ 26% ♀ 31%

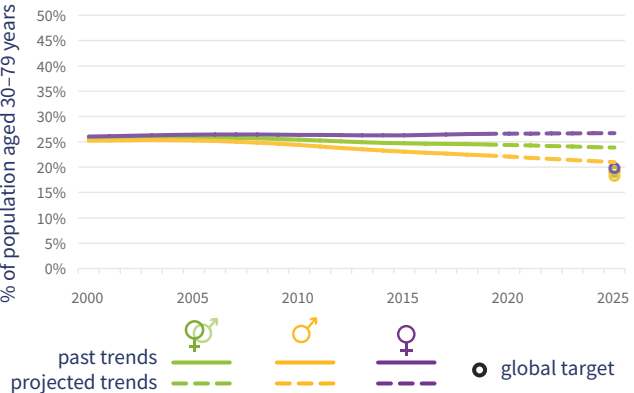
Prevalence of hypertension – global comparison (both sexes)^a



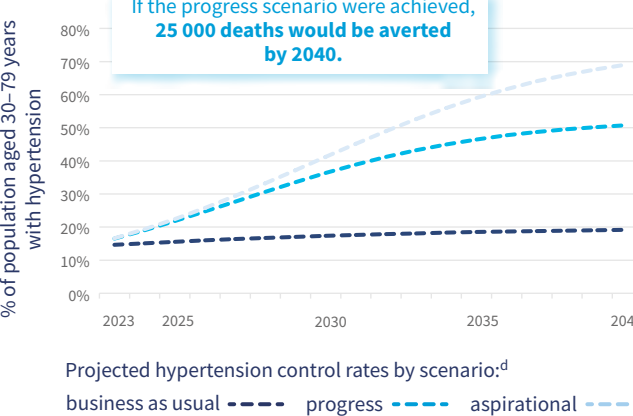
Of the 712 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	27	31	23	2019
Cardiovascular disease deaths	12 500	6500	6100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	55	51	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	10	2019
Current tobacco use, adults aged 15+ years (%)	33	54	11	2019
Obesity, adults aged 18+ years (%)	5	4	7	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	18	5	2019
Physical inactivity, adults aged 18+ years (%)	16	12	21	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Latvia

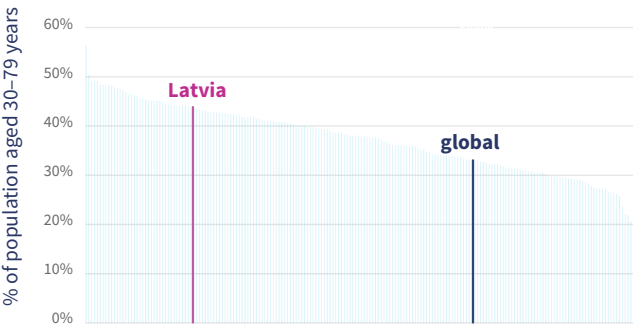
Hypertension profile

Total population (2019): 1 917 000

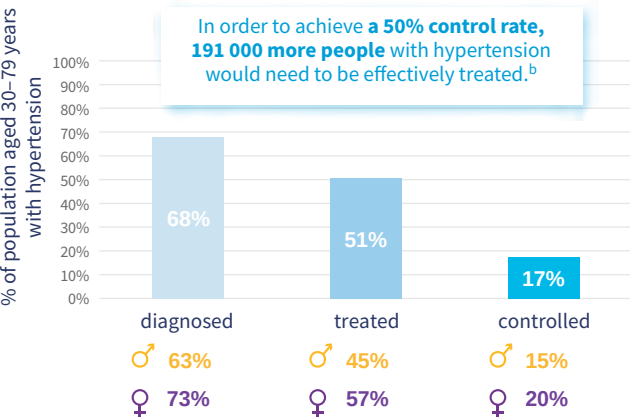
Total deaths (2019): 29 800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 44% ♂ 49% ♀ 39%

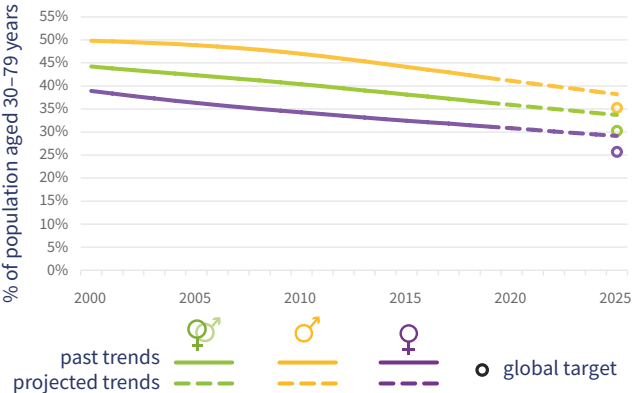
Prevalence of hypertension – global comparison (both sexes)^a



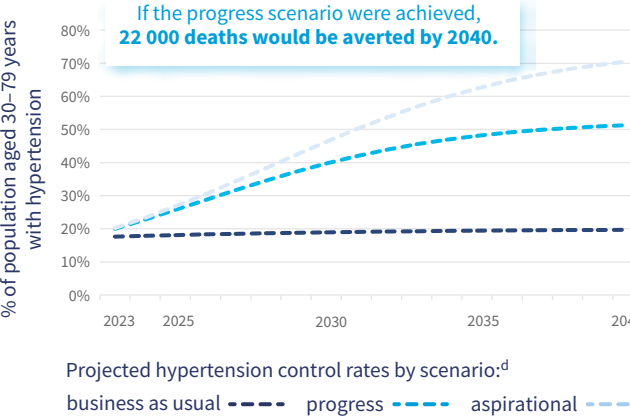
Of the 605 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	31	14	2019
Cardiovascular disease deaths	16 900	6700	10 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	57	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	10	6	2019
Current tobacco use, adults aged 15+ years (%)	37	51	24	2019
Obesity, adults aged 18+ years (%)	24	22	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	13	22	6	2019
Physical inactivity, adults aged 18+ years (%)	30	25	33	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Lebanon

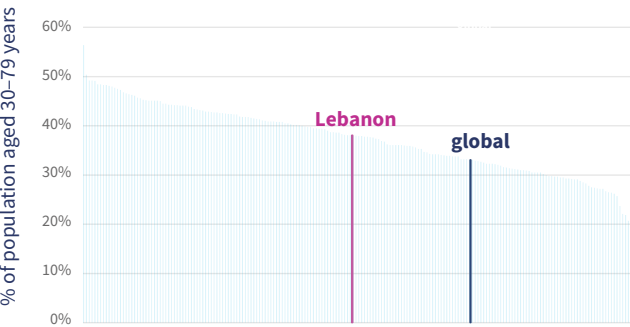
Hypertension profile

Total population (2019): 5 782 000

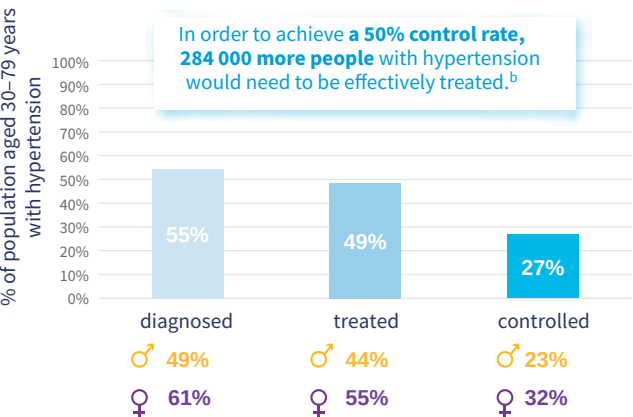
Total deaths (2019): 37 800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 42% ♀ 34%

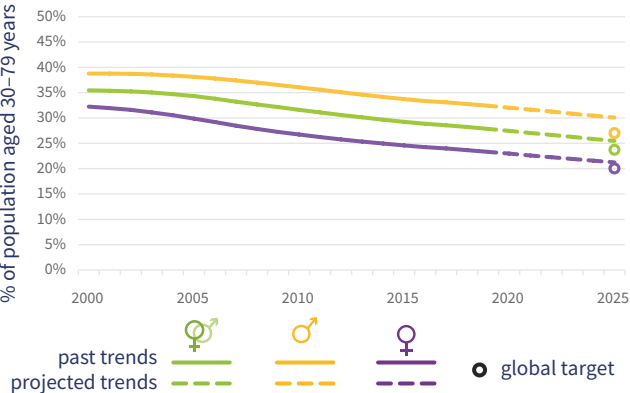
Prevalence of hypertension – global comparison (both sexes)^a



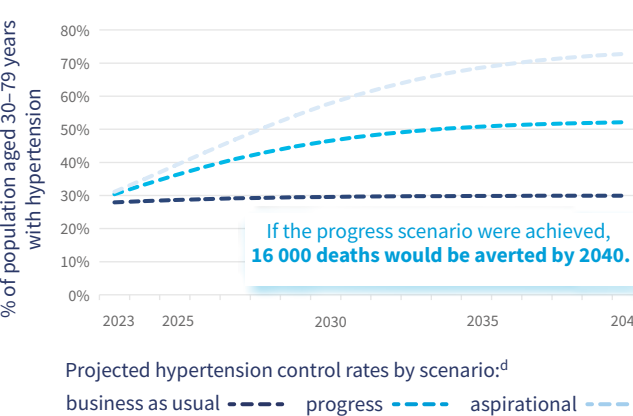
Of the 1.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	20	24	15	2019
Cardiovascular disease deaths	17 500	10 400	7200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	57	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	38	48	29	2019
Obesity, adults aged 18+ years (%)	32	27	37	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	3	0	2019
Physical inactivity, adults aged 18+ years (%)	36	40	33	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Lesotho

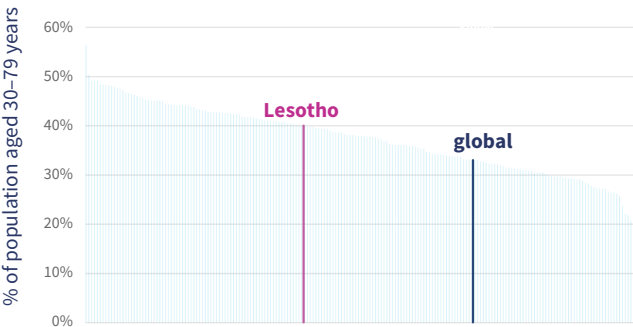
Hypertension profile

Total population (2019): 2 226 000

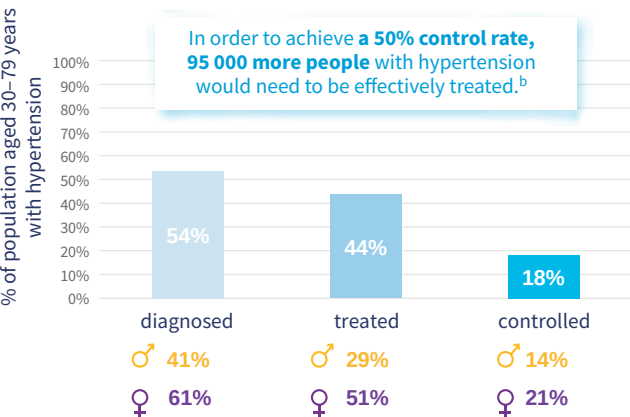
Total deaths (2019): 35 700

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 40% ♂ 32% ♀ 47%

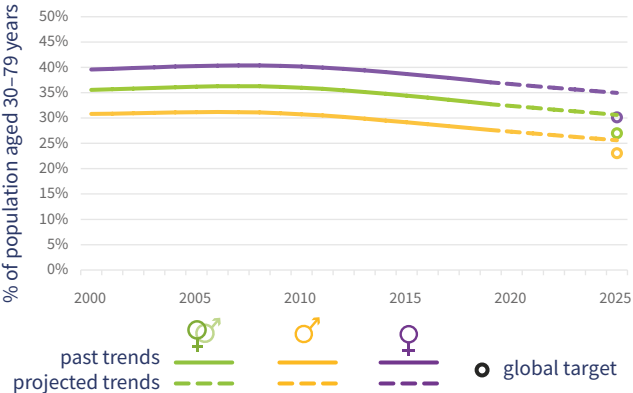
Prevalence of hypertension – global comparison (both sexes)^a



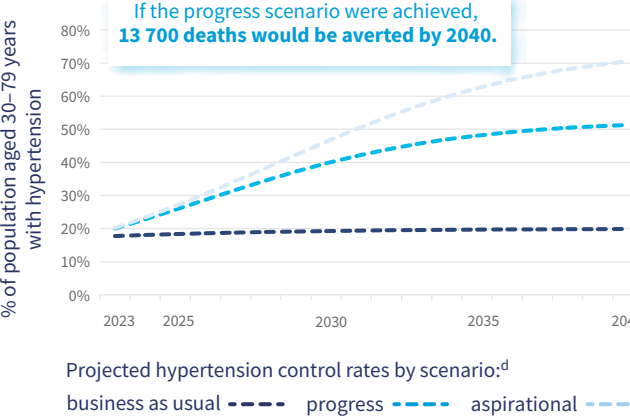
Of the 297 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	43	49	37	2019
Cardiovascular disease deaths	6600	3000	3600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	55	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	7	2019
Current tobacco use, adults aged 15+ years (%)	24	43	6	2019
Obesity, adults aged 18+ years (%)	17	5	27	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	8	1	2019
Physical inactivity, adults aged 18+ years (%)	6	7	6	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Liberia

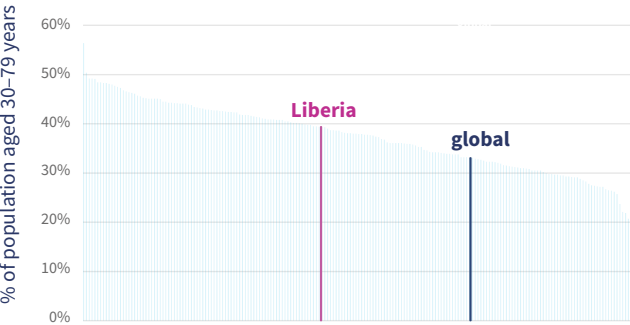
Hypertension profile

Total population (2019): 4 985 000

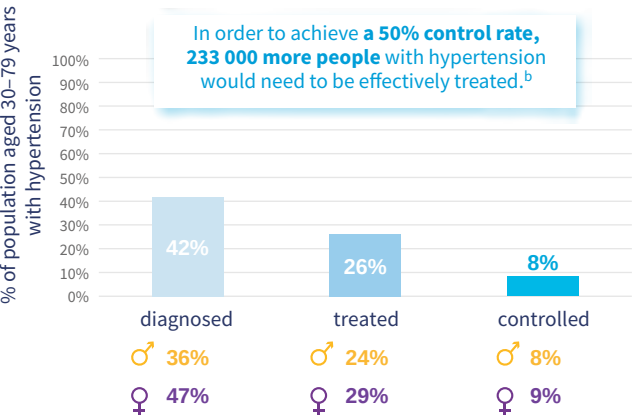
Total deaths (2019): 36 600

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 39% ♂ 37% ♀ 41%

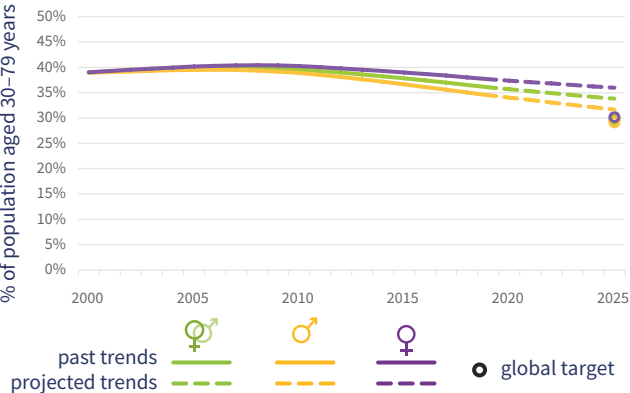
Prevalence of hypertension – global comparison (both sexes)^a



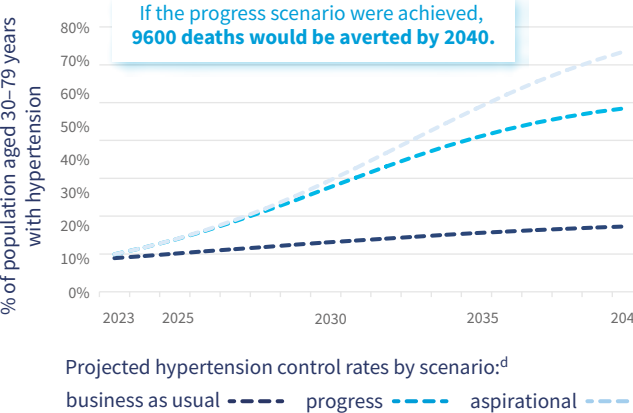
Of the 555 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	18	15	20	2019
Cardiovascular disease deaths	4200	1800	2500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	59	58	61	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	8	15	2	2019
Obesity, adults aged 18+ years (%)	10	6	14	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	6	1	2019
Physical inactivity, adults aged 18+ years (%)	25	22	29	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Libya

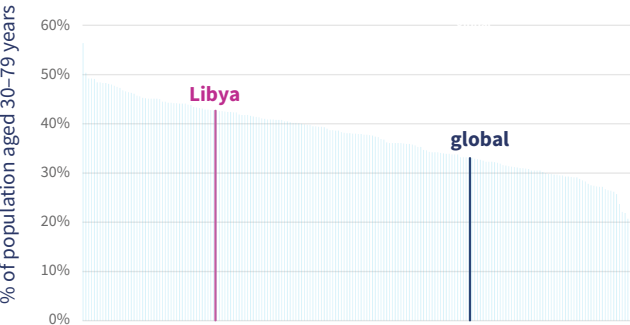
Hypertension profile

Total population (2019): 6 569 000

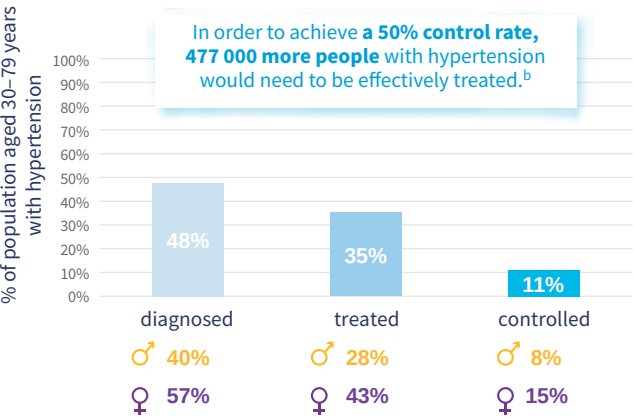
Total deaths (2019): 27 100

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 43% ♂ 46% ♀ 39%

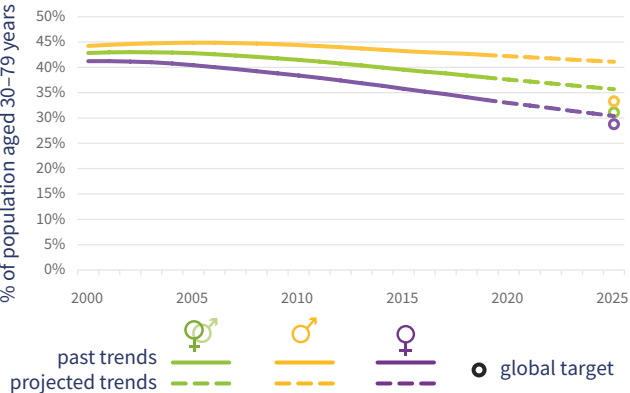
Prevalence of hypertension – global comparison (both sexes)^a



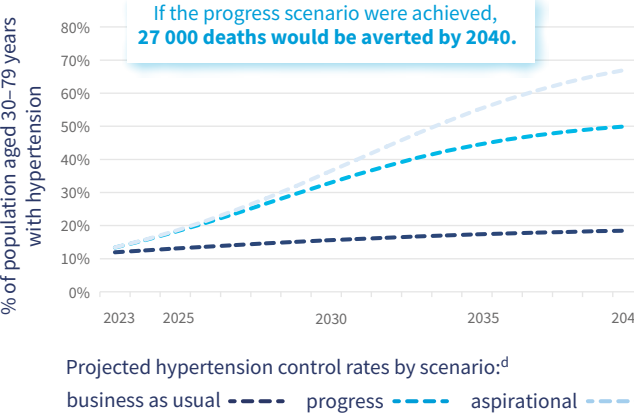
Of the 1.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	19	20	18	2019
Cardiovascular disease deaths	10 700	5200	5500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	61	59	62	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	33	25	40	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	36	31	42	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Lithuania

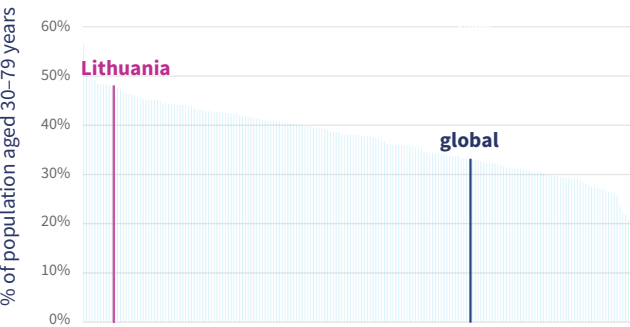
Hypertension profile

Total population (2019): 2 849 000

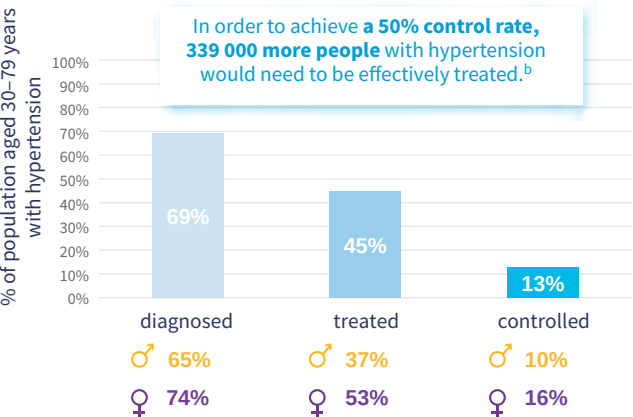
Total deaths (2019): 42 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 48% ♂ 54% ♀ 42%

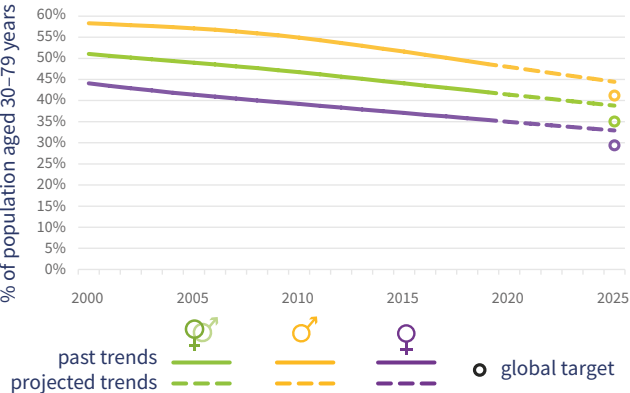
Prevalence of hypertension – global comparison (both sexes)^a



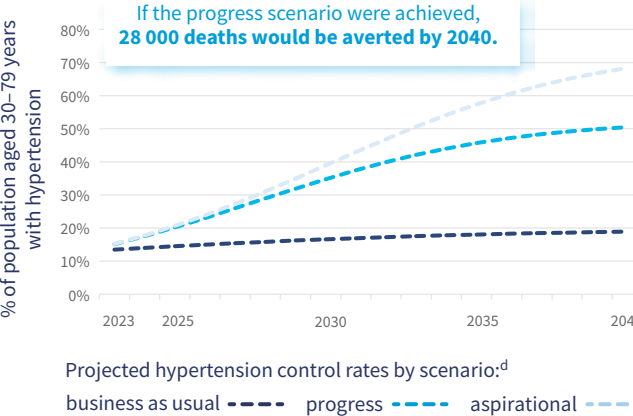
Of the 940 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	19	28	12	2019
Cardiovascular disease deaths	24 000	9400	14 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	60	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	7	2019
Current tobacco use, adults aged 15+ years (%)	32	43	22	2019
Obesity, adults aged 18+ years (%)	26	24	28	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	19	5	2019
Physical inactivity, adults aged 18+ years (%)	27	23	29	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Luxembourg

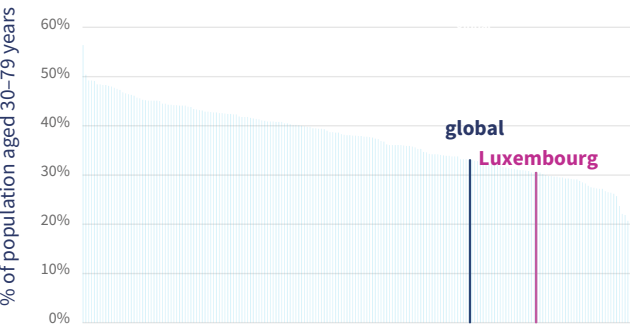
Hypertension profile

Total population (2019): 620 000

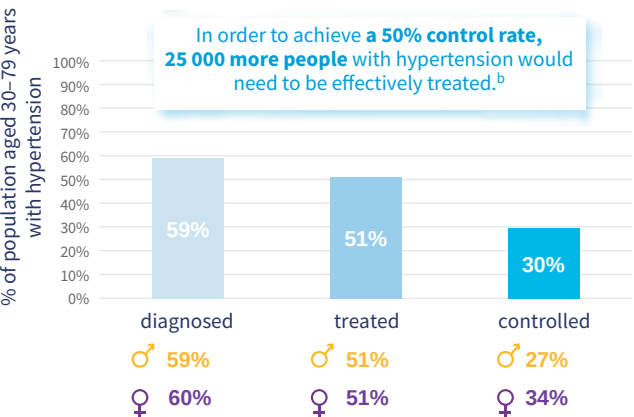
Total deaths (2019): 4250

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 37% ♀ 24%

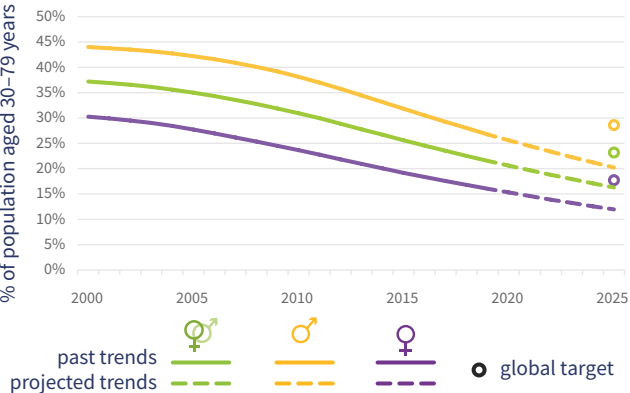
Prevalence of hypertension – global comparison (both sexes)^a



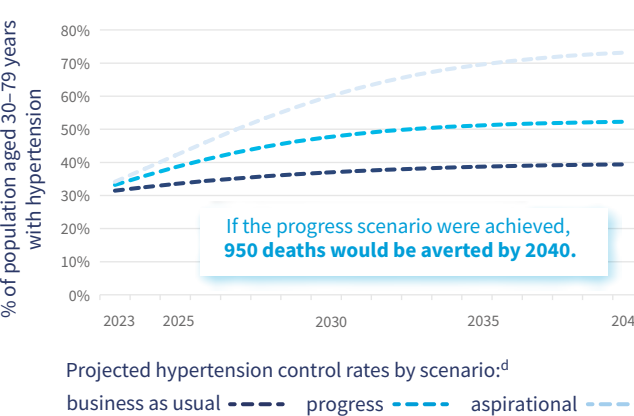
Of the 126 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	12	7	2019
Cardiovascular disease deaths	1300	600	680	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	51	53	49	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	10	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	22	23	20	2019
Obesity, adults aged 18+ years (%)	23	25	21	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	18	5	2019
Physical inactivity, adults aged 18+ years (%)	28	26	30	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Madagascar

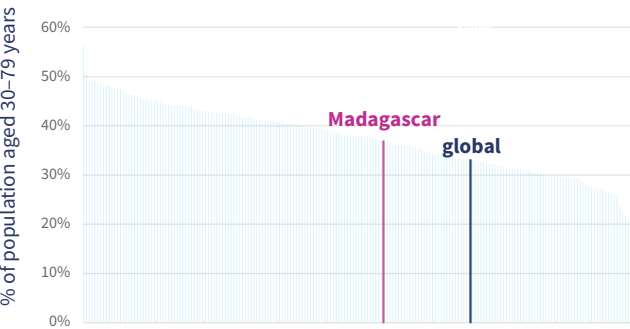
Hypertension profile

Total population (2019): 27 533 000

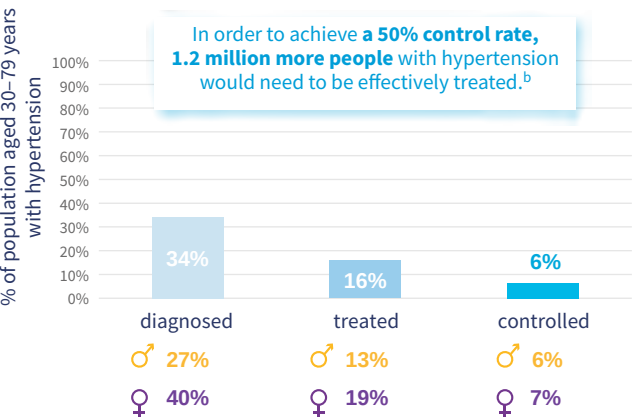
Total deaths (2019): 178 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 37% ♂ 35% ♀ 39%

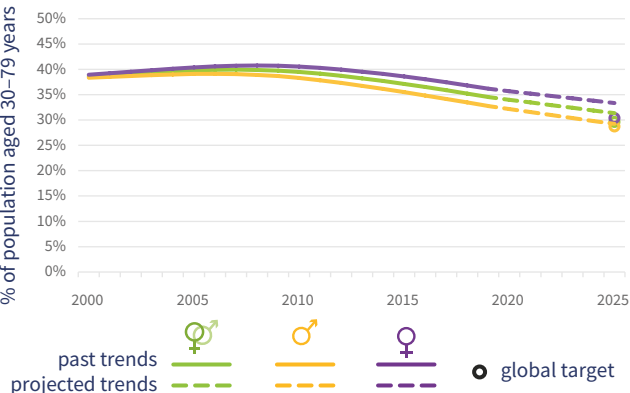
Prevalence of hypertension – global comparison (both sexes)^a



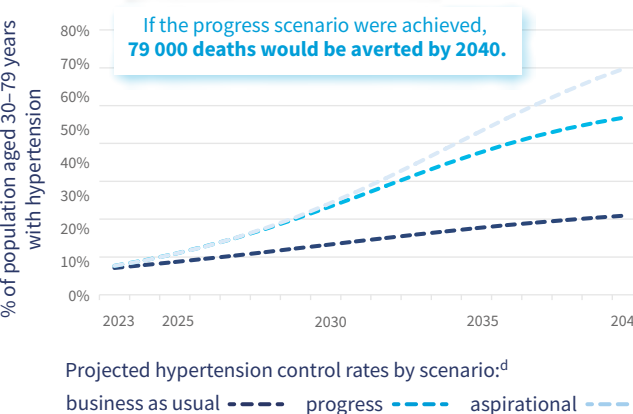
Of the 2.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	26	26	26	2019
Cardiovascular disease deaths	40 500	19 600	20 900	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	61	59	63	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	28	43	14	2019
Obesity, adults aged 18+ years (%)	5	3	8	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	2	0	2019
Physical inactivity, adults aged 18+ years (%)	17	13	21	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Malawi

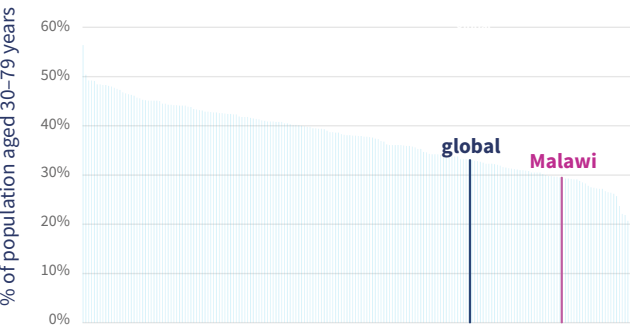
Hypertension profile

Total population (2019): 18 867 000

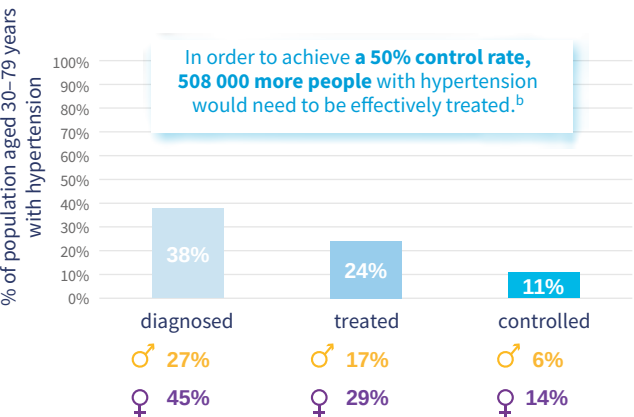
Total deaths (2019): 108 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 30% ♂ 27% ♀ 32%

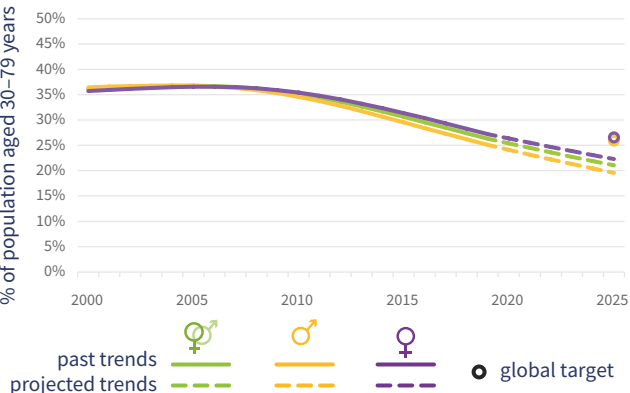
Prevalence of hypertension – global comparison (both sexes)^a



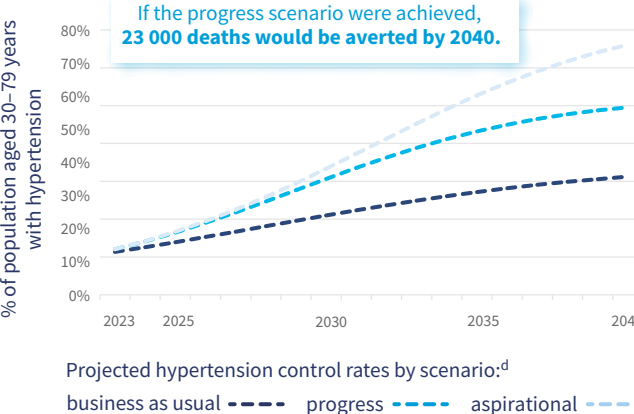
Of the 1.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	26	20	2019
Cardiovascular disease deaths	14 000	7500	6400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	64	64	65	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	11	18	4	2019
Obesity, adults aged 18+ years (%)	6	2	9	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	6	1	2019
Physical inactivity, adults aged 18+ years (%)	16	13	18	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Malaysia

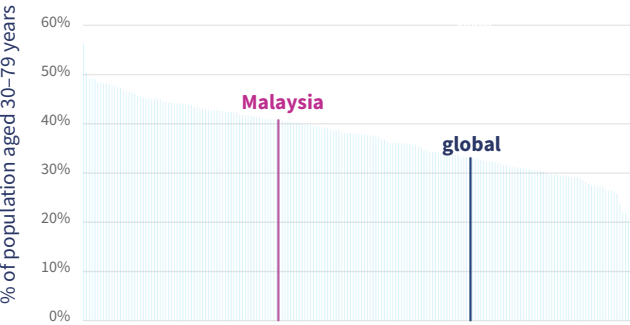
Hypertension profile

Total population (2019): 32 804 000

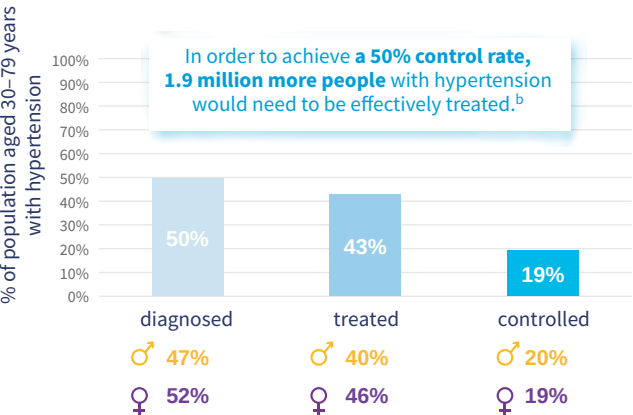
Total deaths (2019): 185 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 41% ♀ 41%

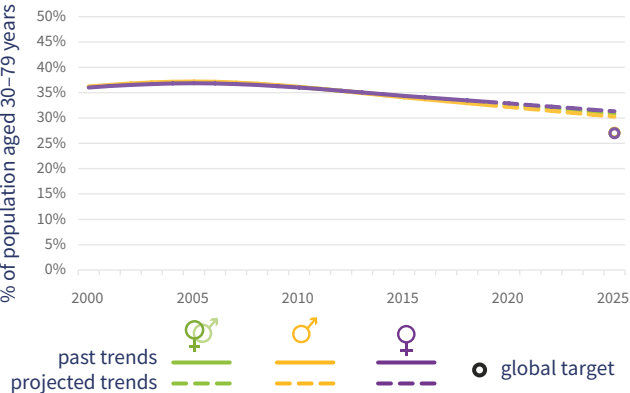
Prevalence of hypertension – global comparison (both sexes)^a



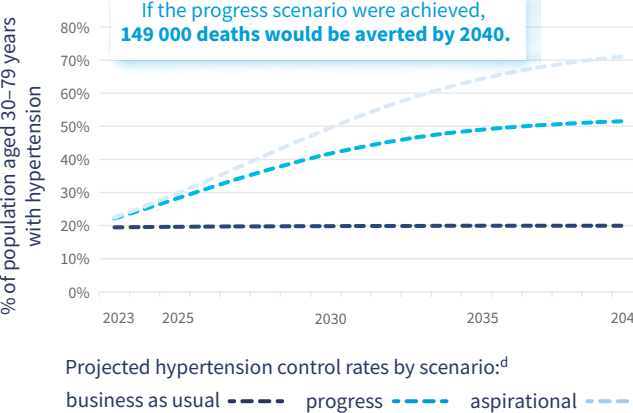
Of the 6.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	18	22	15	2019
Cardiovascular disease deaths	63 600	37 100	26 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	62	62	63	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	10	2019
Current tobacco use, adults aged 15+ years (%)	23	44	1	2019
Obesity, adults aged 18+ years (%)	16	13	18	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	1	0	2019
Physical inactivity, adults aged 18+ years (%)	39	35	43	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Maldives

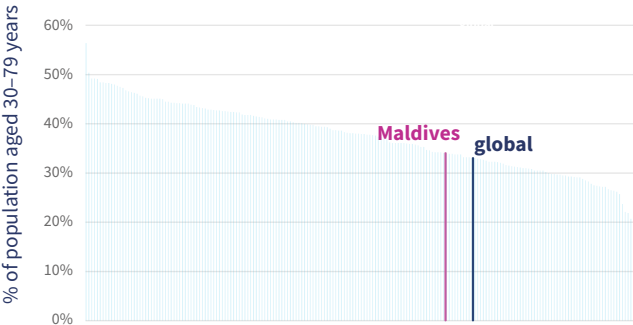
Hypertension profile

Total population (2019): 505 000

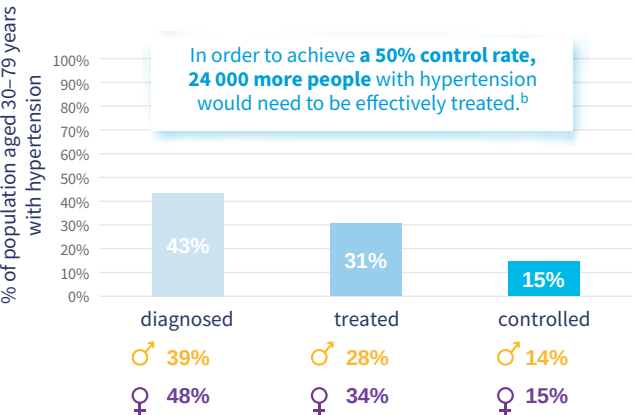
Total deaths (2019): 1330

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 33% ♀ 35%

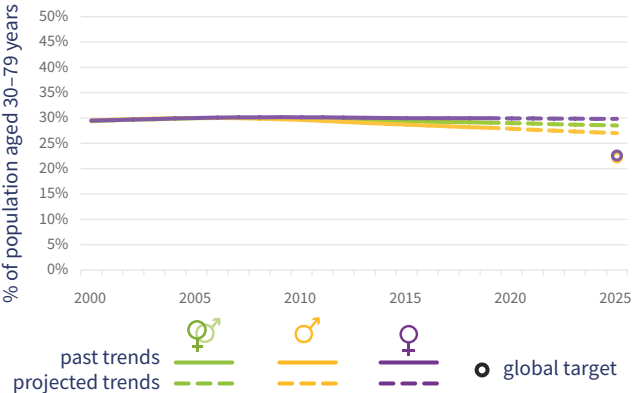
Prevalence of hypertension – global comparison (both sexes)^a



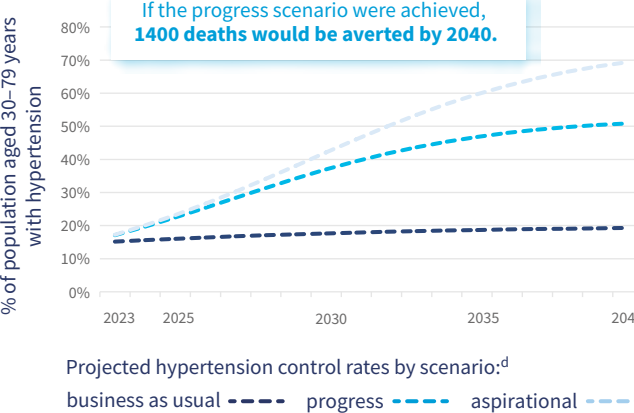
Of the 65 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	12	14	9	2019
Cardiovascular disease deaths	480	300	180	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	53	54	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	10	10	2019
Current tobacco use, adults aged 15+ years (%)	26	45	6	2019
Obesity, adults aged 18+ years (%)	9	6	11	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	2	0	2019
Physical inactivity, adults aged 18+ years (%)	30	26	35	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Mali

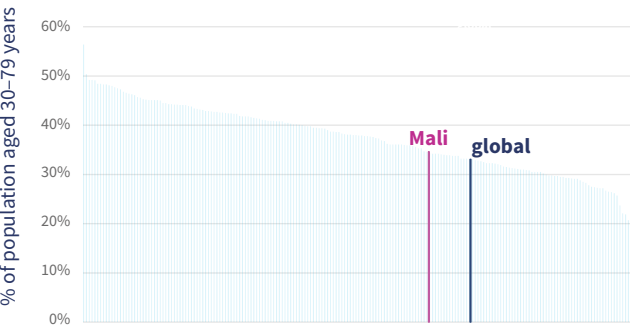
Hypertension profile

Total population (2019): 20 567 000

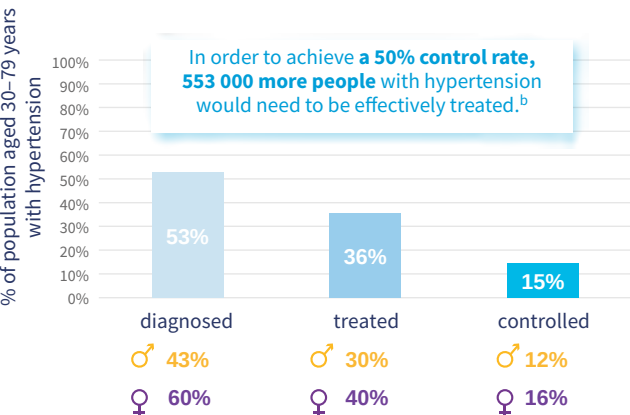
Total deaths (2019): 154 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 35% ♂ 31% ♀ 38%

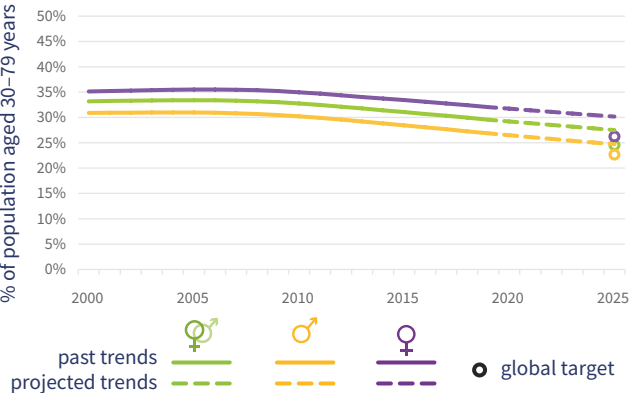
Prevalence of hypertension – global comparison (both sexes)^a



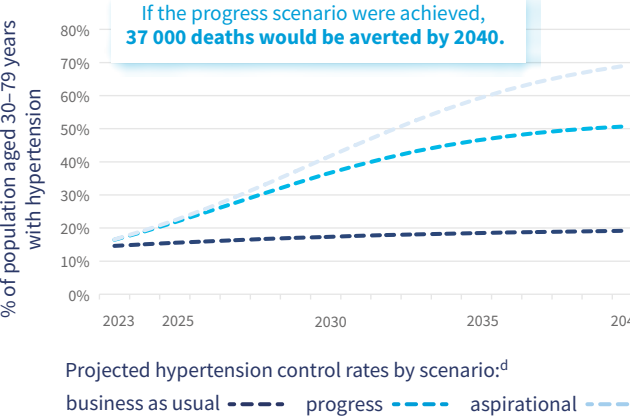
Of the 1.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	20	24	2019
Cardiovascular disease deaths	15 300	6300	9000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	50	45	55	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	9	16	1	2019
Obesity, adults aged 18+ years (%)	9	5	12	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	7	1	2019
Physical inactivity, adults aged 18+ years (%)	40	34	47	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Malta

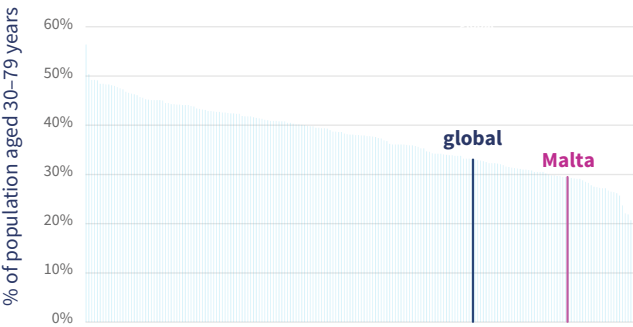
Hypertension profile

Total population (2019): 504 000

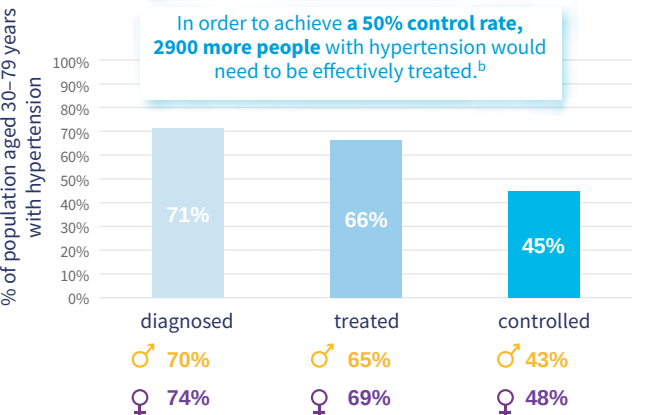
Total deaths (2019): 3800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 30% ♂ 34% ♀ 25%

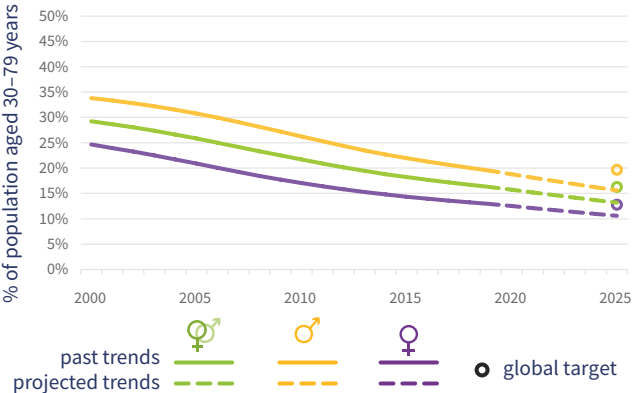
Prevalence of hypertension – global comparison (both sexes)^a



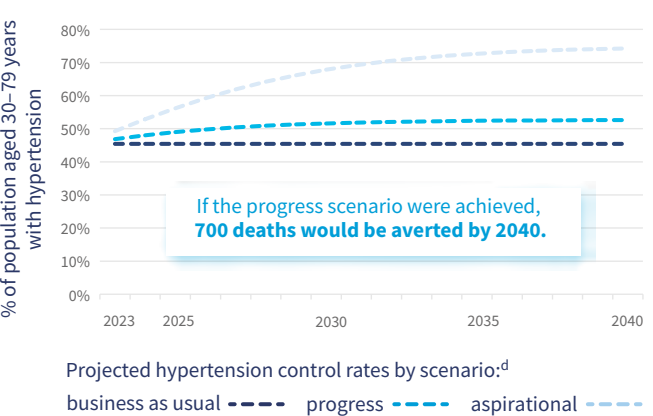
Of the 103 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	13	8	2019
Cardiovascular disease deaths	1400	650	700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	54	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	11	8	2019
Current tobacco use, adults aged 15+ years (%) ^f	25	27	22	2019
Obesity, adults aged 18+ years (%)	29	29	29	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	13	4	2019
Physical inactivity, adults aged 18+ years (%)	42	36	47	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Marshall Islands

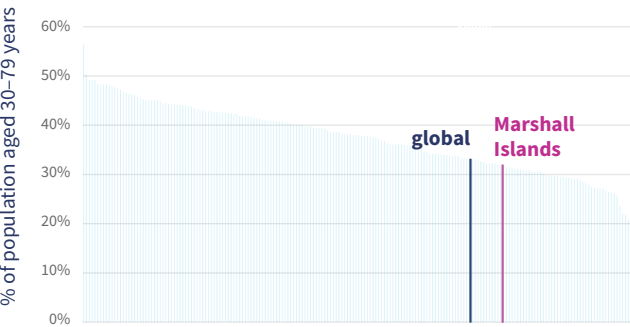
Hypertension profile

Total population (2019): 44 700

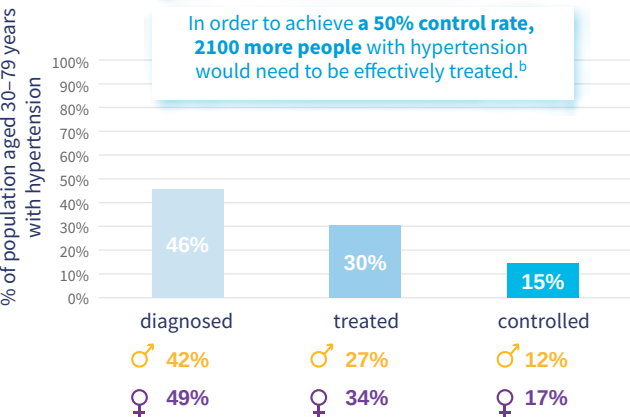
Total deaths (2019): no data

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 32% ♂ 31% ♀ 33%

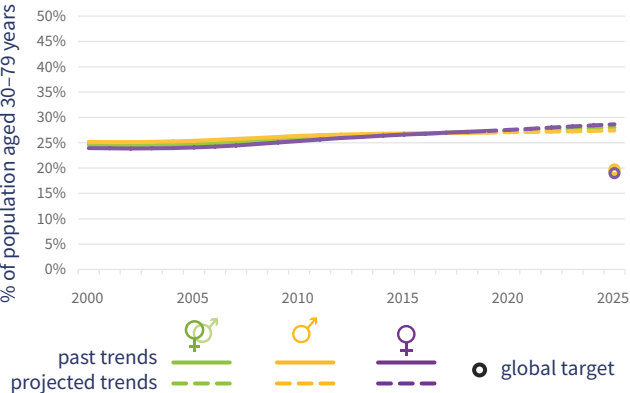
Prevalence of hypertension – global comparison (both sexes)^a



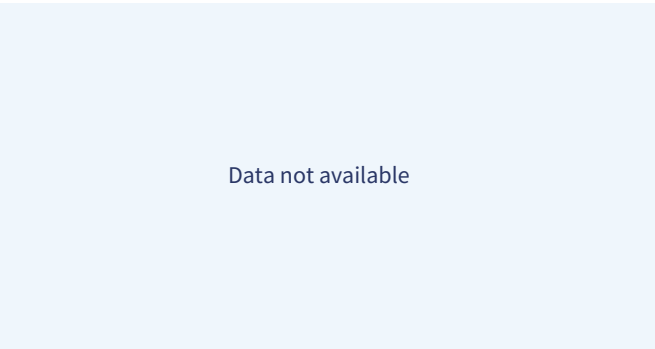
Of the 6000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	48	49	45	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	28	49	8	2019
Obesity, adults aged 18+ years (%)	53	48	57	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	no data	no data	no data	2019
Physical inactivity, adults aged 18+ years (%)	44	37	50	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake.

Mauritania

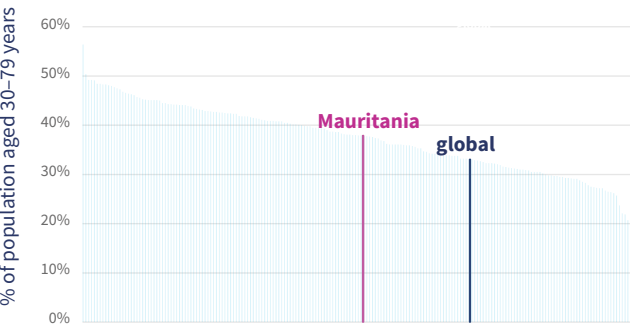
Hypertension profile

Total population (2019): 4 384 000

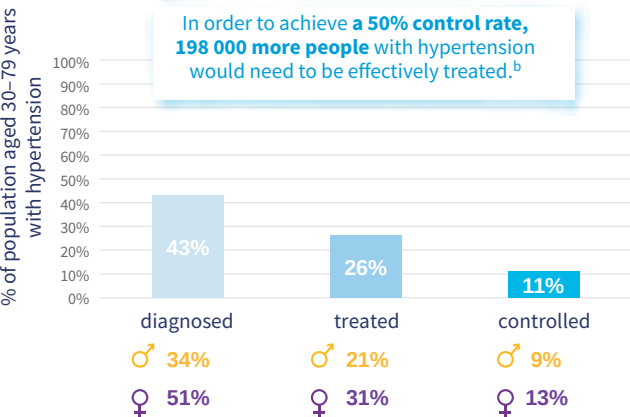
Total deaths (2019): 26 300

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 36% ♀ 40%

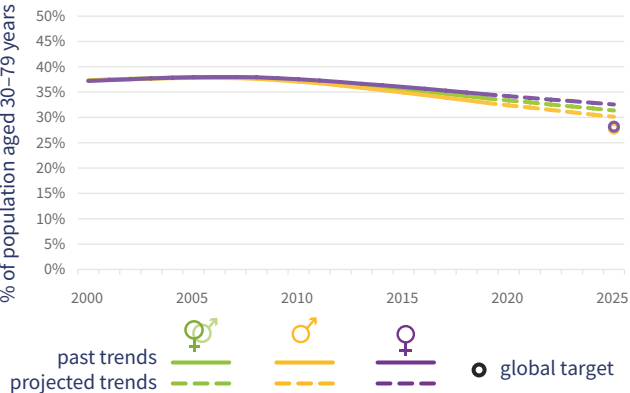
Prevalence of hypertension – global comparison (both sexes)^a



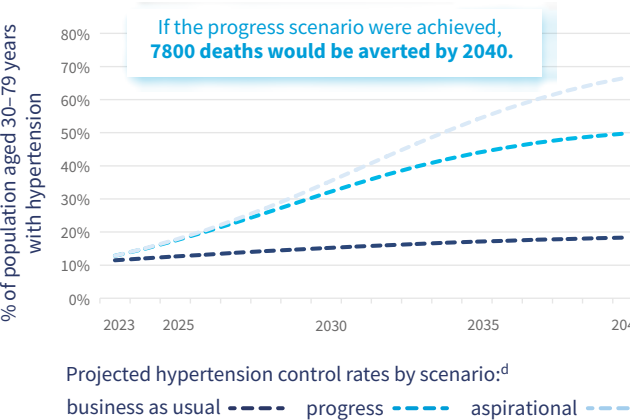
Of the 501 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	14	18	2019
Cardiovascular disease deaths	3600	1400	2200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	55	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	11	20	2	2019
Obesity, adults aged 18+ years (%)	13	7	19	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	41	37	46	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Mauritius

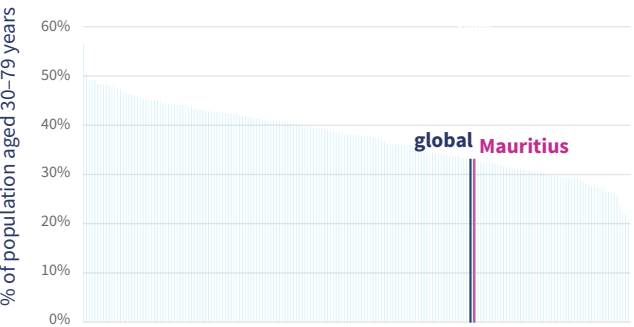
Hypertension profile

Total population (2019): 1 296 000

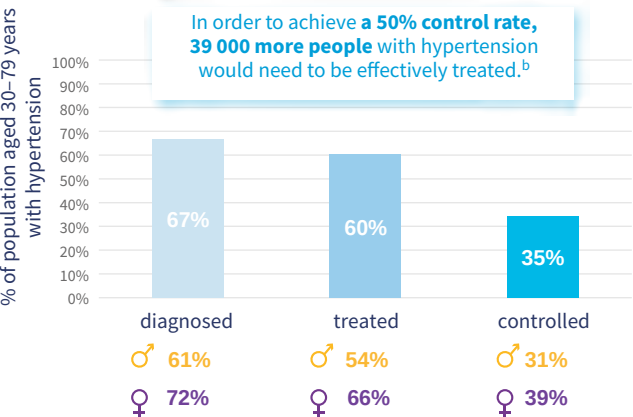
Total deaths (2019): 11 400

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 33% ♂ 34% ♀ 32%

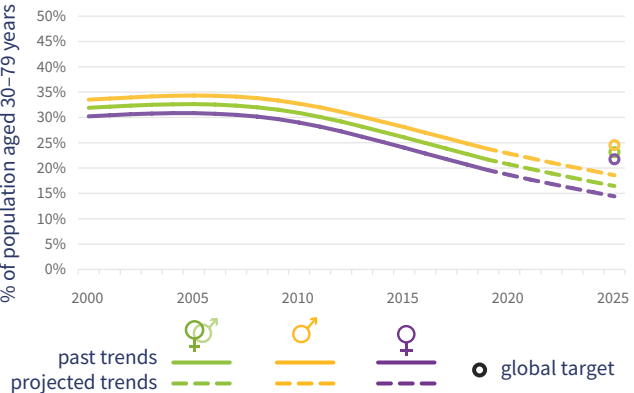
Prevalence of hypertension – global comparison (both sexes)^a



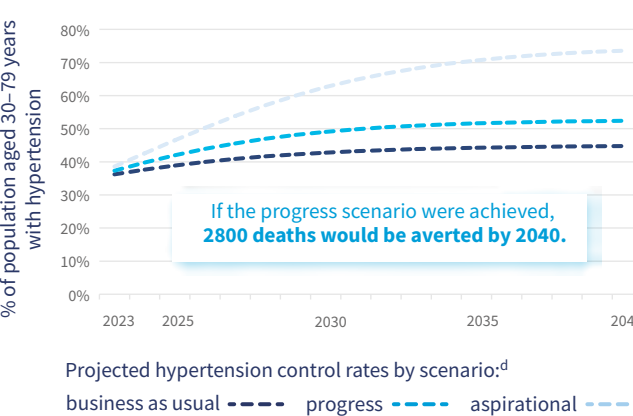
Of the 268 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	28	18	2019
Cardiovascular disease deaths	3600	1900	1700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	56	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	11	11	10	2019
Current tobacco use, adults aged 15+ years (%) ^f	21	38	3	2019
Obesity, adults aged 18+ years (%)	11	6	16	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	12	3	2019
Physical inactivity, adults aged 18+ years (%)	30	28	32	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Mexico

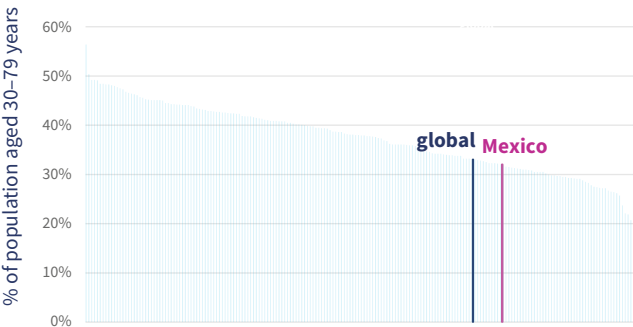
Hypertension profile

Total population (2019): 125 100 000

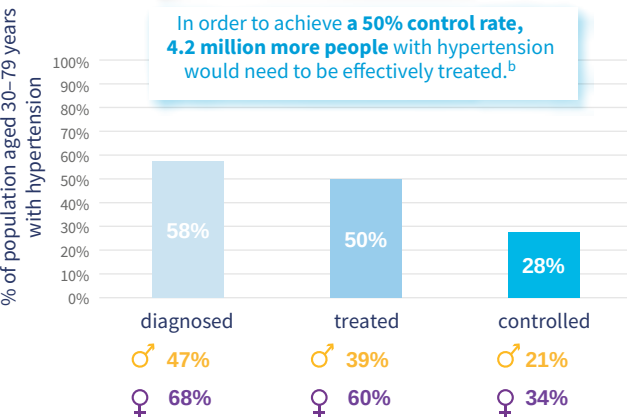
Total deaths (2019): 712 000[!]

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 32% ♂ 33% ♀ 31%

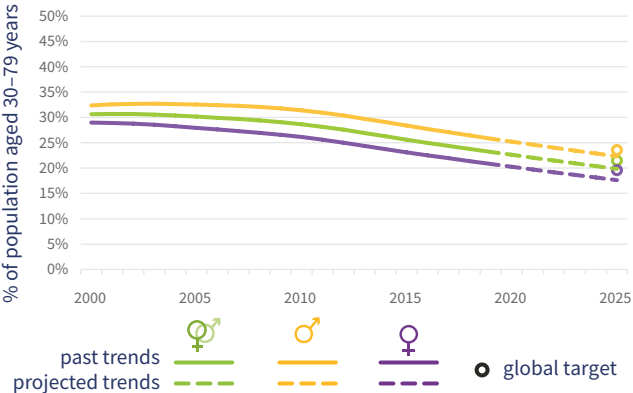
Prevalence of hypertension – global comparison (both sexes)^a



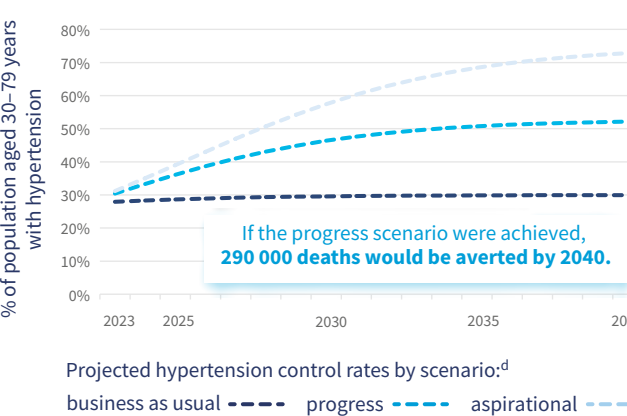
Of the 18.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	18	13	2019
Cardiovascular disease deaths	188 000	98 600	89 300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	52	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	9	8	2019
Current tobacco use, adults aged 15+ years (%)	13	20	6	2019
Obesity, adults aged 18+ years (%)	29	24	33	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	9	2	2019
Physical inactivity, adults aged 18+ years (%)	29	25	32	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Micronesia (Federated States of)

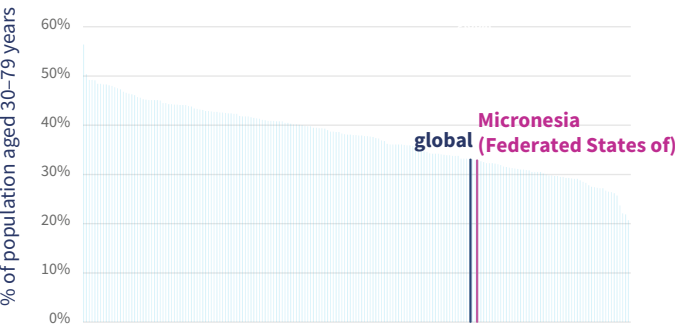
Hypertension profile

Total population (2019): 111 000

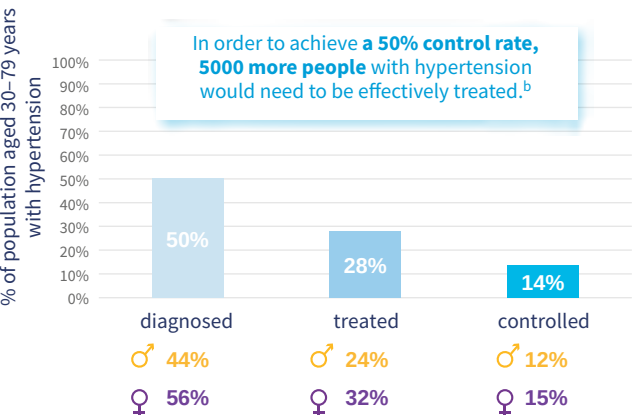
Total deaths (2019): 10301

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 33% ♂ 32% ♀ 34%

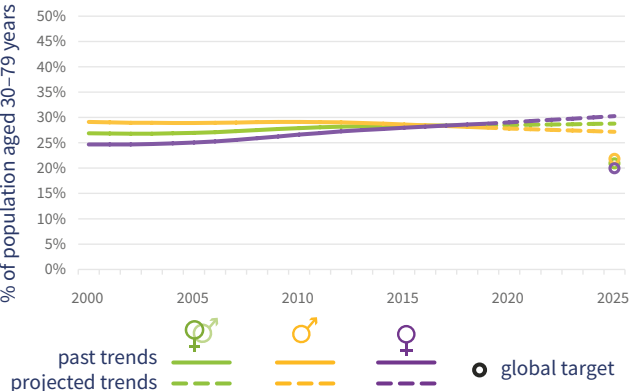
Prevalence of hypertension – global comparison (both sexes)^a



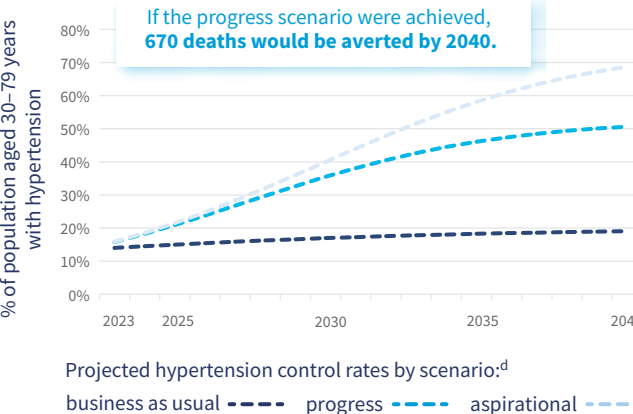
Of the 14 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	46	52	40	2019
Cardiovascular disease deaths	370	220	150	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	46	48	43	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	46	40	52	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	4	1	2019
Physical inactivity, adults aged 18+ years (%)	37	33	40	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Monaco

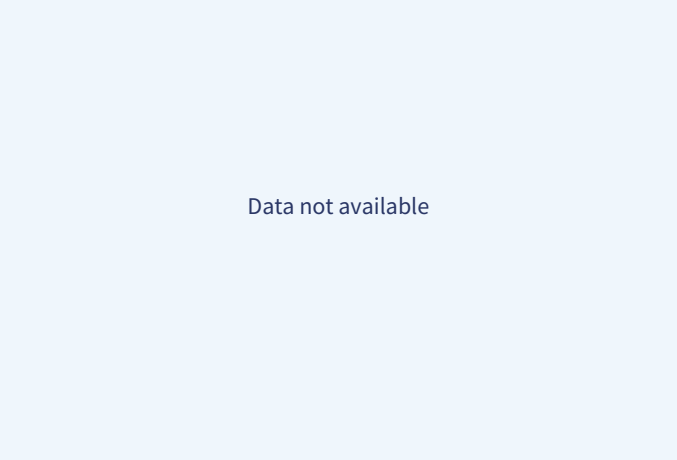
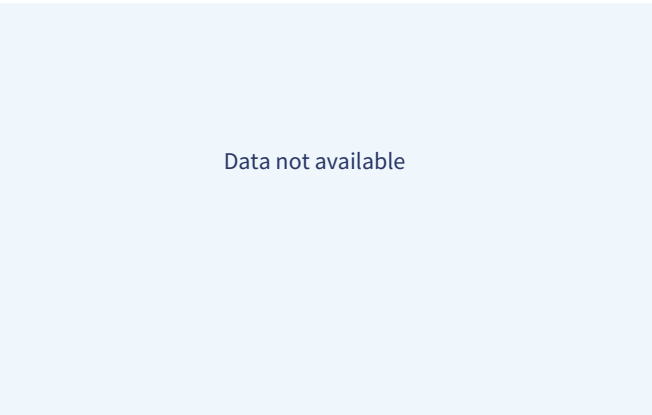
Hypertension profile

Total population (2019): 37 000

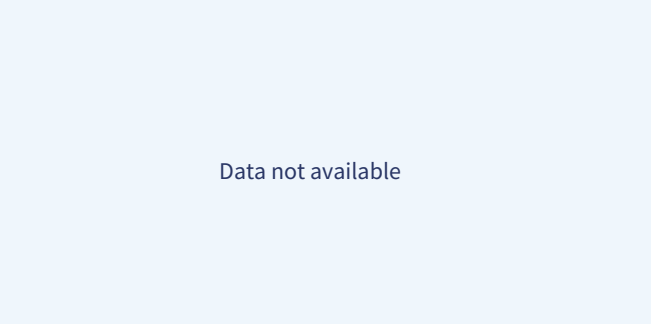
Total deaths (2019): no data

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019) ♀ no data ♂ no data ♀ no data

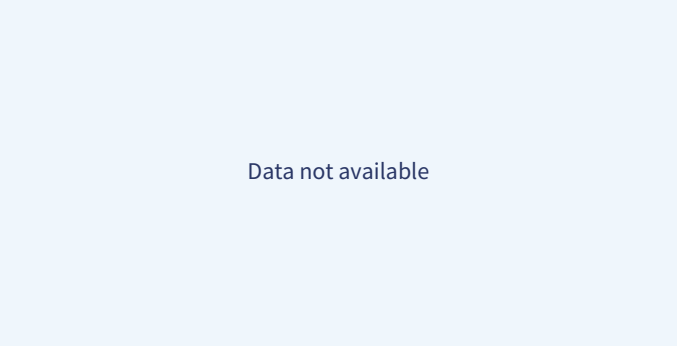
Prevalence of hypertension – global comparison (both sexes)



Trends in uncontrolled hypertension in adults aged 30–79 years



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	51	53	50	2019

Risk factors^a

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	no data	no data	no data	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	no data	no data	no data	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnote: a. Age-standardized estimates are presented for all indicators except salt intake.

Mongolia

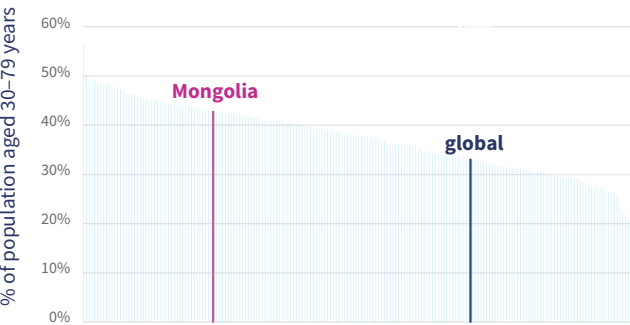
Hypertension profile

Total population (2019): 3 232 000

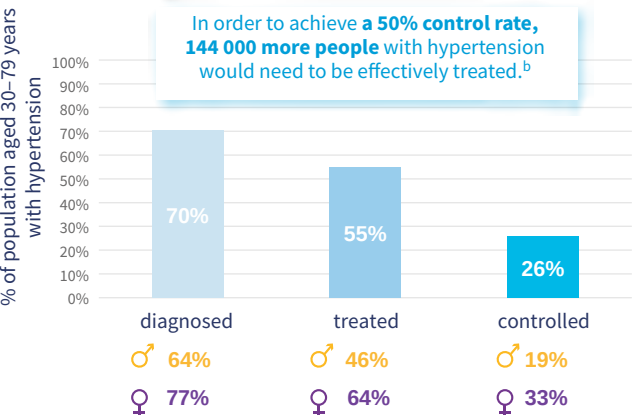
Total deaths (2019): 23 400

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 43% ♂ 45% ♀ 41%

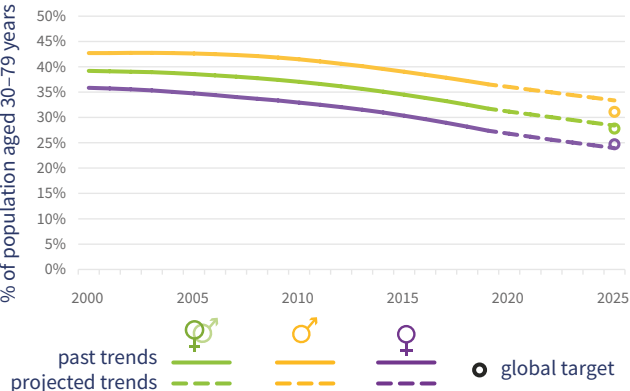
Prevalence of hypertension – global comparison (both sexes)^a



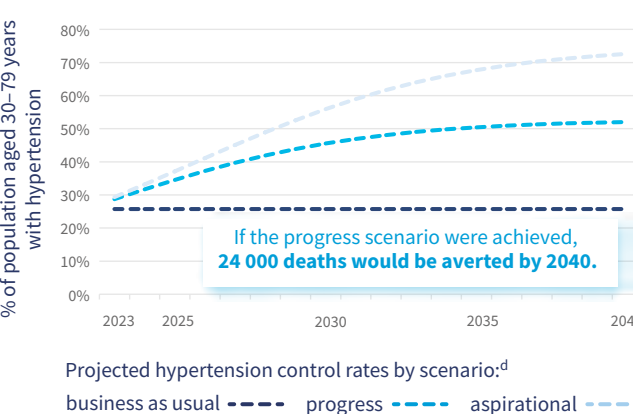
Of the 578 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	35	45	26	2019
Cardiovascular disease deaths	10 400	6000	4400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	62	64	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%)	30	52	7	2019
Obesity, adults aged 18+ years (%)	21	18	23	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	13	3	2019
Physical inactivity, adults aged 18+ years (%)	19	18	19	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Montenegro

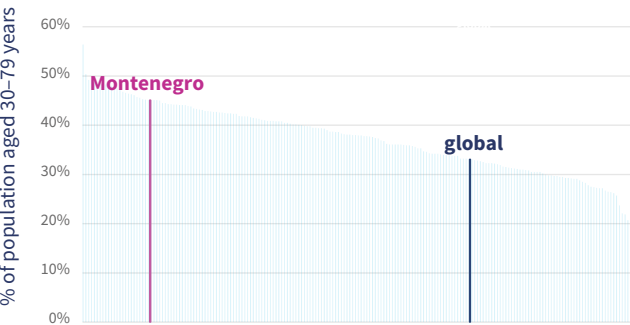
Hypertension profile

Total population (2019): 630 000

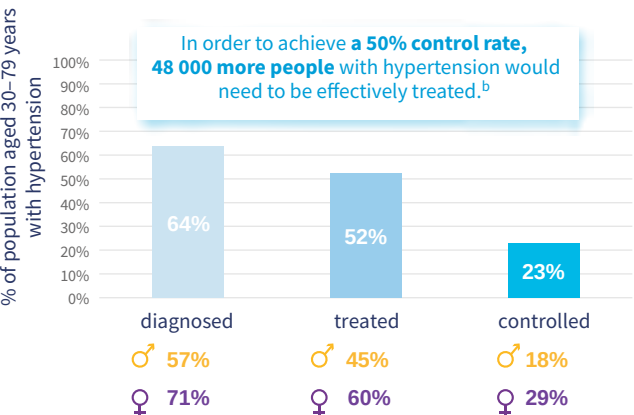
Total deaths (2019): 7280

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 45% ♂ 50% ♀ 41%

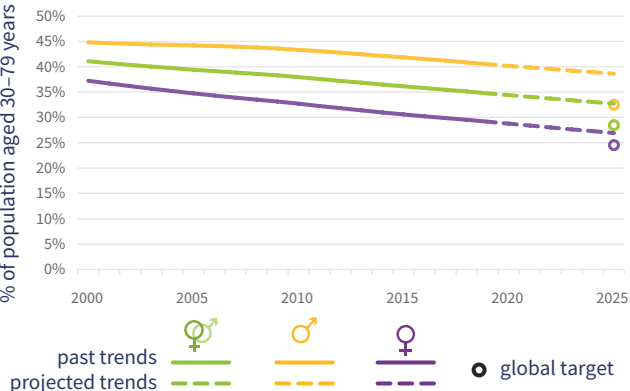
Prevalence of hypertension – global comparison (both sexes)^a



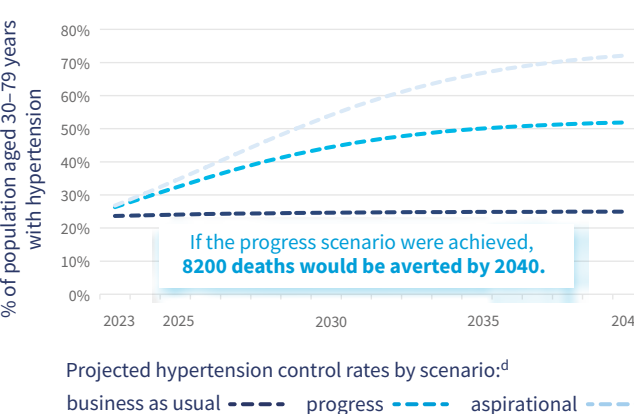
Of the 185 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	29	16	2019
Cardiovascular disease deaths	4600	2300	2300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	56	56	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%)	32	32	31	2019
Obesity, adults aged 18+ years (%)	23	23	23	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	17	4	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Morocco

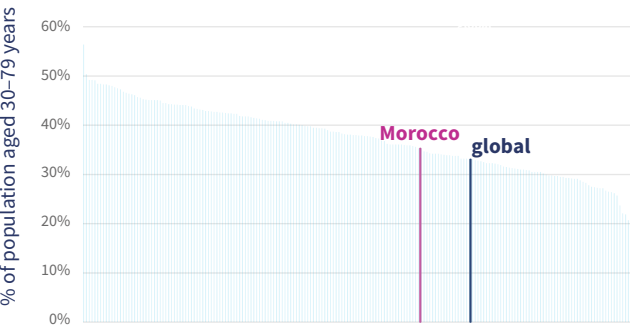
Hypertension profile

Total population (2019): 36 304 000

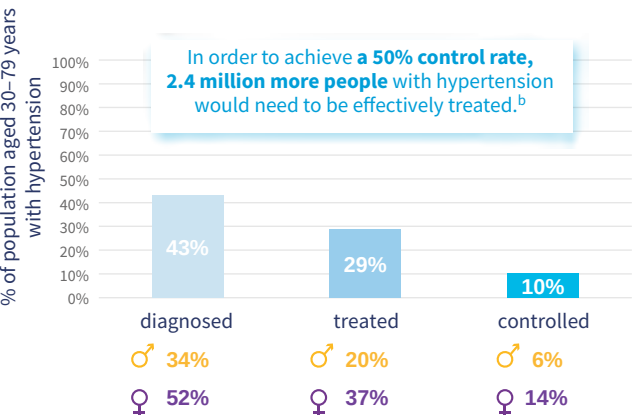
Total deaths (2019): 247 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 35% ♂ 35% ♀ 36%

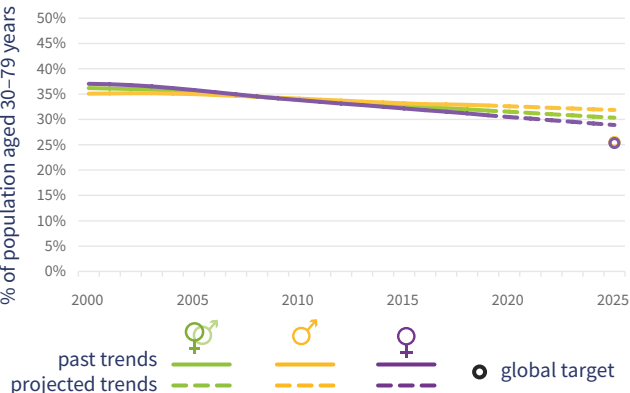
Prevalence of hypertension – global comparison (both sexes)^a



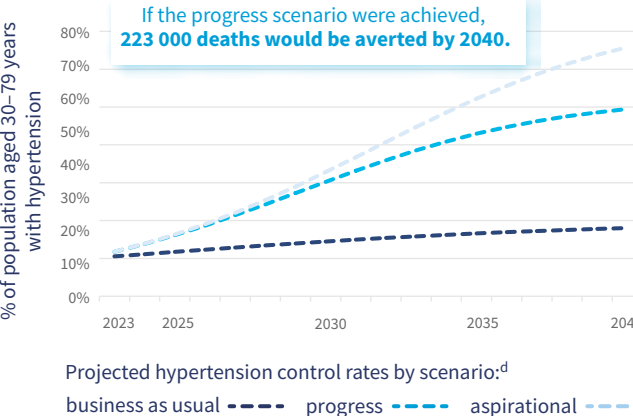
Of the **6.1 million** adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	26	22	2019
Cardiovascular disease deaths	127 000	60 700	65 800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	59	67	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	15	29	1	2019
Obesity, adults aged 18+ years (%)	26	19	32	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	1	0	2019
Physical inactivity, adults aged 18+ years (%)	26	21	31	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Mozambique

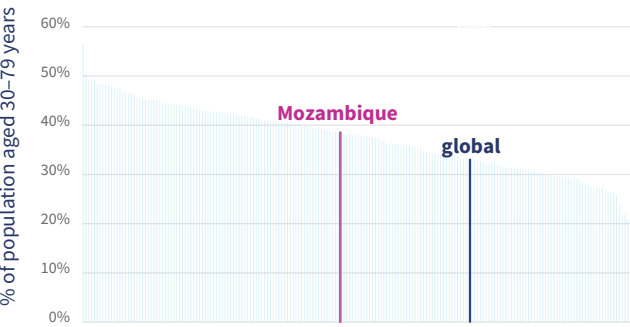
Hypertension profile

Total population (2019): 30 286 000

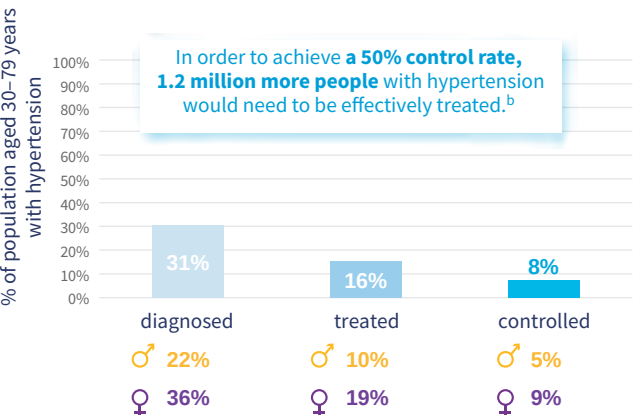
Total deaths (2019): 284 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 39% ♂ 34% ♀ 42%

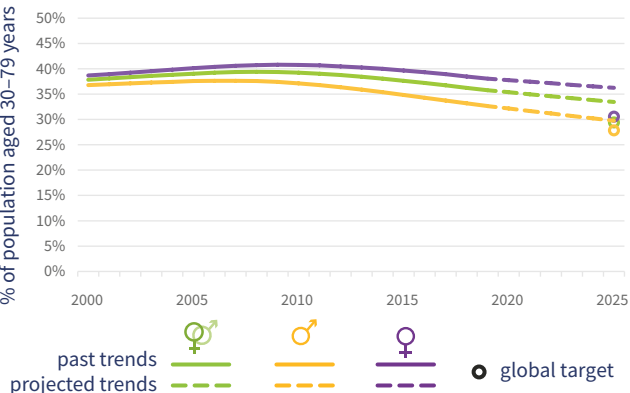
Prevalence of hypertension – global comparison (both sexes)^a



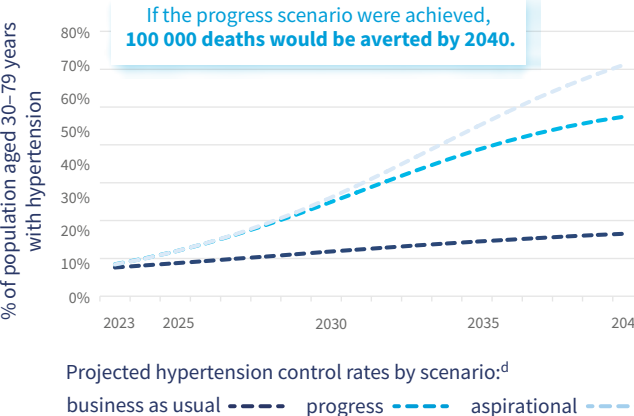
Of the **2.9 million** adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	31	39	24	2019
Cardiovascular disease deaths	45 400	24 000	21 400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	65	66	65	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	15	24	6	2019
Obesity, adults aged 18+ years (%)	7	3	11	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	1	2019
Physical inactivity, adults aged 18+ years (%)	6	5	6	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Myanmar

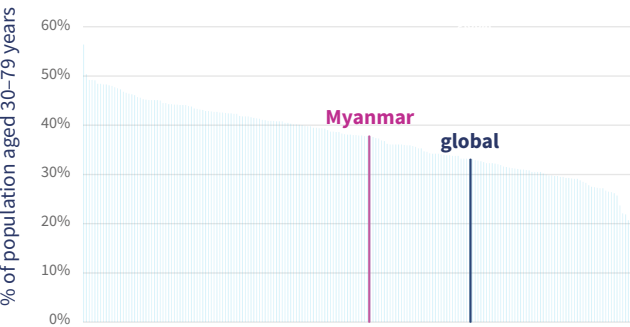
Hypertension profile

Total population (2019): 53 040 000

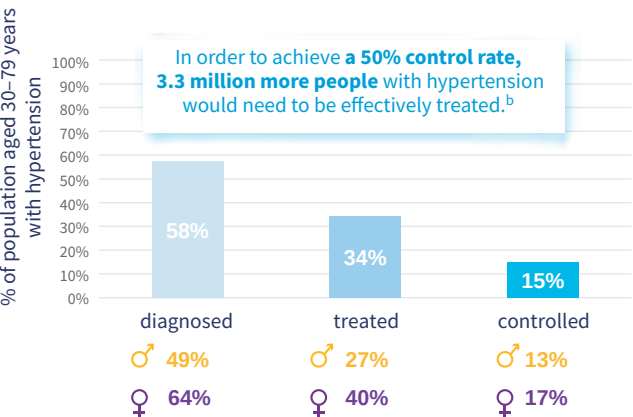
Total deaths (2019): 387 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 35% ♀ 40%

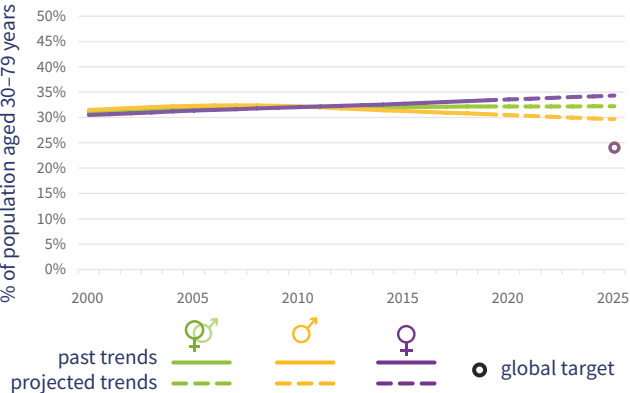
Prevalence of hypertension – global comparison (both sexes)^a



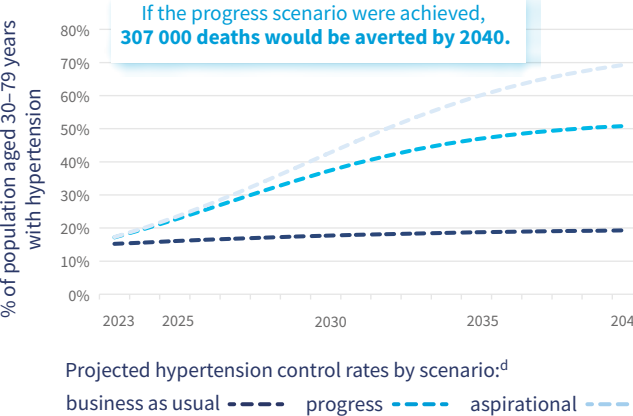
Of the 9.4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	31	20	2019
Cardiovascular disease deaths	119 000	58 500	60 700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	58	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	10	2019
Current tobacco use, adults aged 15+ years (%)	45	69	21	2019
Obesity, adults aged 18+ years (%)	6	4	7	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	4	1	2019
Physical inactivity, adults aged 18+ years (%)	11	8	13	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Namibia

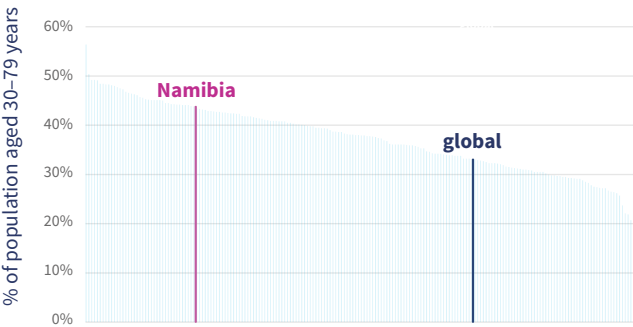
Hypertension profile

Total population (2019): 2 447 000

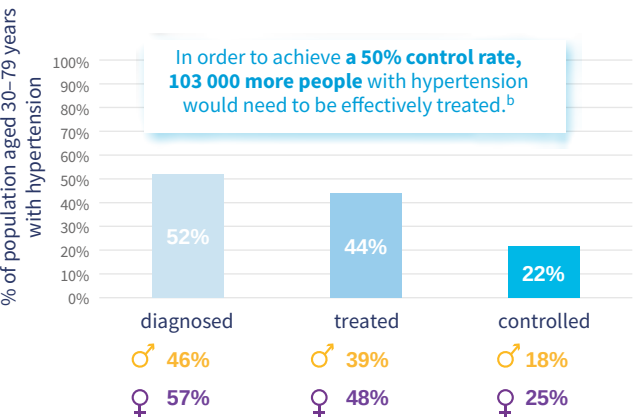
Total deaths (2019): 18 300

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 44% ♂ 43% ♀ 45%

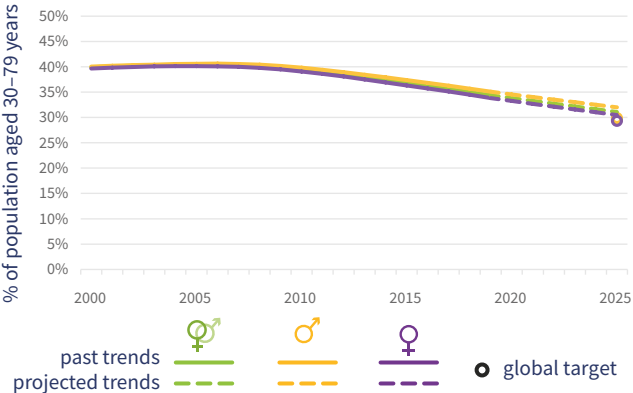
Prevalence of hypertension – global comparison (both sexes)^a



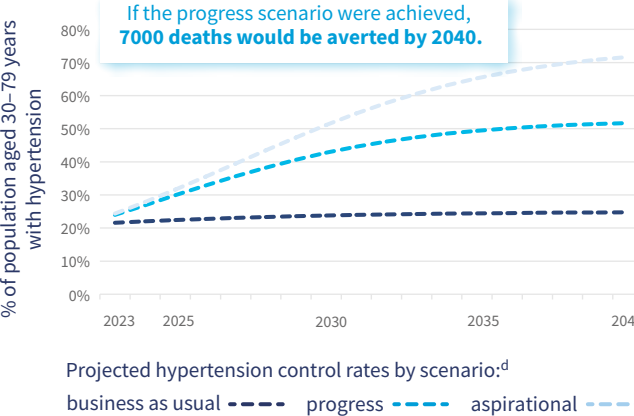
Of the 341 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	30	17	2019
Cardiovascular disease deaths	3300	1600	1700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	58	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	7	2019
Current tobacco use, adults aged 15+ years (%)	16	25	6	2019
Obesity, adults aged 18+ years (%)	17	8	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	10	2	2019
Physical inactivity, adults aged 18+ years (%)	33	29	37	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Nauru

Hypertension profile

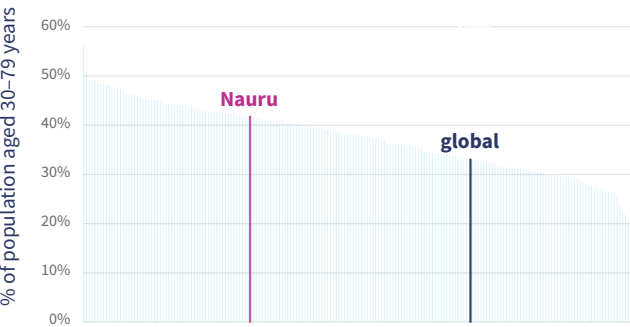
Total population (2019): 12 100

Total deaths (2019): no data

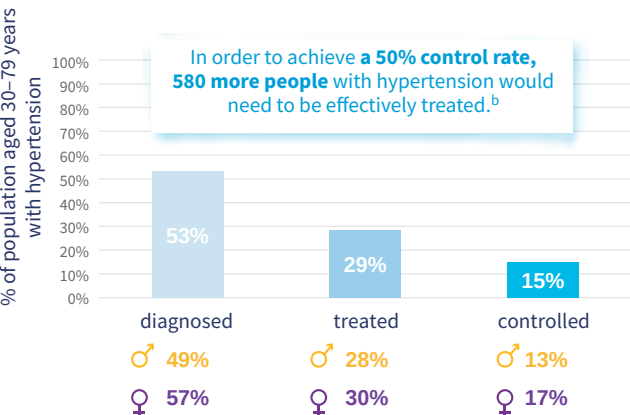
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

42% 44% 40%

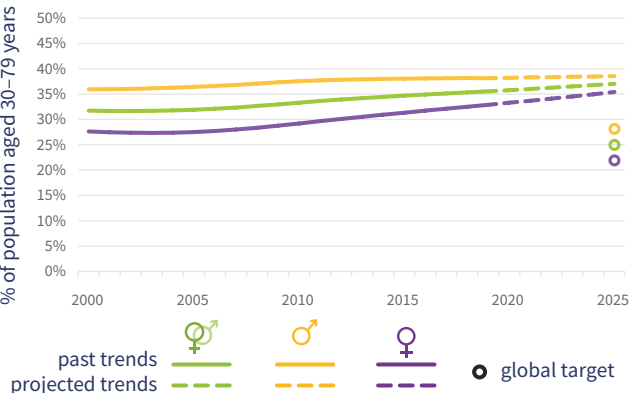
Prevalence of hypertension – global comparison (both sexes)^a



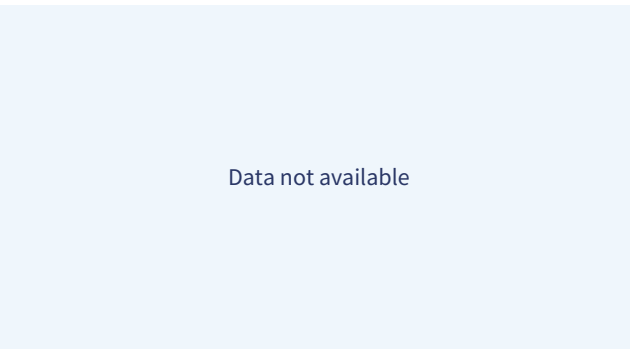
Of the 1600 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	65	62	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	49	48	50	2019
Obesity, adults aged 18+ years (%)	61	59	63	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	42	35	49	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake.

Nepal

Hypertension profile

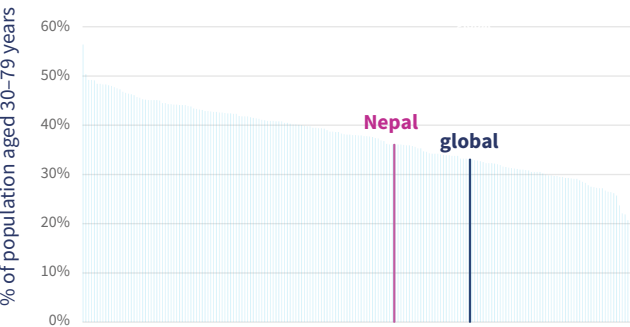
Total population (2019): 28 832 000

Total deaths (2019): 176 000

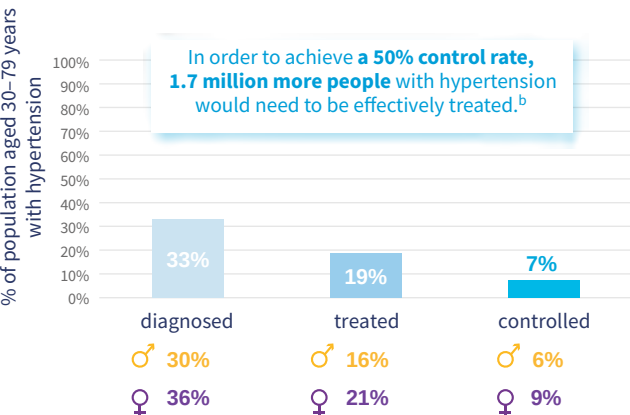
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

36% 40% 34%

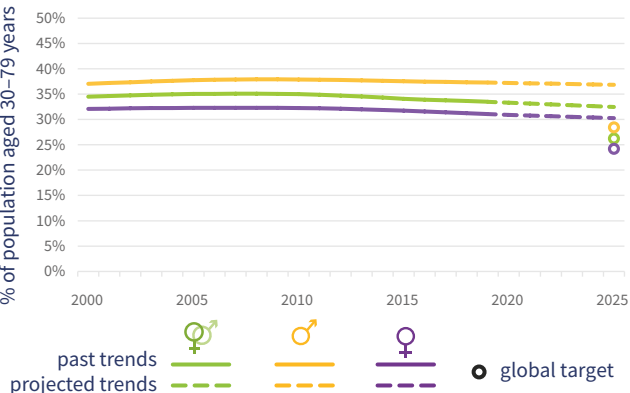
Prevalence of hypertension – global comparison (both sexes)^a



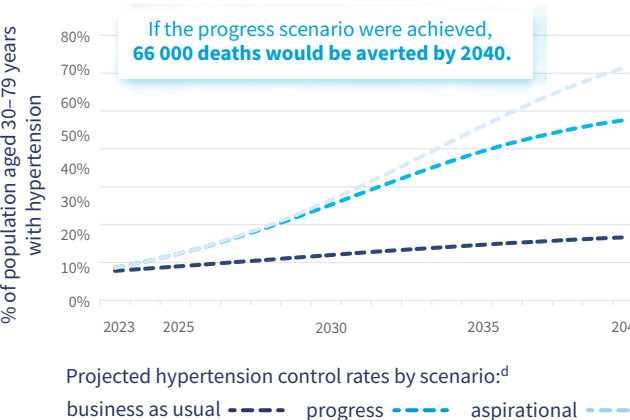
Of the 3.9 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	24	19	2019
Cardiovascular disease deaths	39 000	22 800	16 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	46	50	39	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	9	8	2019
Current tobacco use, adults aged 15+ years (%)	32	49	14	2019
Obesity, adults aged 18+ years (%)	4	3	5	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	2	0	2019
Physical inactivity, adults aged 18+ years (%)	13	12	15	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Netherlands (Kingdom of the)

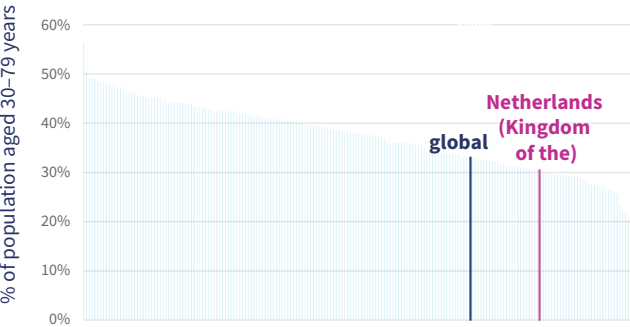
Hypertension profile

Total population (2019): 17 363 000

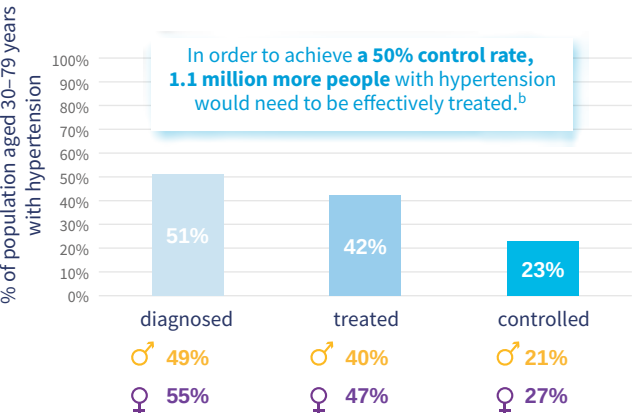
Total deaths (2019): 153 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 36% ♀ 25%

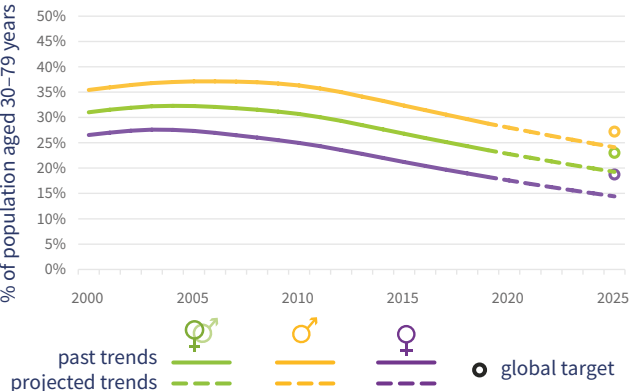
Prevalence of hypertension – global comparison (both sexes)^a



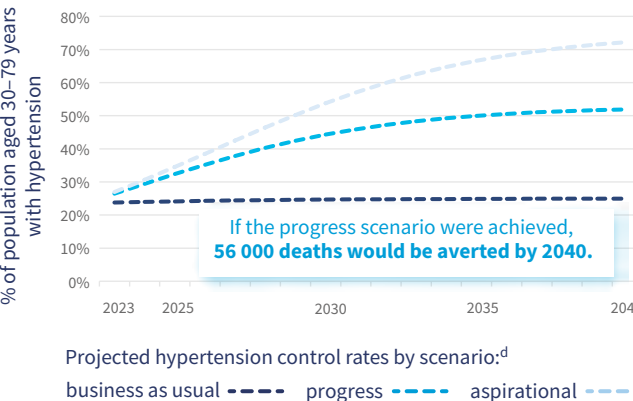
Of the 4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	12	9	2019
Cardiovascular disease deaths	36 400	17 600	18 900	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	48	51	45	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	23	25	20	2019
Obesity, adults aged 18+ years (%)	20	21	20	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	15	4	2019
Physical inactivity, adults aged 18+ years (%)	27	25	29	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

New Zealand

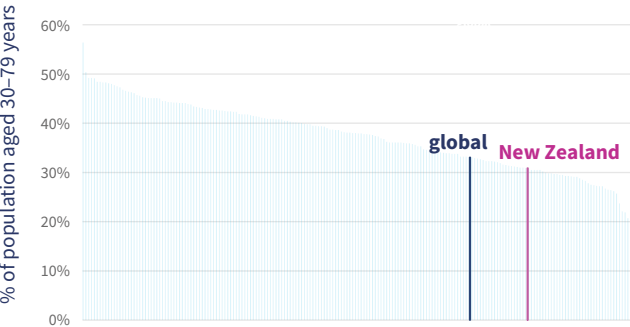
Hypertension profile

Total population (2019): 4 959 000

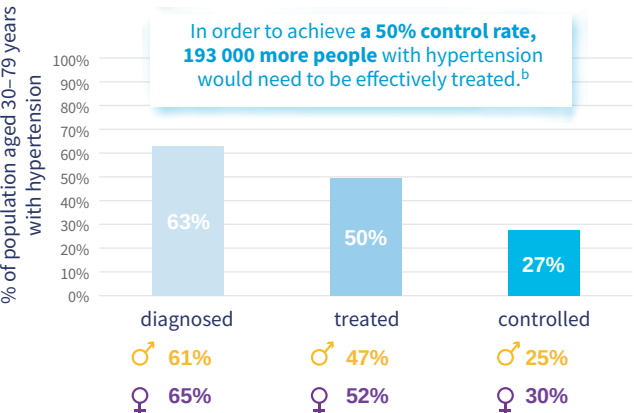
Total deaths (2019): 33 300

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 34% ♀ 28%

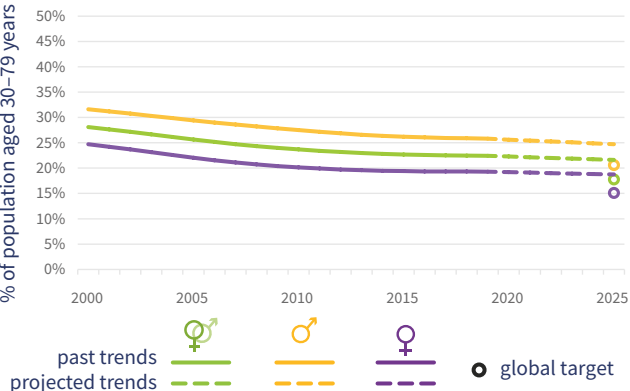
Prevalence of hypertension – global comparison (both sexes)^a



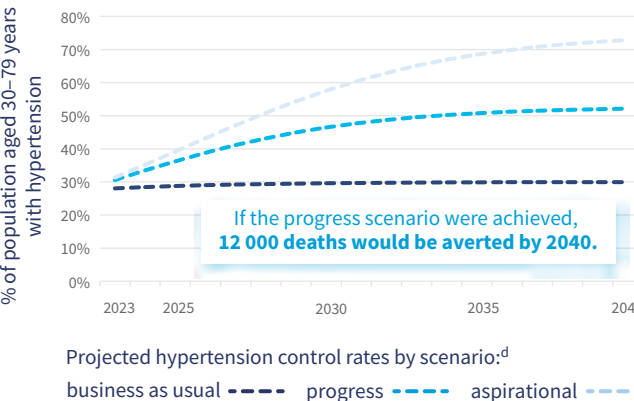
Of the 968 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	12	9	2019
Cardiovascular disease deaths	9700	5000	4800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	45	46	45	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	14	16	13	2019
Obesity, adults aged 18+ years (%)	31	30	31	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	15	5	2019
Physical inactivity, adults aged 18+ years (%)	42	39	45	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Nicaragua

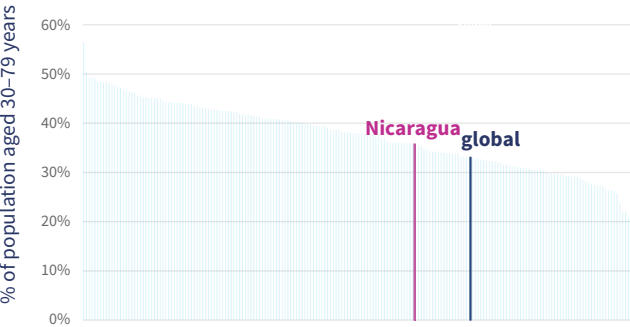
Hypertension profile

Total population (2019): 6 664 000

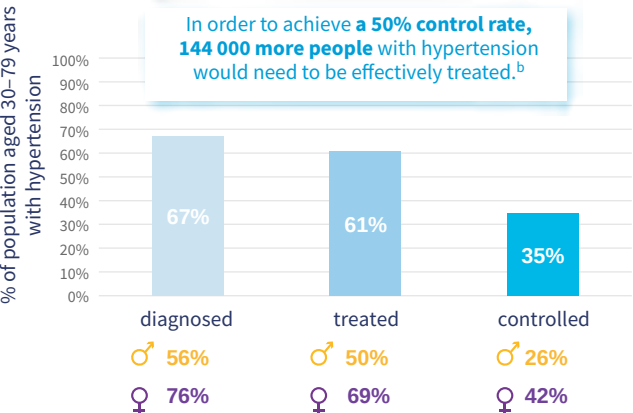
Total deaths (2019): 32 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 36% ♂ 35% ♀ 37%

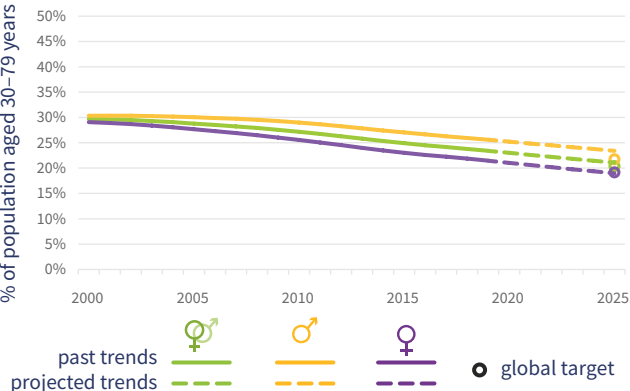
Prevalence of hypertension – global comparison (both sexes)^a



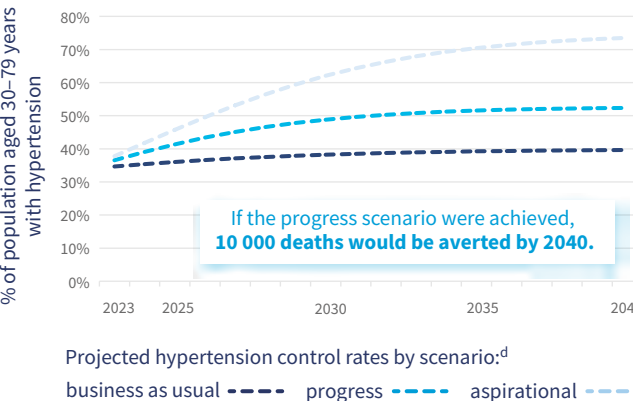
Of the 904 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	15	18	13	2019
Cardiovascular disease deaths	9800	4800	5000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	56	54	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	9	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	24	18	29	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	7	2	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Niger

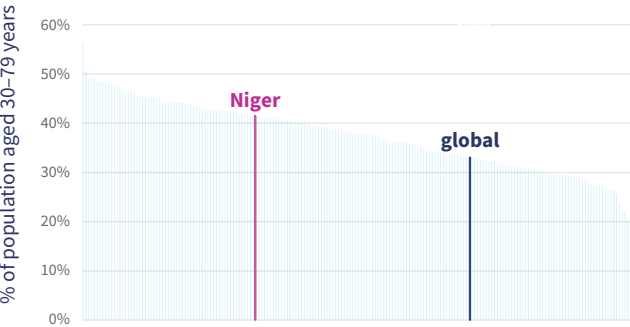
Hypertension profile

Total population (2019): 23 443 000

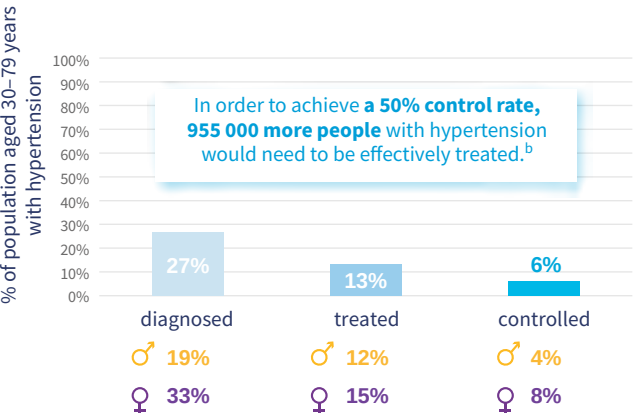
Total deaths (2019): 183 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 42% ♂ 40% ♀ 43%

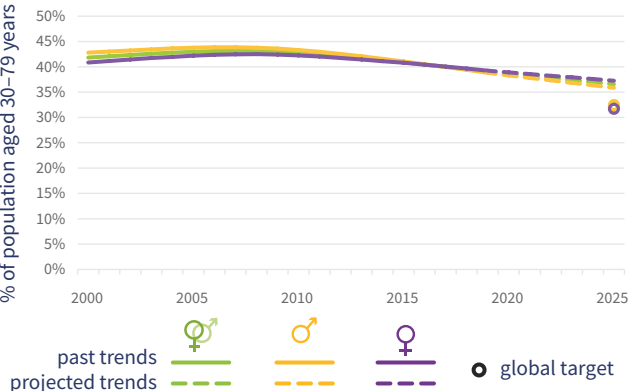
Prevalence of hypertension – global comparison (both sexes)^a



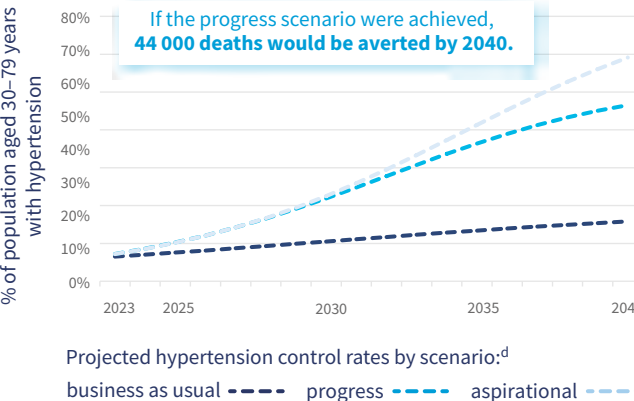
Of the 2.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	21	21	2019
Cardiovascular disease deaths	20 900	10 100	10 800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	53	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	8	14	1	2019
Obesity, adults aged 18+ years (%)	6	3	9	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	22	20	25	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Nigeria

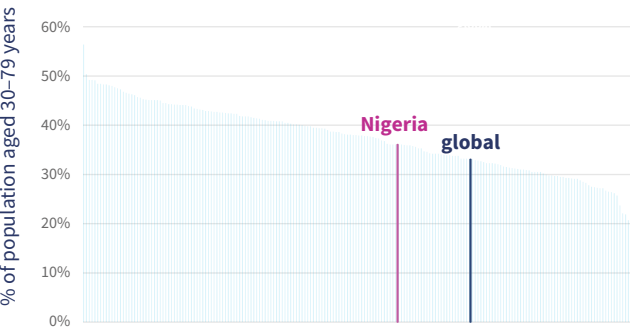
Hypertension profile

Total population (2019): 203 300 000

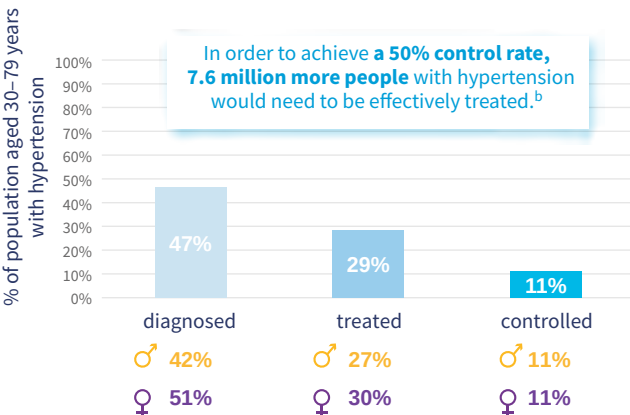
Total deaths (2019): 1 650 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 36% ♂ 33% ♀ 39%

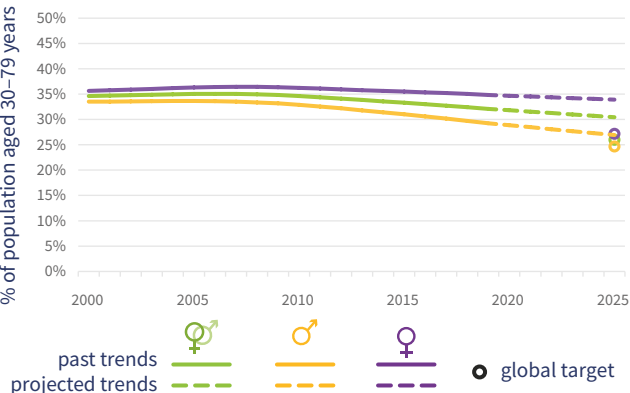
Prevalence of hypertension – global comparison (both sexes)^a



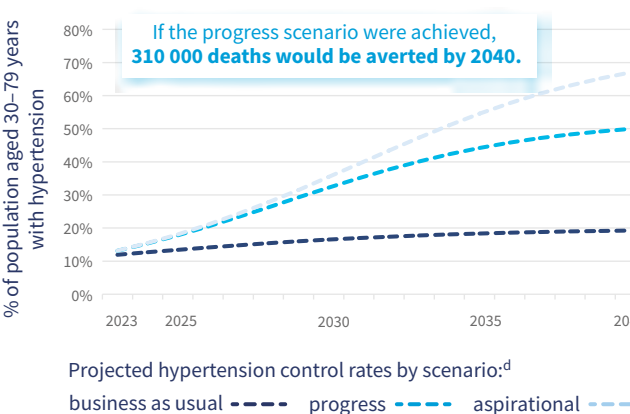
Of the 19.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	17	16	18	2019
Cardiovascular disease deaths	147 000	69 100	77 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	62	61	63	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	6	6	2019
Current tobacco use, adults aged 15+ years (%)	4	7	1	2019
Obesity, adults aged 18+ years (%)	9	5	13	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	7	1	2019
Physical inactivity, adults aged 18+ years (%)	27	25	30	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✗
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Niue

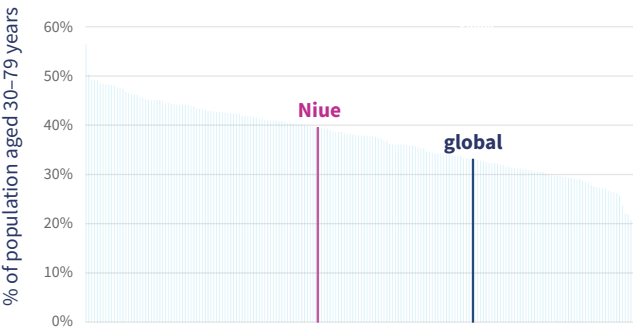
Hypertension profile

Total population (2019): 1900

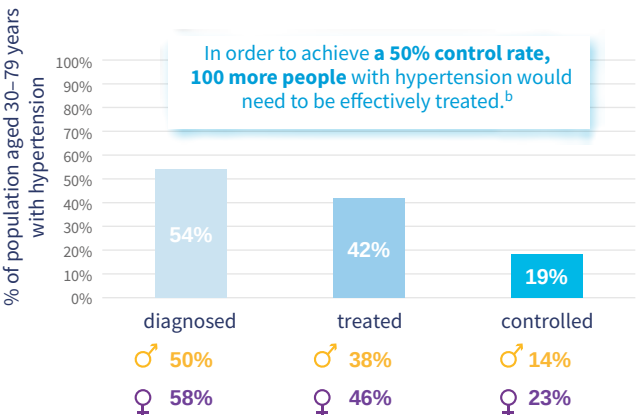
Total deaths (2019): no data

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 40% ♂ 39% ♀ 40%

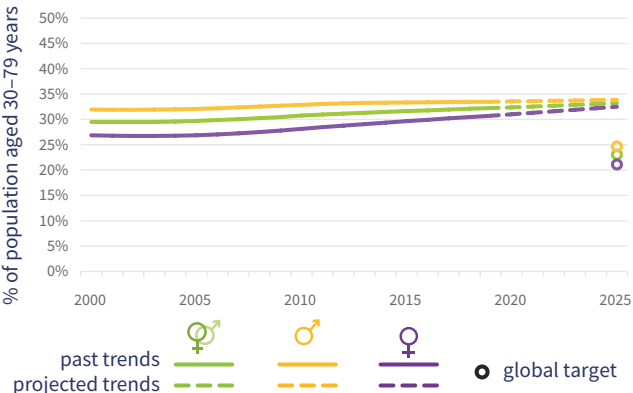
Prevalence of hypertension – global comparison (both sexes)^a



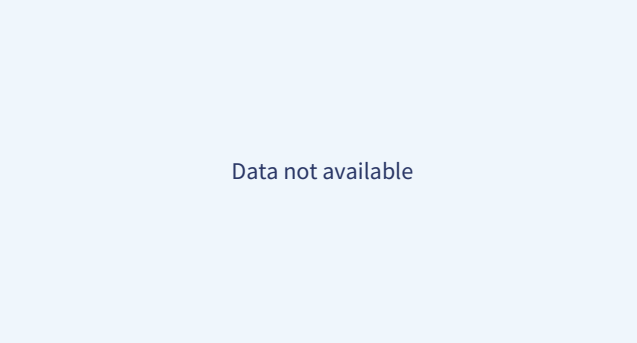
Of the 340 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	59	59	59	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	8	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	50	45	55	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	16	3	2019
Physical inactivity, adults aged 18+ years (%)	7	8	6	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake.

North Macedonia

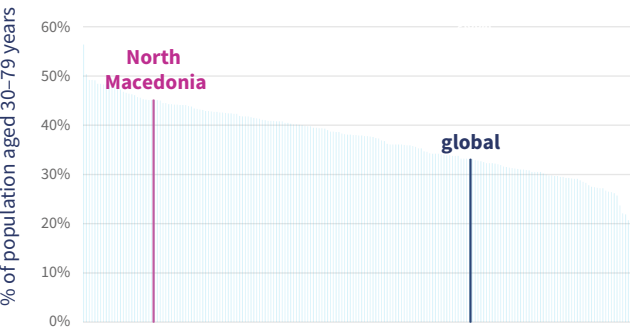
Hypertension profile

Total population (2019): 2 114 000

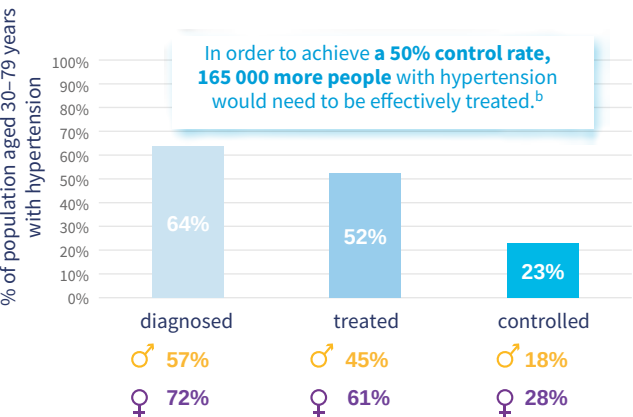
Total deaths (2019): 25 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 45% ♂ 49% ♀ 41%

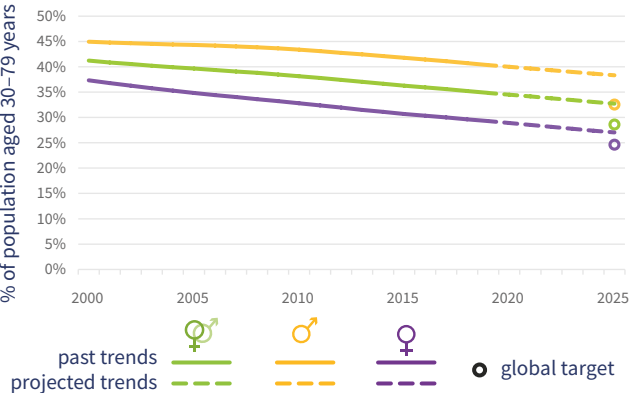
Prevalence of hypertension – global comparison (both sexes)^a



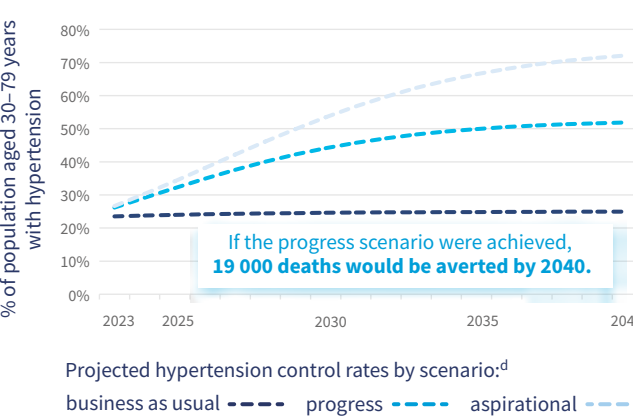
Of the 625 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	28	17	2019
Cardiovascular disease deaths	15 500	7400	8100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	59	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	10	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	22	23	22	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	8	2	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Norway

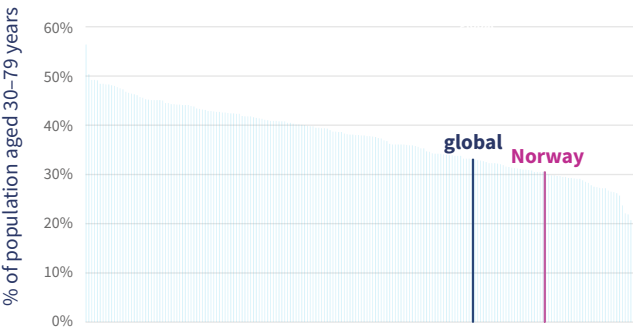
Hypertension profile

Total population (2019): 5 348 000

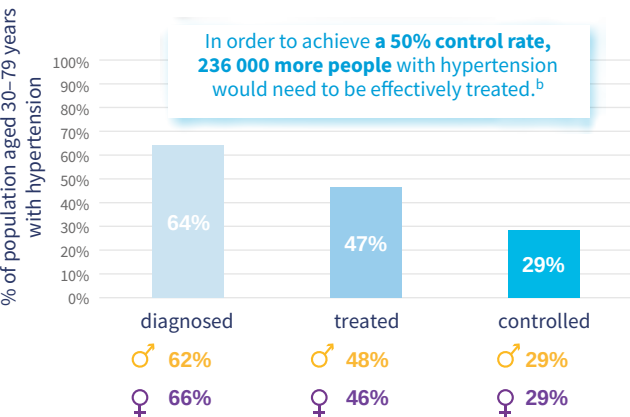
Total deaths (2019): 40 900

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 31% ♂ 35% ♀ 26%

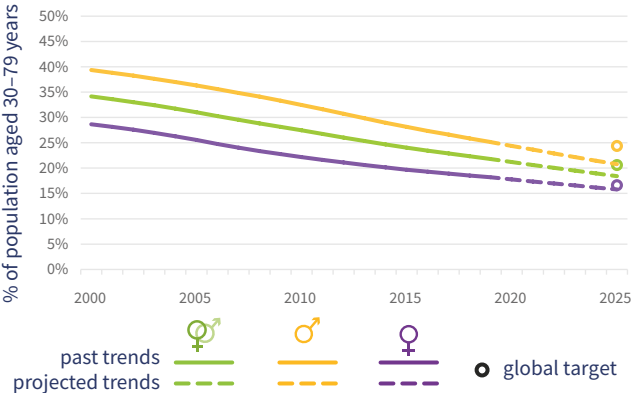
Prevalence of hypertension – global comparison (both sexes)^a



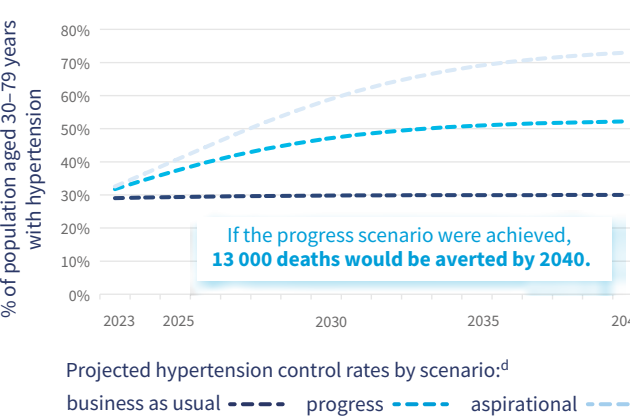
Of the 1.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	9	10	7	2019
Cardiovascular disease deaths	10 300	4900	5400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	49	50	47	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	6	2019
Current tobacco use, adults aged 15+ years (%) ^f	17	18	16	2019
Obesity, adults aged 18+ years (%)	23	24	23	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	10	3	2019
Physical inactivity, adults aged 18+ years (%)	32	30	34	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Oman

Hypertension profile

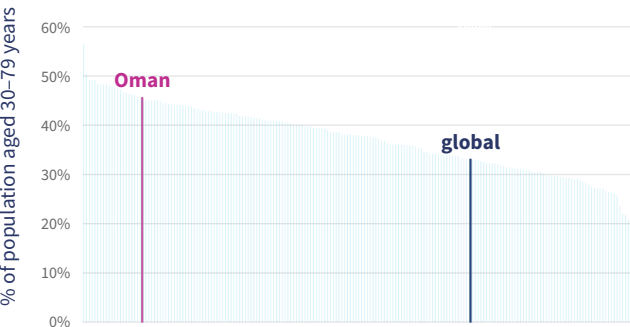
Total population (2019): 4 603 000

Total deaths (2019): 16 300

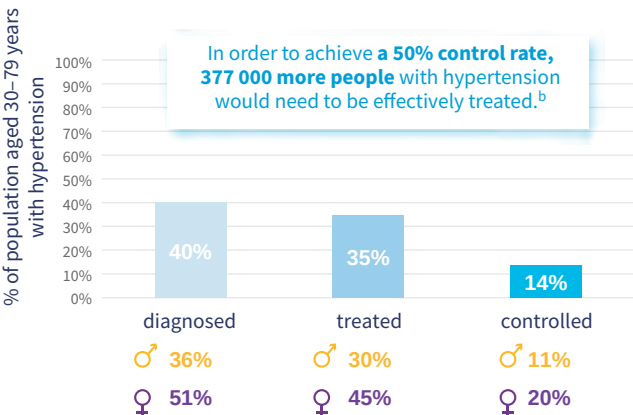
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

46% 48% 39%

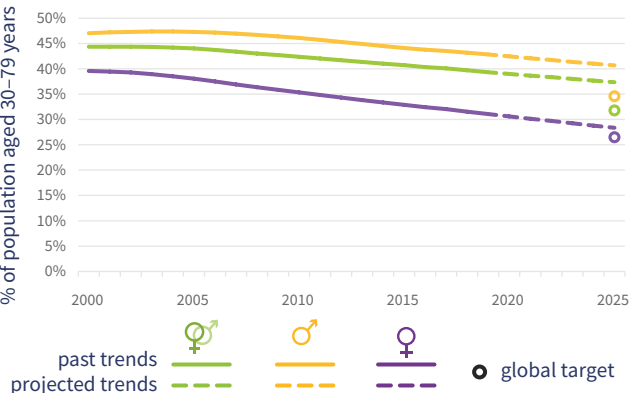
Prevalence of hypertension – global comparison (both sexes)^a



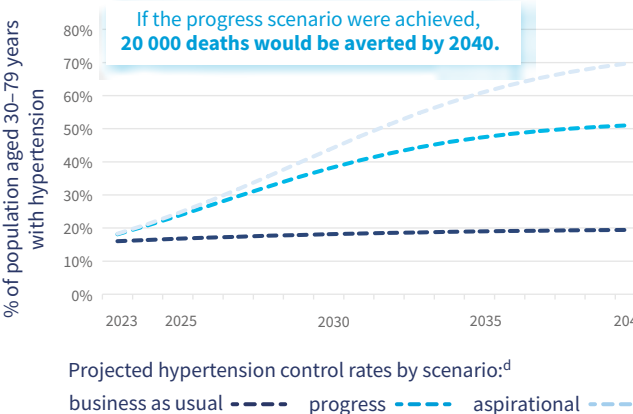
Of the 920 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	22	20	2019
Cardiovascular disease deaths	7800	4600	3300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	56	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	5	2019
Current tobacco use, adults aged 15+ years (%)	8	15	0	2019
Obesity, adults aged 18+ years (%)	27	23	34	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	1	0	2019
Physical inactivity, adults aged 18+ years (%)	33	30	40	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Pakistan

Hypertension profile

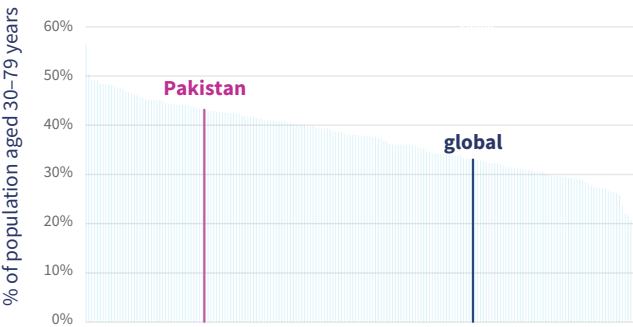
Total population (2019): 223 300 000

Total deaths (2019): 1 628 000

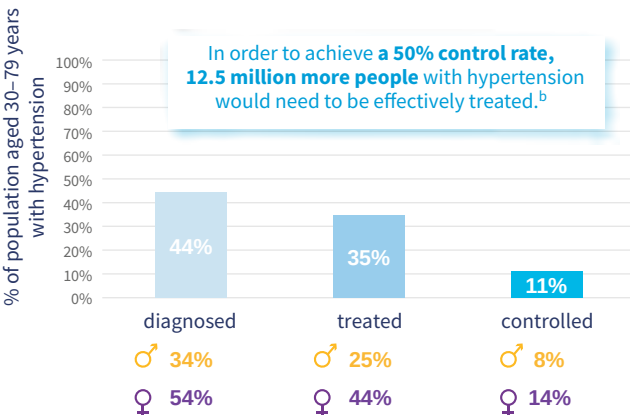
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

43% 42% 45%

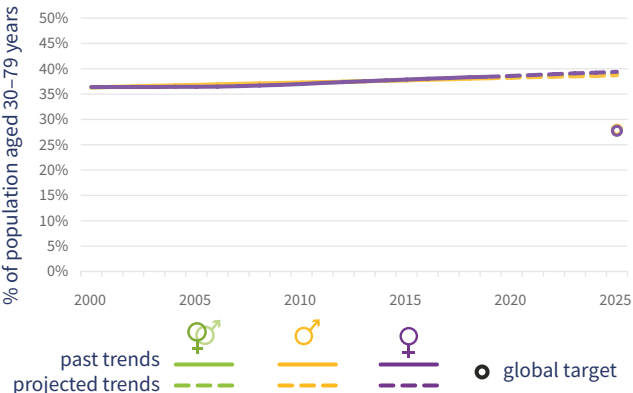
Prevalence of hypertension – global comparison (both sexes)^a



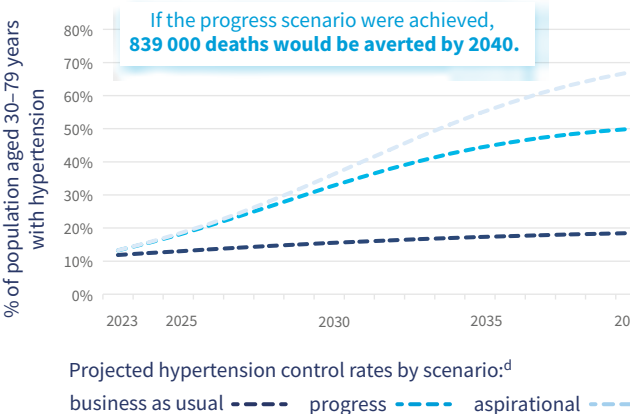
Of the 32.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	29	32	27	2019
Cardiovascular disease deaths	450 000	245 000	205 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	57	58	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	9	8	2019
Current tobacco use, adults aged 15+ years (%)	21	34	8	2019
Obesity, adults aged 18+ years (%)	9	6	11	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	34	24	43	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Palau

Hypertension profile

Total population (2019): 17 900

Total deaths (2019): no data!

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a



44%

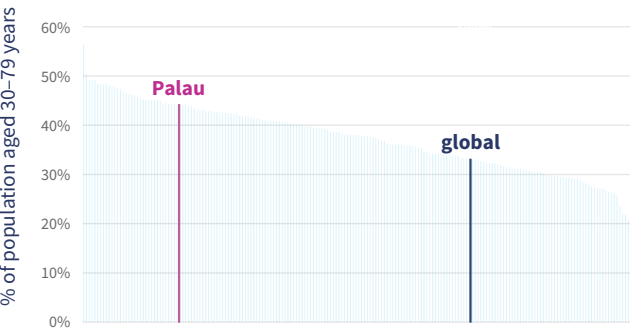


45%

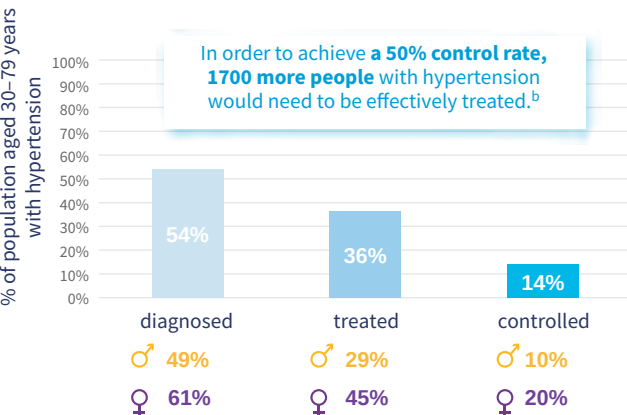


43%

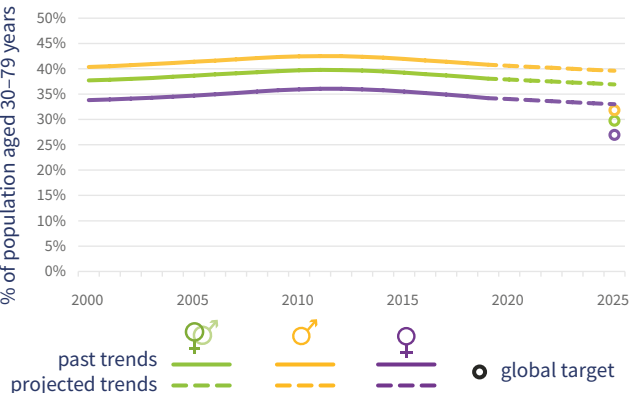
Prevalence of hypertension – global comparison (both sexes)^a



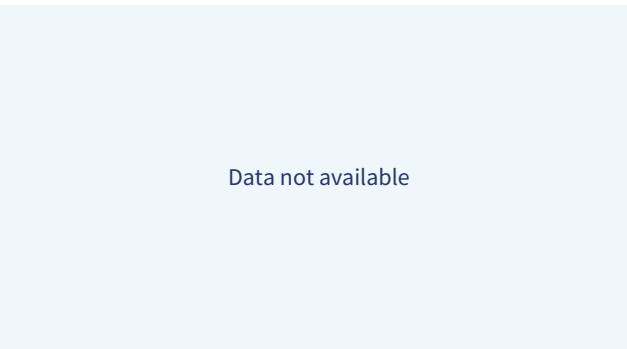
Of the 4900 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	55	53	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	7	2019
Current tobacco use, adults aged 15+ years (%) ^e	18	28	8	2019
Obesity, adults aged 18+ years (%)	55	52	59	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	no data	no data	no data	2019
Physical inactivity, adults aged 18+ years (%)	41	28	53	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake. e. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Panama

Hypertension profile

Total population (2019): 4 233 000

Total deaths (2019): 20 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a



36%

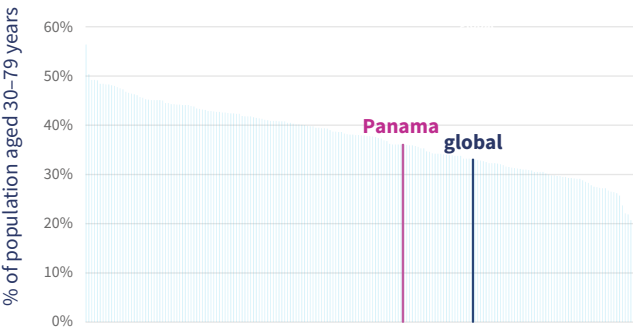


37%

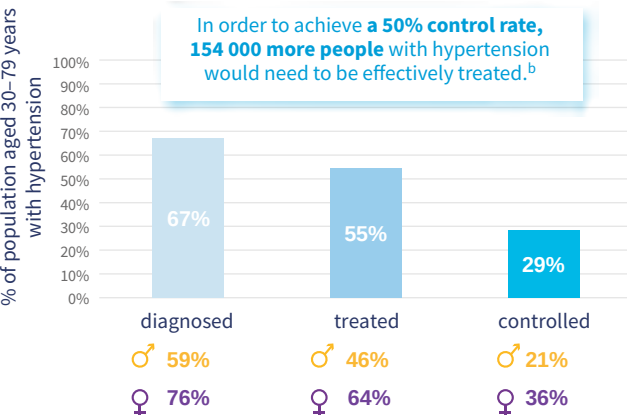


35%

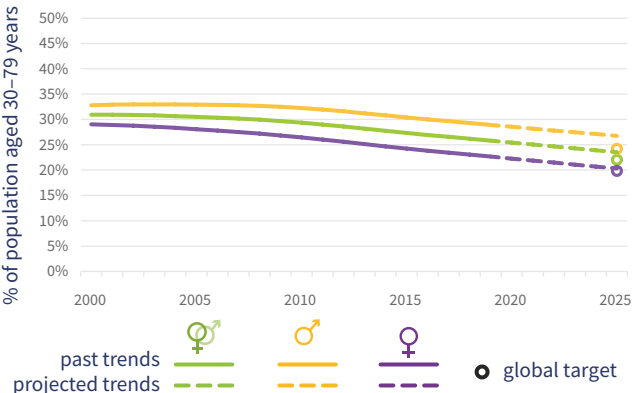
Prevalence of hypertension – global comparison (both sexes)^a



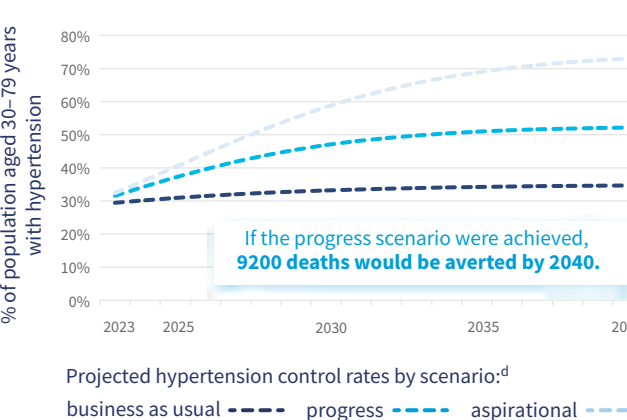
Of the 718 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	13	9	2019
Cardiovascular disease deaths	5800	3200	2600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	53	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	9	2019
Current tobacco use, adults aged 15+ years (%)	5	8	2	2019
Obesity, adults aged 18+ years (%)	23	18	28	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	11	3	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Papua New Guinea

Hypertension profile

Total population (2019): 9 542 000

Total deaths (2019): 60 600¹

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a



28%

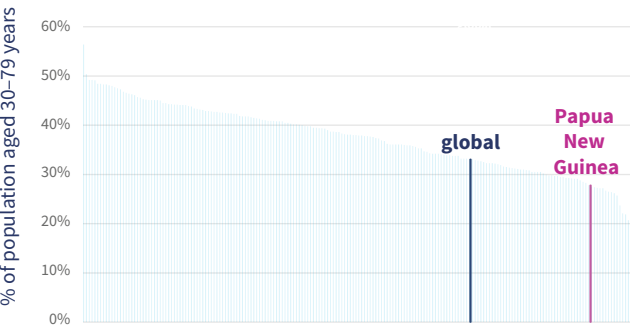


25%

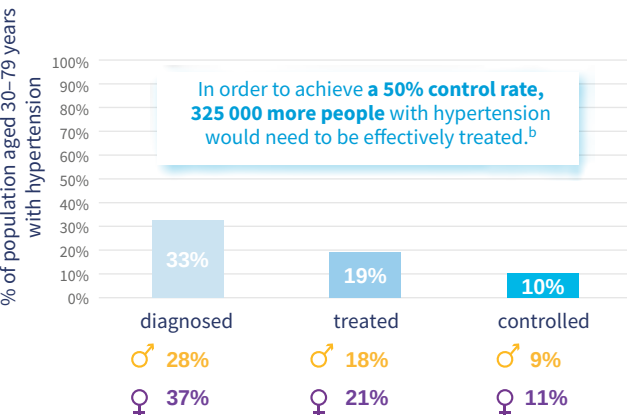


30%

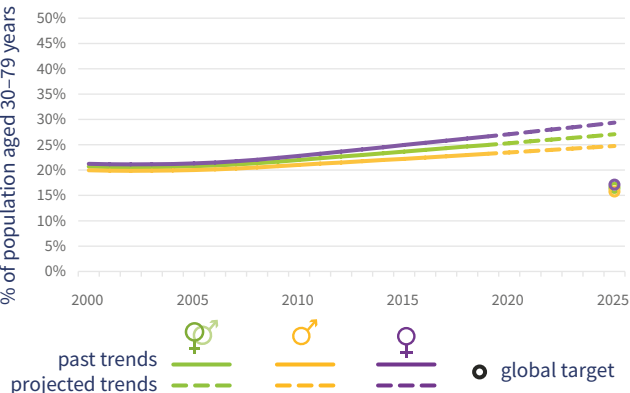
Prevalence of hypertension – global comparison (both sexes)^a



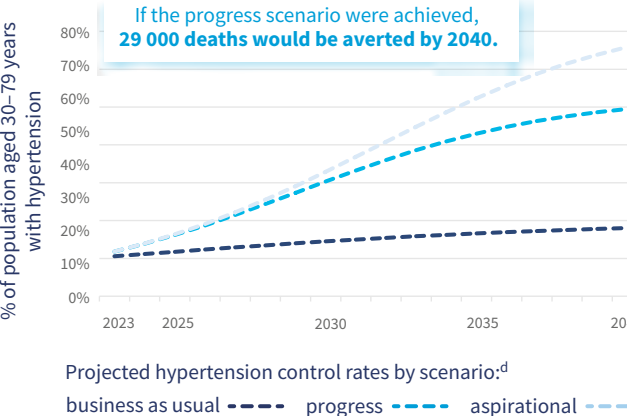
Of the 812 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	36	40	32	2019
Cardiovascular disease deaths	14 800	8800	6000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	41	41	41	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	40	54	26	2019
Obesity, adults aged 18+ years (%)	21	17	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	1	2019
Physical inactivity, adults aged 18+ years (%)	15	11	18	2016

National response

Targets

National target for blood pressure



Treatment

Guidelines for management of hypertension



National target for salt consumption



Surveillance

Conducted recent, national survey measuring raised blood pressure/hypertension



Conducted recent, national survey on salt/sodium intake



Functioning system for generating reliable cause-specific mortality data on a routine basis



Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Paraguay

Hypertension profile

Total population (2019): 6 530 000

Total deaths (2019): 34 400¹

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a



56%

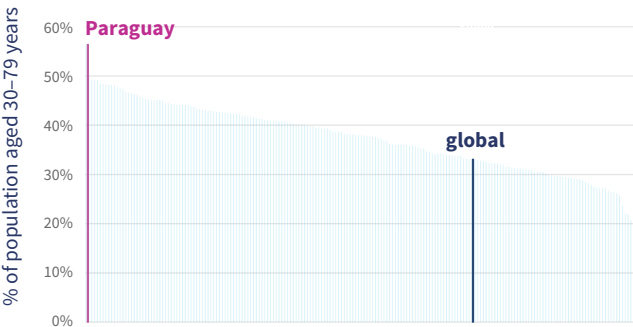


62%

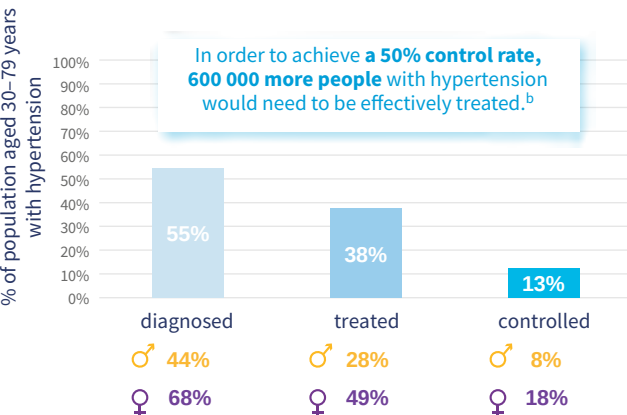


51%

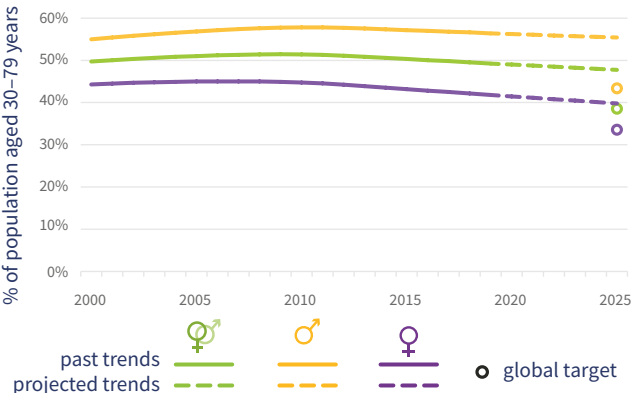
Prevalence of hypertension – global comparison (both sexes)^a



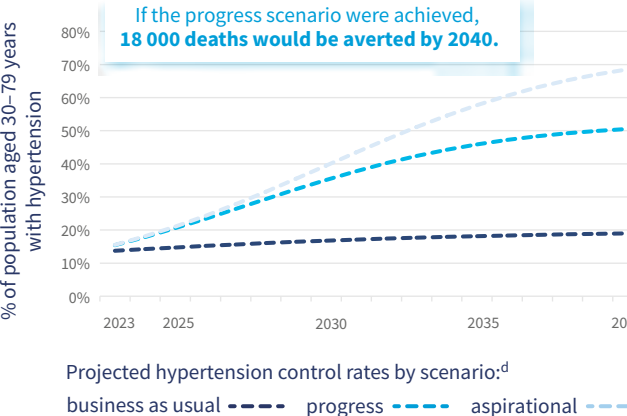
Of the 1.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	19	13	2019
Cardiovascular disease deaths	9200	5200	4000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	52	58	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%) ^f	12	20	5	2019
Obesity, adults aged 18+ years (%)	20	17	23	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	9	2	2019
Physical inactivity, adults aged 18+ years (%)	37	38	37	2016

National response

Targets

National target for blood pressure



National target for salt consumption



Treatment

Guidelines for management of hypertension



Surveillance

Conducted recent, national survey measuring raised blood pressure/hypertension



Conducted recent, national survey on salt/sodium intake



Functioning system for generating reliable cause-specific mortality data on a routine basis



Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Peru

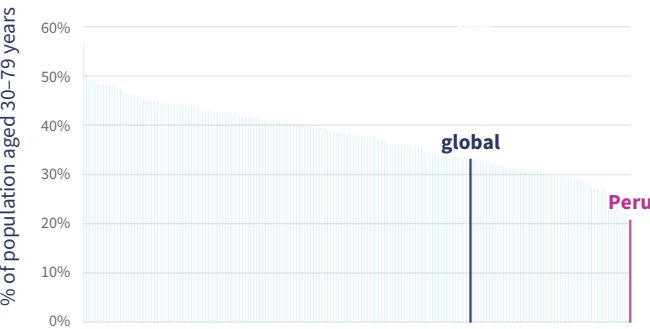
Hypertension profile

Total population (2019): 32 825 000

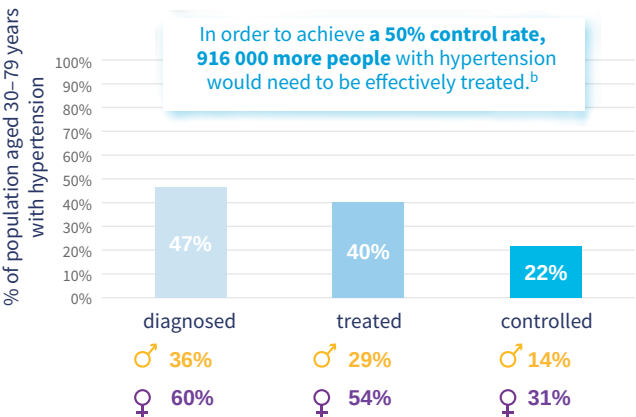
Total deaths (2019): 142 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 21% ♂ 23% ♀ 18%

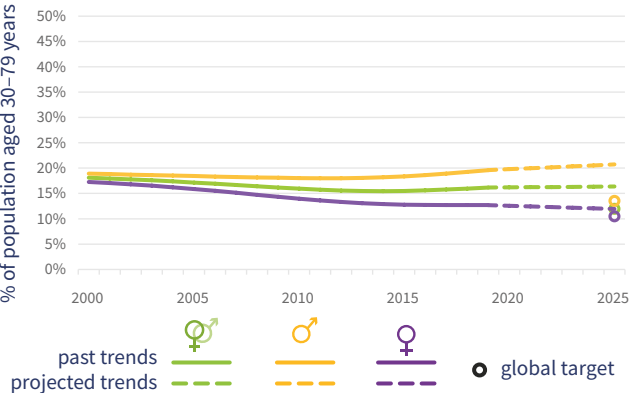
Prevalence of hypertension – global comparison (both sexes)^a



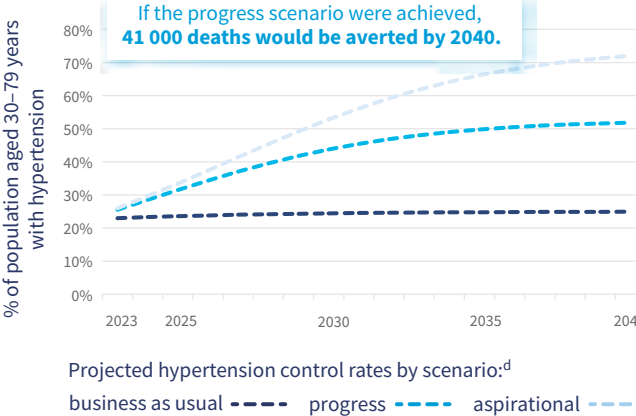
Of the 3.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	10	10	2019
Cardiovascular disease deaths	25 300	12 700	12 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	49	47	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%) ^f	9	14	3	2019
Obesity, adults aged 18+ years (%)	20	15	24	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	12	3	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✓
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Philippines

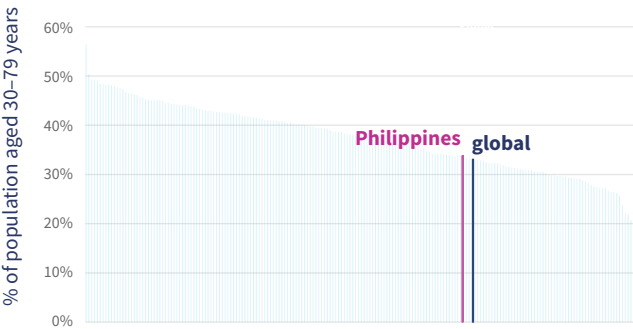
Hypertension profile

Total population (2019): 110 400 000

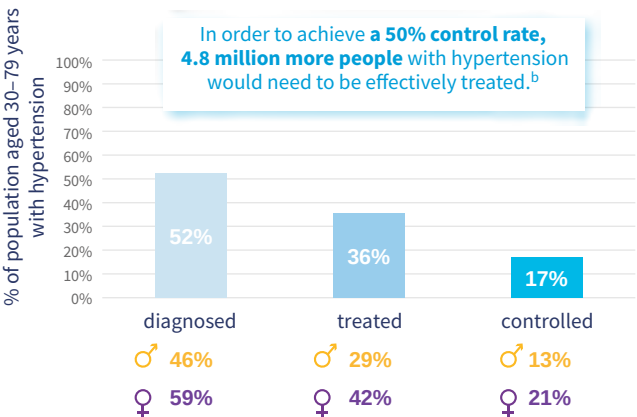
Total deaths (2019): 734 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 35% ♀ 33%

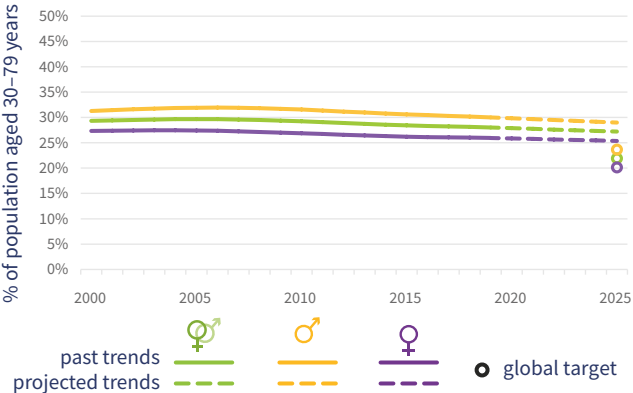
Prevalence of hypertension – global comparison (both sexes)^a



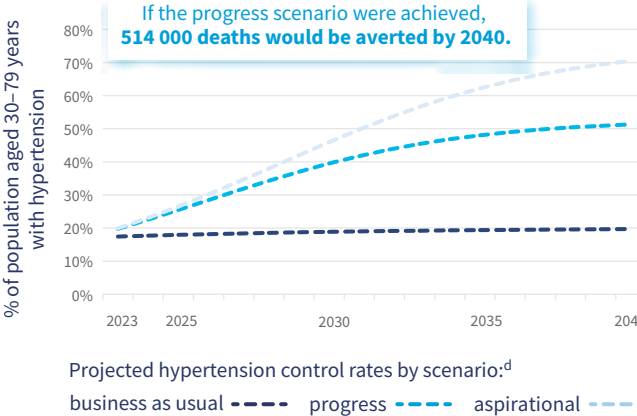
Of the 14.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	30	19	2019
Cardiovascular disease deaths	250 000	132 000	118 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	53	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	10	2019
Current tobacco use, adults aged 15+ years (%) ^f	23	40	7	2019
Obesity, adults aged 18+ years (%)	6	5	8	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	10	2	2019
Physical inactivity, adults aged 18+ years (%)	40	30	49	2016

National response

Targets	Treatment
National target for blood pressure	✓
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Poland

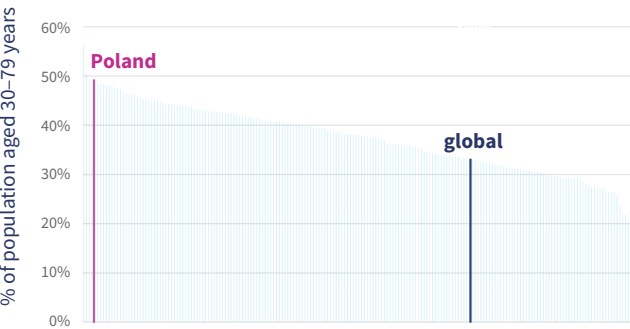
Hypertension profile

Total population (2019): 38 494 000

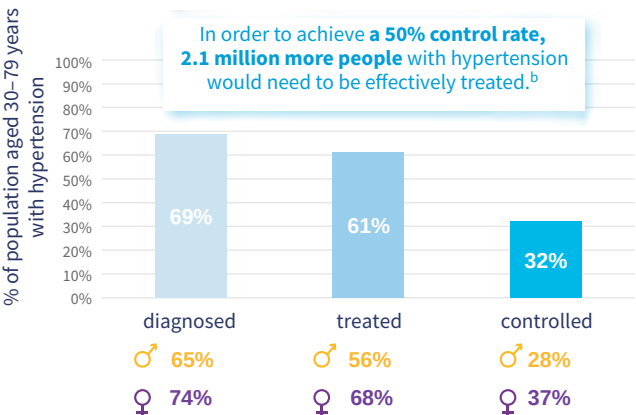
Total deaths (2019): 401 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 49% ♂ 56% ♀ 43%

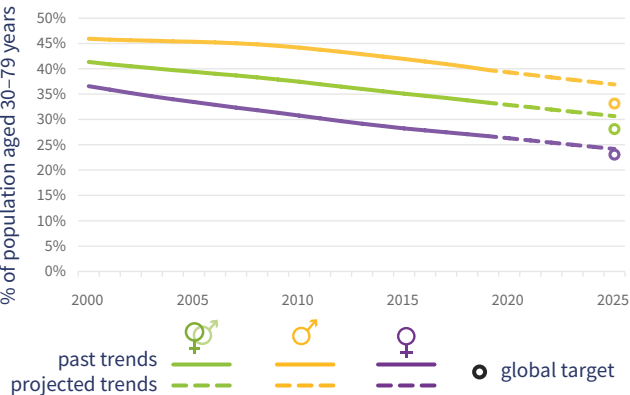
Prevalence of hypertension – global comparison (both sexes)^a



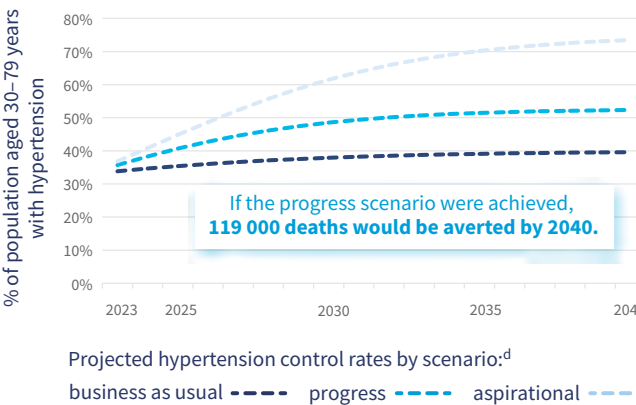
Of the 13 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	17	23	12	2019
Cardiovascular disease deaths	173 000	77 200	95 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	50	52	48	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	11	13	9	2019
Current tobacco use, adults aged 15+ years (%)	25	29	21	2019
Obesity, adults aged 18+ years (%)	23	24	22	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	19	5	2019
Physical inactivity, adults aged 18+ years (%)	32	31	33	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Portugal

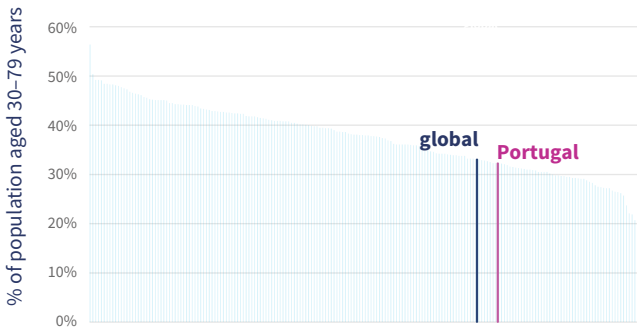
Hypertension profile

Total population (2019): 10 290 000

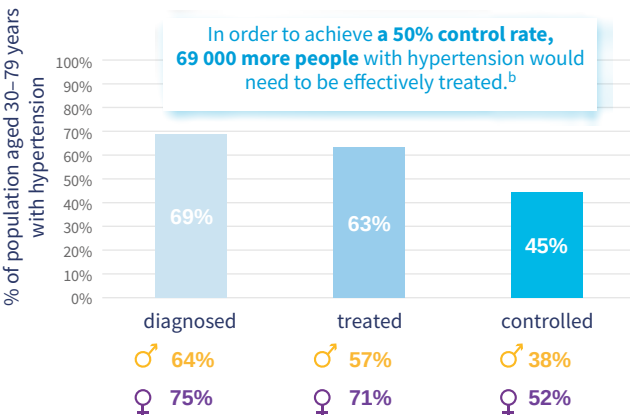
Total deaths (2019): 111 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 32% ♂ 37% ♀ 28%

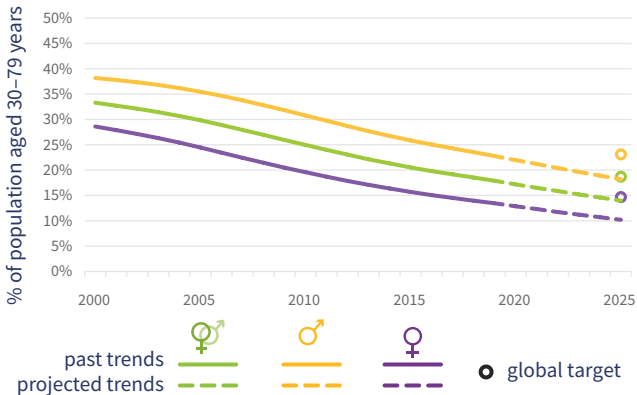
Prevalence of hypertension – global comparison (both sexes)^a



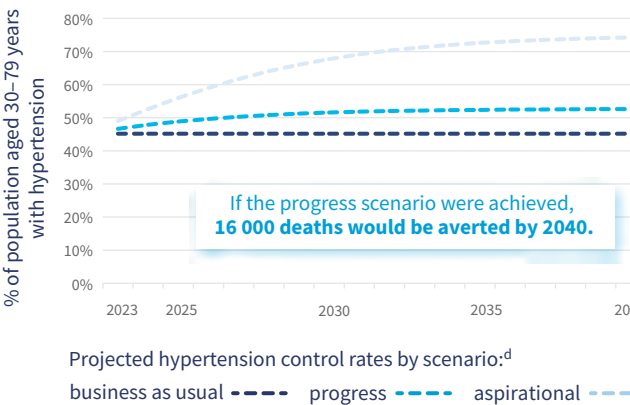
Of the 2.7 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	15	7	2019
Cardiovascular disease deaths	32 600	14 900	17 700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	45	48	44	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%) ^f	25	31	20	2019
Obesity, adults aged 18+ years (%)	21	20	21	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	17	5	2019
Physical inactivity, adults aged 18+ years (%)	43	38	48	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Qatar

Hypertension profile

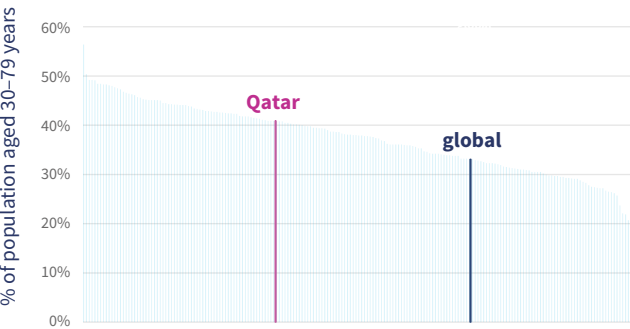
Total population (2019): 2 807 000

Total deaths (2019): 4540

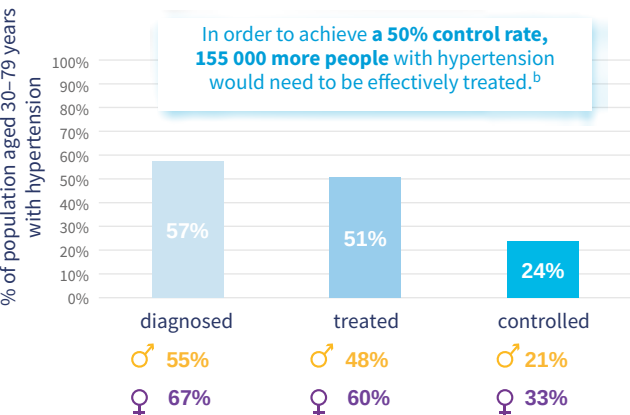
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

41% 42% 38%

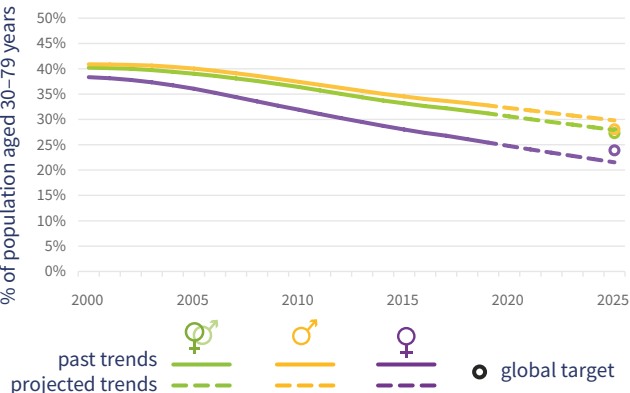
Prevalence of hypertension – global comparison (both sexes)^a



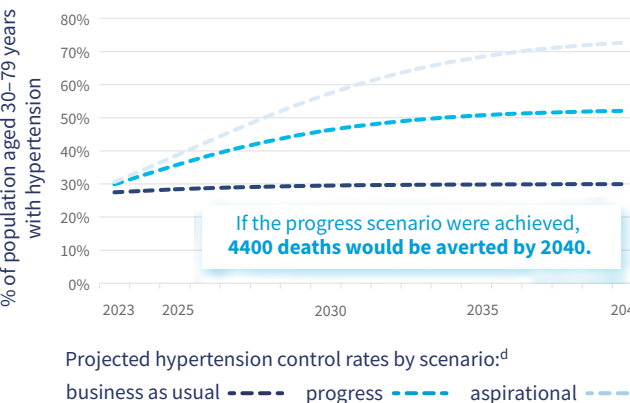
Of the 497 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	10	13	2019
Cardiovascular disease deaths	1500	930	560	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	59	53	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	5	2019
Current tobacco use, adults aged 15+ years (%)	12	22	2	2019
Obesity, adults aged 18+ years (%)	35	33	43	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	1	0	2019
Physical inactivity, adults aged 18+ years (%)	37	33	49	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Republic of Korea

Hypertension profile

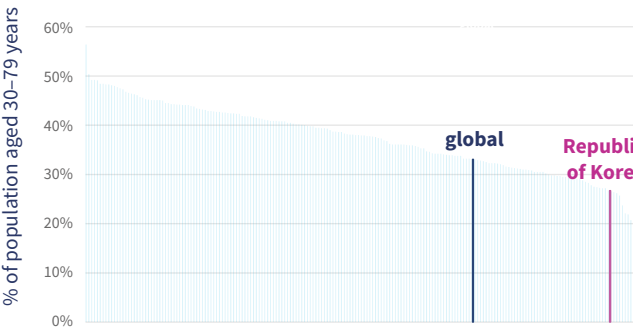
Total population (2019): 51 804 000

Total deaths (2019): 295 000

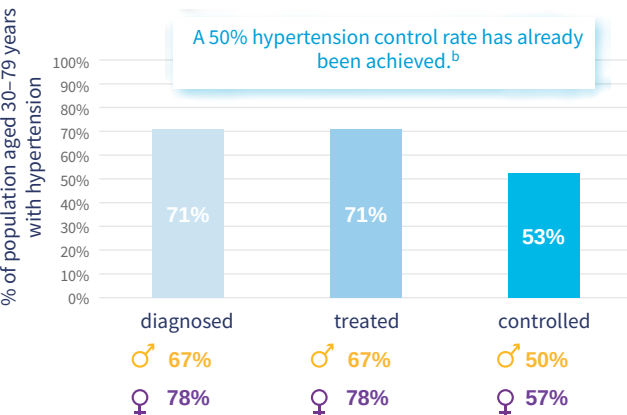
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

27% 32% 21%

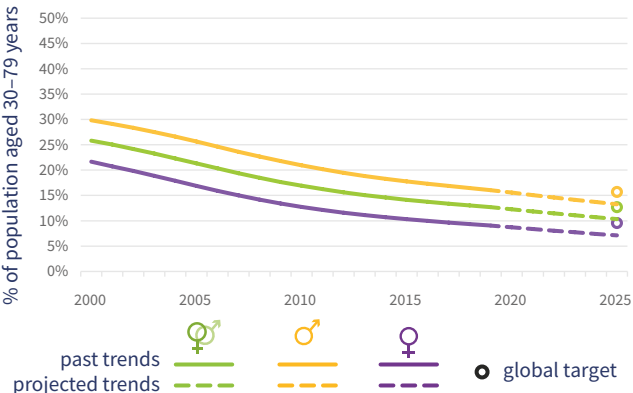
Prevalence of hypertension – global comparison (both sexes)^a



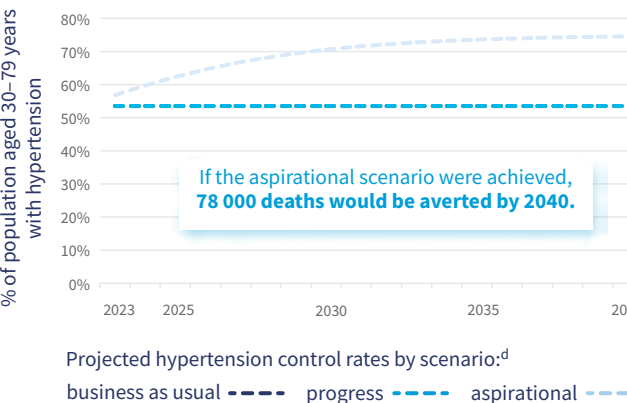
Of the 10.4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	7	10	4	2019
Cardiovascular disease deaths	64 400	29 900	34 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	42	40	43	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	12	13	11	2019
Current tobacco use, adults aged 15+ years (%) ^f	21	37	6	2019
Obesity, adults aged 18+ years (%)	5	4	5	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	13	4	2019
Physical inactivity, adults aged 18+ years (%)	35	30	41	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Republic of Moldova

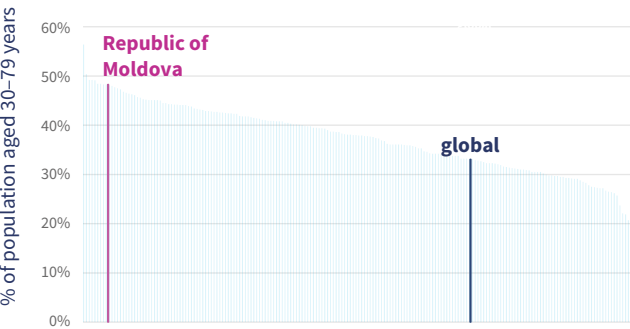
Hypertension profile

Total population (2019): 3 109 000

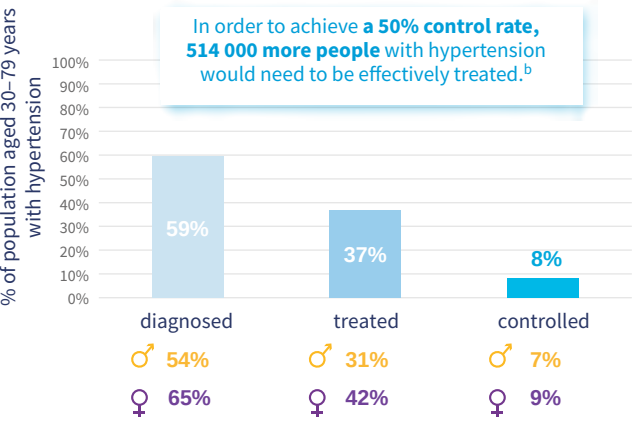
Total deaths (2019): 41 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 48% ♂ 49% ♀ 47%

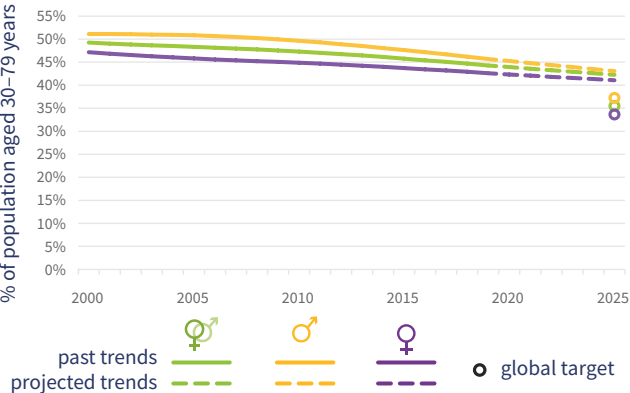
Prevalence of hypertension – global comparison (both sexes)^a



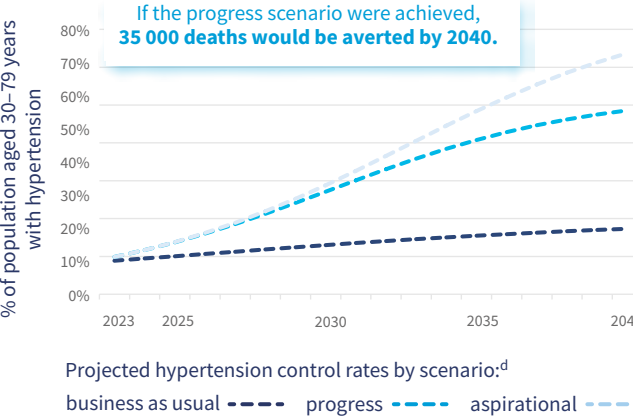
Of the 1.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	34	16	2019
Cardiovascular disease deaths	23 300	10 500	12 900	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	63	64	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	29	51	6	2019
Obesity, adults aged 18+ years (%)	19	16	21	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	18	5	2019
Physical inactivity, adults aged 18+ years (%)	11	12	11	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Romania

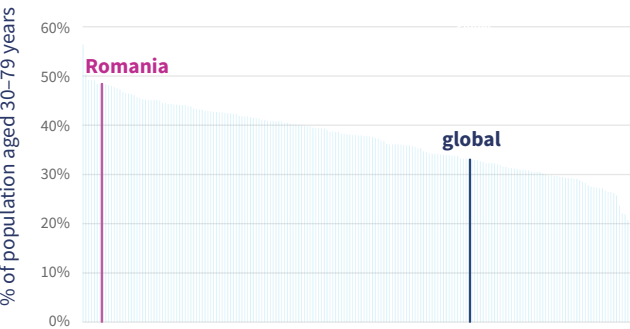
Hypertension profile

Total population (2019): 19 524 000

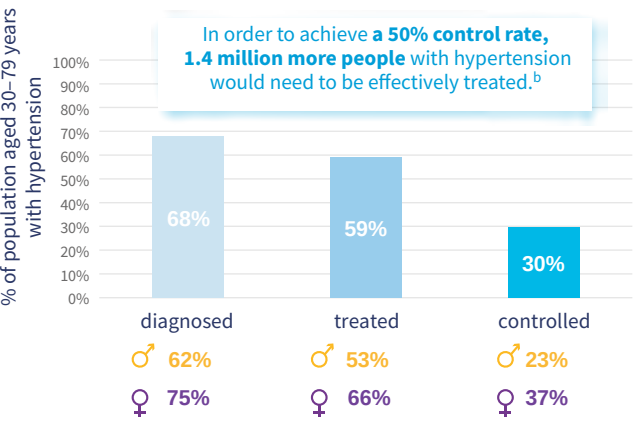
Total deaths (2019): 259 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 48% ♂ 53% ♀ 44%

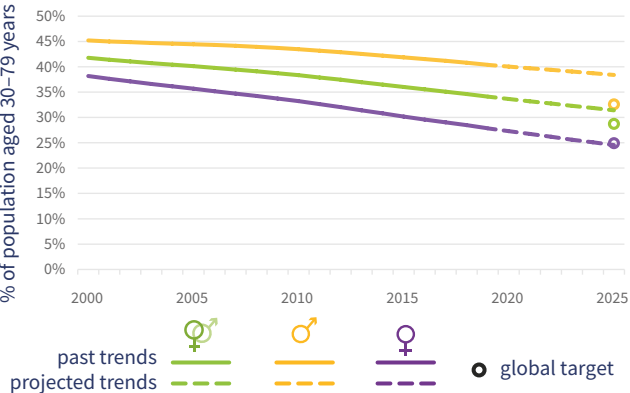
Prevalence of hypertension – global comparison (both sexes)^a



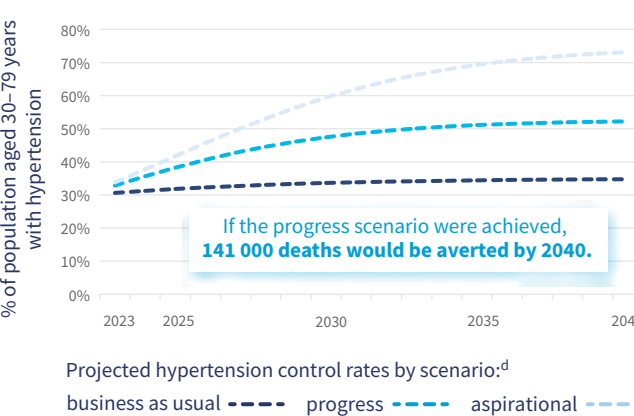
Of the 6.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	29	14	2019
Cardiovascular disease deaths	144 000	66 900	77 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	61	61	61	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%)	28	37	20	2019
Obesity, adults aged 18+ years (%)	23	23	22	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	17	27	8	2019
Physical inactivity, adults aged 18+ years (%)	35	32	38	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Russian Federation

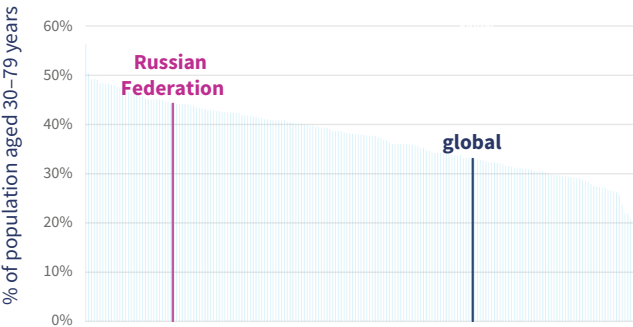
Hypertension profile

Total population (2019): 145 700 000

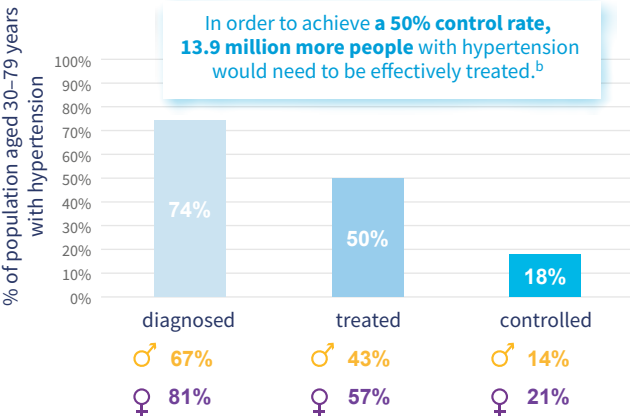
Total deaths (2019): 1 790 000!

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 44% ♂ 47% ♀ 41%

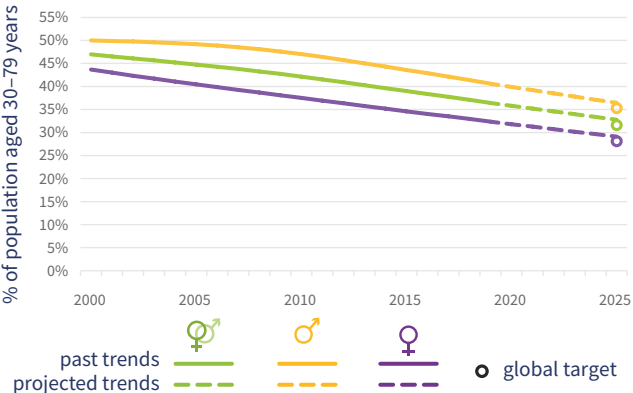
Prevalence of hypertension – global comparison (both sexes)^a



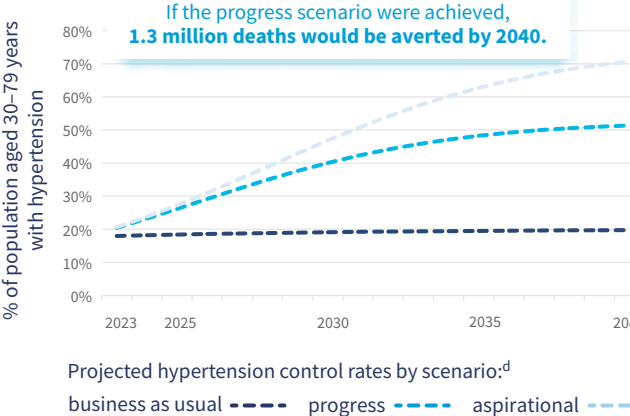
Of the 43.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	35	15	2019
Cardiovascular disease deaths	1 004 000	436 000	568 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	55	55	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	8	2019
Current tobacco use, adults aged 15+ years (%)	27	42	13	2019
Obesity, adults aged 18+ years (%)	23	18	27	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	18	4	2019
Physical inactivity, adults aged 18+ years (%)	17	17	18	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Rwanda

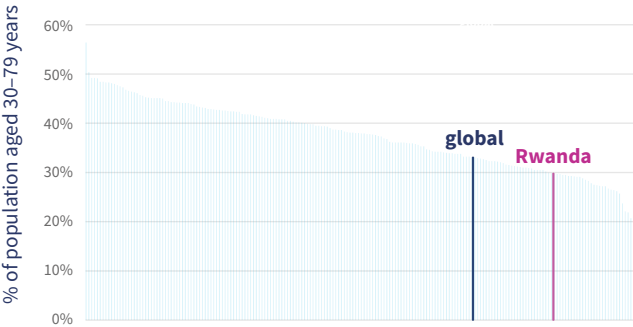
Hypertension profile

Total population (2019): 12 835 000

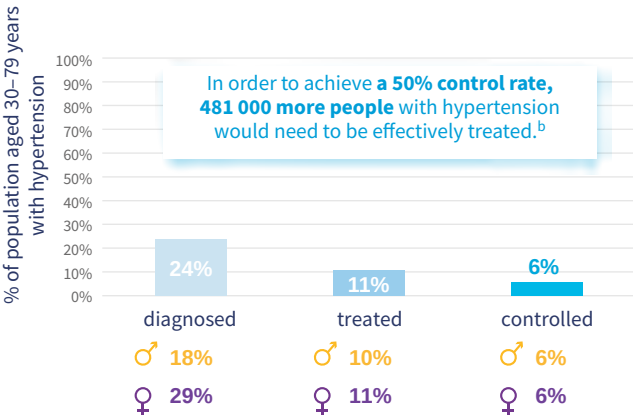
Total deaths (2019): 64 300

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 30% ♂ 29% ♀ 31%

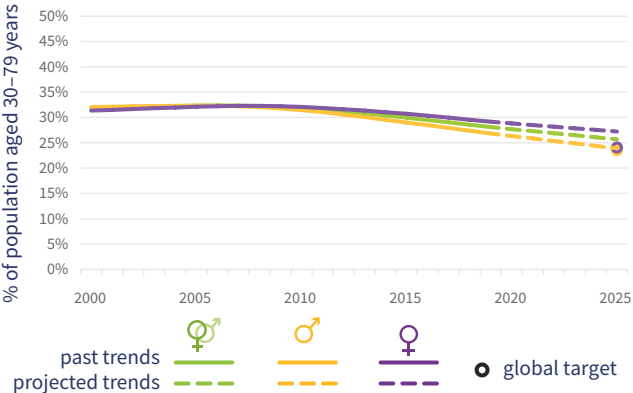
Prevalence of hypertension – global comparison (both sexes)^a



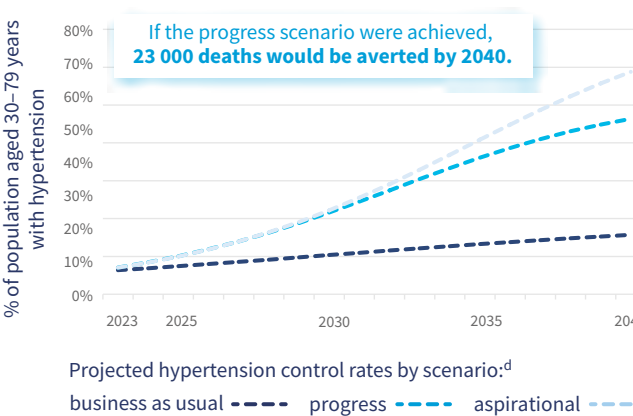
Of the 1.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	20	22	19	2019
Cardiovascular disease deaths	11 200	5400	5900	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	51	58	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	14	21	7	2019
Obesity, adults aged 18+ years (%)	6	2	9	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	1	2019
Physical inactivity, adults aged 18+ years (%)	15	11	18	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Saint Kitts and Nevis

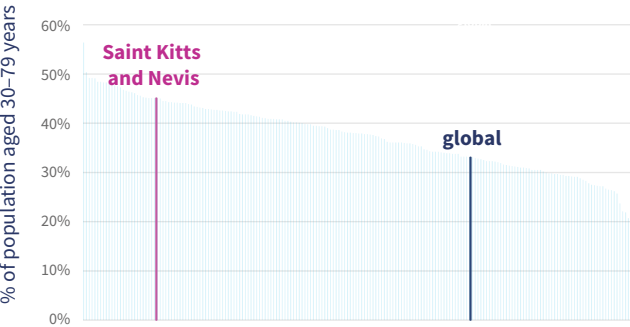
Hypertension profile

Total population (2019): 47 700

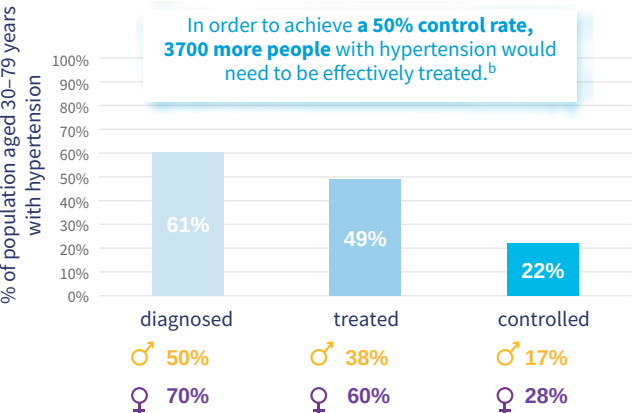
Total deaths (2019): no data

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 45% ♂ 45% ♀ 45%

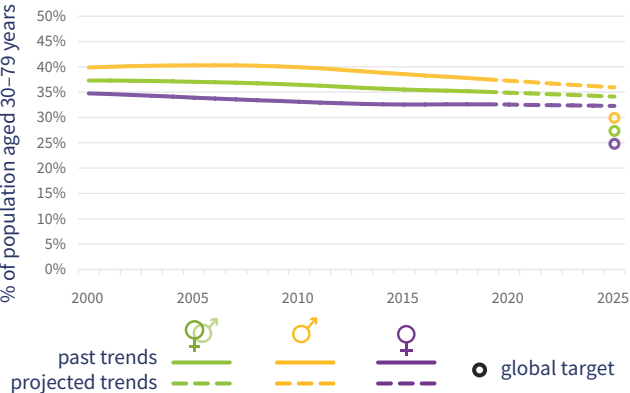
Prevalence of hypertension – global comparison (both sexes)^a



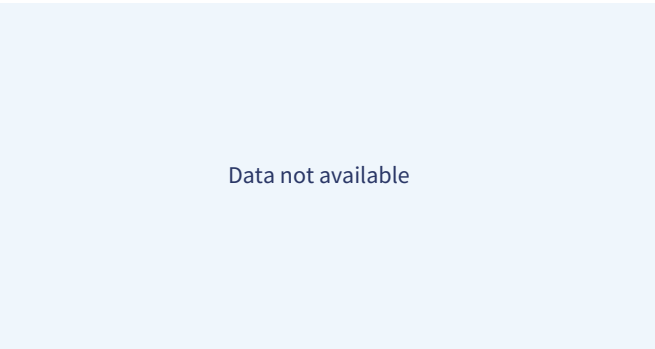
Of the 13 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	61	49	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	23	15	30	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	10	3	2019
Physical inactivity, adults aged 18+ years (%)	32	24	40	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake.

Saint Lucia

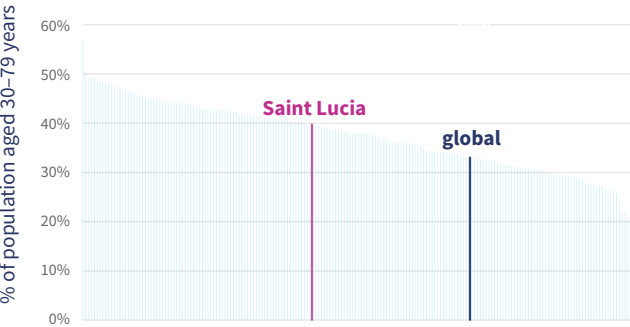
Hypertension profile

Total population (2019): 179 000

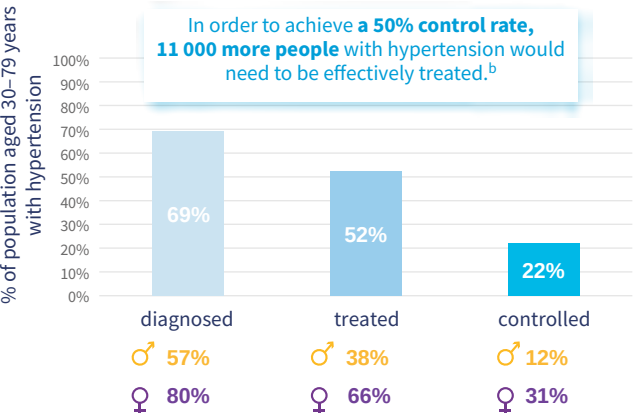
Total deaths (2019): 1500

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 40% ♂ 39% ♀ 41%

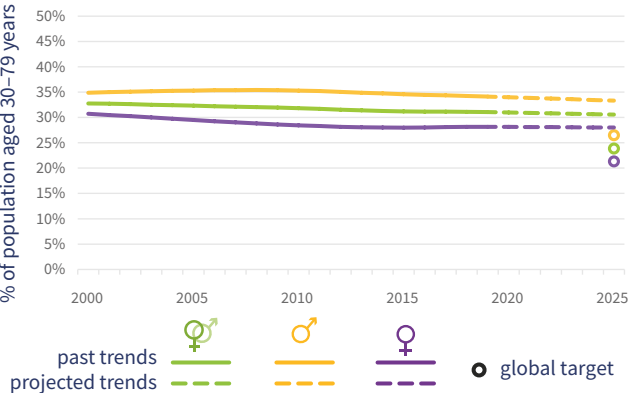
Prevalence of hypertension – global comparison (both sexes)^a



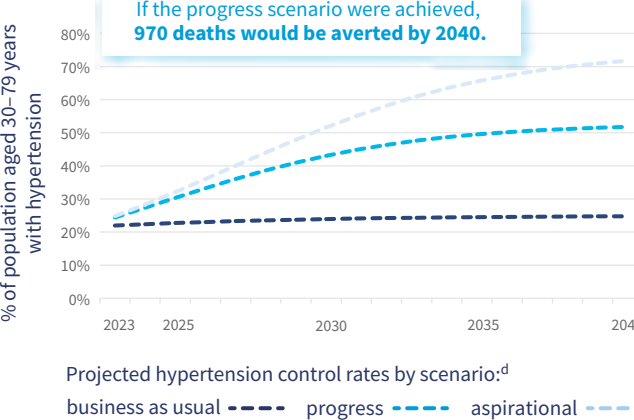
Of the 40 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	18	21	15	2019
Cardiovascular disease deaths	490	280	220	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	53	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	20	12	27	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	15	4	2019
Physical inactivity, adults aged 18+ years (%)	40	27	52	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Saint Vincent and the Grenadines

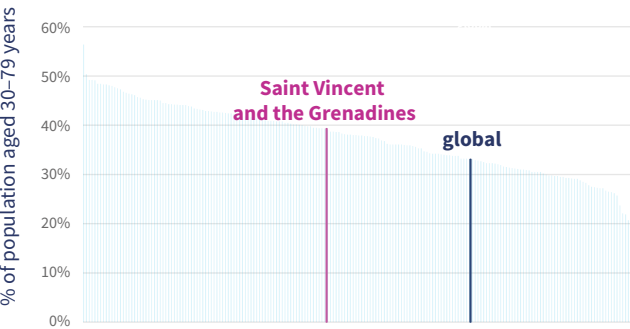
Hypertension profile

Total population (2019): 105 000

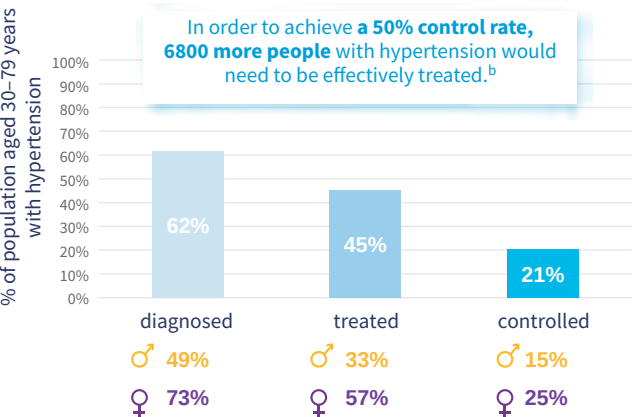
Total deaths (2019): 930

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 39% ♂ 37% ♀ 42%

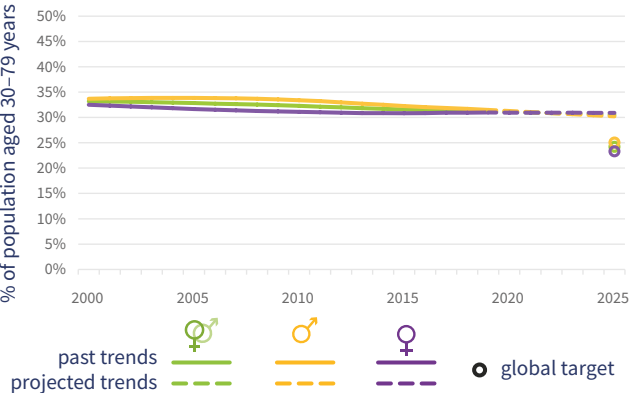
Prevalence of hypertension – global comparison (both sexes)^a



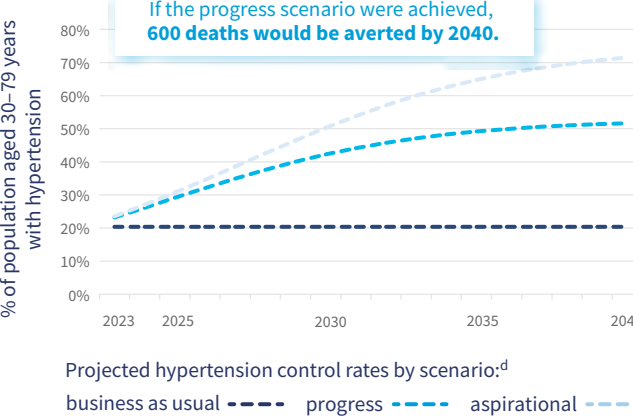
Of the 23 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	23	18	2019
Cardiovascular disease deaths	330	170	160	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	54	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	24	17	31	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	11	3	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of	
National target for salt consumption	✓	hypertension	✗
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Samoa

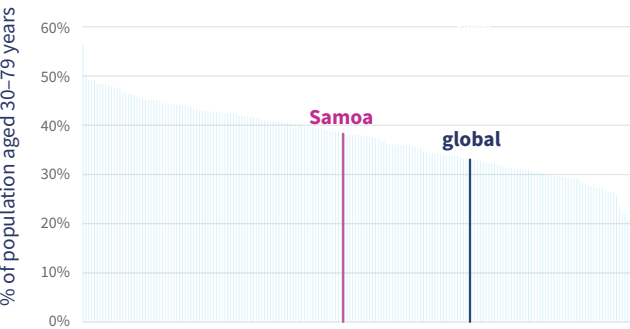
Hypertension profile

Total population (2019): 212 000

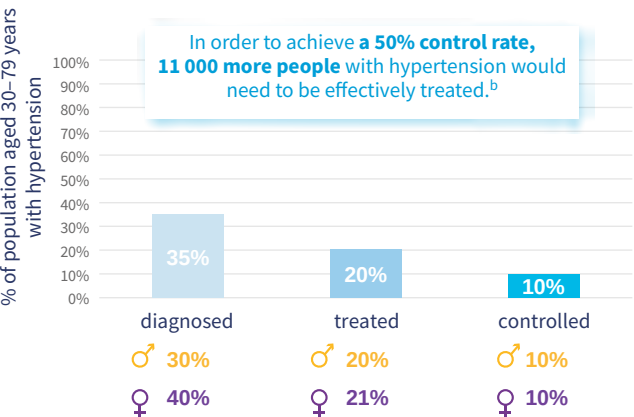
Total deaths (2019): 1210

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 38% ♂ 39% ♀ 38%

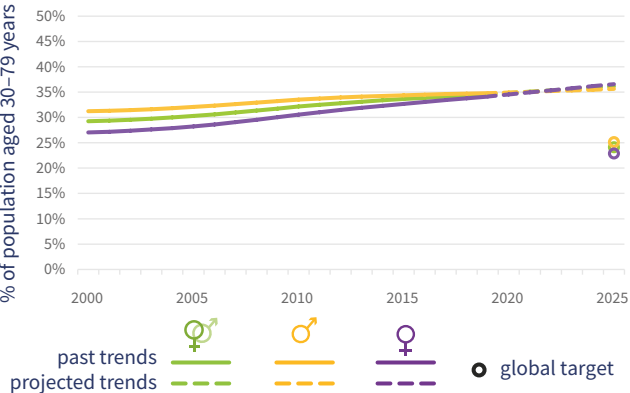
Prevalence of hypertension – global comparison (both sexes)^a



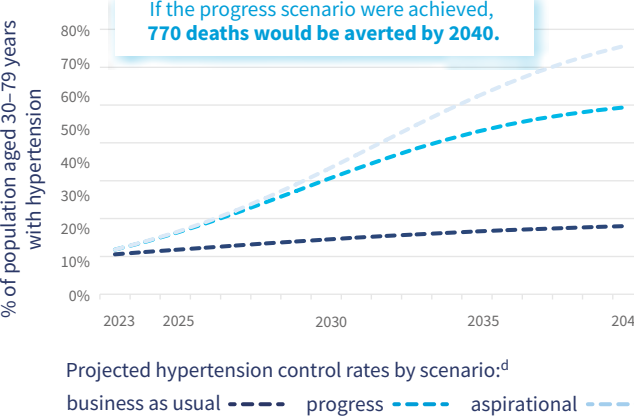
Of the 27 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	31	33	29	2019
Cardiovascular disease deaths	450	230	220	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	54	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	5	5	5	2019
Current tobacco use, adults aged 15+ years (%)	26	37	15	2019
Obesity, adults aged 18+ years (%)	47	40	55	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	4	1	2019
Physical inactivity, adults aged 18+ years (%)	13	8	17	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of	
National target for salt consumption	✗	hypertension	✓
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

San Marino

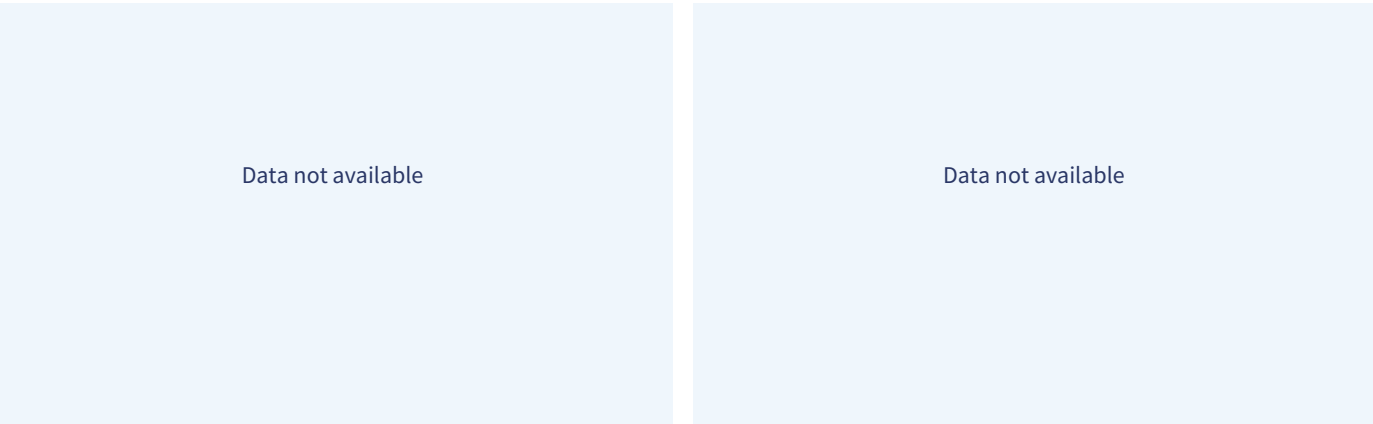
Hypertension profile

Total population (2019): 34 200

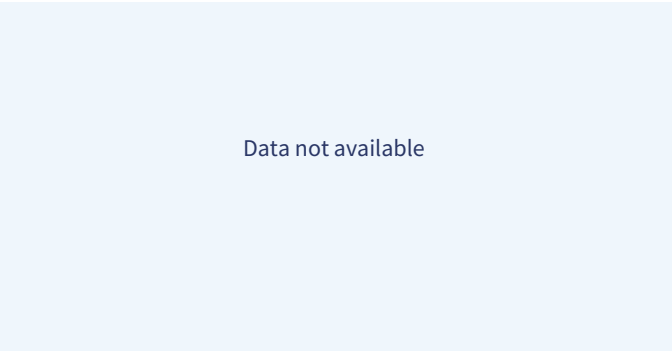
Total deaths (2019): no data

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019) no data

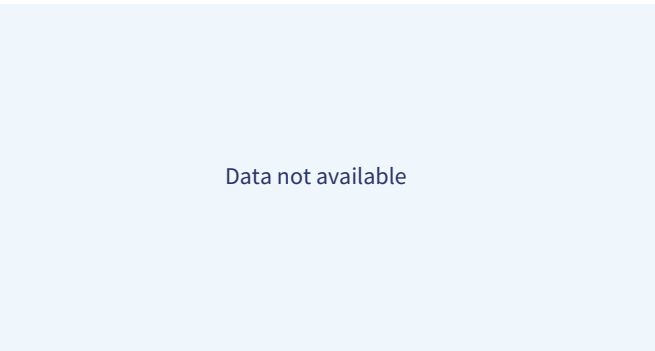
Prevalence of hypertension – global comparison (both sexes)



Trends in uncontrolled hypertension in adults aged 30–79 years



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	50	50	49	2019

Risk factors^a

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	no data	no data	no data	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	no data	no data	no data	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✗
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓		

Footnote: a. Age-standardized estimates are presented for all indicators except salt intake.

Sao Tome and Principe

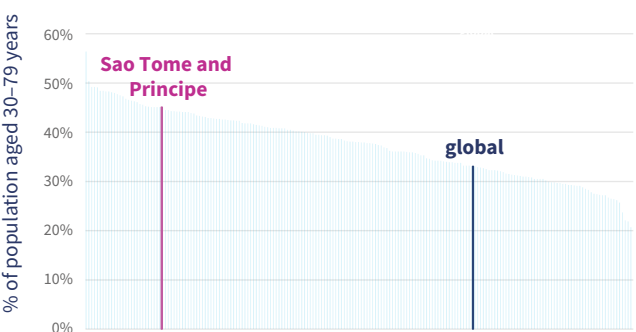
Hypertension profile

Total population (2019): 215 000

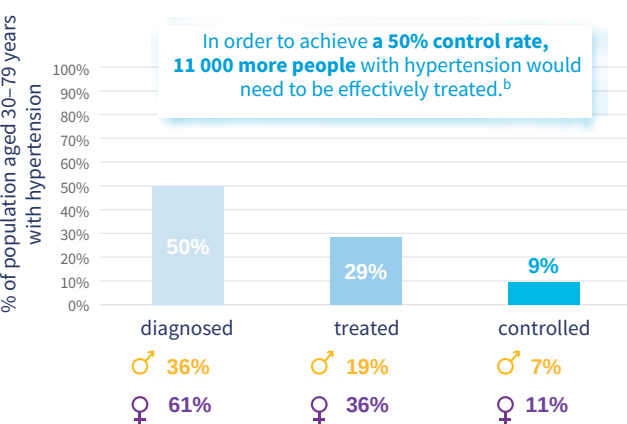
Total deaths (2019): 1000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 45% ♂ 42% ♀ 48%

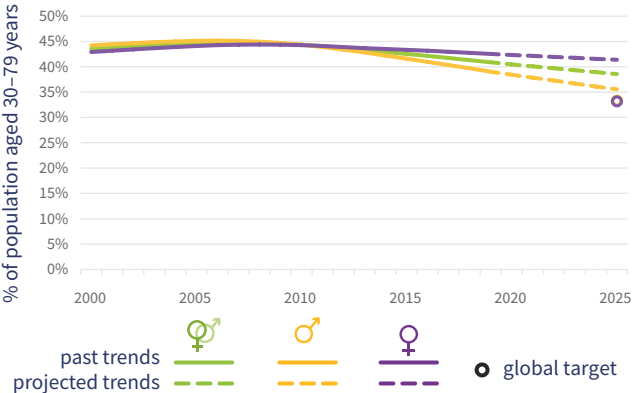
Prevalence of hypertension – global comparison (both sexes)^a



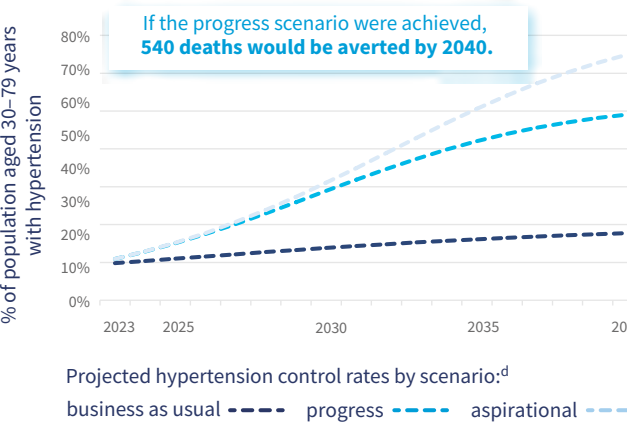
Of the 27 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	22	20	2019
Cardiovascular disease deaths	240	90	150	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	60	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	6	10	1	2019
Obesity, adults aged 18+ years (%)	12	7	17	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	8	2	2019
Physical inactivity, adults aged 18+ years (%)	15	10	21	2016

National response

Targets		Treatment	
National target for blood pressure	✗	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Saudi Arabia

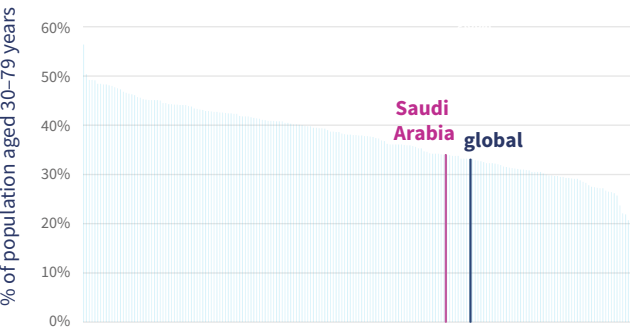
Hypertension profile

Total population (2019): 35 827 000

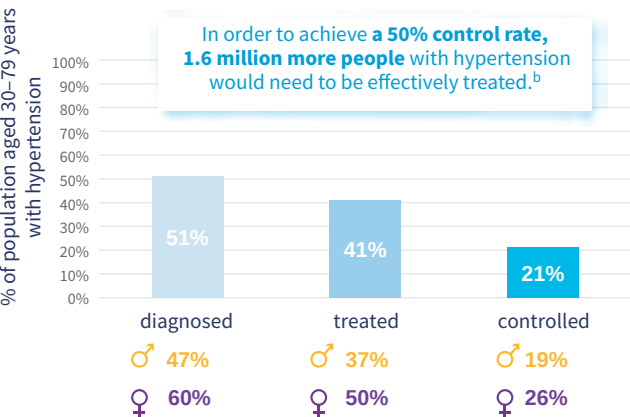
Total deaths (2019): 143 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 36% ♀ 30%

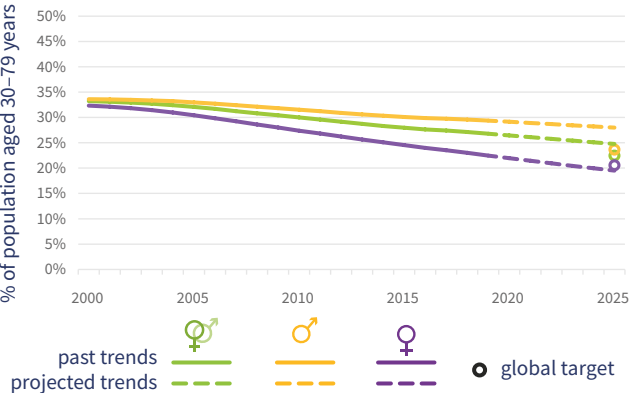
Prevalence of hypertension – global comparison (both sexes)^a



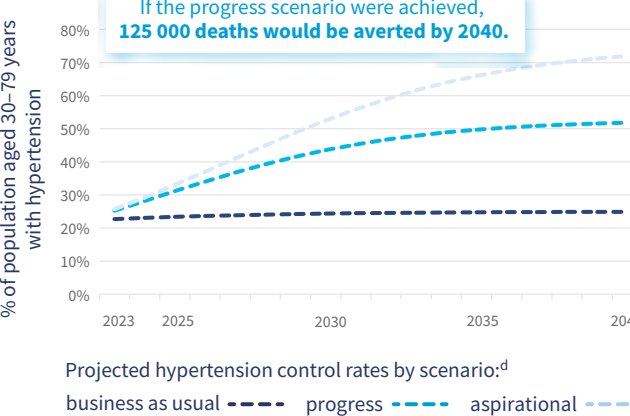
Of the 5.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	22	18	2019
Cardiovascular disease deaths	60 300	38 200	22 100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	57	49	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	5	2019
Current tobacco use, adults aged 15+ years (%)	14	26	2	2019
Obesity, adults aged 18+ years (%)	35	31	42	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	53	45	65	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Senegal

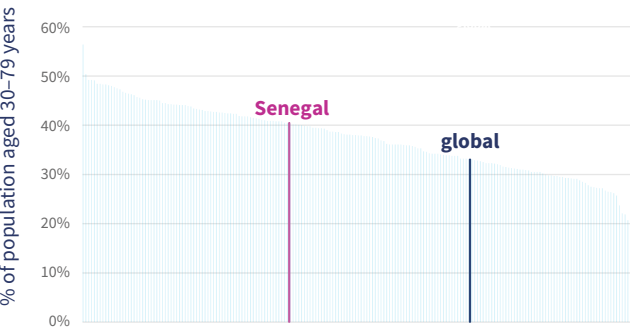
Hypertension profile

Total population (2019): 16 001 000

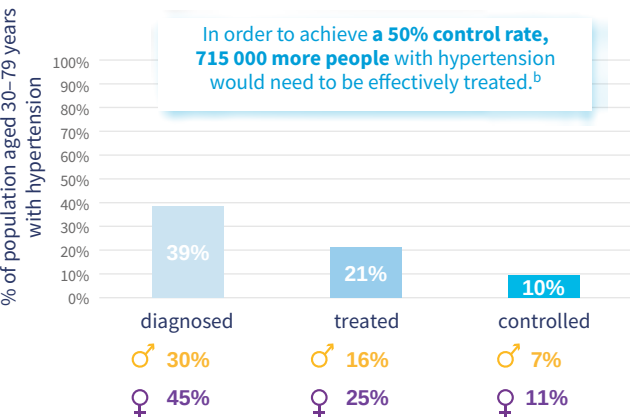
Total deaths (2019): 86 600

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 37% ♀ 43%

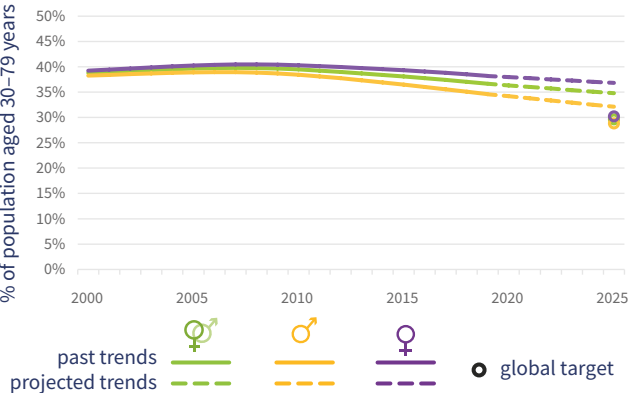
Prevalence of hypertension – global comparison (both sexes)^a



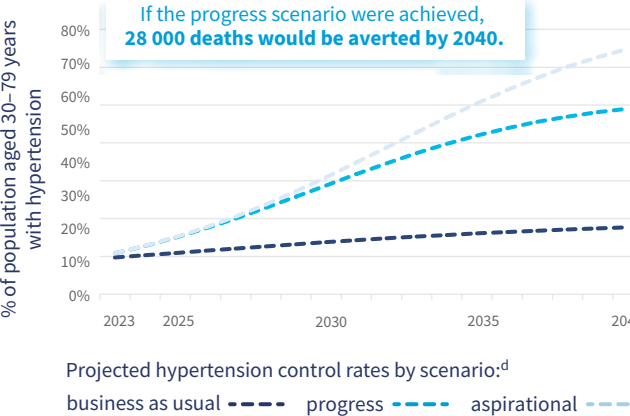
Of the 1.7 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	20	19	20	2019
Cardiovascular disease deaths	14 100	6100	8000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	62	58	66	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	7	14	1	2019
Obesity, adults aged 18+ years (%)	9	4	13	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	1	0	2019
Physical inactivity, adults aged 18+ years (%)	23	18	28	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Serbia

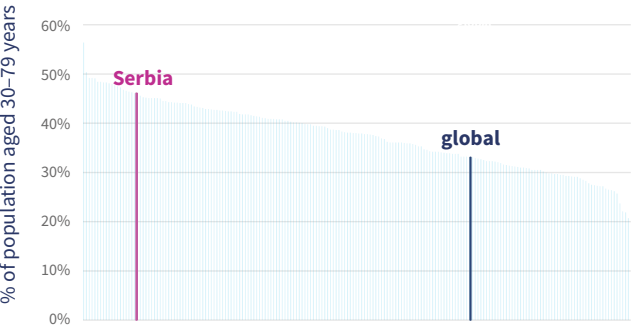
Hypertension profile

Total population (2019): 7 401 000

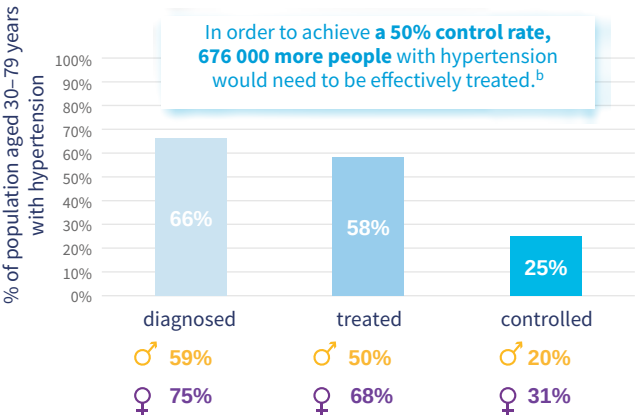
Total deaths (2019): 115 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 46% ♂ 50% ♀ 42%

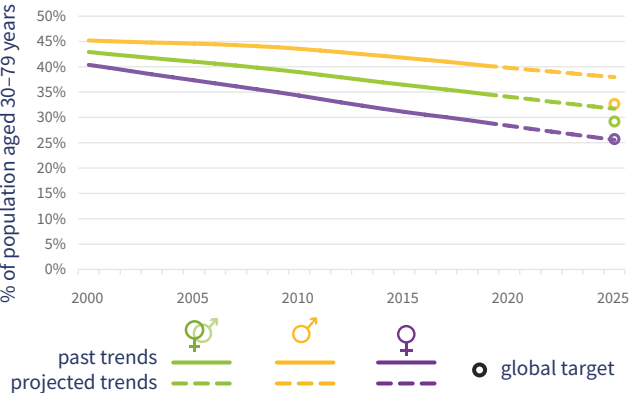
Prevalence of hypertension – global comparison (both sexes)^a



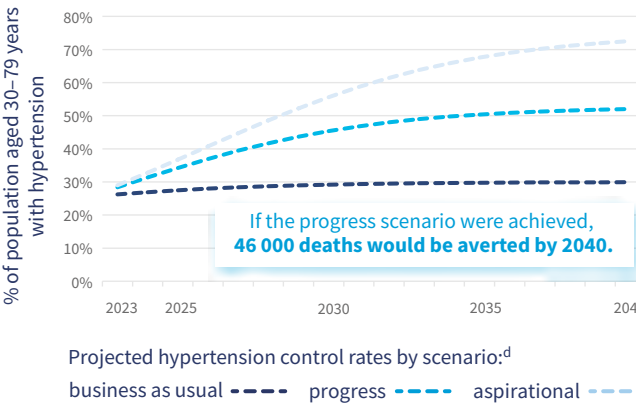
Of the 2.9 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	28	16	2019
Cardiovascular disease deaths	59 300	27 300	32 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	59	59	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%) ^f	40	41	39	2019
Obesity, adults aged 18+ years (%)	22	21	22	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	13	3	2019
Physical inactivity, adults aged 18+ years (%)	39	35	44	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Seychelles

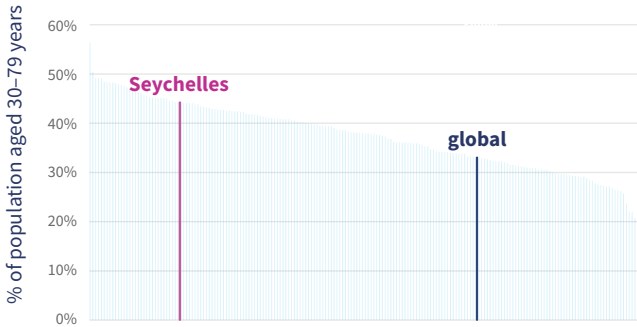
Hypertension profile

Total population (2019): 104 000

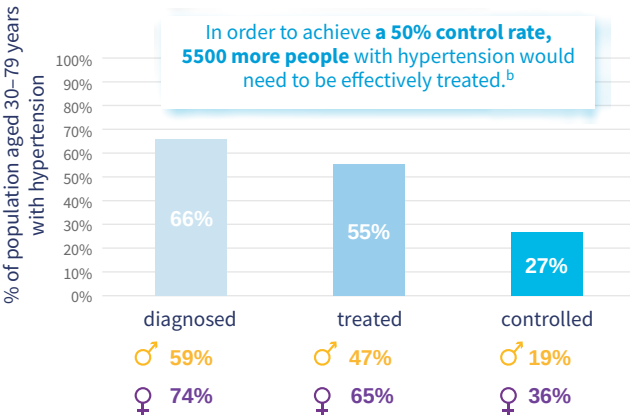
Total deaths (2019): 770

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 44% ♂ 46% ♀ 42%

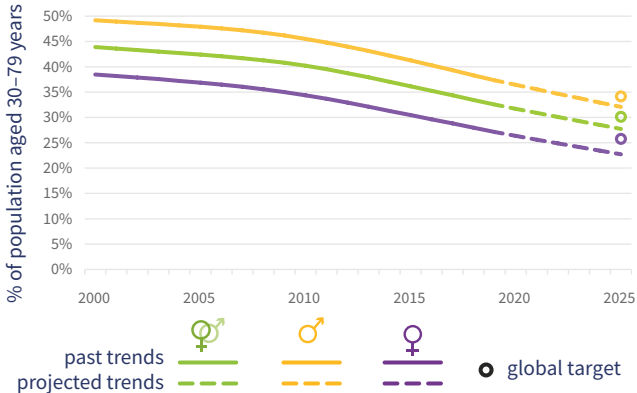
Prevalence of hypertension – global comparison (both sexes)^a



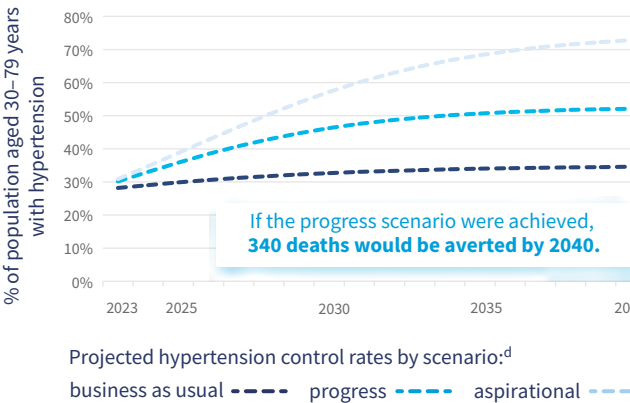
Of the 24 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	27	15	2019
Cardiovascular disease deaths	230	120	110	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	61	63	60	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	21	35	7	2019
Obesity, adults aged 18+ years (%)	14	8	21	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	18	5	2019
Physical inactivity, adults aged 18+ years (%)	19	18	20	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Sierra Leone

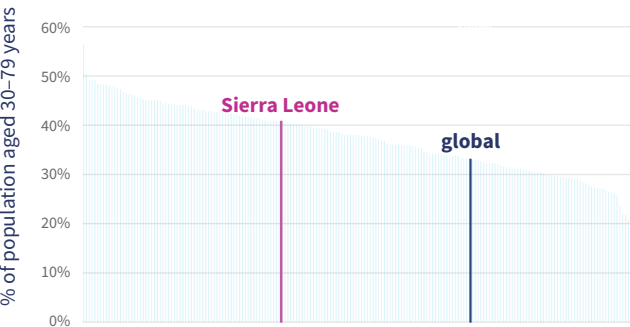
Hypertension profile

Total population (2019): 8 047 000

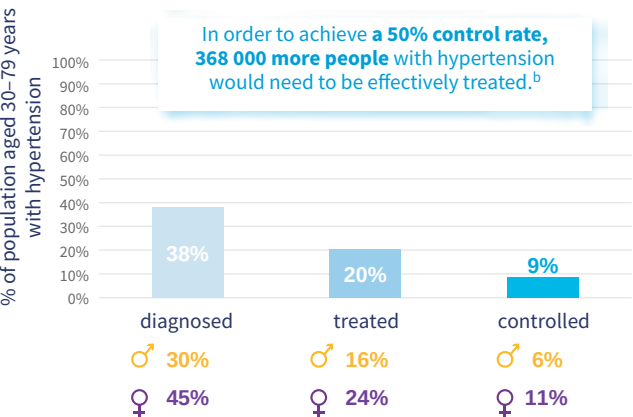
Total deaths (2019): 67 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 38% ♀ 43%

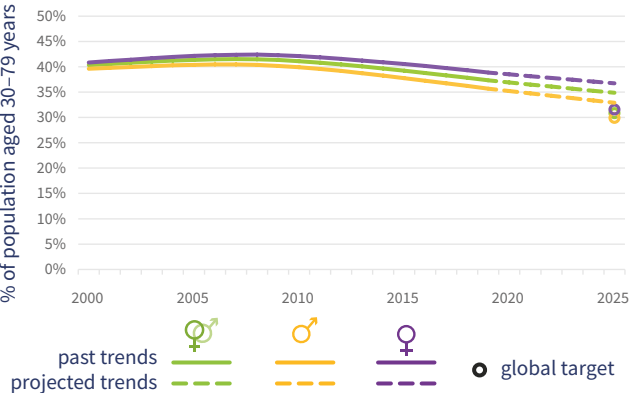
Prevalence of hypertension – global comparison (both sexes)^a



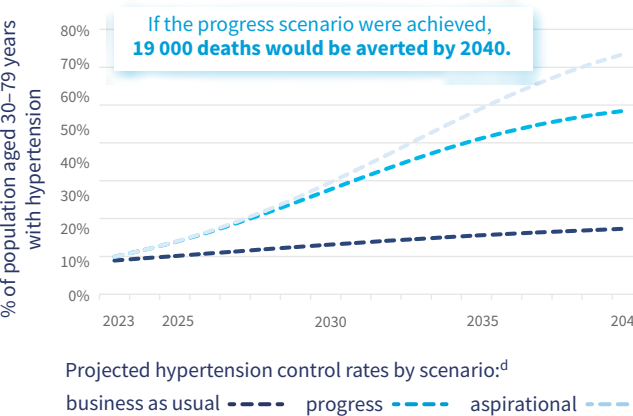
Of the 878 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	22	25	2019
Cardiovascular disease deaths	9400	3900	5400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	65	63	66	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	14	22	7	2019
Obesity, adults aged 18+ years (%)	9	4	13	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	14	10	18	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Singapore

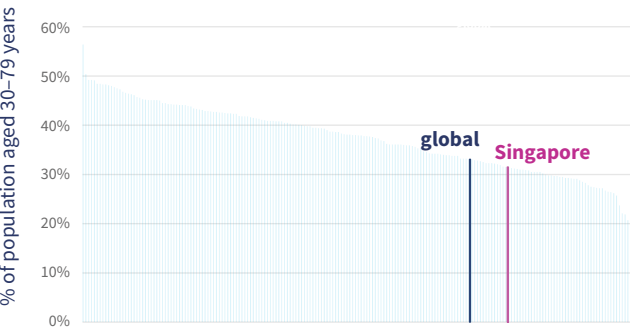
Hypertension profile

Total population (2019): 5 866 000

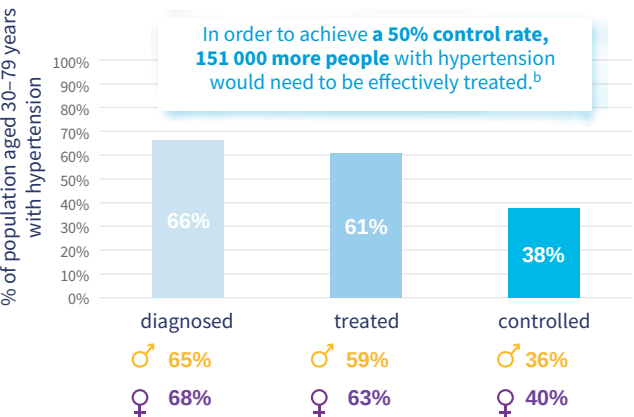
Total deaths (2019): 26 800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 32% ♂ 35% ♀ 27%

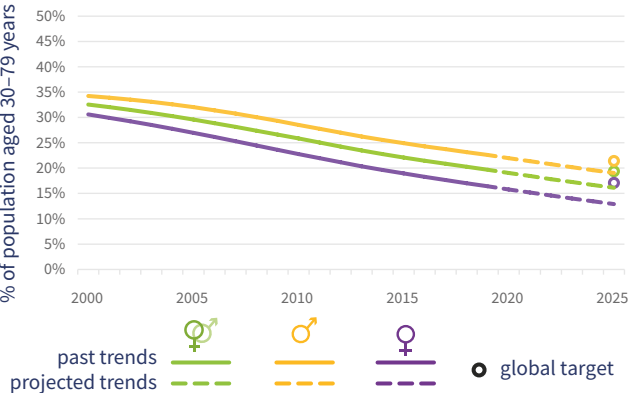
Prevalence of hypertension – global comparison (both sexes)^a



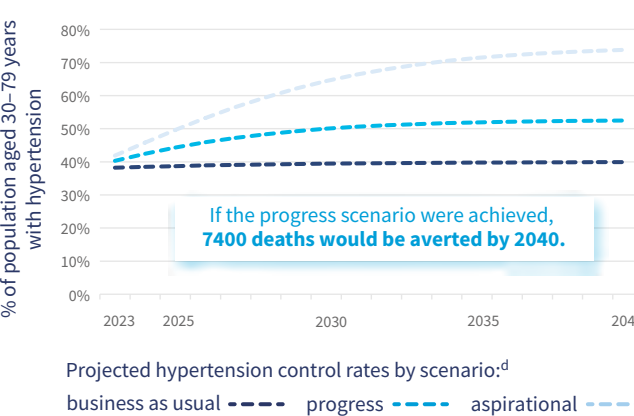
Of the 1.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	9	12	7	2019
Cardiovascular disease deaths	8400	5100	3300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	45	47	42	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	11	12	10	2019
Current tobacco use, adults aged 15+ years (%) ^f	17	28	5	2019
Obesity, adults aged 18+ years (%)	6	6	6	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	1	2019
Physical inactivity, adults aged 18+ years (%)	37	34	39	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Slovakia

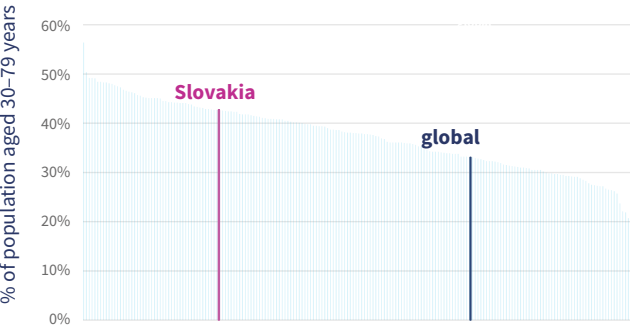
Hypertension profile

Total population (2019): 5 454 000

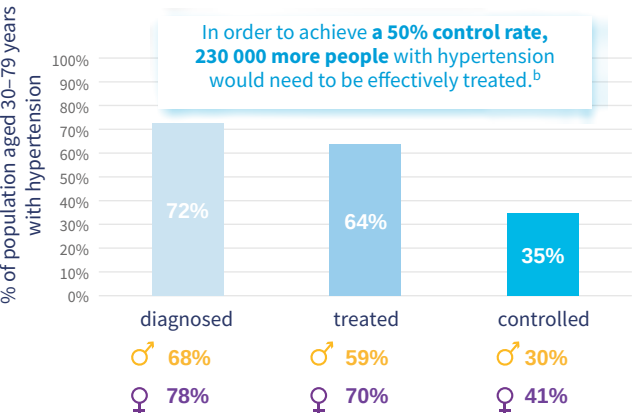
Total deaths (2019): 50 500

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 43% ♂ 47% ♀ 38%

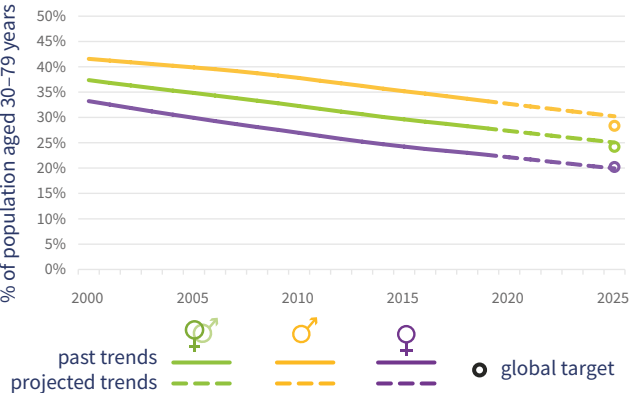
Prevalence of hypertension – global comparison (both sexes)^a



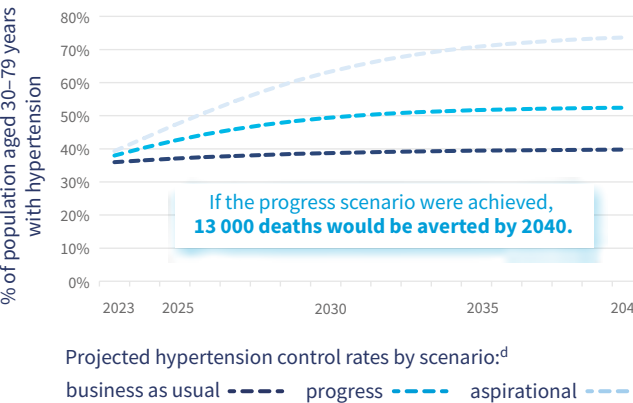
Of the 1.7 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	15	21	10	2019
Cardiovascular disease deaths	16 800	8600	8200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	58	59	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%) ^f	32	38	25	2019
Obesity, adults aged 18+ years (%)	21	21	20	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	17	5	2019
Physical inactivity, adults aged 18+ years (%)	35	31	38	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Slovenia

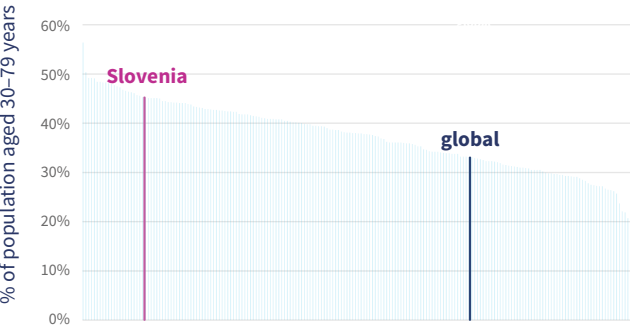
Hypertension profile

Total population (2019): 2 113 000

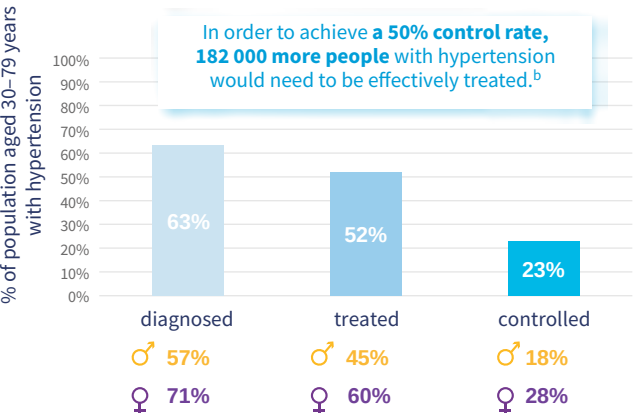
Total deaths (2019): 20 500

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 45% ♂ 50% ♀ 41%

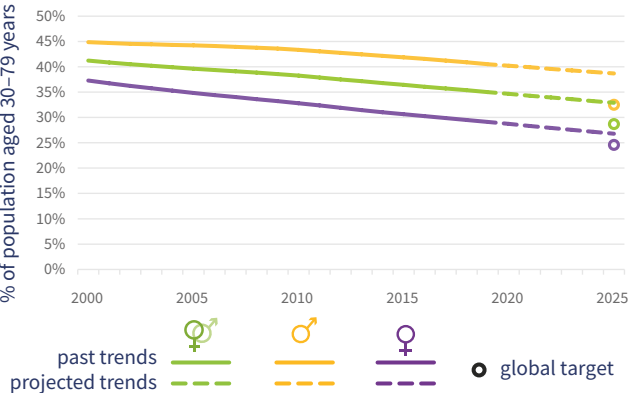
Prevalence of hypertension – global comparison (both sexes)^a



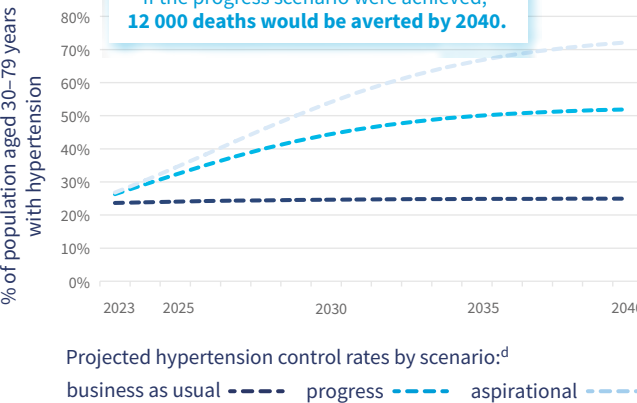
Of the 699 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	11	15	8	2019
Cardiovascular disease deaths	8000	3200	4800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	56	57	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	13	15	11	2019
Current tobacco use, adults aged 15+ years (%) ^f	22	25	20	2019
Obesity, adults aged 18+ years (%)	20	19	21	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	17	5	2019
Physical inactivity, adults aged 18+ years (%)	32	28	37	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Solomon Islands

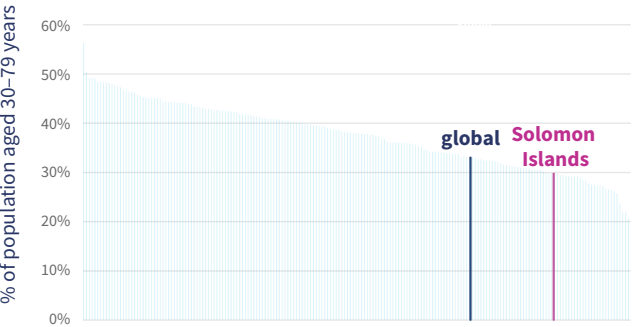
Hypertension profile

Total population (2019): 675 000

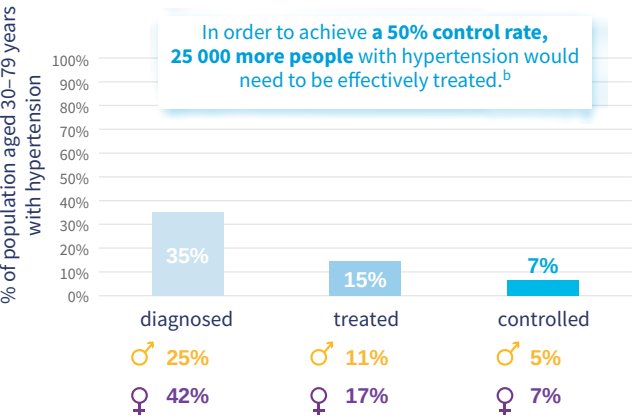
Total deaths (2019): 4780

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 30% ♂ 25% ♀ 35%

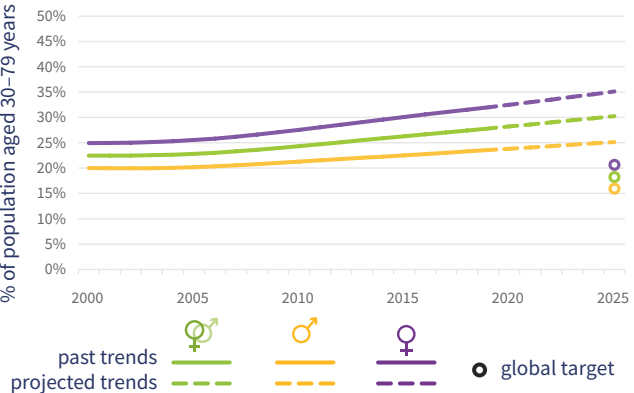
Prevalence of hypertension – global comparison (both sexes)^a



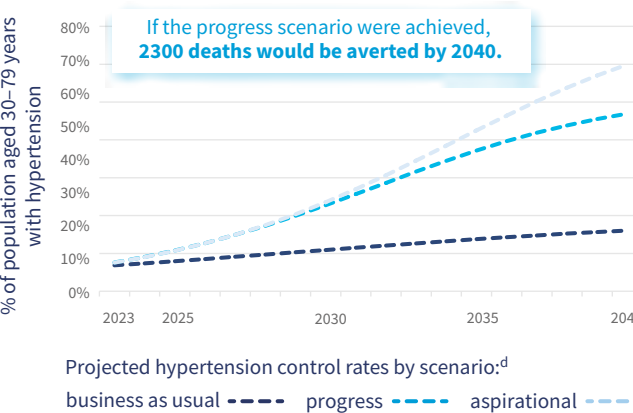
Of the 58 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	39	44	34	2019
Cardiovascular disease deaths	1800	1100	780	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	40	38	43	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	37	54	20	2019
Obesity, adults aged 18+ years (%)	23	18	27	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	1	2019
Physical inactivity, adults aged 18+ years (%)	18	13	23	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Somalia

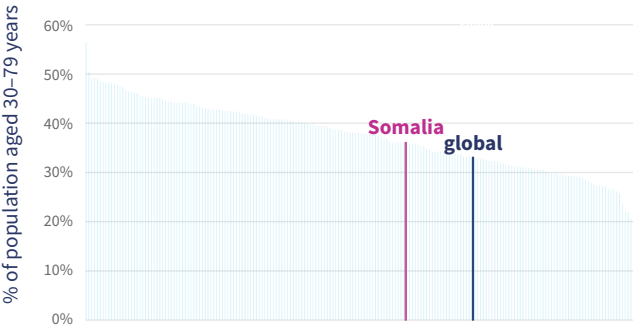
Hypertension profile

Total population (2019): 15 981 000

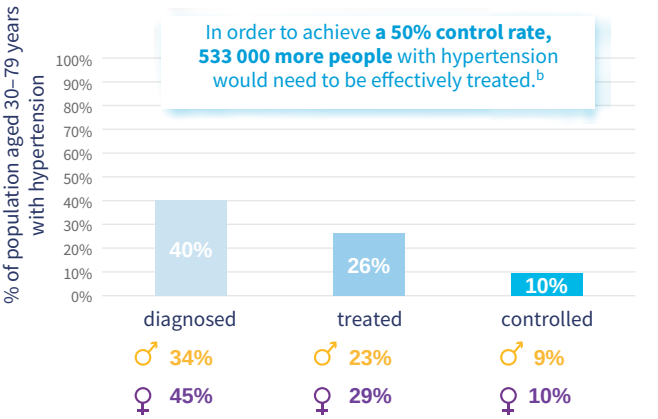
Total deaths (2019): 170 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 36% ♂ 33% ♀ 39%

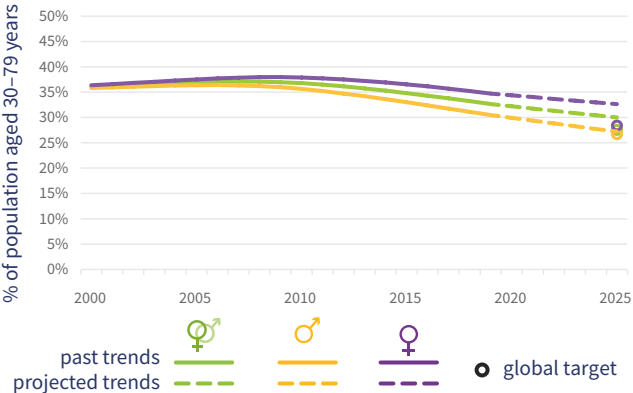
Prevalence of hypertension – global comparison (both sexes)^a



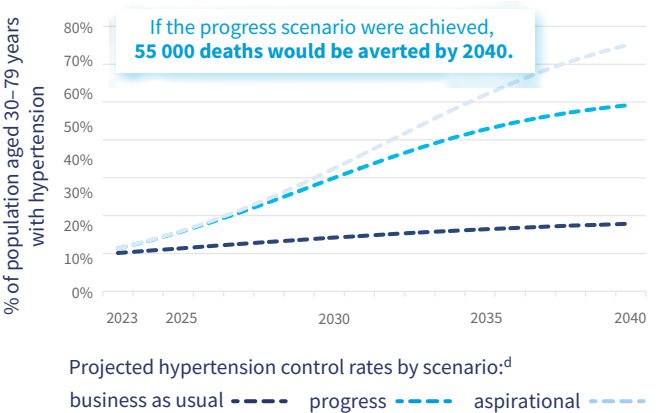
Of the 1.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	30	34	27	2019
Cardiovascular disease deaths	20 300	11 800	8 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	56	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	8	4	12	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

South Africa

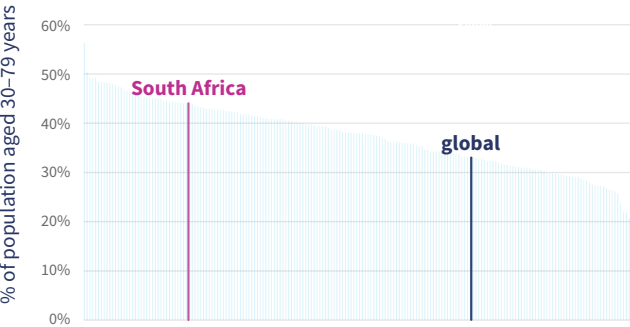
Hypertension profile

Total population (2019): 58 087 000

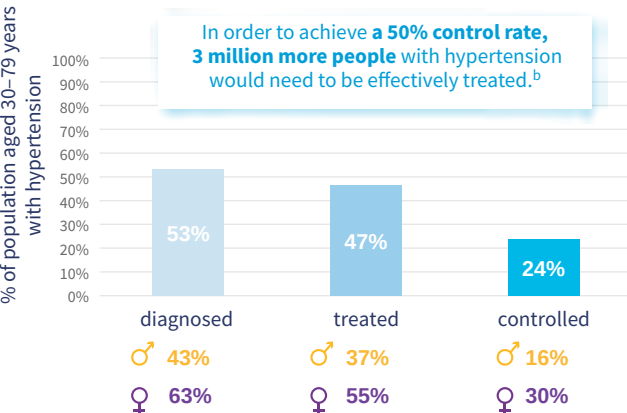
Total deaths (2019): 496 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 44% ♂ 44% ♀ 44%

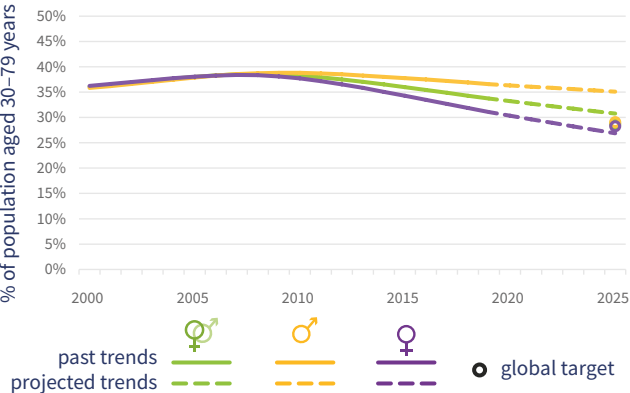
Prevalence of hypertension – global comparison (both sexes)^a



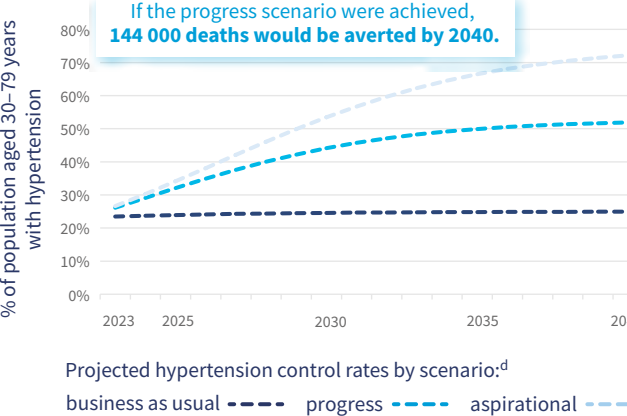
Of the 10.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	28	20	2019
Cardiovascular disease deaths	87 800	42 600	45 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	64	63	65	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	6	2019
Current tobacco use, adults aged 15+ years (%) ^f	20	34	7	2019
Obesity, adults aged 18+ years (%)	28	15	40	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	15	3	2019
Physical inactivity, adults aged 18+ years (%)	38	28	47	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

South Sudan

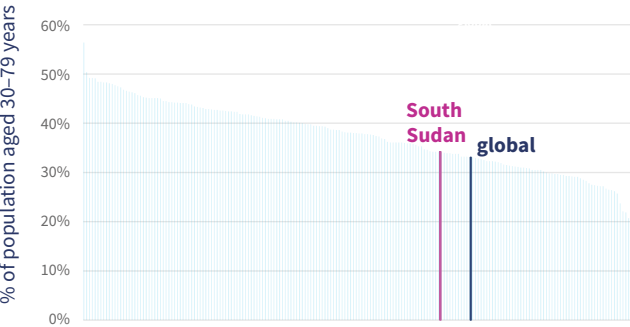
Hypertension profile

Total population (2019): 10 448 000

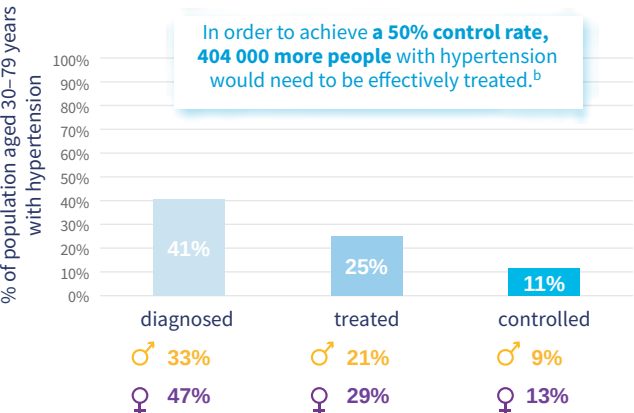
Total deaths (2019): 90 400

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 34% ♂ 32% ♀ 36%

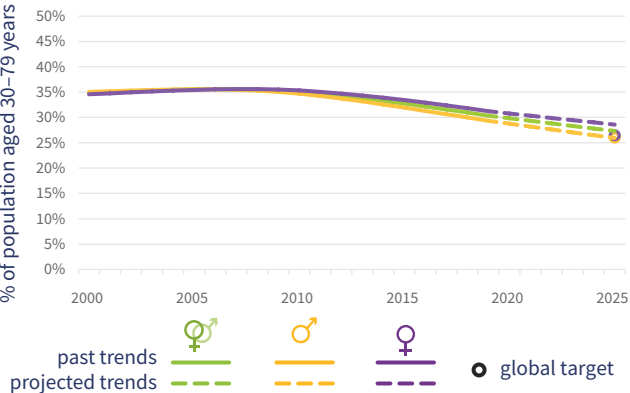
Prevalence of hypertension – global comparison (both sexes)^a



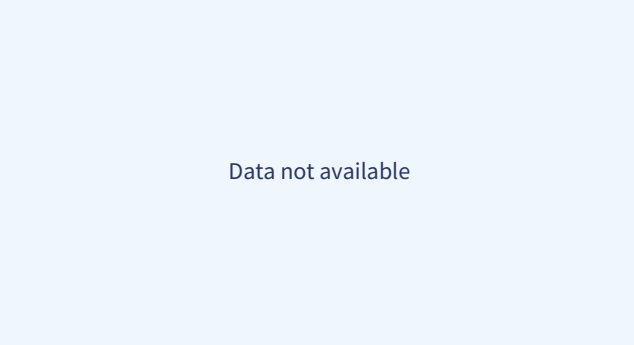
Of the 1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	17	18	16	2019
Cardiovascular disease deaths	8400	4200	4300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	59	58	60	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	8	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	no data	no data	no data	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	no data	no data	no data	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake.

Spain

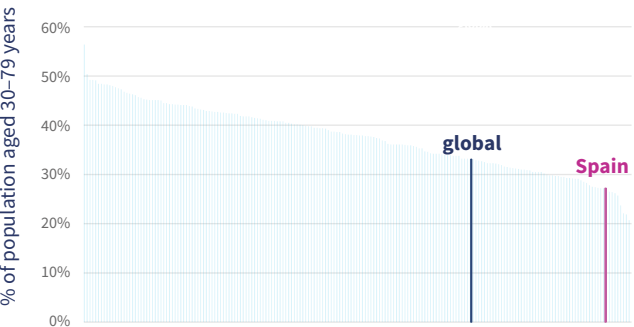
Hypertension profile

Total population (2019): 47 131 000

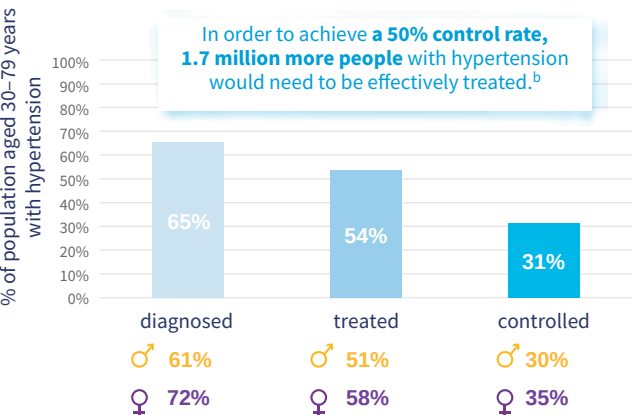
Total deaths (2019): 427 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 27% ♂ 34% ♀ 21%

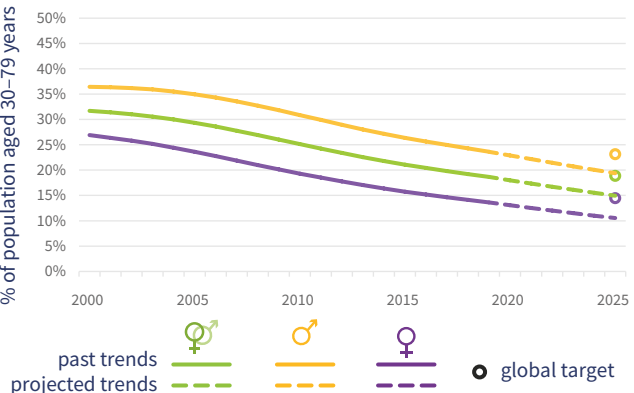
Prevalence of hypertension – global comparison (both sexes)^a



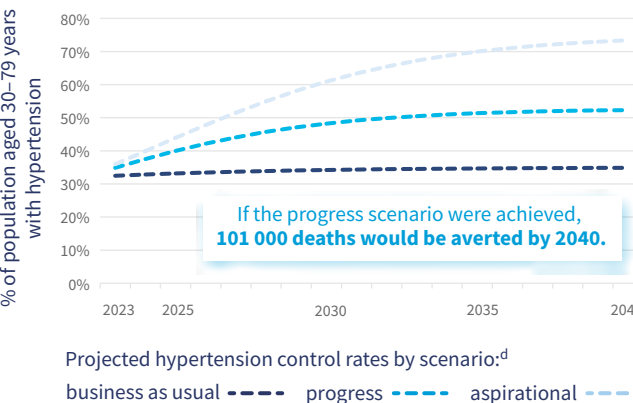
Of the 9.9 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	13	6	2019
Cardiovascular disease deaths	118 000	55 100	62 800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	47	49	46	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	28	29	27	2019
Obesity, adults aged 18+ years (%)	24	25	23	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	17	5	2019
Physical inactivity, adults aged 18+ years (%)	27	23	31	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Sri Lanka

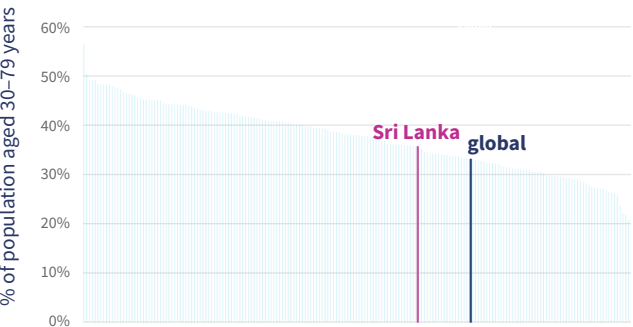
Hypertension profile

Total population (2019): 21 650 000

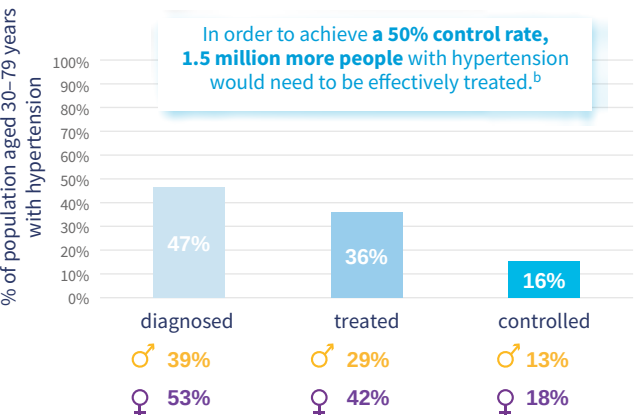
Total deaths (2019): 146 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 36% ♂ 34% ♀ 37%

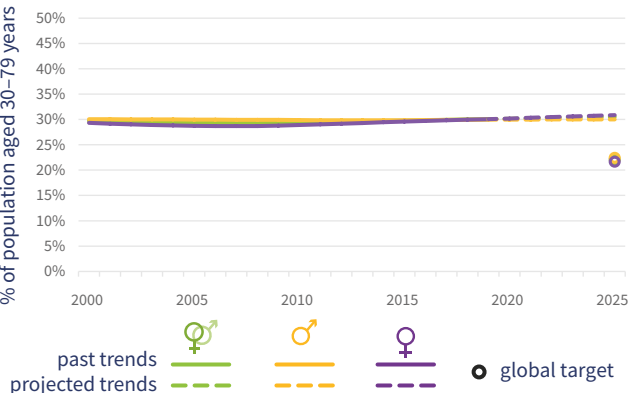
Prevalence of hypertension – global comparison (both sexes)^a



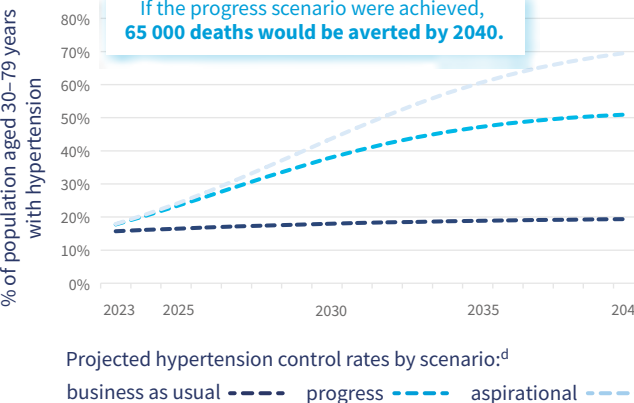
Of the 4.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	13	17	9	2019
Cardiovascular disease deaths	35 900	19 600	16 300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	56	56	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	11	11	10	2019
Current tobacco use, adults aged 15+ years (%)	22	42	3	2019
Obesity, adults aged 18+ years (%)	5	3	7	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	29	20	37	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Sudan

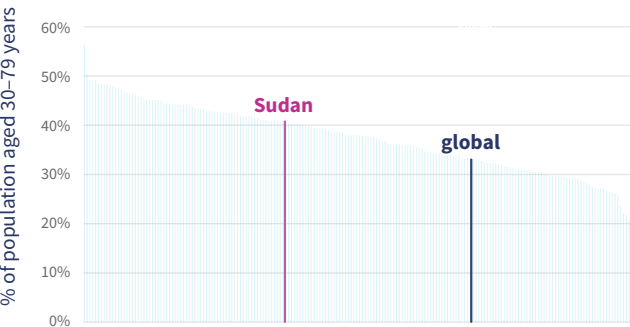
Hypertension profile

Total population (2019): 43 232 000

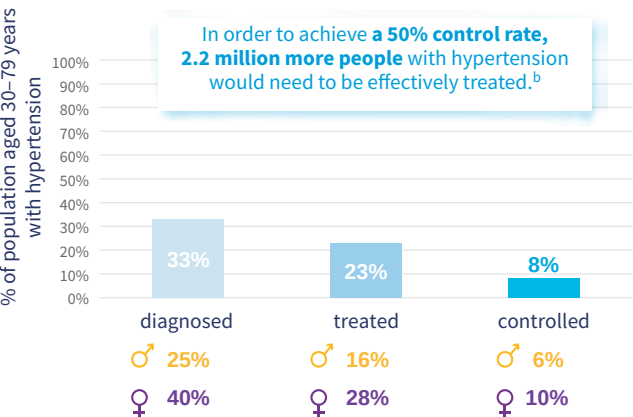
Total deaths (2019): 241 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 38% ♀ 44%

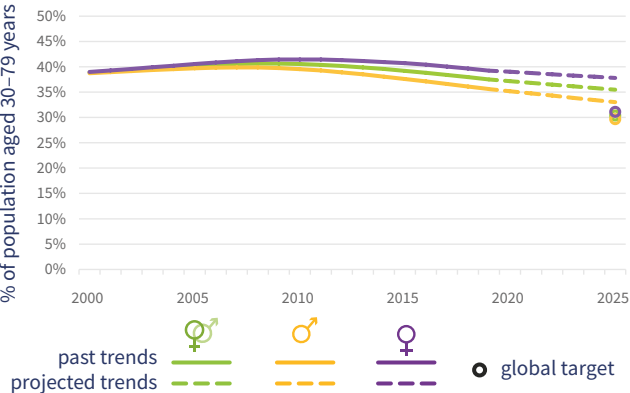
Prevalence of hypertension – global comparison (both sexes)^a



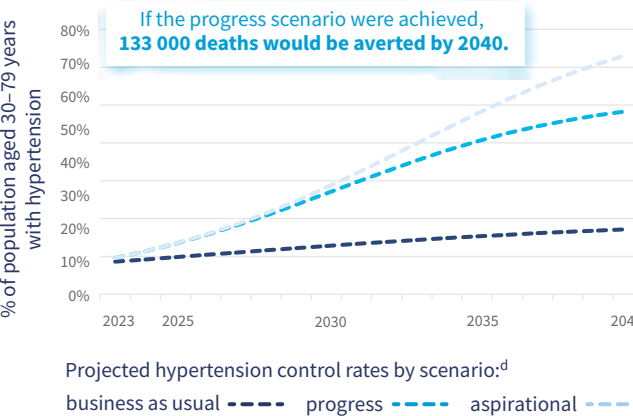
Of the 5.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	24	21	2019
Cardiovascular disease deaths	76 800	38 500	38 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	64	62	66	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	no data	no data	no data	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Suriname

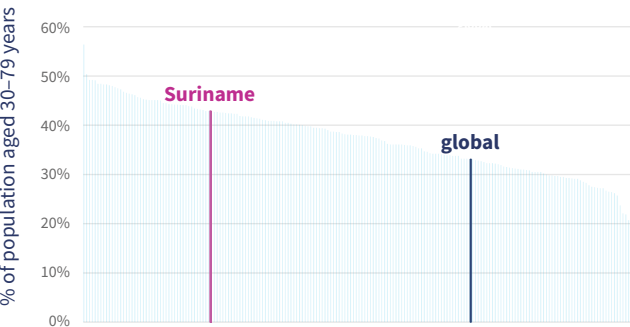
Hypertension profile

Total population (2019): 600 000

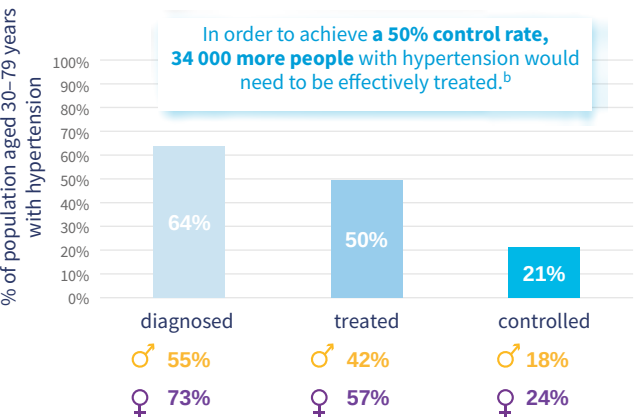
Total deaths (2019): 4490

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 43% ♂ 42% ♀ 43%

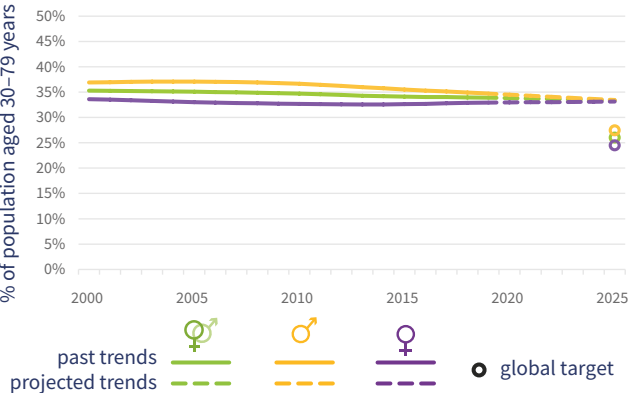
Prevalence of hypertension – global comparison (both sexes)^a



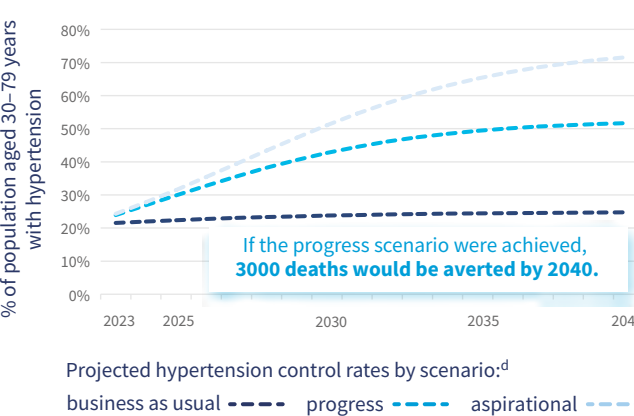
Of the 116 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	23	28	18	2019
Cardiovascular disease deaths	1500	810	710	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	49	49	48	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	26	19	34	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	7	11	3	2019
Physical inactivity, adults aged 18+ years (%)	44	38	51	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Sweden

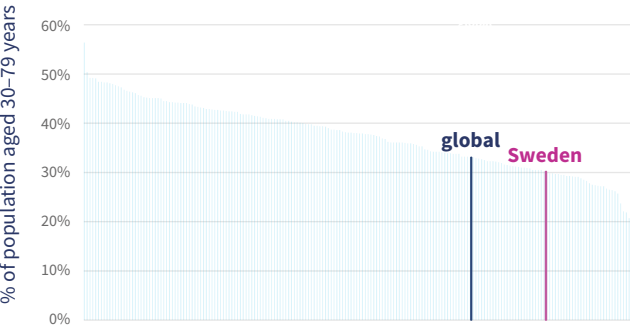
Hypertension profile

Total population (2019): 10 268 000

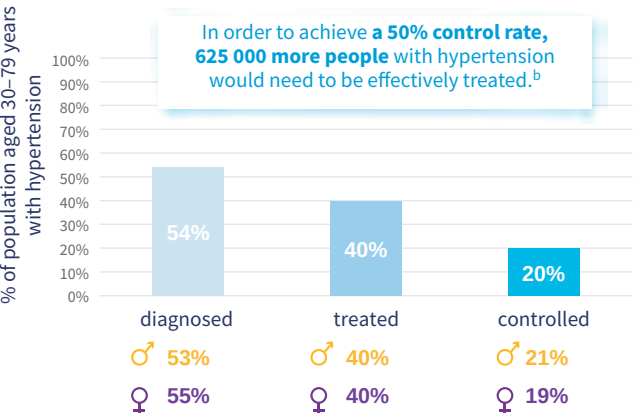
Total deaths (2019): 92 200

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 30% ♂ 36% ♀ 25%

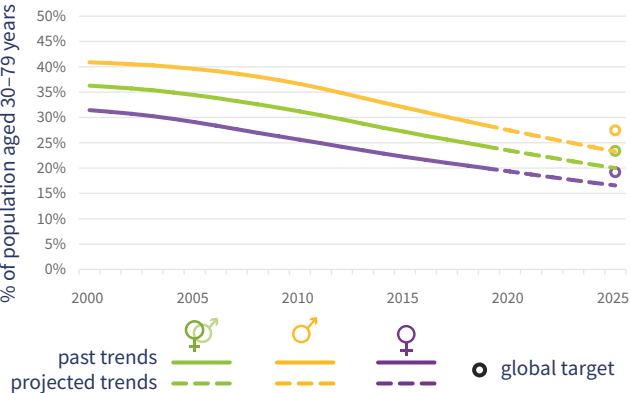
Prevalence of hypertension – global comparison (both sexes)^a



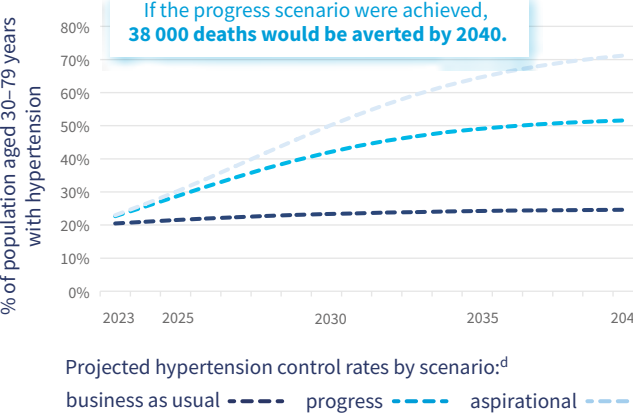
Of the 2.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	8	10	7	2019
Cardiovascular disease deaths	29 700	14 600	15 100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	50	51	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%)	25	31	19	2019
Obesity, adults aged 18+ years (%)	21	23	18	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	14	4	2019
Physical inactivity, adults aged 18+ years (%)	23	22	25	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	no data
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Switzerland

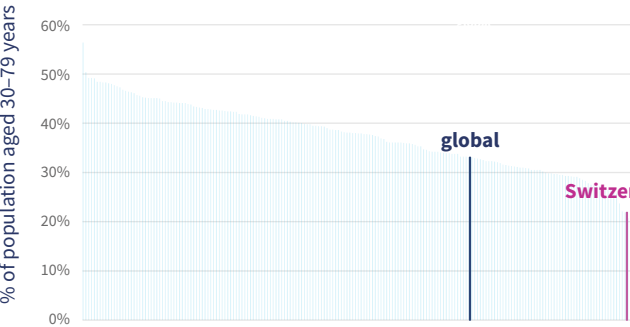
Hypertension profile

Total population (2019): 8 576 000

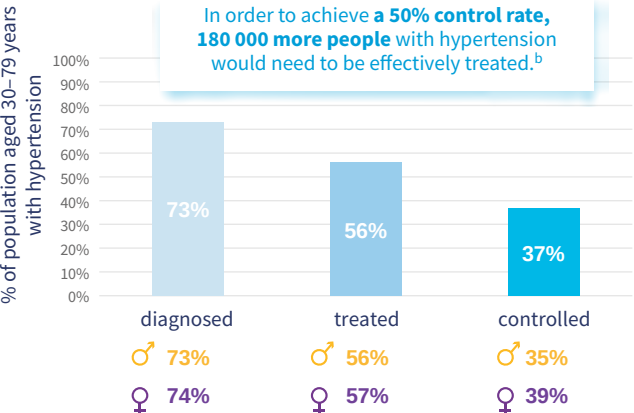
Total deaths (2019): 69 100

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 22% ♂ 26% ♀ 18%

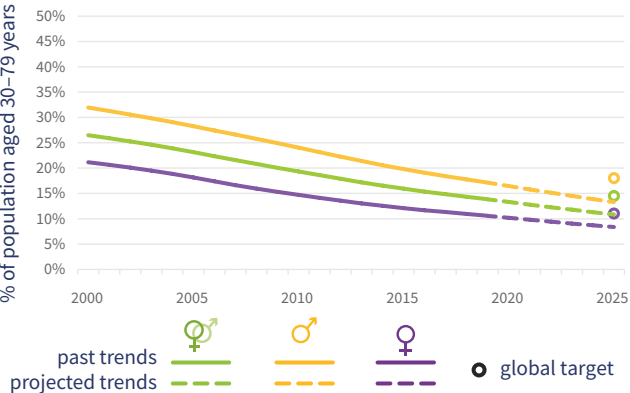
Prevalence of hypertension – global comparison (both sexes)^a



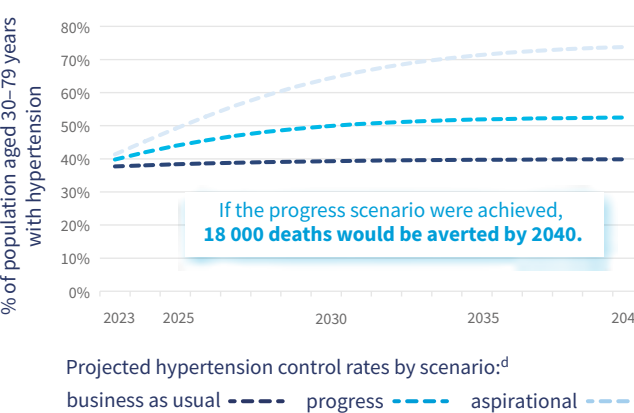
Of the 1.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	8	10	6	2019
Cardiovascular disease deaths	21 300	9700	11 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	46	47	46	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	9	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	26	28	23	2019
Obesity, adults aged 18+ years (%)	20	22	17	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	16	5	2019
Physical inactivity, adults aged 18+ years (%)	24	22	26	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Syrian Arab Republic

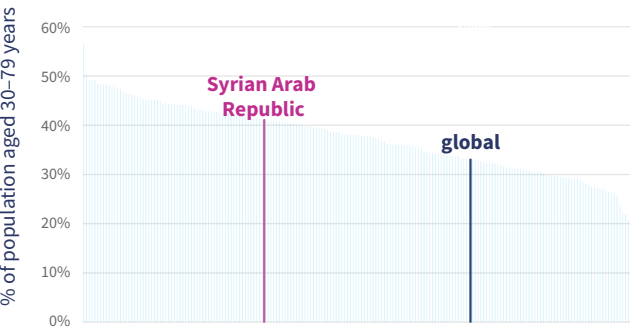
Hypertension profile

Total population (2019): 20 098 000

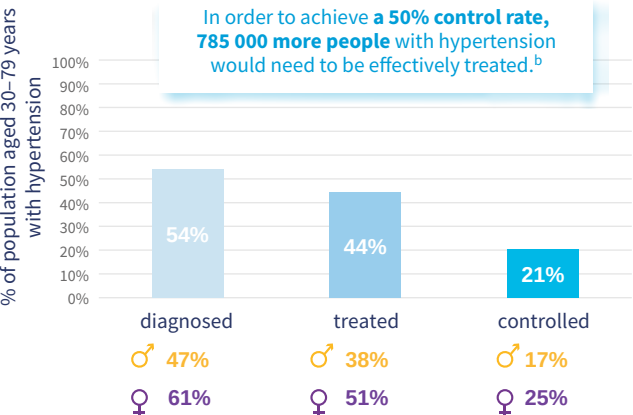
Total deaths (2019): 89 300

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 41% ♂ 42% ♀ 40%

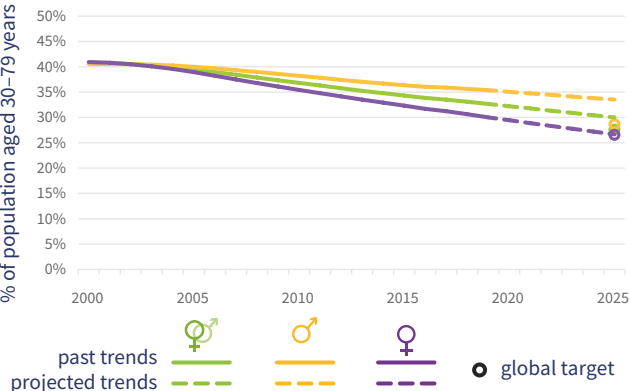
Prevalence of hypertension – global comparison (both sexes)^a



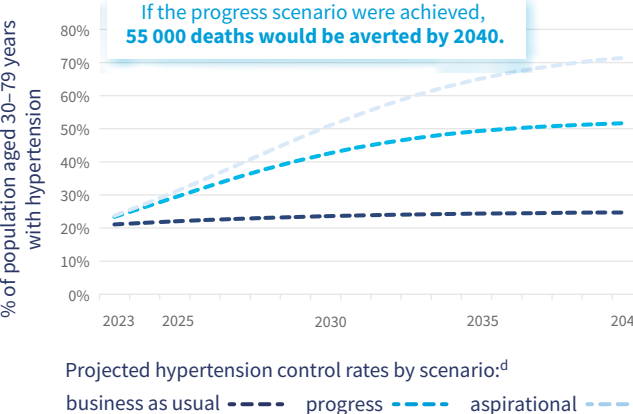
Of the 2.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	22	26	18	2019
Cardiovascular disease deaths	39 000	19 200	19 800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	52	53	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	28	21	35	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Tajikistan

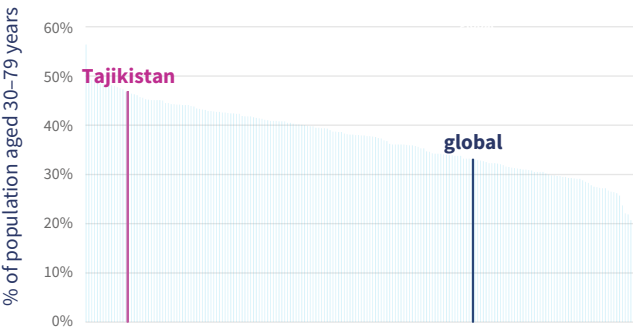
Hypertension profile

Total population (2019): 9 337 000

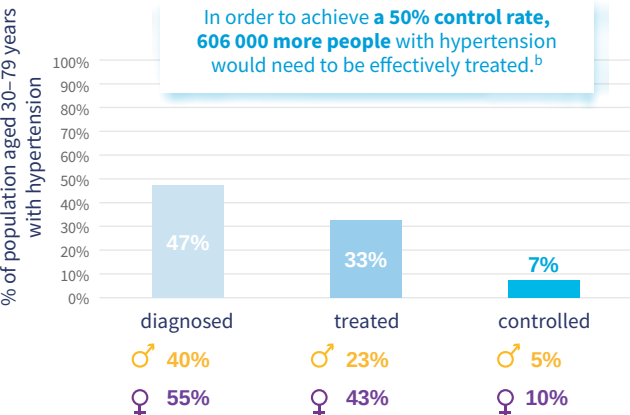
Total deaths (2019): 50 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 47% ♂ 51% ♀ 43%

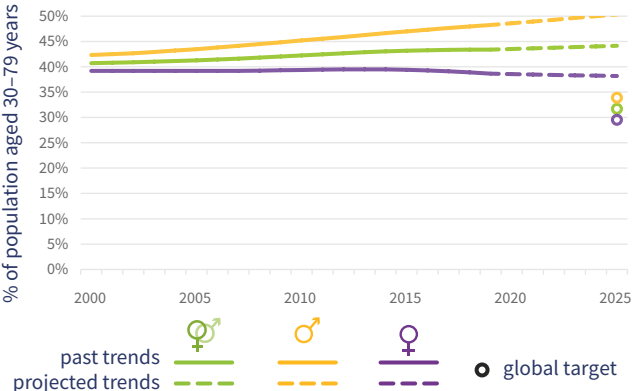
Prevalence of hypertension – global comparison (both sexes)^a



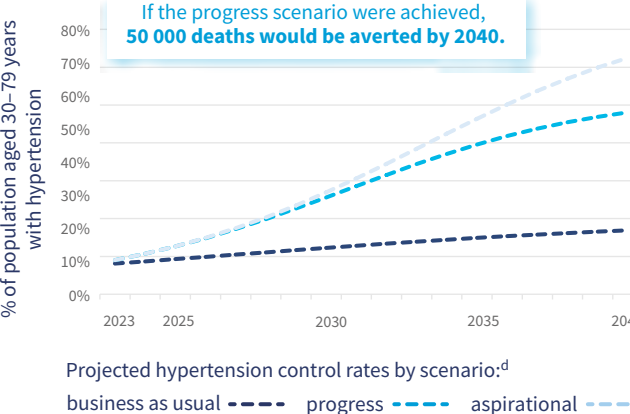
Of the 1.4 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	28	32	25	2019
Cardiovascular disease deaths	23 100	12 400	10 700	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	59	60	58	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	14	12	17	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	2	0	2019
Physical inactivity, adults aged 18+ years (%)	29	20	39	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Thailand

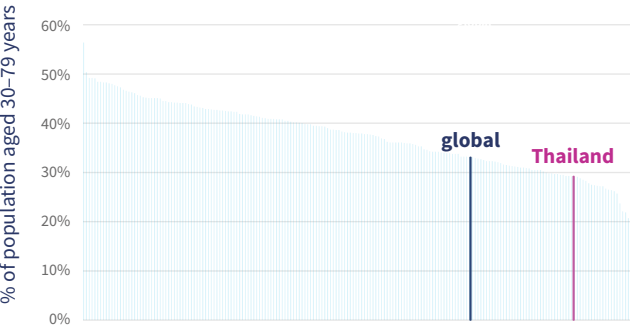
Hypertension profile

Total population (2019): 71 308 000

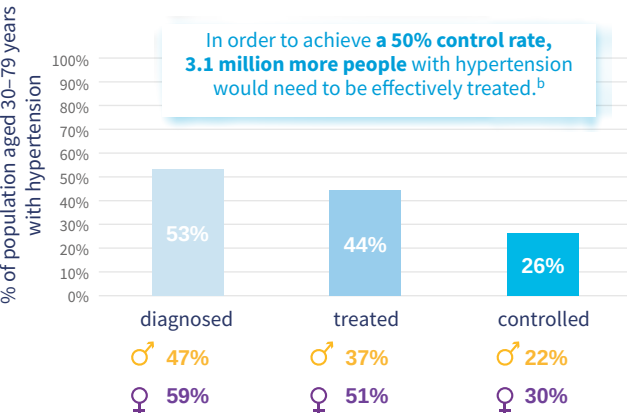
Total deaths (2019): 497 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 29% ♂ 29% ♀ 29%

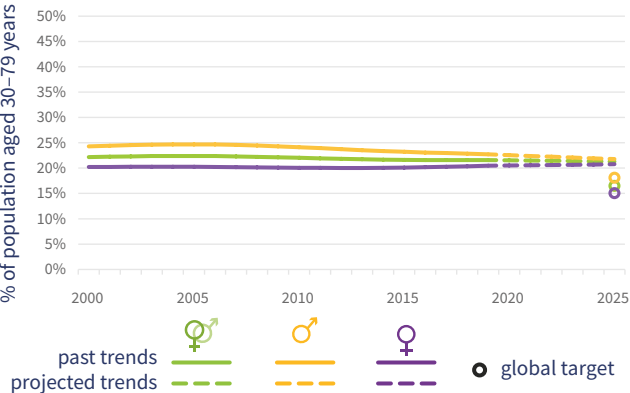
Prevalence of hypertension – global comparison (both sexes)^a



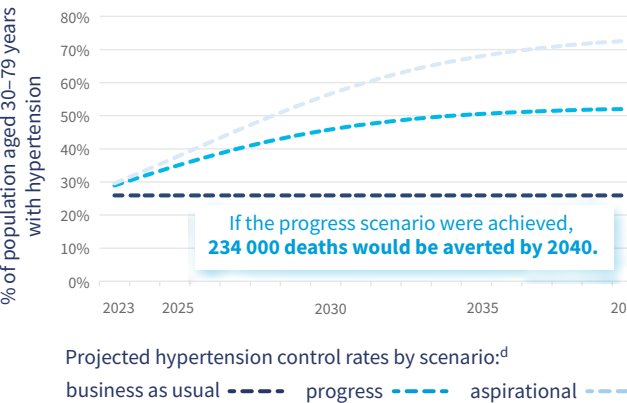
Of the 13.5 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	14	17	11	2019
Cardiovascular disease deaths	115 000	59 800	55 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	44	46	42	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	11	11	10	2019
Current tobacco use, adults aged 15+ years (%)	23	42	3	2019
Obesity, adults aged 18+ years (%)	10	7	13	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	8	13	3	2019
Physical inactivity, adults aged 18+ years (%)	25	22	27	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Timor-Leste

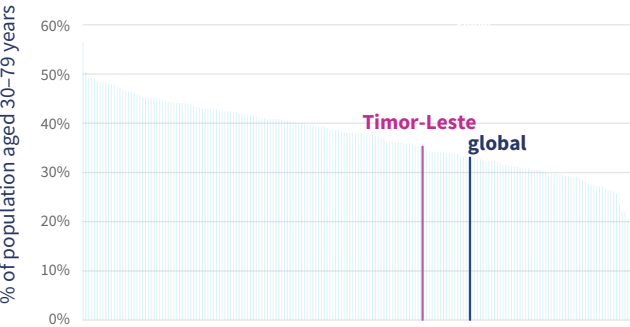
Hypertension profile

Total population (2019): 1 280 000

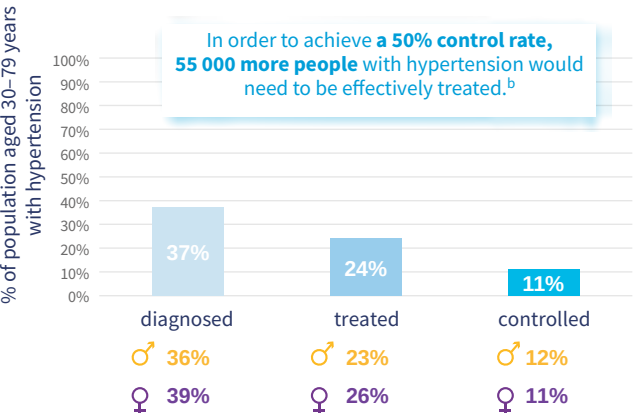
Total deaths (2019): 7550

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 35% ♂ 34% ♀ 37%

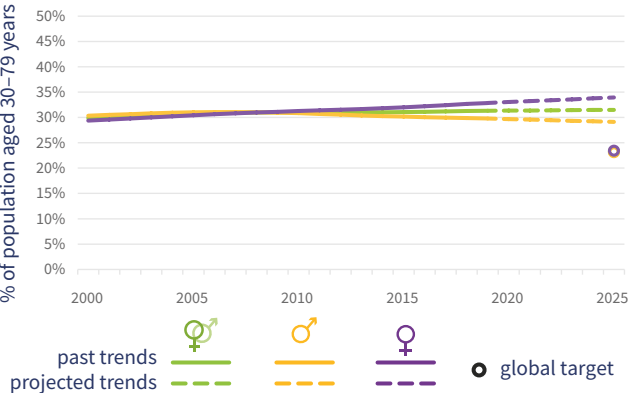
Prevalence of hypertension – global comparison (both sexes)^a



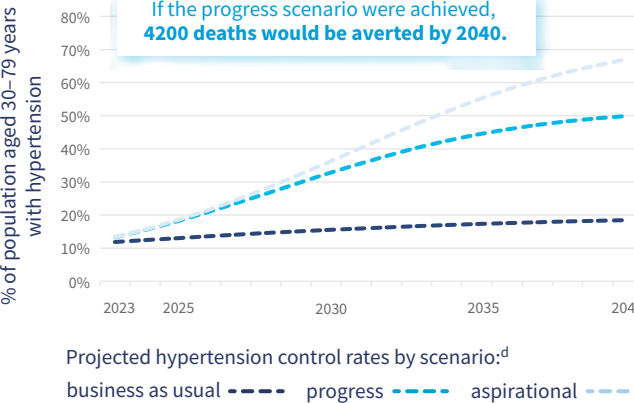
Of the 141 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	20	21	18	2019
Cardiovascular disease deaths	2000	970	1000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	62	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	10	2019
Current tobacco use, adults aged 15+ years (%)	40	68	11	2019
Obesity, adults aged 18+ years (%)	4	3	5	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	1	0	2019
Physical inactivity, adults aged 18+ years (%)	18	10	25	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Togo

Hypertension profile

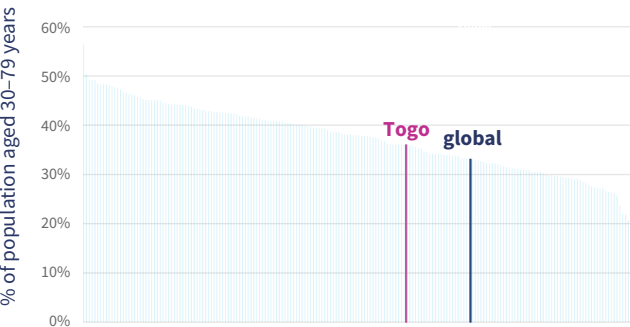
Total population (2019): 8 243 000

Total deaths (2019): 55 200

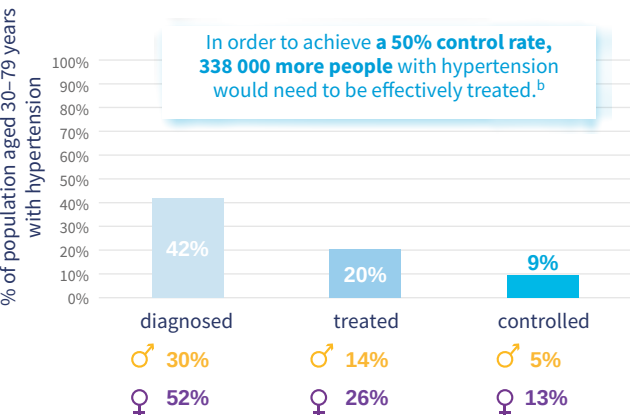
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

36% 34% 38%

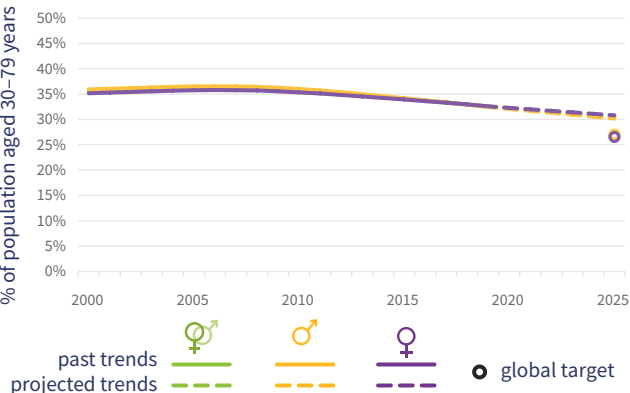
Prevalence of hypertension – global comparison (both sexes)^a



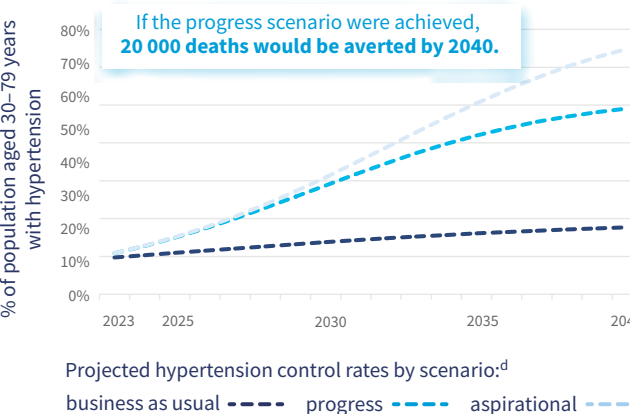
Of the 822 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	24	27	21	2019
Cardiovascular disease deaths	9300	5100	4200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	61	59	63	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%)	7	13	1	2019
Obesity, adults aged 18+ years (%)	8	4	13	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	2	1	2019
Physical inactivity, adults aged 18+ years (%)	10	9	10	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Tonga

Hypertension profile

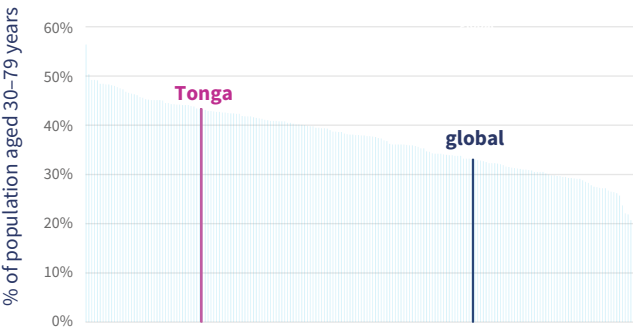
Total population (2019): 105 000

Total deaths (2019): 620

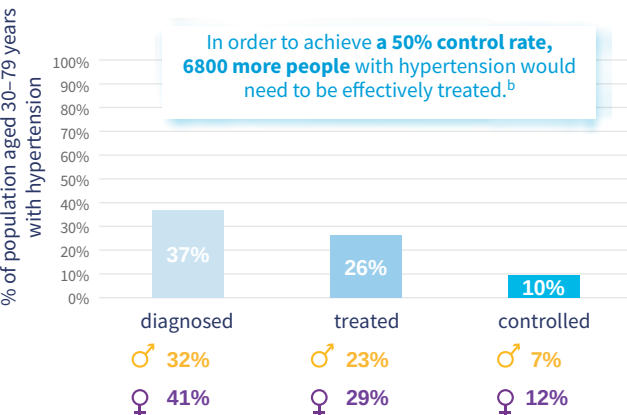
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

43% 40% 47%

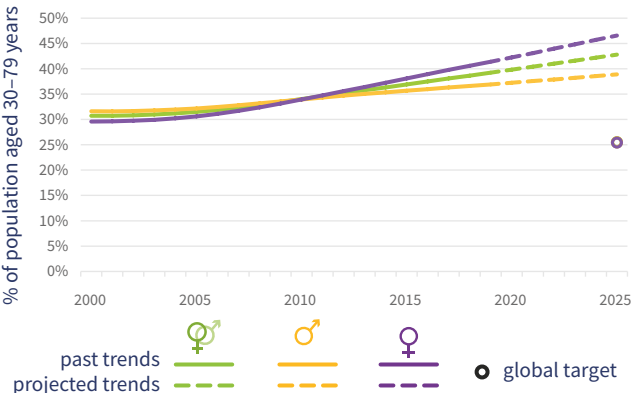
Prevalence of hypertension – global comparison (both sexes)^a



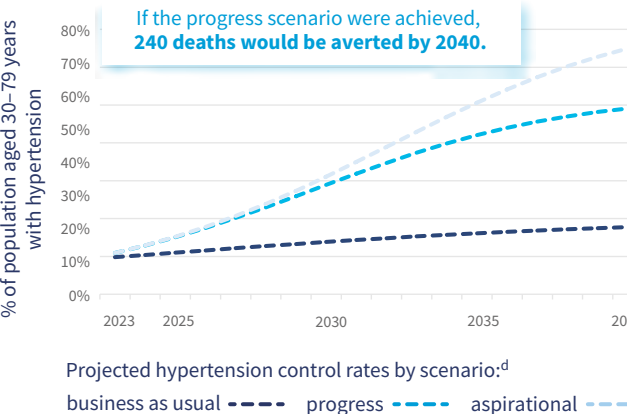
Of the 17 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	30	20	2019
Cardiovascular disease deaths	150	90	60	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	54	55	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	31	47	15	2019
Obesity, adults aged 18+ years (%)	48	41	55	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	1	0	2019
Physical inactivity, adults aged 18+ years (%)	17	8	26	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Trinidad and Tobago

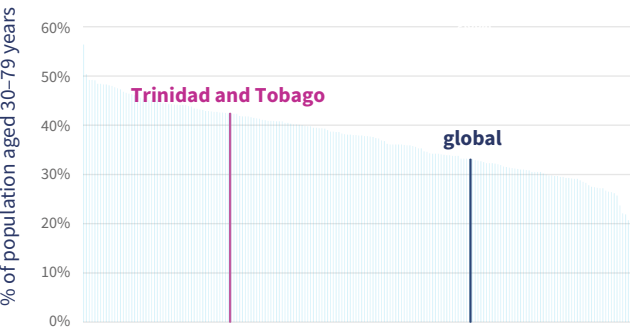
Hypertension profile

Total population (2019): 1 520 000

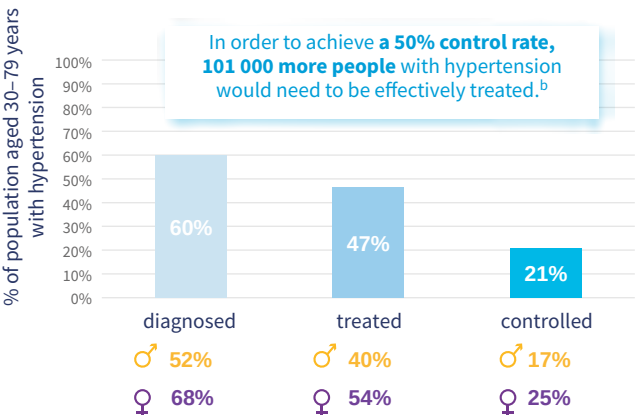
Total deaths (2019): 9440

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 42% ♂ 43% ♀ 42%

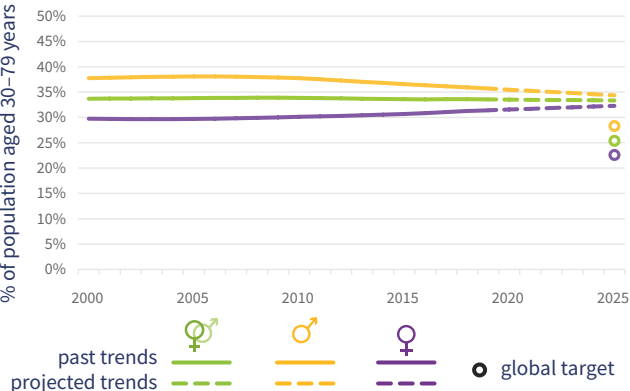
Prevalence of hypertension – global comparison (both sexes)^a



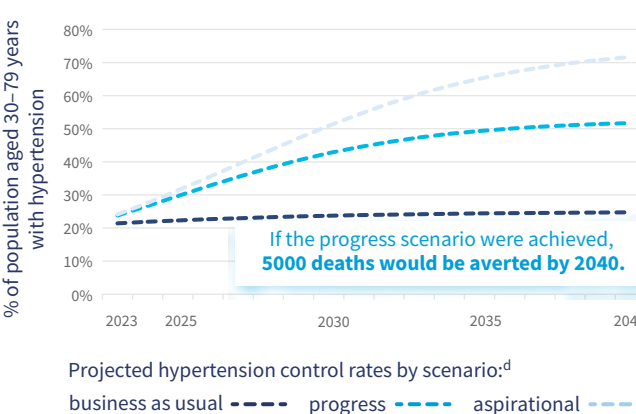
Of the 354 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	17	20	14	2019
Cardiovascular disease deaths	2800	1600	1200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	64	50	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	19	11	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	6	10	2	2019
Physical inactivity, adults aged 18+ years (%)	38	27	49	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Tunisia

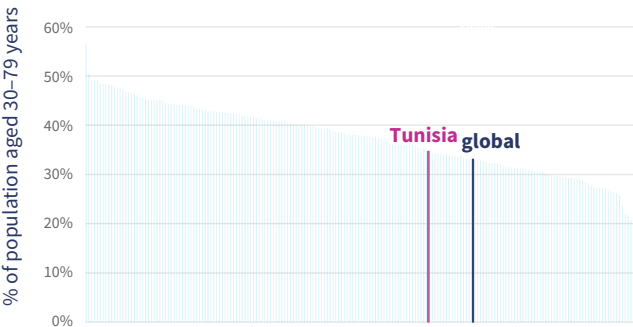
Hypertension profile

Total population (2019): 12 049 000

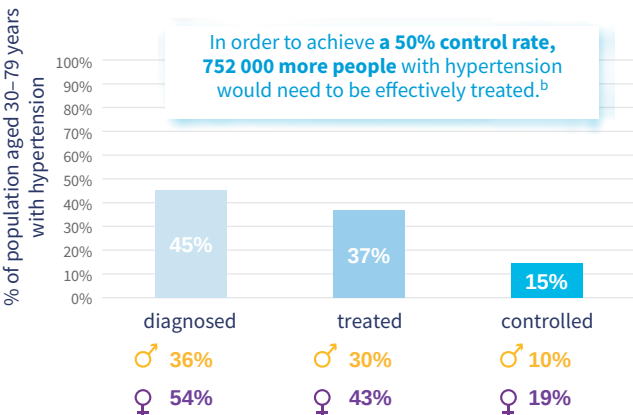
Total deaths (2019): 67 600

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 35% ♂ 35% ♀ 35%

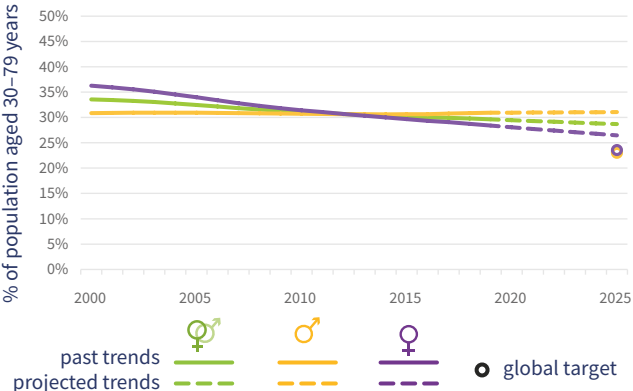
Prevalence of hypertension – global comparison (both sexes)^a



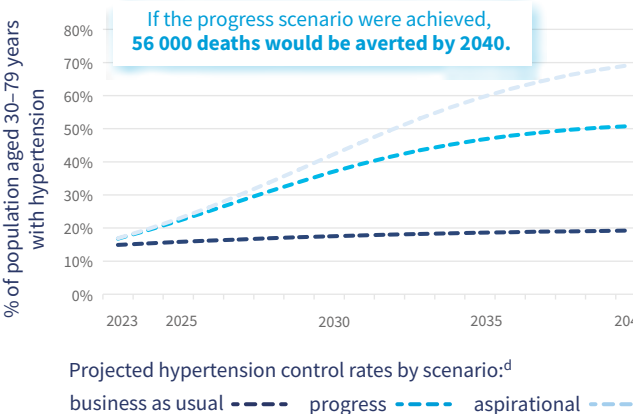
Of the 2.1 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	19	12	2019
Cardiovascular disease deaths	33 900	17 300	16 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	51	50	53	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	25	48	2	2019
Obesity, adults aged 18+ years (%)	27	19	34	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	4	0	2019
Physical inactivity, adults aged 18+ years (%)	30	26	34	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Türkiye

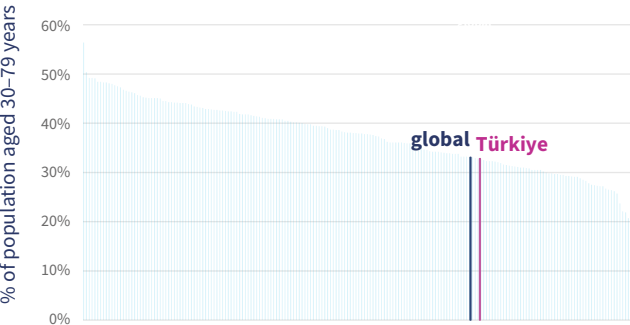
Hypertension profile

Total population (2019): 83 482 000

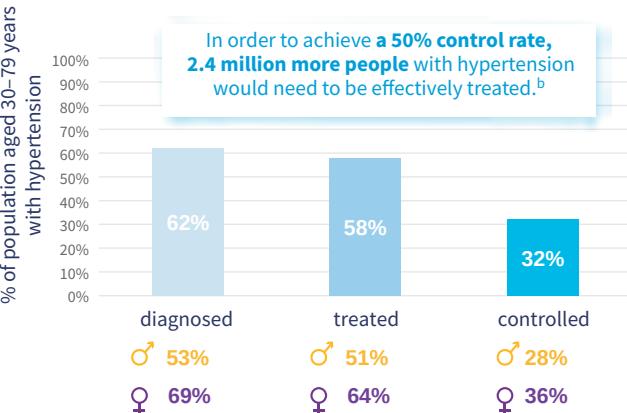
Total deaths (2019): 432 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 33% ♂ 31% ♀ 34%

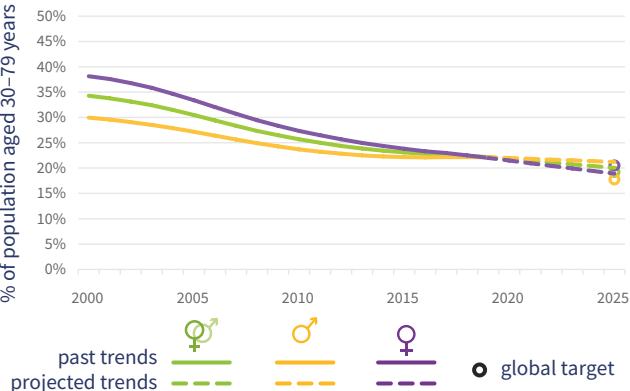
Prevalence of hypertension – global comparison (both sexes)^a



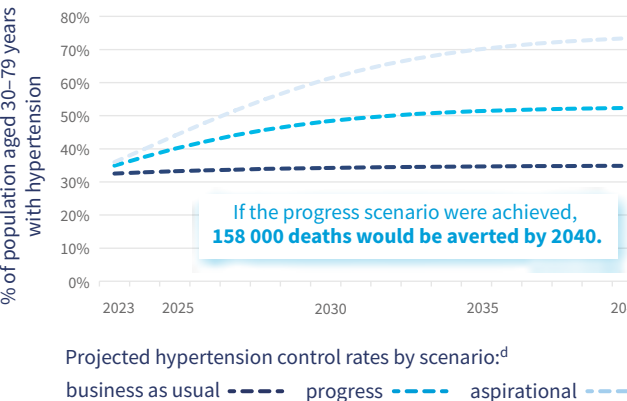
Of the 13.8 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	21	11	2019
Cardiovascular disease deaths	155 000	67 000	88 500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	55	53	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	5	6	4	2019
Current tobacco use, adults aged 15+ years (%)	31	43	19	2019
Obesity, adults aged 18+ years (%)	32	24	39	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	0	2019
Physical inactivity, adults aged 18+ years (%)	31	22	39	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Turkmenistan

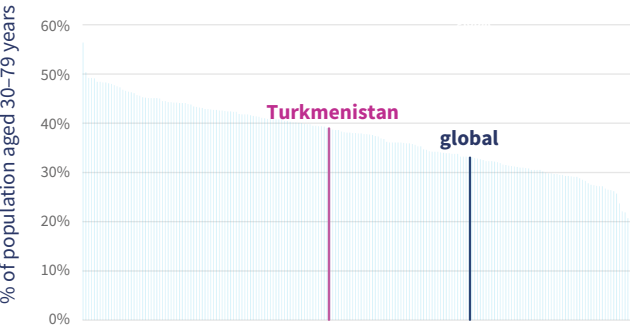
Hypertension profile

Total population (2019): 6 158 000

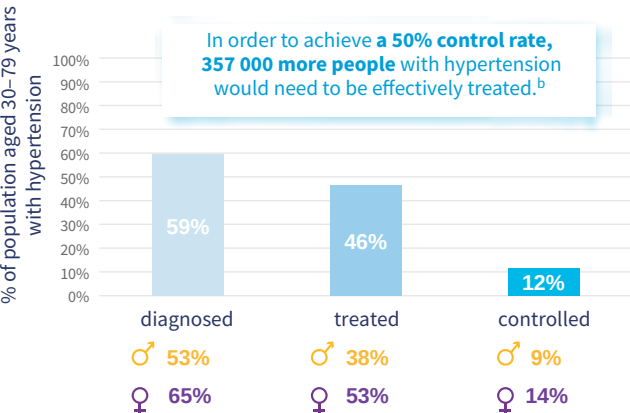
Total deaths (2019): 36 900

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 39% ♂ 38% ♀ 40%

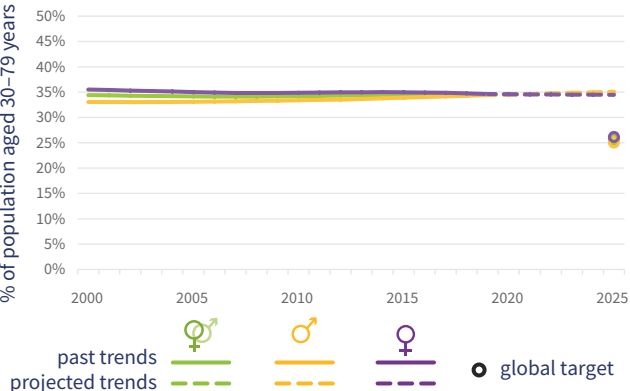
Prevalence of hypertension – global comparison (both sexes)^a



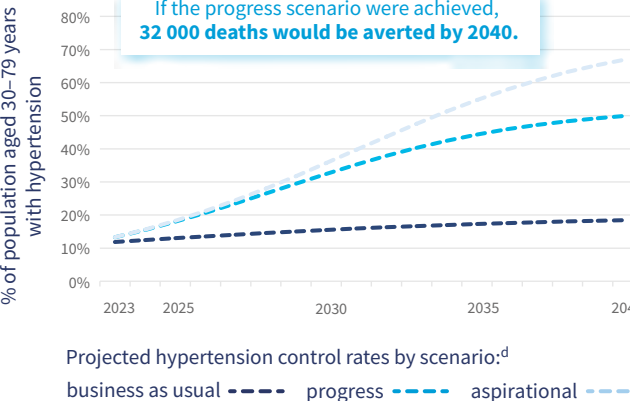
Of the 917 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	28	34	22	2019
Cardiovascular disease deaths	16 300	8500	7800	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	61	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%)	6	11	1	2019
Obesity, adults aged 18+ years (%)	19	16	21	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Tuvalu

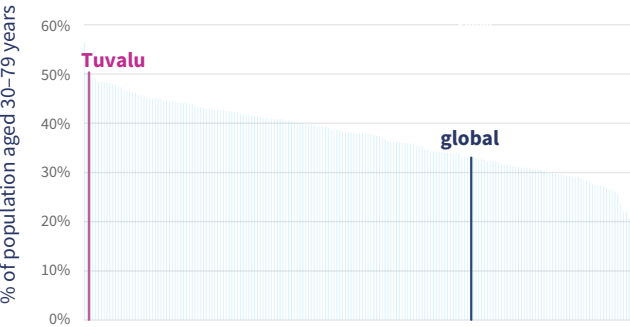
Hypertension profile

Total population (2019): 11 000

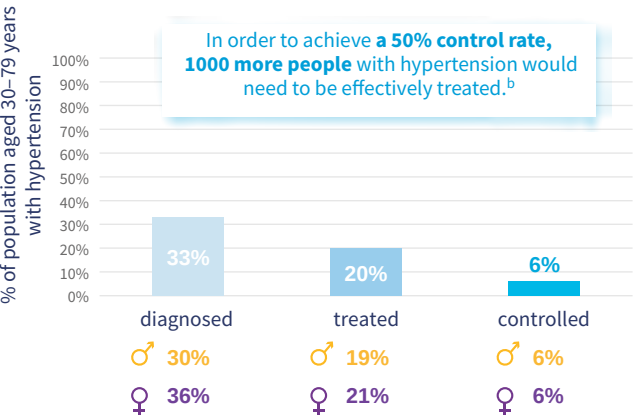
Total deaths (2019): no data

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 50% ♂ 49% ♀ 51%

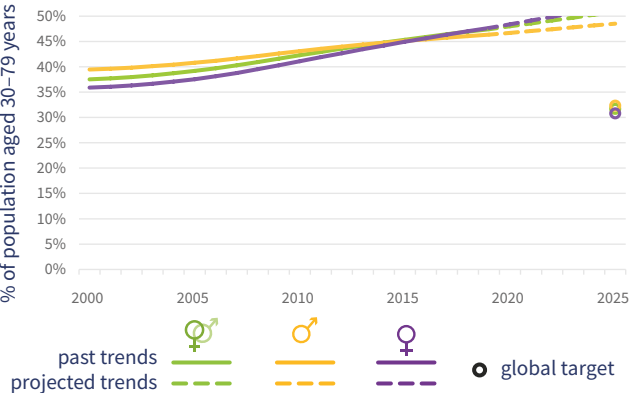
Prevalence of hypertension – global comparison (both sexes)^a



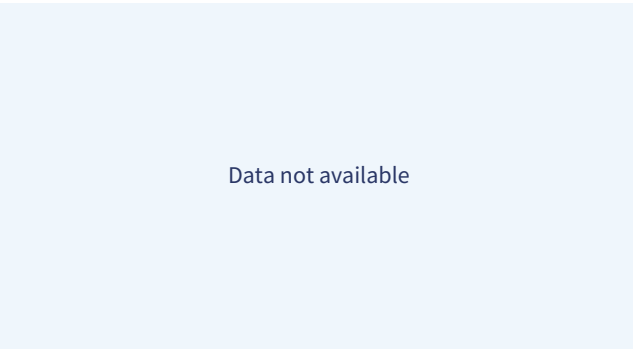
Of the 2300 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	no data	no data	no data	2019
Cardiovascular disease deaths	no data	no data	no data	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	53	52	2019

Risk factors^d

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	7	2019
Current tobacco use, adults aged 15+ years (%) ^e	36	50	22	2019
Obesity, adults aged 18+ years (%)	52	47	56	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	1	2	0	2019
Physical inactivity, adults aged 18+ years (%)	27	18	37	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✗		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✗		
Conducted recent, national survey on salt/sodium intake	✗		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Age-standardized estimates are presented for all indicators except salt intake. e. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Uganda

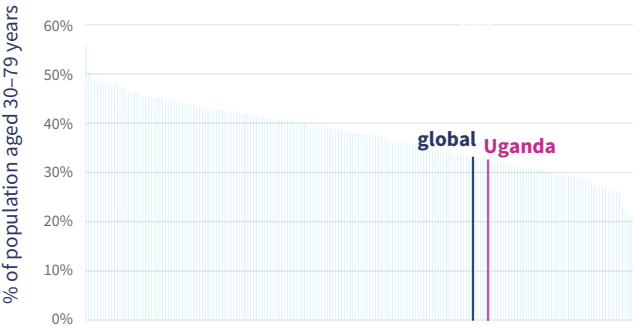
Hypertension profile

Total population (2019): 42 949 000

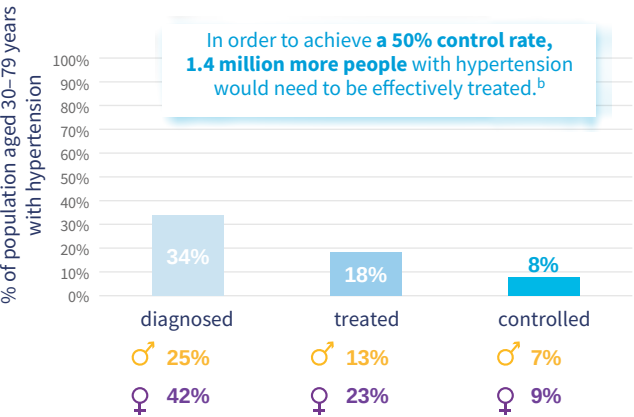
Total deaths (2019): 228 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 33% ♂ 31% ♀ 34%

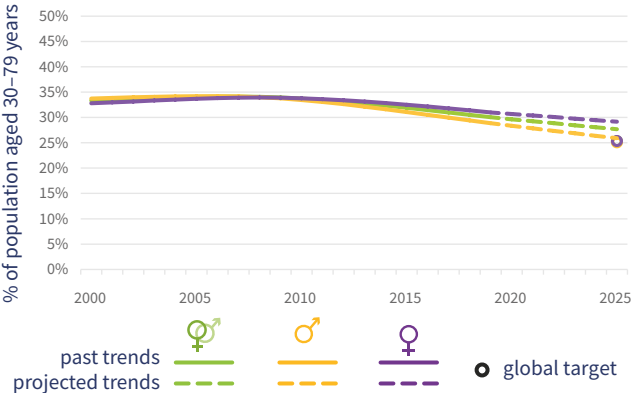
Prevalence of hypertension – global comparison (both sexes)^a



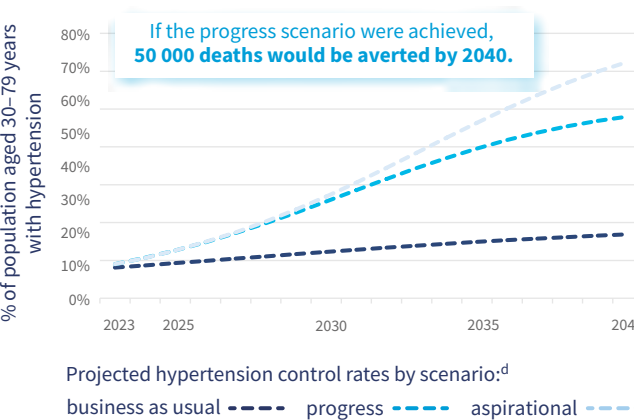
Of the 3.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	25	18	2019
Cardiovascular disease deaths	25 900	13 300	12 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	59	55	63	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	9	14	4	2019
Obesity, adults aged 18+ years (%)	5	2	9	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	12	20	5	2019
Physical inactivity, adults aged 18+ years (%)	6	5	6	2016

National response

Targets		Treatment	
National target for blood pressure	✓	Guidelines for management of hypertension	✓
National target for salt consumption	✓		
Surveillance			
Conducted recent, national survey measuring raised blood pressure/hypertension	✓		
Conducted recent, national survey on salt/sodium intake	✓		
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗		

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Ukraine

Hypertension profile

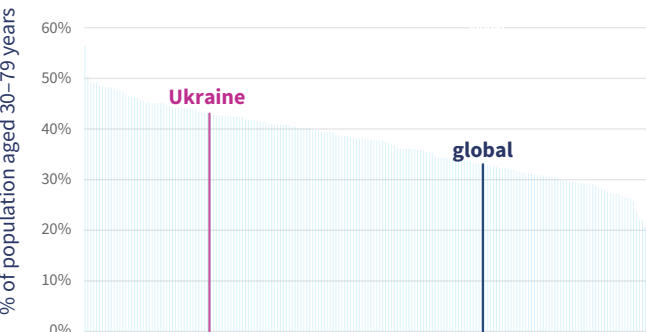
Total population (2019): 44 211 000

Total deaths (2019): 594 000

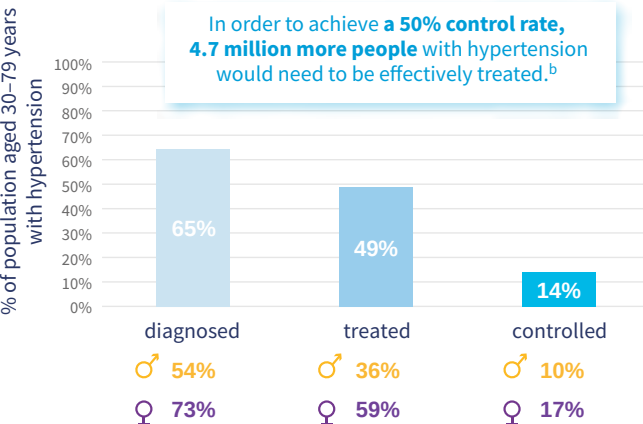
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

♀ 43% ♂ 45% ♀ 42%

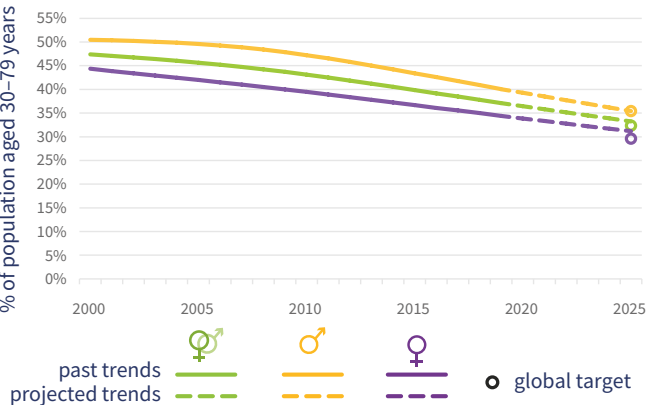
Prevalence of hypertension – global comparison (both sexes)^a



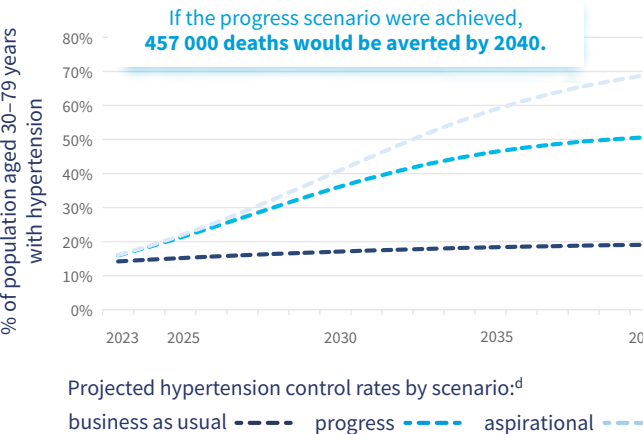
Of the **13.2 million** adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	26	37	16	2019
Cardiovascular disease deaths	383 000	168 000	214 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	53	55	51	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%)	26	41	12	2019
Obesity, adults aged 18+ years (%)	24	22	26	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	15	4	2019
Physical inactivity, adults aged 18+ years (%)	20	19	20	2016

National response

Targets	Treatment
National target for blood pressure	✗
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

United Arab Emirates

Hypertension profile

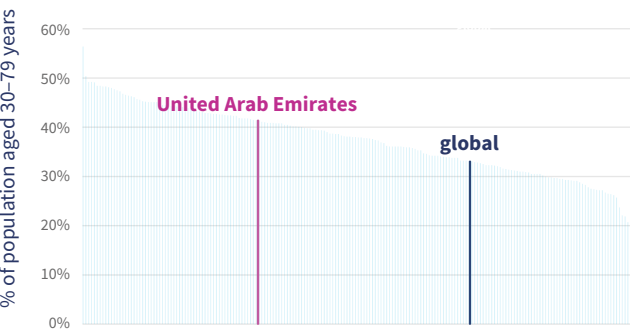
Total population (2019): 9 212 000

Total deaths (2019): 20 900

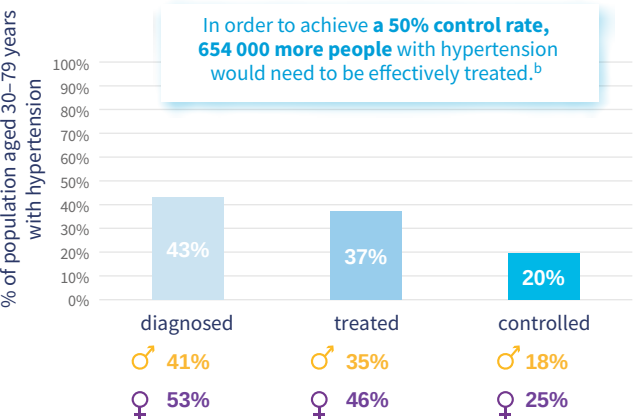
Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a

♀ 41% ♂ 44% ♀ 35%

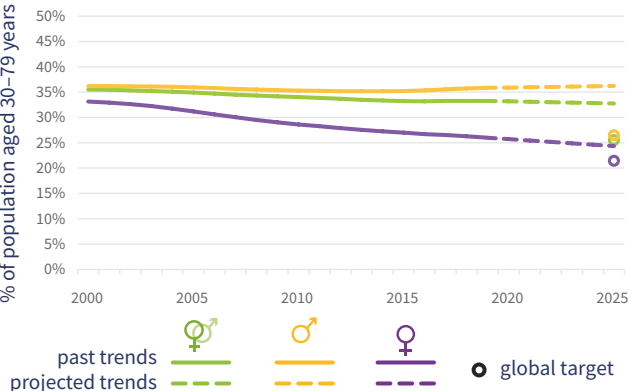
Prevalence of hypertension – global comparison (both sexes)^a



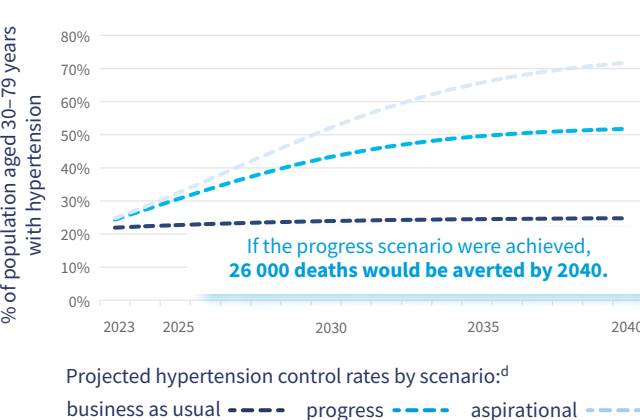
Of the **1.8 million** adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	19	20	15	2019
Cardiovascular disease deaths	7600	6100	1500	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	61	61	59	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	5	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	32	28	41	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	0	2019
Physical inactivity, adults aged 18+ years (%)	41	39	49	2016

National response

Targets	Treatment
National target for blood pressure	✓
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

United Kingdom of Great Britain and Northern Ireland

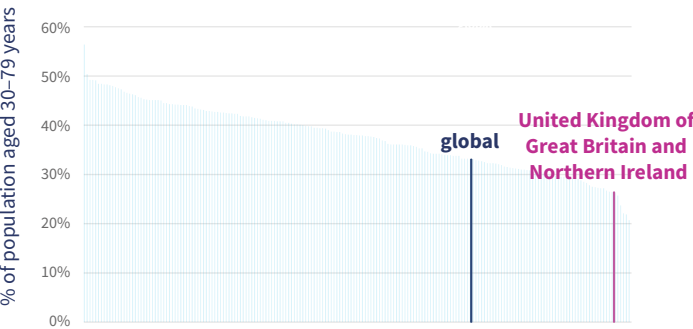
Total population (2019): 66 779 000

Hypertension profile

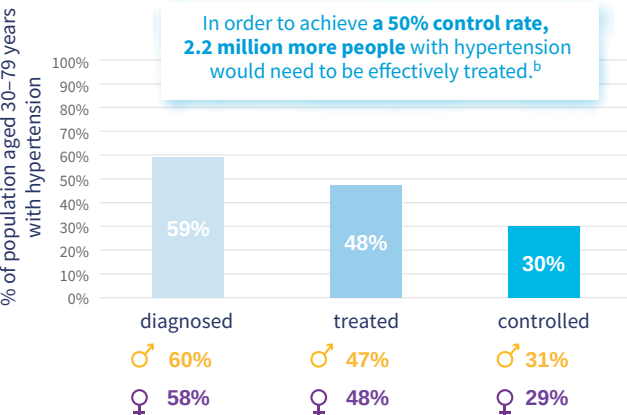
Total deaths (2019): 615 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 26% ♂ 30% ♀ 23%

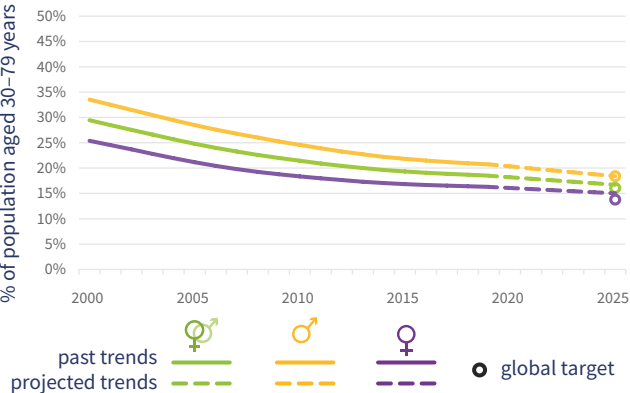
Prevalence of hypertension – global comparison (both sexes)^a



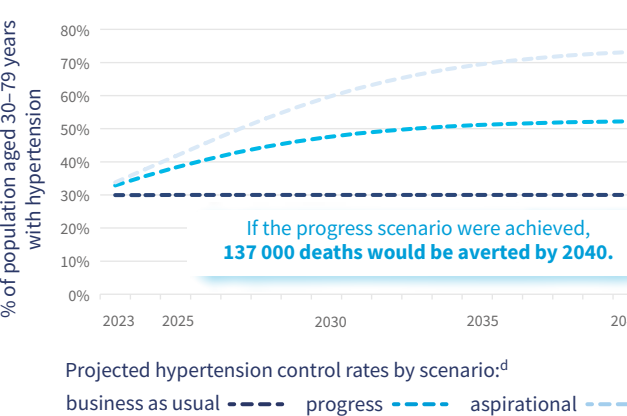
Of the 12.6 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	10	12	9	2019
Cardiovascular disease deaths	146 000	76 600	69 300	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	44	45	43	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	8	6	2019
Current tobacco use, adults aged 15+ years (%) ^f	16	18	14	2019
Obesity, adults aged 18+ years (%)	28	27	29	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	11	17	5	2019
Physical inactivity, adults aged 18+ years (%)	36	32	40	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

United Republic of Tanzania

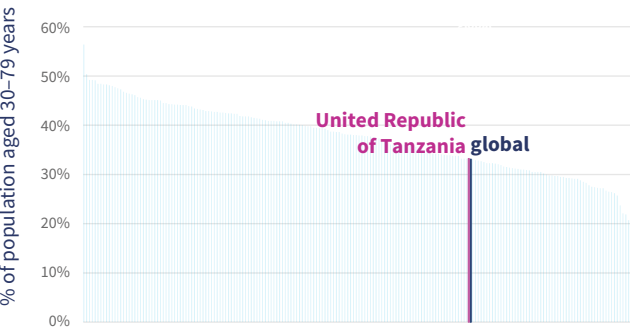
Total population (2019): 59 873 000

Hypertension profile

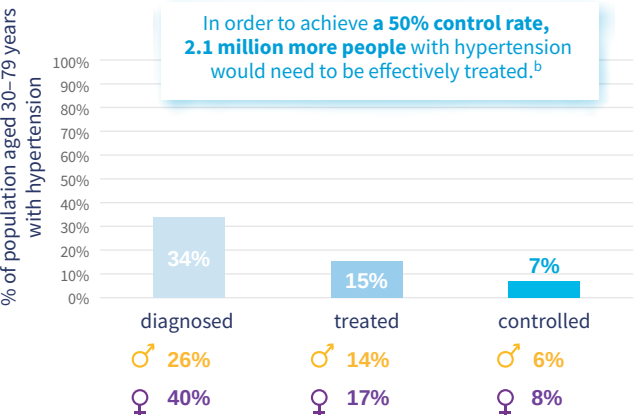
Total deaths (2019): 322 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 33% ♂ 31% ♀ 35%

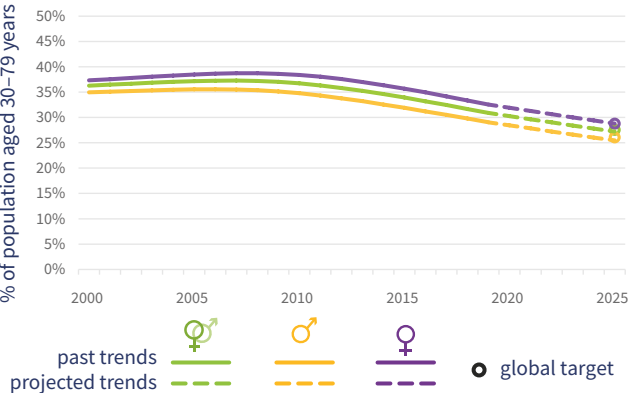
Prevalence of hypertension – global comparison (both sexes)^a



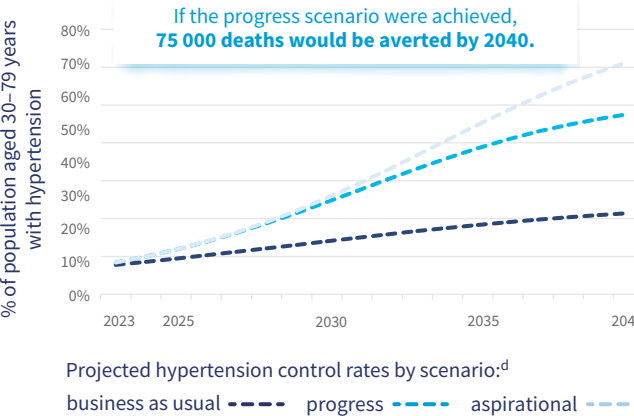
Of the 4.9 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	17	18	17	2019
Cardiovascular disease deaths	39 500	19 100	20 400	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	65	71	58	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	8	2019
Current tobacco use, adults aged 15+ years (%)	9	15	4	2019
Obesity, adults aged 18+ years (%)	8	4	13	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	18	4	2019
Physical inactivity, adults aged 18+ years (%)	6	6	7	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

United States of America

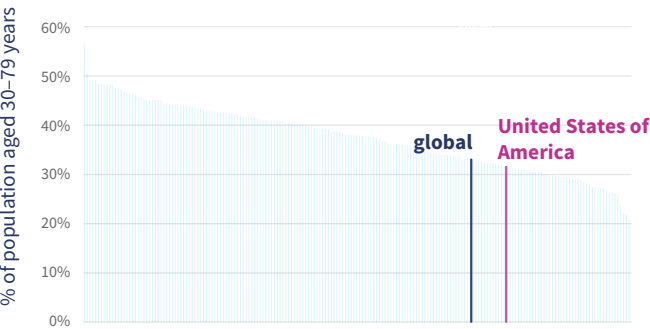
Hypertension profile

Total population (2019): 334 300 000

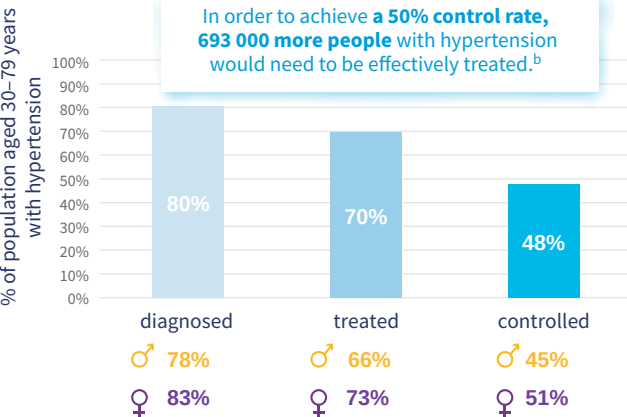
Total deaths (2019): 2 949 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 32% ♂ 34% ♀ 29%

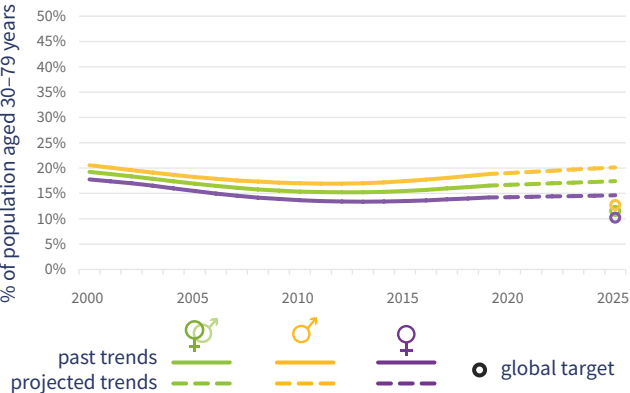
Prevalence of hypertension – global comparison (both sexes)^a



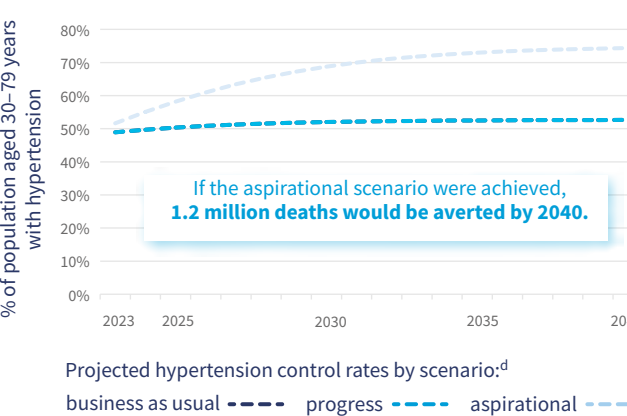
Of the 69.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	14	16	11	2019
Cardiovascular disease deaths	873 000	449 000	425 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	44	43	46	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%)	23	29	18	2019
Obesity, adults aged 18+ years (%)	36	36	37	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	10	15	4	2019
Physical inactivity, adults aged 18+ years (%)	40	32	48	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Uruguay

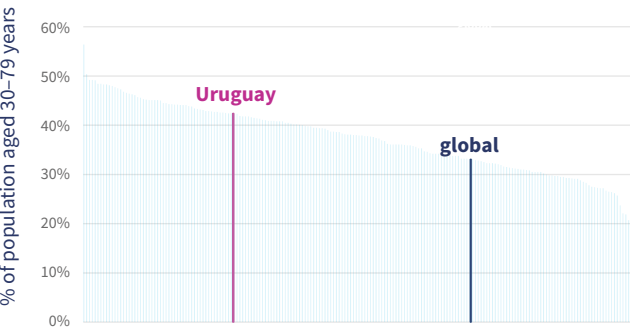
Hypertension profile

Total population (2019): 3 428 000

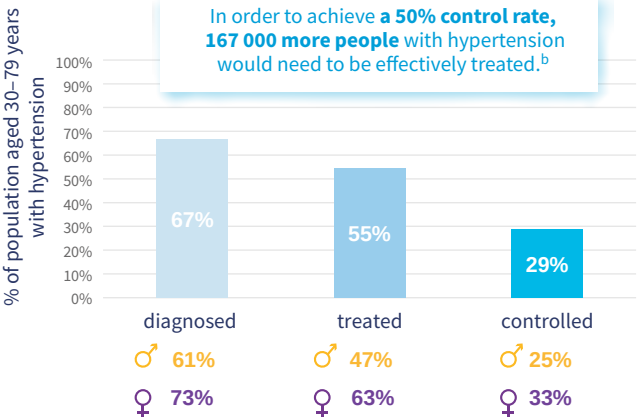
Total deaths (2019): 34 800

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 42% ♂ 46% ♀ 39%

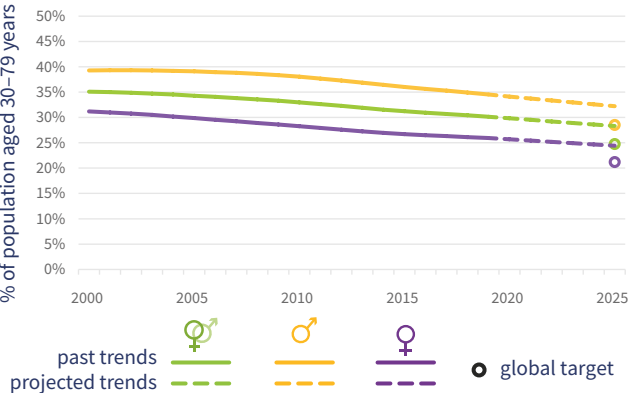
Prevalence of hypertension – global comparison (both sexes)^a



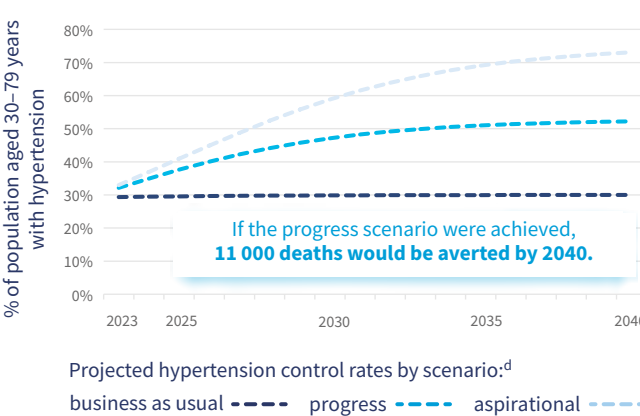
Of the 857 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	16	21	13	2019
Cardiovascular disease deaths	9500	4400	5000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	48	49	48	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	8	2019
Current tobacco use, adults aged 15+ years (%)	22	25	19	2019
Obesity, adults aged 18+ years (%)	28	25	31	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	5	9	2	2019
Physical inactivity, adults aged 18+ years (%)	22	19	26	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Uzbekistan

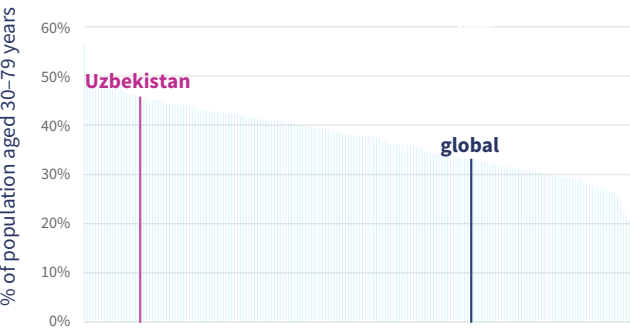
Hypertension profile

Total population (2019): 32 977 000

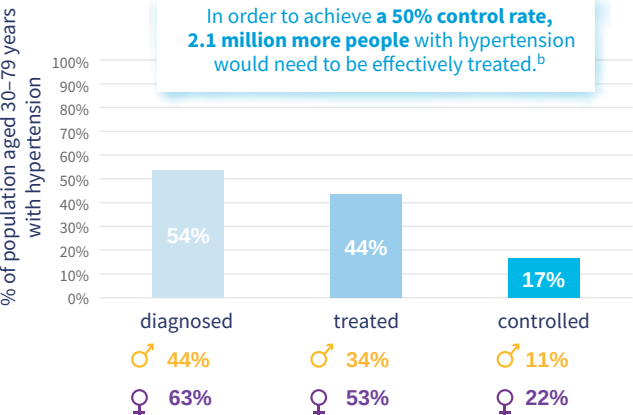
Total deaths (2019): 170 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 46% ♂ 47% ♀ 45%

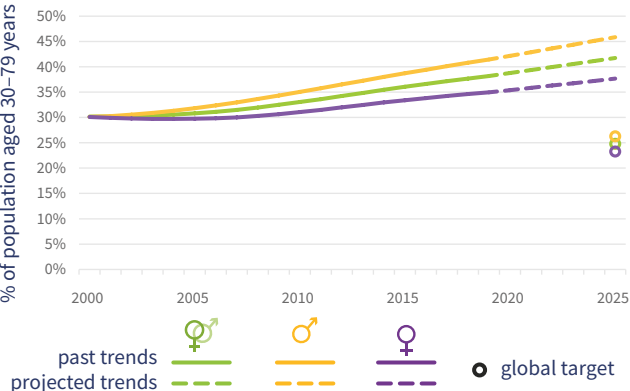
Prevalence of hypertension – global comparison (both sexes)^a



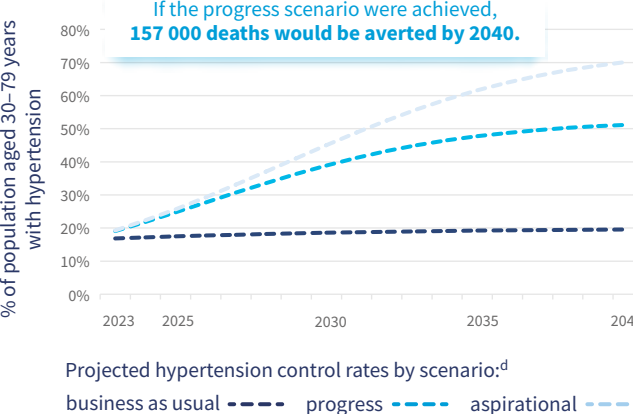
Of the 6.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	30	21	2019
Cardiovascular disease deaths	97 100	51 000	46 200	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	57	58	55	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	9	10	7	2019
Current tobacco use, adults aged 15+ years (%)	18	35	1	2019
Obesity, adults aged 18+ years (%)	17	14	19	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	19	13	24	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Vanuatu

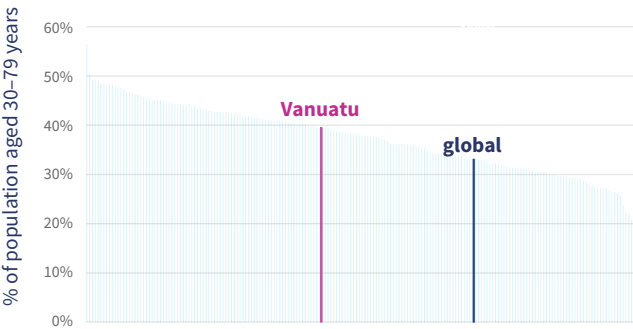
Hypertension profile

Total population (2019): 304 000

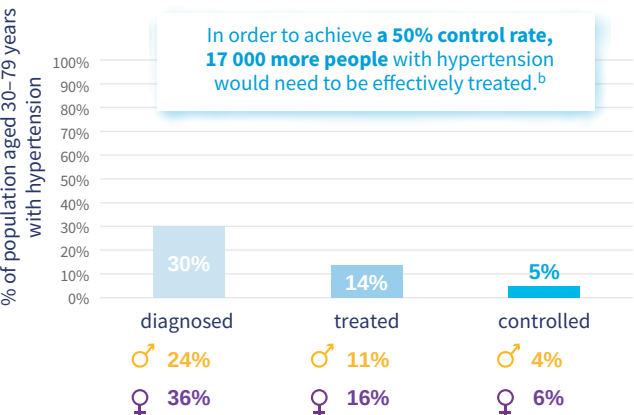
Total deaths (2019): 2130

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 40% ♂ 37% ♀ 42%

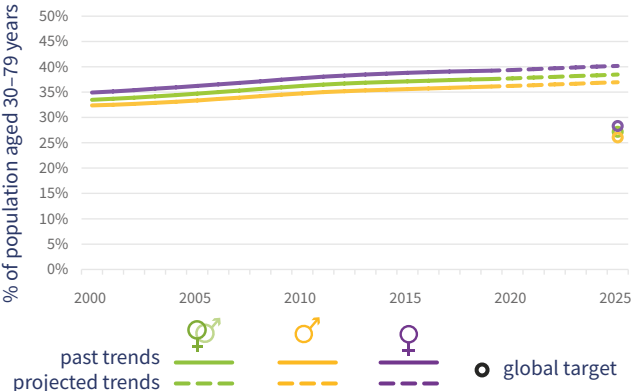
Prevalence of hypertension – global comparison (both sexes)^a



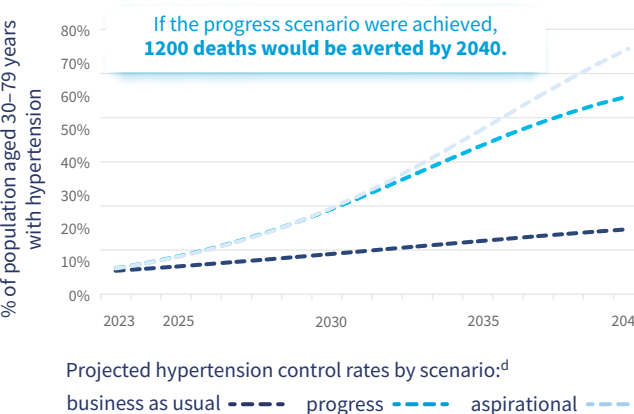
Of the 37 000 adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	40	45	34	2019
Cardiovascular disease deaths	860	530	330	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	63	62	65	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	7	7	2019
Current tobacco use, adults aged 15+ years (%) ^f	18	34	3	2019
Obesity, adults aged 18+ years (%)	25	20	30	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	2	3	1	2019
Physical inactivity, adults aged 18+ years (%)	8	7	9	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake. f. Data refer to tobacco smoking only, in the absence of sufficient data on all tobacco use.

Venezuela (Bolivarian Republic of)

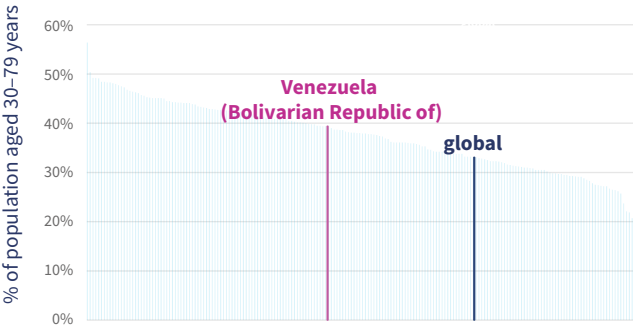
Total population (2019): 28 972 000

Hypertension profile

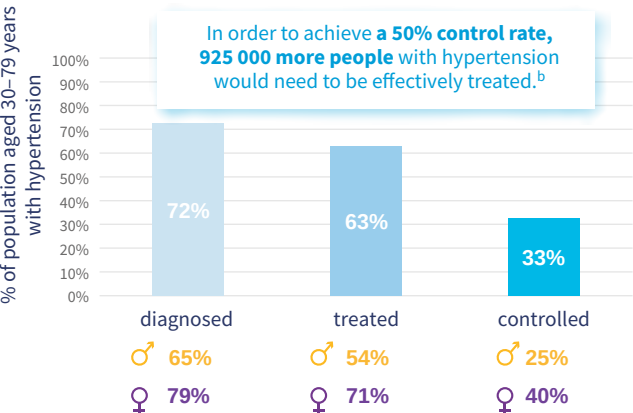
Total deaths (2019): 172 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 39% ♂ 40% ♀ 39%

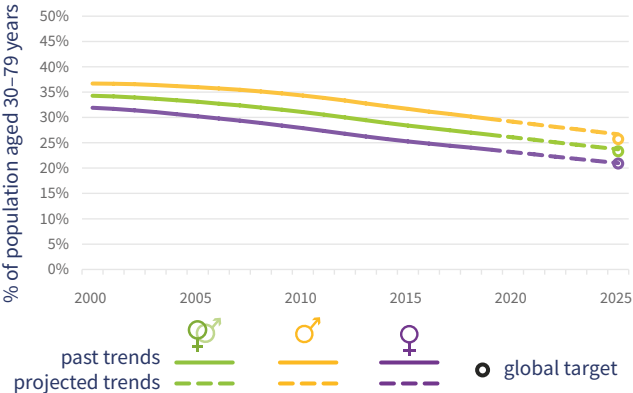
Prevalence of hypertension – global comparison (both sexes)^a



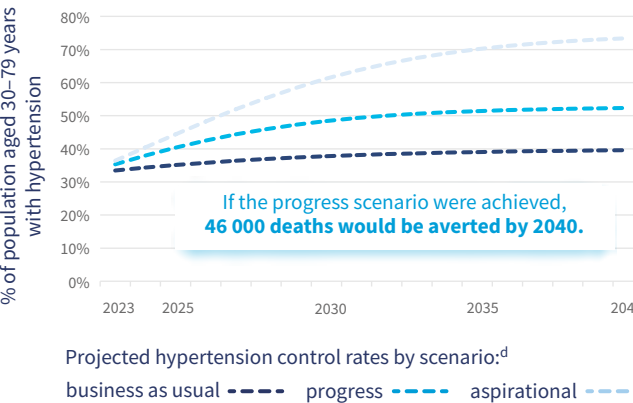
Of the 5.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	15	17	13	2019
Cardiovascular disease deaths	47 000	24 900	22 100	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	62	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	9	2019
Current tobacco use, adults aged 15+ years (%)	no data	no data	no data	2019
Obesity, adults aged 18+ years (%)	26	22	29	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	3	5	1	2019
Physical inactivity, adults aged 18+ years (%)	31	29	33	2016

National response

Targets	Treatment
National target for blood pressure	✗ Guidelines for management of hypertension
National target for salt consumption	✗
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✗
Conducted recent, national survey on salt/sodium intake	✗
Functioning system for generating reliable cause-specific mortality data on a routine basis	✓

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Viet Nam

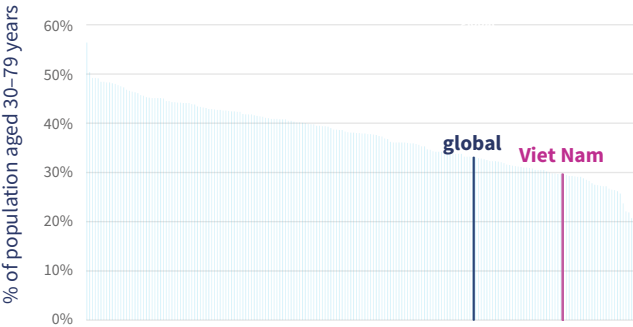
Total population (2019): 95 777 000

Hypertension profile

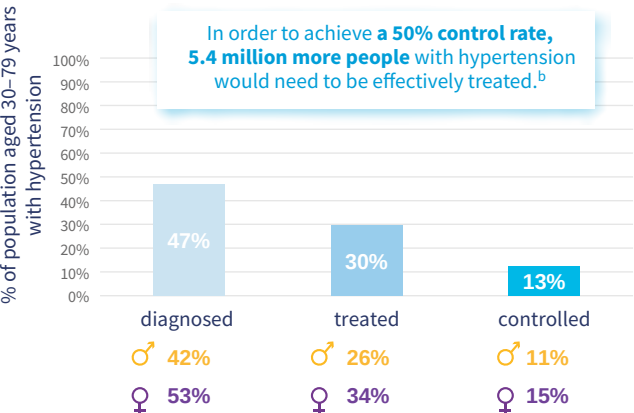
Total deaths (2019): 728 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 30% ♂ 33% ♀ 26%

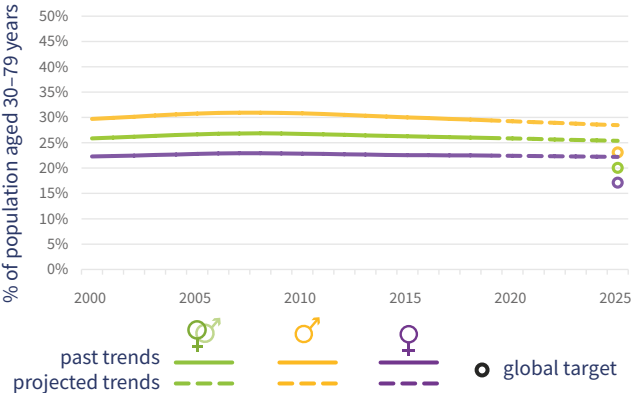
Prevalence of hypertension – global comparison (both sexes)^a



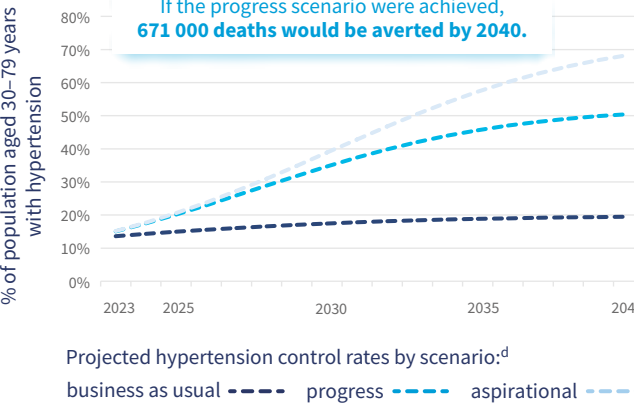
Of the 14.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	21	29	14	2019
Cardiovascular disease deaths	287 000	154 000	133 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	62	57	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	10	11	10	2019
Current tobacco use, adults aged 15+ years (%)	25	48	2	2019
Obesity, adults aged 18+ years (%)	2	2	3	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	9	15	4	2019
Physical inactivity, adults aged 18+ years (%)	25	20	31	2016

National response

Targets	Treatment
National target for blood pressure	✓ Guidelines for management of hypertension
National target for salt consumption	✓
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	✓
Conducted recent, national survey on salt/sodium intake	✓
Functioning system for generating reliable cause-specific mortality data on a routine basis	✗

Footnotes: a. SBP ≥140 mmHg or DBP ≥90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP ≥140 mmHg or DBP ≥90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Yemen

Hypertension profile

Total population (2019): 31 547 000

Total deaths (2019): 170 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a



29%

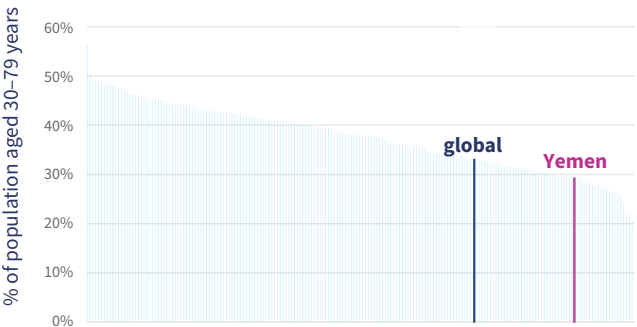


29%

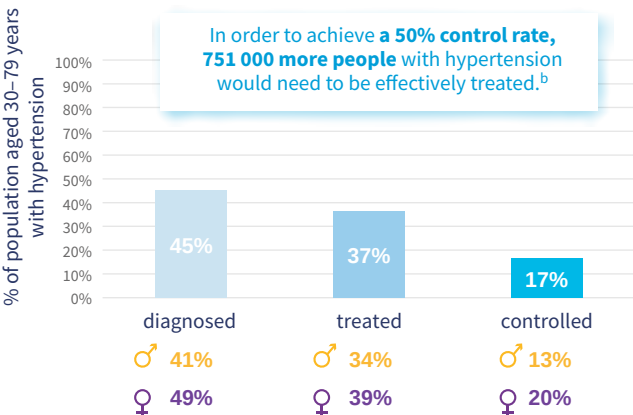


30%

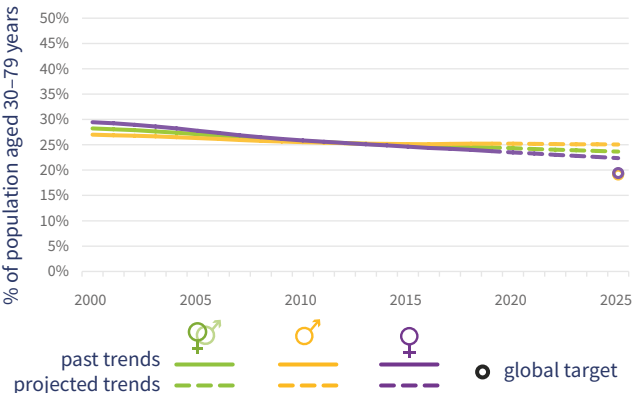
Prevalence of hypertension – global comparison (both sexes)^a



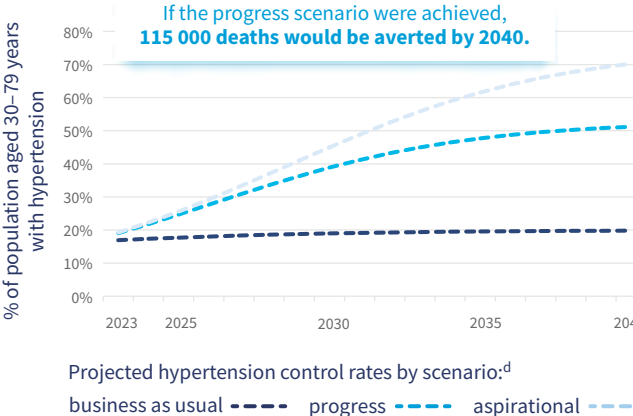
Of the 2.2 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	28	31	25	2019
Cardiovascular disease deaths	52 600	27 100	25 600	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	49	56	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	6	7	5	2019
Current tobacco use, adults aged 15+ years (%)	21	33	8	2019
Obesity, adults aged 18+ years (%)	17	12	22	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	0	0	0	2019
Physical inactivity, adults aged 18+ years (%)	no data	no data	no data	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP $\geq$ 140 mmHg or DBP $\geq$ 90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP $\geq$ 140 mmHg or DBP $\geq$ 90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Zambia

Hypertension profile

Total population (2019): 18 380 000

Total deaths (2019): 121 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a



32%

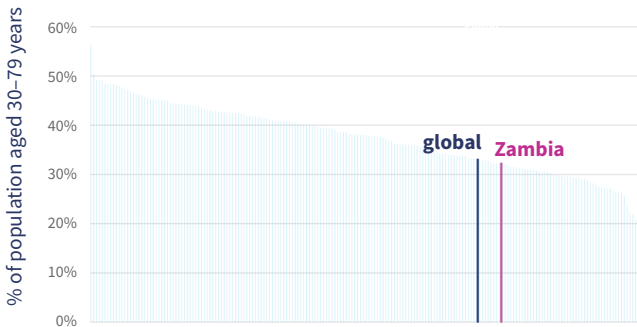


30%

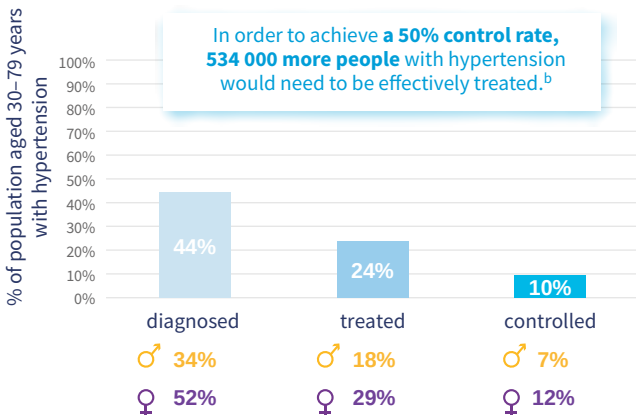


34%

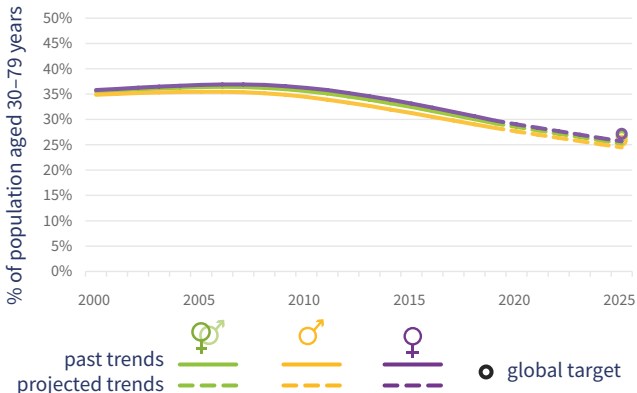
Prevalence of hypertension – global comparison (both sexes)^a



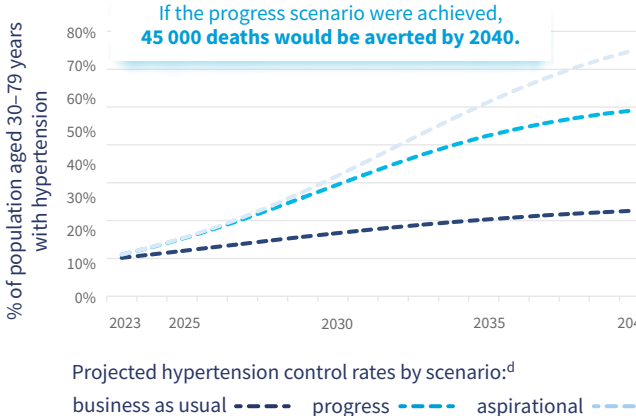
Of the 1.3 million adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	25	29	21	2019
Cardiovascular disease deaths	16 700	8800	7900	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	52	52	52	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	7	6	8	2019
Current tobacco use, adults aged 15+ years (%)	15	25	4	2019
Obesity, adults aged 18+ years (%)	8	4	12	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	6	1	2019
Physical inactivity, adults aged 18+ years (%)	22	19	25	2016

National response

Targets	Treatment
National target for blood pressure	Guidelines for management of hypertension
National target for salt consumption	
Surveillance	
Conducted recent, national survey measuring raised blood pressure/hypertension	
Conducted recent, national survey on salt/sodium intake	
Functioning system for generating reliable cause-specific mortality data on a routine basis	

Footnotes: a. SBP $\geq$ 140 mmHg or DBP $\geq$ 90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP <140 mmHg and DBP <90 mmHg. c. SBP $\geq$ 140 mmHg or DBP $\geq$ 90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Zimbabwe

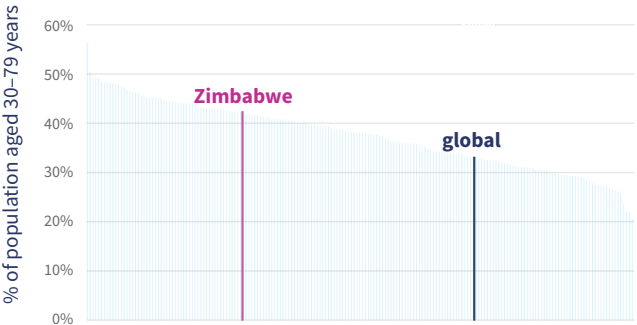
Hypertension profile

Total population (2019): 15 355 000

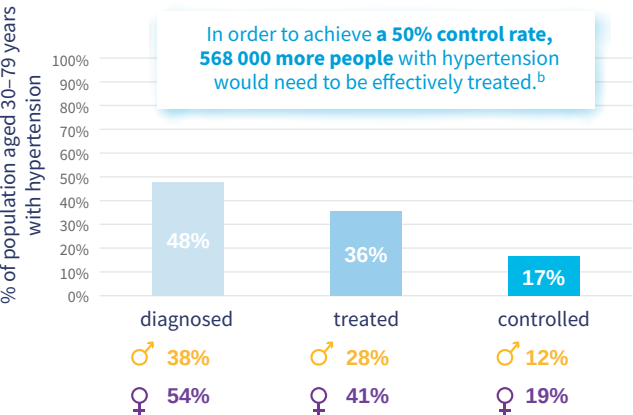
Total deaths (2019): 118 000

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019)^a ♀ 42% ♂ 37% ♀ 46%

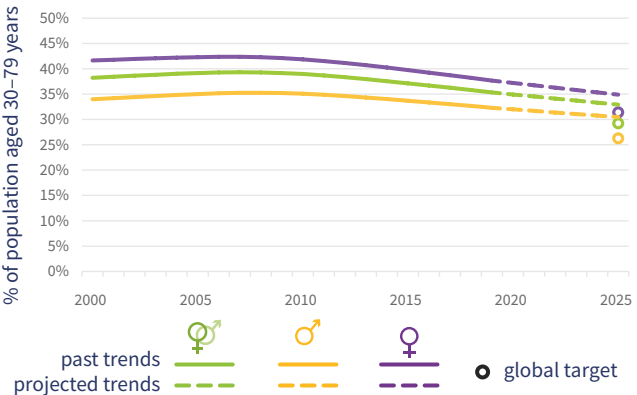
Prevalence of hypertension – global comparison (both sexes)^a



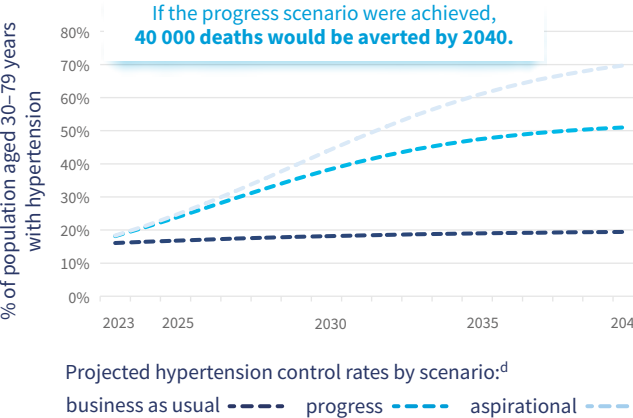
Of the **1.6 million** adults aged 30–79 years with hypertension:



Trends in uncontrolled hypertension in adults aged 30–79 years^c



Hypertension control rate scenarios



Mortality

	both sexes	males	females	year
Probability of premature mortality from NCDs (%)	28	30	27	2019
Cardiovascular disease deaths	17 300	7300	10 000	2019
Cardiovascular disease deaths attributable to high systolic blood pressure (%)	60	55	64	2019

Risk factors^e

	both sexes	males	females	year
Mean population salt intake, adults aged 25+ years (g/day)	8	8	8	2019
Current tobacco use, adults aged 15+ years (%)	12	23	2	2019
Obesity, adults aged 18+ years (%)	16	5	25	2016
Total alcohol per capita consumption, adults aged 15+ years (litres)	4	7	1	2019
Physical inactivity, adults aged 18+ years (%)	27	23	31	2016

National response

Targets

National target for blood pressure
National target for salt consumption

Surveillance

Conducted recent, national survey measuring raised blood pressure/hypertension
Conducted recent, national survey on salt/sodium intake
Functioning system for generating reliable cause-specific mortality data on a routine basis

Treatment

Guidelines for management of hypertension

Footnotes: a. SBP ≥ 140 mmHg or DBP ≥ 90 mmHg or taking medication for hypertension. b. Control rate: adults aged 30–79 years receiving treatment, with blood pressure SBP < 140 mmHg and DBP < 90 mmHg. c. SBP ≥ 140 mmHg or DBP ≥ 90 mmHg. d. Progress and aspirational scenarios reflect a theoretical scaling up of treatment and control. e. Age-standardized estimates are presented for all indicators except salt intake.

Explanatory notes

Profile header

Total population (2019): The total population estimates for the year 2019 were taken from the most recent United Nations Population Division World Population Prospects (1). Data were rounded to two to four significant digits depending on the value.

Total deaths (2019): Mortality estimates for all deaths were taken from the WHO Global Health Estimates 2019 (2). Values were rounded to two to four significant digits depending on the value.

Hypertension prevalence and related graphs and statements

Age-standardized prevalence of hypertension among adults aged 30–79 years (2019): Prevalence of hypertension (defined as having systolic blood pressure (SBP) ≥ 140 mmHg, diastolic blood pressure (DBP) ≥ 90 mmHg, or taking medication for hypertension) among adults aged 30–79. A total of 1201 population-based studies that included measured blood pressure and data on blood pressure treatment in 104 million individuals aged 30–79 years were used to estimate trends in hypertension and hypertension diagnosis, treatment and control from 1990 to 2019. Full details of input and data methods are available in the related publication (3). Age-standardized estimates were produced by applying the age-specific estimates to the WHO Standard Population.

Number of adults aged 30–79 years with hypertension (2019): The number of adults aged 30–79 with hypertension was computed by multiplying the crude prevalence of hypertension by the number of adults aged 30–79 from the 2019 United Nations Population Division World Population Prospects (4).

Coverage of previous diagnosis, treatment and control of hypertension among adults aged 30–79 years with hypertension (2019): Coverage of previous diagnosis, treatment (taking medication) and control (or effective treatment, defined as taking medication for hypertension and having SBP < 140 mmHg and DBP < 90 mmHg) among adults aged 30–79 years with hypertension (as defined above). Full details of input and data methods are available in the related publication (3). Age-standardized estimates were computed by weighting age-specific coverage by the product of the WHO Standard Population weight and the prevalence of hypertension for the age group as defined above.

Additional people with hypertension needing treatment to achieve 50% control rate (2019): For countries who have not yet attained a 50% control rate, these figures were calculated by subtracting the crude estimates of the proportion of people aged 30–79 years with hypertension whose hypertension was controlled from 0.5. This difference was then multiplied by the number of people with hypertension in the country (calculated as described above) to arrive at the number of people who would need to be effectively treated (i.e. have their hypertension controlled) to attain a 50% control rate. Values were rounded to two to four significant digits depending on value.

Trends in uncontrolled hypertension in adults aged 30–79 years (2019): Uncontrolled hypertension, also referred to as raised blood pressure, is defined as having systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg, regardless of diagnosis or treatment status, among adults aged 30–79 years. It was computed by subtracting the prevalence of controlled hypertension (out of all adults aged 30–79) from the prevalence of hypertension. Full details of input and data methods used to estimate the prevalence of hypertension and proportion of hypertension that is controlled are available in the related publication (3). Age-standardized estimates were produced using the WHO Standard Population. Uncontrolled hypertension was projected to 2025 assuming that rates of change 2010–2019 continue to 2025, using methods described elsewhere (5). A 2025 global voluntary target on blood pressure, endorsed by the World Health Assembly in 2013, envisages a 25% reduction in uncontrolled hypertension by 2025 against a 2010 baseline (6). For each country, the global target is depicted as a 25% relative reduction from the country's 2010 baseline prevalence.

Hypertension control rate scenarios (2023–2040): Three hypertension control scenarios from a related publication are shown (7). These were designed to reflect realistic scale-up patterns for hypertension care. The assumptions underlying the scenarios were informed by rates of change in blood pressure treatment and control observed in high-performing countries, as well as recent experiences implementing the HEARTS technical package in several countries. Specifically:

- Business as usual scenario: the proportion of adults with hypertension who were treated and whose hypertension was controlled continues to increase at the rate of change observed in the recent past;
- Progress scenario: increases in hypertension treatment and control accelerate to rates historically observed in high-performing countries, until around 50% effective coverage is reached;
- Aspirational scenario: hypertension treatment and control rates increase more rapidly than observed in best-performing countries, but similar to increases observed in specific settings and for other interventions, until around 75% effective coverage is reached.

For all scenarios, annual increases in hypertension coverage were modeled using a parabolic function, which implies that improvements in hypertension coverage occur faster at intermediate coverage levels as compared to low or high coverage levels. This approach is intended to represent the slowing of progress that countries inevitably experience as they expand health care to harder-to-reach populations.

Deaths averted if progress or aspirational scenario achieved by 2040: Holding all else equal, expected deaths from ischaemic heart disease (IHD), hypertensive heart disease (HHD), ischaemic stroke, and haemorrhagic stroke in each year 2023–2040 under each of the three scenarios above were computed. Methods for computing expected deaths are described in Annex 1. Averted deaths are the expected deaths (summed over the years 2023–2040) under the business as usual scenario less the expected deaths under the progress/aspirational scenario (summed over the years 2023–2040).

Mortality

Probability of premature mortality from NCDs (2019): Percentage of 30-year-old-people who would die before their 70th birthday from any of cardiovascular diseases, cancer, diabetes or chronic respiratory diseases, assuming that s/he would experience current mortality rates at every age and s/he would not die from any other cause of death (e.g. injuries or HIV/AIDS). Probability of death between exact age 30 and exact age 70 was calculated using cause-specific mortality rates in each five-year age group and standard life table methods. The estimates were derived from the WHO Global Health Estimates 2019 (2).

Cardiovascular disease deaths (2019): Mortality estimates for cardiovascular diseases were taken from the WHO Global Health Estimates 2019 (2). Values were rounded to two to four significant digits depending on the value.

Cardiovascular disease deaths attributable to high systolic blood pressure (2019): Percentage of cardiovascular deaths that would have been averted in 2019 if all adults had theoretical minimum risk level of systolic blood pressure, estimated to be in the range 110–115 mmHg. Estimates are from the Global Burden of Disease 2019 study (8), and additional details on methods can be found in a related publication (9).

Risk factors

Mean population salt intake, adults aged 25+ years (g/day): Mean daily population salt intake, in grams/day, among adults aged 25+ years. Estimates for mean population sodium intake were calculated by the Institute for Health Metrics and Evaluation (IHME). More information is available on their website: https://www.healthdata.org/results/gbd_summaries/2019/diet-high-sodium-level-3-risk Estimates were converted to salt intake by multiplying by 2.5.

Current tobacco use, adults aged 15+ (2019): Age-standardized estimates for tobacco use prevalence for the population aged 15 years or older. Estimates were based on a modelling process published elsewhere (10). Estimates were produced only for countries who had at least two nationally representative population-based surveys with prevalence rates broken down by age and sex, completed from 1990–2020 (and if only two, they were at least two years apart), one of which was in the field since 2013. Twenty-nine countries did not meet these criteria and thus do not have any estimates. For 56 countries, the estimate only reflects tobacco smoking as there was insufficient survey data on all tobacco use. These are noted with a footnote on these countries' profiles (11).

Obesity, adults aged 18+ years (2016): Age-standardized prevalence of obesity (defined as having a body mass index ≥ 30 kg/m²) among adults aged 18+ years. Based on measured height and weight. Input data and methods are described in the related publication (12). Age-standardized estimates were produced by applying the crude estimates to the WHO Standard Population.

Total alcohol per capita consumption, adults aged 15+ years (litres) (2019): Total alcohol per capita consumption (APC) is defined as the total (sum of three-year average recorded and three-year average unrecorded APC, adjusted for three-year average tourist consumption) amount of alcohol consumed per adult (15+ years) over a calendar year, in litres of pure alcohol. Recorded alcohol consumption refers to official statistics (production, import, export, and sales or taxation data), while the unrecorded alcohol consumption refers to alcohol which is not taxed and is outside the usual system of governmental control. Tourist consumption accounts for tourists visiting the country and inhabitants visiting other countries. Positive figures denote alcohol consumption of outbound tourists being greater than alcohol consumption by inbound tourists, negative numbers the opposite. Tourist consumption is based on UN tourist statistics. Recorded alcohol per capita (15+) consumption of pure alcohol is calculated as the sum of beverage-specific alcohol consumption of pure alcohol (beer, wine, spirits, other) from different sources: the first priority in the decision tree is given to government statistics; second are country-specific alcohol industry statistics in the public domain based on interviews or field work (GlobalData, formerly Canadean; IWSR-International Wine and Spirit Research; Wine Institute, historically World Drink Trends), or data from the International Organisation of Vine and Wine (OIV); third is the Food and Agriculture Organization of the United Nations' statistical database (FAOSTAT); and fourth is data from alcohol industry statistics in the public domain based on desk review. For countries, where the data source is FAOSTAT the unrecorded consumption may be included in the recorded consumption. As from the introduction of the "Other" beverage-specific category, beer includes malt beers, wine includes wine made from grapes, spirits include all distilled beverages, and other includes one or several other alcoholic beverages, such as fermented beverages made from sorghum, maize, millet, rice, or cider, fruit wine, fortified wine.

Physical inactivity, adults aged 18+ (2016): Percentage of adults aged 18+ years attaining less than 150 minutes of moderate-intensity physical activity per week, or less than 75 minutes of vigorous-intensity physical activity per week, or equivalent. Full details of methods have been previously published (13). The estimates were based on self-reported physical activity captured using the GPAQ (Global Physical Activity Questionnaire), the IPAQ (International Physical Activity Questionnaire) or a similar questionnaire covering activity at work/in the household, for transport, and during leisure time. Where necessary, adjustments were made for the reported definition (in case it was different to the indicator definition), for known over-reporting of activity of the IPAQ, for survey coverage (in case a survey only covered urban areas), and for age coverage (in case the survey age range was narrower than 18+ years). No estimates were produced for countries with no data. Age-standardized estimates were produced by applying the crude estimates to the WHO Standard Population.

National response

National targets for salt, blood pressure (2021): Data were drawn from country responses to the WHO 2021 NCD country capacity survey (NCD CCS) (14). The NCD CCS is a web-based survey of NCD focal points or designated colleagues within the ministry of health or a national institute or agency in all WHO Member States. While all Member States responded to the 2021 NCD CCS, it is possible for countries to leave questions blank or respond "don't know", resulting in "no data" or "don't know" values on the profiles. For the question on national targets, respondents were asked to provide documentation of national targets which are based on the WHO Global monitoring framework for NCDs (15). Countries reported as having national targets for either salt or blood pressure on their profiles have submitted evidence of these targets to WHO.

National guidelines on hypertension management (2021): Data were drawn from country responses to the WHO 2021 NCD country capacity survey (14). Respondents indicated whether evidence-based national guidelines/protocols/ standards are available for the management (diagnosis and treatment) of cardiovascular diseases through a primary care approach recognized/ approved by government or competent authorities and could submit multiple guidelines addressing cardiovascular diseases. Supporting documents submitted by countries for this question were reviewed for evidence of guidelines specifically addressing hypertension.

Conducted recent, national survey measuring raised blood pressure/hypertension, salt/sodium intake (2021): Data were drawn from country responses to the WHO 2021 NCD Country Capacity Survey (14). Countries with a positive value on their profile had to have indicated that they have conducted a recent (i.e. in the past five years since the 2021 NCD country capacity survey), national adult risk factor survey covering the risk factor and provided documentation of

the survey. Data for those countries who have implemented an NCD risk factor survey covering blood pressure or salt under WHO’s technical guidance since 2021 have been updated accordingly.

Functioning system for generating reliable cause-specific mortality data on a routine basis (2023): Countries with a functioning system for generating reliable cause-specific mortality data on a routine basis ensure that the International Form of Medical Certificate of the Cause of Death is completed by certifiers, the International Classification of Diseases (ICD) is used to code the causes of death, and that data compiled are made available to policy-makers and researchers. The WHO collects mortality data, including cause of death, from civil registration systems in the WHO mortality database through a routine annual call for data. Data are considered to generate reliable cause-specific mortality data on a routine basis if: 1. Data from the five most recent reporting years are, on average, at least 70% usable (usability is calculated as: (Completeness (%)) * (1- Proportion Garbage)), 2. At least five years of cause-of-death data have been reported to the WHO in the last 10 years, and 3. The most recent year of data reported to the WHO is no more than five years old.

Annex 2 references

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noncommunicable-diseases](https://www.who.int/teams/noncommunicable-diseases)



Hypertension Overview & Clinical Recommendations

Additional section appended to the HEARTS & HYPERTENSION combined reference. Content is based on the supplied text file and formatted for readability.

1.1 Measuring blood pressure

Training, technique and device maintenance

1.1.1

Ensure that healthcare professionals taking blood pressure measurements have adequate initial training and periodic review of their performance. [2004]

1.1.2

Because automated devices may not measure blood pressure accurately if there is pulse irregularity (for example, due to atrial fibrillation), palpate the radial or brachial pulse before measuring blood pressure. If pulse irregularity is present, measure blood pressure manually using direct auscultation over the brachial artery. [2011]

1.1.3

Healthcare providers must ensure that devices for measuring blood pressure are properly validated, maintained and regularly recalibrated according to manufacturers' instructions. See the British and Irish Hypertension Society's website for a list of validated blood pressure monitoring devices. [2004]

1.1.4

When measuring blood pressure in the clinic or in the home, standardise the environment and provide a relaxed, temperate setting, with the person quiet and seated, and their arm outstretched and supported. Use an appropriate cuff size for the person's arm. [2011, amended 2019]

Postural hypotension

1.1.5

In people with symptoms of postural hypotension, including falls or postural dizziness:

measure blood pressure with the person lying on their back (or consider a seated position, if it is inconvenient to measure blood pressure with the person lying down)

measure blood pressure again after the person has been standing for at least 1 minute. [2004, amended 2023]

1.1.6

If the person's systolic blood pressure falls by 20 mmHg or more, or their diastolic blood pressure falls by 10 mmHg or more, after the person has been standing for at least 1 minute:

consider likely causes, including reviewing their current medication

manage appropriately (for example, for advice on preventing falls in older people, see NICE's guideline on falls in older people: assessing risk and prevention)

measure subsequent blood pressures with the person standing

consider referral to specialist care if symptoms of postural hypotension persist despite addressing likely causes. [2004, amended 2023]

1.1.7

If the blood pressure drop is less than the thresholds in recommendation 1.1.6 despite suggestive symptoms and the baseline measurement was previously taken from a seated position, repeat the measurements this time starting with the person lying on their back. [2023]

1.1.8

Consider referring the person for further specialist assessment if blood pressure measurements do not confirm postural hypotension despite suggestive symptoms. [2023]

1.2 Diagnosing hypertension

1.2.1

When considering a diagnosis of hypertension, measure blood pressure in both arms:

If the difference in readings between arms is more than 15 mmHg, repeat the measurements.

If the difference in readings between arms remains more than 15 mmHg on the second measurement, measure subsequent blood pressures in the arm with the higher reading. [2019]

1.2.2

If blood pressure measured in the clinic is 140/90 mmHg or higher:

Take a second measurement during the consultation.

If the second measurement is substantially different from the first, take a third measurement.

Record the lower of the last 2 measurements as the clinic blood pressure. [2019]

1.2.3

If clinic blood pressure is between 140/90 mmHg and 180/120 mmHg, offer ambulatory blood pressure monitoring (ABPM) to confirm the diagnosis of hypertension. See the section on identifying who to refer for people with a clinic blood pressure 180/120 mmHg or higher. [2019]

1.2.4

If ABPM is unsuitable or the person is unable to tolerate it, offer home blood pressure monitoring (HBPM) to confirm the diagnosis of hypertension. [2019]

1.2.5

While waiting for confirmation of a diagnosis of hypertension, carry out:

investigations for target organ damage (see recommendation 1.3.3), followed by

formal assessment of cardiovascular risk using a cardiovascular risk assessment tool (see the section on full formal risk assessment in NICE's guideline on cardiovascular disease). [2019]

1.2.6

When using ABPM to confirm a diagnosis of hypertension, ensure that at least 2 measurements per hour are taken during the person's usual waking hours (for example, between 08:00 and 22:00). Use the average value of at least 14 measurements taken during the person's usual waking hours to confirm a diagnosis of hypertension. [2011]

1.2.7

When using HBPM to confirm a diagnosis of hypertension, ensure that:

for each blood pressure recording, 2 consecutive measurements are taken, at least 1 minute apart and with the person seated and

blood pressure is recorded twice daily, ideally in the morning and evening and

blood pressure recording continues for at least 4 days, ideally for 7 days.

Discard the measurements taken on the first day and use the average value of all the remaining measurements to confirm a diagnosis of hypertension. [2011]

1.2.8

Confirm diagnosis of hypertension in people with a:

clinic blood pressure of 140/90 mmHg or higher and

ABPM daytime average or HBPM average of 135/85 mmHg or higher. [2019]

1.2.9

If hypertension is not diagnosed but there is evidence of target organ damage, consider carrying out investigations for alternative causes of the target organ damage (for information on investigations, see NICE's guidelines on chronic kidney disease and chronic heart failure). [2011]

1.2.10

If hypertension is not diagnosed, measure the person's clinic blood pressure at least every 5 years subsequently, and consider measuring it more frequently if the person's clinic blood pressure is less than 140/90 mmHg but still raised. [2011]

1.2.11

Offer advice on healthy living in line with the NHS information on healthy living to people who have raised blood pressure but have not been diagnosed with hypertension. [2026]

Annual blood pressure measurement for people with type 2 diabetes

1.2.12

Measure blood pressure at least annually in an adult with type 2 diabetes without previously diagnosed hypertension or renal disease. Offer and reinforce preventive lifestyle advice. [2009]

Specialist investigations for possible secondary causes of hypertension

1.2.13

Consider the need for specialist investigations in people with signs and symptoms suggesting a secondary cause of hypertension. [2004, amended 2011]

For a short explanation of why the committee made the 2019 recommendations and how they might affect practice, see the rationale and impact section on diagnosing hypertension .

Full details of the evidence and the committee's discussion are in evidence review A: diagnosis.

1.3 Assessing cardiovascular risk and target organ damage

For guidance on the early identification and management of chronic kidney disease, see NICE's guideline on chronic kidney disease.

1.3.1

Use a formal estimation of cardiovascular risk to discuss prognosis and healthcare options with people with hypertension, both for raised blood pressure and other modifiable risk factors. [2004]

1.3.2

Estimate cardiovascular risk in line with the recommendations on identifying and assessing cardiovascular disease risk in NICE's guideline on cardiovascular disease. Use clinic blood pressure measurements to calculate cardiovascular risk. [2008]

1.3.3

For all people with hypertension offer to:

test for the presence of protein in the urine by sending a urine sample for estimation of the albumin:creatinine ratio and test for haematuria using a reagent strip

take a blood sample to measure glycated haemoglobin (HbA1C), electrolytes, creatinine, estimated glomerular filtration rate, total cholesterol and HDL cholesterol

examine the fundi for the presence of hypertensive retinopathy

arrange for a 12-lead electrocardiograph to be performed. [2011, amended 2019]

1.4 Treating and monitoring hypertension

Lifestyle interventions

For guidance on the prevention of obesity and cardiovascular disease, see NICE's guidelines on overweight and obesity management and cardiovascular disease prevention.

1.4.1

Offer lifestyle advice to people with suspected or diagnosed hypertension, and continue to offer it periodically. [2004]

1.4.2

Ask about people's diet and exercise patterns because a healthy diet and regular exercise can reduce blood pressure. Offer appropriate guidance and written or audiovisual materials to promote lifestyle changes. [2004]

1.4.3

Ask about people's alcohol consumption and encourage a reduced intake if they drink excessively, because this can reduce blood pressure and has broader health benefits. See the recommendations for practice in NICE's guideline on alcohol-use disorders. [2004, amended 2019]

1.4.4

Discourage excessive consumption of coffee and other caffeine-rich products. [2004]

1.4.5

Encourage people to keep their dietary sodium intake low, either by reducing or substituting sodium salt, as this can reduce blood pressure. Note that salt substitutes containing potassium chloride should not be used by older people, people with diabetes, pregnant women, people with kidney disease and people taking some antihypertensive drugs, such as ACE inhibitors and angiotensin II receptor blockers. Encourage salt reduction in these groups. [2004, amended 2019]

1.4.6

Do not offer calcium, magnesium or potassium supplements as a method for reducing blood pressure. [2004]

1.4.7

Offer advice and help to smokers to stop smoking. See NICE's guideline on tobacco. [2004]

1.4.8

Inform people about local initiatives by, for example, healthcare teams or patient organisations that provide support and promote healthy lifestyle change, especially those that include group work for motivating lifestyle change. [2004]

For a short explanation of why the committee deleted the recommendation on relaxation therapies and how this might affect practice, see the rationale and impact section on relaxation therapies .

Full details of the evidence and the committee's discussion are in evidence review H: relaxation therapies.

Starting antihypertensive drug treatment

NICE has produced a patient decision aid on treatment options for hypertension to help people and their healthcare professionals discuss the different types of treatment and make a decision that is right for each person.

For advice on shared decision making for medicines, see the information on patient decision aids in NICE's guideline on medicines optimisation.

To support adherence and ensure that people with hypertension make the most effective use of their medicines, see NICE's guideline on medicines adherence.

1.4.9

Offer antihypertensive drug treatment in addition to lifestyle advice to adults of any age with persistent stage 2 hypertension. Use clinical judgement for people of any age with frailty or multimorbidity (see also NICE's guideline on multimorbidity). [2019]

1.4.10

Discuss starting antihypertensive drug treatment, in addition to lifestyle advice, with adults aged under 80 with persistent stage 1 hypertension who have 1 or more of the following:

target organ damage

established cardiovascular disease

renal disease

diabetes

an estimated 10-year risk of cardiovascular disease of 10% or more.

Use clinical judgement for people with frailty or multimorbidity (see also NICE's guideline on multimorbidity). [2019]

1.4.11

Discuss with the person their individual cardiovascular disease risk and their preferences for treatment, including no treatment, and explain the risks and benefits before starting antihypertensive drug treatment. Continue to offer lifestyle advice and support them to make lifestyle changes (see the section on lifestyle interventions), whether or not they choose to start antihypertensive drug treatment. [2019]

1.4.12

Consider antihypertensive drug treatment in addition to lifestyle advice for adults aged under 60 with stage 1 hypertension and an estimated 10-year risk below 10%. Bear in mind that 10-year cardiovascular risk may underestimate the lifetime probability of developing cardiovascular disease. [2019]

1.4.13

Consider antihypertensive drug treatment in addition to lifestyle advice for people aged over 80 with stage 1 hypertension if their clinic blood pressure is over 150/90 mmHg. Use clinical judgement for people with frailty or multimorbidity (see also NICE's guideline on multimorbidity). [2019]

1.4.14

For adults aged under 40 with hypertension, consider seeking specialist evaluation of secondary causes of hypertension and a more detailed assessment of the long-term balance of treatment benefit and risks. [2019]

For a short explanation of why the committee made the 2019 recommendations and how they might affect practice, see the rationale and impact section on starting antihypertensive drug treatment .

Full details of the evidence and the committee's discussion are in evidence review C: initiating treatment.

Monitoring treatment and blood pressure targets

For specific recommendations on blood pressure control in people with other conditions or who are pregnant, see NICE's guidelines on chronic kidney disease, type 1 diabetes and hypertension in pregnancy.

See also table 1 for clinic blood pressure targets for people aged under 80 and table 2 for clinic blood pressure targets for people aged 80 and over. The tables cover people with hypertension (with or without type 2 diabetes) as well as people with chronic kidney disease or type 1 diabetes.

Table 1: Clinic blood pressure targets for people aged under 80

Person under 80 with: Clinic blood pressure target Source

hypertension (with or without type 2 diabetes) or

type 1 diabetes plus albumin to creatinine ratio less than 70 mg/mmol or

chronic kidney disease plus albumin to creatinine ratio less than 70 mg/mmol

Below 140/90

Recommendation 1.4.20

NICE's guideline on type 1 diabetes in adults (recommendation 1.13.8)

NICE's guideline on chronic kidney disease (recommendation 1.6.1)

type 1 diabetes plus albumin to creatinine ratio of 70 mg/mmol or more or
chronic kidney disease plus albumin to creatinine ratio of 70 mg/mmol or more

Below 130/80

NICE's guideline on type 1 diabetes in adults (recommendation 1.13.8)

NICE's guideline on chronic kidney disease (recommendation 1.6.2)

Table 2: Clinic blood pressure targets for people aged 80 and over

Person aged 80 and over with: Clinic blood pressure target Source

hypertension (with or without type 2 diabetes) or
type 1 diabetes (regardless of albumin to creatinine ratio)

Below 150/90

Recommendation 1.4.21

NICE's guideline on type 1 diabetes in adults (recommendation 1.13.8)

chronic kidney disease plus albumin to creatinine ratio less than 70 mg/mmol

Below 140/90

NICE's guideline on chronic kidney disease (recommendation 1.6.1)

chronic kidney disease plus albumin to creatinine ratio of 70 mg/mmol or more

Below 130/80

NICE's guideline on chronic kidney disease (recommendation 1.6.2)

1.4.15

Use clinic blood pressure measurements to monitor the response to lifestyle changes or drug treatment in people with hypertension. [2019]

1.4.16

Check for postural hypotension (see recommendation 1.1.5) in people with hypertension and:
type 2 diabetes or
symptoms of postural hypotension (see also recommendation 1.1.7) or
aged 80 and over.

In people with a significant postural drop or symptoms of postural hypotension, treat to a blood pressure target based on standing blood pressure. [2019, amended 2023]

1.4.17

Advise people with hypertension who choose to self-monitor their blood pressure to use HBPM. (NHS England is supporting the use of HBPM through the blood pressure@home scheme.) [2019]

1.4.18

Consider ABPM or HBPM, in addition to clinic blood pressure measurements, for people with hypertension identified as having a white-coat effect or masked hypertension (in which clinic and non-clinic blood pressure results are conflicting). Be aware that the corresponding measurements for ABPM and HBPM are 5 mmHg lower than for clinic measurements (see recommendation 1.2.8 for diagnostic thresholds). [2019]

1.4.19

For people who choose to use HBPM, provide:

training and advice on using home blood pressure monitors

information about what to do if they are not achieving their target blood pressure.

Be aware that the corresponding measurements for HBPM are 5 mmHg lower than for clinic measurements (see recommendation 1.2.8 for diagnostic thresholds). [2019]

1.4.20

For adults with hypertension aged under 80, reduce clinic blood pressure to below 140/90 mmHg and ensure that it is maintained below that level. See also table 1 for guidance on clinic blood pressure targets for people aged under 80 with type 1 diabetes or severe chronic kidney disease. [2019, amended 2022]

1.4.21

For adults with hypertension aged 80 and over, reduce clinic blood pressure to below 150/90 mmHg and ensure that it is maintained below that level. Use clinical judgement for people with frailty or multimorbidity (see NICE's guideline on multimorbidity). See also table 2 for guidance on clinic blood pressure targets for people aged 80 and over with type 1 diabetes or severe chronic kidney disease. [2019, amended 2022]

1.4.22

When using ABPM or HBPM to monitor the response to treatment in adults with hypertension, use the average blood pressure level taken during the person's usual waking hours (see recommendations 1.2.6 and 1.2.7). Reduce blood pressure and ensure that it is maintained:

below 135/85 mmHg for adults aged under 80

below 145/85 mmHg for adults aged 80 and over.

Use clinical judgement for people with frailty or multimorbidity (see also NICE's guideline on multimorbidity). [2019, amended 2022]

For a short explanation of why the committee made the 2019 recommendations and how they might affect practice, see the rationale and impact section on monitoring treatment and blood pressure targets for people without cardiovascular disease .

Full details of the evidence and the committee's discussion are in evidence review B: monitoring the response to treatment.

1.4.23

Use the same blood pressure targets for people with and without cardiovascular disease. [2022]

For a short explanation of why the committee made the recommendation on blood pressure targets for people with cardiovascular disease and how this might affect practice, see the rationale and impact section on monitoring treatment and blood pressure targets for people with cardiovascular disease .

Full details of the evidence and the committee's discussion are in the evidence review J: blood pressure targets.

1.4.24

Provide an annual review of care for adults with hypertension to monitor blood pressure, provide people with support, and discuss their lifestyle, symptoms and medication. [2004]

Treatment review when type 2 diabetes is diagnosed

1.4.25

For an adult with type 2 diabetes on antihypertensive drug treatment when diabetes is diagnosed, review blood pressure control and medications used. Make changes only if there is poor control or if current drug treatment is not appropriate because of microvascular complications or metabolic problems. [2009]

Choosing antihypertensive drug treatment (for people with or without type 2 diabetes)

The recommendations in this section apply to people with hypertension with or without type 2 diabetes. They replace the recommendations on diagnosing and managing hypertension in NICE's guideline on type 2 diabetes in adults. For guidance on choosing antihypertensive drug treatment in people with type 1 diabetes, see also the section on control of cardiovascular risk in NICE's guideline on type 1 diabetes.

Note that ACE inhibitors and angiotensin II receptor antagonists should not be used in pregnant or breastfeeding women or women planning pregnancy unless absolutely necessary, in which case the potential risks and benefits should be discussed. Follow the MHRA safety advice on ACE inhibitors and angiotensin II receptor antagonists: not for use in pregnancy, recommendations on how to use for breastfeeding and the related clarification on breastfeeding.

1.4.26

For guidance on choice of antihypertensive medicine in people with chronic kidney disease, see NICE's guideline on chronic kidney disease. If possible, offer treatment with drugs taken only once a day. [2004]

1.4.27

Prescribe non-proprietary drugs if these are appropriate and minimise cost. [2004]

1.4.28

Offer people with isolated systolic hypertension (systolic blood pressure 160 mmHg or more) the same treatment as people with both raised systolic and diastolic blood pressure. [2004]

1.4.29

Offer antihypertensive drug treatment to women of childbearing potential with diagnosed hypertension in line with the recommendations in this guideline. For women considering pregnancy or who are pregnant or breastfeeding, manage hypertension in line with the recommendations on management of pregnancy with chronic hypertension, and on antihypertensive treatment while breastfeeding in NICE's guideline on hypertension in pregnancy. [2010, amended 2019]

1.4.30

When choosing antihypertensive drug treatment for adults of Black African or African-Caribbean family origin, consider an angiotensin II receptor blocker (ARB), in preference to an angiotensin-converting enzyme (ACE) inhibitor. [2019]

Follow the MHRA safety advice on ACE inhibitors and angiotensin II receptor antagonists: not for use in pregnancy, how to use for breastfeeding and clarification on breastfeeding.

1.4.31

For people with cardiovascular disease:

Follow the recommendations for disease-specific indications in the NICE guideline on their condition (for example, when prescribing an ACE inhibitor or an ARB for secondary prevention of myocardial infarction). Relevant recommendations include:

drug therapy for secondary prevention in NICE's guideline on acute coronary syndromes

treatment after stabilisation in NICE's guideline on acute heart failure

treating heart failure with reduced ejection fraction in NICE's guideline on chronic heart failure

drugs for secondary prevention of cardiovascular disease in NICE's guideline on stable angina

blood pressure management in NICE's guideline on type 1 diabetes in adults.

If their blood pressure remains uncontrolled, offer antihypertensive drug treatment in line with the recommendations in this section. [2022]

For a short explanation of why the committee made the recommendation on choosing antihypertensive drug treatment for people with cardiovascular disease and how this might affect practice, see the rationale and impact section on choosing antihypertensive drug treatment for people with cardiovascular disease .

Full details of the evidence and the committee's discussion are in the evidence review K: pharmacological treatment in cardiovascular disease.

Step 1 treatment

1.4.32

Offer an ACE inhibitor or an ARB to adults starting step 1 antihypertensive treatment who:

have type 2 diabetes and are of any age or family origin (see also recommendation 1.4.30 for adults of Black African or African–Caribbean family origin) or

are aged under 55 but not of Black African or African–Caribbean family origin. [2019]

Follow the MHRA safety advice on ACE inhibitors and angiotensin II receptor antagonists: not for use in pregnancy, how to use for breastfeeding and clarification on breastfeeding.

1.4.33

If an ACE inhibitor is not tolerated, for example because of cough, offer an ARB to treat hypertension. [2019]

Follow the MHRA safety advice on ACE inhibitors and angiotensin II receptor antagonists: not for use in pregnancy, how to use for breastfeeding and clarification on breastfeeding.

1.4.34

Do not combine an ACE inhibitor with an ARB to treat hypertension. [2019]

1.4.35

Offer a calcium-channel blocker (CCB) to adults starting step 1 antihypertensive treatment who:

are aged 55 or over and do not have type 2 diabetes or

are of Black African or African–Caribbean family origin and do not have type 2 diabetes (of any age). [2019]

1.4.36

If a CCB is not tolerated, for example because of oedema, offer a thiazide-like diuretic to treat hypertension. [2019]

1.4.37

If there is evidence of heart failure, offer a thiazide-like diuretic and follow NICE's guideline on chronic heart failure. [2019]

1.4.38

If starting or changing diuretic treatment for hypertension, offer a thiazide-like diuretic, such as indapamide in preference to a conventional thiazide diuretic such as bendroflumethiazide or hydrochlorothiazide. [2019]

1.4.39

For adults with hypertension already having treatment with bendroflumethiazide or hydrochlorothiazide, who have stable, well-controlled blood pressure, continue with their current treatment. [2019]

For a short explanation of why the committee made the 2019 recommendations and how they might affect practice, see the rationale and impact section on step 1 treatment .

Full details of the evidence and the committee's discussion are in evidence review E: step 1 treatment.

Step 2 treatment

1.4.40

Before considering next step treatment for hypertension discuss with the person if they are taking their medicine as prescribed and support adherence in line with NICE's guideline on medicines adherence. [2019]

1.4.41

If hypertension is not controlled in adults taking step 1 treatment of an ACE inhibitor or ARB, offer the choice of 1 of the following drugs in addition to step 1 treatment:

a CCB or

a thiazide-like diuretic. [2019]

1.4.42

If hypertension is not controlled in adults taking step 1 treatment of a CCB, offer the choice of 1 of the following drugs in addition to step 1 treatment:

an ACE inhibitor or

an ARB or

a thiazide-like diuretic. [2019]

1.4.43

If hypertension is not controlled in adults of Black African or African-Caribbean family origin who do not have type 2 diabetes taking step 1 treatment, consider an ARB, in preference to an ACE inhibitor, in addition to step 1 treatment. [2019]

Step 3 treatment

1.4.44

Before considering next step treatment for hypertension:

review the person's medications to ensure they are being taken at the optimal tolerated doses and discuss adherence (see recommendation 1.4.40). [2019]

1.4.45

If hypertension is not controlled in adults taking step 2 treatment, offer a combination of:

an ACE inhibitor or ARB (see also recommendation 1.4.30 for people of Black African or African-Caribbean family origin) and

a CCB and

a thiazide-like diuretic. [2019]

For a short explanation of why the committee made the 2019 recommendations and how they might affect practice, see the rationale and impact section on step 2 and 3 treatment .

Full details of the evidence and the committee's discussion are in evidence review F: step 2 and step 3 treatment.

Step 4 treatment

1.4.46

If hypertension is not controlled in adults taking the optimal tolerated doses of an ACE inhibitor or an ARB plus a CCB and a thiazide-like diuretic, regard them as having resistant hypertension. [2019]

1.4.47

Before considering further treatment for a person with resistant hypertension:

Confirm elevated clinic blood pressure measurements using ambulatory or home blood pressure recordings.

Assess for postural hypotension.

Discuss adherence (see recommendation 1.4.40). [2019]

1.4.48

For people with confirmed resistant hypertension, consider adding a fourth antihypertensive drug as step 4 treatment or seeking specialist advice. [2019]

Follow the MHRA safety advice on ACE inhibitors and angiotensin II receptor antagonists: not for use in pregnancy, how to use for breastfeeding and clarification on breastfeeding.

1.4.49

Consider further diuretic therapy with low-dose spironolactone for adults with resistant hypertension starting step 4 treatment who have a blood potassium level of 4.5 mmol/l or less. Use particular caution in people with a reduced estimated glomerular filtration rate because they have an increased risk of hyperkalaemia. [2019]

In March 2019, this was an off-label use of some preparations of spironolactone. See NICE's information on prescribing medicines.

1.4.50

When using further diuretic therapy for step 4 treatment of resistant hypertension, monitor blood sodium and potassium and renal function within 1 month of starting treatment and repeat as needed thereafter. [2019]

1.4.51

Consider an alpha-blocker or beta-blocker for adults with resistant hypertension starting step 4 treatment who have a blood potassium level of more than 4.5 mmol/l. [2019]

1.4.52

If blood pressure remains uncontrolled in people with resistant hypertension taking the optimal tolerated doses of 4 drugs, seek specialist advice. [2019]

For a short explanation of why the committee made the 2019 recommendations and how they might affect practice, see the rationale and impact section on step 4 treatment .

Full details of the evidence and the committee's discussion are in evidence review G: step 4 treatment.

1.5 Identifying who to refer for same-day specialist review

1.5.1

If a person has severe hypertension (clinic blood pressure of 180/120 mmHg or higher), but no symptoms or signs indicating same-day referral (see recommendation 1.5.2), carry out investigations for target organ damage (see recommendation 1.3.3) as soon as possible:

If target organ damage is identified, consider starting antihypertensive drug treatment immediately, without waiting for the results of ABPM or HBPM.

If no target organ damage is identified, confirm diagnosis by:

repeating clinic blood pressure measurement within 7 days, or

considering monitoring using ABPM (or HBPM if ABPM is not suitable or not tolerated), following recommendations 1.2.6 and 1.2.7, and ensuring a clinical review within 7 days. [2019]

1.5.2

Refer people for specialist assessment, carried out on the same day, if they have a clinic blood pressure of 180/120 mmHg and higher with:

signs of retinal haemorrhage or papilloedema (accelerated hypertension) or

life-threatening symptoms such as new onset confusion, chest pain, signs of heart failure, or acute kidney injury. [2019]

1.5.3

Refer people for specialist assessment, carried out on the same day, if they have suspected pheochromocytoma (for example, labile or postural hypotension, headache, palpitations, pallor, abdominal pain or diaphoresis). [2019]

For a short explanation of why the committee made the 2019 recommendations and how they might affect practice, see the rationale and impact section on identifying who to refer for same-day specialist review .

Full details of the evidence and the committee's discussion are in evidence review I: same-day specialist review.

Terms used in this guideline

This section defines terms that have been used in a particular way for this guideline. For other definitions see the NICE glossary.

Accelerated hypertension

A severe increase in blood pressure to 180/120 mmHg or higher (and often over 220/120 mmHg) with signs of retinal haemorrhage and/or papilloedema (swelling of the optic nerve). It is usually associated with new or progressive target organ damage and is also known as malignant hypertension.

Established cardiovascular disease

Medical history of ischaemic heart disease, cerebrovascular disease, peripheral vascular disease, aortic aneurysm or heart failure. Cardiovascular disease is a general term for conditions affecting the heart or blood vessels. It is usually associated with a build-up of fatty deposits inside the arteries (atherosclerosis) and an increased risk of blood clots. It can also be associated with damage to arteries in organs such as the brain, heart, kidneys and eyes through deposition of glassy material within the artery walls (arteriosclerosis). Cardiovascular disease is 1 of the main causes of death and disability in the UK, but it can often largely be prevented by leading a healthy lifestyle.

Masked hypertension

Clinic blood pressure measurements are normal (less than 140/90 mmHg), but blood pressure measurements are higher when taken outside the clinic using average daytime ambulatory blood pressure monitoring (ABPM) or average home blood pressure monitoring (HBPM) blood pressure measurements.

Persistent hypertension

High blood pressure at repeated clinical encounters.

Stage 1 hypertension

Clinic blood pressure ranging from 140/90 mmHg to 159/99 mmHg and subsequent ABPM daytime average or HBPM average blood pressure ranging from 135/85 mmHg to 149/94 mmHg.

Stage 2 hypertension

Clinic blood pressure of 160/100 mmHg or higher but less than 180/120 mmHg and subsequent ABPM daytime average or HBPM average blood pressure of 150/95 mmHg or higher.

Stage 3 or severe hypertension

Clinic systolic blood pressure of 180 mmHg or higher or clinic diastolic blood pressure of 120 mmHg or higher.

Target organ damage

Damage to organs such as the heart, brain, kidneys and eyes. Examples are left ventricular hypertrophy, chronic kidney disease, hypertensive retinopathy or increased urine albumin:creatinine ratio.

White-coat effect

A discrepancy of more than 20/10 mmHg between clinic and average daytime ABPM or average HBPM blood pressure measurements at the time of diagnosis.